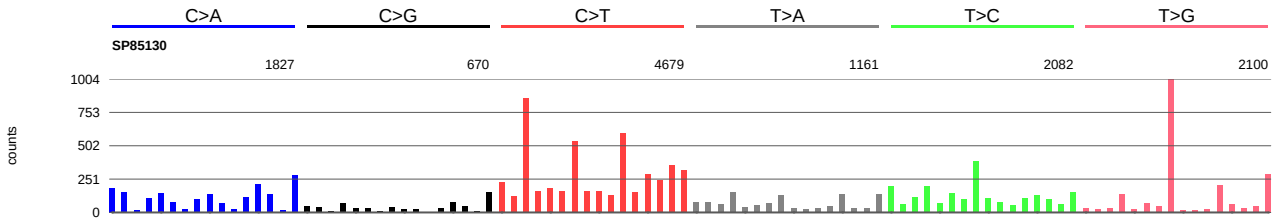
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SP85130
1827
670
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2082
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counts
1004
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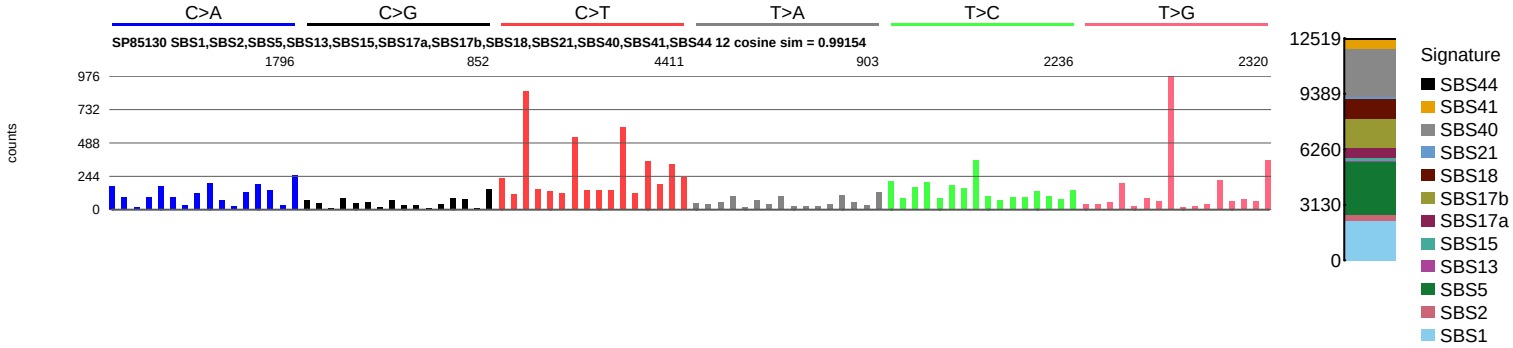
502
251
0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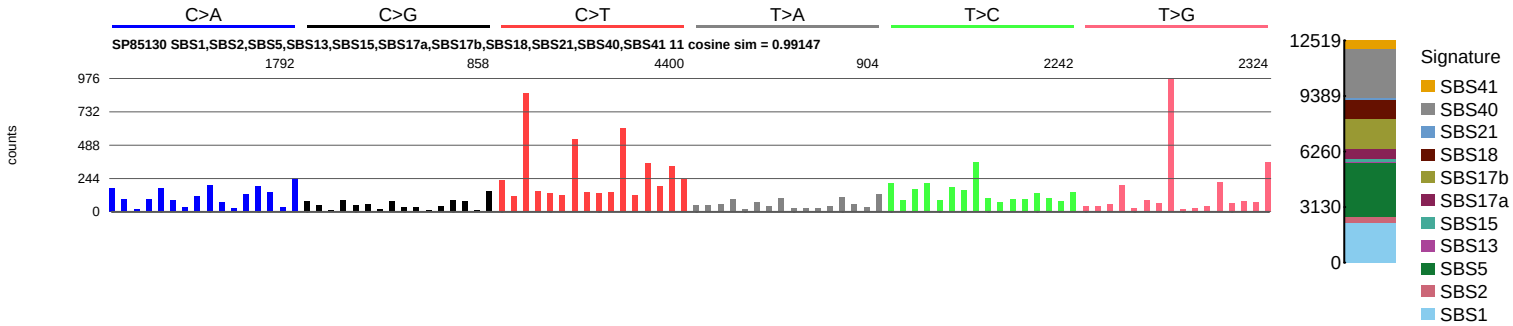
T>G
SP85130 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 12 cosine sim = 0.99154
1796
852
4411
903
2236
2320
counts
976
732
488
244
0
12519
9389
6260
3130
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1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SP85130 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 11 cosine sim = 0.99147
1792
858
4400
904
2242
2324
counts
976
732
488
244
0
12519
9389
6260
3130
0
Signature
SBS41
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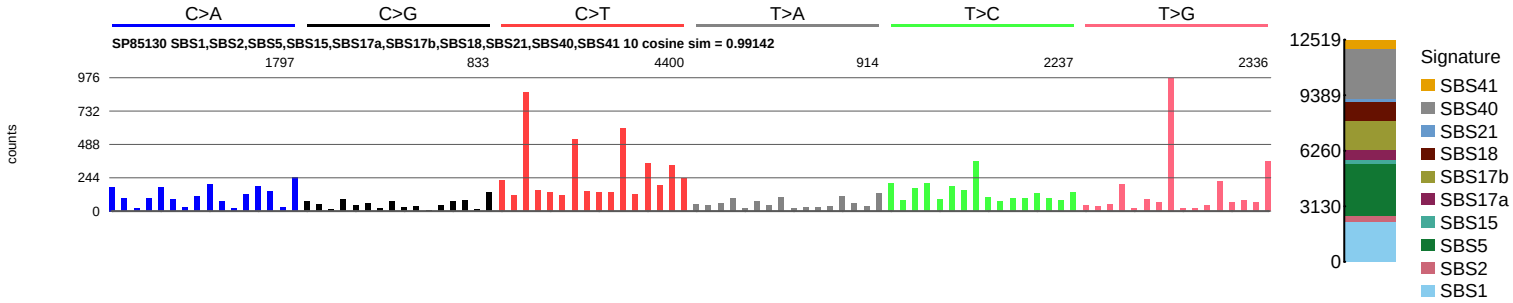
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1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SP85130 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 10 cosine sim = 0.99142
1797
833
4400
914
2237
2336
counts
976
732
488
244
0
12519
9389
6260
3130
0
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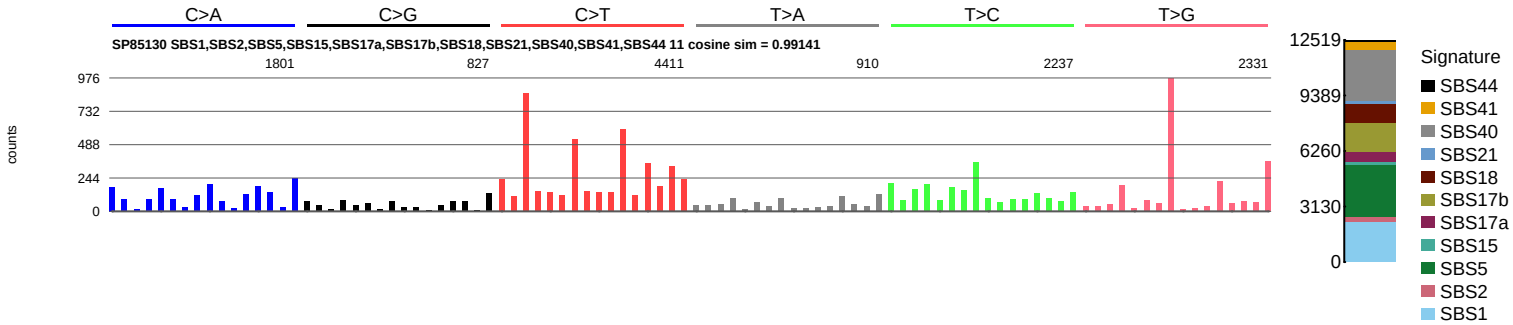
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1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C>G
C>T
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T>C
T>G
SP85130 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 11 cosine sim = 0.99141
1801
827
4411
910
2237
2331
counts
976
732
488
244
0
12519
9389
6260
3130
0
Signature
SBS44
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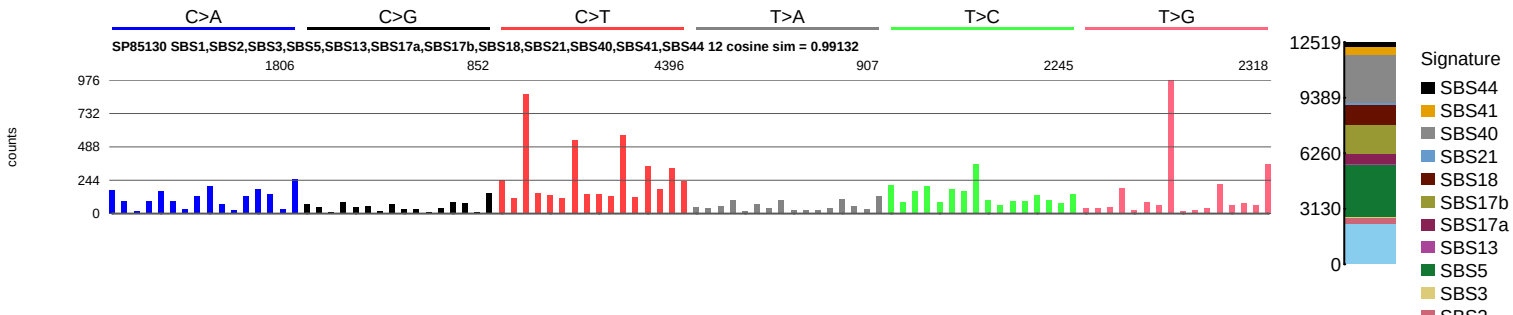
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1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C>G
C>T
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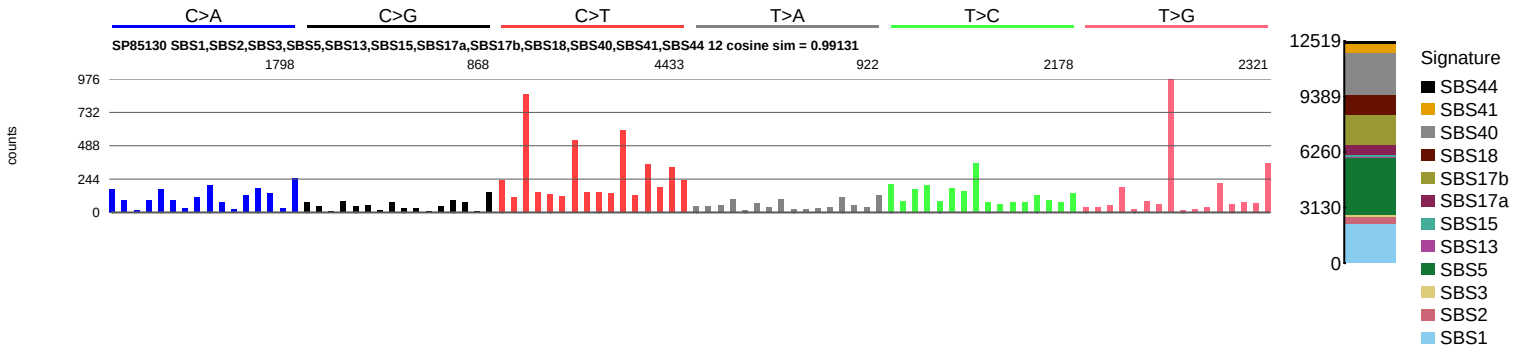
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 12 cosine sim = 0.99132
1806
852
4396
907
2245
2318
counts
976
732
488
244
0
12519
9389
6260
3130
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3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C>G
C>T
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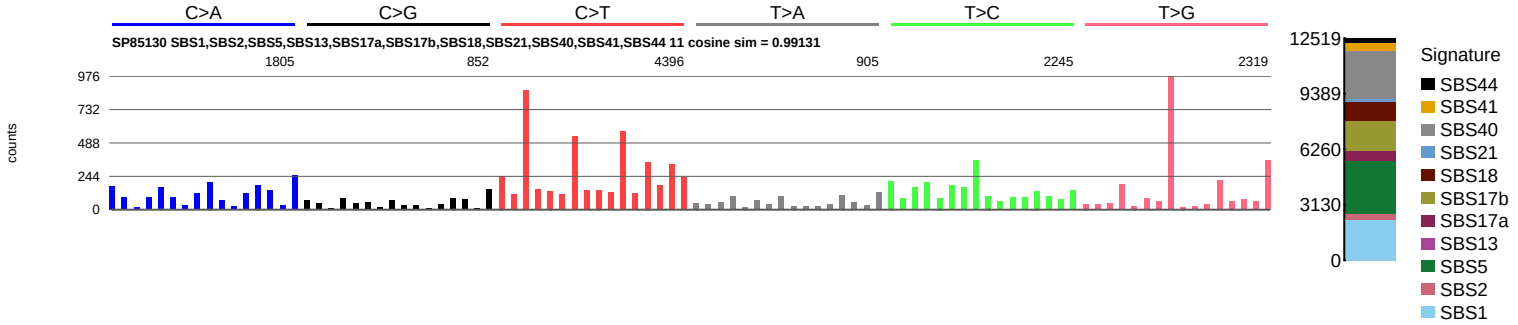
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 12 cosine sim = 0.99131
1798
868
4433
922
2178
2321
counts
976
732
488
244
0
12519
9389
6260
3130
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1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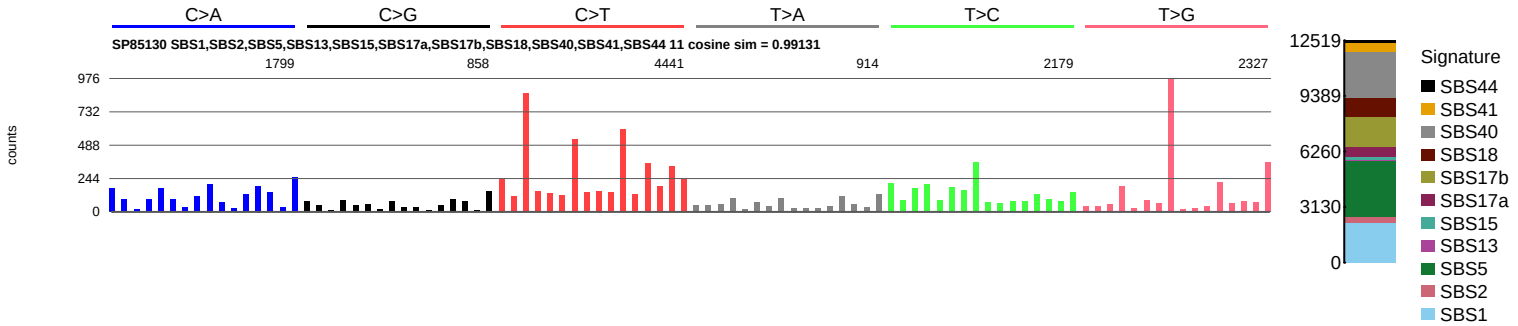
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 11 cosine sim = 0.99131
1805
852
4396
905
2245
2319
counts
976
732
488
244
0
12519
9389
6260
3130
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1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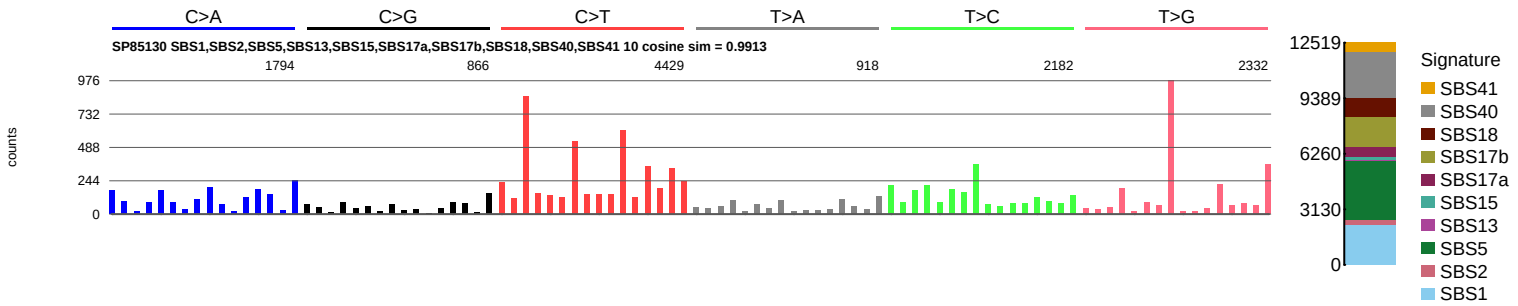
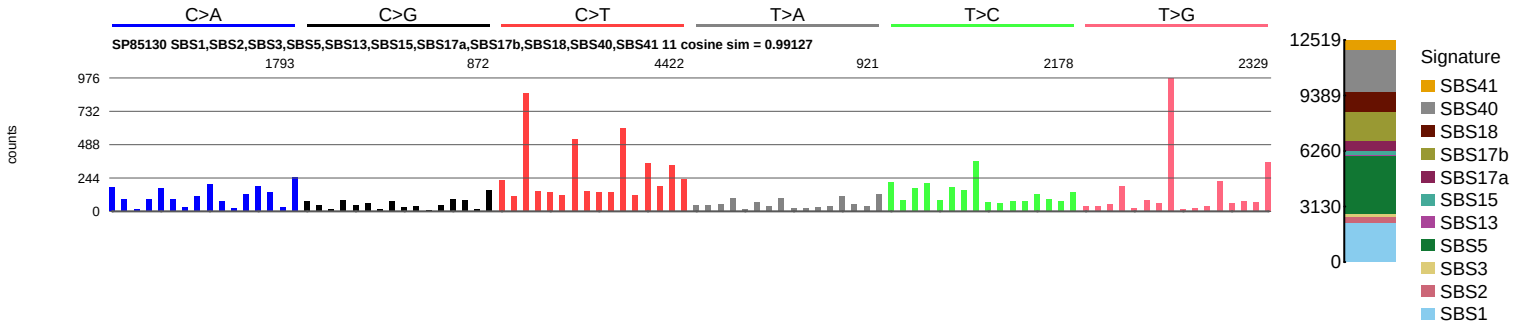
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 11 cosine sim = 0.99131
1799
858
4441
914
2179
2327
counts
976
732
488
244
0
12519
9389
6260
3130
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41 10 cosine sim = 0.9913
1794
866
4429
918
2182
2332
counts
976
732
488
244
0
12519
9389
6260
3130
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1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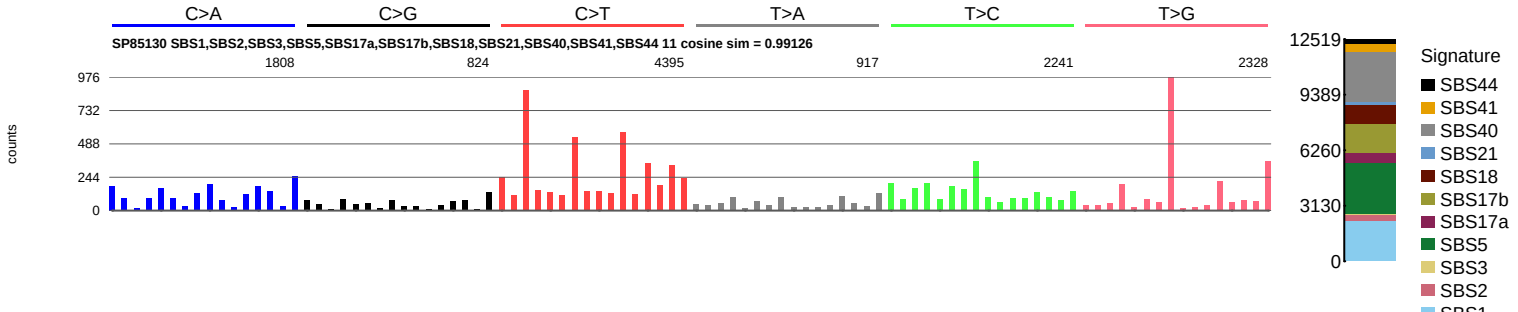
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41 11 cosine sim = 0.99127
1793
872
4422
921
2178
2329
counts
976
732
488
244
0
12519
9389
6260
3130
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1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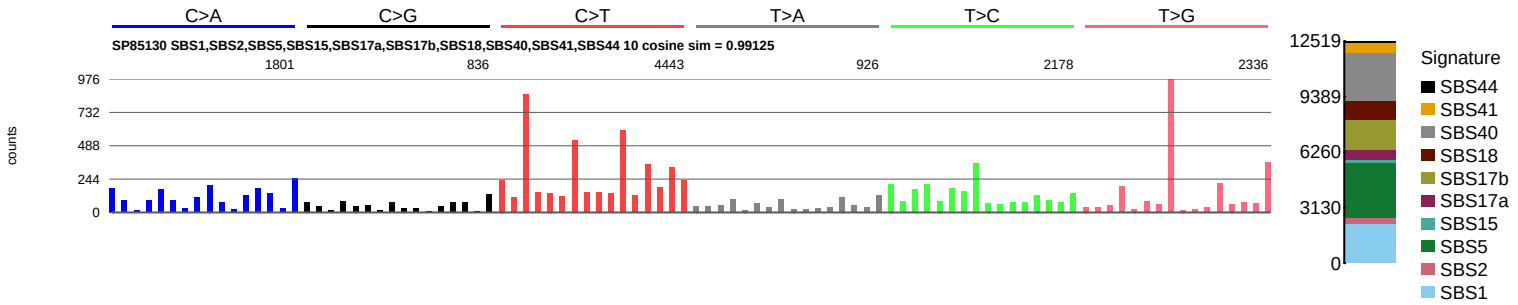
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 11 cosine sim = 0.99126
1808
824
4395
917
2241
2328
counts
976
732
488
244
0
12519
9389
6260
3130
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2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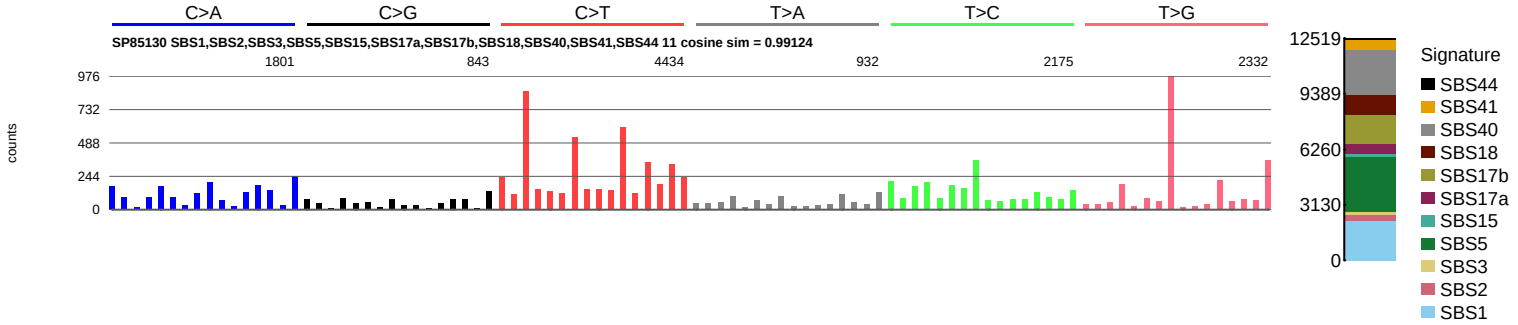
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 10 cosine sim = 0.99125
1801
836
4443
926
2178
2336
counts
976
732
488
244
0
12519
9389
6260
3130
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1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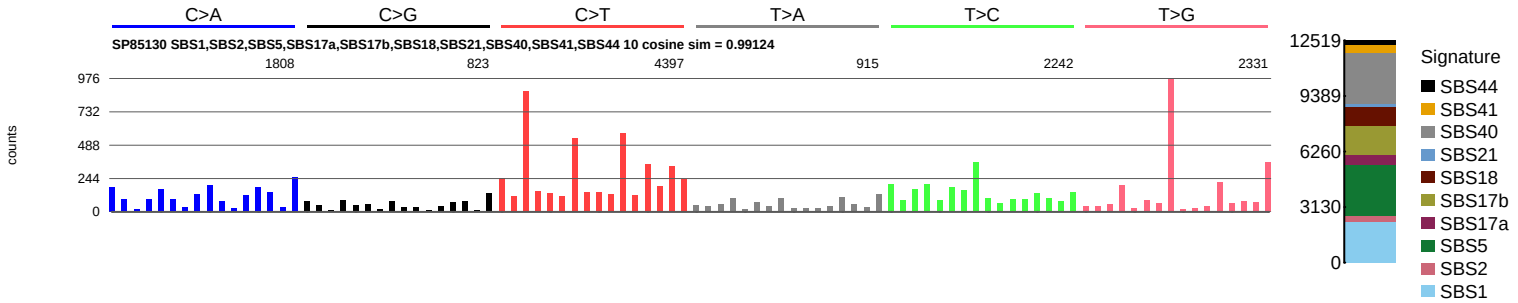
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 11 cosine sim = 0.99124
1801
843
4434
932
2175
2332
counts
976
732
488
244
0
12519
9389
6260
3130
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1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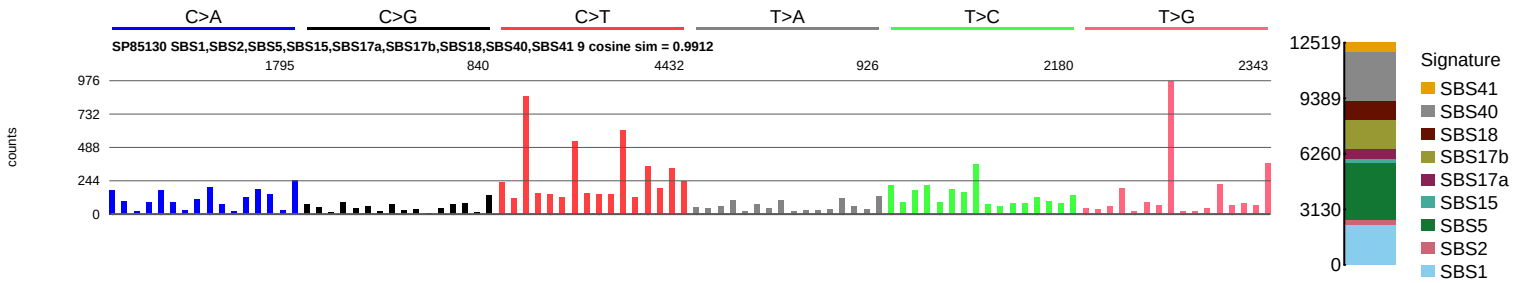
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 10 cosine sim = 0.99124
1808
823
4397
915
2242
2331
counts
976
732
488
244
0
12519
9389
6260
3130
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1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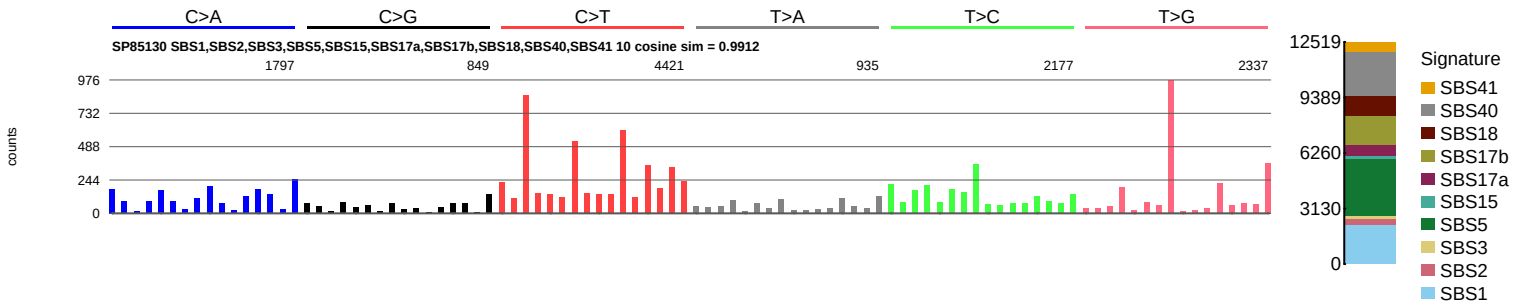
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41 9 cosine sim = 0.9912
1795
840
4432
926
2180
2343
counts
976
732
488
244
0
12519
9389
6260
3130
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1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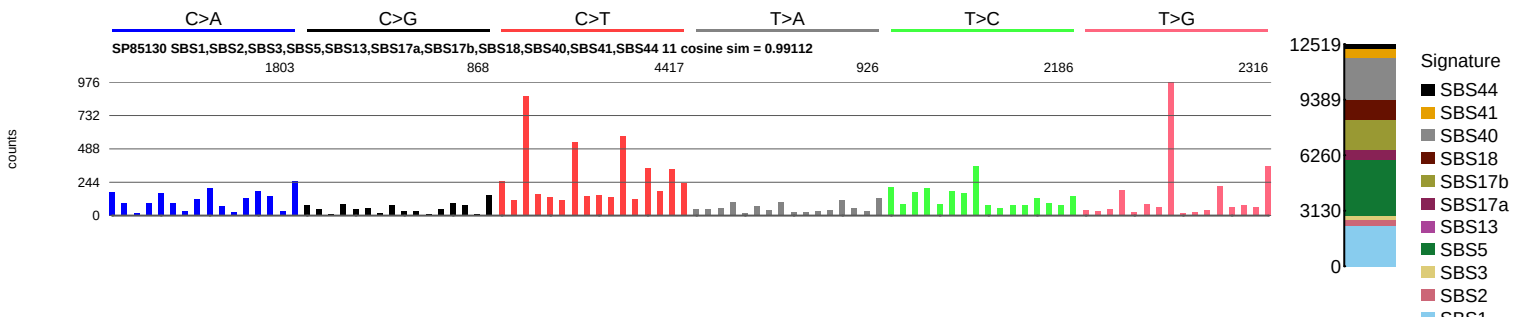
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41 10 cosine sim = 0.9912
1797
849
4421
935
2177
2337
counts
976
732
488
244
0
12519
9389
6260
3130
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1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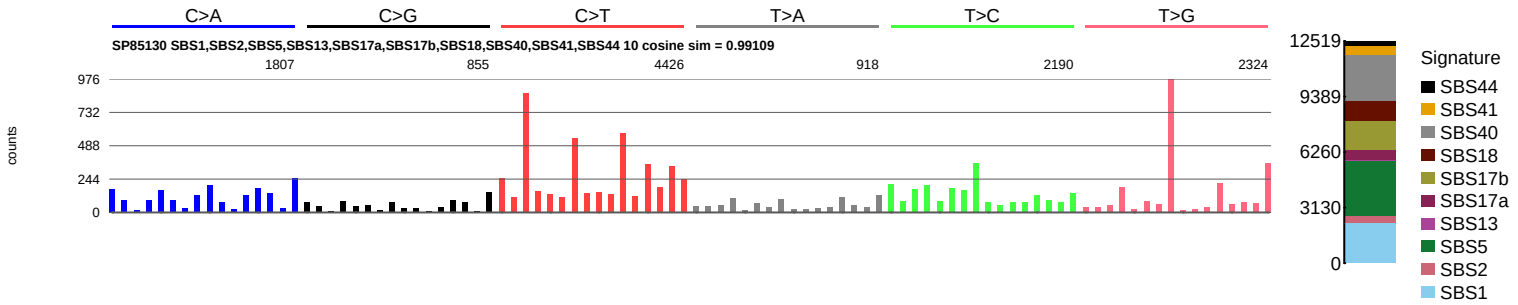
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 11 cosine sim = 0.99112
1803
868
4417
926
2186
2316
counts
976
732
488
244
0
12519
9389
6260
3130
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2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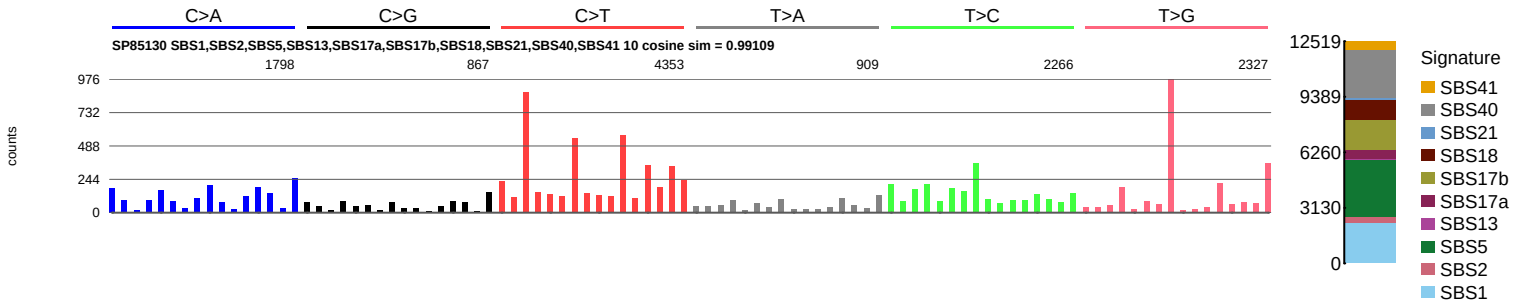
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 10 cosine sim = 0.99109
1807
855
4426
918
2190
2324
counts
976
732
488
244
0
12519
9389
6260
3130
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 10 cosine sim = 0.99109
1798
867
4353
909
2266
2327
counts
976
732
488
244
0
12519
9389
6260
3130
0
Signature
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SBS40
SBS21
SBS18
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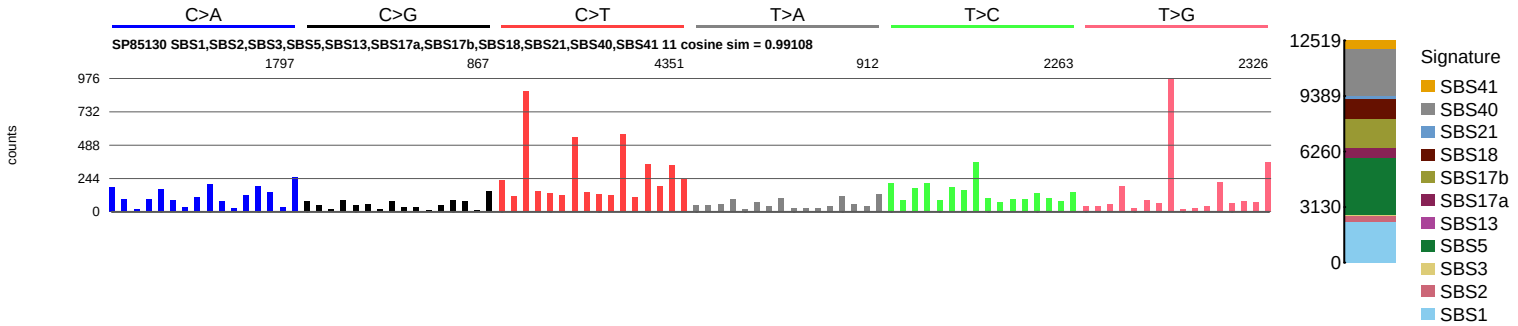
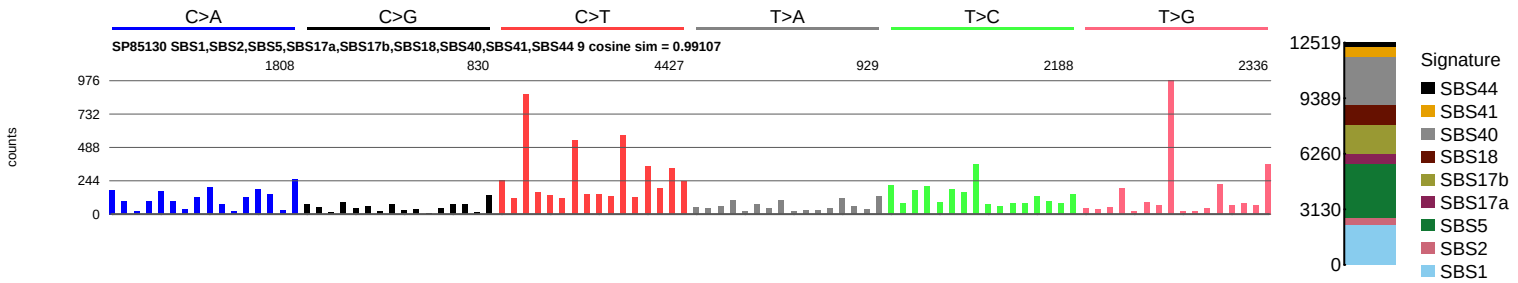
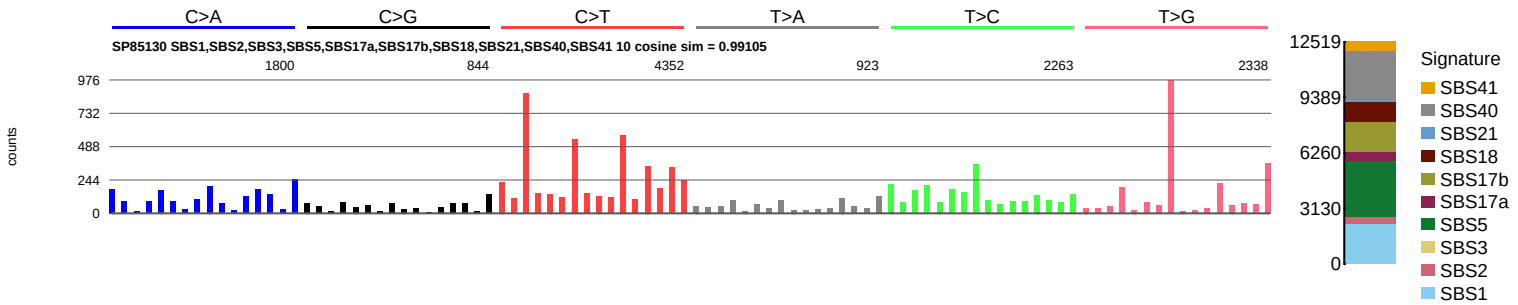
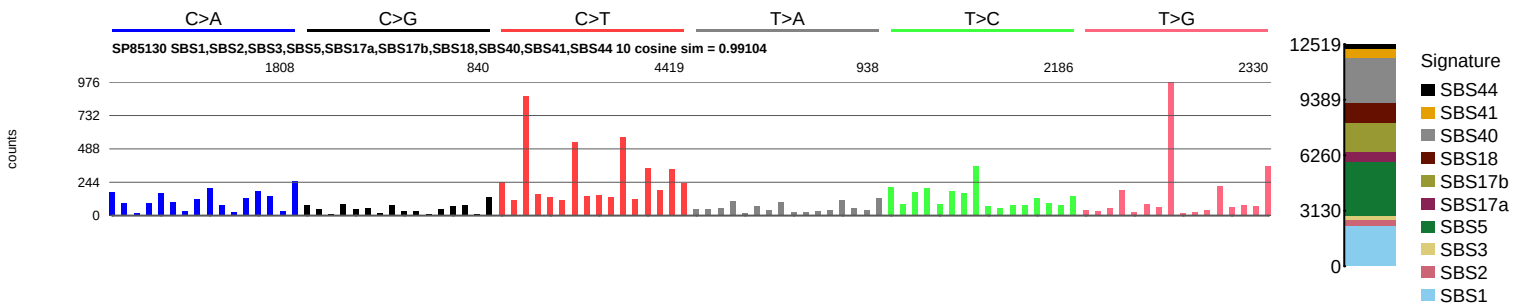
SBS17a
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 11 cosine sim = 0.99108
1797
867
4351
912
2263
2326
counts
976
732
488
244
0
12519
9389
6260
3130
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 9 cosine sim = 0.99107
1808
830
4427
929
2188
2336
counts
976
732
488
244
0
12519
9389
6260
3130
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 10 cosine sim = 0.99105
1800
844
4352
923
2263
2338
counts
976
732
488
244
0
12519
9389
6260
3130
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 10 cosine sim = 0.99104
1808
840
4419
938
2186
2330
counts
976
732
488
244
0
12519
9389
6260
3130
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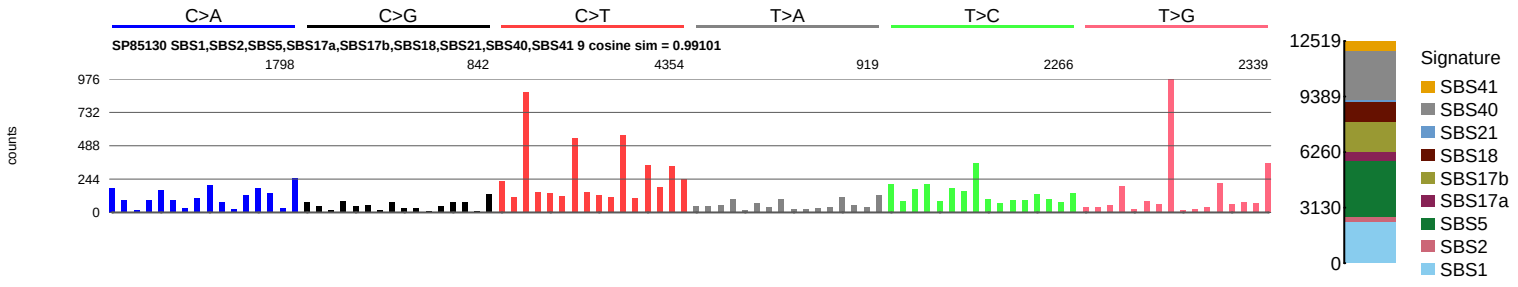
SBS1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SP85130 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 9 cosine sim = 0.99101
1798
842
4354
919
2266
2339
counts
976
732
488
244
0
12519
9389
6260
3130
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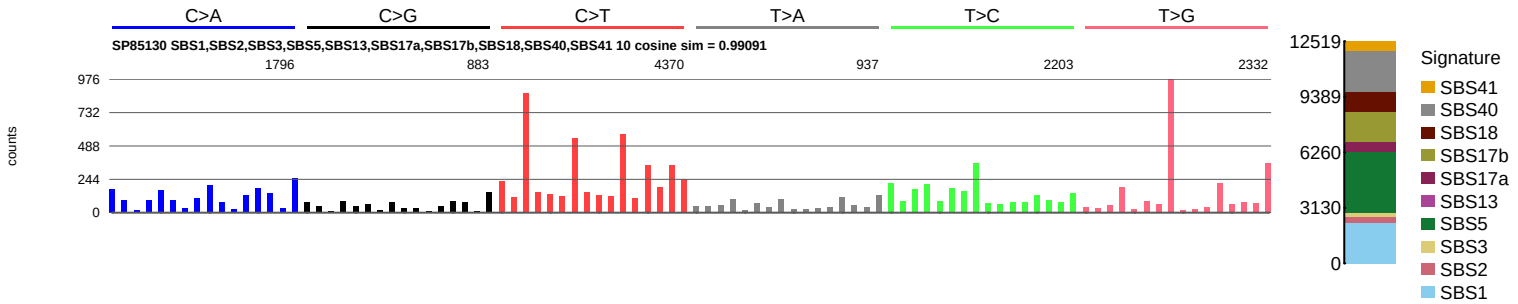
SBS5
SBS2
SBS1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SP85130 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS40,SBS41 10 cosine sim = 0.99091
1796
883
4370
937
2203
2332
counts
976
732
488
244
0
12519
9389
6260
3130
0
Signature
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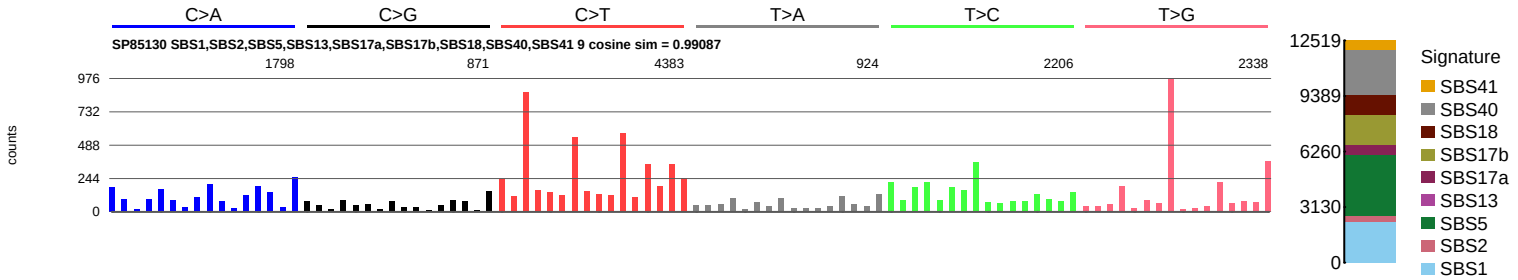
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1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C>T
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T>C
T>G
SP85130 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS40,SBS41 9 cosine sim = 0.99087
1798
871
4383
924
2206
2338
counts
976
732
488
244
0
12519
9389
6260
3130
0
Signature
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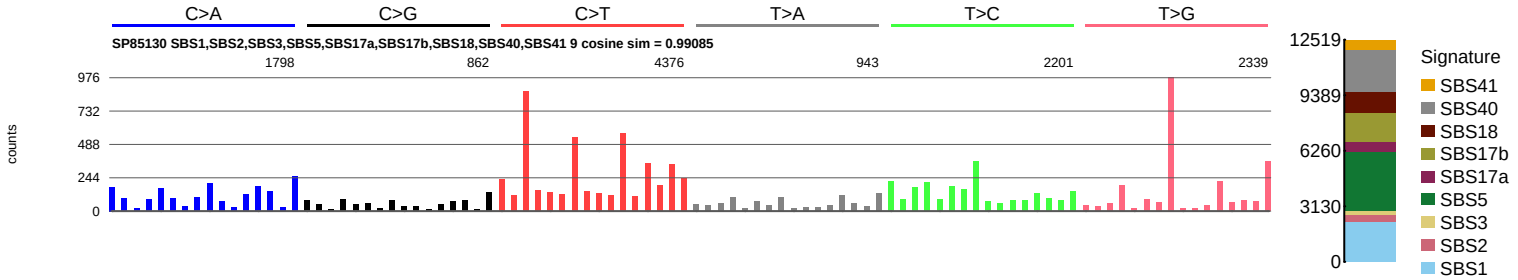
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1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T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS40,SBS41 9 cosine sim = 0.99085
1798
862
4376
943
2201
2339
counts
976
732
488
244
0
12519
9389
6260
3130
0
Signature
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SBS40
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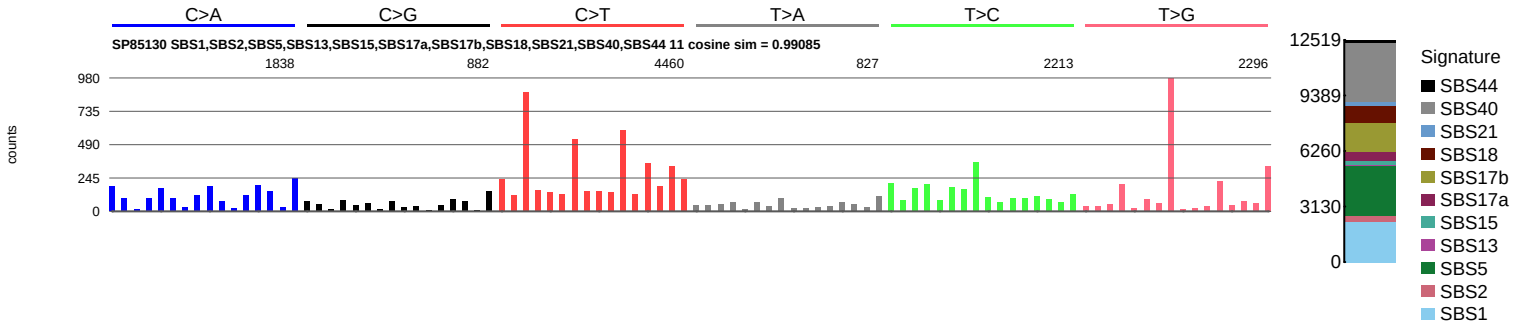
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1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T>C
T>G
SP85130 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 11 cosine sim = 0.99085
1838
882
4460
827
2213
2296
counts
980
735
490
245
0
12519
9389
6260
3130
0
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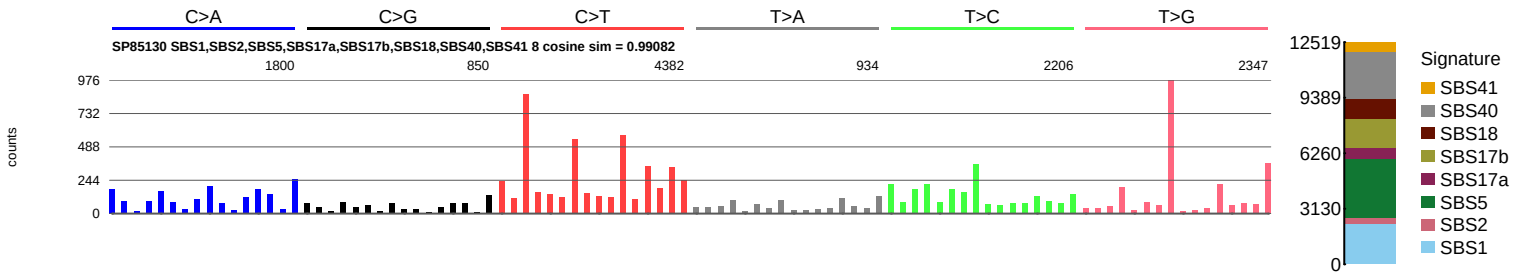
SBS2
SBS1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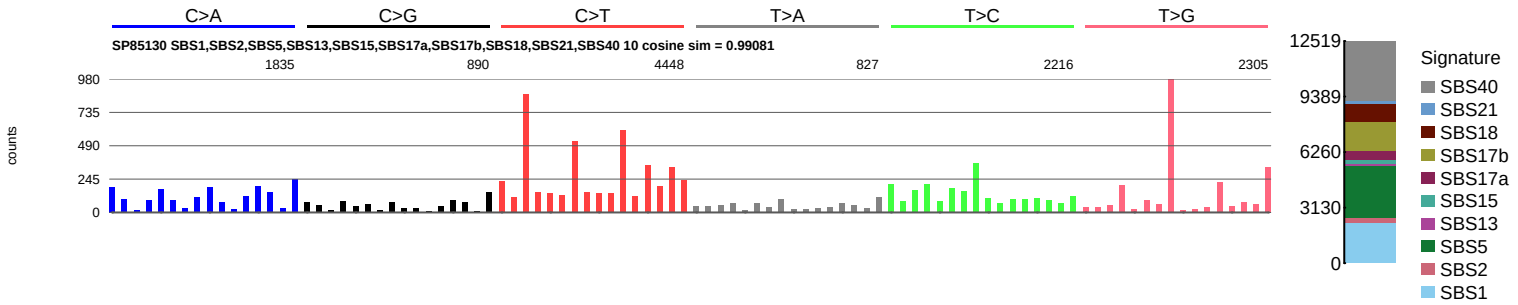
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS40,SBS41 8 cosine sim = 0.99082
1800
850
4382
934
2206
2347
counts
976
732
488
244
0
12519
9389
6260
3130
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1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T>G
SP85130 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40 10 cosine sim = 0.99081
1835
890
4448
827
2216
2305
counts
980
735
490
245
0
12519
9389
6260
3130
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
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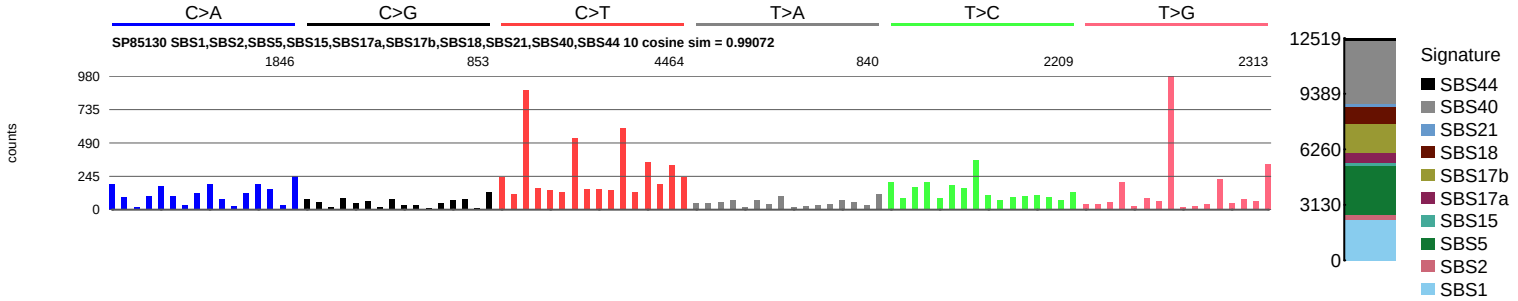
SBS2
SBS1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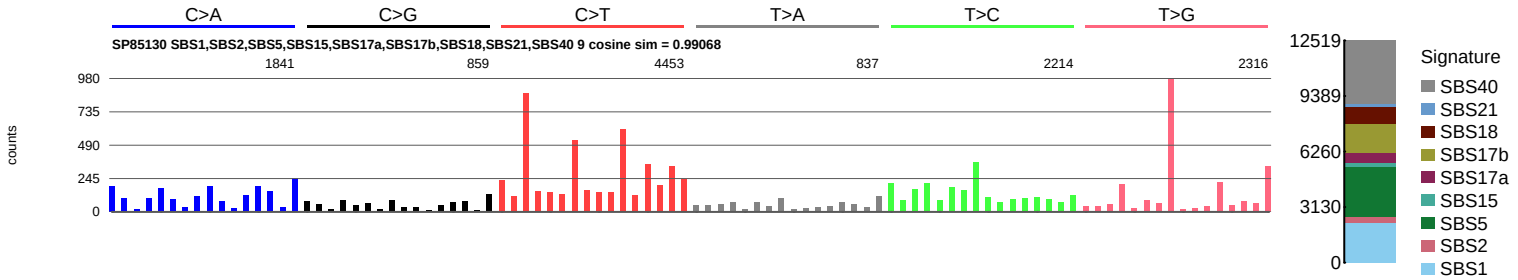
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 10 cosine sim = 0.99072
1846
853
4464
840
2209
2313
counts
980
735
490
245
0
12519
9389
6260
3130
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1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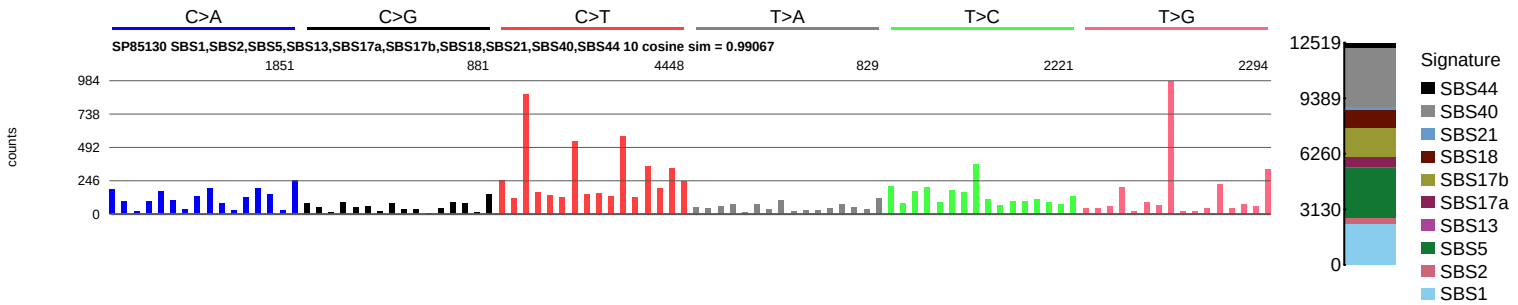
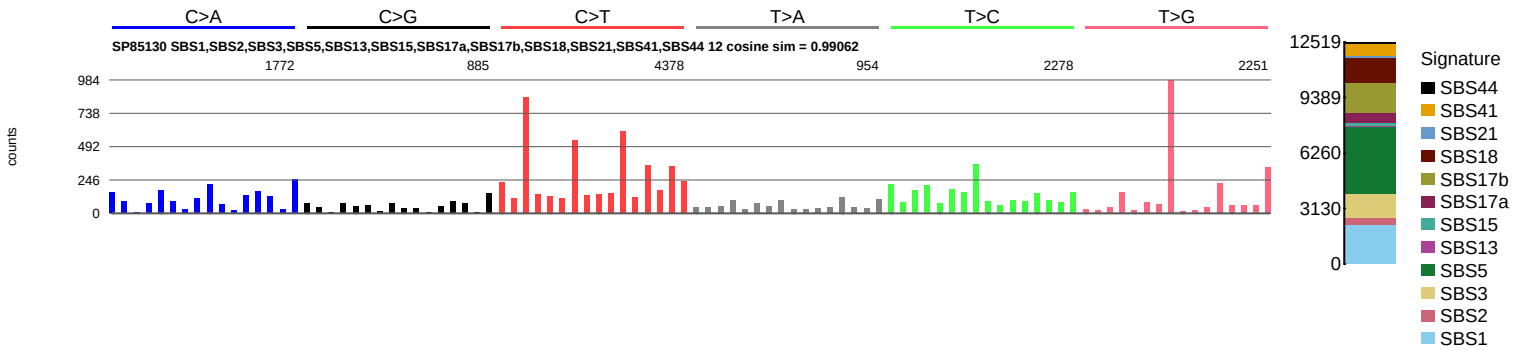
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40 9 cosine sim = 0.99068
1841
859
4453
837
2214
2316
counts
980
735
490
245
0
12519
9389
6260
3130
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 10 cosine sim = 0.99067
1851
881
4448
829
2221
2294
counts
984
738
492
246
0
12519
9389
6260
3130
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1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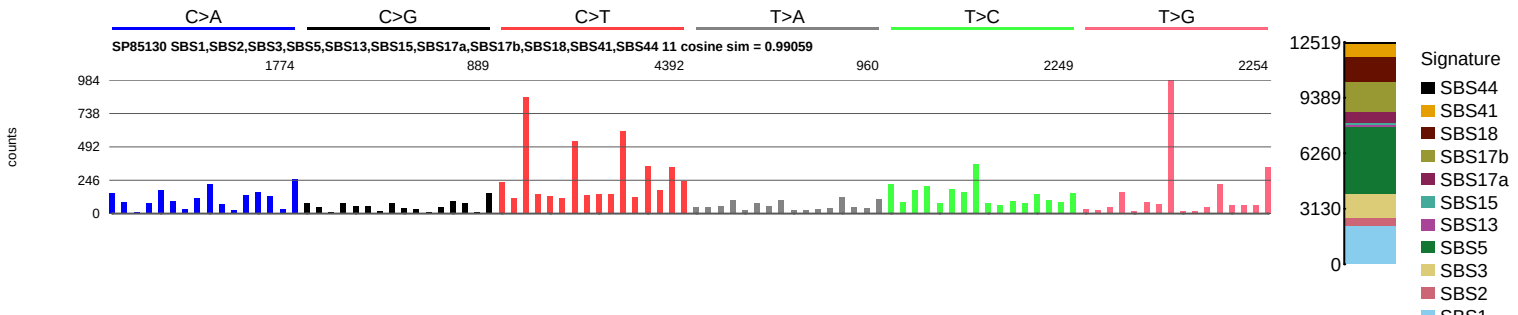
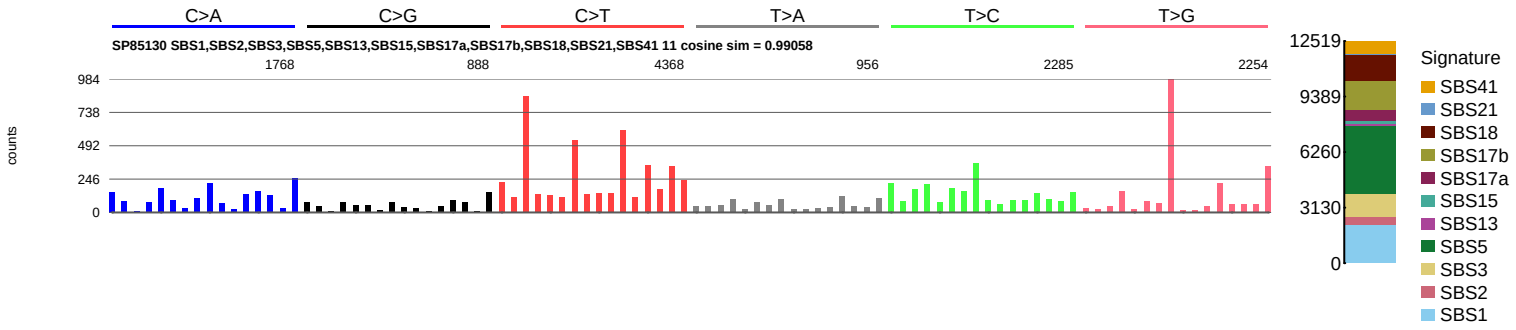
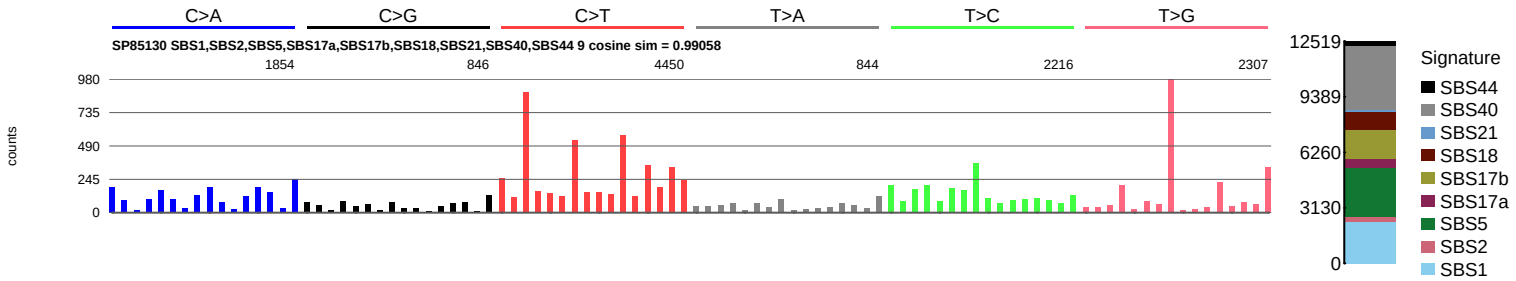
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 12 cosine sim = 0.99062
1772
885
4378
954
2278
2251
counts
984
738
492
246
0
12519
9389
6260
3130
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS41,SBS44 11 cosine sim = 0.99059
1774
889
4392
960
2249
2254
counts
984
738
492
246
0
12519
9389
6260
3130
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 11 cosine sim = 0.99058
1768
888
4368
956
2285
2254
counts
984
738
492
246
0
12519
9389
6260
3130
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1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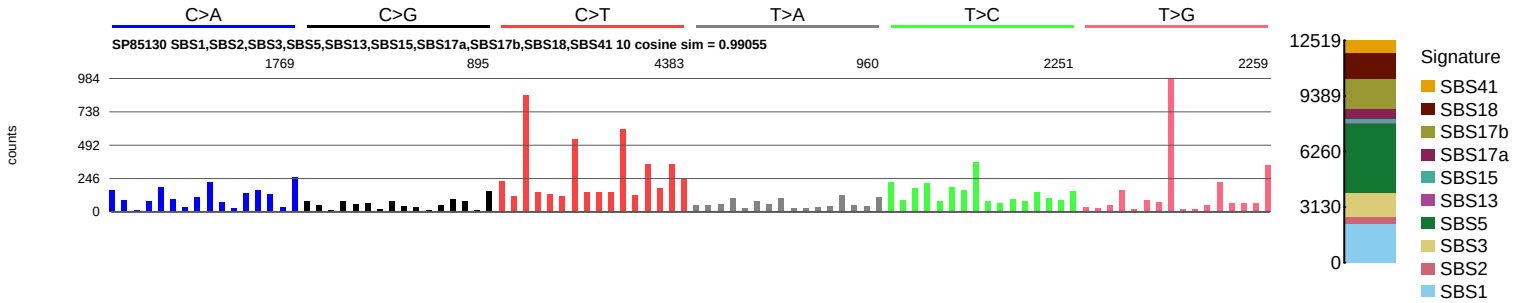
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 9 cosine sim = 0.99058
1854
846
4450
844
2216
2307
counts
980
735
490
245
0
12519
9389
6260
3130
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS41 10 cosine sim = 0.99055
1769
895
4383
960
2251
2259
counts
984
738
492
246
0
12519
9389
6260
3130
0
Signature
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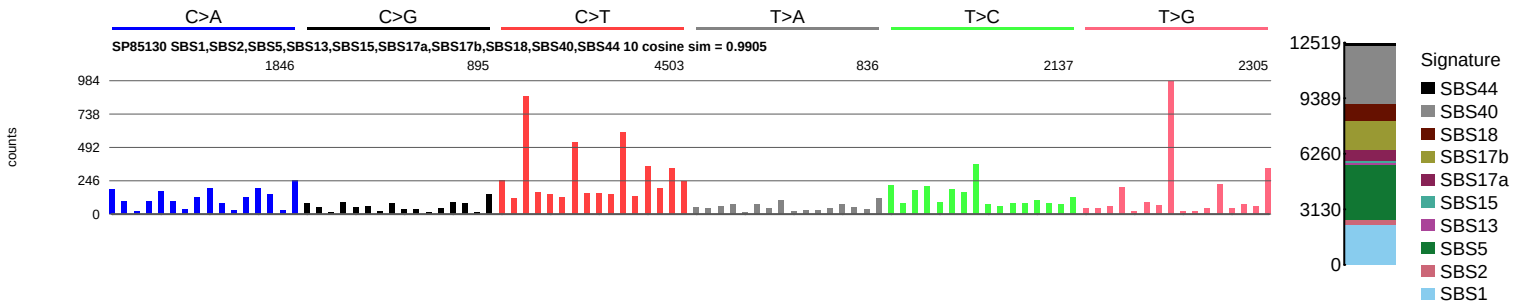
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS44 10 cosine sim = 0.9905
1846
895
4503
836
2137
2305
counts
984
738
492
246
0
12519
9389
6260
3130
0
Signature
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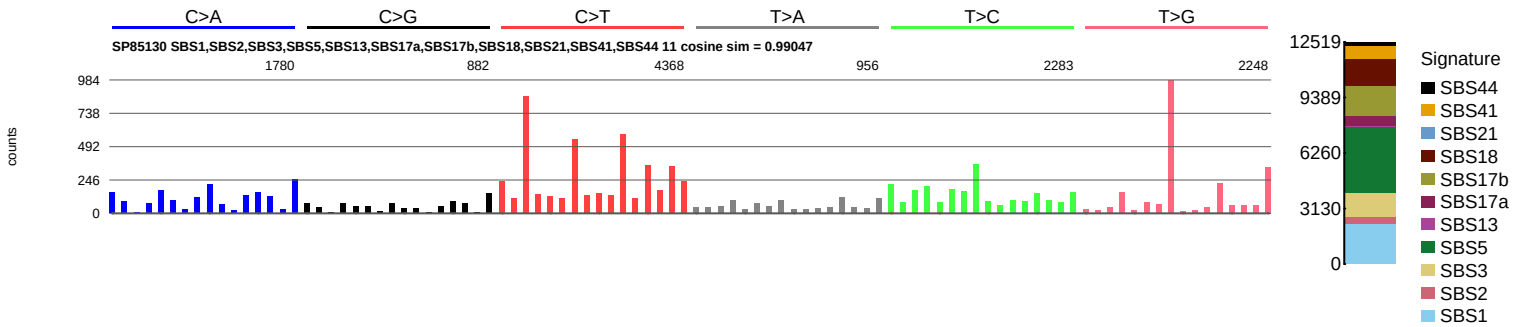
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 11 cosine sim = 0.99047
1780
882
4368
956
2283
2248
counts
984
738
492
246
0
12519
9389
6260
3130
0
Signature
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SBS41
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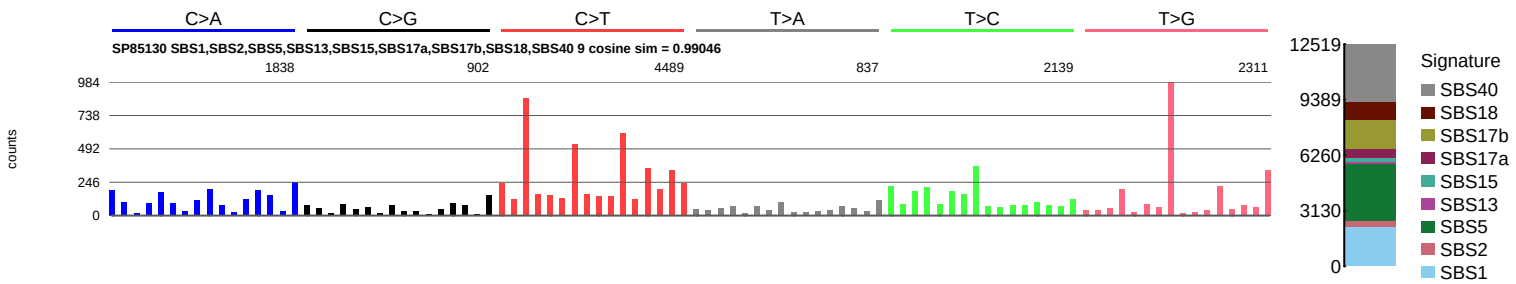
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40 9 cosine sim = 0.99046
1838
902
4489
837
2139
2311
counts
984
738
492
246
0
12519
9389
6260
3130
0
Signature
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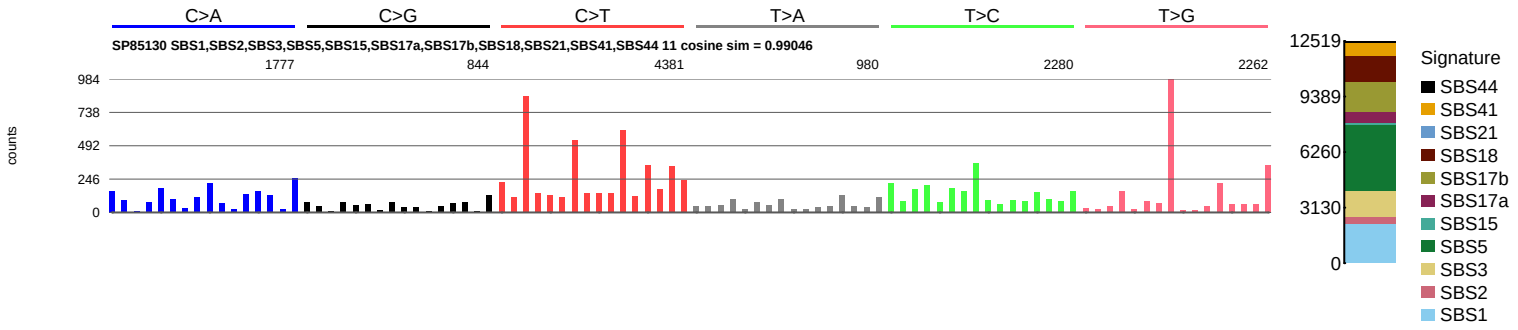
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1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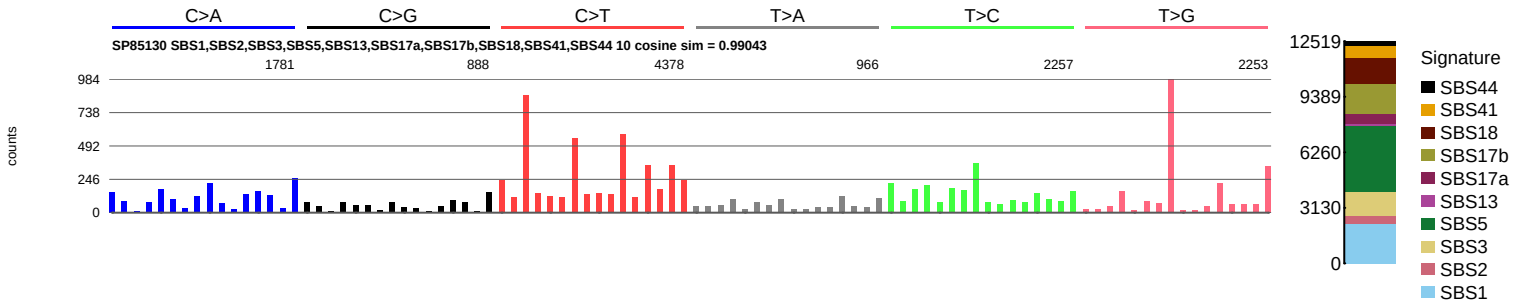
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 11 cosine sim = 0.99046
1777
844
4381
980
2280
2262
counts
984
738
492
246
0
12519
9389
6260
3130
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1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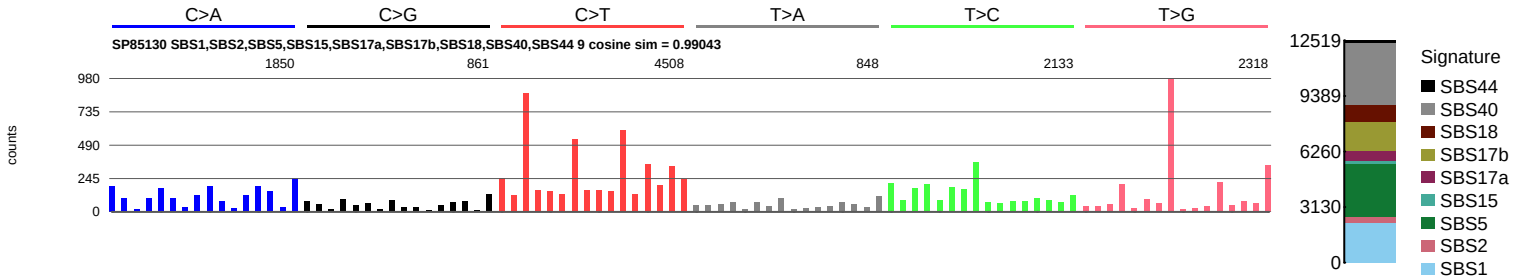
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS41,SBS44 10 cosine sim = 0.99043
1781
888
4378
966
2257
2253
counts
984
738
492
246
0
12519
9389
6260
3130
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS44 9 cosine sim = 0.99043
1850
861
4508
848
2133
2318
counts
980
735
490
245
0
12519
9389
6260
3130
0
Signature
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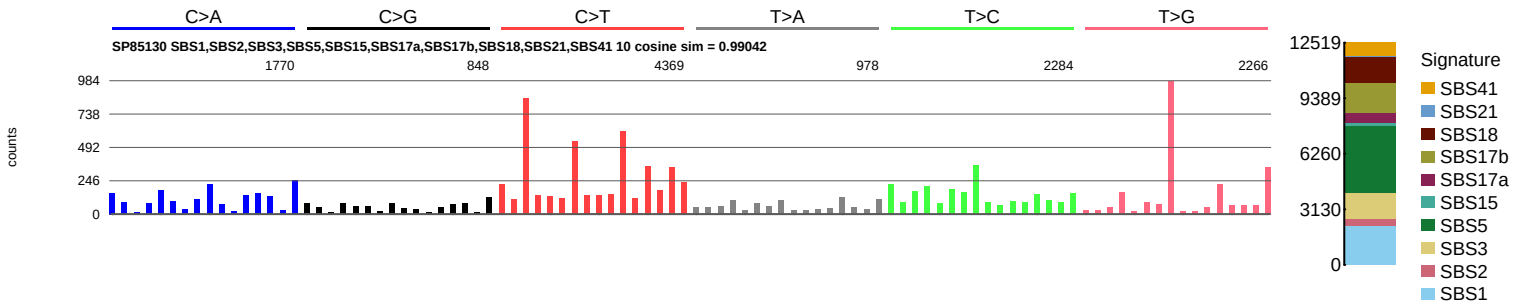
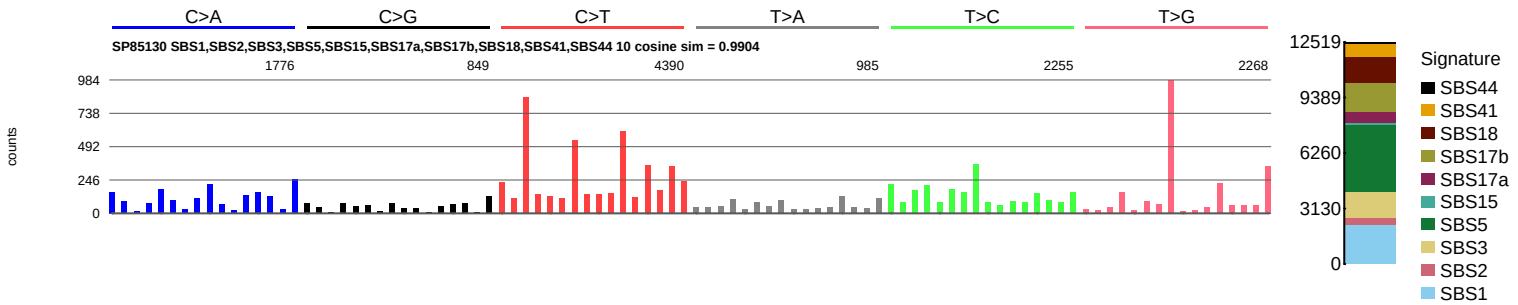
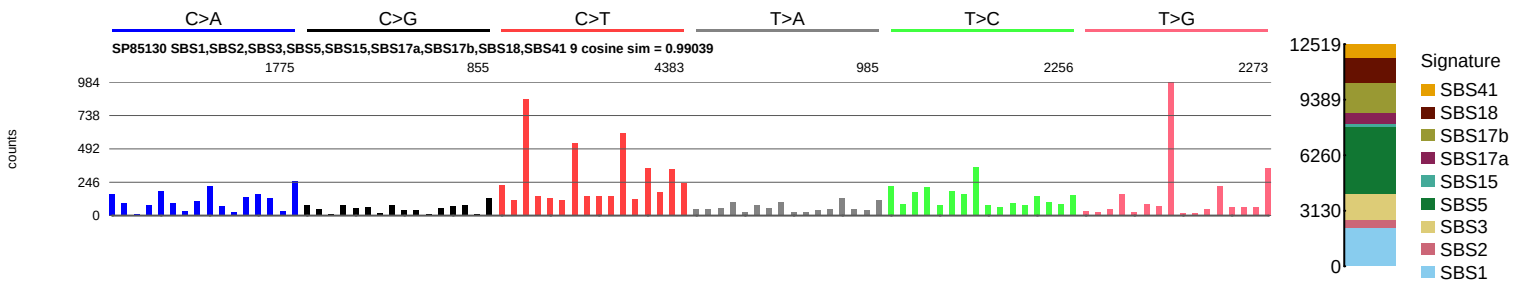
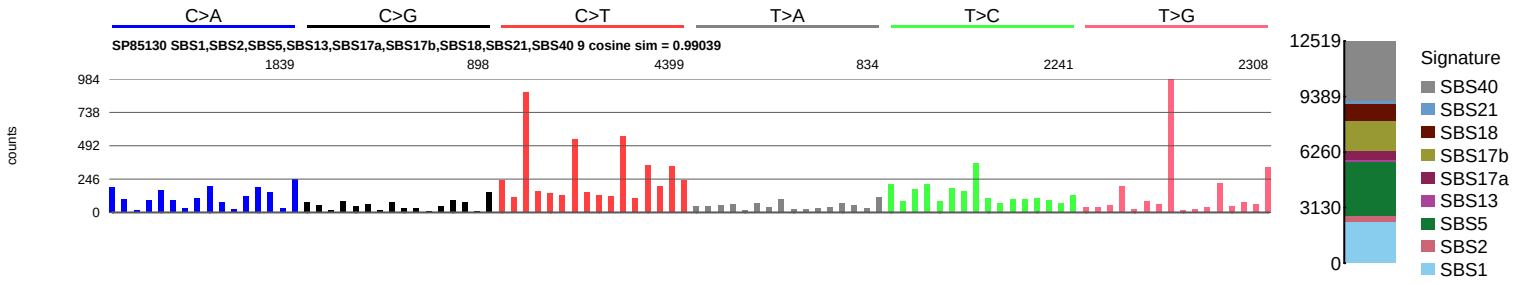
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 10 cosine sim = 0.99042
1770
848
4369
978
2284
2266
counts
984
738
492
246
0
12519
9389
6260
3130
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS41,SBS44 10 cosine sim = 0.9904
1776
849
4390
985
2255
2268
counts
984
738
492
246
0
12519
9389
6260
3130
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS41 9 cosine sim = 0.99039
1775
855
4383
985
2256
2273
counts
984
738
492
246
0
12519
9389
6260
3130
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1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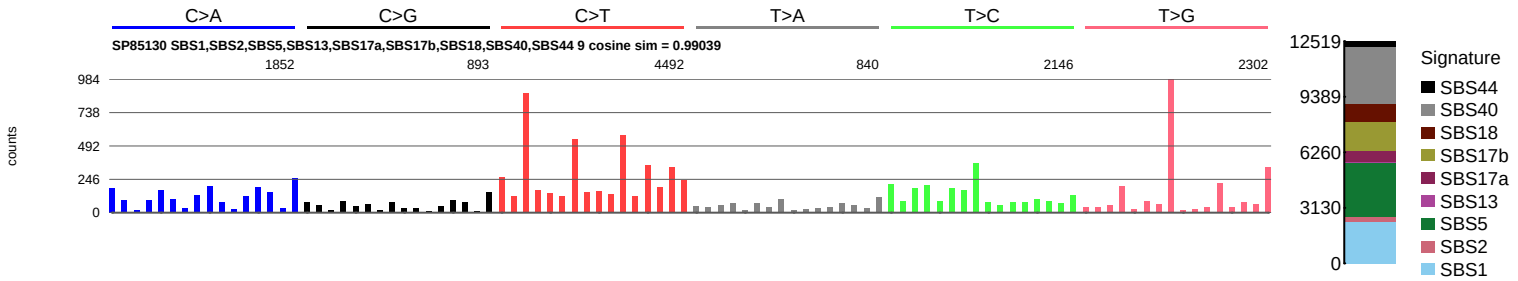
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40 9 cosine sim = 0.99039
1839
898
4399
834
2241
2308
counts
984
738
492
246
0
12519
9389
6260
3130
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1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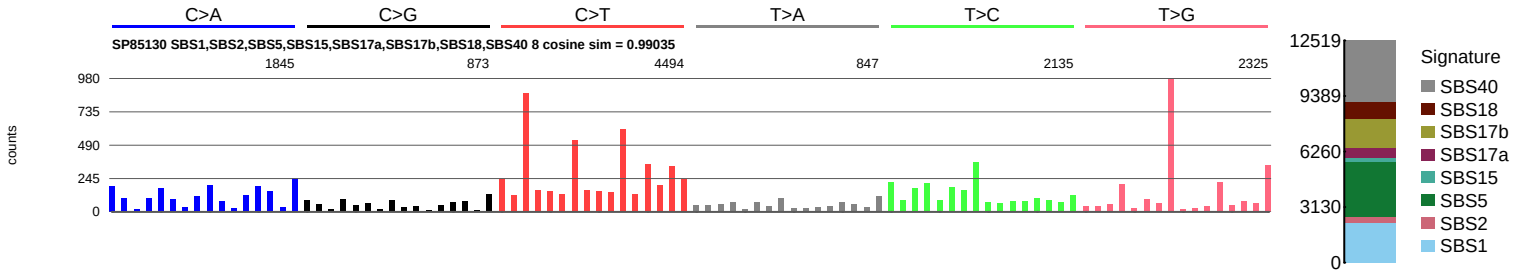
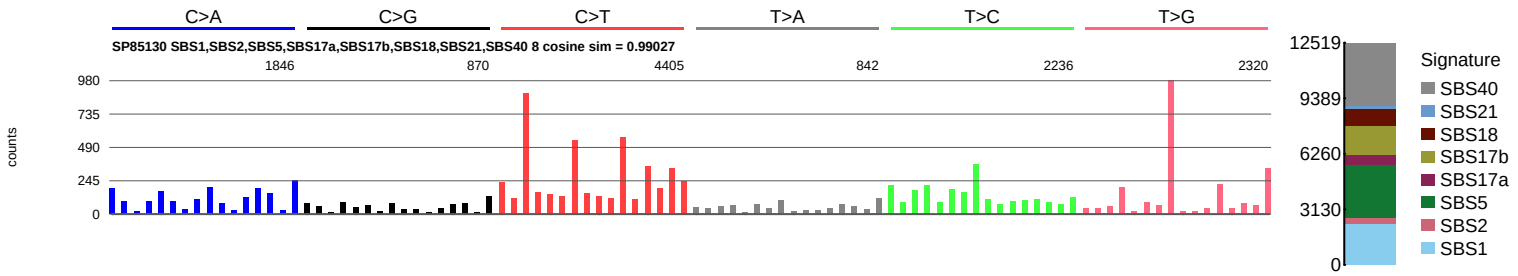
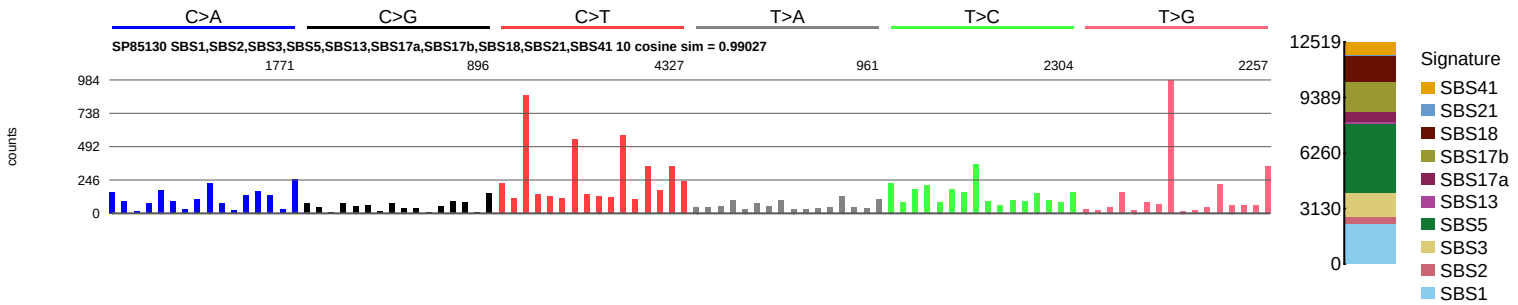
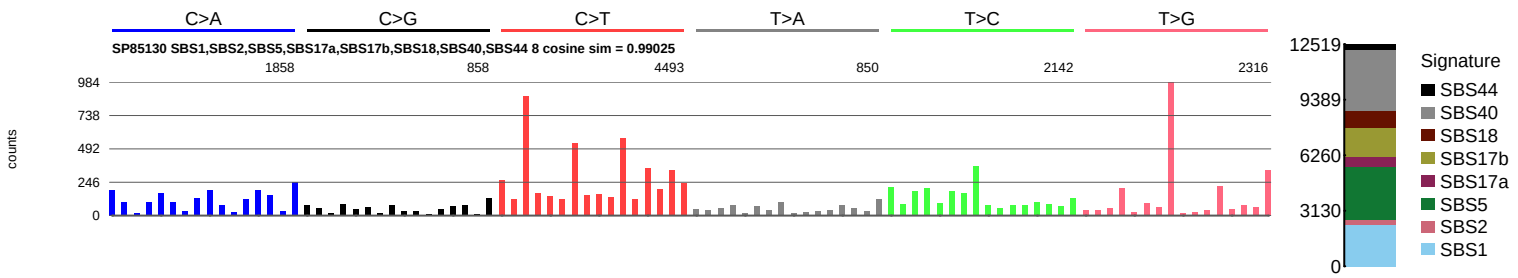
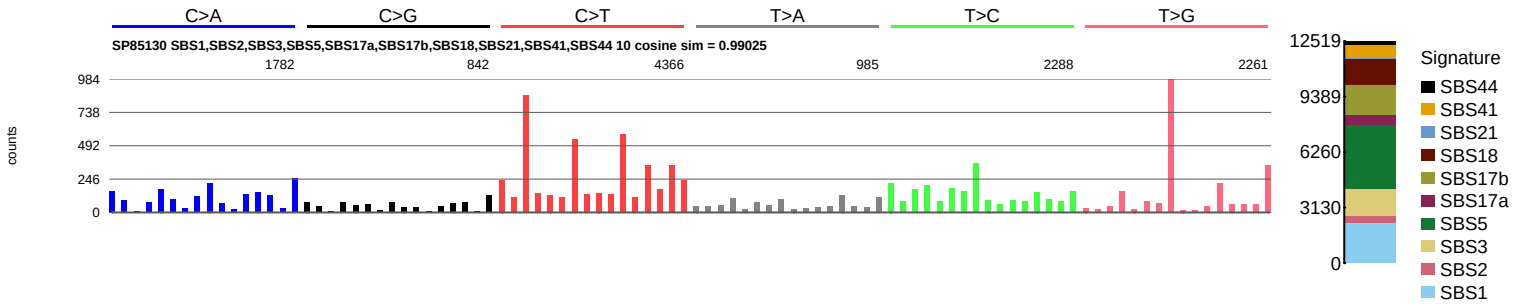
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS40,SBS44 9 cosine sim = 0.99039
1852
893
4492
840
2146
2302
counts
984
738
492
246
0
12519
9389
6260
3130
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1

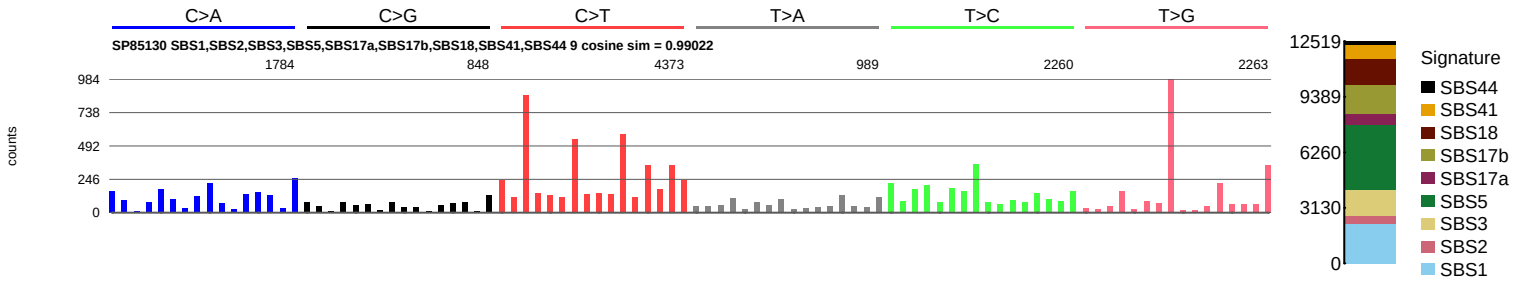

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS40 8 cosine sim = 0.99035
1845
873
4494
847
2135
2325
counts
980
735
490
245
0
12519
9389
6260
3130
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1

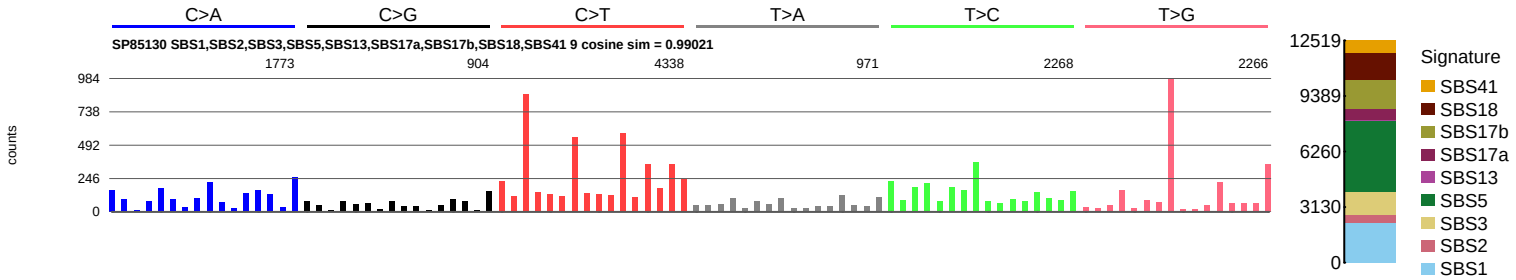

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS40 8 cosine sim = 0.99027
1846
870
4405
842
2236
2320
counts
980
735
490
245
0
12519
9389
6260
3130
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1

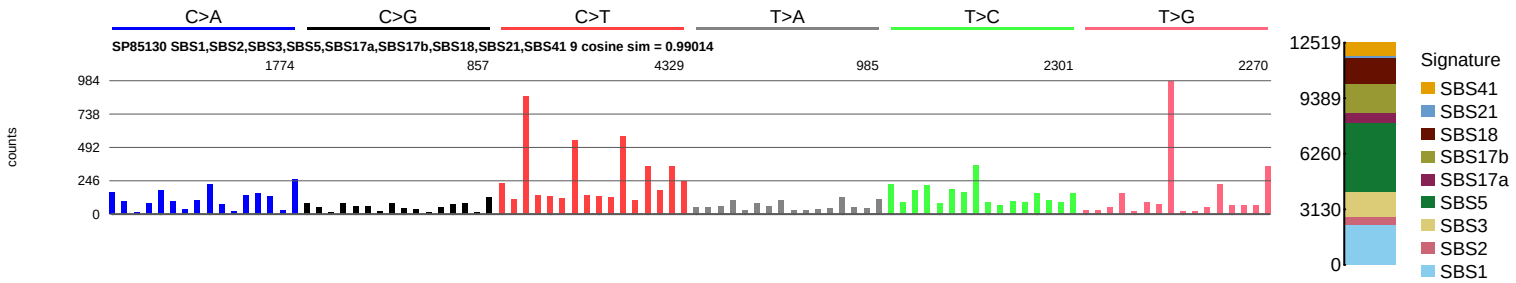

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS41 10 cosine sim = 0.99027
1771
896
4327
961
2304
2257
counts
984
738
492
246
0
12519
9389
6260
3130
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1

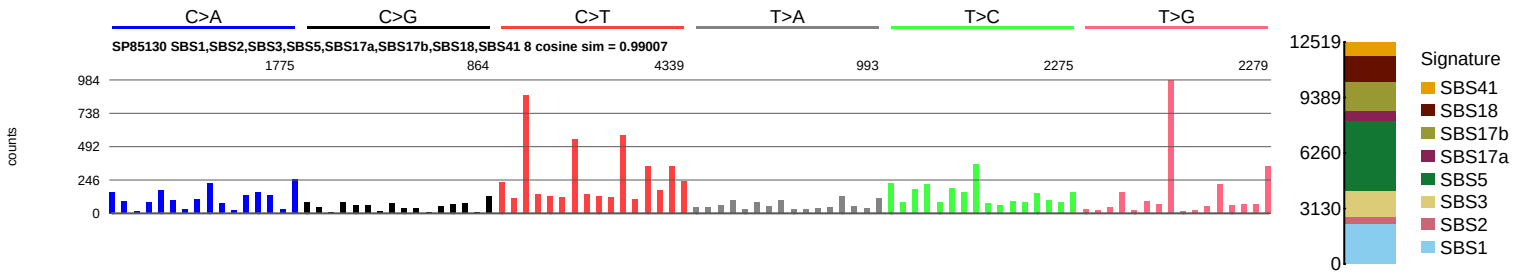

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS40,SBS44 8 cosine sim = 0.99025
1858
858
4493
850
2142
2316
counts
984
738
492
246
0
12519
9389
6260
3130
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1

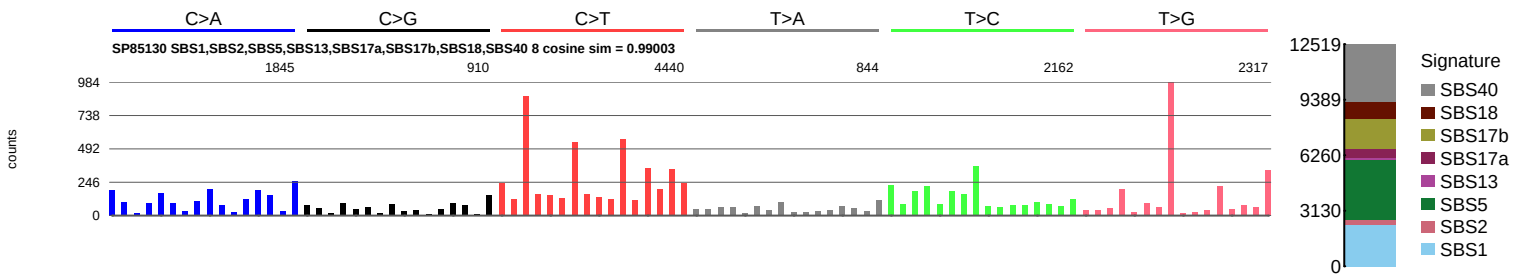

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 10 cosine sim = 0.99025
1782
842
4366
985
2288
2261
counts
984
738
492
246
0
12519
9389
6260
3130
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1

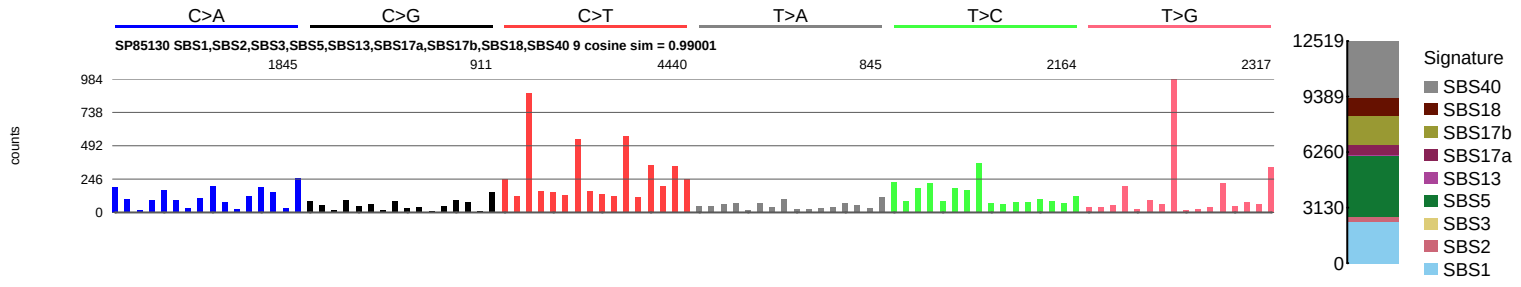

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS41,SBS44 9 cosine sim = 0.99022
1784
848
4373
989
2260
2263
counts
984
738
492
246
0
12519
9389
6260
3130
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1

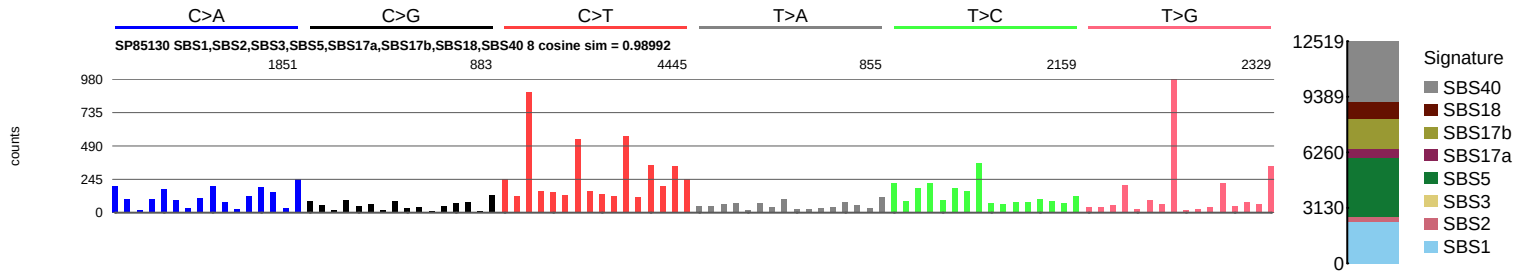

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS41 9 cosine sim = 0.99021
1773
904
4338
971
2268
2266
counts
984
738
492
246
0
12519
9389
6260
3130
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1

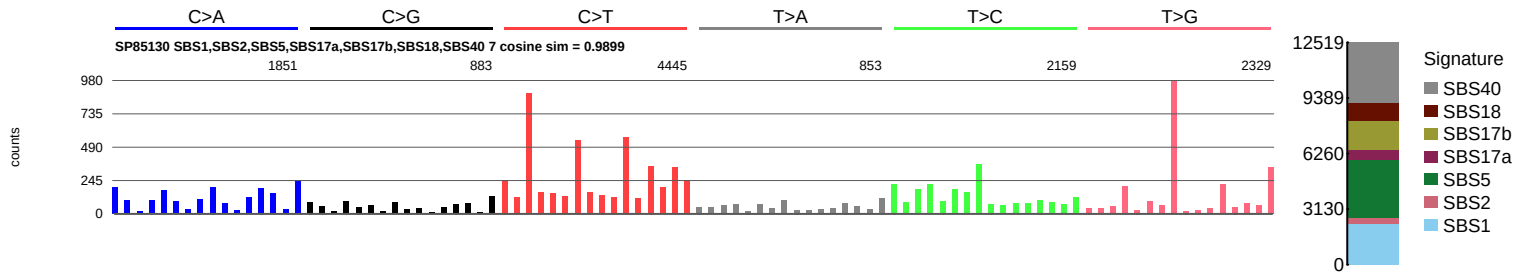

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS41 9 cosine sim = 0.99014
1774
857
4329
985
2301
2270
counts
984
738
492
246
0
12519
9389
6260
3130
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS41 8 cosine sim = 0.99007
1775
864
4339
993
2275
2279
counts
984
738
492
246
0
12519
9389
6260
3130
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS40 8 cosine sim = 0.99003
1845
910
4440
844
2162
2317
counts
984
738
492
246
0
12519
9389
6260
3130
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS40 9 cosine sim = 0.99001
1845
911
4440
845
2164
2317
984
738
492
246
0
counts
12519
9389
6260
3130
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS40 8 cosine sim = 0.98992
1851
883
4445
855
2159
2329
980
735
490
245
0
counts
12519
9389
6260
3130
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1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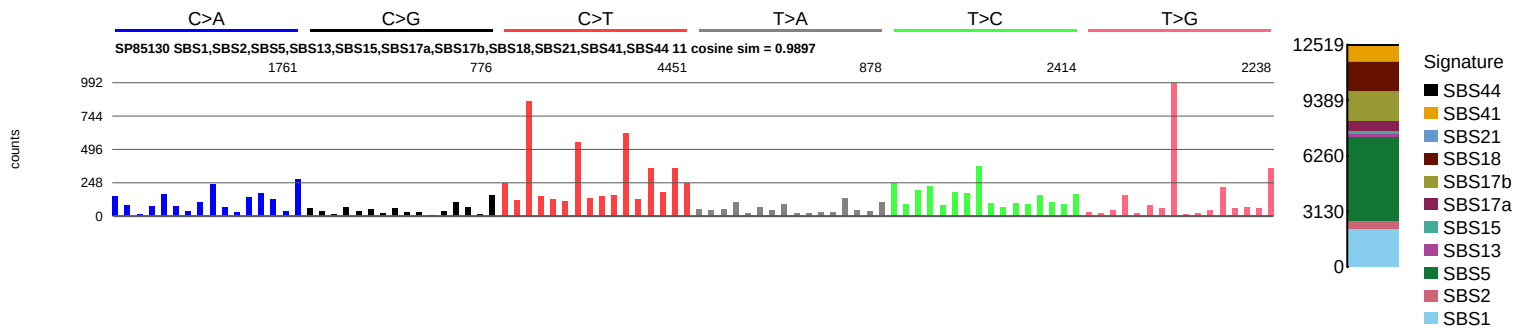
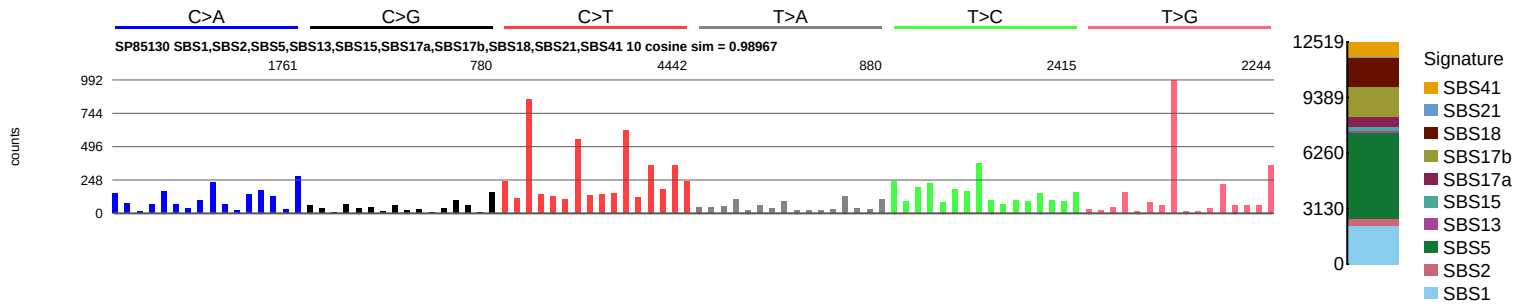
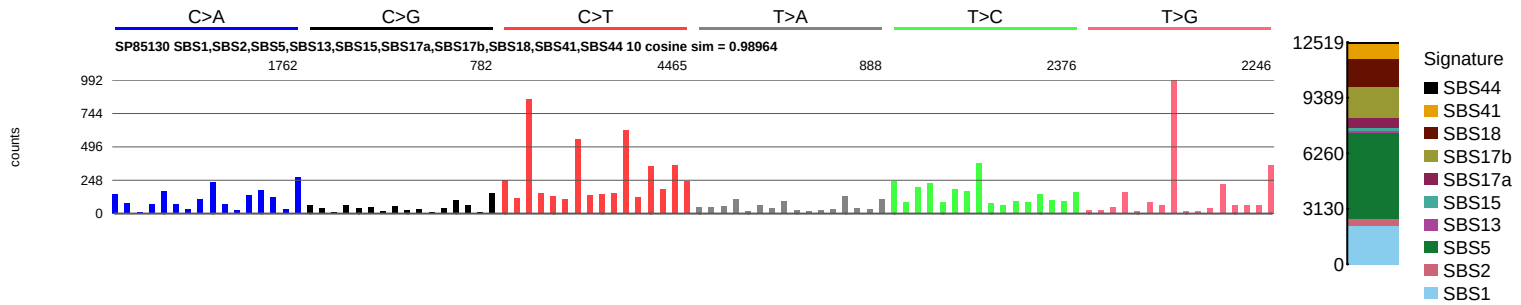
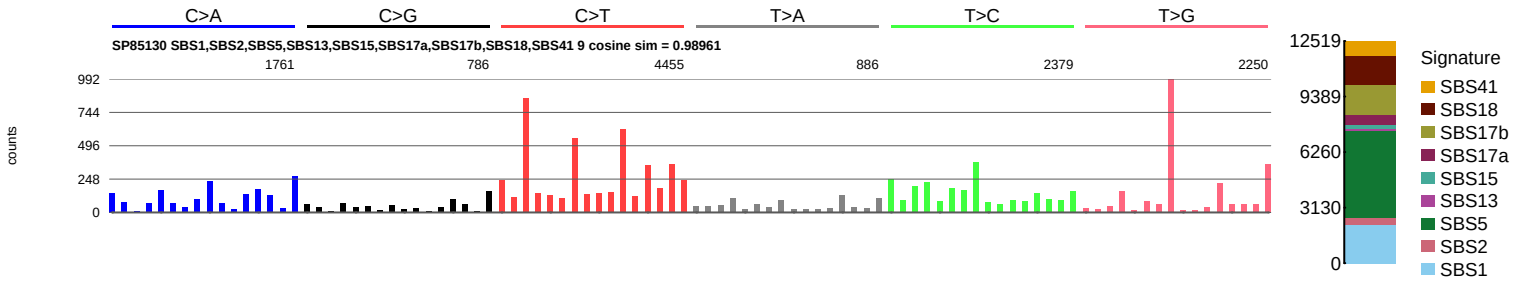
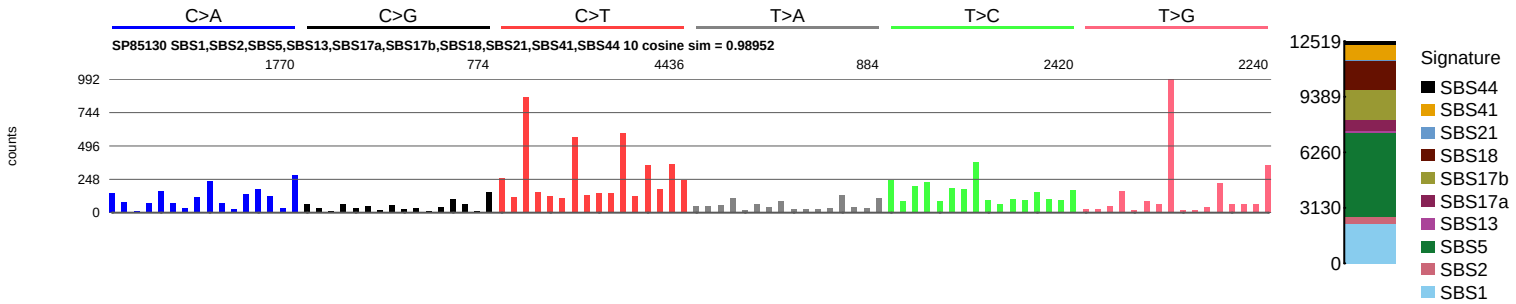
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS40 7 cosine sim = 0.9899
1851
883
4445
853
2159
2329
980
735
490
245
0
counts
12519
9389
6260
3130
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1

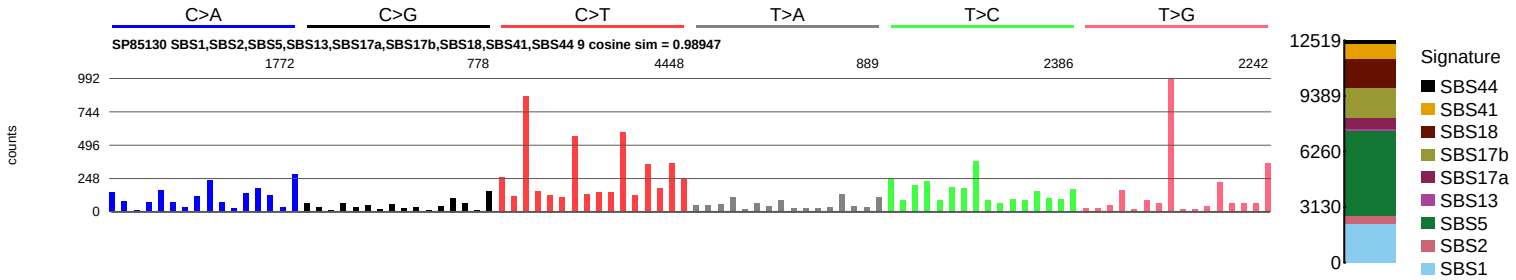

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 11 cosine sim = 0.9897
1761
776
4451
878
2414
2238
992
744
496
248
0
counts
12519
9389
6260
3130
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1

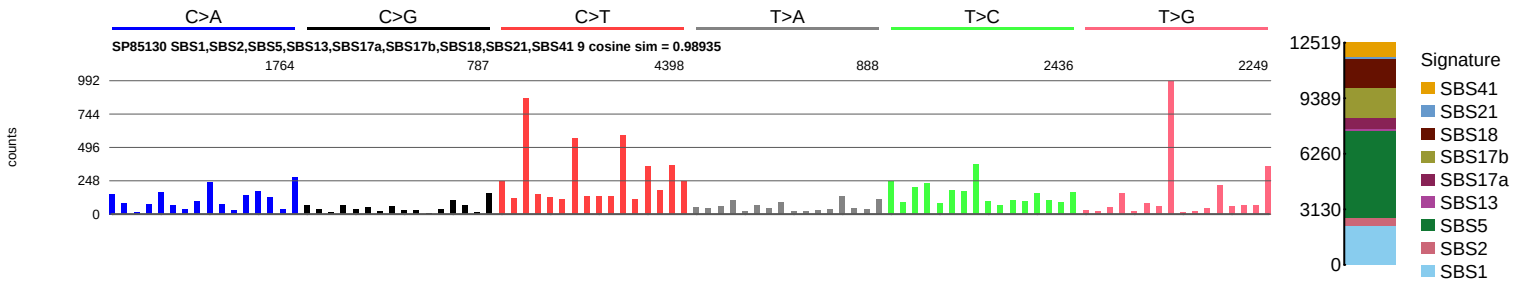

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 10 cosine sim = 0.98967
1761
780
4442
880
2415
2244
992
744
496
248
0
counts
12519
9389
6260
3130
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS41,SBS44 10 cosine sim = 0.98964
1762
782
4465
888
2376
2246
992
744
496
248
0
counts
12519
9389
6260
3130
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS41 9 cosine sim = 0.98961
1761
786
4455
886
2379
2250
counts
992
744
496
248
0
12519
9389
6260
3130
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 10 cosine sim = 0.98952
1770
774
4436
884
2420
2240
counts
992
744
496
248
0
12519
9389
6260
3130
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS41,SBS44 9 cosine sim = 0.98947
1772
778
4448
889
2386
2242
counts
992
744
496
248
0
12519
9389
6260
3130
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS41 9 cosine sim = 0.98935
1764
787
4398
888
2436
2249
counts
992
744
496
248
0
12519
9389
6260
3130
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
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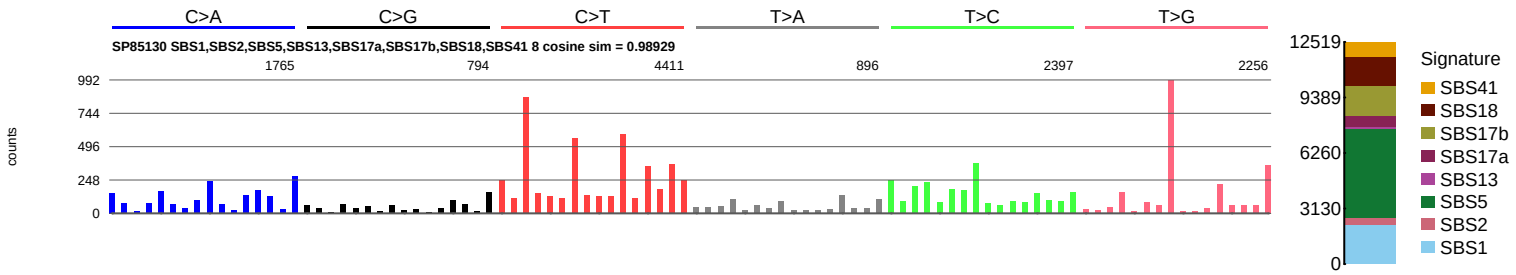
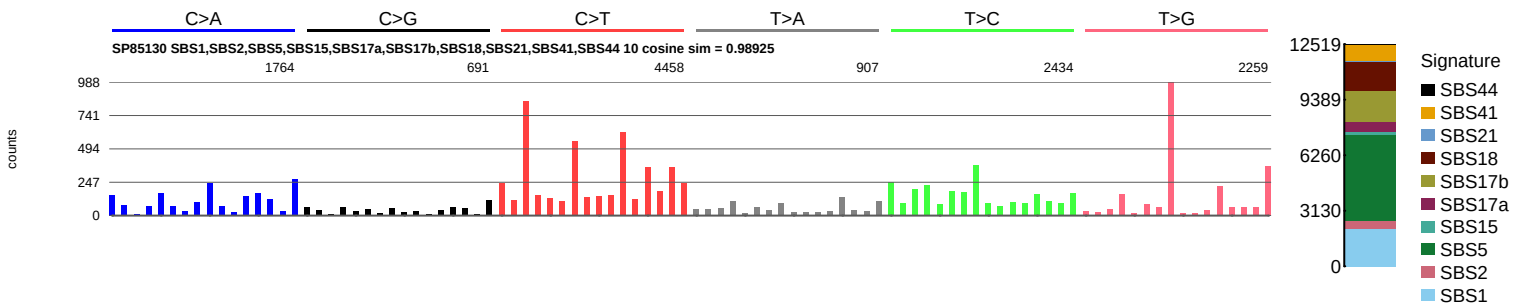
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS41 8 cosine sim = 0.98929
1765
794
4411
896
2397
2256
counts
992
744
496
248
0
12519
9389
6260
3130
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 10 cosine sim = 0.98925
1764
691
4458
907
2434
2259
counts
988
741
494
247
0
12519
9389
6260
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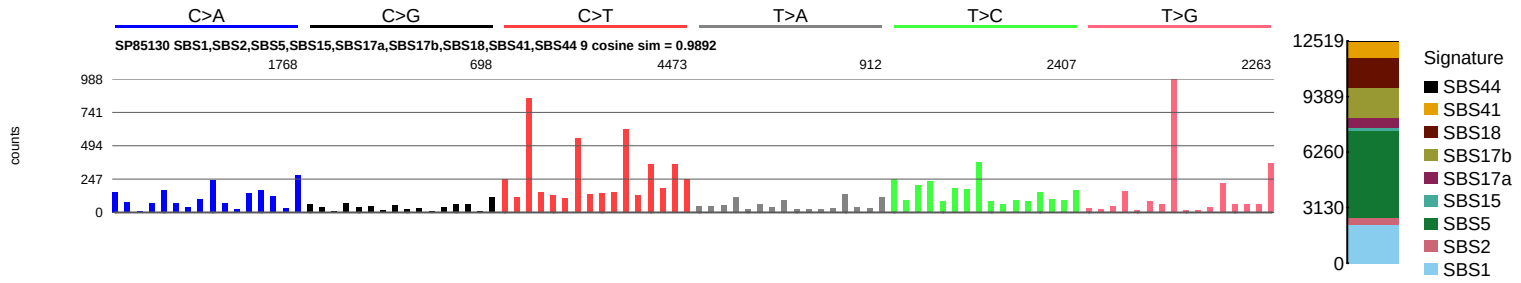
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SP85130 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS41,SBS44 9 cosine sim = 0.9892
1768
698
4473
912
2407
2263
counts
988
741
494
247
0
12519
9389
6260
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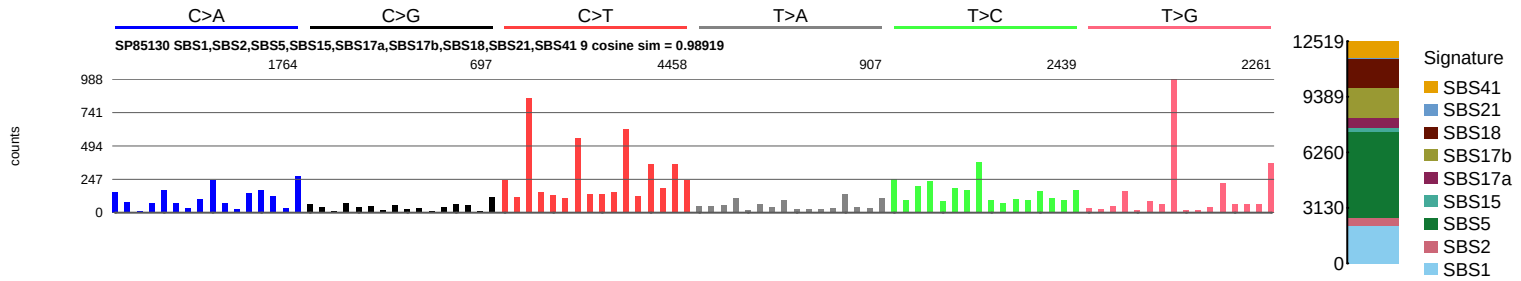
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SP85130 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 9 cosine sim = 0.98919
1764
697
4458
907
2439
2261
counts
988
741
494
247
0
12519
9389
6260
3130
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SBS17a
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SBS2
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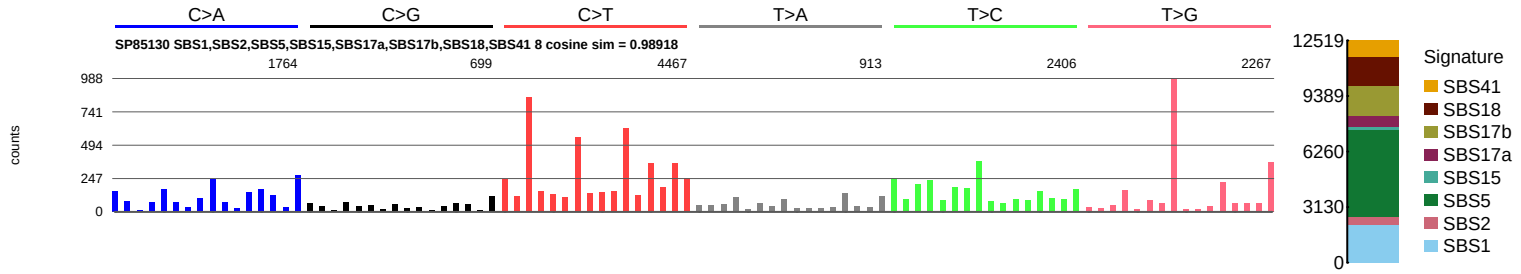
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T>G
SP85130 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS41 8 cosine sim = 0.98918
1764
699
4467
913
2406
2267
counts
988
741
494
247
0
12519
9389
6260
3130
0
Signature
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SBS17a
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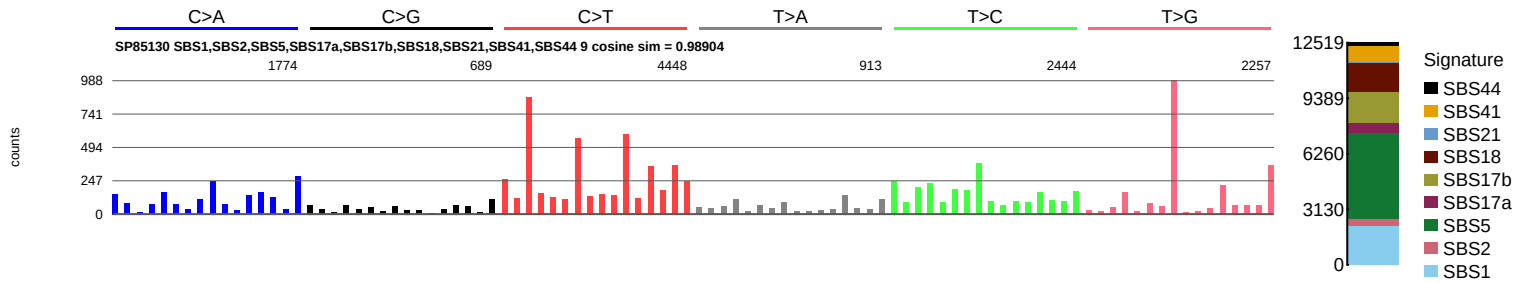
SBS1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SP85130 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 9 cosine sim = 0.98904
1774
689
4448
913
2444
2257
counts
988
741
494
247
0
12519
9389
6260
3130
0
Signature
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SBS41
SBS21
SBS18
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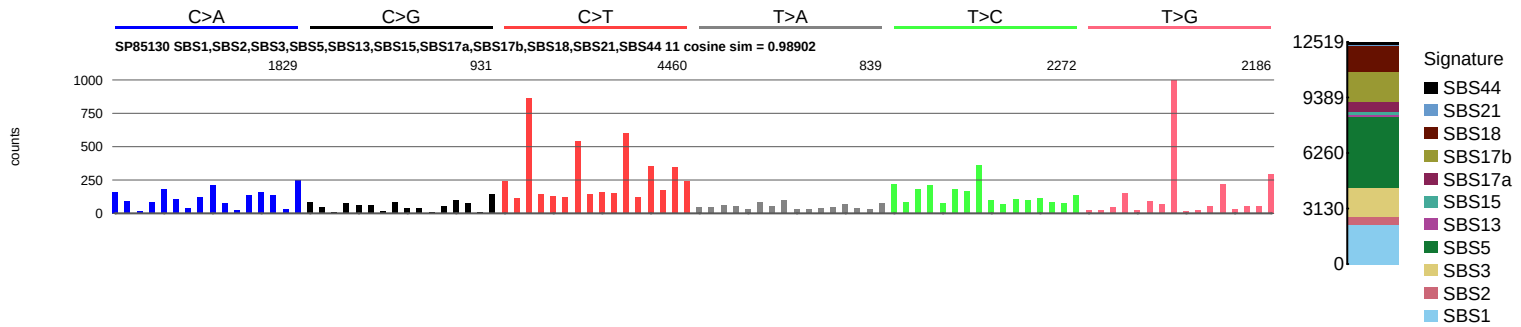
SBS2
SBS1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C>T
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SP85130 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS44 11 cosine sim = 0.98902
1829
931
4460
839
2272
2186
counts
1000
750
500
250
0
12519
9389
6260
3130
0
Signature
SBS44
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SBS17b
SBS17a
SBS15
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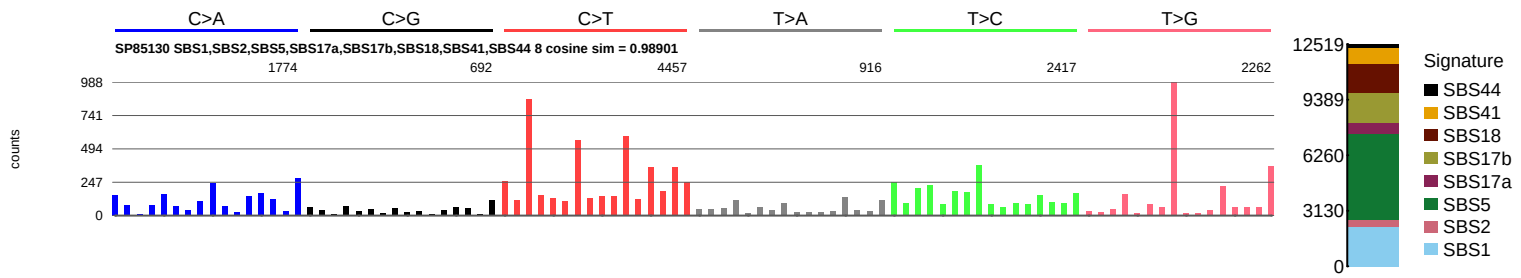
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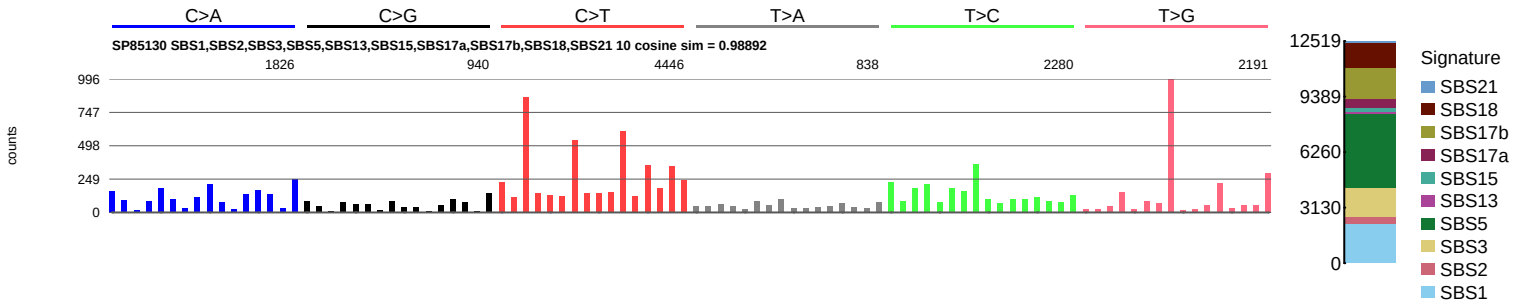
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS41,SBS44 8 cosine sim = 0.98901
1774
692
4457
916
2417
2262
counts
988
741
494
247
0
12519
9389
6260
3130
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1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C>T
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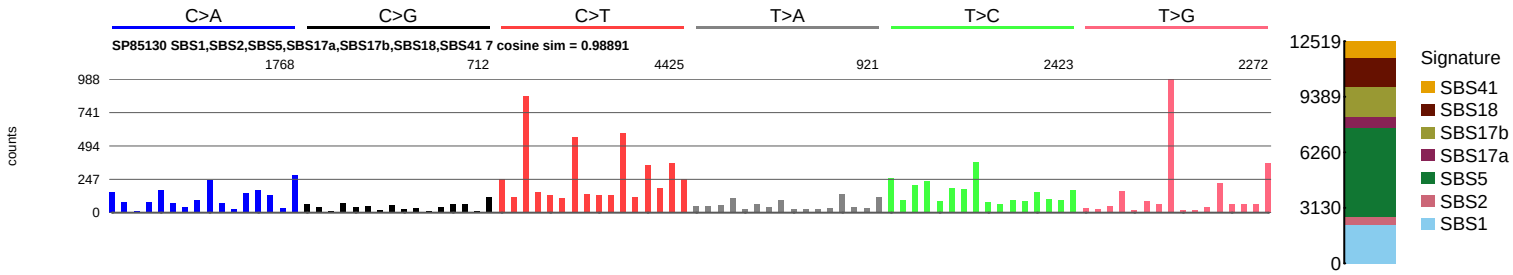
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21 10 cosine sim = 0.98892
1826
940
4446
838
2280
2191
counts
996
747
498
249
0
12519
9389
6260
3130
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS41 7 cosine sim = 0.98891
1768
712
4425
921
2423
2272
counts
988
741
494
247
0
12519
9389
6260
3130
0
Signature
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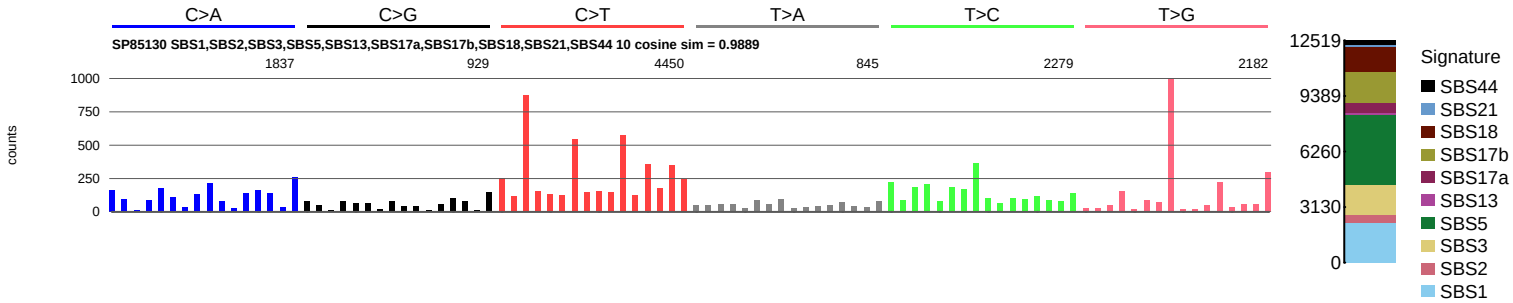
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1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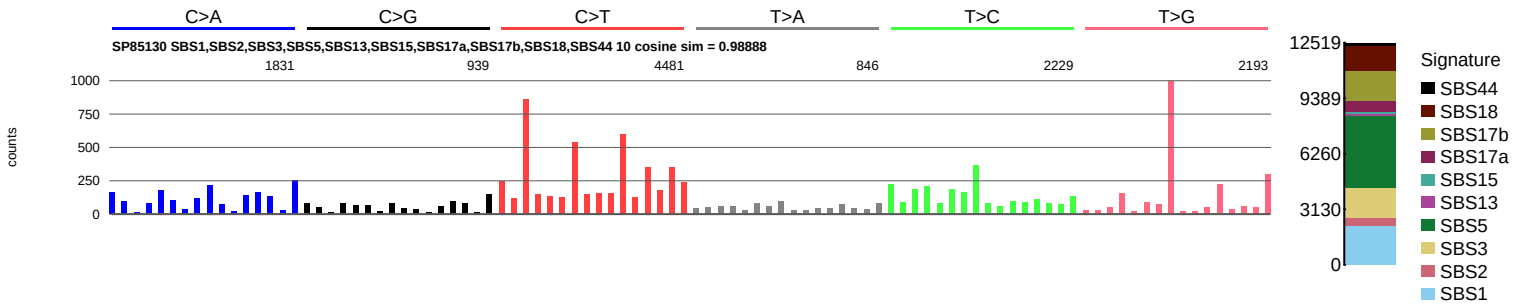
T>G
SP85130 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS44 10 cosine sim = 0.9889
1837
929
4450
845
2279
2182
counts
1000
750
500
250
0
12519
9389
6260
3130
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1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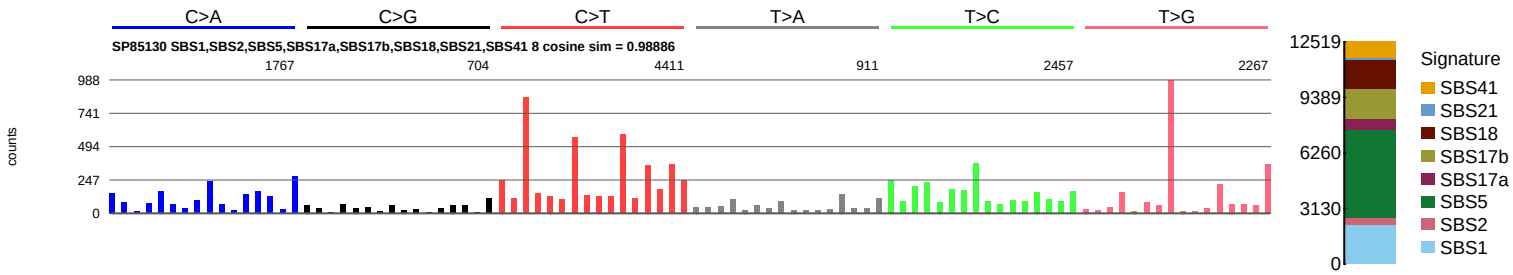
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS44 10 cosine sim = 0.98888
1831
939
4481
846
2229
2193
counts
1000
750
500
250
0
12519
9389
6260
3130
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1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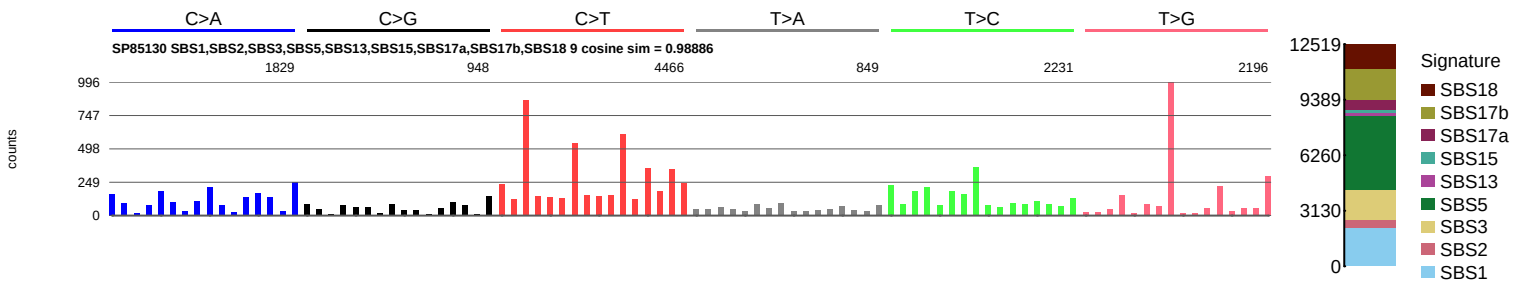
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS41 8 cosine sim = 0.98886
1767
704
4411
911
2457
2267
counts
988
741
494
247
0
12519
9389
6260
3130
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1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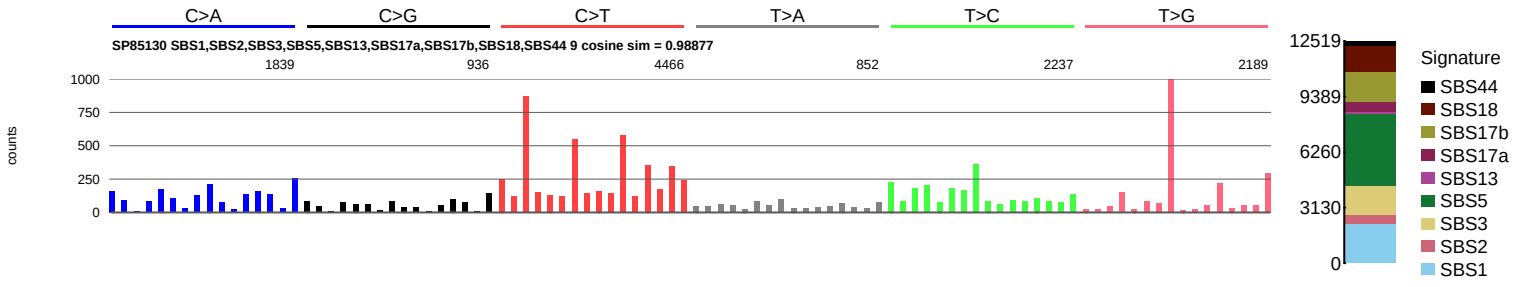
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18 9 cosine sim = 0.98886
1829
948
4466
849
2231
2196
counts
996
747
498
249
0
12519
9389
6260
3130
0
Signature
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1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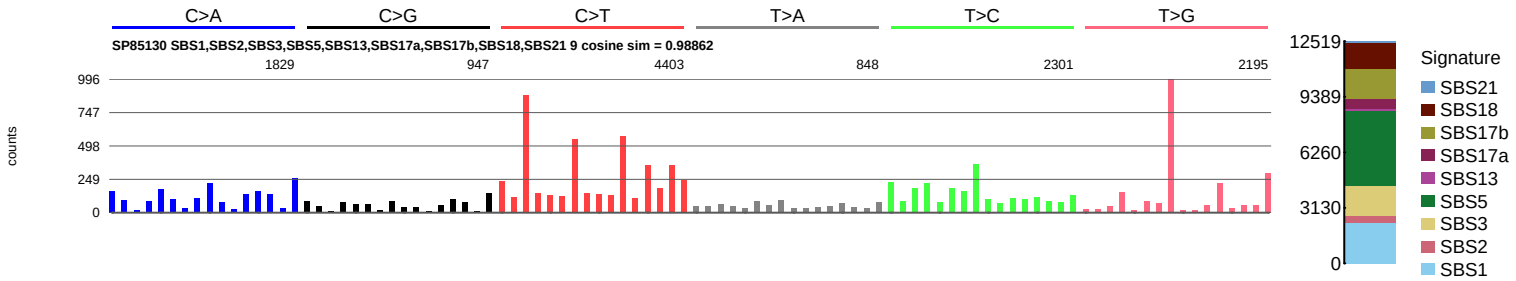
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS44 9 cosine sim = 0.98877
1839
936
4466
852
2237
2189
counts
1000
750
500
250
0
12519
9389
6260
3130
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1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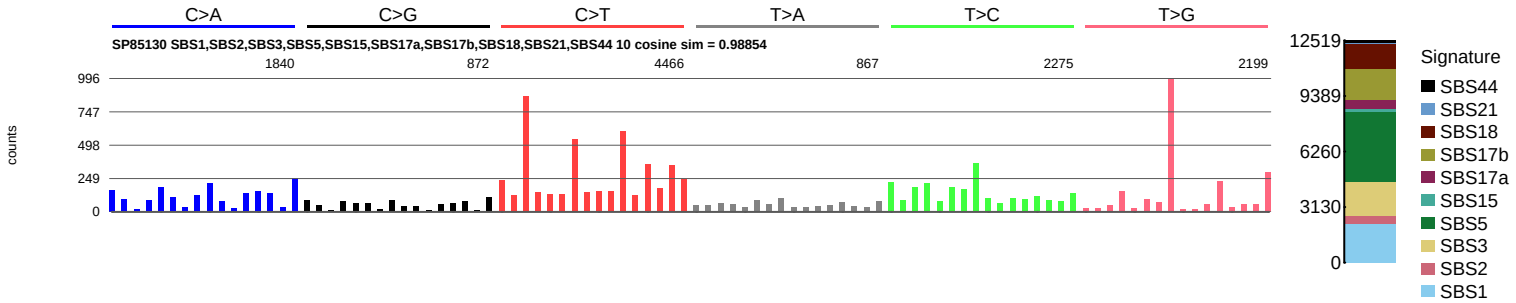
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21 9 cosine sim = 0.98862
1829
947
4403
848
2301
2195
counts
996
747
498
249
0
12519
9389
6260
3130
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS44 10 cosine sim = 0.98854
1840
872
4466
867
2275
2199
counts
996
747
498
249
0
12519
9389
6260
3130
0
Signature
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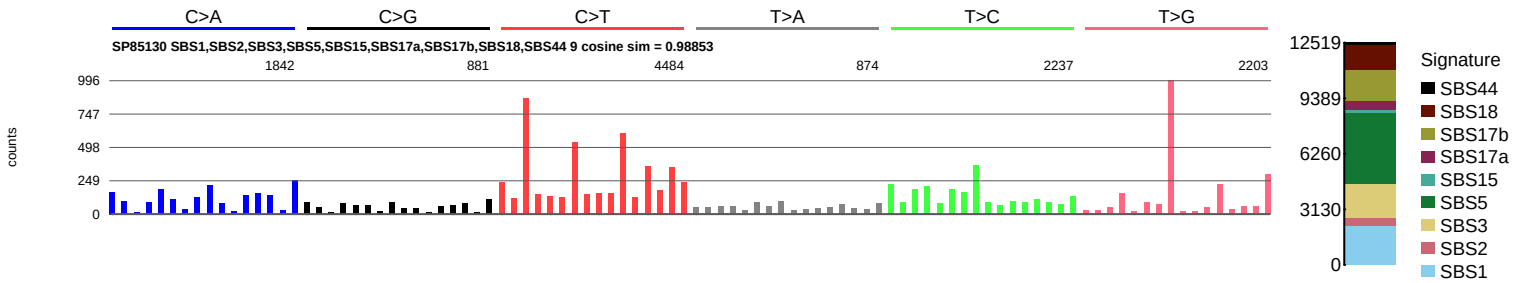
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1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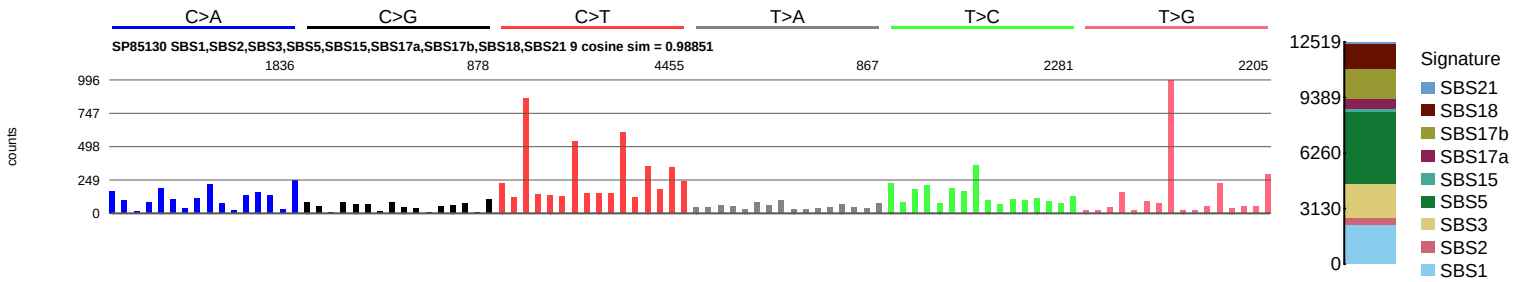
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS44 9 cosine sim = 0.98853
1842
881
4484
874
2237
2203
counts
996
747
498
249
0
12519
9389
6260
3130
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1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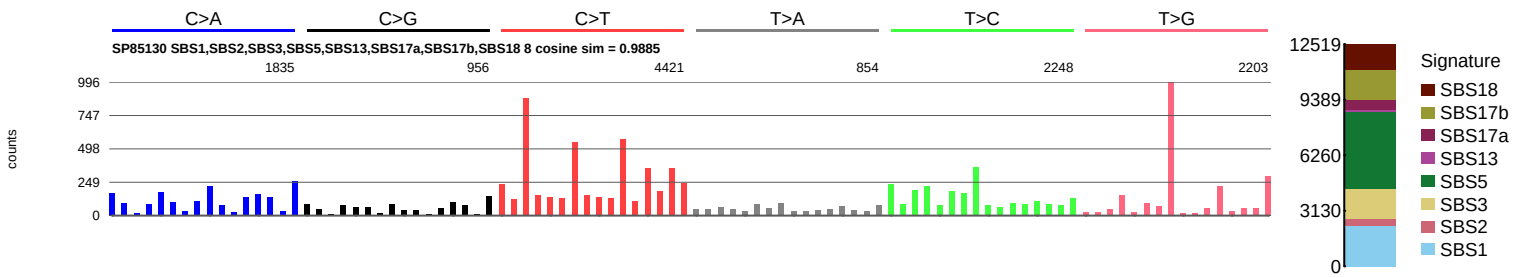
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21 9 cosine sim = 0.98851
1836
878
4455
867
2281
2205
counts
996
747
498
249
0
12519
9389
6260
3130
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18 8 cosine sim = 0.9885
1835
956
4421
854
2248
2203
counts
996
747
498
249
0
12519
9389
6260
3130
0
Signature
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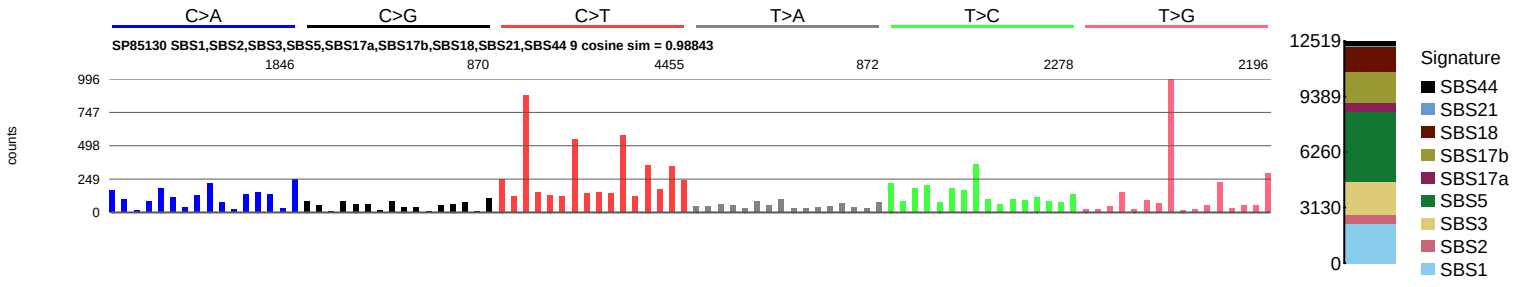
SBS2
SBS1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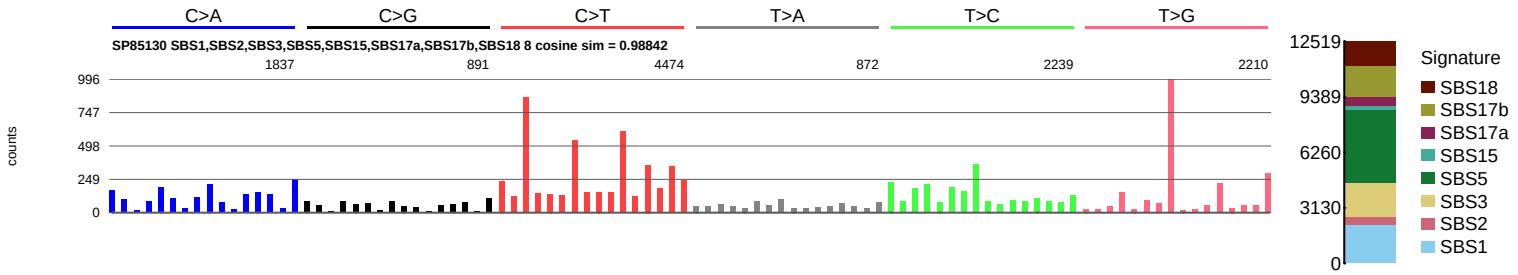
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS44 9 cosine sim = 0.98843
1846
870
4455
872
2278
2196
counts
996
747
498
249
0
12519
9389
6260
3130
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18 8 cosine sim = 0.98842
1837
891
4474
872
2239
2210
counts
996
747
498
249
0
12519
9389
6260
3130
0
Signature
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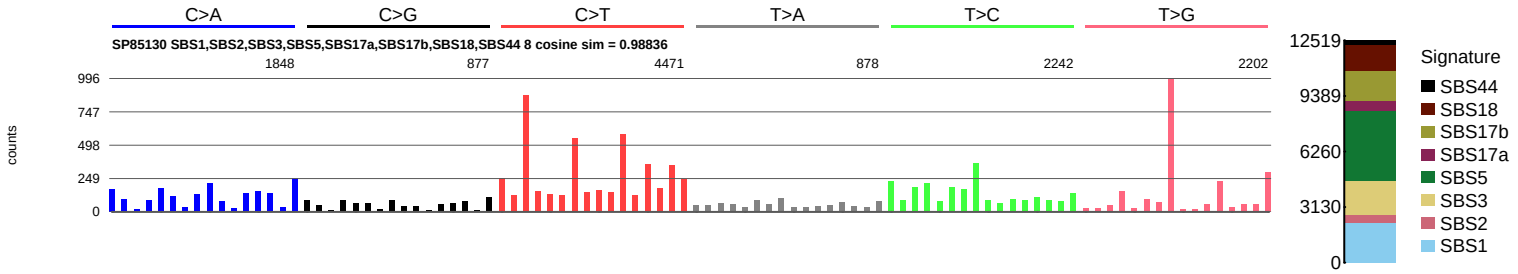
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS44 8 cosine sim = 0.98836
1848
877
4471
878
2242
2202
counts
996
747
498
249
0
12519
9389
6260
3130
0
Signature
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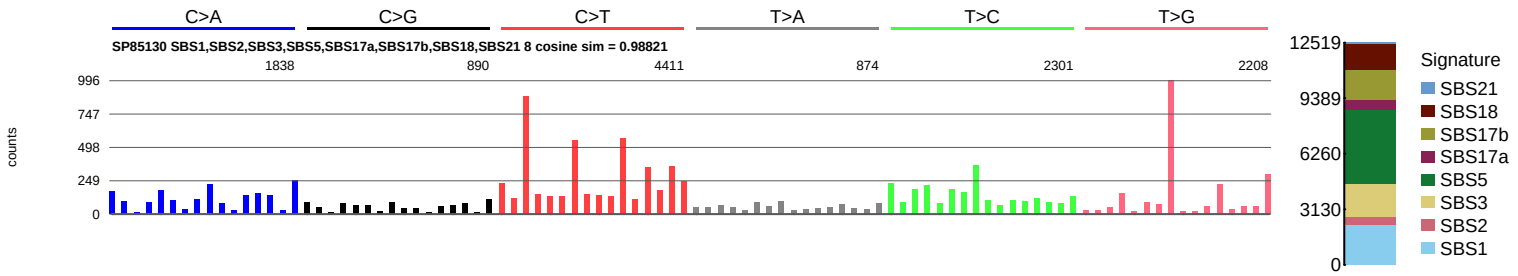
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS21 8 cosine sim = 0.98821
1838
890
4411
874
2301
2208
counts
996
747
498
249
0
12519
9389
6260
3130
0
Signature
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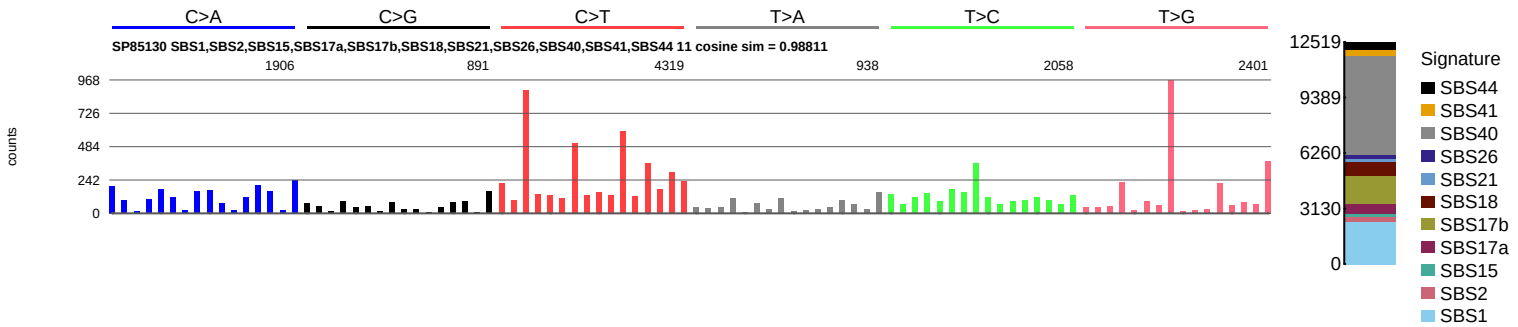
SBS2
SBS1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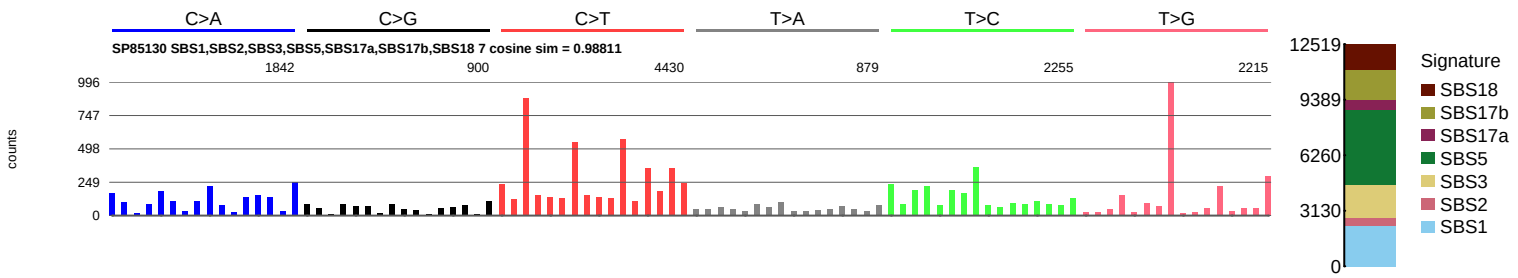
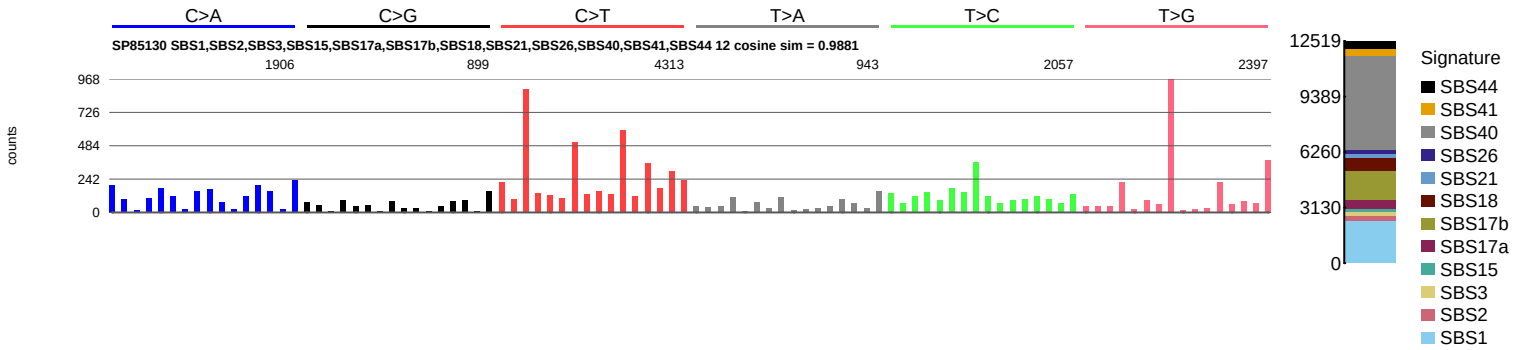
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41,SBS44 11 cosine sim = 0.98811
1906
891
4319
938
2058
2401
counts
968
726
484
242
0
12519
9389
6260
3130
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18 7 cosine sim = 0.98811
1842
900
4430
879
2255
2215
counts
996
747
498
249
0
12519
9389
6260
3130
0
Signature
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1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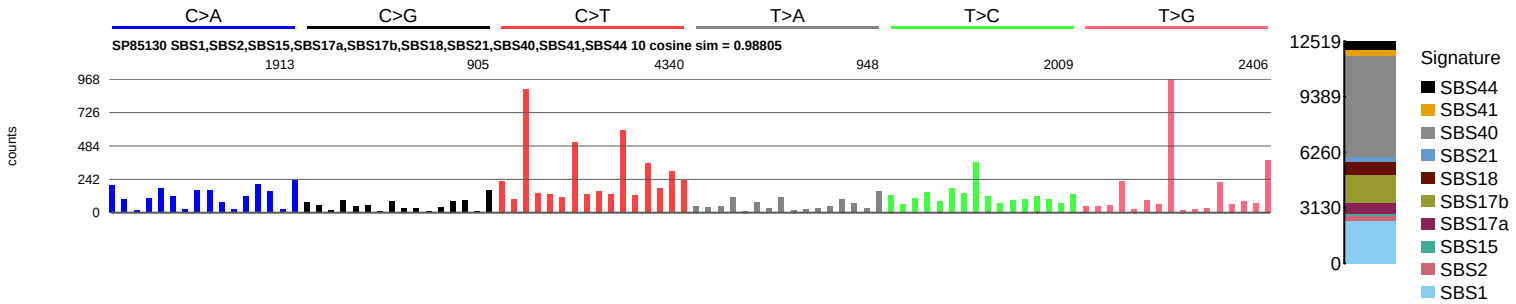
SP85130 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41,SBS44 12 cosine sim = 0.9881
1906
899
4313
943
2057
2397
counts
968
726
484
242
0
12519
9389
6260
3130
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1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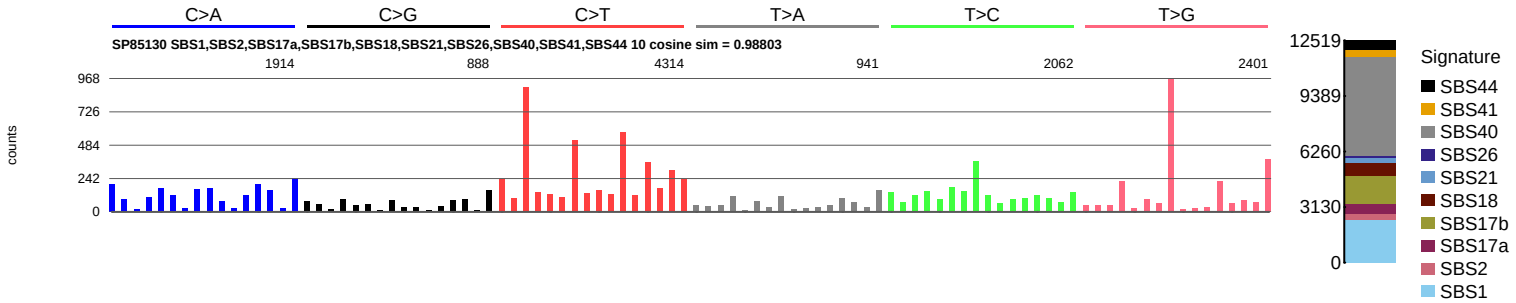
SP85130 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 10 cosine sim = 0.98805
1913
905
4340
948
2009
2406
counts
968
726
484
242
0
12519
9389
6260
3130
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1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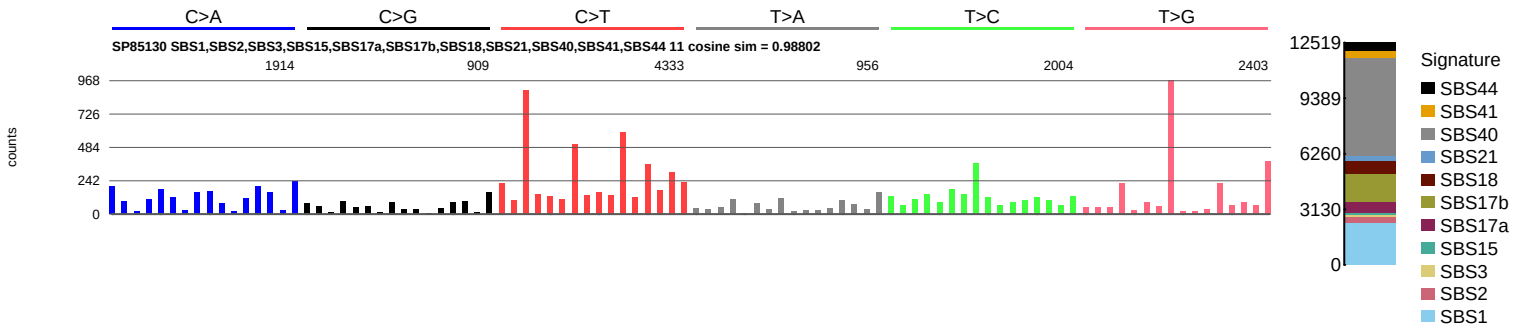
SP85130 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.98803
1914
888
4314
941
2062
2401
counts
968
726
484
242
0
12519
9389
6260
3130
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS2
SBS1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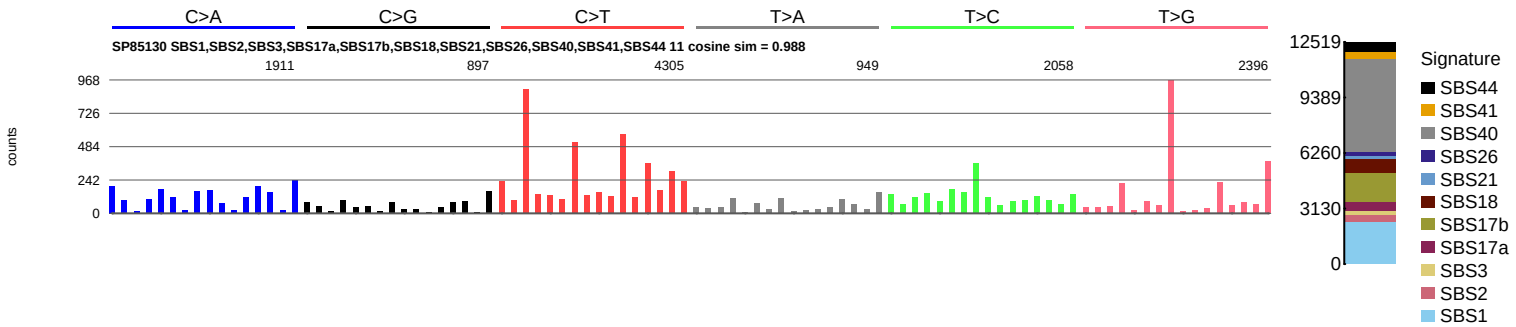
SP85130 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 11 cosine sim = 0.98802
1914
909
4333
956
2004
2403
counts
968
726
484
242
0
12519
9389
6260
3130
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1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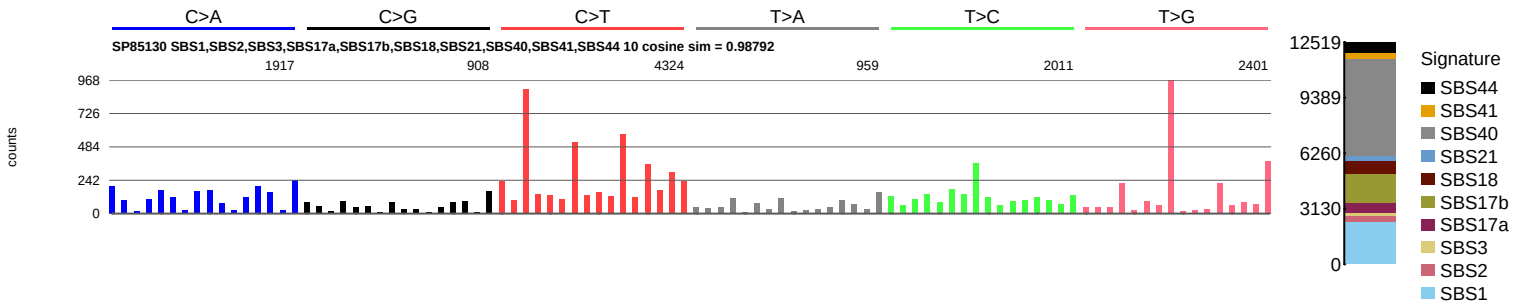
T>G
SP85130 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41,SBS44 11 cosine sim = 0.988
1911
897
4305
949
2058
2396
counts
968
726
484
242
0
12519
9389
6260
3130
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1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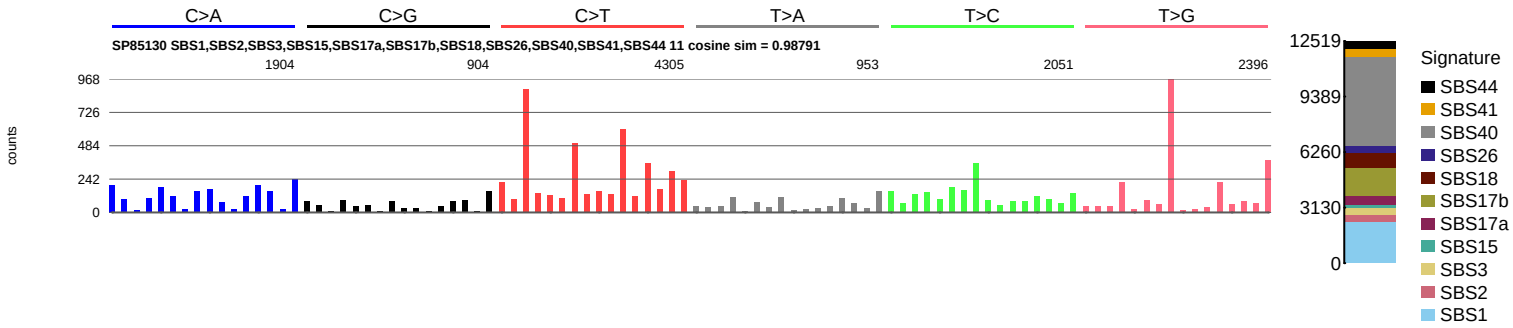
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 10 cosine sim = 0.98792
1917
908
4324
959
2011
2401
counts
968
726
484
242
0
12519
9389
6260
3130
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1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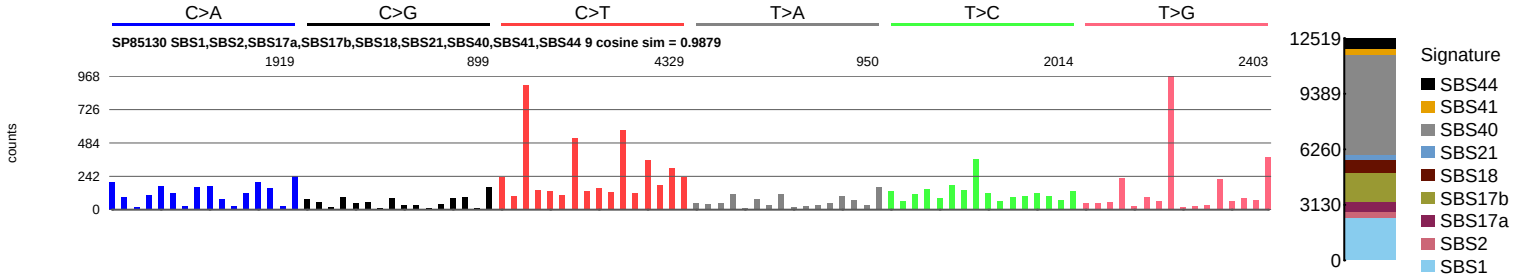
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 11 cosine sim = 0.98791
1904
904
4305
953
2051
2396
counts
968
726
484
242
0
12519
9389
6260
3130
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1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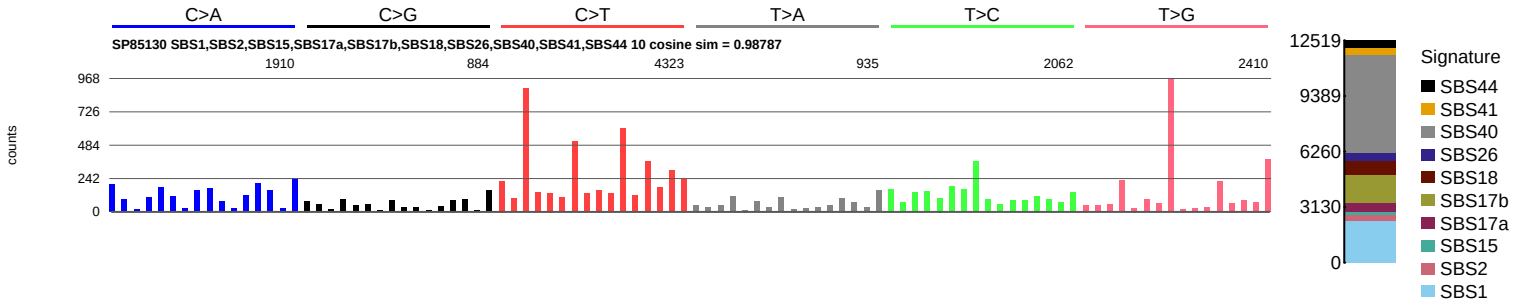
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 9 cosine sim = 0.9879
1919
899
4329
950
2014
2403
counts
968
726
484
242
0
12519
9389
6260
3130
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS2
SBS1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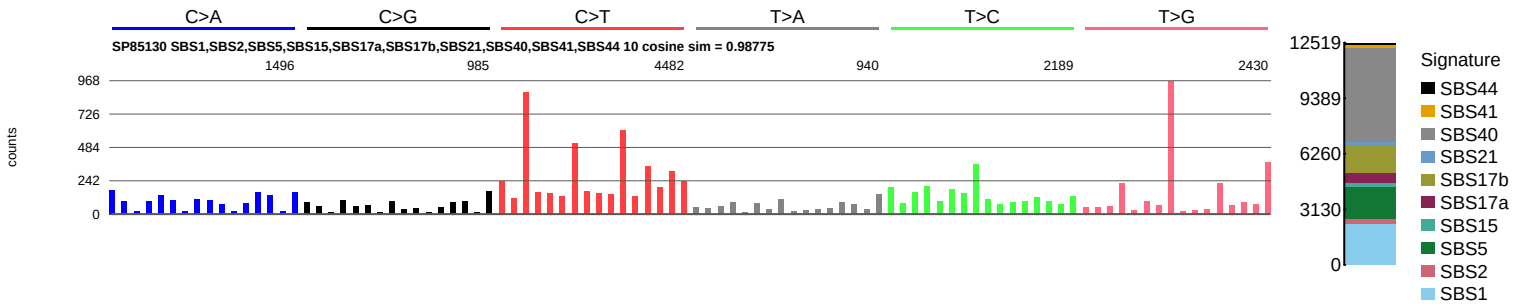
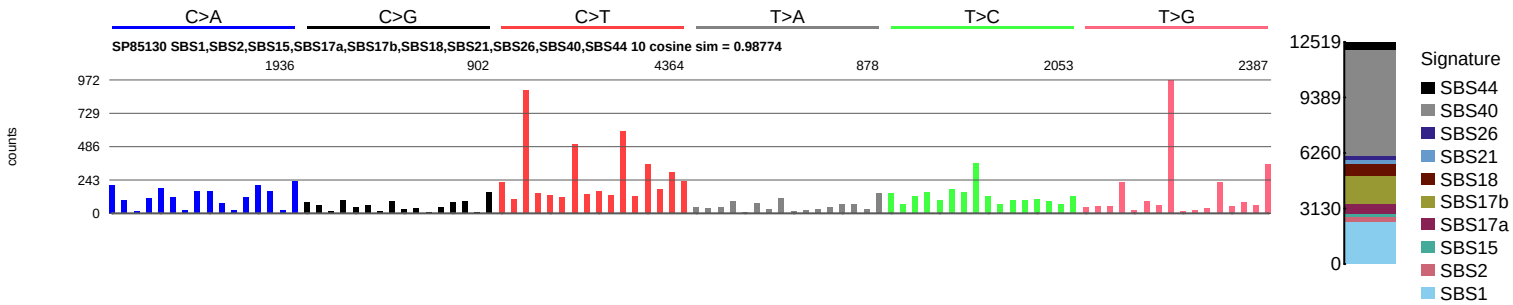
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.98787
1910
884
4323
935
2062
2410
counts
968
726
484
242
0
12519
9389
6260
3130
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS21,SBS40,SBS41,SBS44 10 cosine sim = 0.98775
1496
985
4482
940
2189
2430
counts
968
726
484
242
0
12519
9389
6260
3130
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1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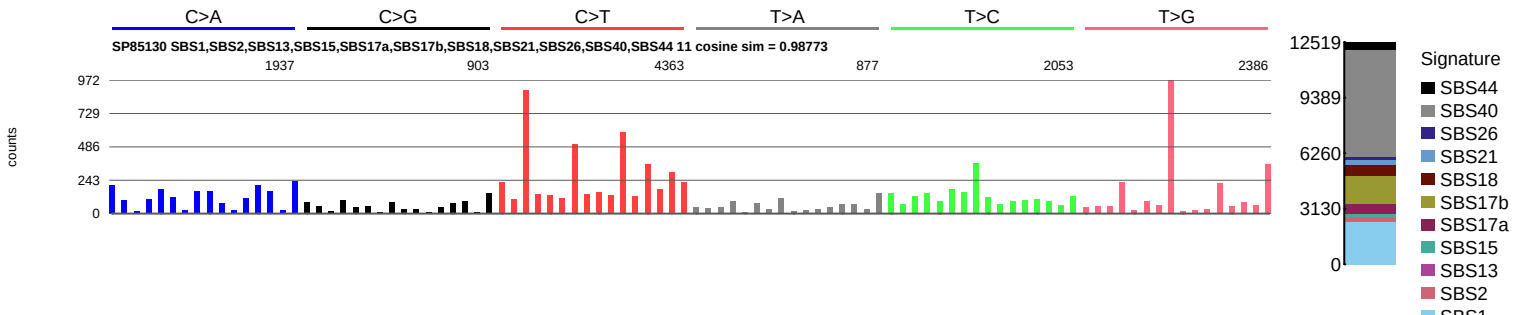
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS44 10 cosine sim = 0.98774
1936
902
4364
878
2053
2387
counts
972
729
486
243
0
12519
9389
6260
3130
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1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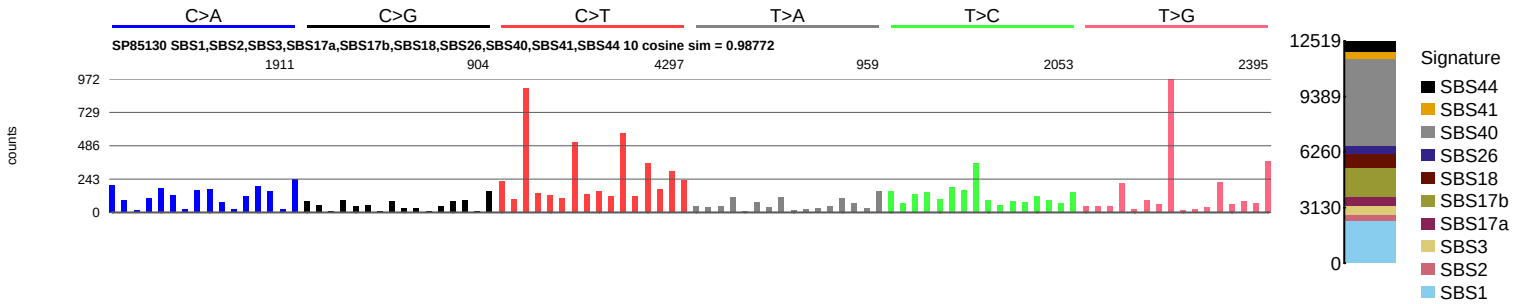
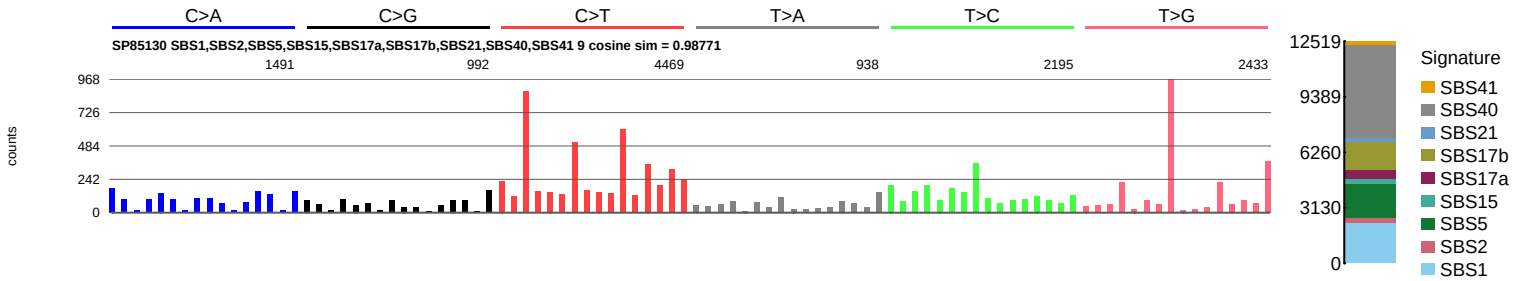
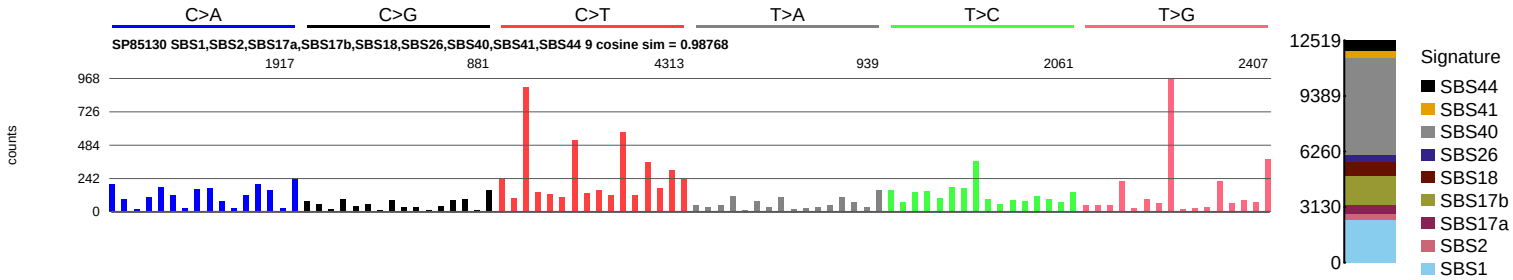
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS44 11 cosine sim = 0.98773
1937
903
4363
877
2053
2386
counts
972
729
486
243
0
12519
9389
6260
3130
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.98772
1911
904
4297
959
2053
2395
counts
972
729
486
243
0
12519
9389
6260
3130
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS21,SBS40,SBS41 9 cosine sim = 0.98771
1491
992
4469
938
2195
2433
counts
968
726
484
242
0
12519
9389
6260
3130
0
Signature
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1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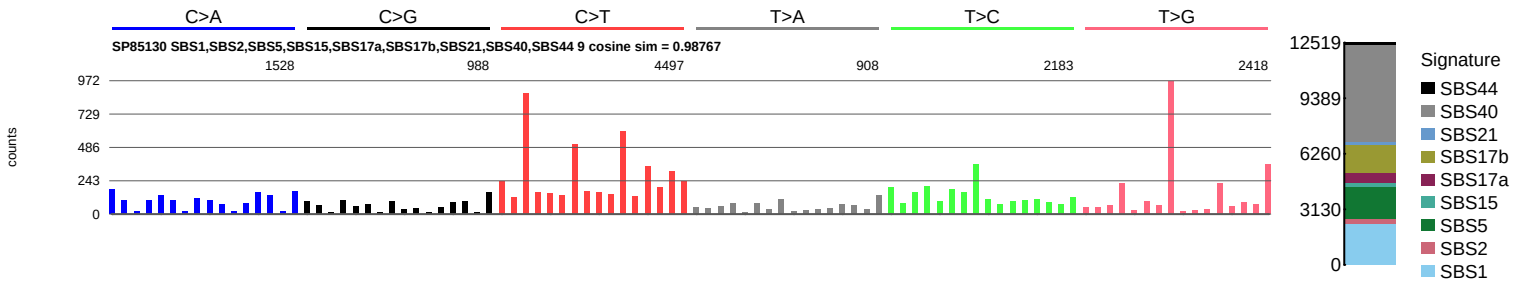
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 9 cosine sim = 0.98768
1917
881
4313
939
2061
2407
counts
968
726
484
242
0
12519
9389
6260
3130
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS2
SBS1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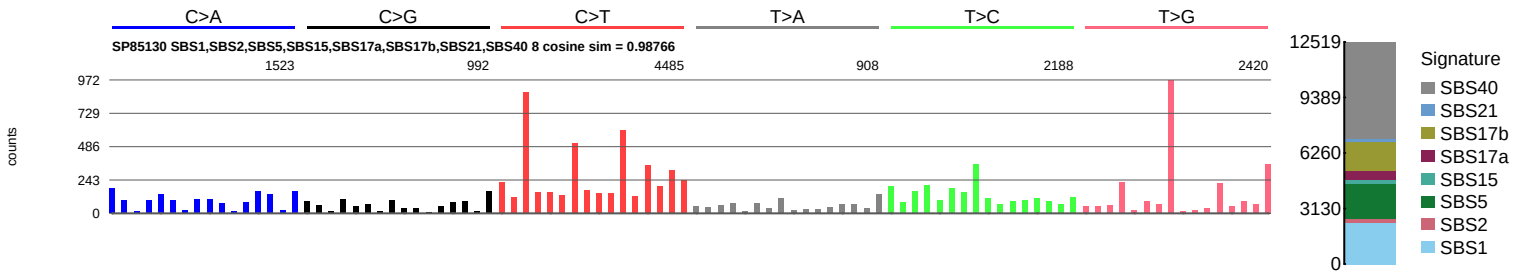
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS21,SBS40,SBS44 9 cosine sim = 0.98767
1528
988
4497
908
2183
2418
counts
972
729
486
243
0
12519
9389
6260
3130
0
Signature
SBS44
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1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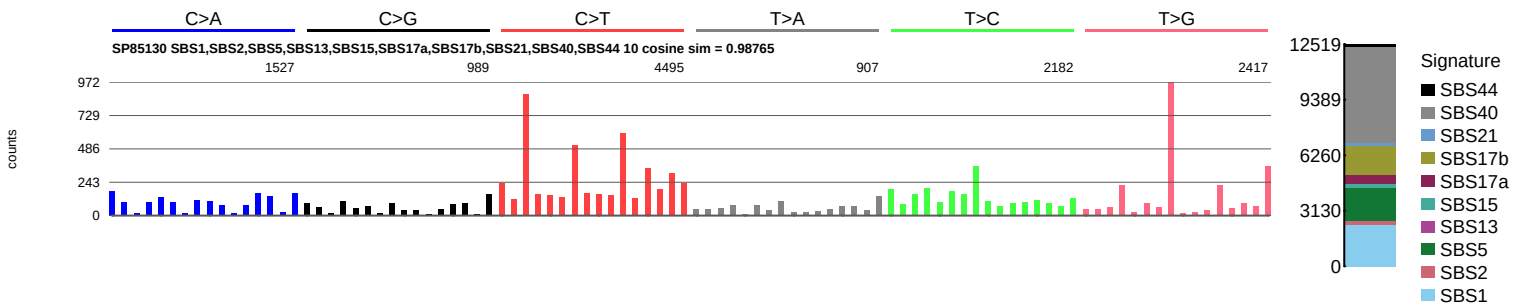
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS21,SBS40 8 cosine sim = 0.98766
1523
992
4485
908
2188
2420
counts
972
729
486
243
0
12519
9389
6260
3130
0
Signature
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1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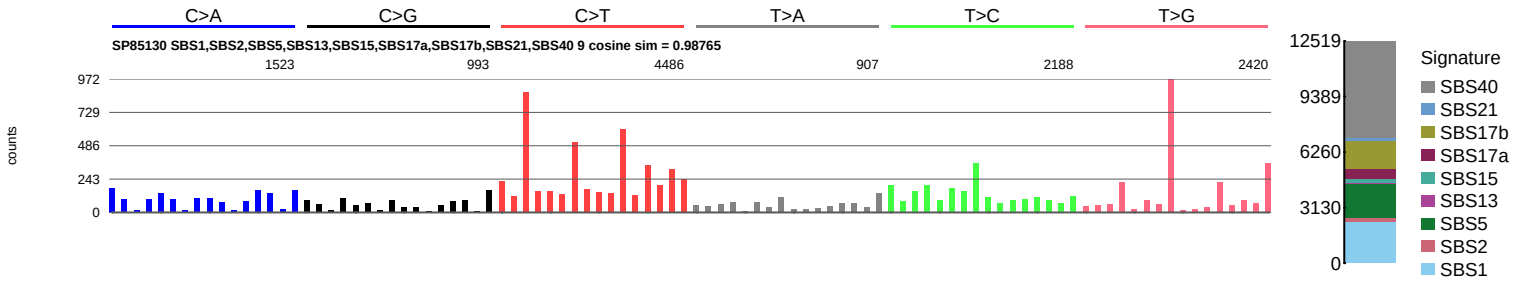
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS40,SBS44 10 cosine sim = 0.98765
1527
989
4495
907
2182
2417
counts
972
729
486
243
0
12519
9389
6260
3130
0
Signature
SBS44
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1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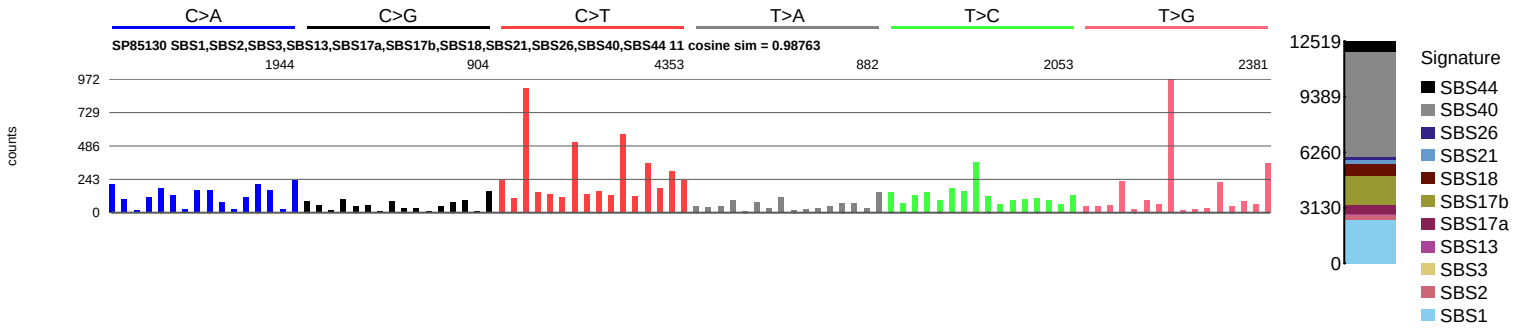
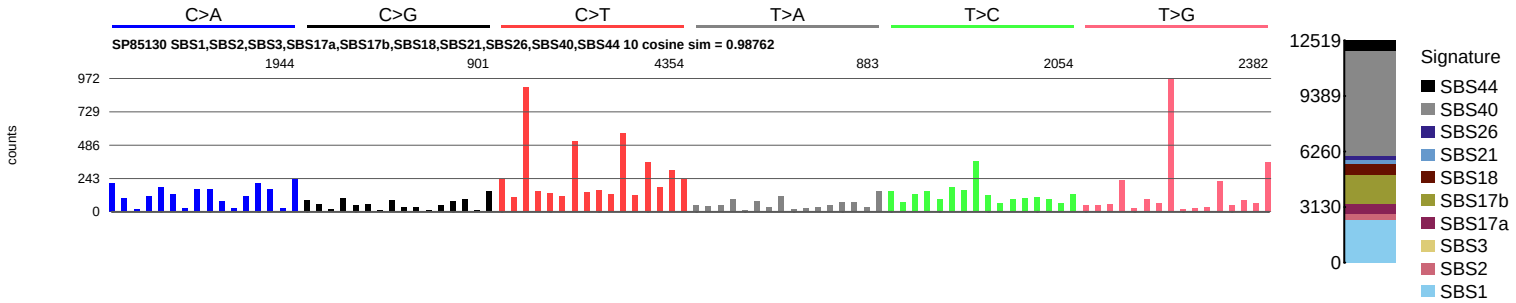
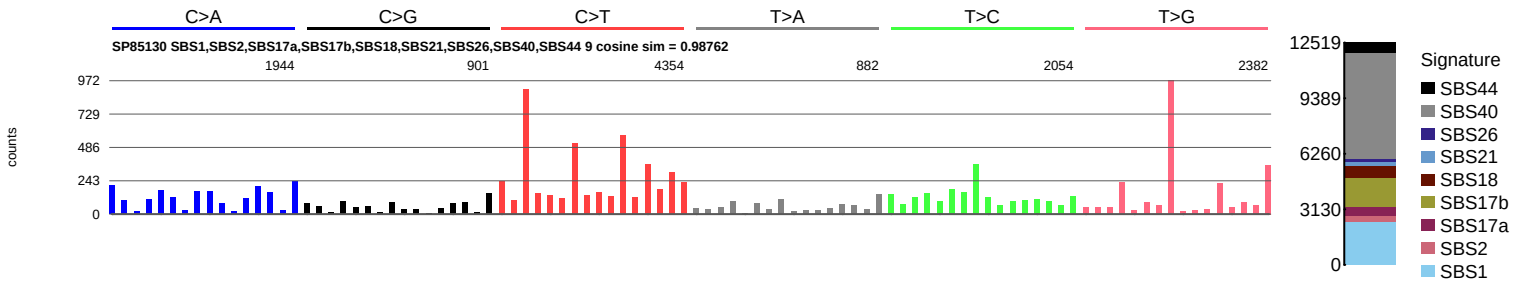
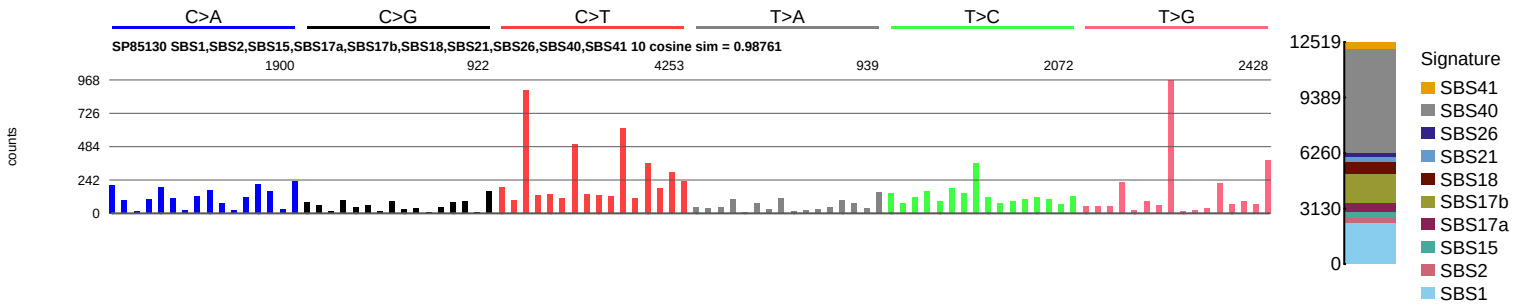
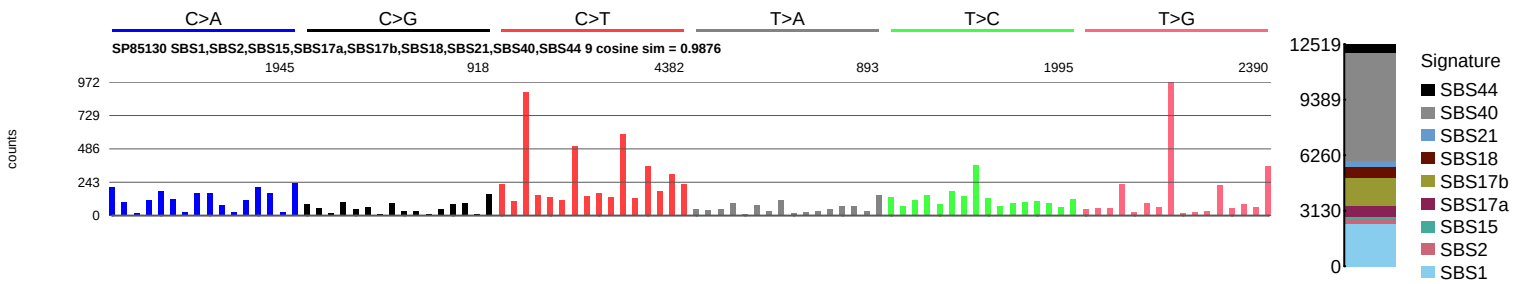
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS40 9 cosine sim = 0.98765
1523
993
4486
907
2188
2420
counts
972
729
486
243
0
12519
9389
6260
3130
0
Signature
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS44 11 cosine sim = 0.98763
1944
904
4353
882
2053
2381
counts
972
729
486
243
0
12519
9389
6260
3130
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS44 10 cosine sim = 0.98762
1944
901
4354
883
2054
2382
counts
972
729
486
243
0
12519
9389
6260
3130
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS44 9 cosine sim = 0.98762
1944
901
4354
882
2054
2382
counts
972
729
486
243
0
12519
9389
6260
3130
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41 10 cosine sim = 0.98761
1900
922
4253
939
2072
2428
counts
968
726
484
242
0
12519
9389
6260
3130
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 9 cosine sim = 0.9876
1945
918
4382
893
1995
2390
counts
972
729
486
243
0
12519
9389
6260
3130
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Signature
SBS44
SBS40
SBS21
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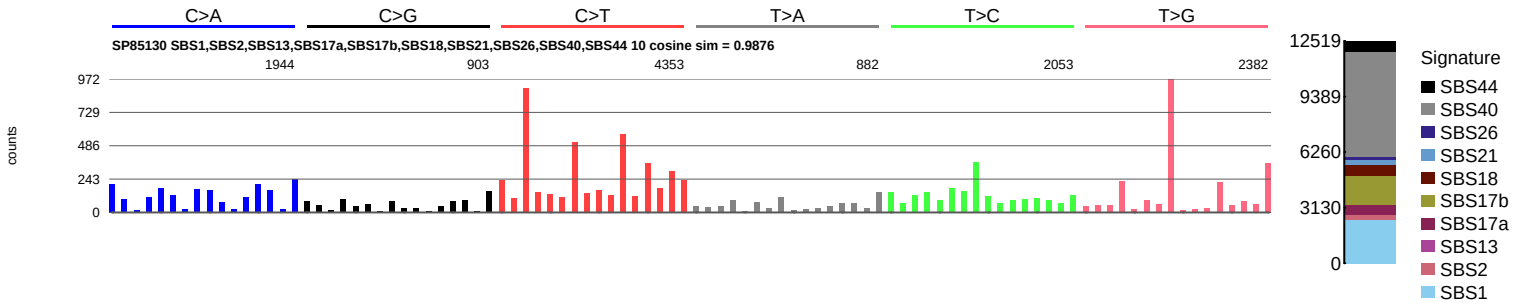
SBS1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T>C
T>G
SP85130 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS44 10 cosine sim = 0.9876
1944
903
4353
882
2053
2382
counts
972
729
486
243
0
12519
9389
6260
3130
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Signature
SBS44
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SBS2
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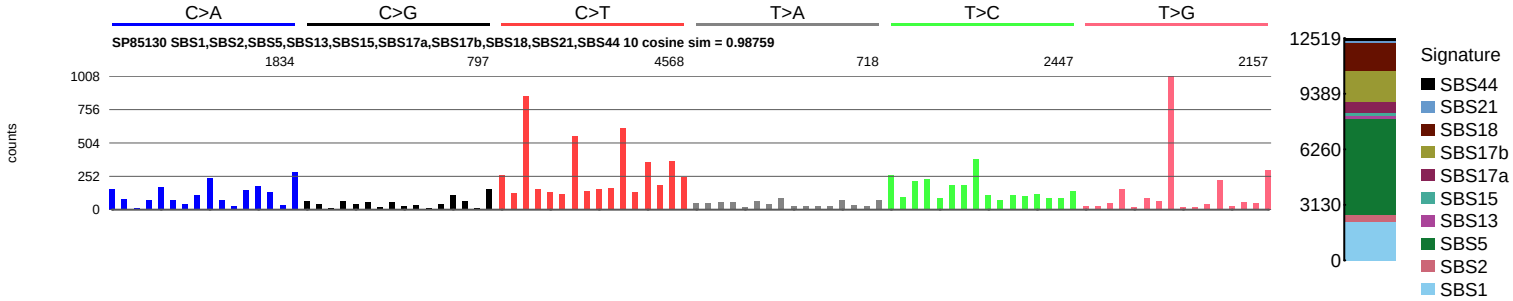
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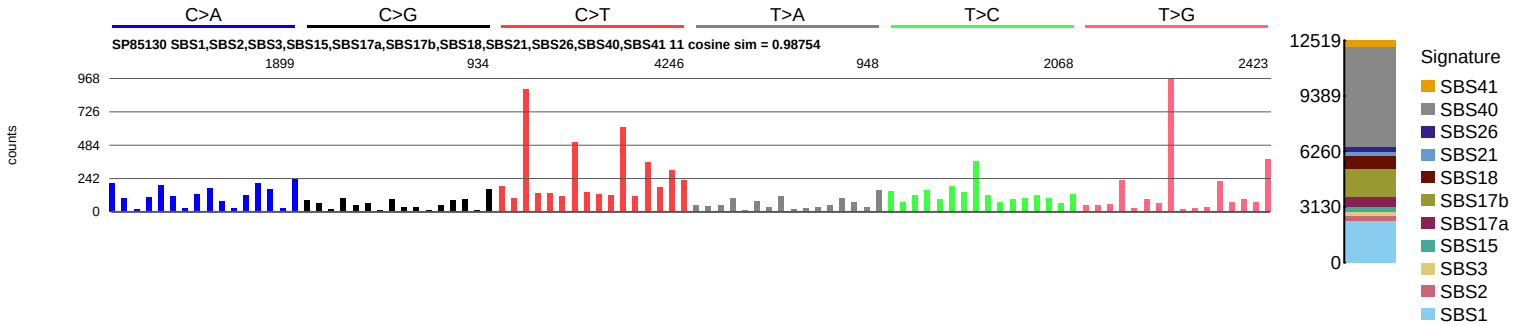
T>G
SP85130 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS44 10 cosine sim = 0.98759
1834
797
4568
718
2447
2157
counts
1008
756
504
252
0
12519
9389
6260
3130
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1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C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41 11 cosine sim = 0.98754
1899
934
4246
948
2068
2423
counts
968
726
484
242
0
12519
9389
6260
3130
0
Signature
SBS41
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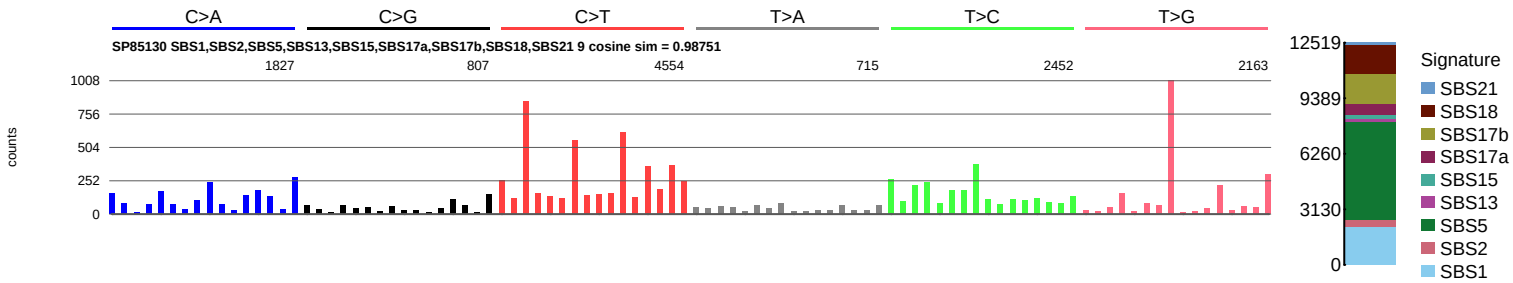
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1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C>T
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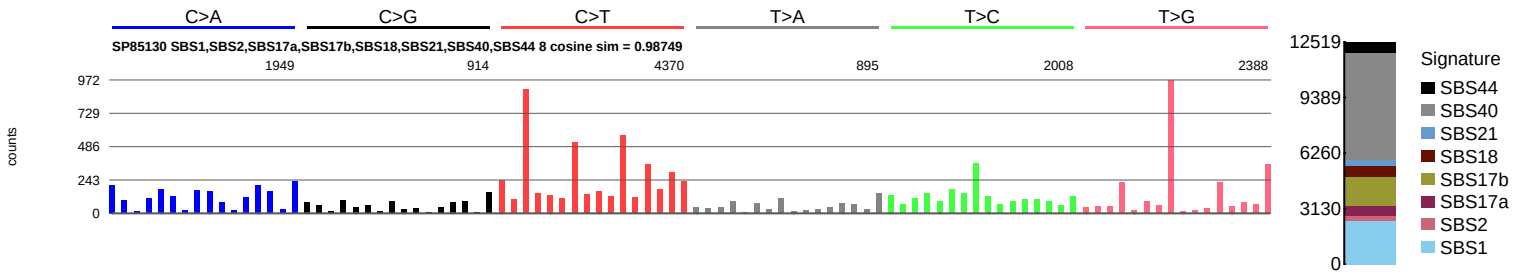
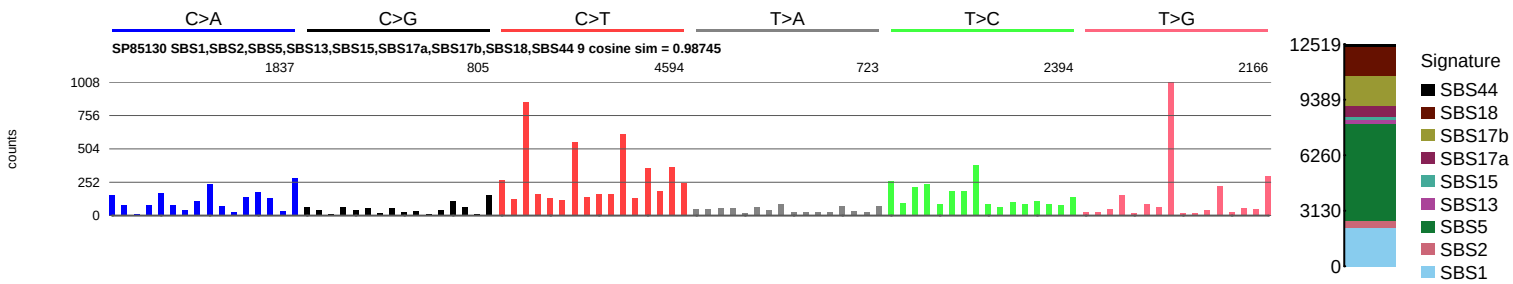
T>C
T>G
SP85130 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21 9 cosine sim = 0.98751
1827
807
4554
715
2452
2163
counts
1008
756
504
252
0
12519
9389
6260
3130
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 8 cosine sim = 0.98749
1949
914
4370
895
2008
2388
counts
972
729
486
243
0
12519
9389
6260
3130
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS2
SBS1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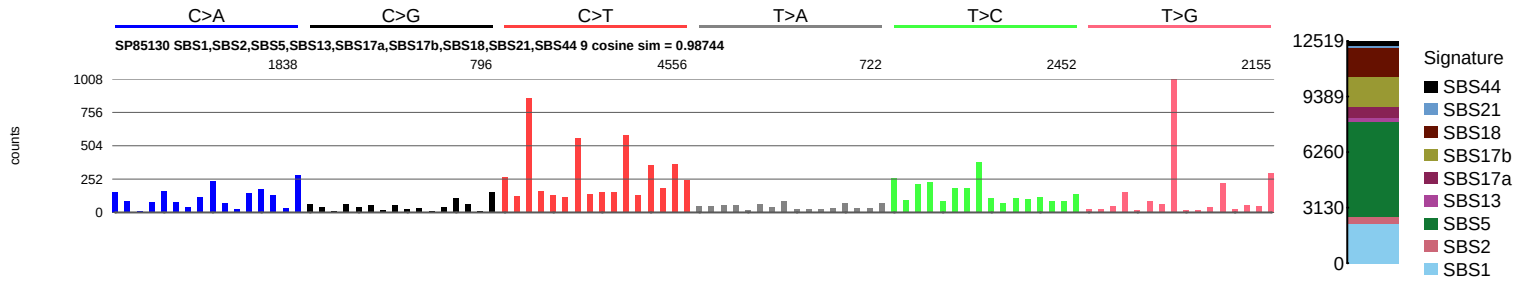
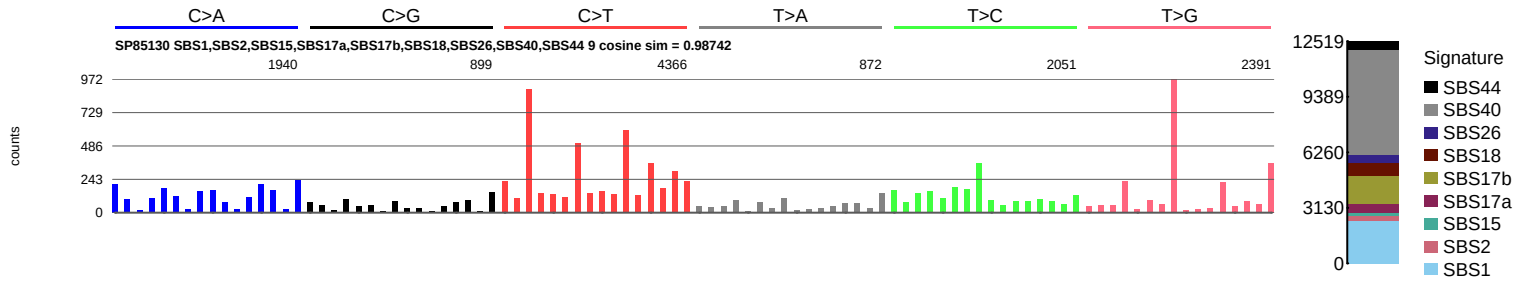
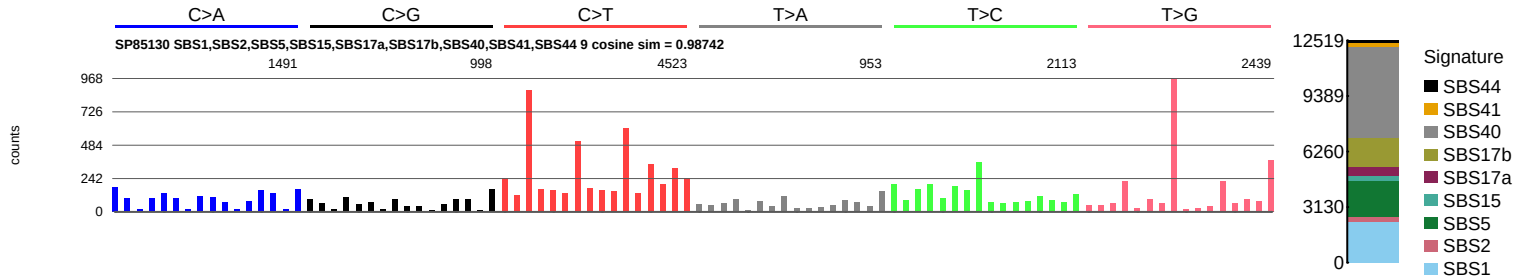
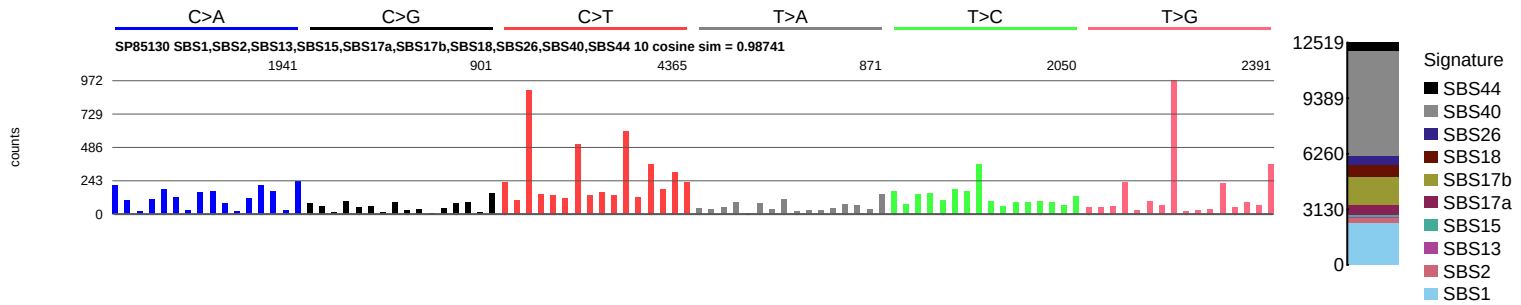
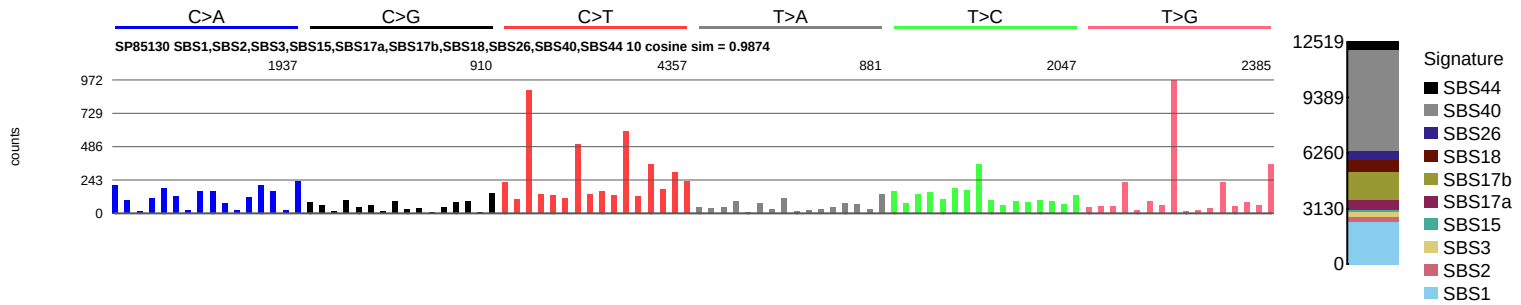
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS44 9 cosine sim = 0.98745
1837
805
4594
723
2394
2166
counts
1008
756
504
252
0
12519
9389
6260
3130
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1

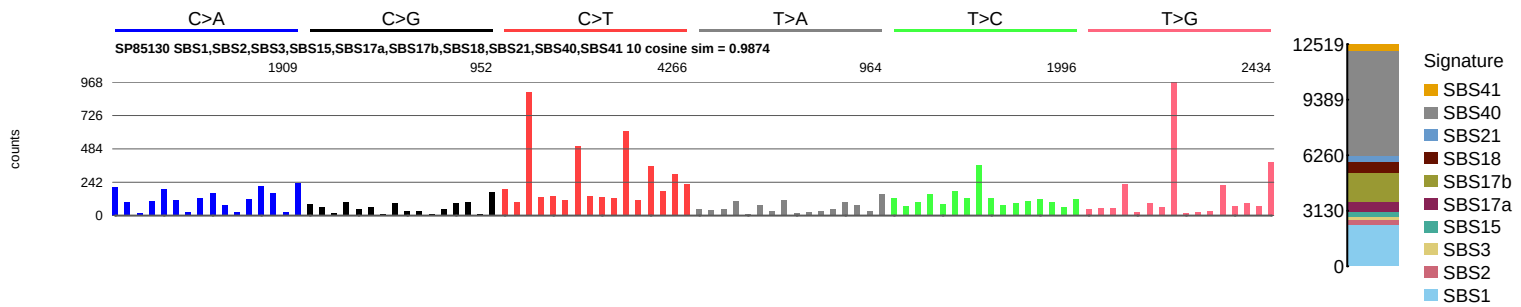

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS44 9 cosine sim = 0.98744
1838
796
4556
722
2452
2155
counts
1008
756
504
252
0
12519
9389
6260
3130
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1

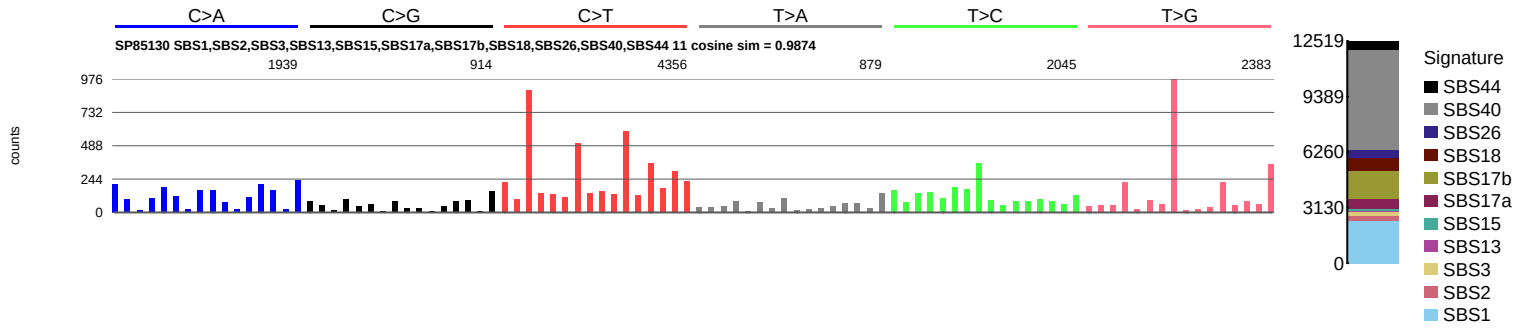

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 9 cosine sim = 0.98742
1940
899
4366
872
2051
2391
counts
972
729
486
243
0
12519
9389
6260
3130
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1

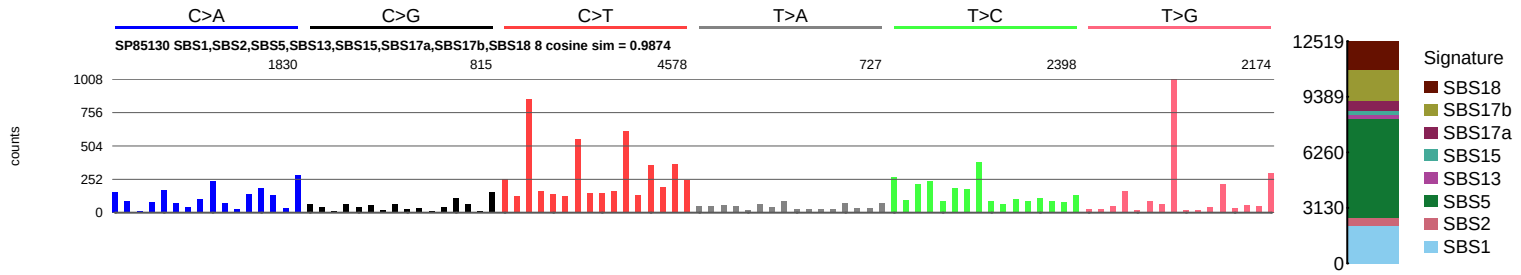

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS40,SBS41,SBS44 9 cosine sim = 0.98742
1491
998
4523
953
2113
2439
counts
968
726
484
242
0
12519
9389
6260
3130
0
Signature
SBS44
SBS41
SBS40
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 10 cosine sim = 0.98741
1941
901
4365
871
2050
2391
counts
972
729
486
243
0
12519
9389
6260
3130
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 10 cosine sim = 0.9874
1937
910
4357
881
2047
2385
counts
972
729
486
243
0
12519
9389
6260
3130
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 10 cosine sim = 0.9874
1909
952
4266
964
1996
2434
counts
968
726
484
242
0
12519
9389
6260
3130
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 11 cosine sim = 0.9874
1939
914
4356
879
2045
2383
counts
976
732
488
244
0
12519
9389
6260
3130
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1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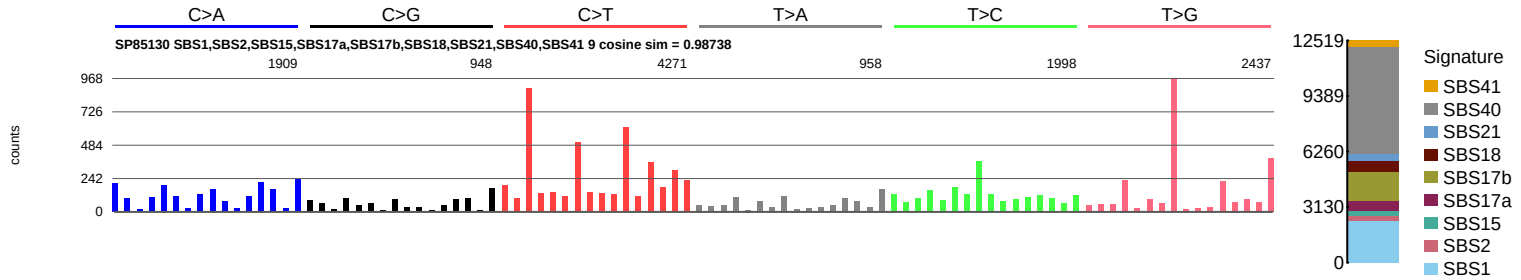
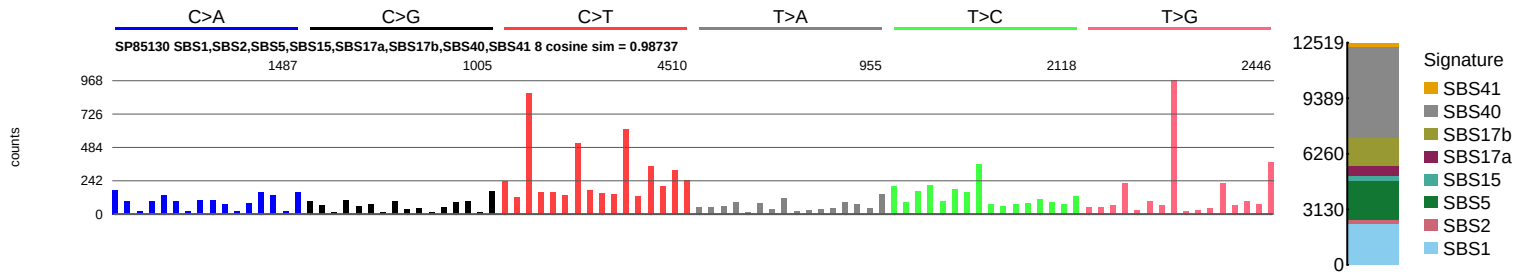
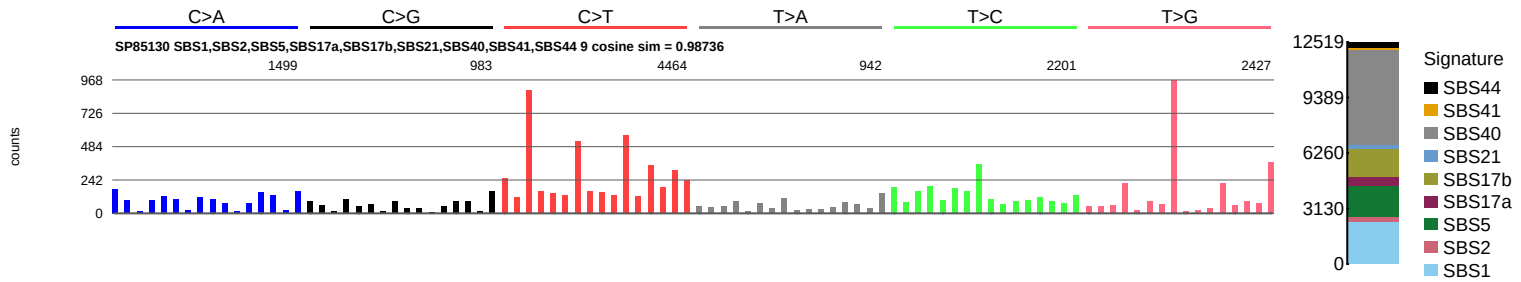
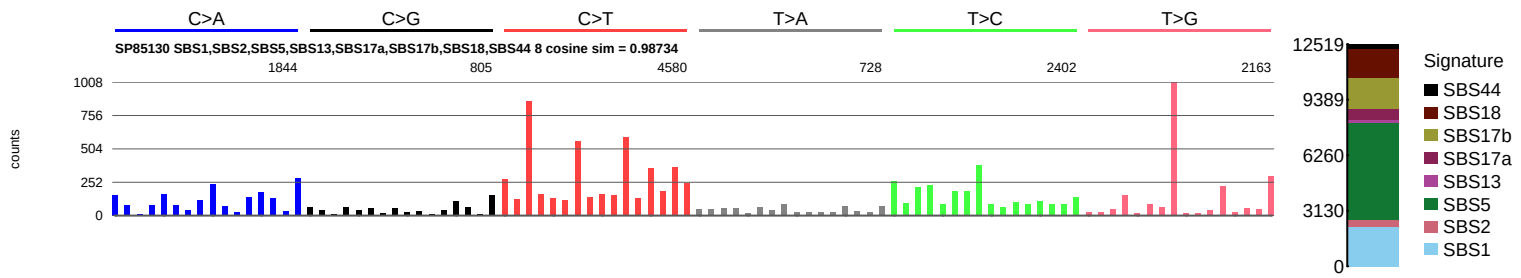
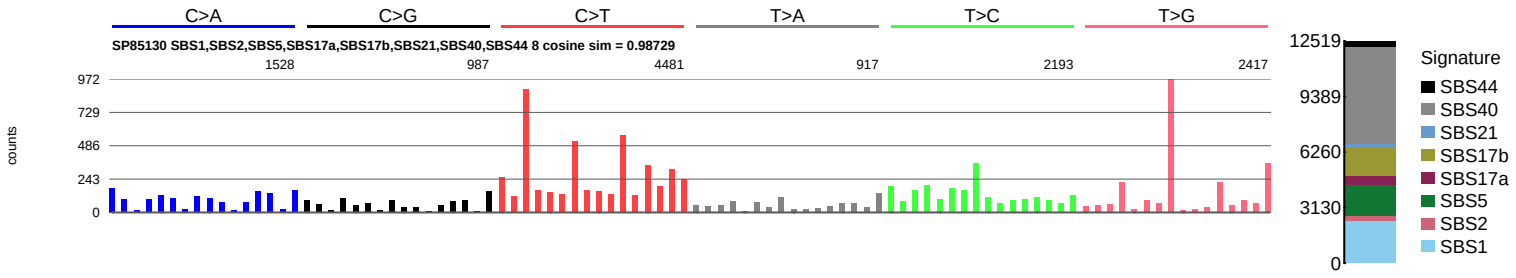
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18 8 cosine sim = 0.9874
1830
815
4578
727
2398
2174
counts
1008
756
504
252
0
12519
9389
6260
3130
0
Signature
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1

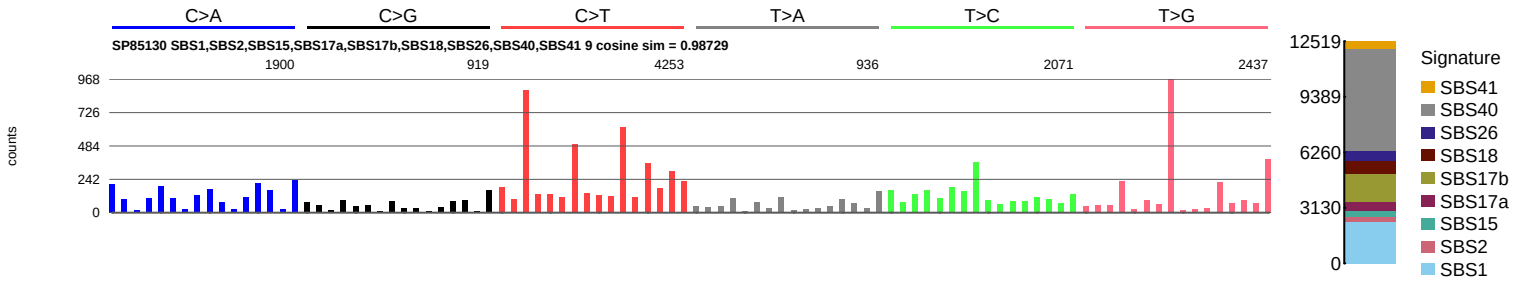

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 9 cosine sim = 0.98738
1909
948
4271
958
1998
2437
counts
968
726
484
242
0
12519
9389
6260
3130
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1

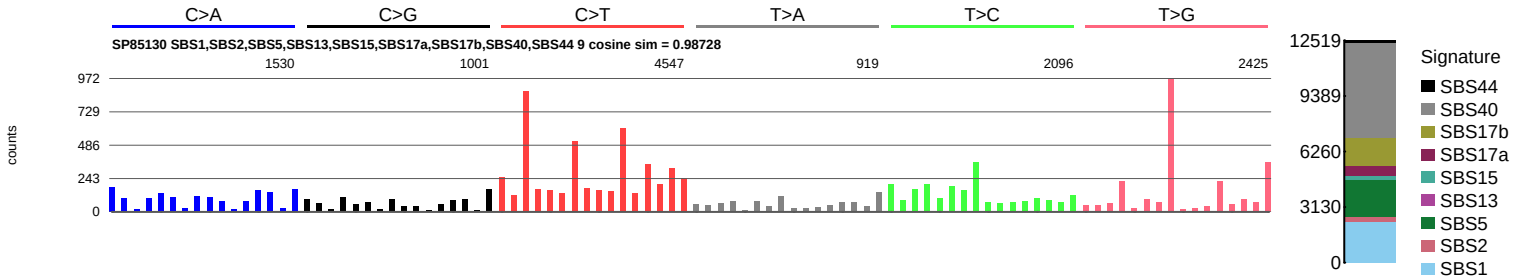

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS40,SBS41 8 cosine sim = 0.98737
1487
1005
4510
955
2118
2446
counts
968
726
484
242
0
12519
9389
6260
3130
0
Signature
SBS41
SBS40
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1

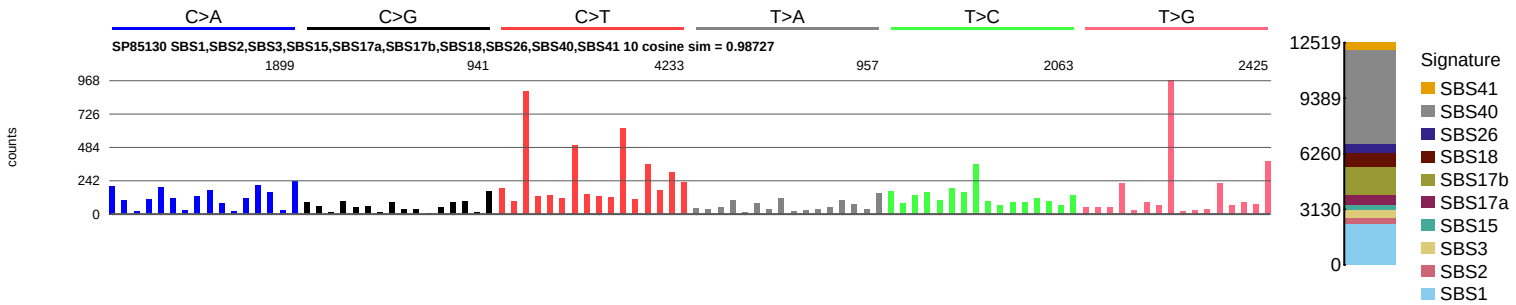

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS21,SBS40,SBS41,SBS44 9 cosine sim = 0.98736
1499
983
4464
942
2201
2427
counts
968
726
484
242
0
12519
9389
6260
3130
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS5
SBS2
SBS1

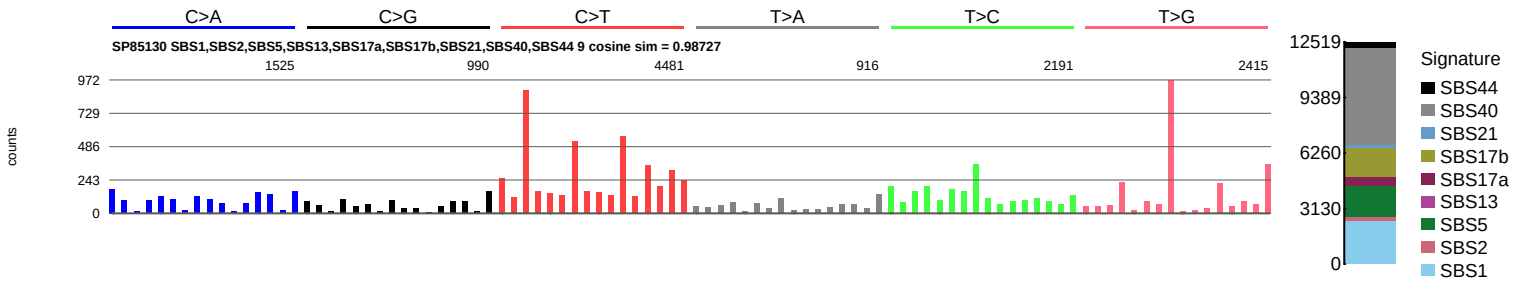

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS44 8 cosine sim = 0.98734
1844
805
4580
728
2402
2163
counts
1008
756
504
252
0
12519
9389
6260
3130
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1

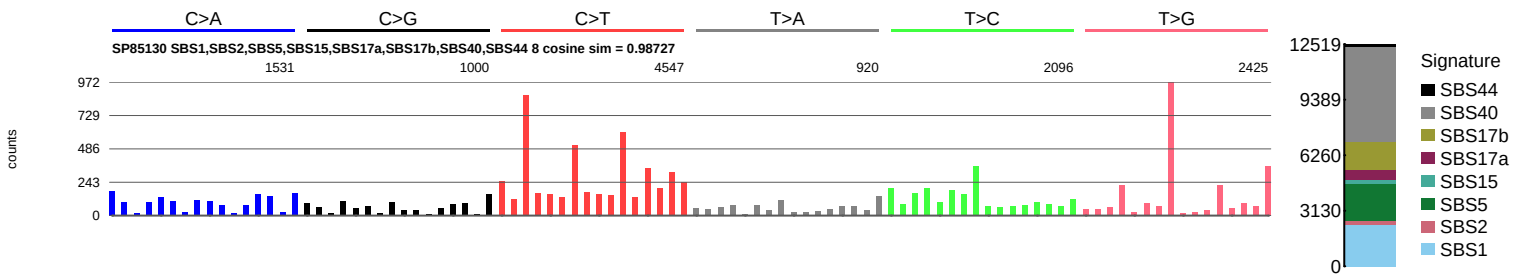

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS21,SBS40,SBS44 8 cosine sim = 0.98729
1528
987
4481
917
2193
2417
counts
972
729
486
243
0
12519
9389
6260
3130
0
Signature
SBS44
SBS40
SBS21
SBS17b
SBS17a
SBS5
SBS2
SBS1

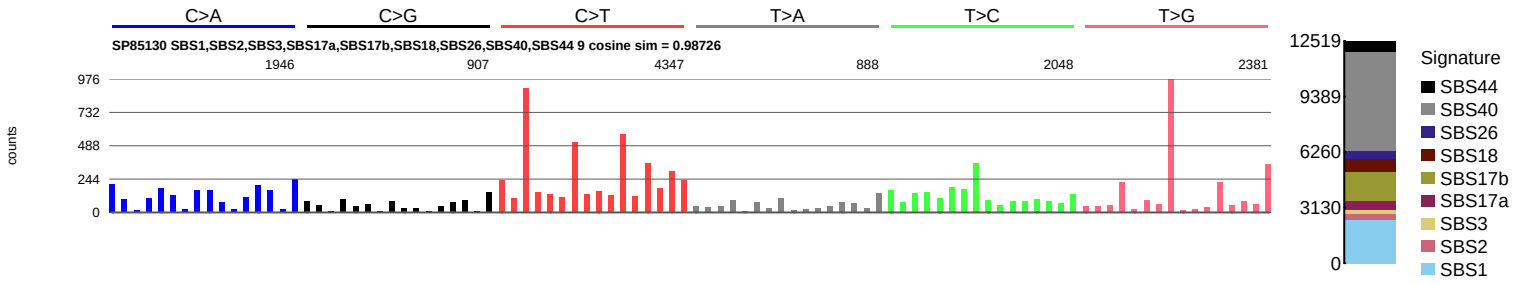

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 9 cosine sim = 0.98729
1900
919
4253
936
2071
2437
counts
968
726
484
242
0
12519
9389
6260
3130
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS40,SBS44 9 cosine sim = 0.98728
1530
1001
4547
919
2096
2425
counts
972
729
486
243
0
12519
9389
6260
3130
0
Signature
SBS44
SBS40
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 10 cosine sim = 0.98727
1899
941
4233
957
2063
2425
counts
968
726
484
242
0
12519
9389
6260
3130
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS21,SBS40,SBS44 9 cosine sim = 0.98727
1525
990
4481
916
2191
2415
counts
972
729
486
243
0
12519
9389
6260
3130
0
Signature
SBS44
SBS40
SBS21
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS40,SBS44 8 cosine sim = 0.98727
1531
1000
4547
920
2096
2425
counts
972
729
486
243
0
12519
9389
6260
3130
0
Signature
SBS44
SBS40
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1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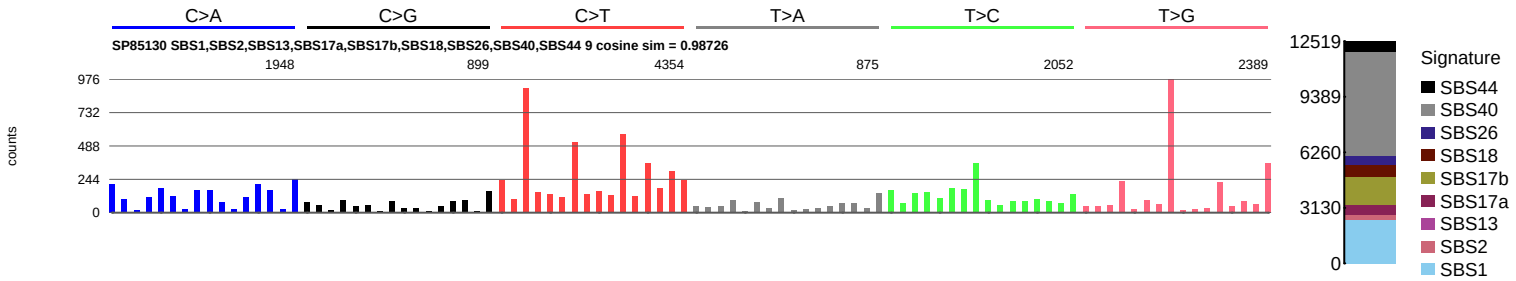
T>G
SP85130 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 9 cosine sim = 0.98726
1946
907
4347
888
2048
2381
976
732
488
244
0
counts
12519
9389
6260
3130
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1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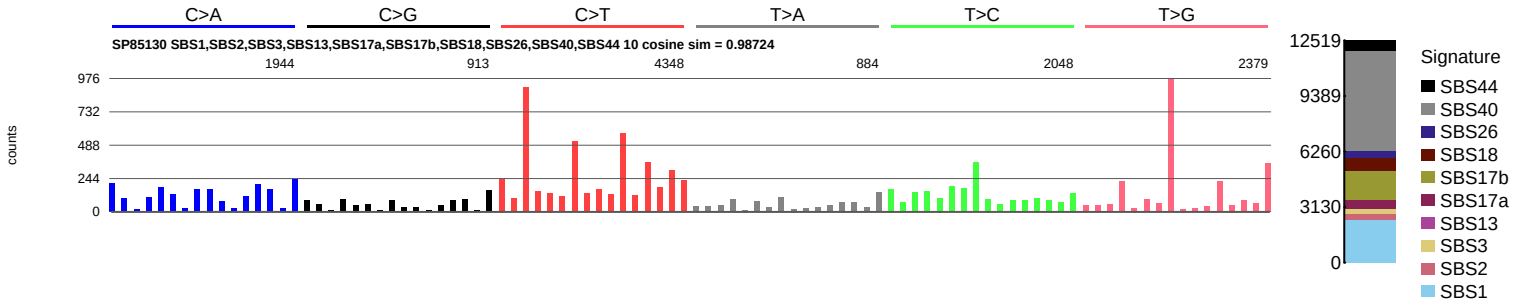
T>C
T>G
SP85130 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 9 cosine sim = 0.98726
1948
899
4354
875
2052
2389
976
732
488
244
0
counts
12519
9389
6260
3130
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1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SP85130 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 10 cosine sim = 0.98724
1944
913
4348
884
2048
2379
976
732
488
244
0
counts
12519
9389
6260
3130
0
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SBS44
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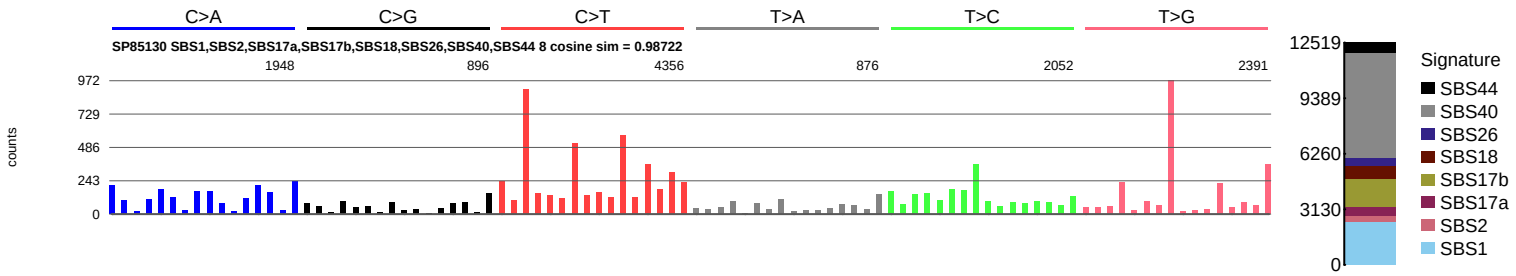
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1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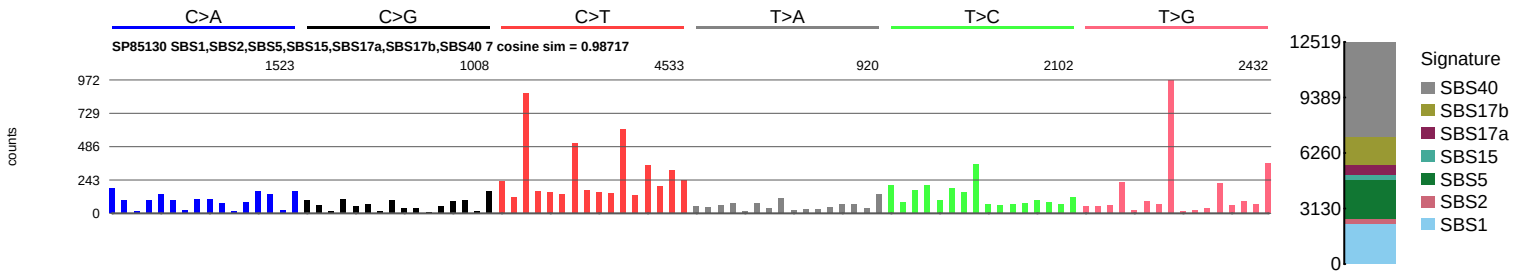
SP85130 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 8 cosine sim = 0.98722
1948
896
4356
876
2052
2391
972
729
486
243
0
counts
12519
9389
6260
3130
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS2
SBS1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C>G
C>T
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T>C
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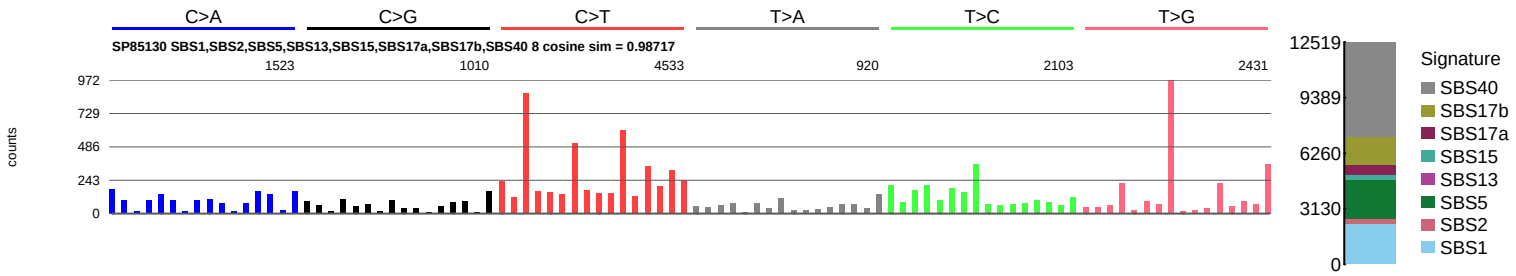
SP85130 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS40 7 cosine sim = 0.98717
1523
1008
4533
920
2102
2432
972
729
486
243
0
counts
12519
9389
6260
3130
0
Signature
SBS40
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1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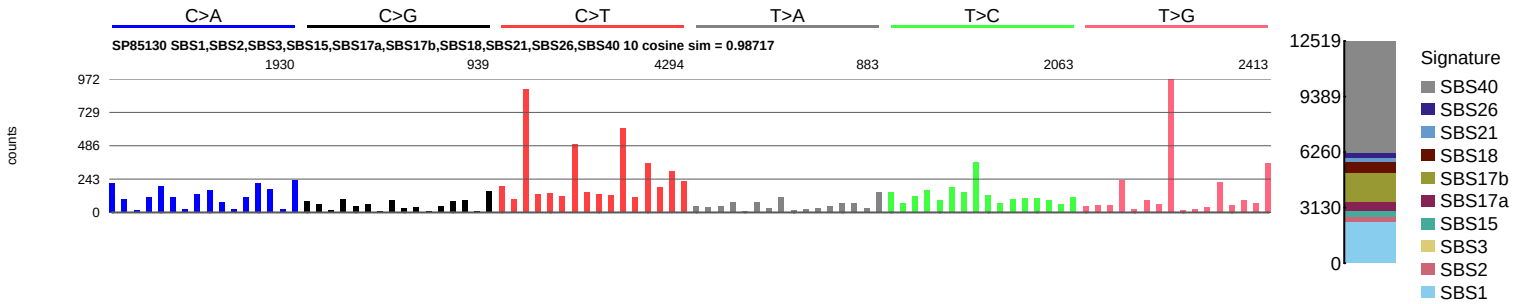
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS40 8 cosine sim = 0.98717
1523
1010
4533
920
2103
2431
972
729
486
243
0
counts
12519
9389
6260
3130
0
Signature
SBS40
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1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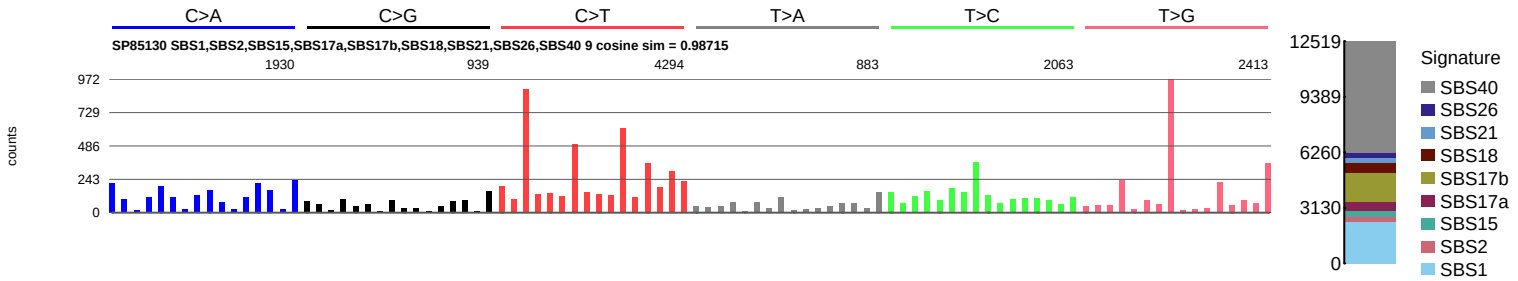
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40 10 cosine sim = 0.98717
1930
939
4294
883
2063
2413
counts
972
729
486
243
0
12519
9389
6260
3130
0
Signature
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1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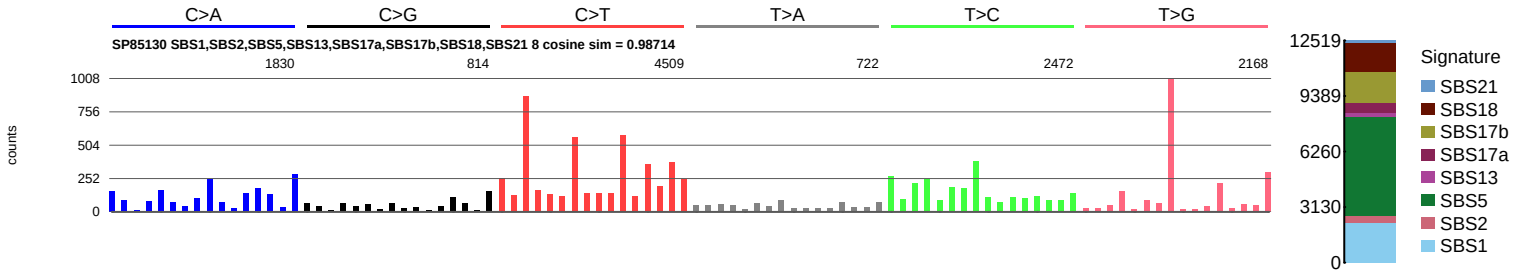
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40 9 cosine sim = 0.98715
1930
939
4294
883
2063
2413
counts
972
729
486
243
0
12519
9389
6260
3130
0
Signature
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1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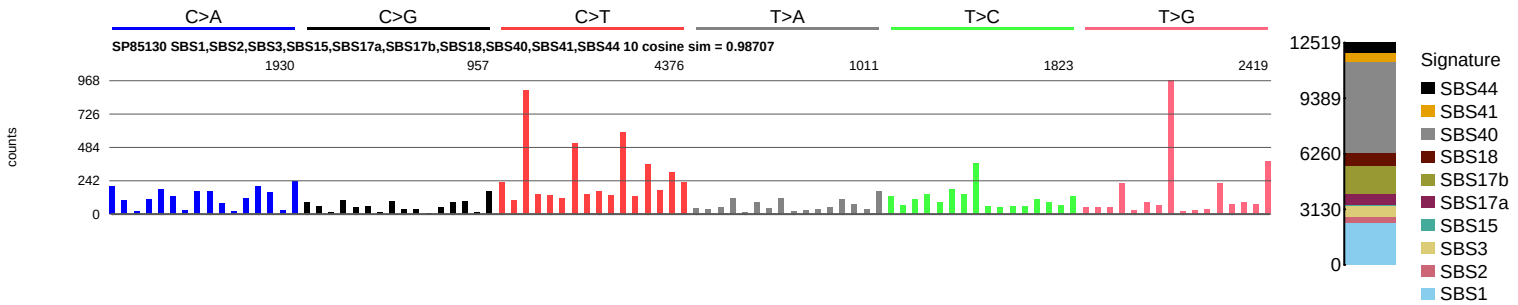
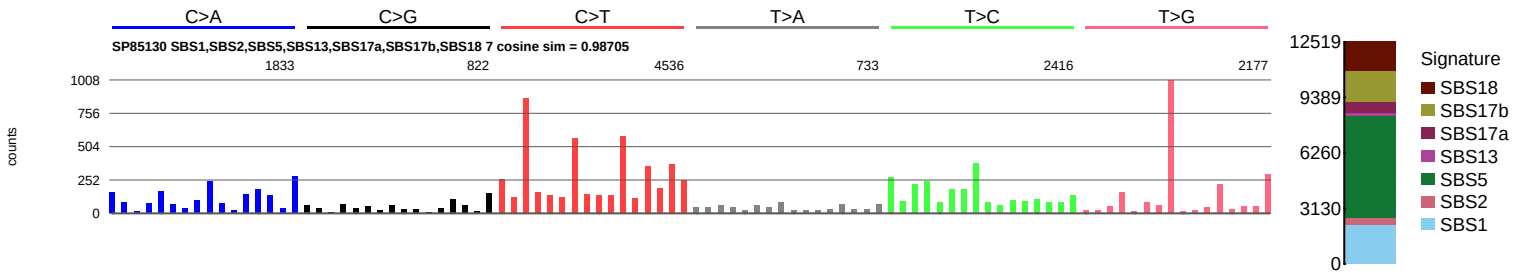
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21 8 cosine sim = 0.98714
1830
814
4509
722
2472
2168
counts
1008
756
504
252
0
12519
9389
6260
3130
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 10 cosine sim = 0.98707
1930
957
4376
1011
1823
2419
counts
968
726
484
242
0
12519
9389
6260
3130
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1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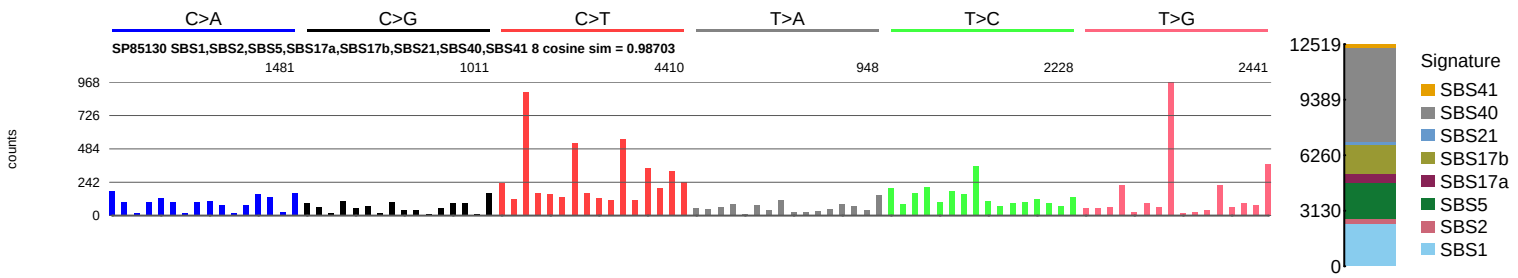
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18 7 cosine sim = 0.98705
1833
822
4536
733
2416
2177
counts
1008
756
504
252
0
12519
9389
6260
3130
0
Signature
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1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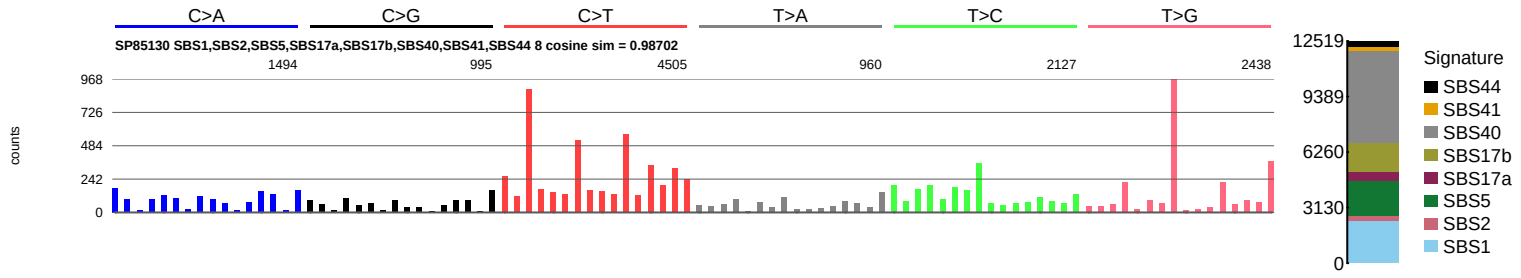
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS21,SBS40,SBS41 8 cosine sim = 0.98703
1481
1011
4410
948
2228
2441
counts
968
726
484
242
0
12519
9389
6260
3130
0
Signature
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS5
SBS2
SBS1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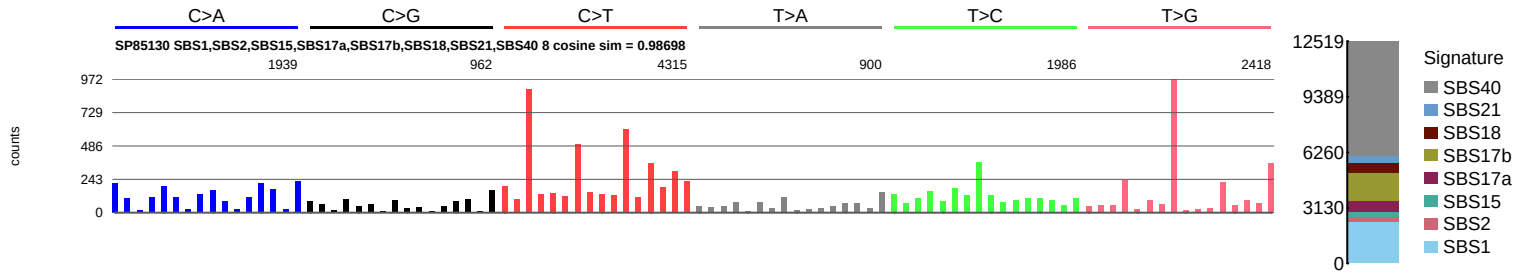
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS40,SBS41,SBS44 8 cosine sim = 0.98702
1494
995
4505
960
2127
2438
counts
968
726
484
242
0
12519
9389
6260
3130
0
Signature
SBS44
SBS41
SBS40
SBS17b
SBS17a
SBS5
SBS2
SBS1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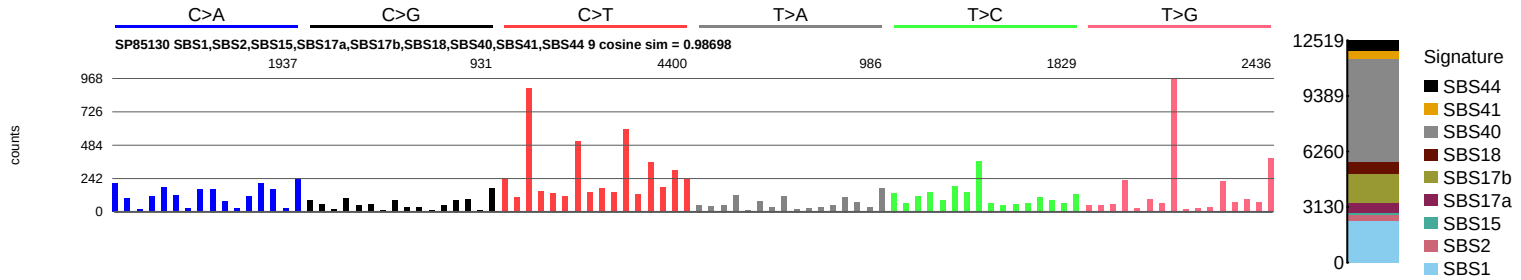
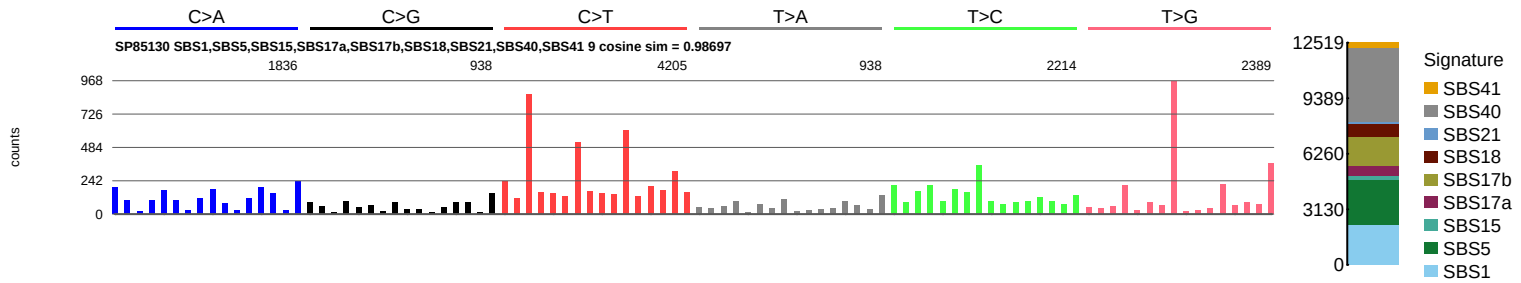
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40 8 cosine sim = 0.98698
1939
962
4315
900
1986
2418
counts
972
729
486
243
0
12519
9389
6260
3130
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 9 cosine sim = 0.98698
1937
931
4400
986
1829
2436
counts
968
726
484
242
0
12519
9389
6260
3130
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1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C>T
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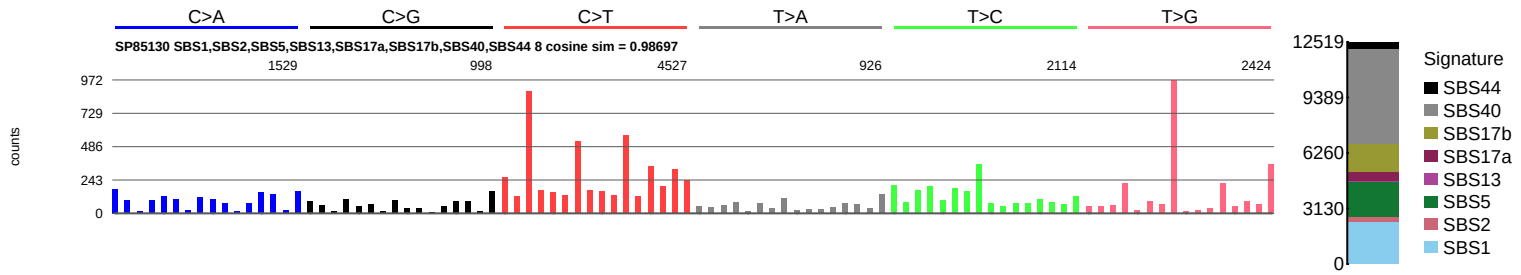
T>C
T>G
SP85130 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 9 cosine sim = 0.98697
1836
938
4205
938
2214
2389
counts
968
726
484
242
0
12519
9389
6260
3130
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS1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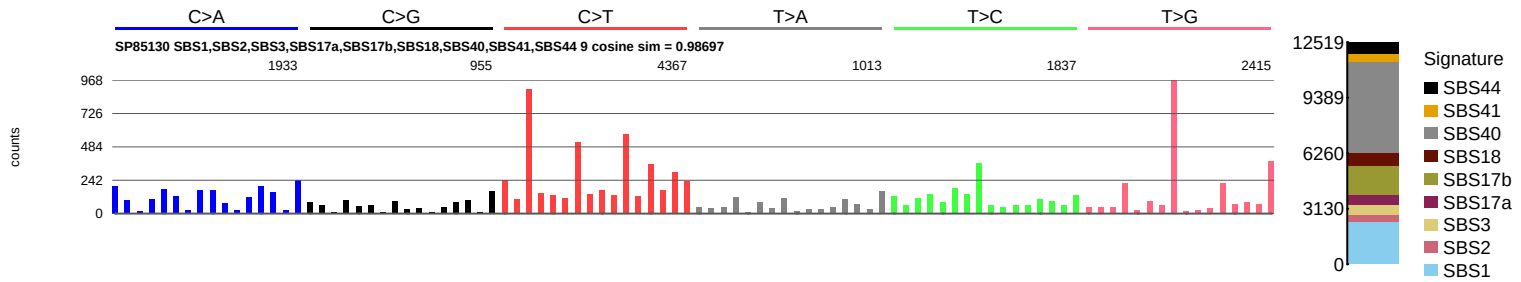
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS40,SBS44 8 cosine sim = 0.98697
1529
998
4527
926
2114
2424
counts
972
729
486
243
0
12519
9389
6260
3130
0
Signature
SBS44
SBS40
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1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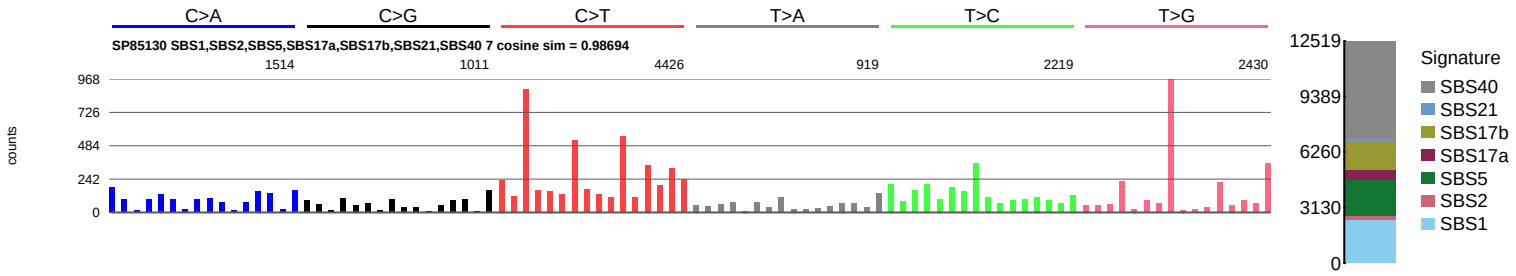
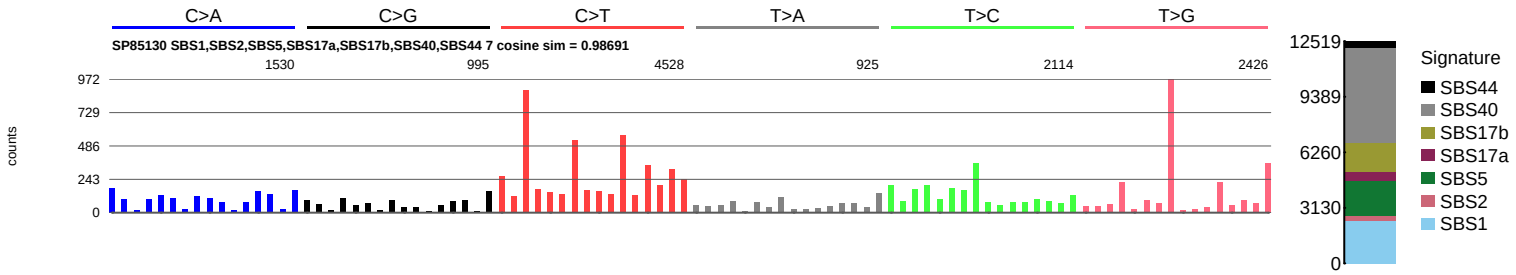
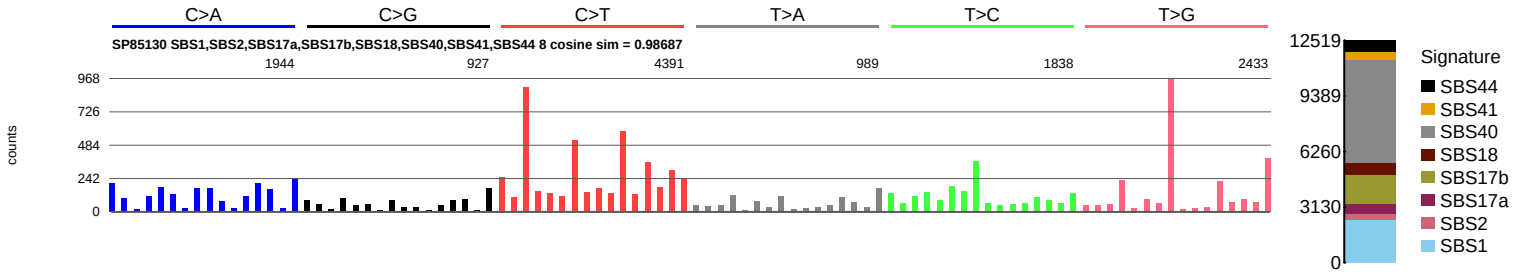
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 9 cosine sim = 0.98697
1933
955
4367
1013
1837
2415
counts
968
726
484
242
0
12519
9389
6260
3130
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS21,SBS40 7 cosine sim = 0.98694
1514
1011
4426
919
2219
2430
counts
968
726
484
242
0
12519
9389
6260
3130
0
Signature
SBS40
SBS21
SBS17b
SBS17a
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS40,SBS44 7 cosine sim = 0.98691
1530
995
4528
925
2114
2426
counts
972
729
486
243
0
12519
9389
6260
3130
0
Signature
SBS44
SBS40
SBS17b
SBS17a
SBS5
SBS2
SBS1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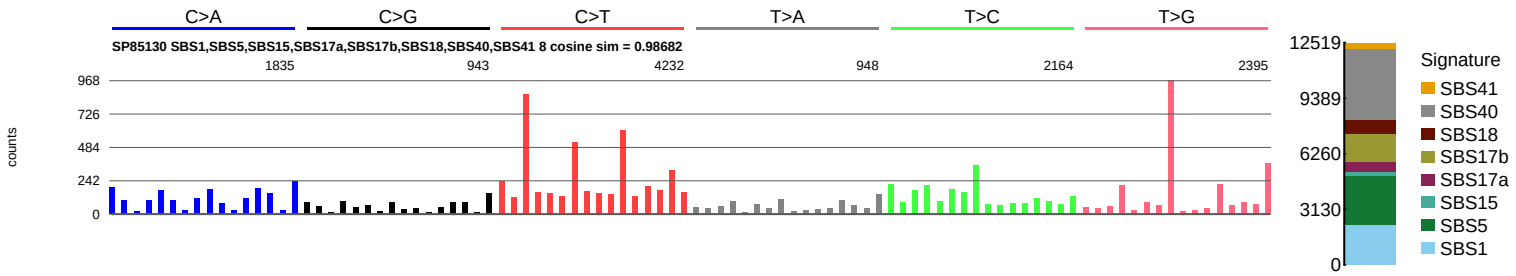
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 8 cosine sim = 0.98687
1944
927
4391
989
1838
2433
counts
968
726
484
242
0
12519
9389
6260
3130
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS2
SBS1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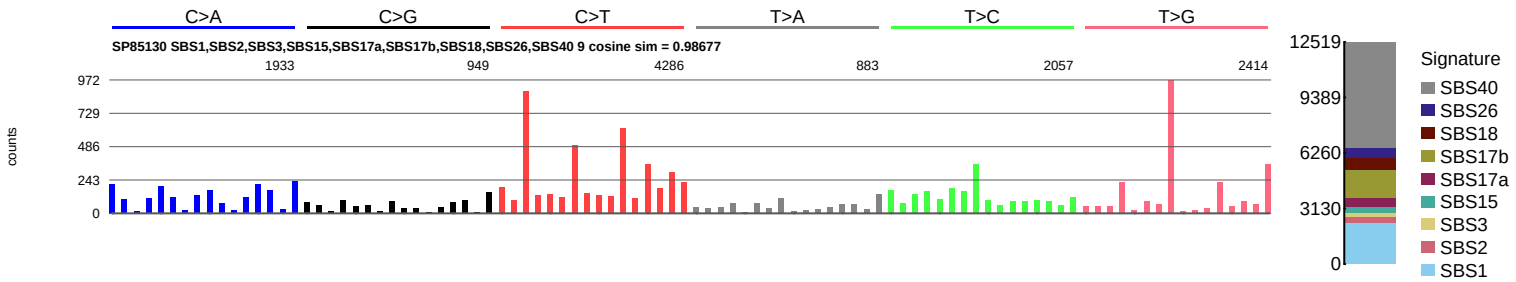
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41 8 cosine sim = 0.98682
1835
943
4232
948
2164
2395
counts
968
726
484
242
0
12519
9389
6260
3130
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40 9 cosine sim = 0.98677
1933
949
4286
883
2057
2414
counts
972
729
486
243
0
12519
9389
6260
3130
0
Signature
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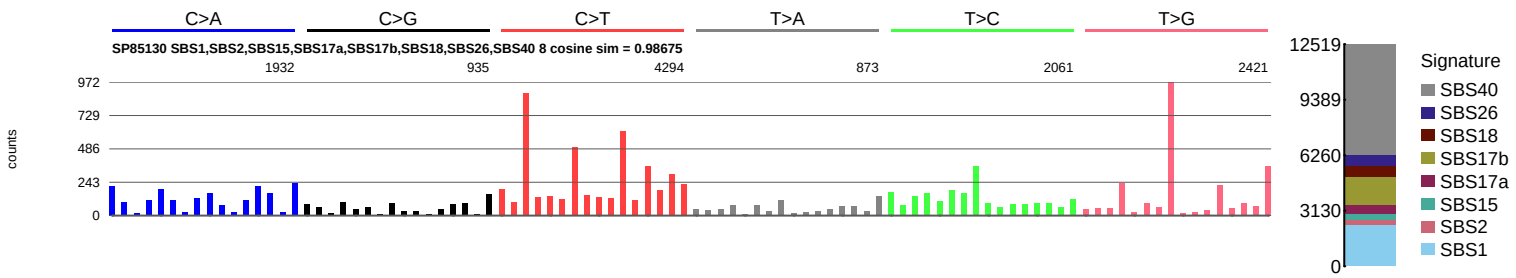
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40 8 cosine sim = 0.98675
1932
935
4294
873
2061
2421
counts
972
729
486
243
0
12519
9389
6260
3130
0
Signature
SBS40
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SBS17b
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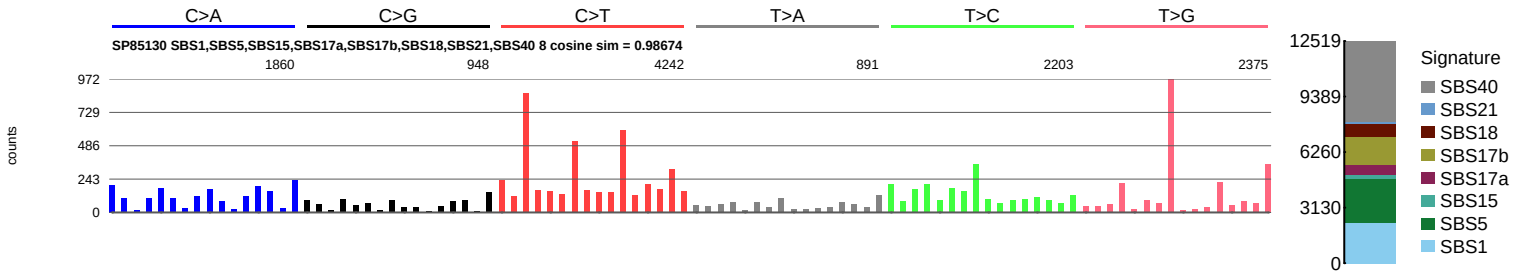
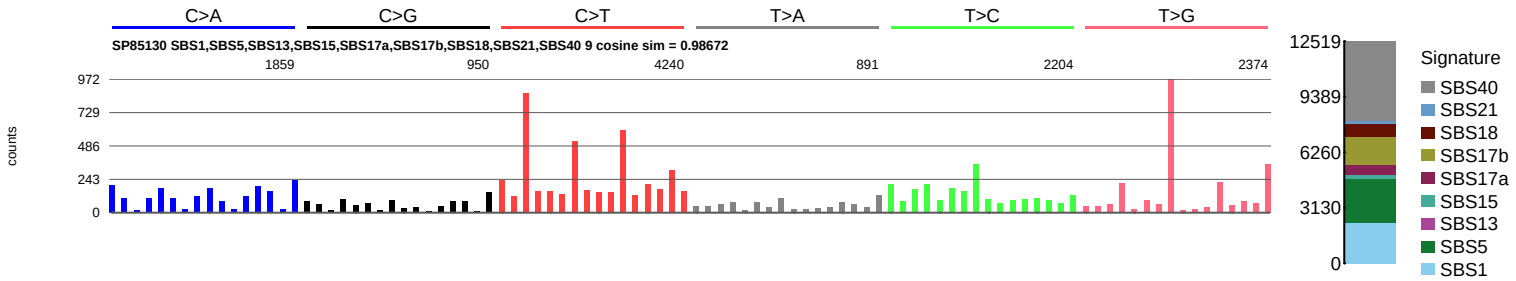
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40 8 cosine sim = 0.98674
1860
948
4242
891
2203
2375
counts
972
729
486
243
0
12519
9389
6260
3130
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS1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C>T
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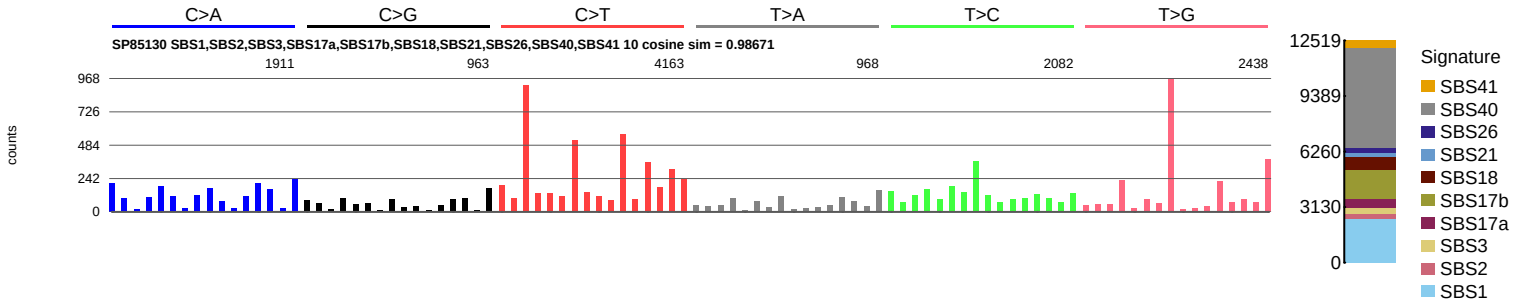
T>C
T>G
SP85130 SBS1,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40 9 cosine sim = 0.98672
1859
950
4240
891
2204
2374
counts
972
729
486
243
0
12519
9389
6260
3130
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS1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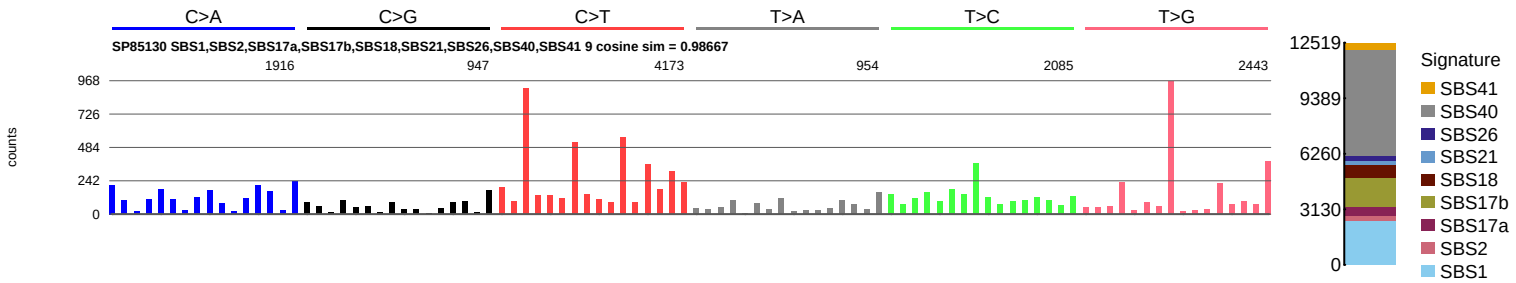
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41 10 cosine sim = 0.98671
1911
963
4163
968
2082
2438
counts
968
726
484
242
0
12519
9389
6260
3130
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1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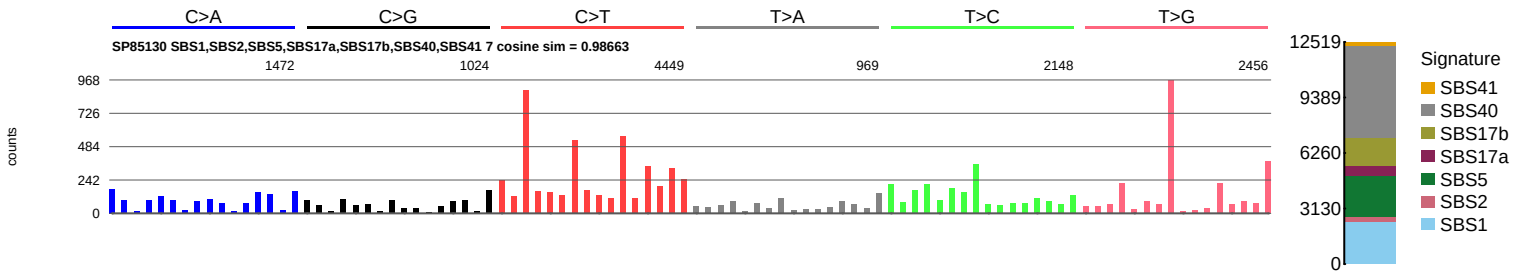
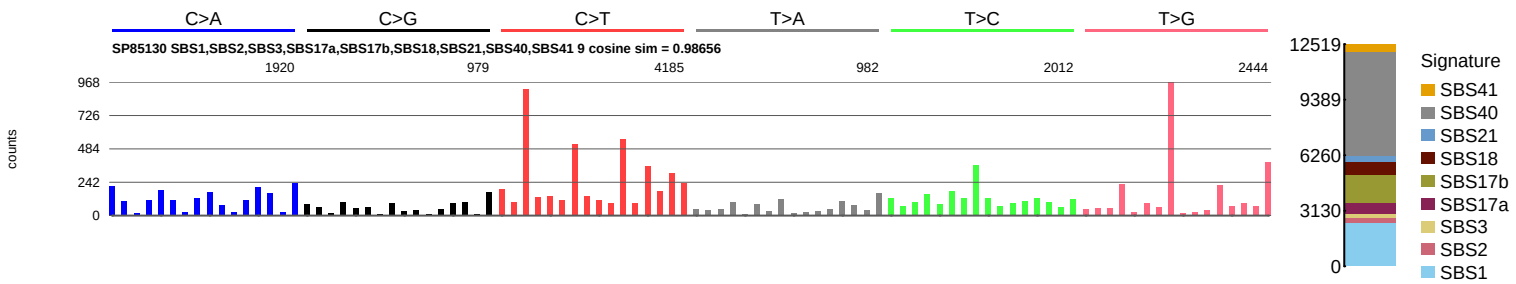
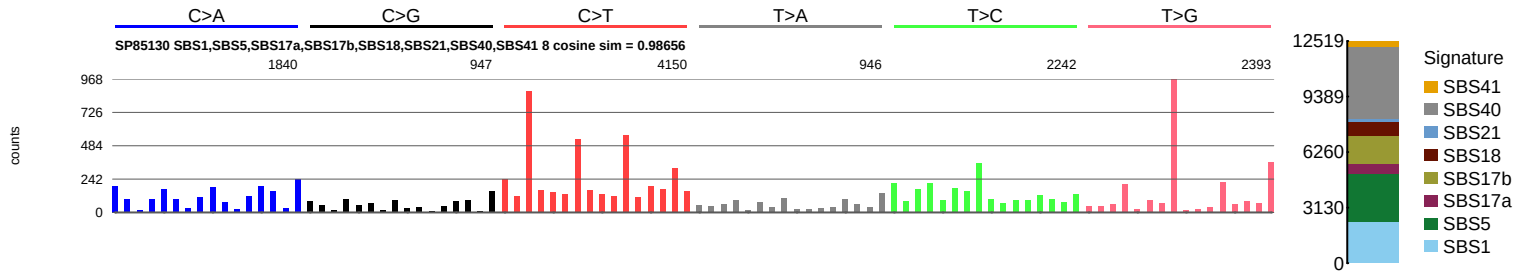
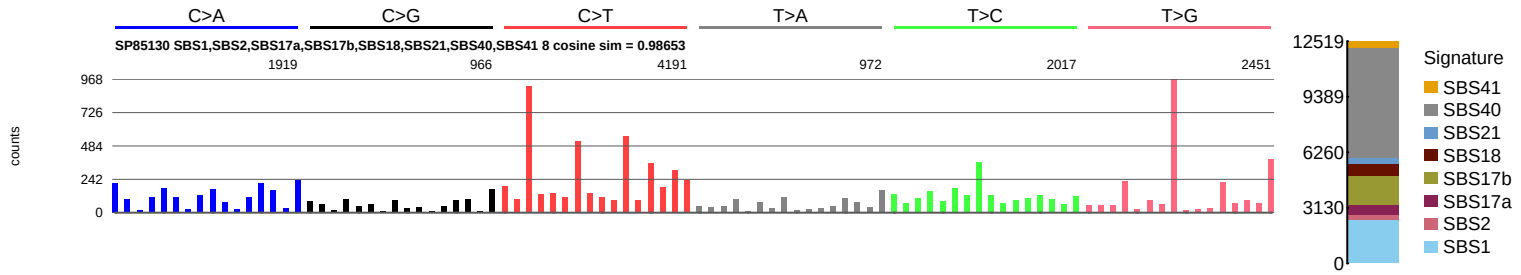
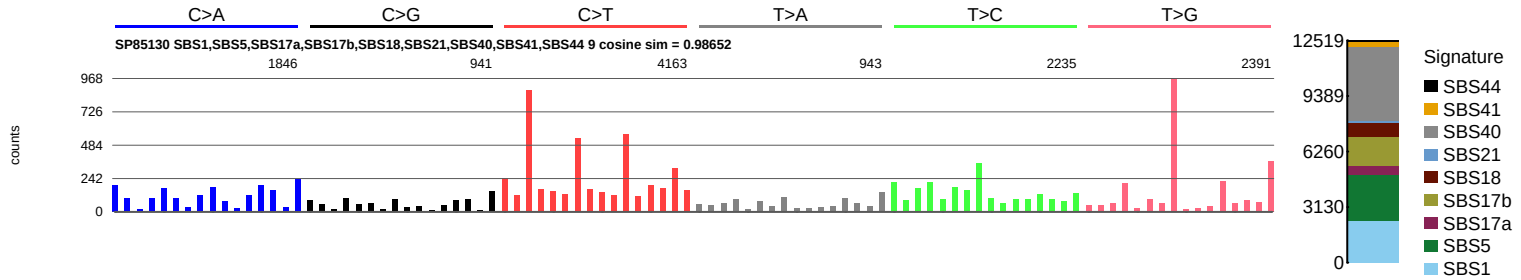
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41 9 cosine sim = 0.98667
1916
947
4173
954
2085
2443
counts
968
726
484
242
0
12519
9389
6260
3130
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS2
SBS1

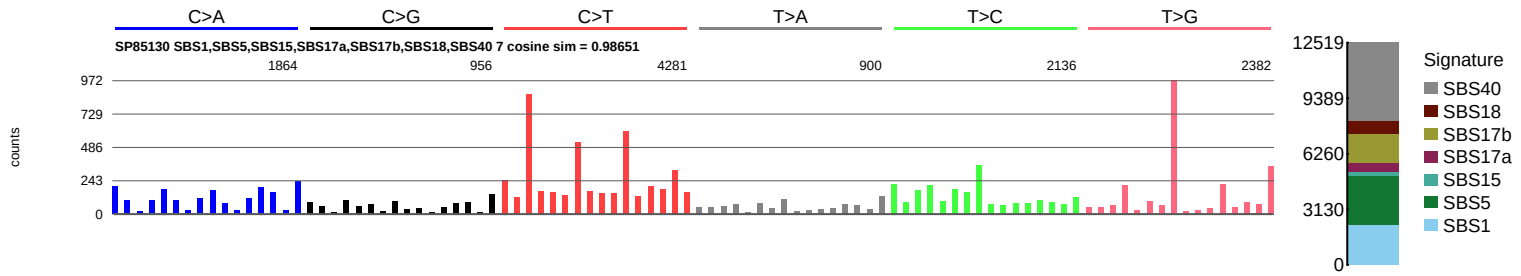

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS40,SBS41 7 cosine sim = 0.98663
1472
1024
4449
969
2148
2456
counts
968
726
484
242
0
12519
9389
6260
3130
0
Signature
SBS41
SBS40
SBS17b
SBS17a
SBS5
SBS2
SBS1

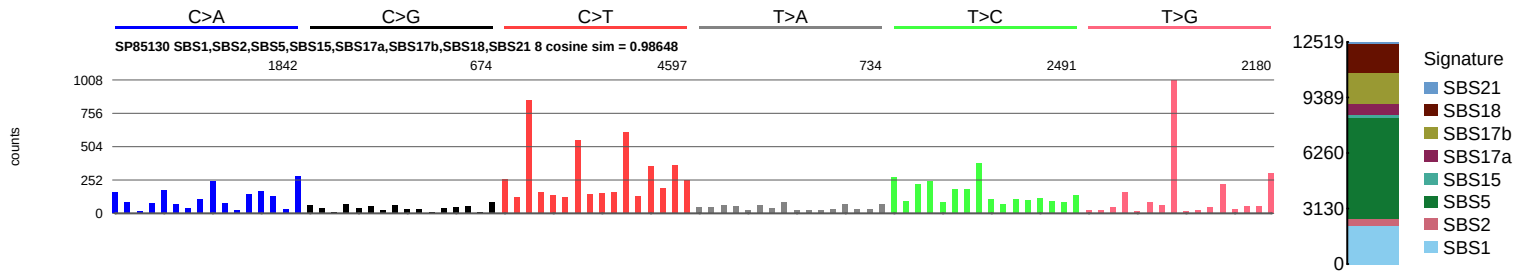

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 9 cosine sim = 0.98656
1920
979
4185
982
2012
2444
counts
968
726
484
242
0
12519
9389
6260
3130
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1

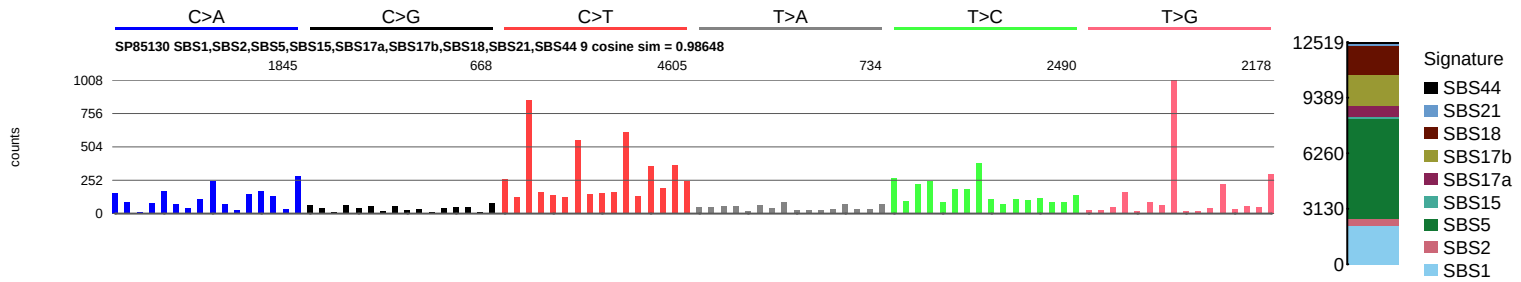

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 8 cosine sim = 0.98656
1840
947
4150
946
2242
2393
counts
968
726
484
242
0
12519
9389
6260
3130
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS1

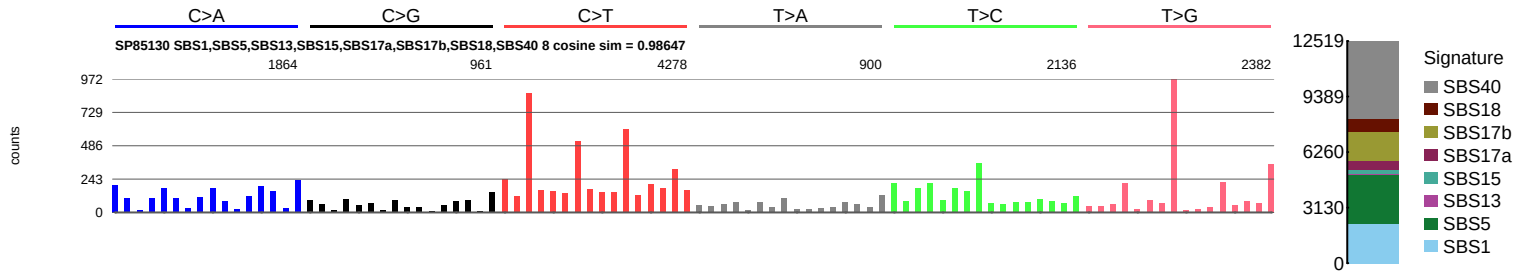

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 8 cosine sim = 0.98653
1919
966
4191
972
2017
2451
counts
968
726
484
242
0
12519
9389
6260
3130
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS2
SBS1

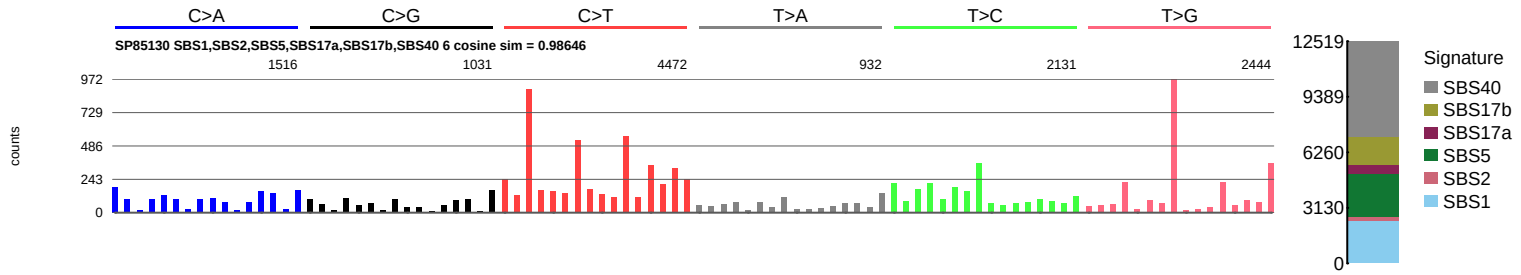

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 9 cosine sim = 0.98652
1846
941
4163
943
2235
2391
counts
968
726
484
242
0
12519
9389
6260
3130
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS1

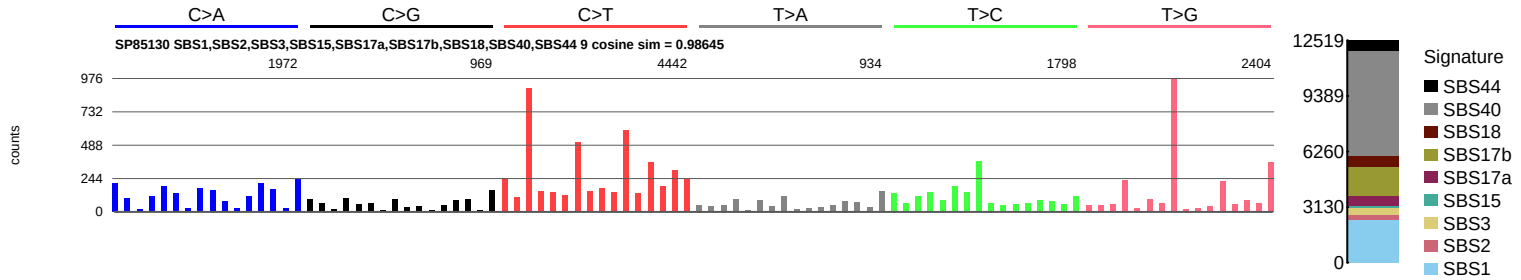

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS40 7 cosine sim = 0.98651
1864
956
4281
900
2136
2382
counts
972
729
486
243
0
12519
9389
6260
3130
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS1

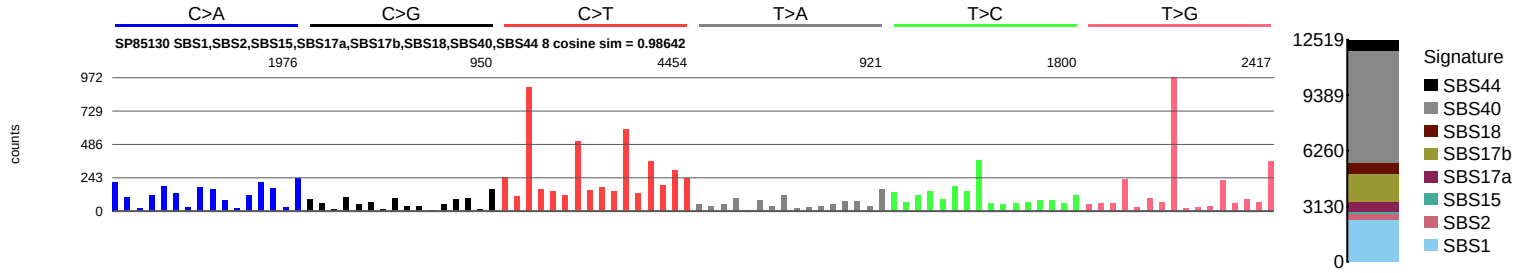

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21 8 cosine sim = 0.98648
1842
674
4597
734
2491
2180
counts
1008
756
504
252
0
12519
9389
6260
3130
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1

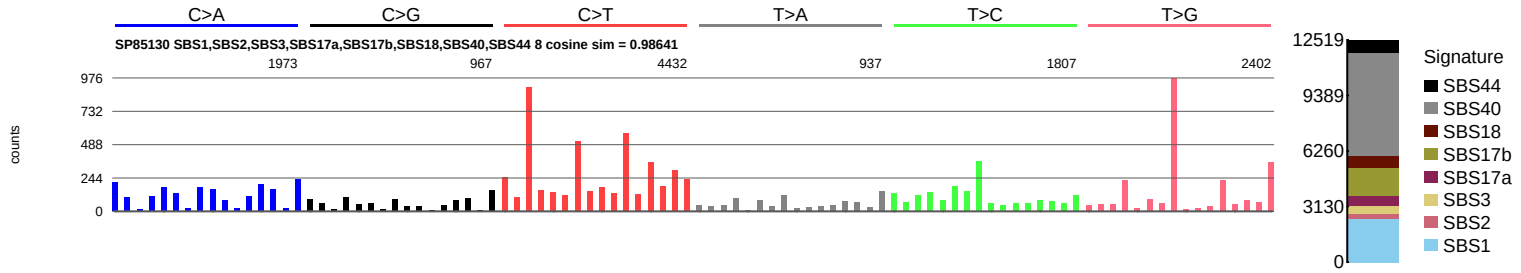

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS44 9 cosine sim = 0.98648
1845
668
4605
734
2490
2178
counts
1008
756
504
252
0
12519
9389
6260
3130
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1

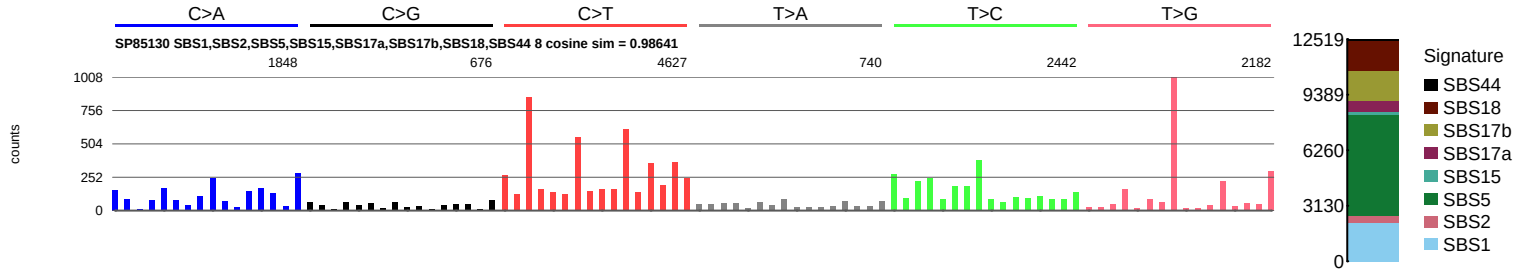

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40 8 cosine sim = 0.98647
1864
961
4278
900
2136
2382
counts
972
729
486
243
0
12519
9389
6260
3130
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS1

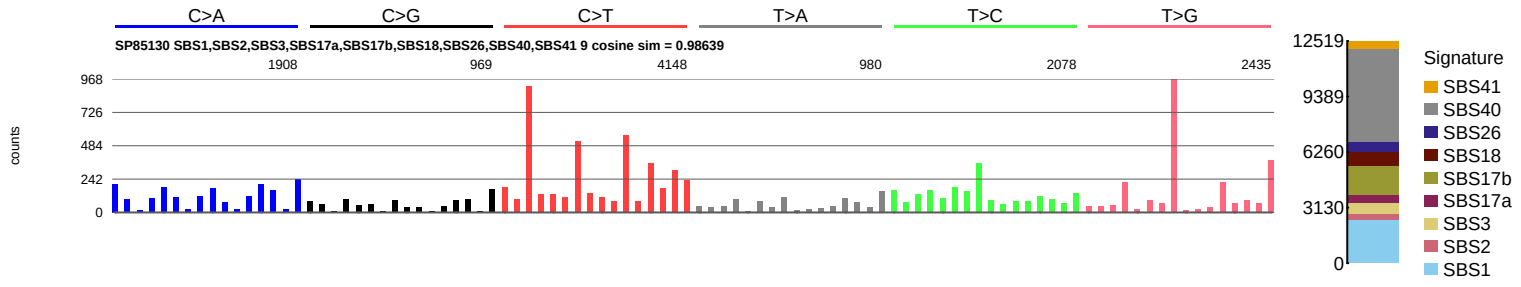

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS40 6 cosine sim = 0.98646
1516
1031
4472
932
2131
2444
counts
972
729
486
243
0
12519
9389
6260
3130
0
Signature
SBS40
SBS17b
SBS17a
SBS5
SBS2
SBS1

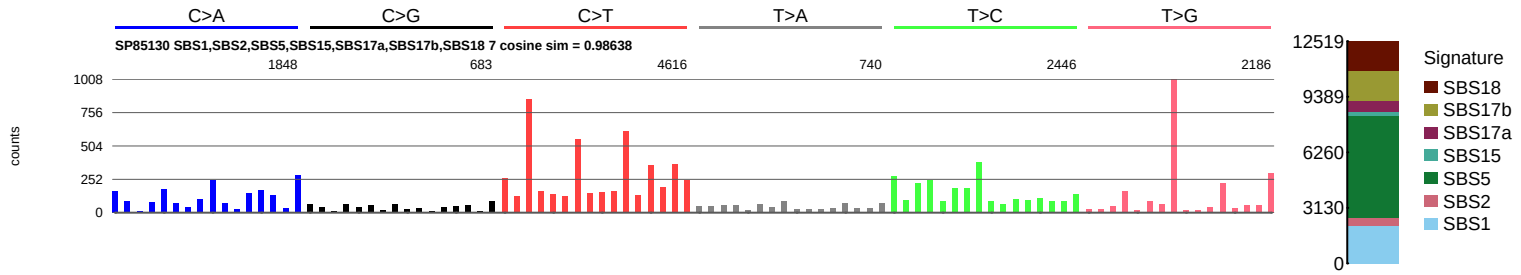

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS44 9 cosine sim = 0.98645
1972
969
4442
934
1798
2404
counts
976
732
488
244
0
12519
9389
6260
3130
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1

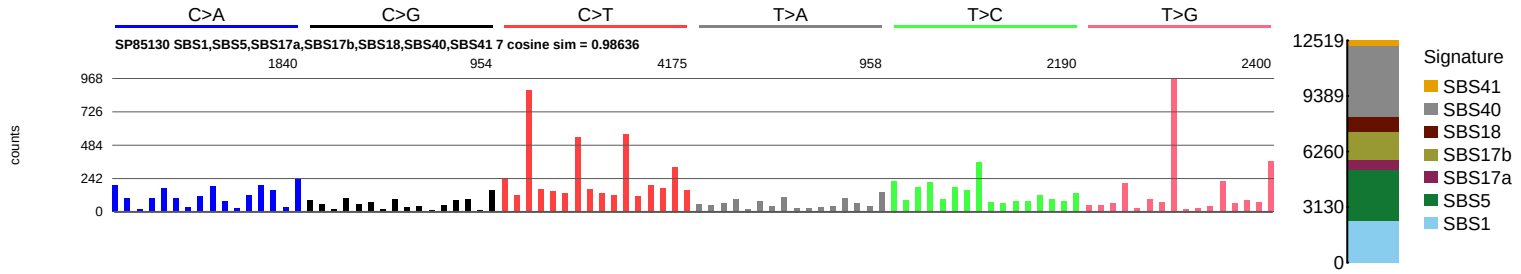

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS44 8 cosine sim = 0.98642
1976
950
4454
921
1800
2417
counts
972
729
486
243
0
12519
9389
6260
3130
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1

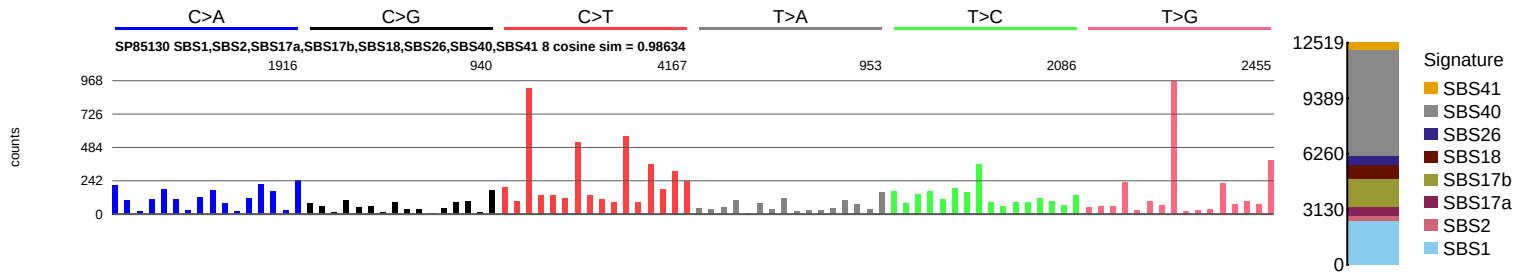

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS40,SBS44 8 cosine sim = 0.98641
1973
967
4432
937
1807
2402
counts
976
732
488
244
0
12519
9389
6260
3130
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1

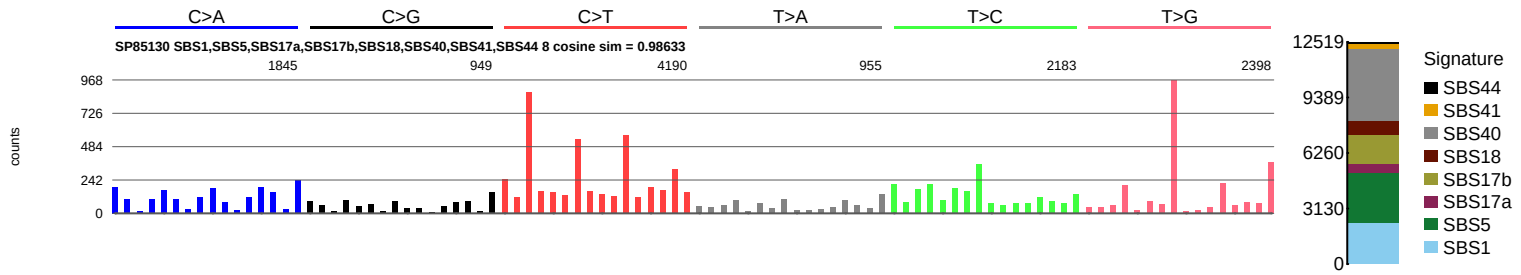

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS44 8 cosine sim = 0.98641
1848
676
4627
740
2442
2182
counts
1008
756
504
252
0
12519
9389
6260
3130
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1

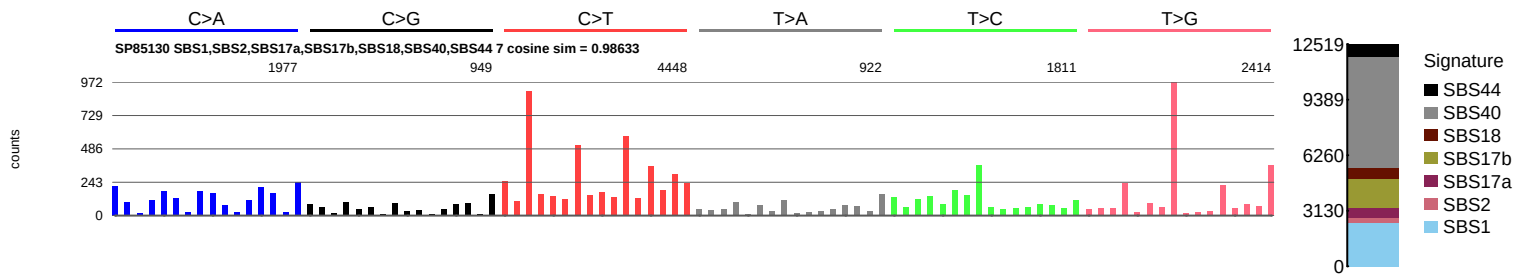

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 9 cosine sim = 0.98639
1908
969
4148
980
2078
2435
counts
968
726
484
242
0
12519
9389
6260
3130
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1

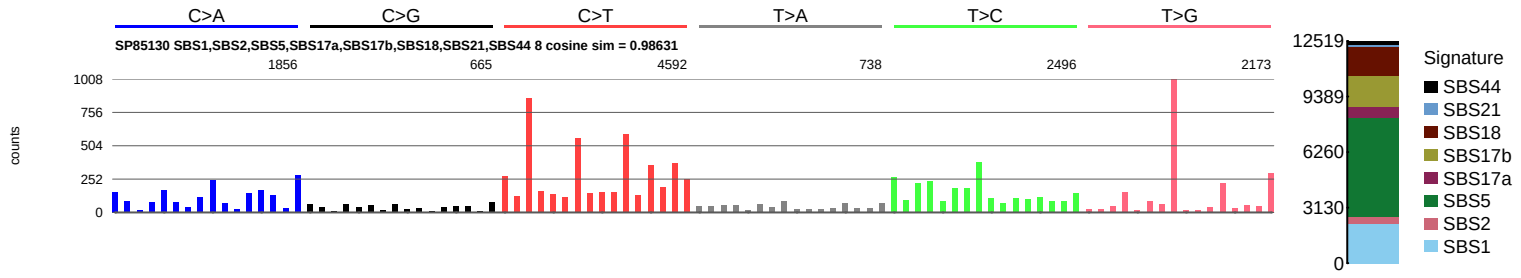

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18 7 cosine sim = 0.98638
1848
683
4616
740
2446
2186
counts
1008
756
504
252
0
12519
9389
6260
3130
0
Signature
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1

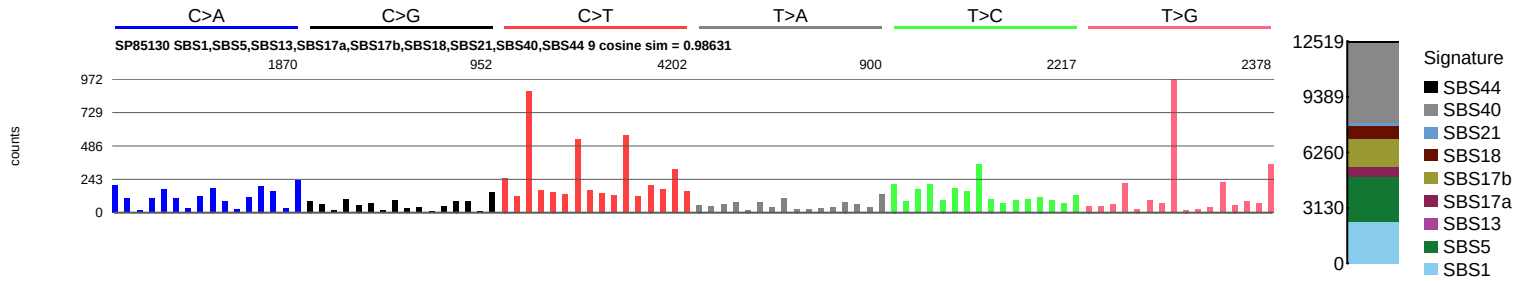

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS5,SBS17a,SBS17b,SBS18,SBS40,SBS41 7 cosine sim = 0.98636
1840
954
4175
958
2190
2400
counts
968
726
484
242
0
12519
9389
6260
3130
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS5
SBS1

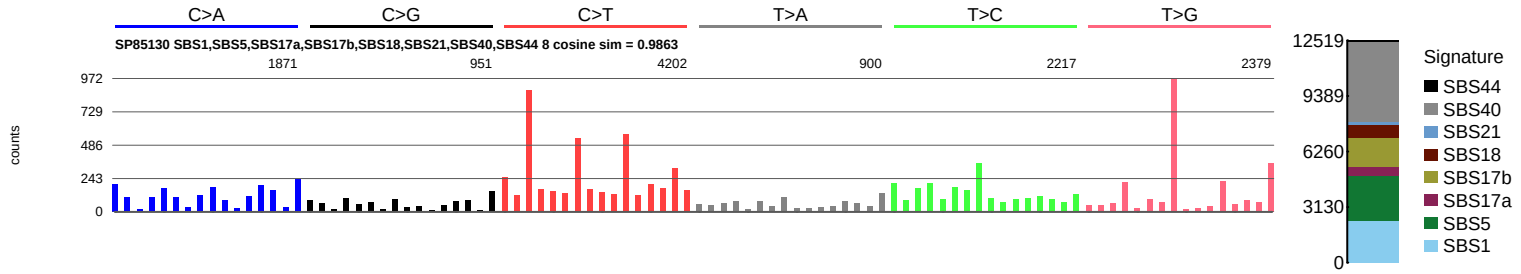

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 8 cosine sim = 0.98634
1916
940
4167
953
2086
2455
counts
968
726
484
242
0
12519
9389
6260
3130
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS5,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 8 cosine sim = 0.98633
1845
949
4190
955
2183
2398
counts
968
726
484
242
0
12519
9389
6260
3130
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS40,SBS44 7 cosine sim = 0.98633
1977
949
4448
922
1811
2414
counts
972
729
486
243
0
12519
9389
6260
3130
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS44 8 cosine sim = 0.98631
1856
665
4592
738
2496
2173
counts
1008
756
504
252
0
12519
9389
6260
3130
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 9 cosine sim = 0.98631
1870
952
4202
900
2217
2378
counts
972
729
486
243
0
12519
9389
6260
3130
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS1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C>G
C>T
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T>C
T>G
SP85130 SBS1,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 8 cosine sim = 0.9863
1871
951
4202
900
2217
2379
counts
972
729
486
243
0
12519
9389
6260
3130
0
Signature
SBS44
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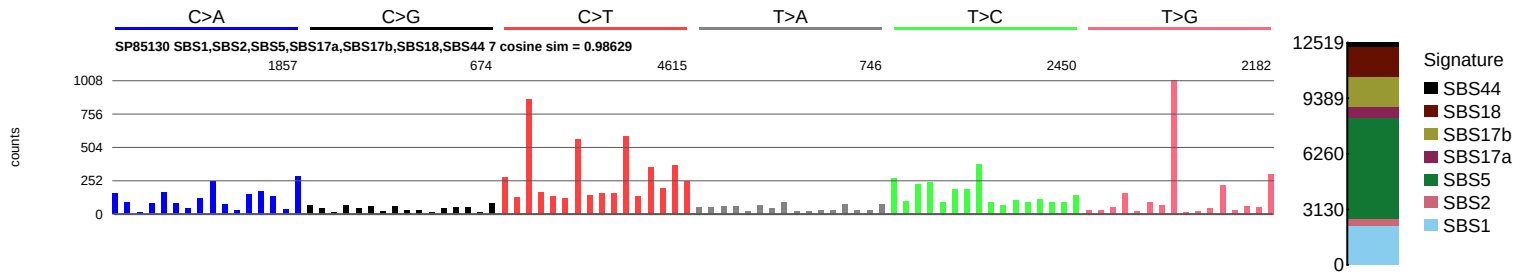
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS1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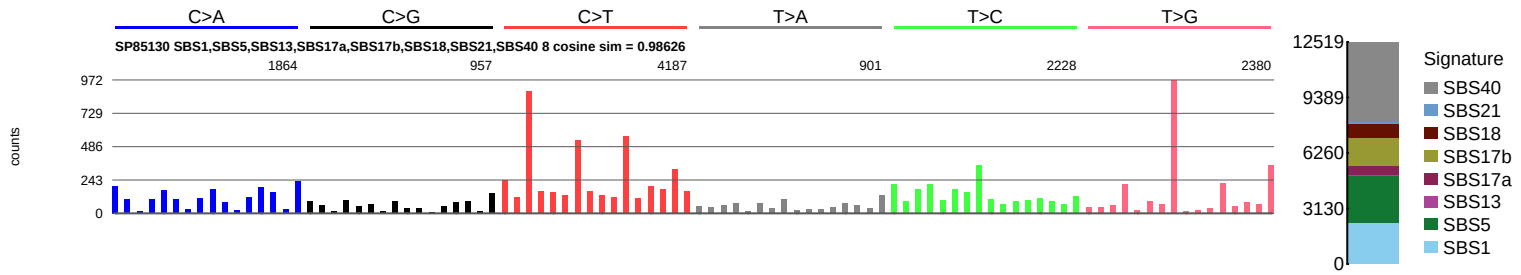
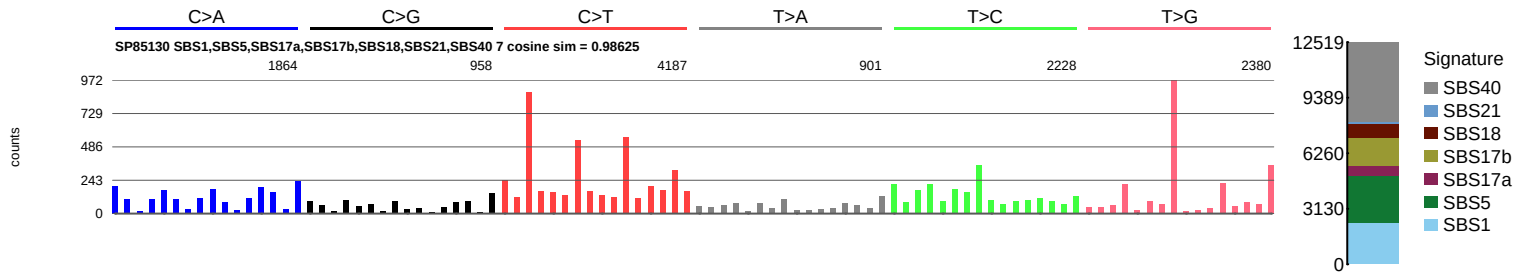
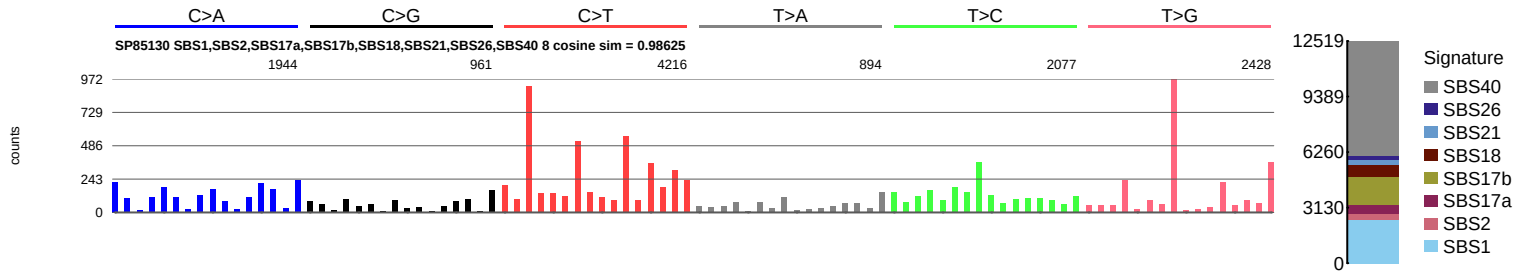
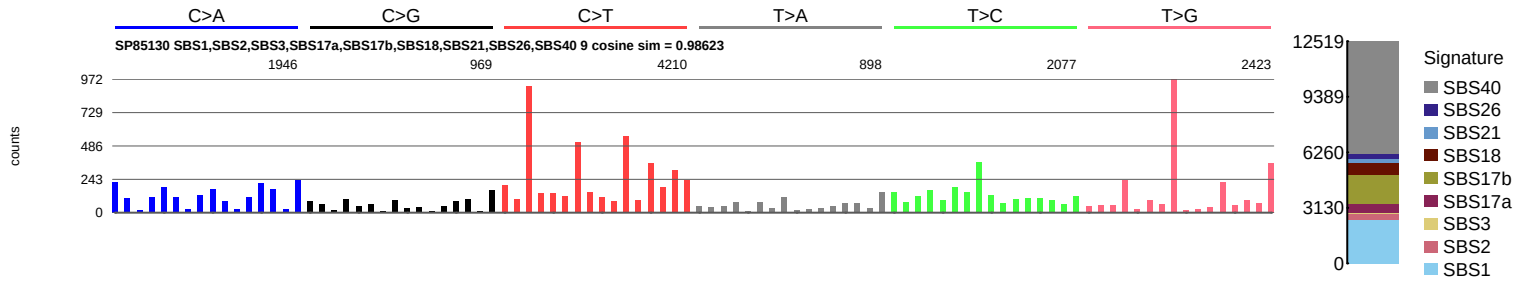
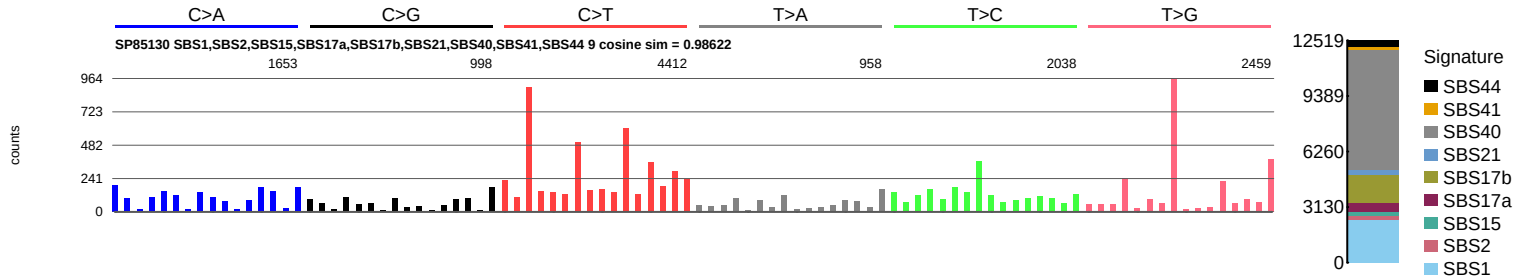
T>G
SP85130 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS44 7 cosine sim = 0.98629
1857
674
4615
746
2450
2182
counts
1008
756
504
252
0
12519
9389
6260
3130
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1

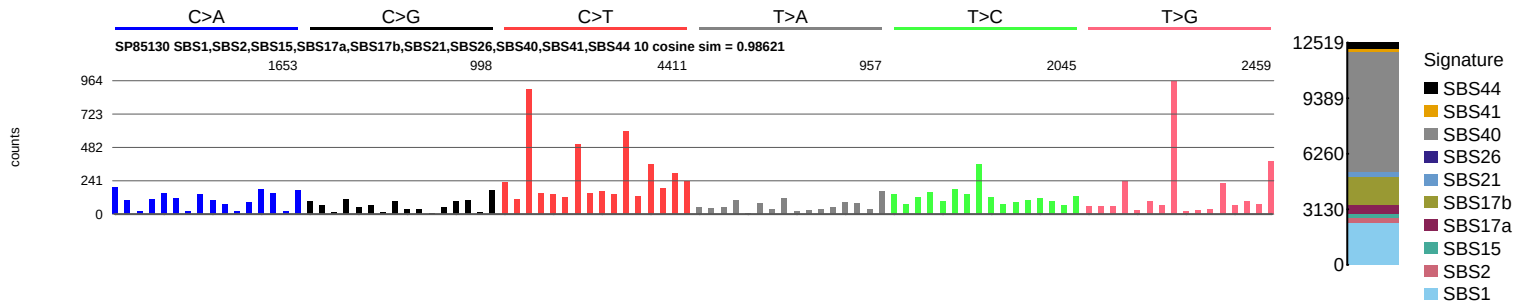

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40 8 cosine sim = 0.98626
1864
957
4187
901
2228
2380
counts
972
729
486
243
0
12519
9389
6260
3130
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS1

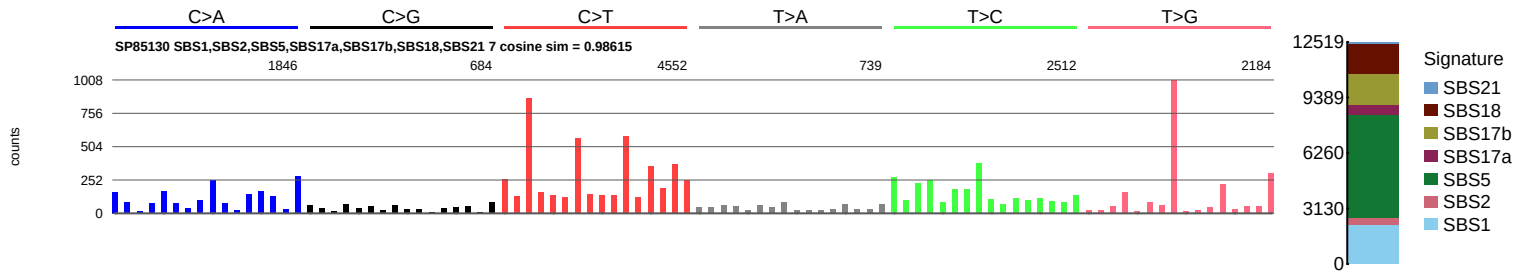

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS40 7 cosine sim = 0.98625
1864
958
4187
901
2228
2380
counts
972
729
486
243
0
12519
9389
6260
3130
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS1

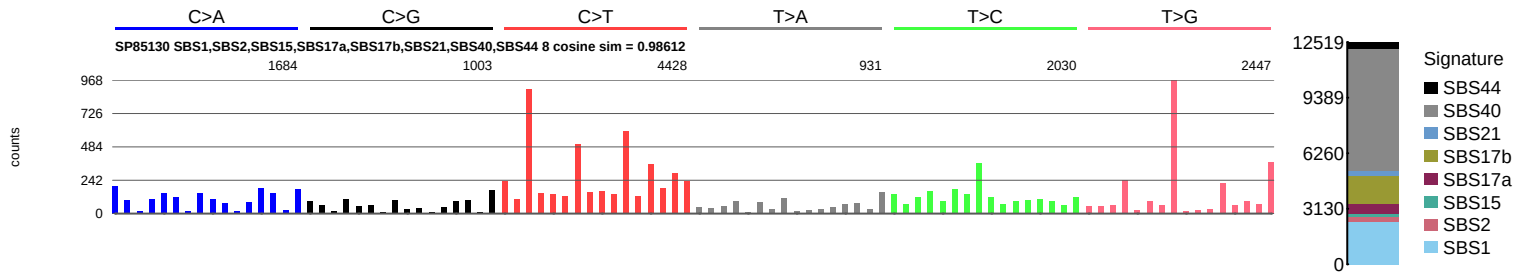

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40 8 cosine sim = 0.98625
1944
961
4216
894
2077
2428
counts
972
729
486
243
0
12519
9389
6260
3130
0
Signature
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS2
SBS1

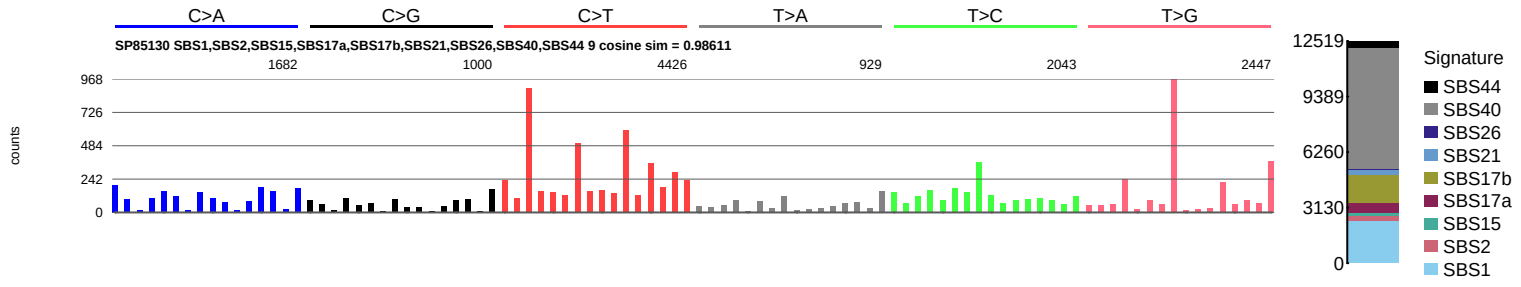

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40 9 cosine sim = 0.98623
1946
969
4210
898
2077
2423
counts
972
729
486
243
0
12519
9389
6260
3130
0
Signature
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1

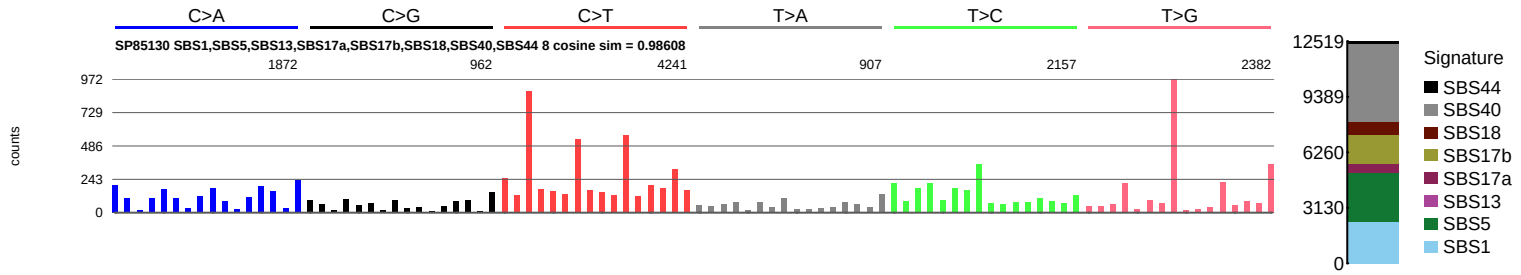

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS21,SBS40,SBS41,SBS44 9 cosine sim = 0.98622
1653
998
4412
958
2038
2459
counts
964
723
482
241
0
12519
9389
6260
3130
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS2
SBS1

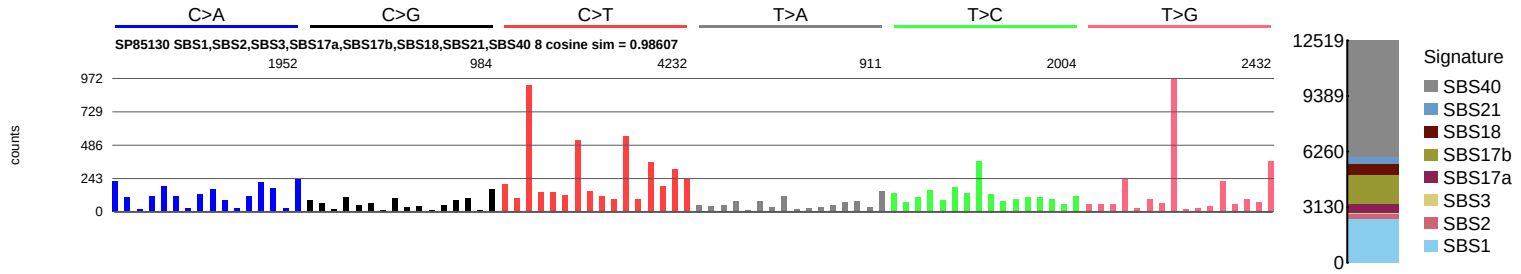

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS21,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.98621
1653
998
4411
957
2045
2459
counts
964
723
482
241
0
12519
9389
6260
3130
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS17b
SBS17a
SBS15
SBS2
SBS1

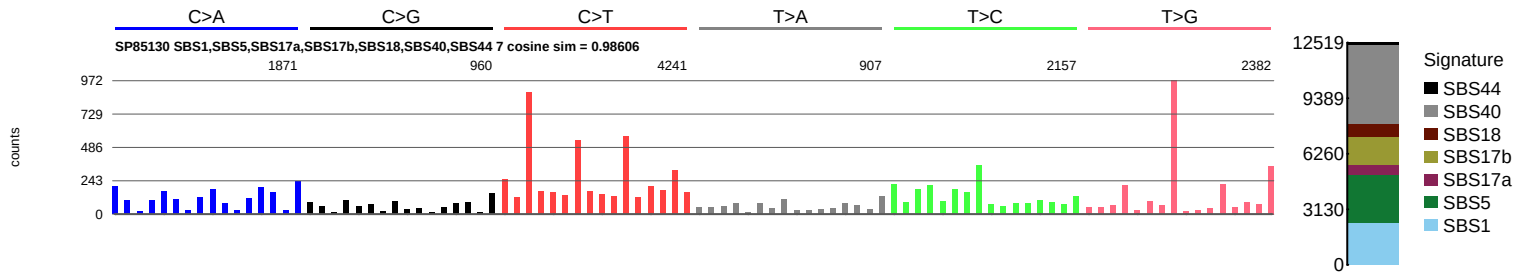

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS21 7 cosine sim = 0.98615
1846
684
4552
739
2512
2184
counts
1008
756
504
252
0
12519
9389
6260
3130
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1

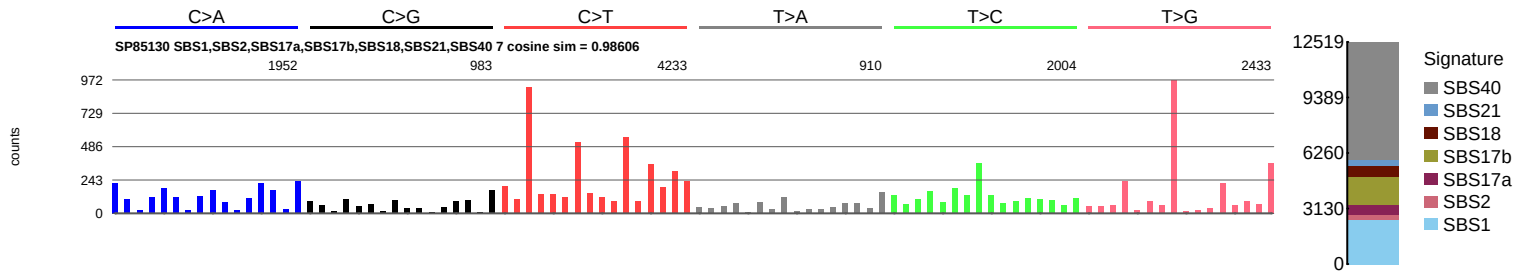

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS21,SBS40,SBS44 8 cosine sim = 0.98612
1684
1003
4428
931
2030
2447
counts
968
726
484
242
0
12519
9389
6260
3130
0
Signature
SBS44
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS2
SBS1

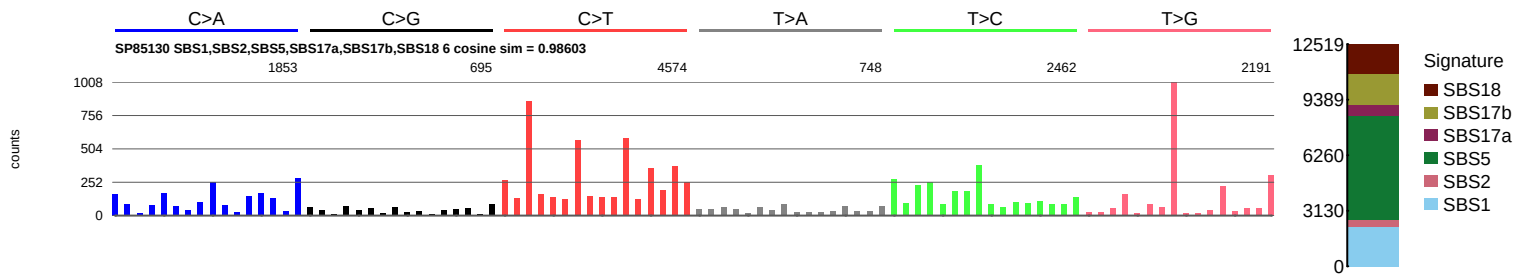

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS21,SBS26,SBS40,SBS44 9 cosine sim = 0.98611
1682
1000
4426
929
2043
2447
counts
968
726
484
242
0
12519
9389
6260
3130
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS17b
SBS17a
SBS15
SBS2
SBS1

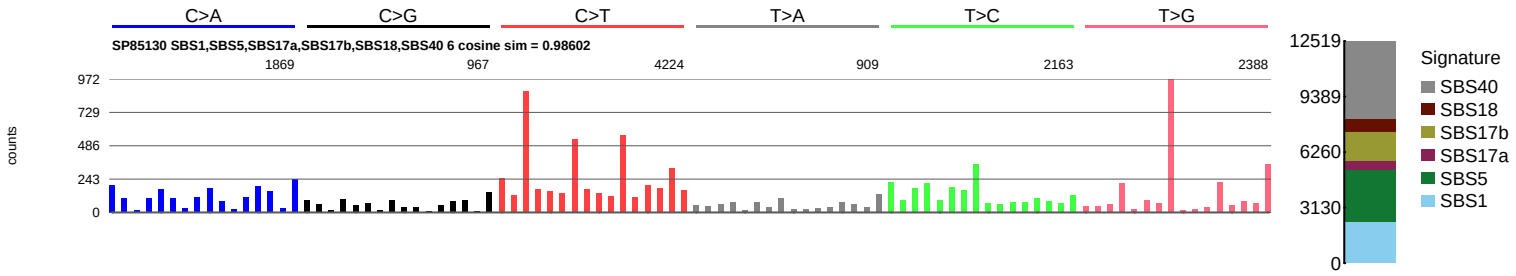

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS40,SBS44 8 cosine sim = 0.98608
1872
962
4241
907
2157
2382
counts
972
729
486
243
0
12519
9389
6260
3130
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS1

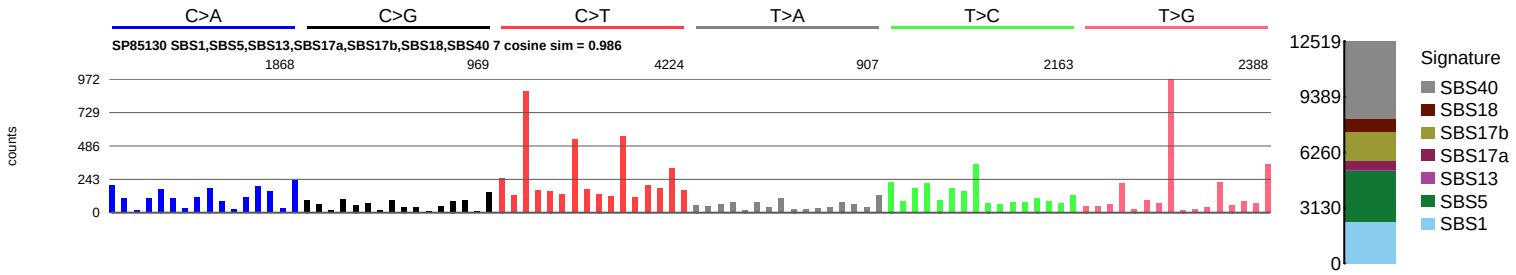

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS40 8 cosine sim = 0.98607
1952
984
4232
911
2004
2432
counts
972
729
486
243
0
12519
9389
6260
3130
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1

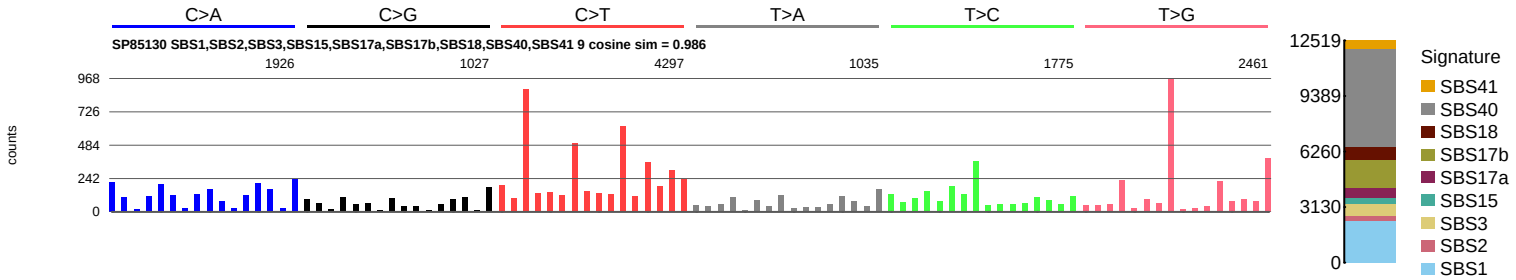

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS5,SBS17a,SBS17b,SBS18,SBS40,SBS44 7 cosine sim = 0.98606
1871
960
4241
907
2157
2382
counts
972
729
486
243
0
12519
9389
6260
3130
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS5
SBS1

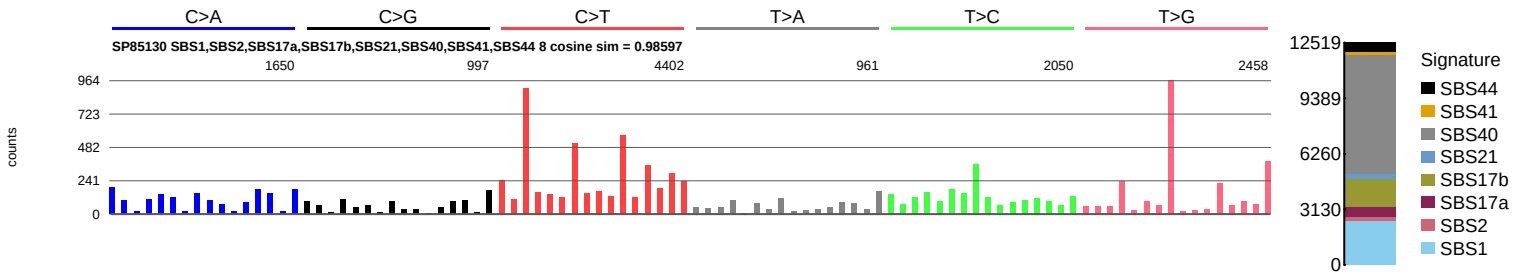

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS21,SBS40 7 cosine sim = 0.98606
1952
983
4233
910
2004
2433
counts
972
729
486
243
0
12519
9389
6260
3130
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS2
SBS1

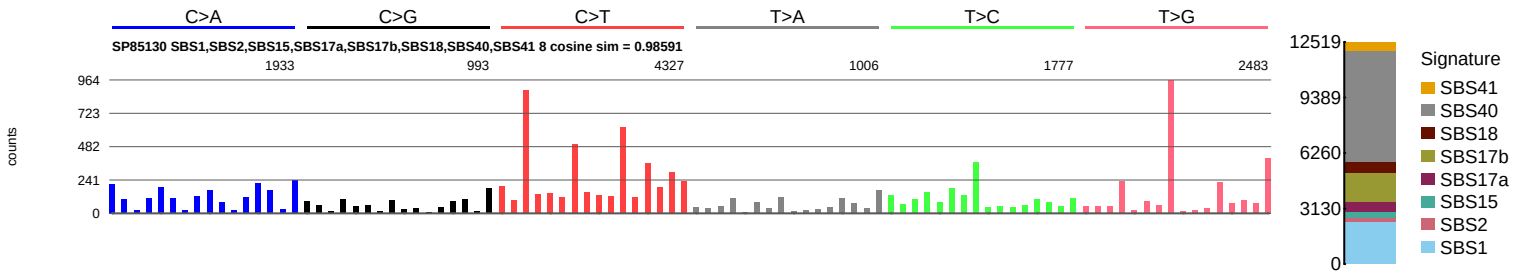

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18 6 cosine sim = 0.98603
1853
695
4574
748
2462
2191
counts
1008
756
504
252
0
12519
9389
6260
3130
0
Signature
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1

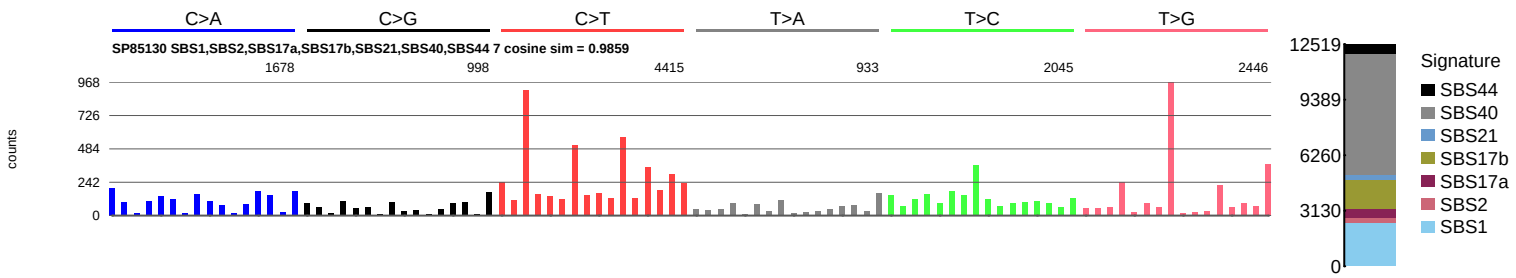

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS5,SBS17a,SBS17b,SBS18,SBS40 6 cosine sim = 0.98602
1869
967
4224
909
2163
2388
counts
972
729
486
243
0
12519
9389
6260
3130
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS5
SBS1

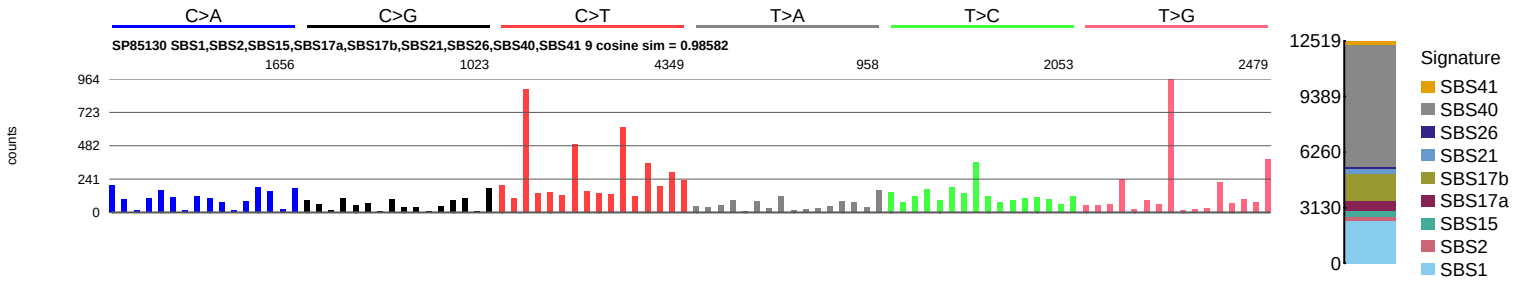

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS40 7 cosine sim = 0.986
1868
969
4224
907
2163
2388
counts
972
729
486
243
0
12519
9389
6260
3130
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS1

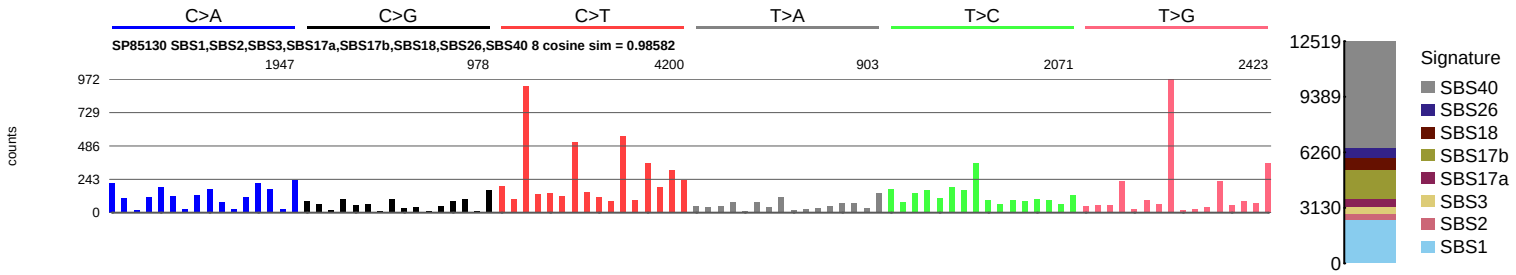

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41 9 cosine sim = 0.986
1926
1027
4297
1035
1775
2461
counts
968
726
484
242
0
12519
9389
6260
3130
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1

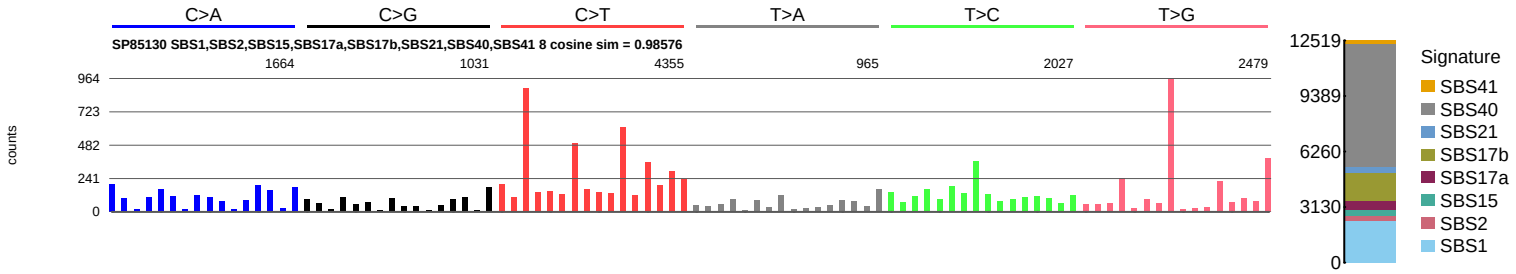

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS17a,SBS17b,SBS21,SBS40,SBS41,SBS44 8 cosine sim = 0.98597
1650
997
4402
961
2050
2458
counts
964
723
482
241
0
12519
9389
6260
3130
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS2
SBS1

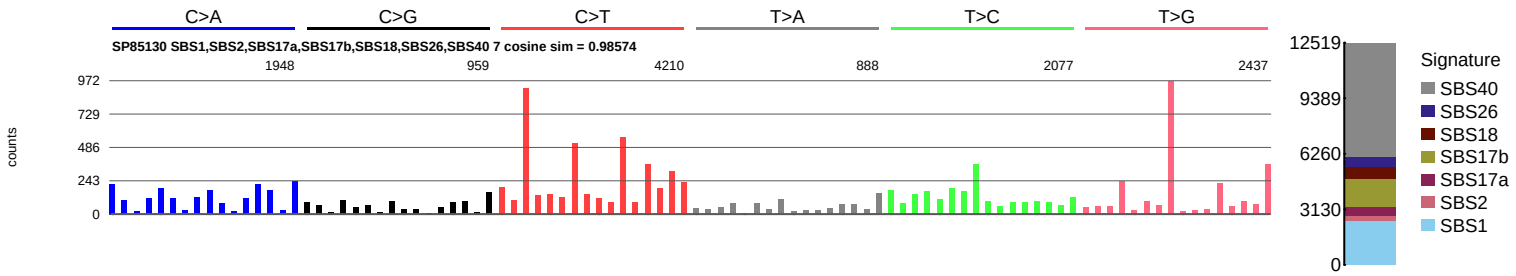

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41 8 cosine sim = 0.98591
1933
993
4327
1006
1777
2483
counts
964
723
482
241
0
12519
9389
6260
3130
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1

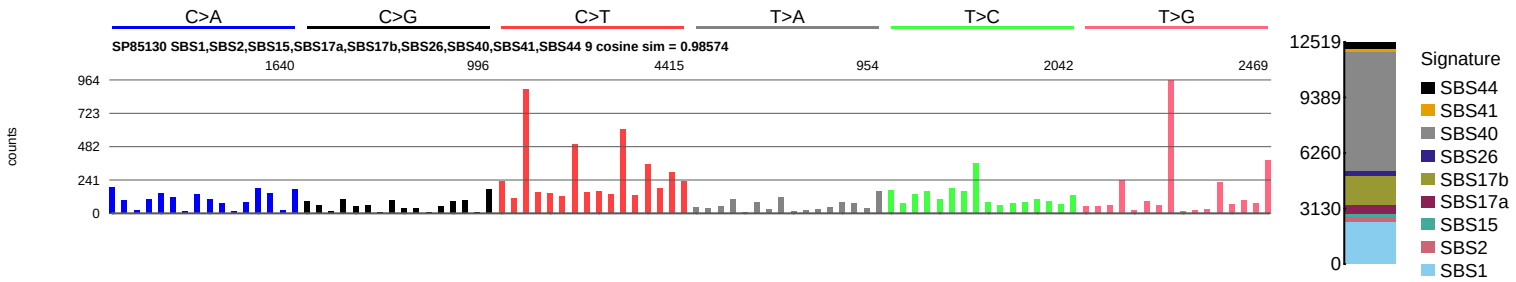

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS17a,SBS17b,SBS21,SBS40,SBS44 7 cosine sim = 0.9859
1678
998
4415
933
2045
2446
counts
968
726
484
242
0
12519
9389
6260
3130
0
Signature
SBS44
SBS40
SBS21
SBS17b
SBS17a
SBS2
SBS1

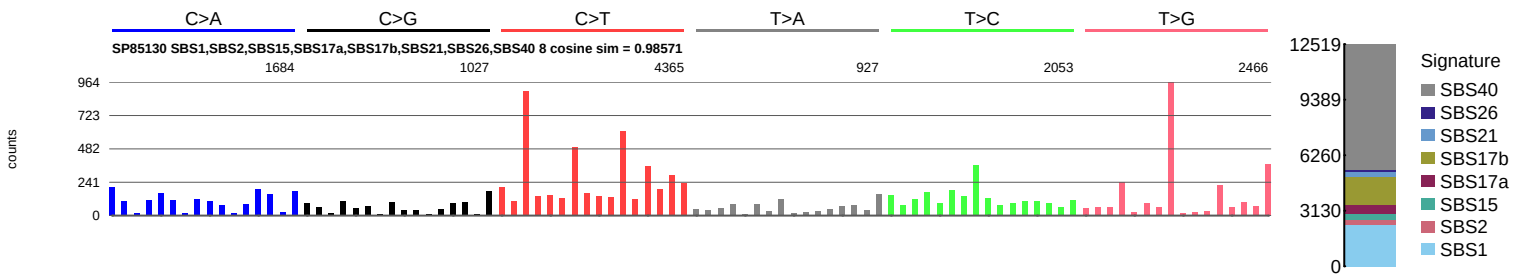

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS21,SBS26,SBS40,SBS41 9 cosine sim = 0.98582
1656
1023
4349
958
2053
2479
counts
964
723
482
241
0
12519
9389
6260
3130
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS17b
SBS17a
SBS15
SBS2
SBS1

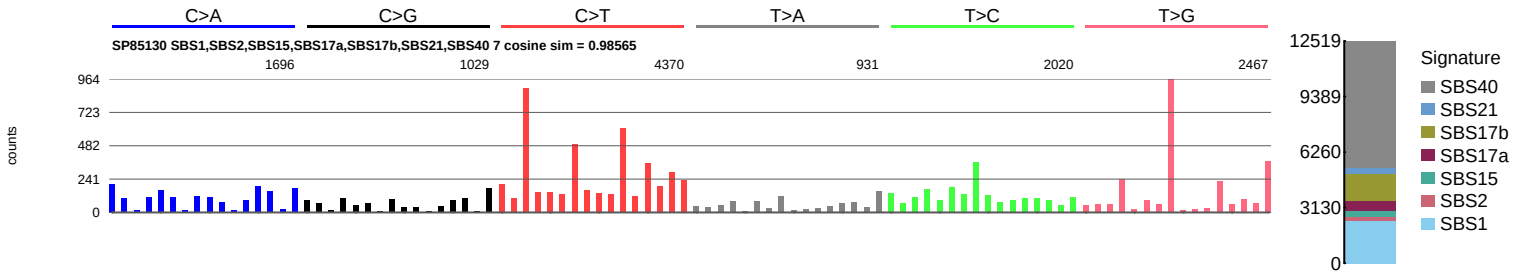

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS26,SBS40 8 cosine sim = 0.98582
1947
978
4200
903
2071
2423
counts
972
729
486
243
0
12519
9389
6260
3130
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS21,SBS40,SBS41 8 cosine sim = 0.98576
1664
1031
4355
965
2027
2479
counts
964
723
482
241
0
12519
9389
6260
3130
0
Signature
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS26,SBS40 7 cosine sim = 0.98574
1948
959
4210
888
2077
2437
counts
972
729
486
243
0
12519
9389
6260
3130
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS26,SBS40,SBS41,SBS44 9 cosine sim = 0.98574
1640
996
4415
954
2042
2469
counts
964
723
482
241
0
12519
9389
6260
3130
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS21,SBS26,SBS40 8 cosine sim = 0.98571
1684
1027
4365
927
2053
2466
counts
964
723
482
241
0
12519
9389
6260
3130
0
Signature
SBS40
SBS26
SBS21
SBS17b
SBS17a
SBS15
SBS2
SBS1


C>A
C>G
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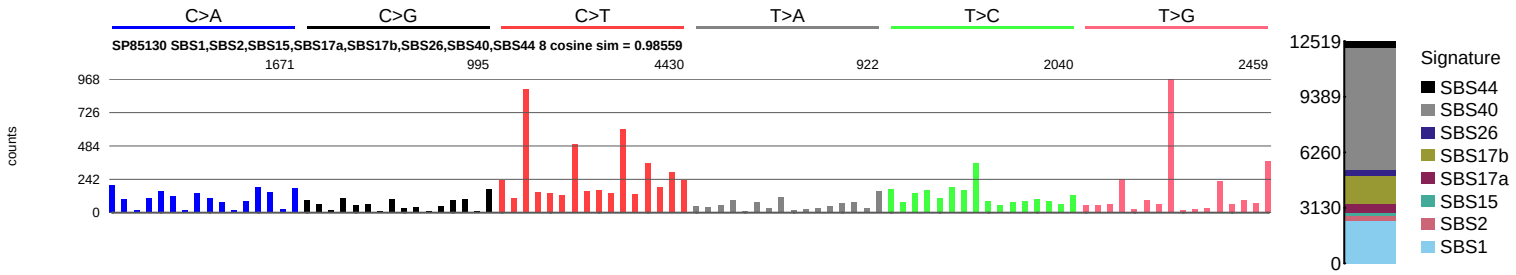
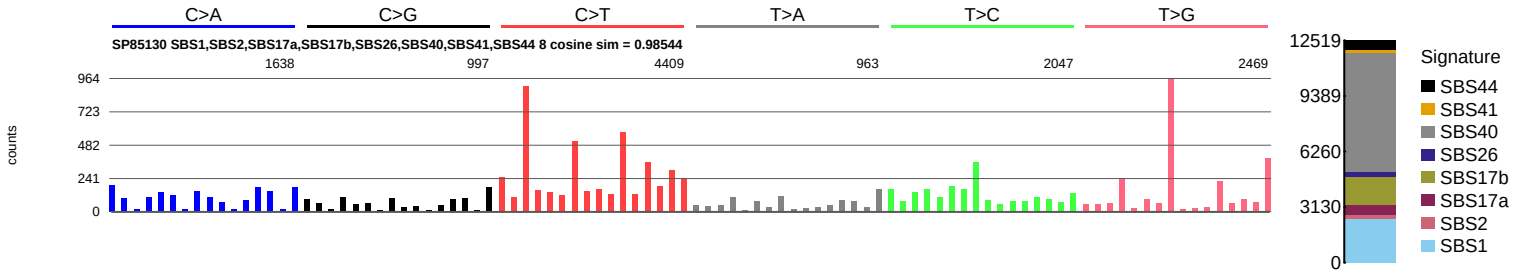
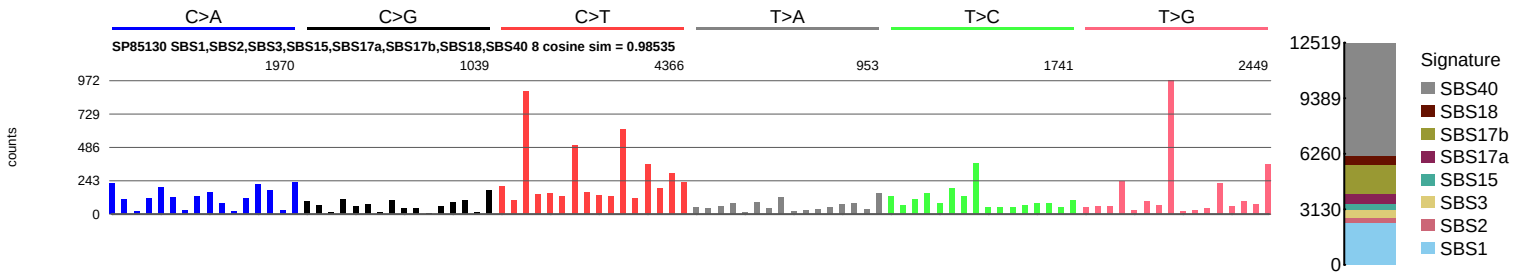
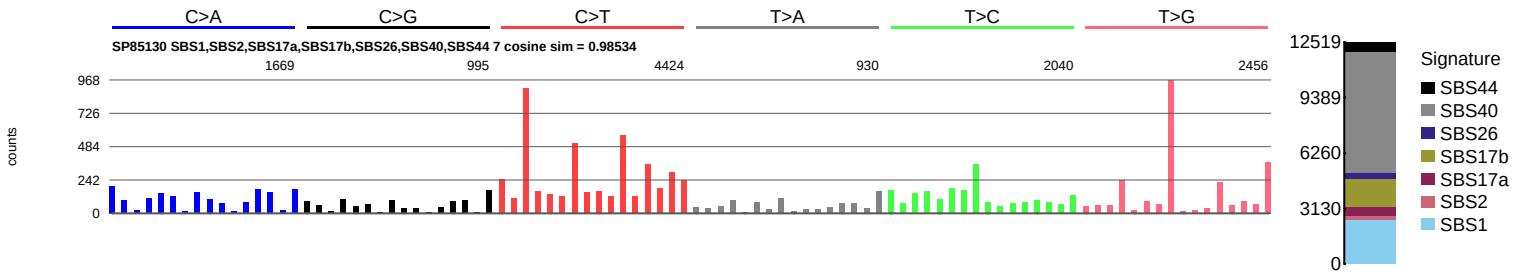
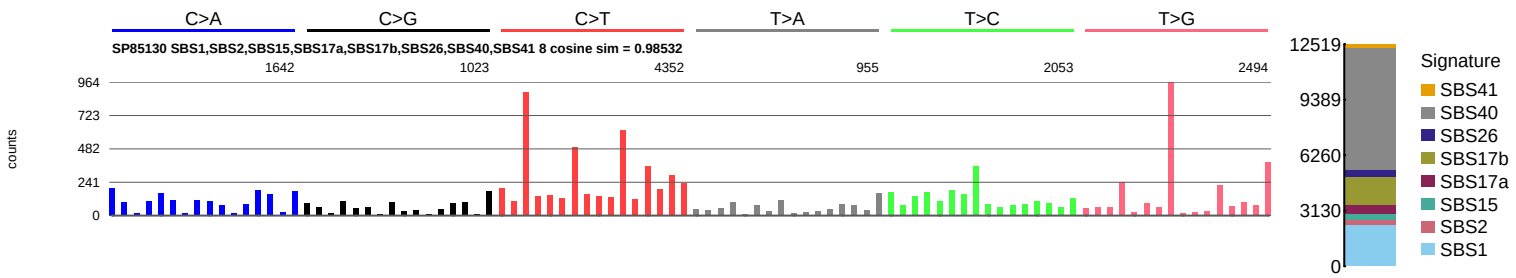
T>C
T>G
SP85130 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS21,SBS40 7 cosine sim = 0.98565
1696
1029
4370
931
2020
2467
counts
964
723
482
241
0
12519
9389
6260
3130
0
Signature
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS2
SBS1

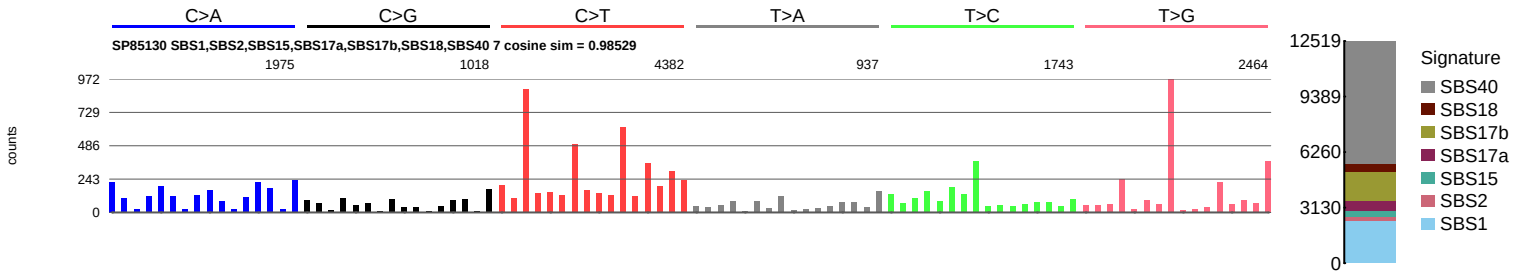

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS26,SBS40,SBS44 8 cosine sim = 0.98559
1671
995
4430
922
2040
2459
counts
968
726
484
242
0
12519
9389
6260
3130
0
Signature
SBS44
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS2
SBS1

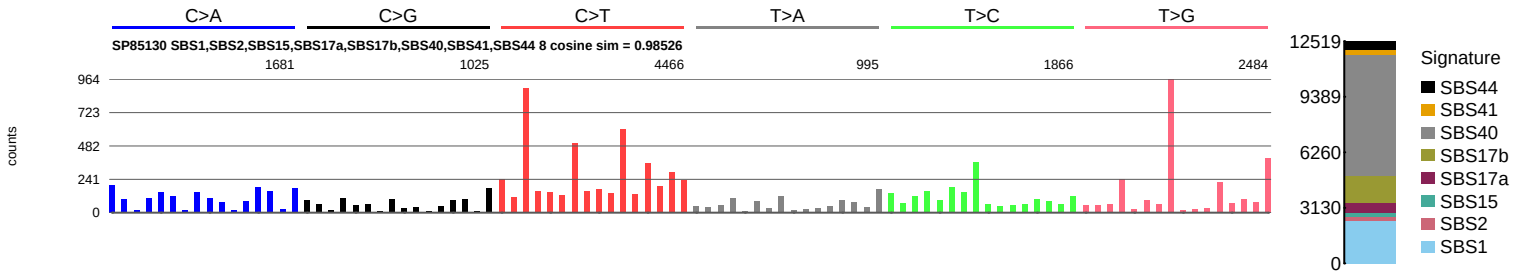

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS17a,SBS17b,SBS26,SBS40,SBS41,SBS44 8 cosine sim = 0.98544
1638
997
4409
963
2047
2469
counts
964
723
482
241
0
12519
9389
6260
3130
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS2
SBS1

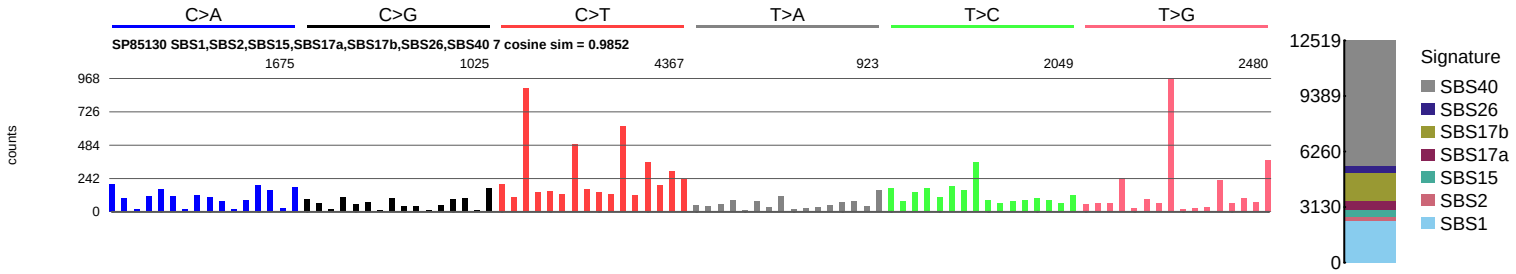

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS40 8 cosine sim = 0.98535
1970
1039
4366
953
1741
2449
counts
972
729
486
243
0
12519
9389
6260
3130
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1

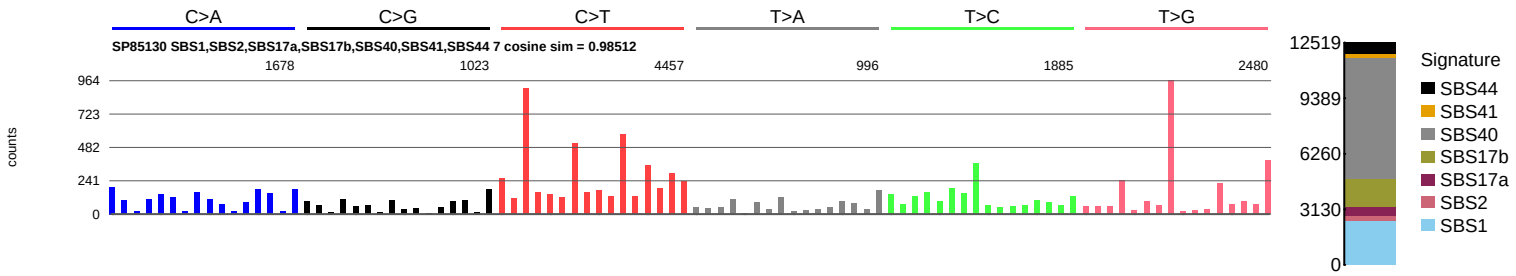

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS17a,SBS17b,SBS26,SBS40,SBS44 7 cosine sim = 0.98534
1669
995
4424
930
2040
2456
counts
968
726
484
242
0
12519
9389
6260
3130
0
Signature
SBS44
SBS40
SBS26
SBS17b
SBS17a
SBS2
SBS1

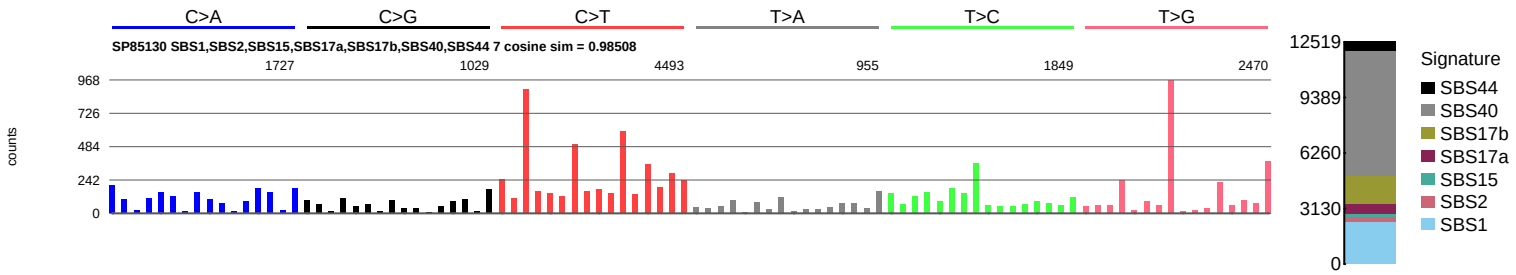

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS26,SBS40,SBS41 8 cosine sim = 0.98532
1642
1023
4352
955
2053
2494
counts
964
723
482
241
0
12519
9389
6260
3130
0
Signature
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS2
SBS1

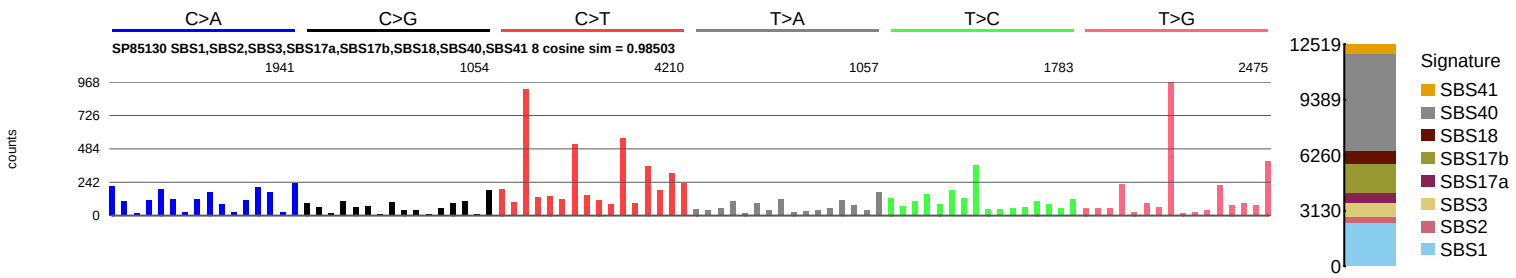

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS40 7 cosine sim = 0.98529
1975
1018
4382
937
1743
2464
counts
972
729
486
243
0
12519
9389
6260
3130
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1

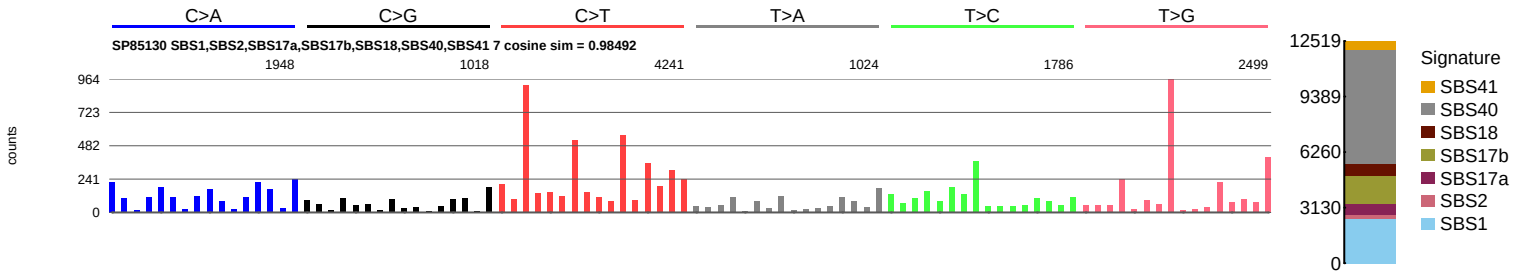

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS40,SBS41,SBS44 8 cosine sim = 0.98526
1681
1025
4466
995
1866
2484
counts
964
723
482
241
0
12519
9389
6260
3130
0
Signature
SBS44
SBS41
SBS40
SBS17b
SBS17a
SBS15
SBS2
SBS1

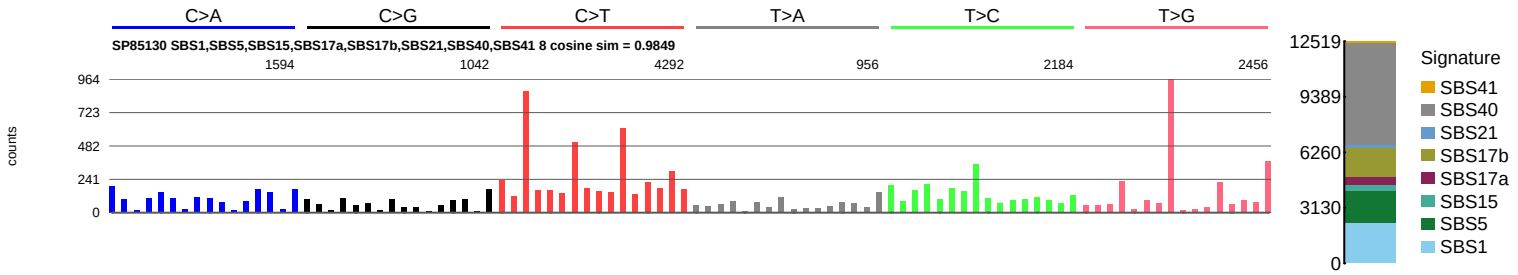

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS26,SBS40 7 cosine sim = 0.9852
1675
1025
4367
923
2049
2480
counts
968
726
484
242
0
12519
9389
6260
3130
0
Signature
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS2
SBS1

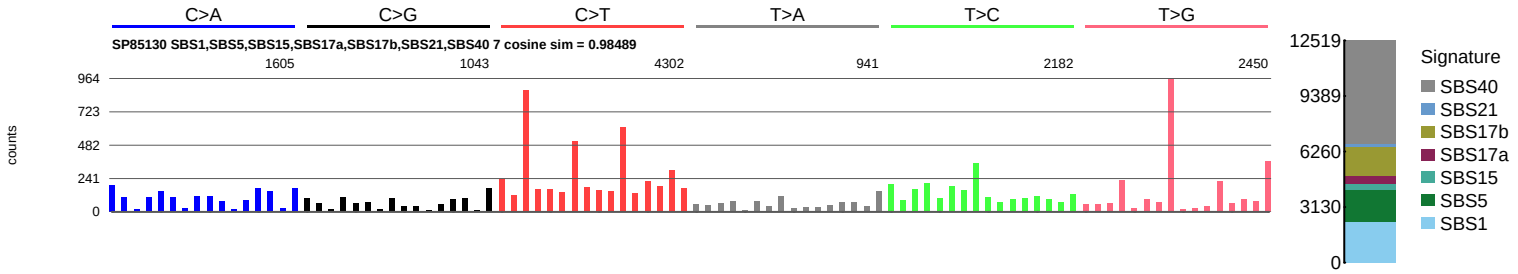

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS17a,SBS17b,SBS40,SBS41,SBS44 7 cosine sim = 0.98512
1678
1023
4457
996
1885
2480
counts
964
723
482
241
0
12519
9389
6260
3130
0
Signature
SBS44
SBS41
SBS40
SBS17b
SBS17a
SBS2
SBS1

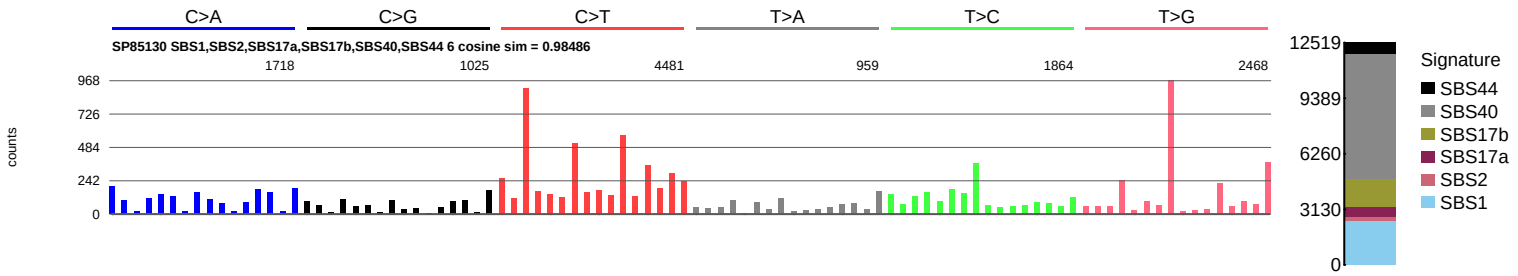

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS40,SBS44 7 cosine sim = 0.98508
1727
1029
4493
955
1849
2470
counts
968
726
484
242
0
12519
9389
6260
3130
0
Signature
SBS44
SBS40
SBS17b
SBS17a
SBS15
SBS2
SBS1

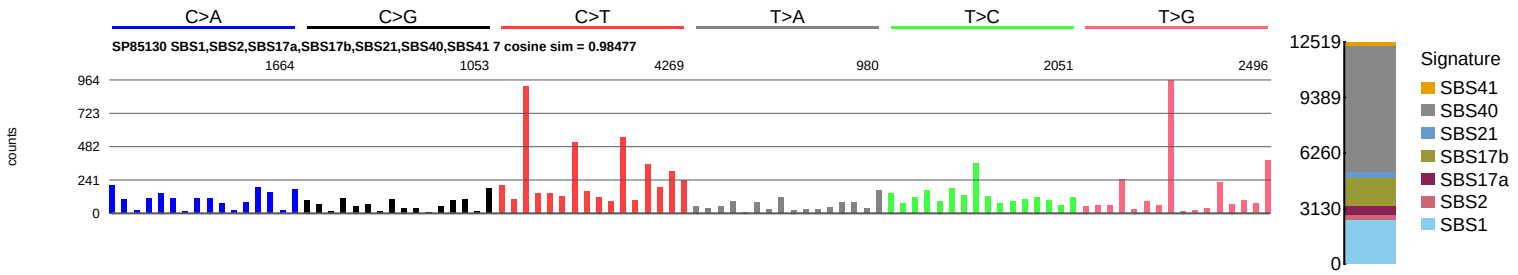

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS40,SBS41 8 cosine sim = 0.98503
1941
1054
4210
1057
1783
2475
counts
968
726
484
242
0
12519
9389
6260
3130
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1

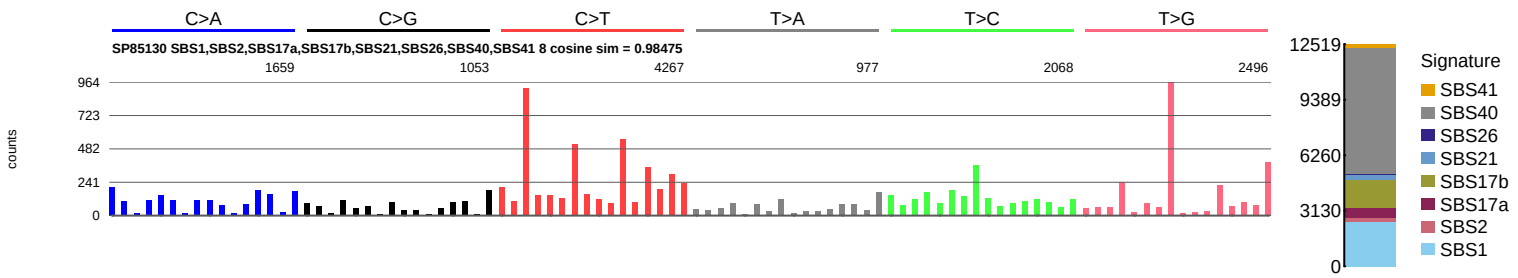

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS40,SBS41 7 cosine sim = 0.98492
1948
1018
4241
1024
1786
2499
counts
964
723
482
241
0
12519
9389
6260
3130
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS2
SBS1

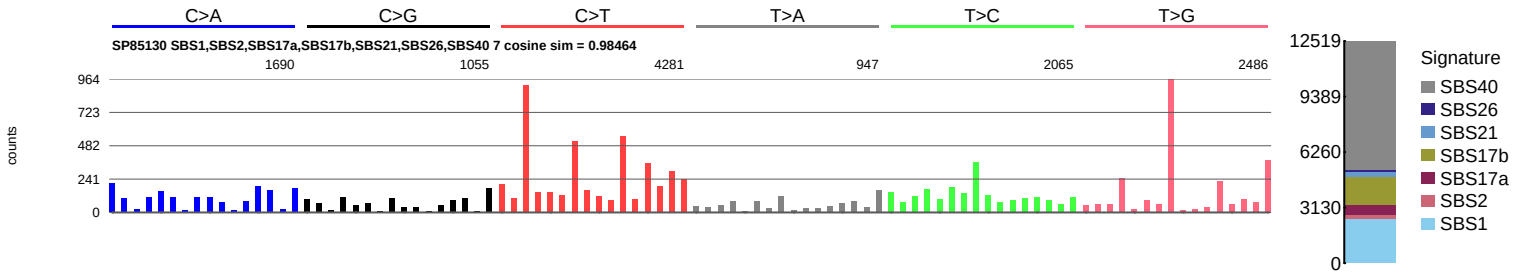

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS21,SBS40,SBS41 8 cosine sim = 0.9849
1594
1042
4292
956
2184
2456
counts
964
723
482
241
0
12519
9389
6260
3130
0
Signature
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS5
SBS1

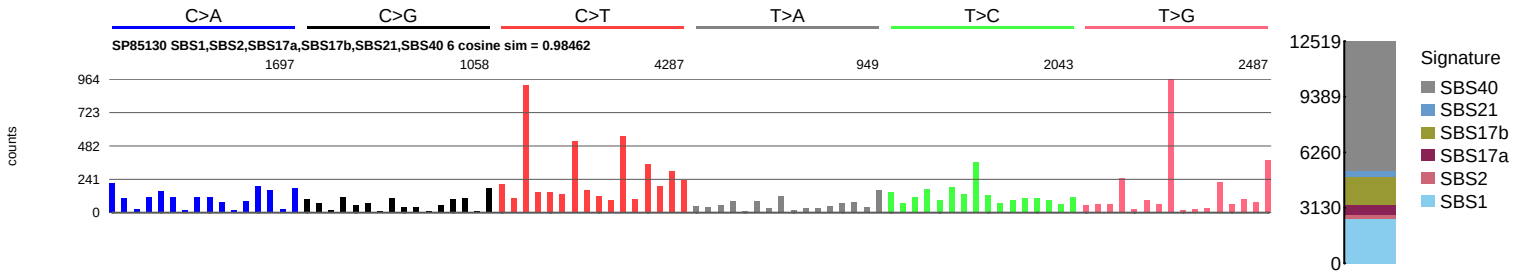

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS21,SBS40 7 cosine sim = 0.98489
1605
1043
4302
941
2182
2450
counts
964
723
482
241
0
12519
9389
6260
3130
0
Signature
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS17a,SBS17b,SBS40,SBS44 6 cosine sim = 0.98486
1718
1025
4481
959
1864
2468
counts
968
726
484
242
0
12519
9389
6260
3130
0
Signature
SBS44
SBS40
SBS17b
SBS17a
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS17a,SBS17b,SBS21,SBS40,SBS41 7 cosine sim = 0.98477
1664
1053
4269
980
2051
2496
counts
964
723
482
241
0
12519
9389
6260
3130
0
Signature
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS17a,SBS17b,SBS21,SBS26,SBS40,SBS41 8 cosine sim = 0.98475
1659
1053
4267
977
2068
2496
counts
964
723
482
241
0
12519
9389
6260
3130
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS17b
SBS17a
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS17a,SBS17b,SBS21,SBS26,SBS40 7 cosine sim = 0.98464
1690
1055
4281
947
2065
2486
counts
964
723
482
241
0
12519
9389
6260
3130
0
Signature
SBS40
SBS26
SBS21
SBS17b
SBS17a
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS17a,SBS17b,SBS21,SBS40 6 cosine sim = 0.98462
1697
1058
4287
949
2043
2487
counts
964
723
482
241
0
12519
9389
6260
3130
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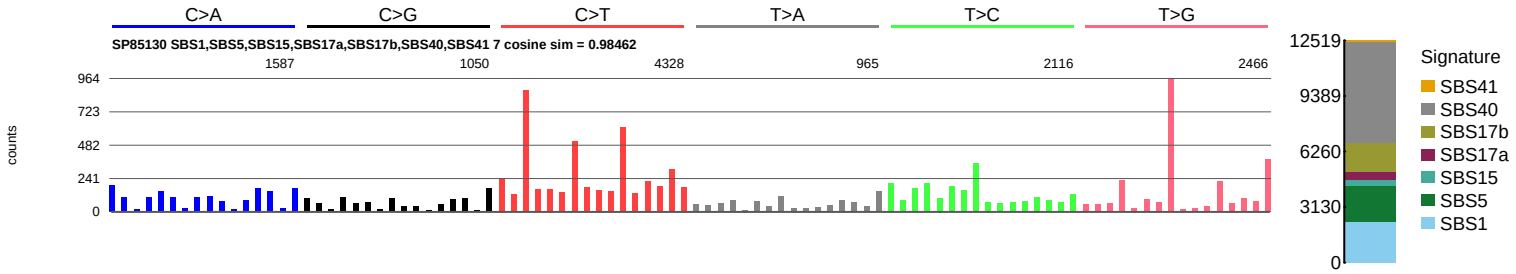
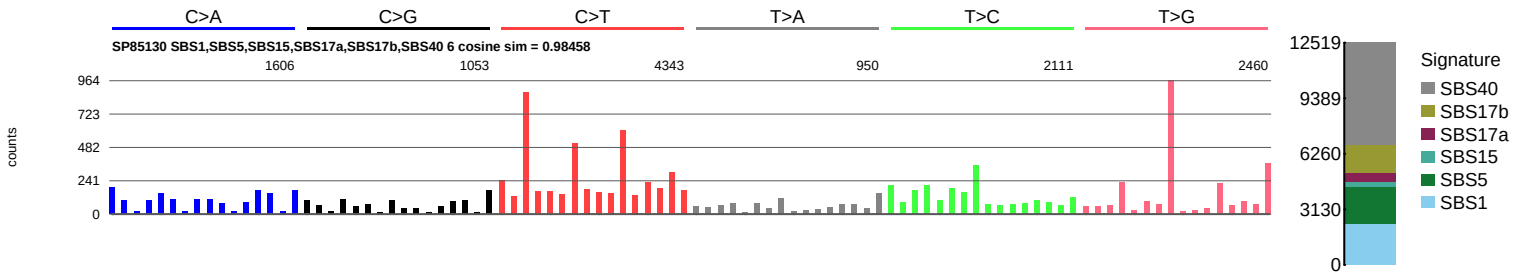
Signature
SBS40
SBS21
SBS17b
SBS17a
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS40,SBS41 7 cosine sim = 0.98462
1587
1050
4328
965
2116
2466
counts
964
723
482
241
0
12519
9389
6260
3130
0
Signature
SBS41
SBS40
SBS17b
SBS17a
SBS15
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS40 6 cosine sim = 0.98458
1606
1053
4343
950
2111
2460
counts
964
723
482
241
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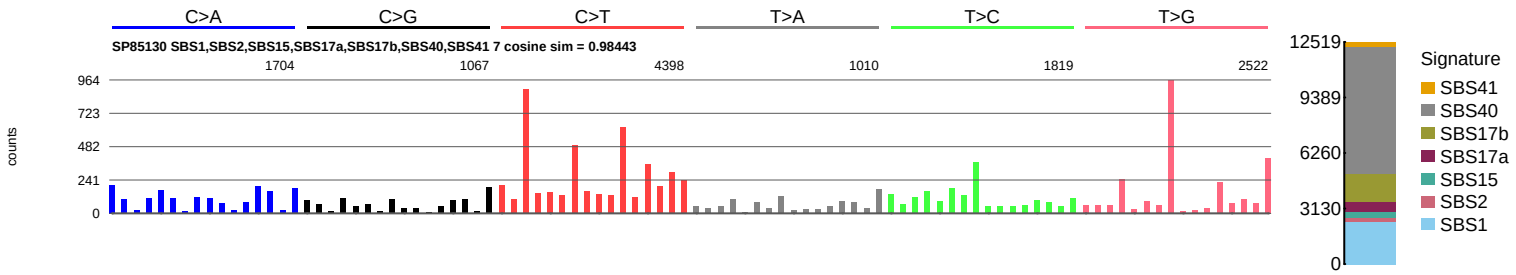
SBS1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SP85130 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS40,SBS41 7 cosine sim = 0.98443
1704
1067
4398
1010
1819
2522
counts
964
723
482
241
0
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9389
6260
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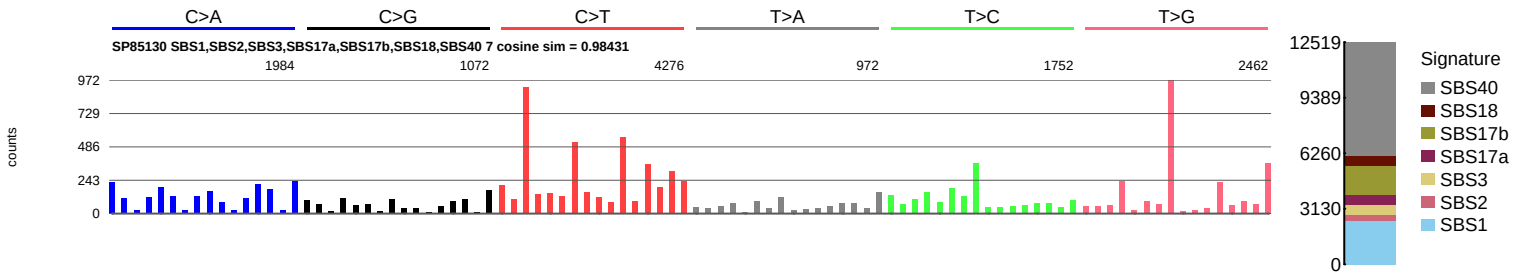
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SP85130 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS40 7 cosine sim = 0.98431
1984
1072
4276
972
1752
2462
counts
972
729
486
243
0
12519
9389
6260
3130
0
Signature
SBS40
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SBS17b
SBS17a
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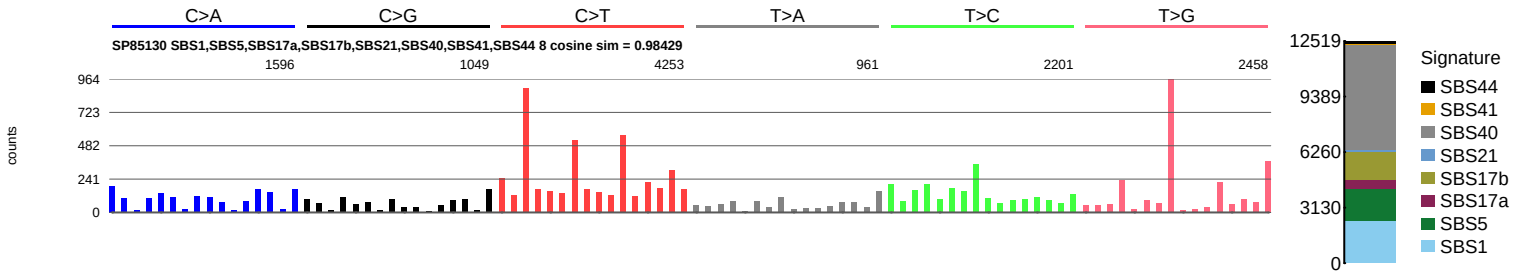
SBS2
SBS1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SP85130 SBS1,SBS5,SBS17a,SBS17b,SBS21,SBS40,SBS41,SBS44 8 cosine sim = 0.98429
1596
1049
4253
961
2201
2458
counts
964
723
482
241
0
12519
9389
6260
3130
0
Signature
SBS44
SBS41
SBS40
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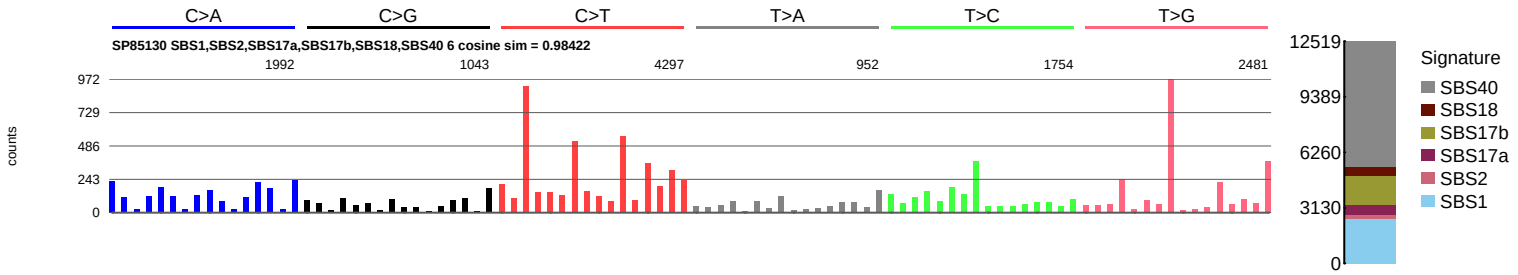
SBS17b
SBS17a
SBS5
SBS1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T>C
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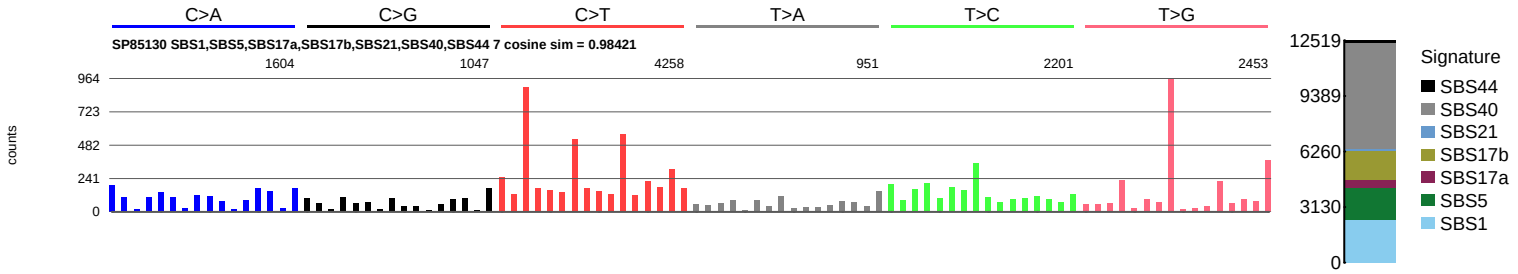
SP85130 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS40 6 cosine sim = 0.98422
1992
1043
4297
952
1754
2481
counts
972
729
486
243
0
12519
9389
6260
3130
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS2
SBS1


C>A
C>G
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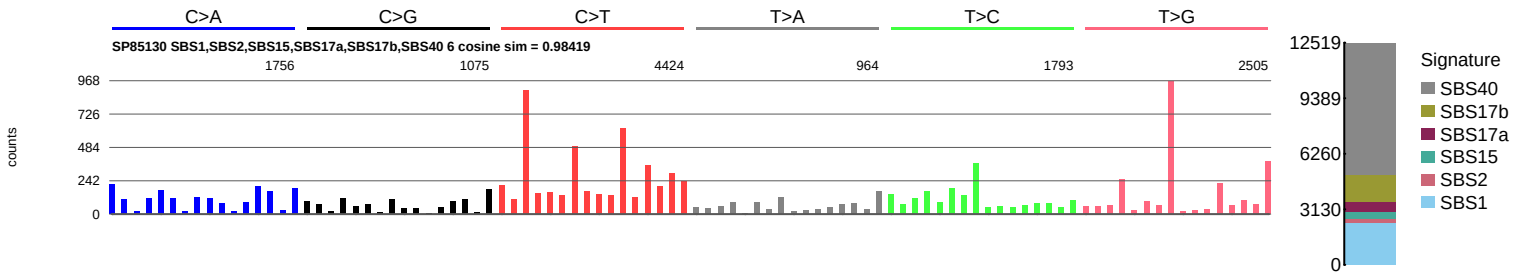
T>C
T>G
SP85130 SBS1,SBS5,SBS17a,SBS17b,SBS21,SBS40,SBS44 7 cosine sim = 0.98421
1604
1047
4258
951
2201
2453
counts
964
723
482
241
0
12519
9389
6260
3130
0
Signature
SBS44
SBS40
SBS21
SBS17b
SBS17a
SBS5
SBS1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C>G
C>T
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T>C
T>G
SP85130 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS40 6 cosine sim = 0.98419
1756
1075
4424
964
1793
2505
counts
968
726
484
242
0
12519
9389
6260
3130
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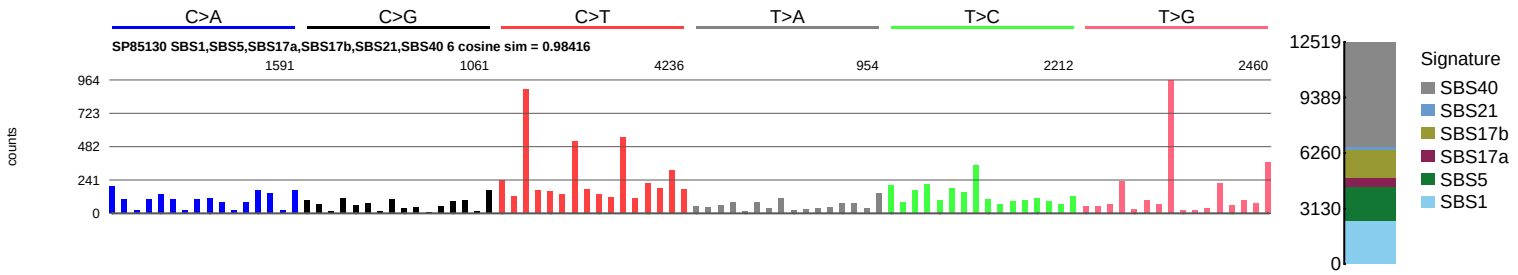
SBS2
SBS1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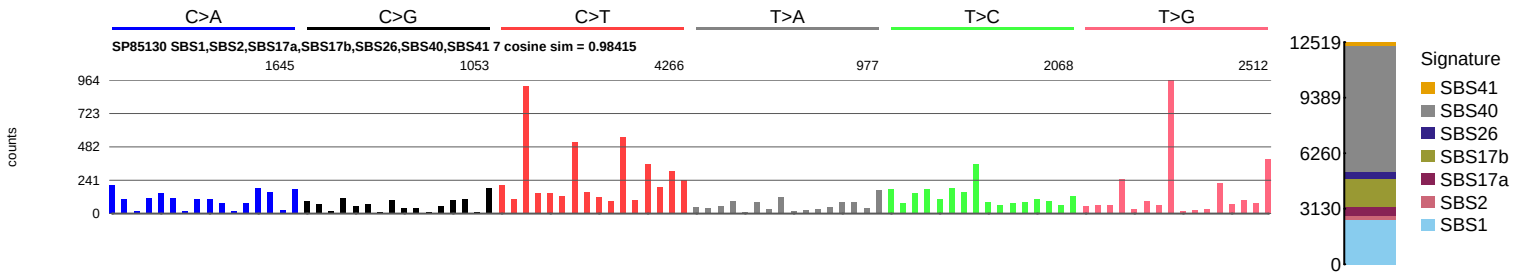
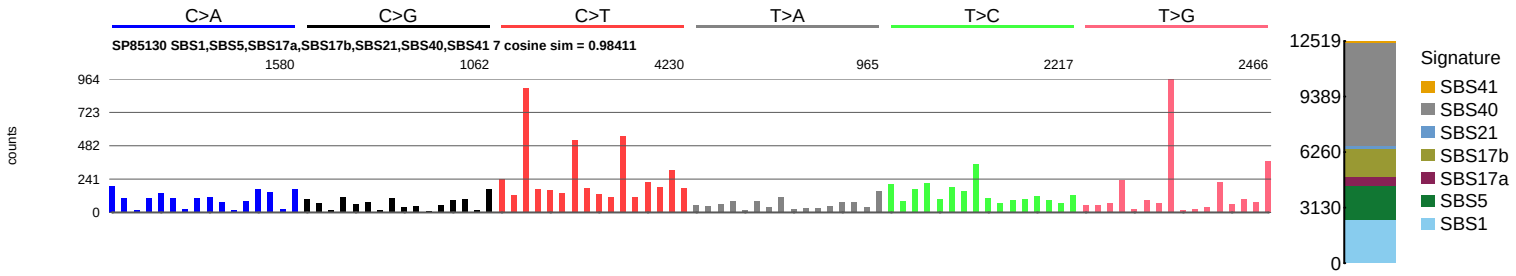
T>A
T>C
T>G
SP85130 SBS1,SBS5,SBS17a,SBS17b,SBS21,SBS40 6 cosine sim = 0.98416
1591
1061
4236
954
2212
2460
counts
964
723
482
241
0
12519
9389
6260
3130
0
Signature
SBS40
SBS21
SBS17b
SBS17a
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS17a,SBS17b,SBS26,SBS40,SBS41 7 cosine sim = 0.98415
1645
1053
4266
977
2068
2512
counts
964
723
482
241
0
12519
9389
6260
3130
0
Signature
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS2
SBS1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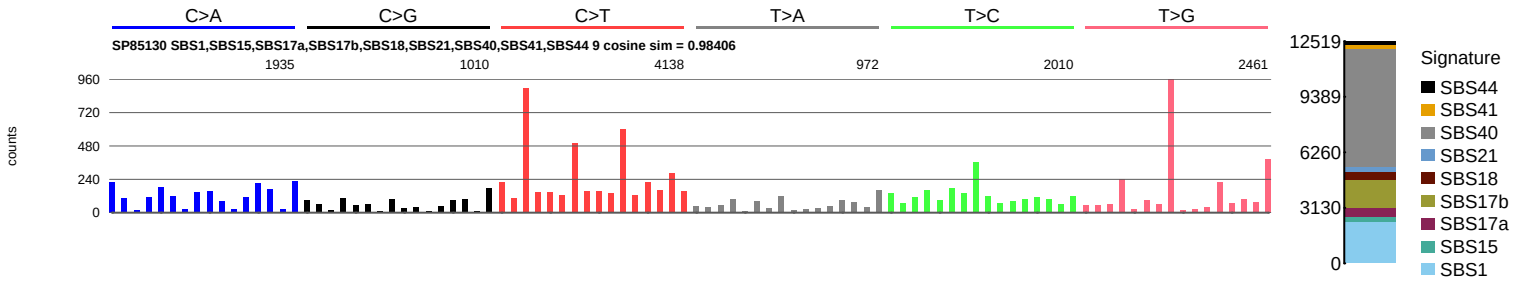
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS5,SBS17a,SBS17b,SBS21,SBS40,SBS41 7 cosine sim = 0.98411
1580
1062
4230
965
2217
2466
counts
964
723
482
241
0
12519
9389
6260
3130
0
Signature
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS5
SBS1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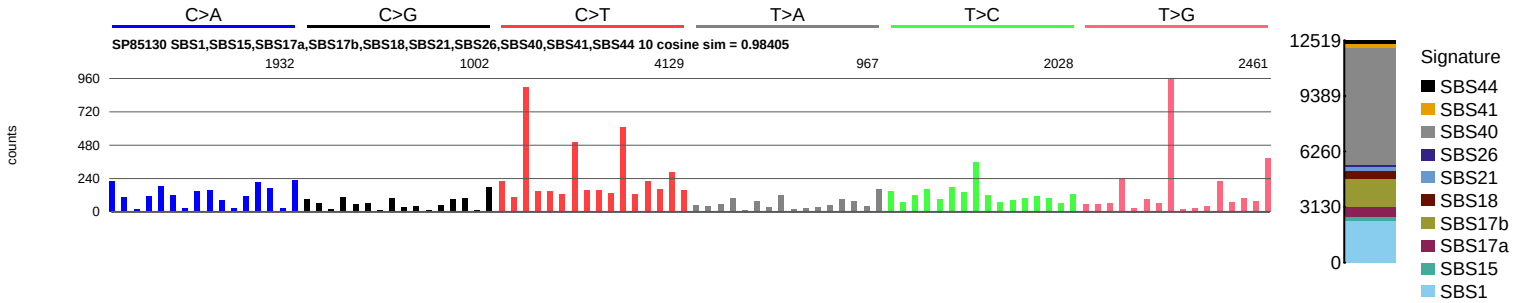
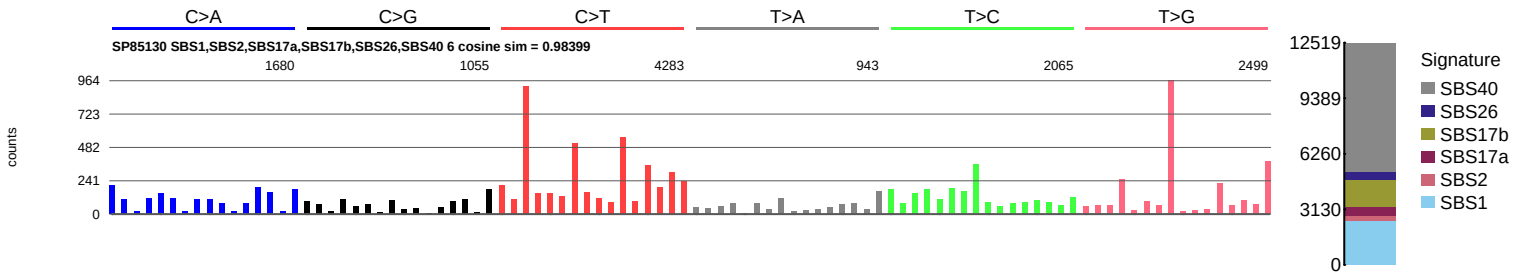
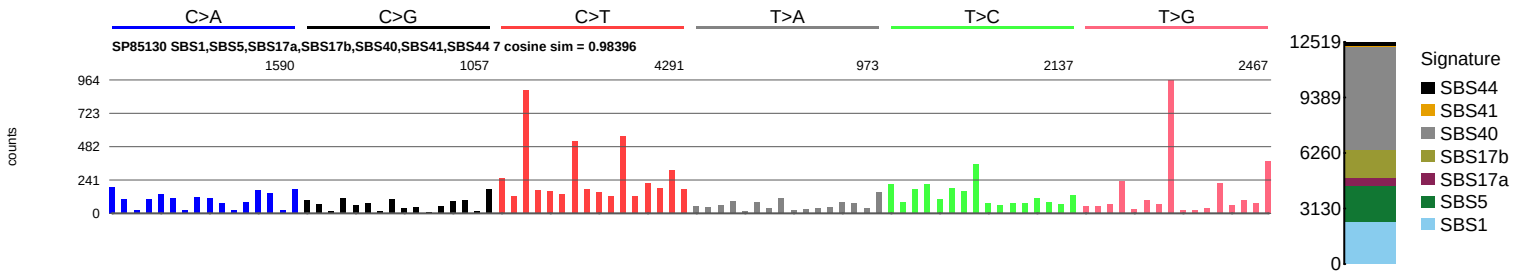
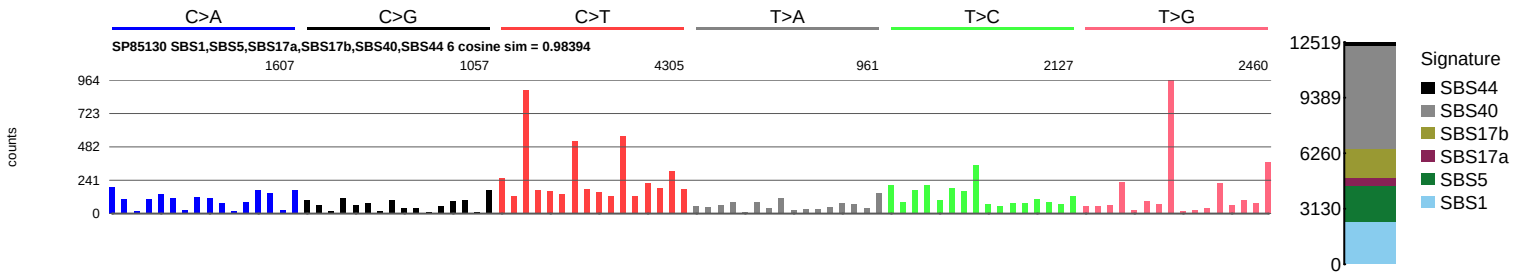
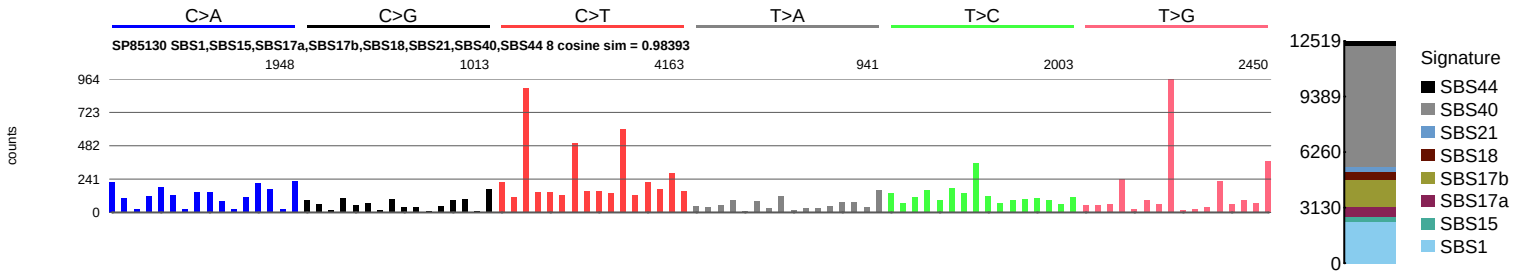
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 9 cosine sim = 0.98406
1935
1010
4138
972
2010
2461
counts
960
720
480
240
0
12519
9389
6260
3130
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS1

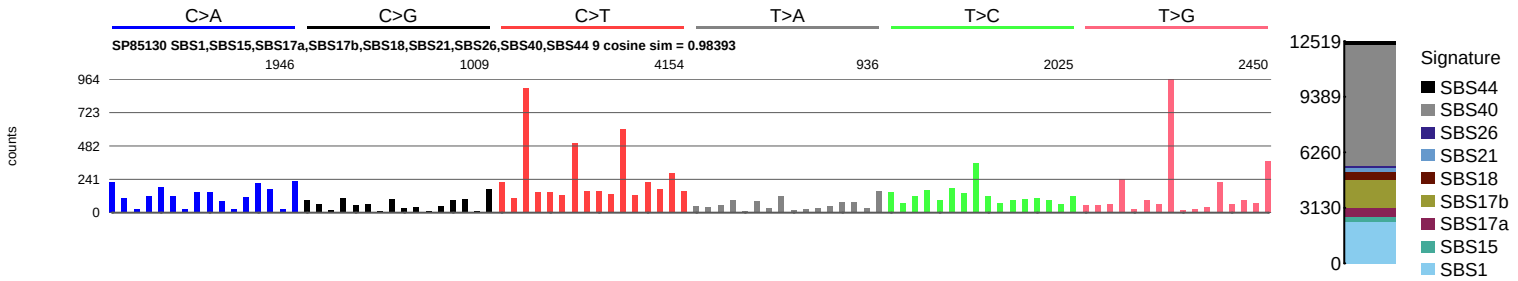

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.98405
1932
1002
4129
967
2028
2461
counts
960
720
480
240
0
12519
9389
6260
3130
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS17a,SBS17b,SBS26,SBS40 6 cosine sim = 0.98399
1680
1055
4283
943
2065
2499
counts
964
723
482
241
0
12519
9389
6260
3130
0
Signature
SBS40
SBS26
SBS17b
SBS17a
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS5,SBS17a,SBS17b,SBS40,SBS41,SBS44 7 cosine sim = 0.98396
1590
1057
4291
973
2137
2467
counts
964
723
482
241
0
12519
9389
6260
3130
0
Signature
SBS44
SBS41
SBS40
SBS17b
SBS17a
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS5,SBS17a,SBS17b,SBS40,SBS44 6 cosine sim = 0.98394
1607
1057
4305
961
2127
2460
counts
964
723
482
241
0
12519
9389
6260
3130
0
Signature
SBS44
SBS40
SBS17b
SBS17a
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 8 cosine sim = 0.98393
1948
1013
4163
941
2003
2450
counts
964
723
482
241
0
12519
9389
6260
3130
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS1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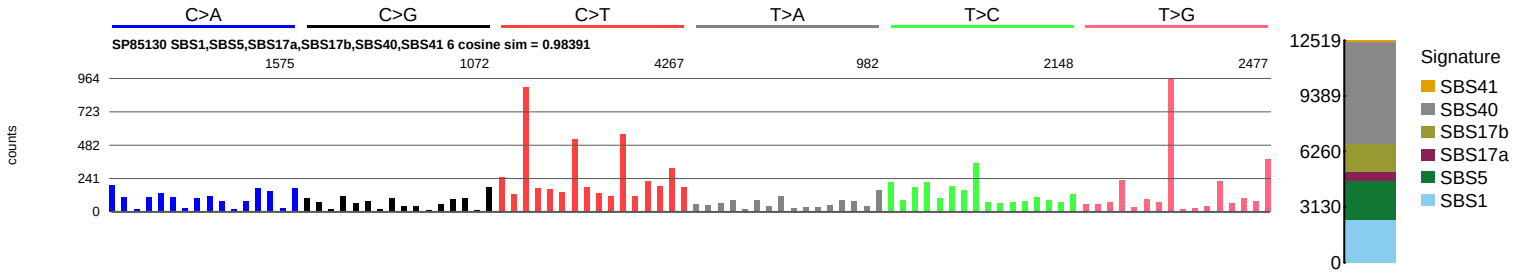
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS44 9 cosine sim = 0.98393
1946
1009
4154
936
2025
2450
counts
964
723
482
241
0
12519
9389
6260
3130
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS1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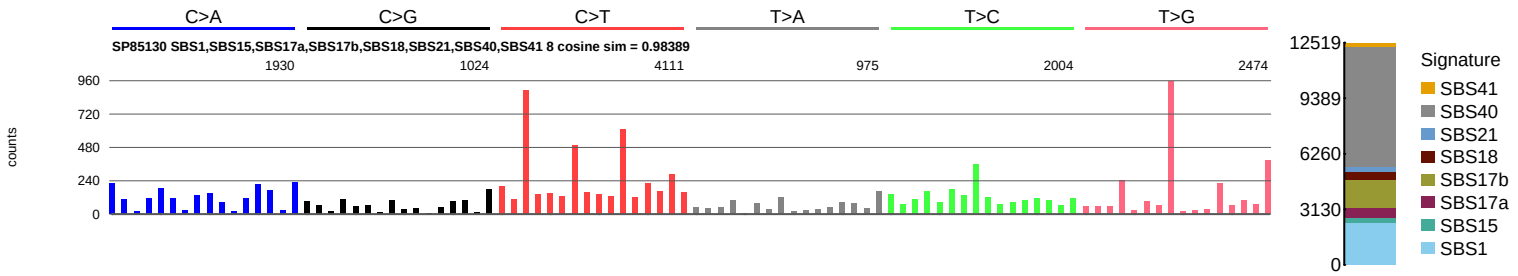
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS5,SBS17a,SBS17b,SBS40,SBS41 6 cosine sim = 0.98391
1575
1072
4267
982
2148
2477
counts
964
723
482
241
0
12519
9389
6260
3130
0
Signature
SBS41
SBS40
SBS17b
SBS17a
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 8 cosine sim = 0.98389
1930
1024
4111
975
2004
2474
counts
960
720
480
240
0
12519
9389
6260
3130
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
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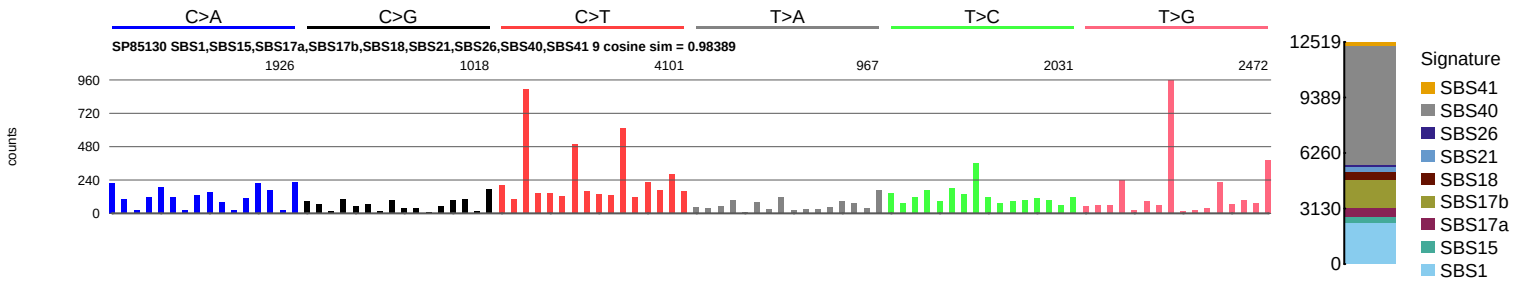
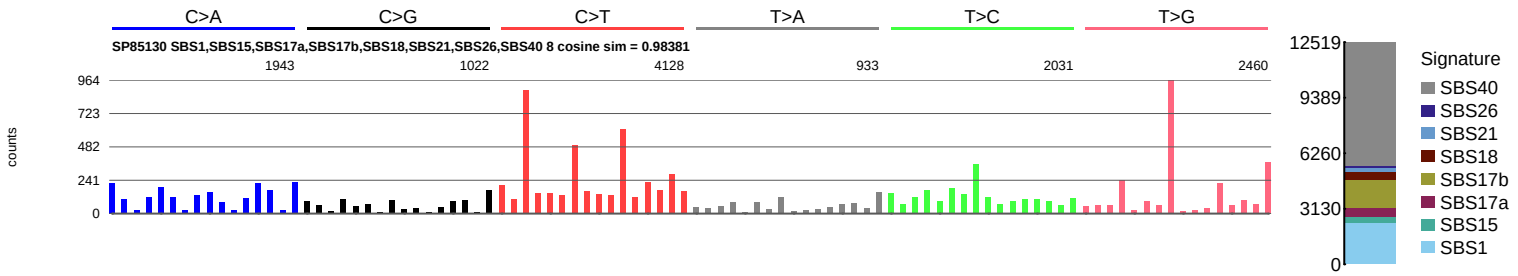
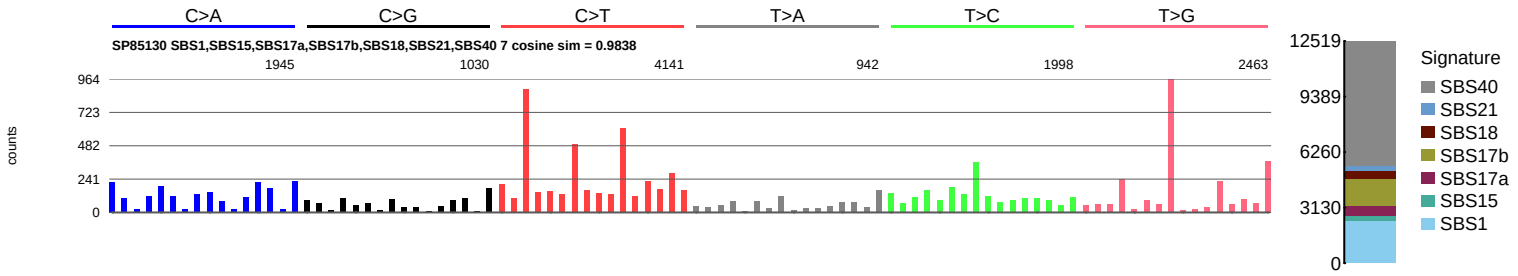
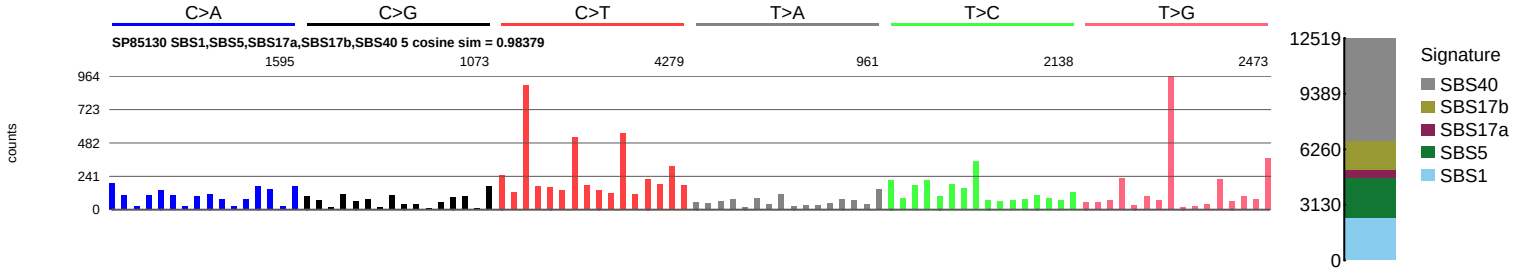
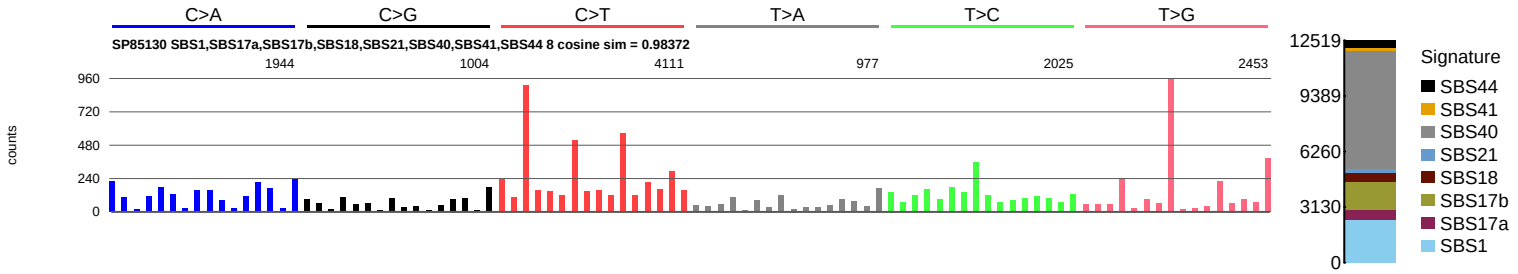
SBS15
SBS1

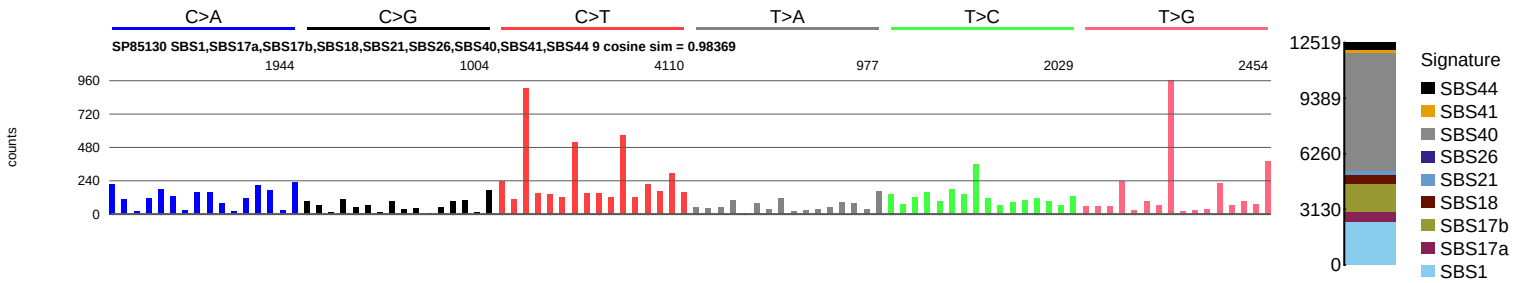

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41 9 cosine sim = 0.98389
1926
1018
4101
967
2031
2472
counts
960
720
480
240
0
12519
9389
6260
3130
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS1

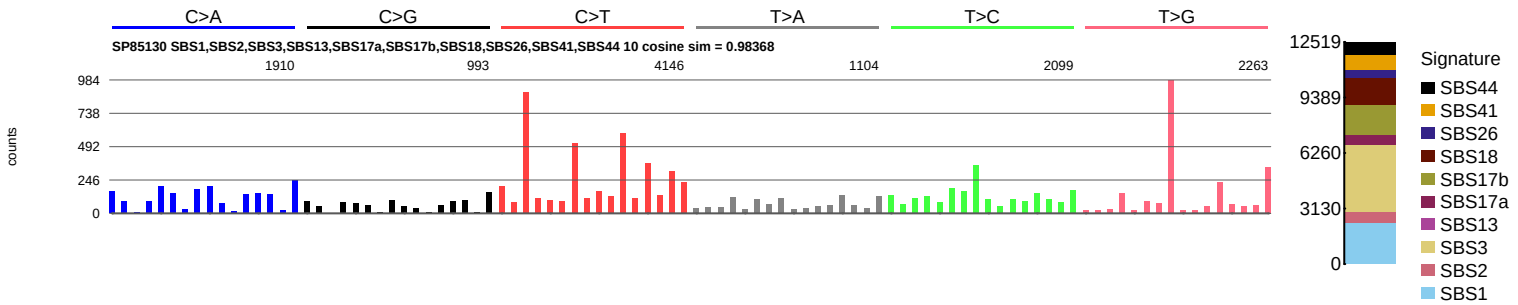

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40 8 cosine sim = 0.98381
1943
1022
4128
933
2031
2460
counts
964
723
482
241
0
12519
9389
6260
3130
0
Signature
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS1

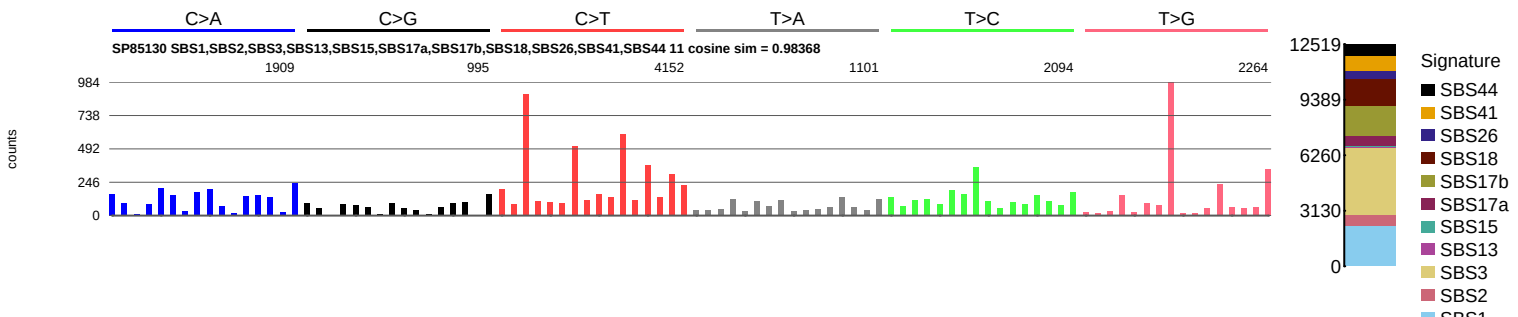

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40 7 cosine sim = 0.9838
1945
1030
4141
942
1998
2463
counts
964
723
482
241
0
12519
9389
6260
3130
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS5,SBS17a,SBS17b,SBS40 5 cosine sim = 0.98379
1595
1073
4279
961
2138
2473
counts
964
723
482
241
0
12519
9389
6260
3130
0
Signature
SBS40
SBS17b
SBS17a
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 8 cosine sim = 0.98372
1944
1004
4111
977
2025
2453
counts
960
720
480
240
0
12519
9389
6260
3130
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41,SBS44 9 cosine sim = 0.98369
1944
1004
4110
977
2029
2454
counts
960
720
480
240
0
12519
9389
6260
3130
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 10 cosine sim = 0.98368
1910
993
4146
1104
2099
2263
counts
984
738
492
246
0
12519
9389
6260
3130
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1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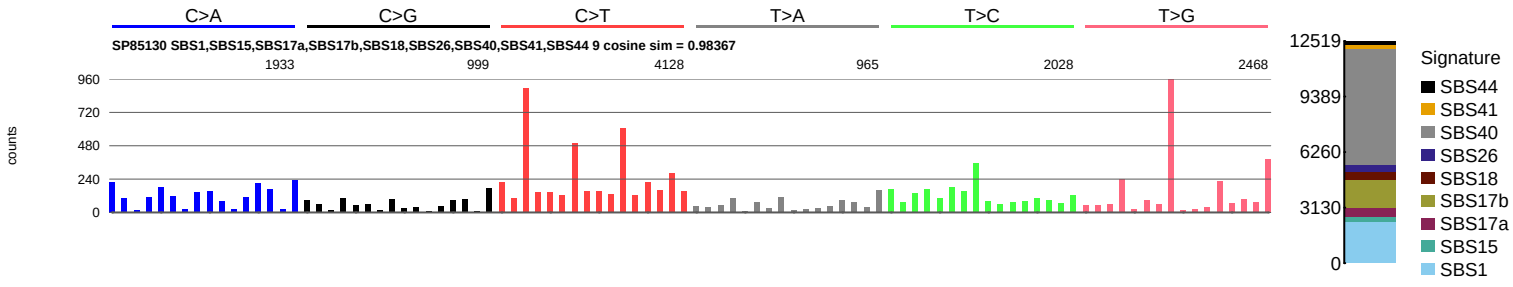
SP85130 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 11 cosine sim = 0.98368
1909
995
4152
1101
2094
2264
counts
984
738
492
246
0
12519
9389
6260
3130
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2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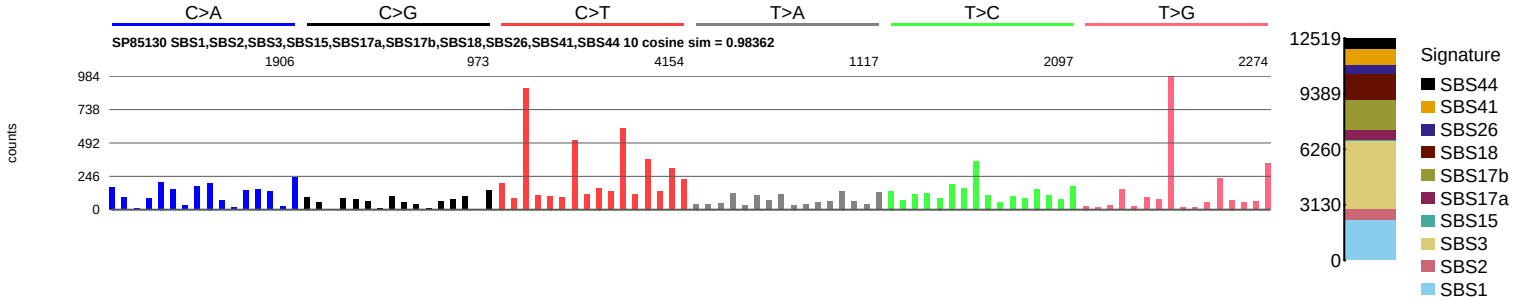
T>G
SP85130 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 9 cosine sim = 0.98367
1933
999
4128
965
2028
2468
counts
960
720
480
240
0
12519
9389
6260
3130
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS1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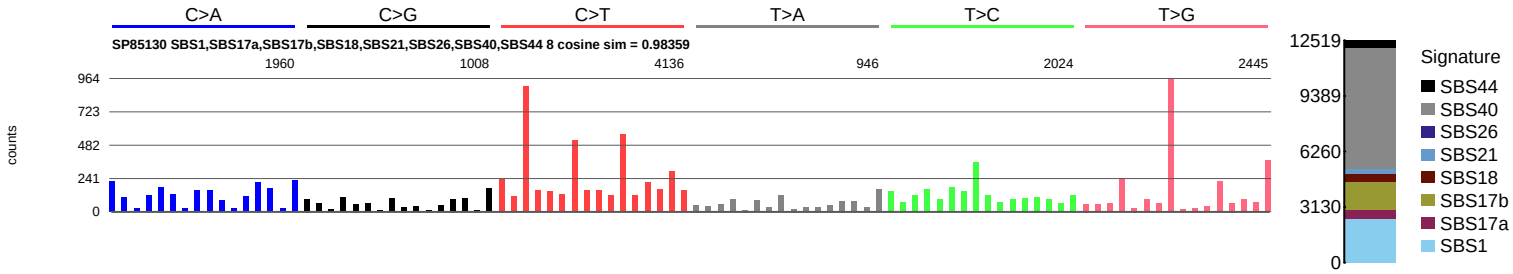
SP85130 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 10 cosine sim = 0.98362
1906
973
4154
1117
2097
2274
counts
984
738
492
246
0
12519
9389
6260
3130
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1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C>G
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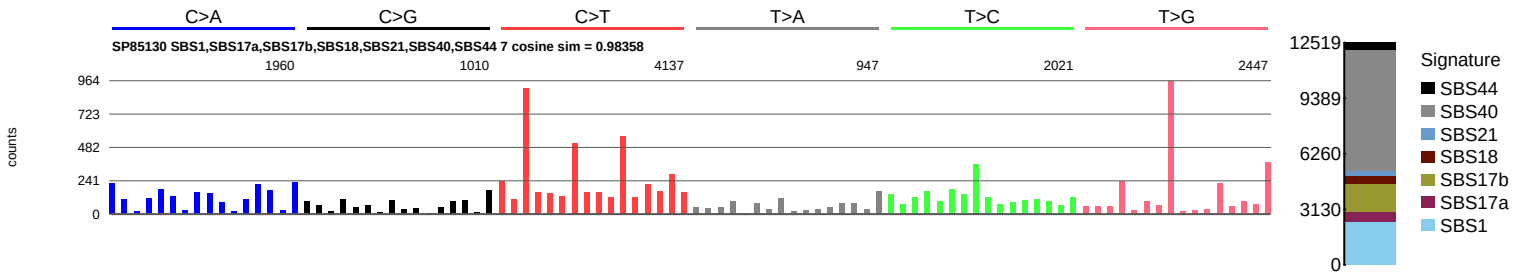
SP85130 SBS1,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS44 8 cosine sim = 0.98359
1960
1008
4136
946
2024
2445
counts
964
723
482
241
0
12519
9389
6260
3130
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS1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C>G
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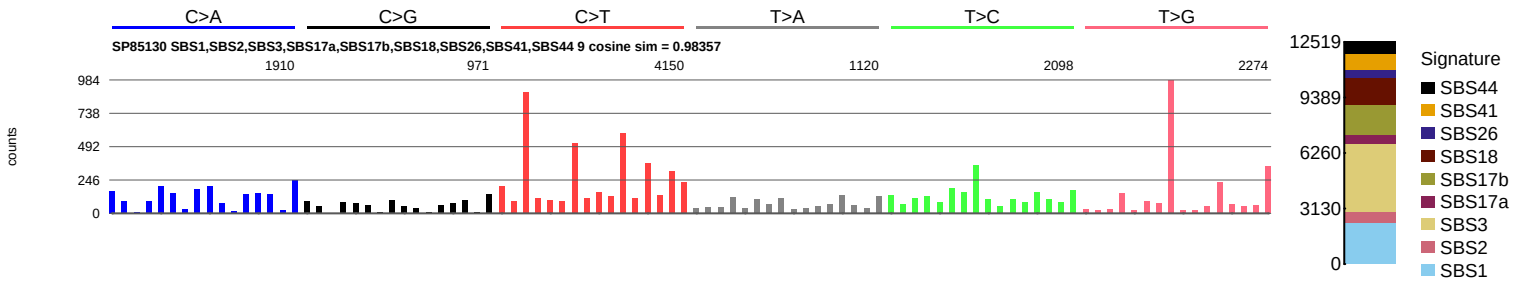
T>G
SP85130 SBS1,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 7 cosine sim = 0.98358
1960
1010
4137
947
2021
2447
counts
964
723
482
241
0
12519
9389
6260
3130
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS1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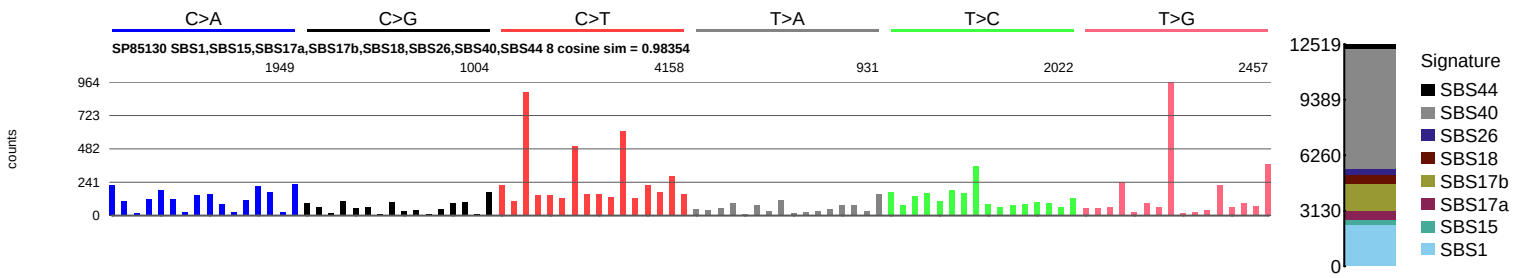
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 9 cosine sim = 0.98357
1910
971
4150
1120
2098
2274
counts
984
738
492
246
0
12519
9389
6260
3130
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1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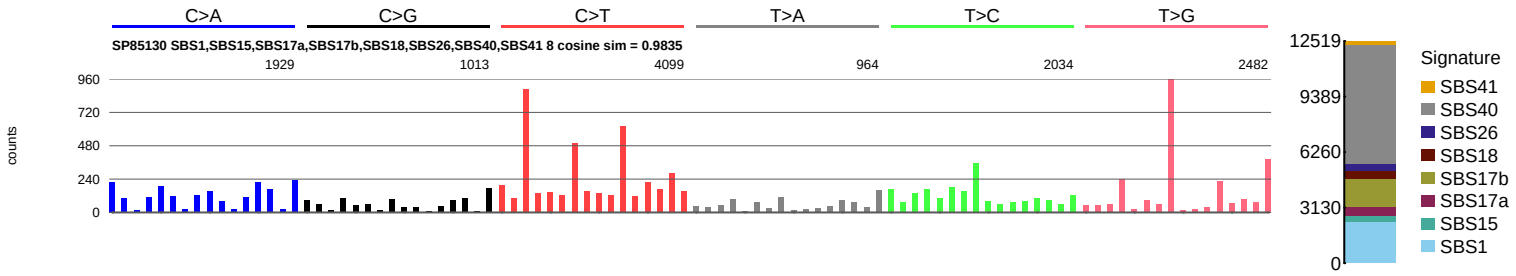
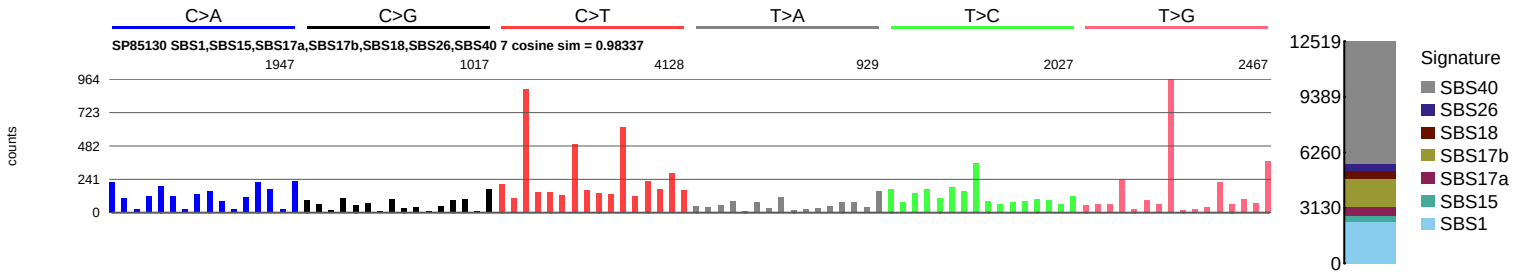
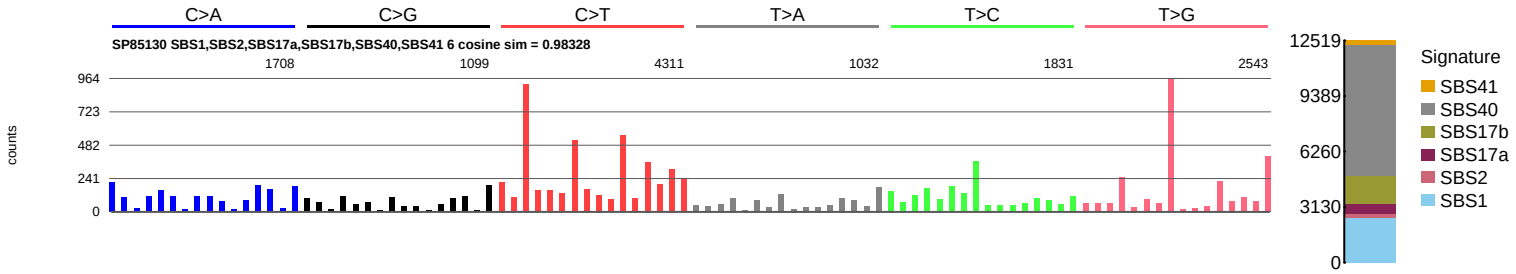
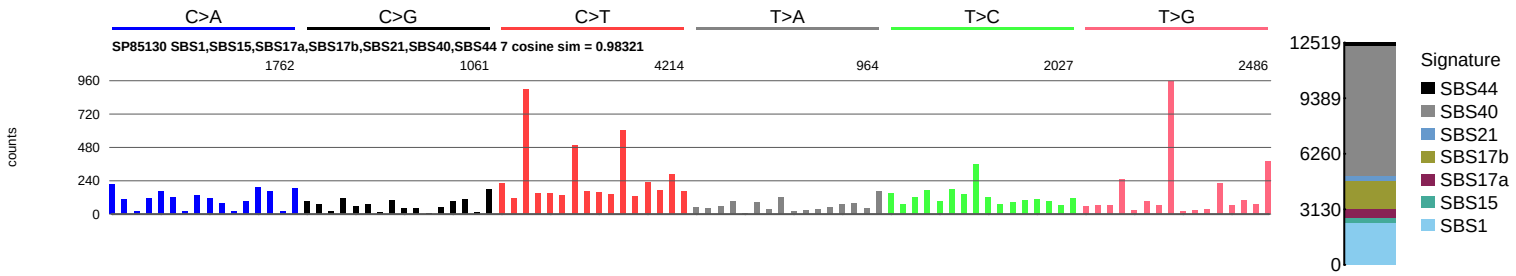
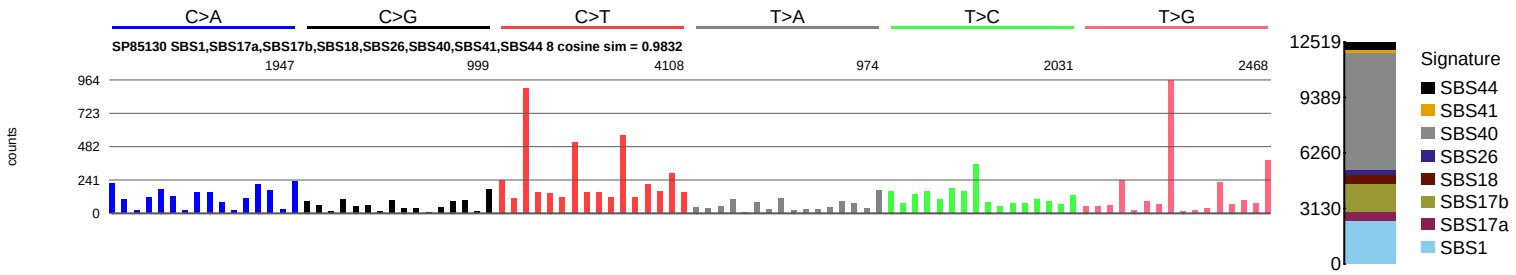
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 8 cosine sim = 0.98354
1949
1004
4158
931
2022
2457
counts
964
723
482
241
0
12519
9389
6260
3130
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS1

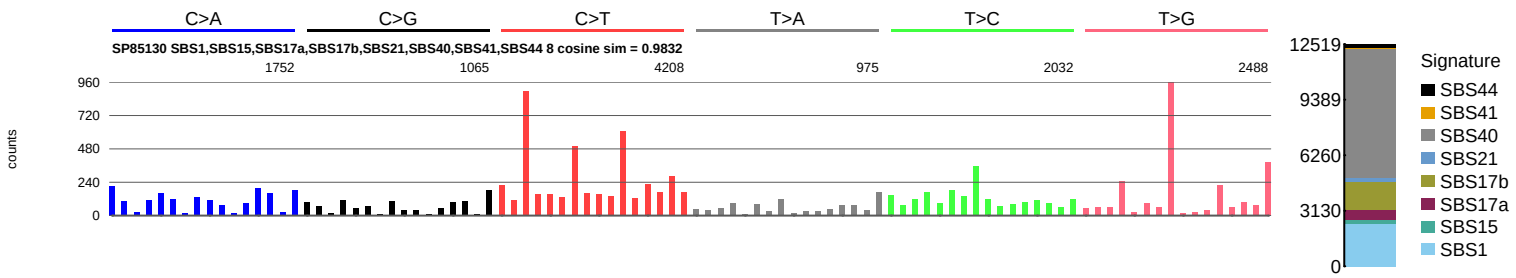

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 8 cosine sim = 0.9835
1929
1013
4099
964
2034
2482
counts
960
720
480
240
0
12519
9389
6260
3130
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS1

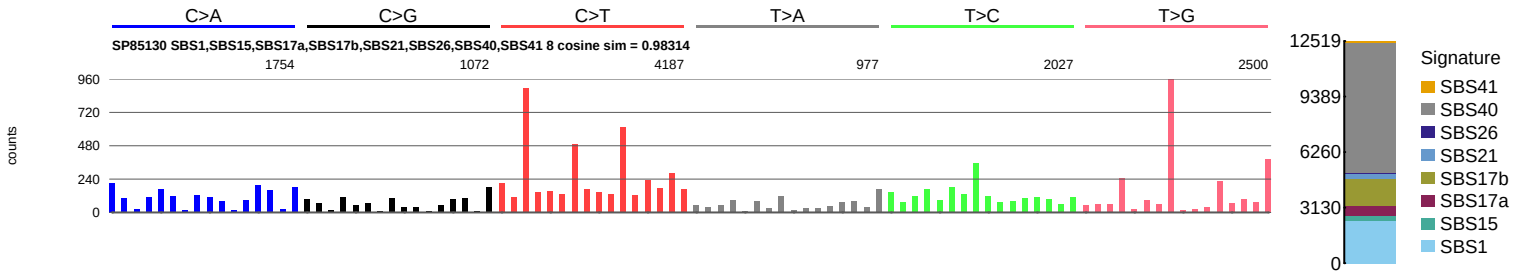

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40 7 cosine sim = 0.98337
1947
1017
4128
929
2027
2467
counts
964
723
482
241
0
12519
9389
6260
3130
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS1

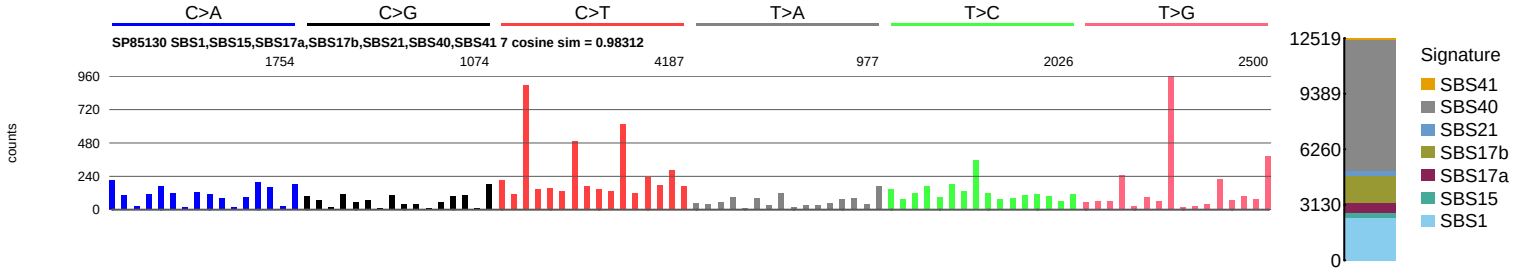

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS17a,SBS17b,SBS40,SBS41 6 cosine sim = 0.98328
1708
1099
4311
1032
1831
2543
counts
964
723
482
241
0
12519
9389
6260
3130
0
Signature
SBS41
SBS40
SBS17b
SBS17a
SBS2
SBS1

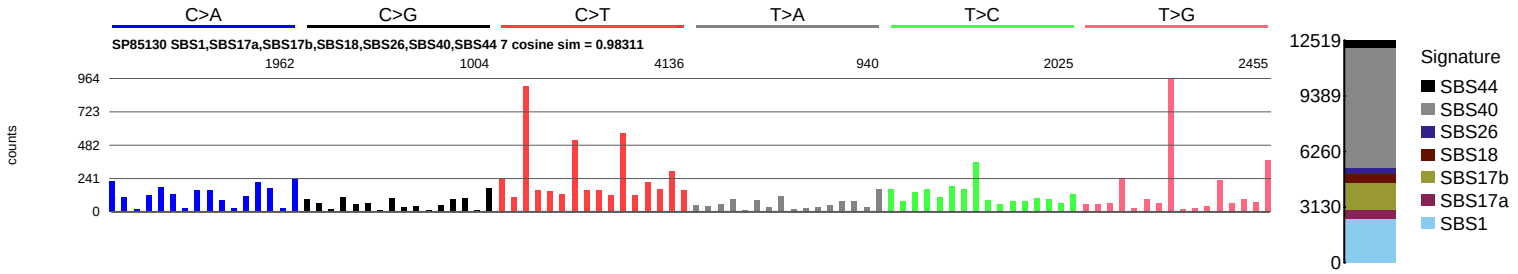

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS15,SBS17a,SBS17b,SBS21,SBS40,SBS44 7 cosine sim = 0.98321
1762
1061
4214
964
2027
2486
counts
960
720
480
240
0
12519
9389
6260
3130
0
Signature
SBS44
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS1

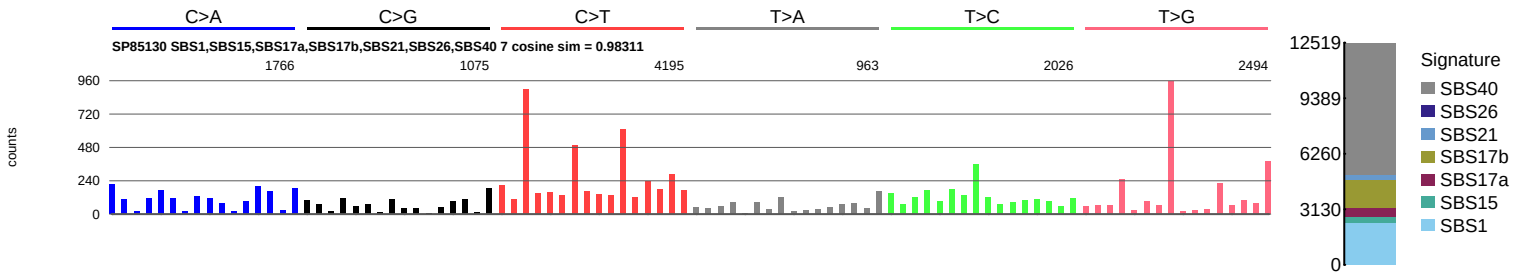

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 8 cosine sim = 0.9832
1947
999
4108
974
2031
2468
counts
964
723
482
241
0
12519
9389
6260
3130
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS1

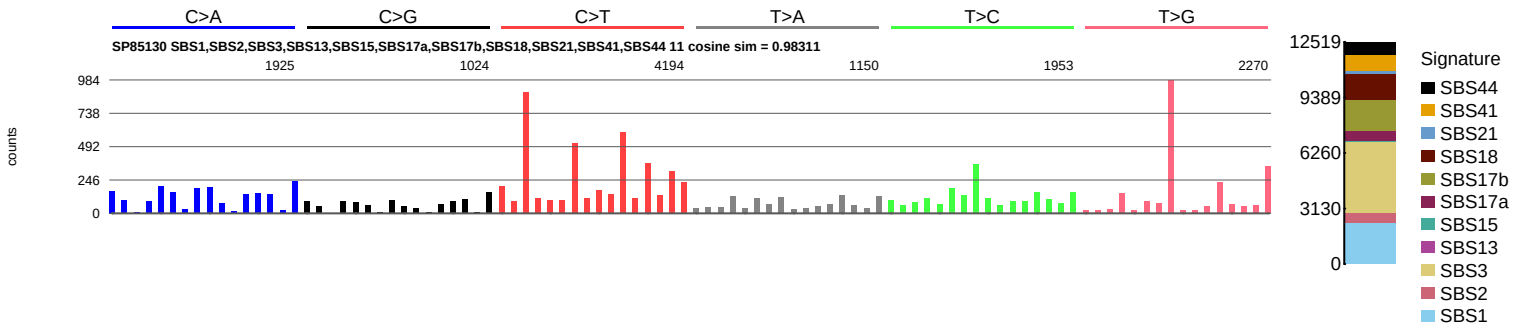

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS15,SBS17a,SBS17b,SBS21,SBS40,SBS41,SBS44 8 cosine sim = 0.9832
1752
1065
4208
975
2032
2488
counts
960
720
480
240
0
12519
9389
6260
3130
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS1

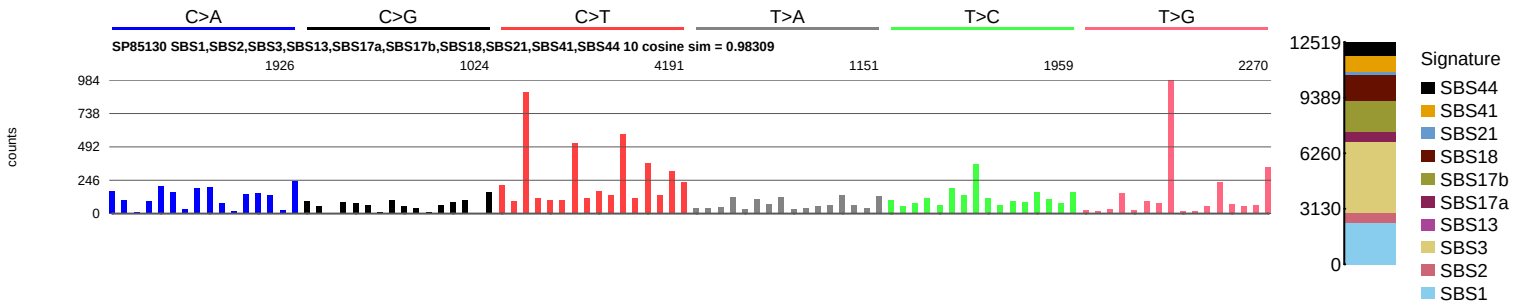

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS15,SBS17a,SBS17b,SBS21,SBS26,SBS40,SBS41 8 cosine sim = 0.98314
1754
1072
4187
977
2027
2500
960
720
480
240
0
counts
12519
9389
6260
3130
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS17b
SBS17a
SBS15
SBS1

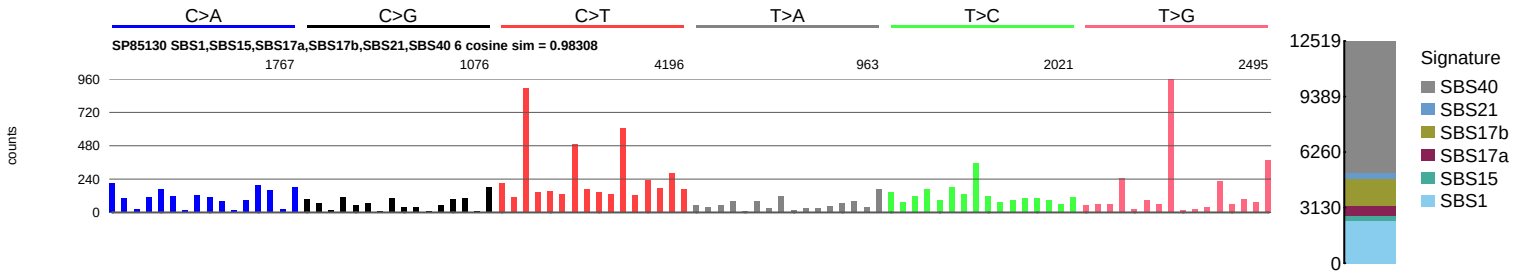

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS15,SBS17a,SBS17b,SBS21,SBS40,SBS41 7 cosine sim = 0.98312
1754
1074
4187
977
2026
2500
960
720
480
240
0
counts
12519
9389
6260
3130
0
Signature
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS1

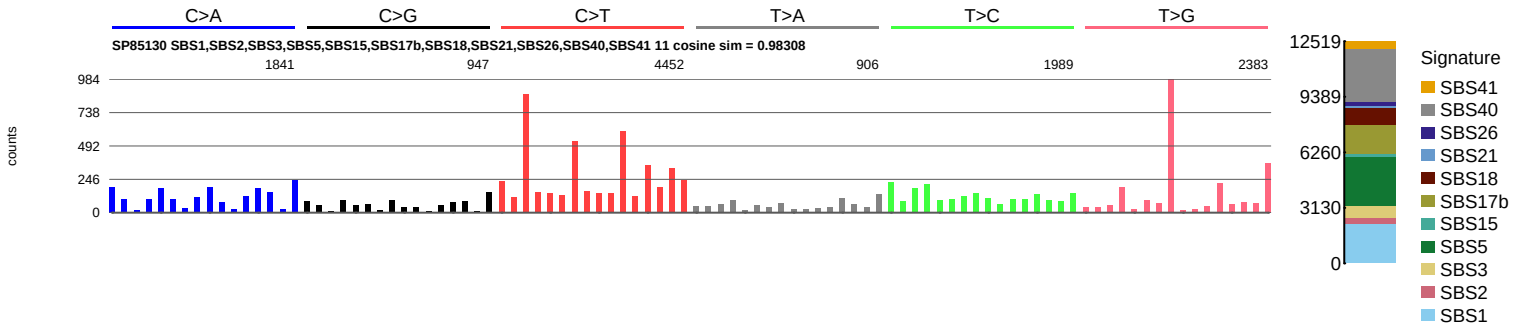

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 7 cosine sim = 0.98311
1962
1004
4136
940
2025
2455
964
723
482
241
0
counts
12519
9389
6260
3130
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS1

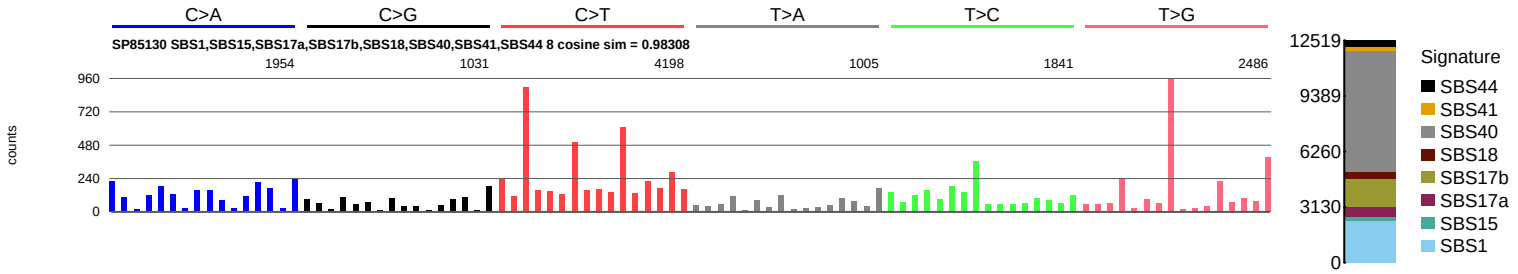

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS15,SBS17a,SBS17b,SBS21,SBS26,SBS40 7 cosine sim = 0.98311
1766
1075
4195
963
2026
2494
960
720
480
240
0
counts
12519
9389
6260
3130
0
Signature
SBS40
SBS26
SBS21
SBS17b
SBS17a
SBS15
SBS1

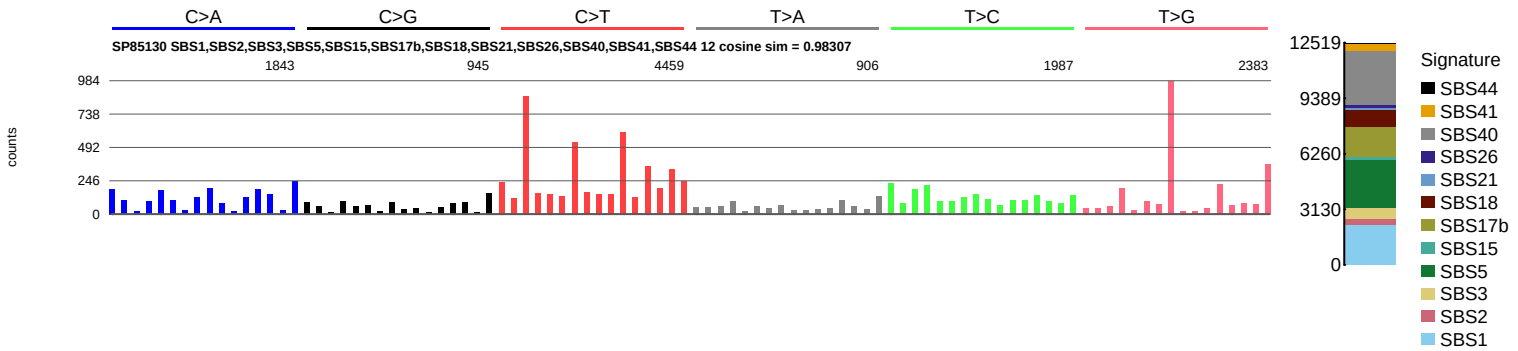

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 11 cosine sim = 0.98311
1925
1024
4194
1150
1953
2270
984
738
492
246
0
counts
12519
9389
6260
3130
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

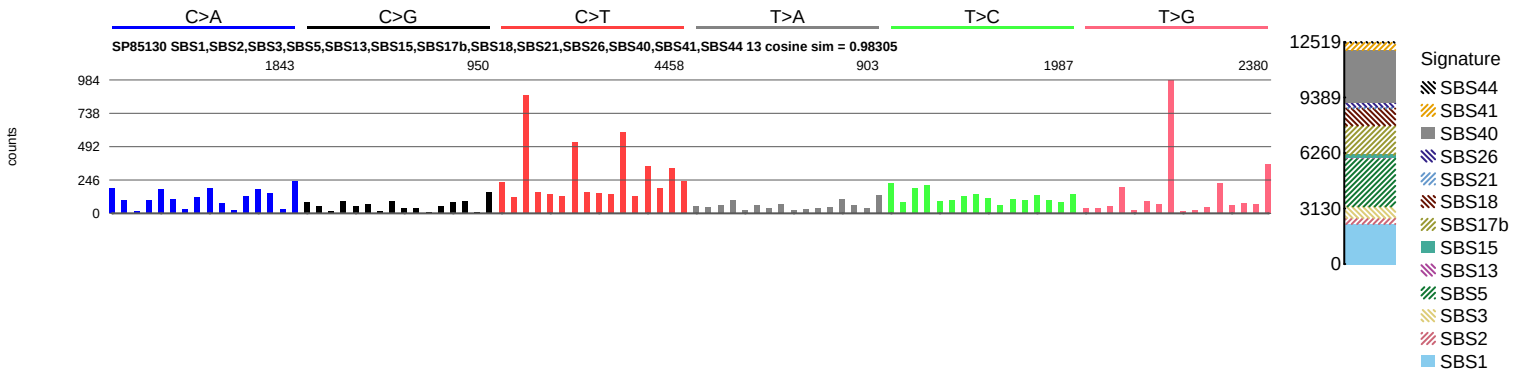

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 10 cosine sim = 0.98309
1926
1024
4191
1151
1959
2270
984
738
492
246
0
counts
12519
9389
6260
3130
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1

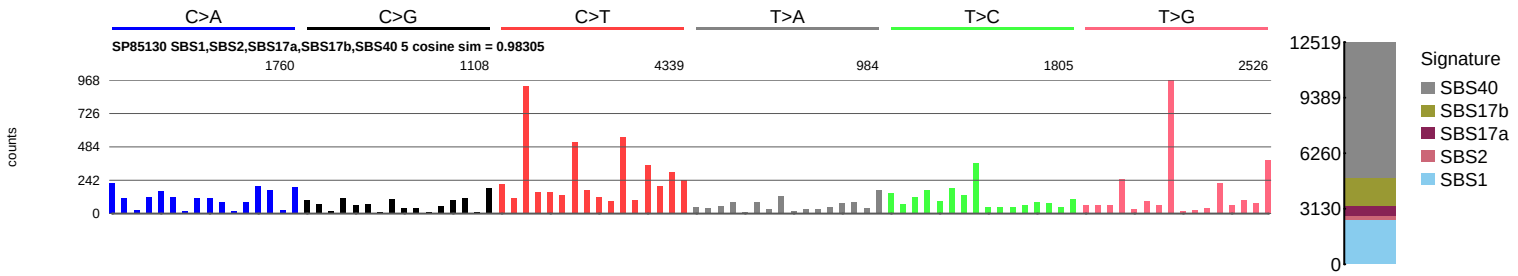

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS15,SBS17a,SBS17b,SBS21,SBS40 6 cosine sim = 0.98308
1767
1076
4196
963
2021
2495
counts
960
720
480
240
0
12519
9389
6260
3130
0
Signature
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS1

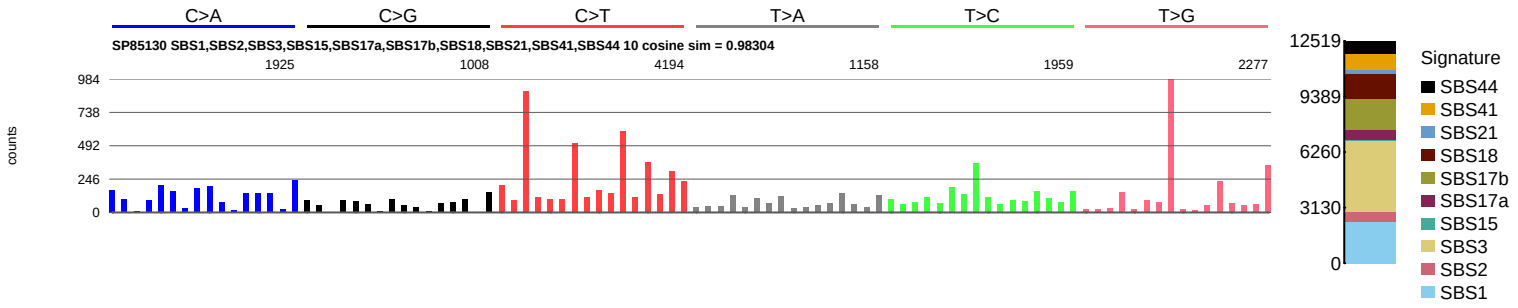

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41 11 cosine sim = 0.98308
1841
947
4452
906
1989
2383
counts
984
738
492
246
0
12519
9389
6260
3130
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS15
SBS5
SBS3
SBS2
SBS1

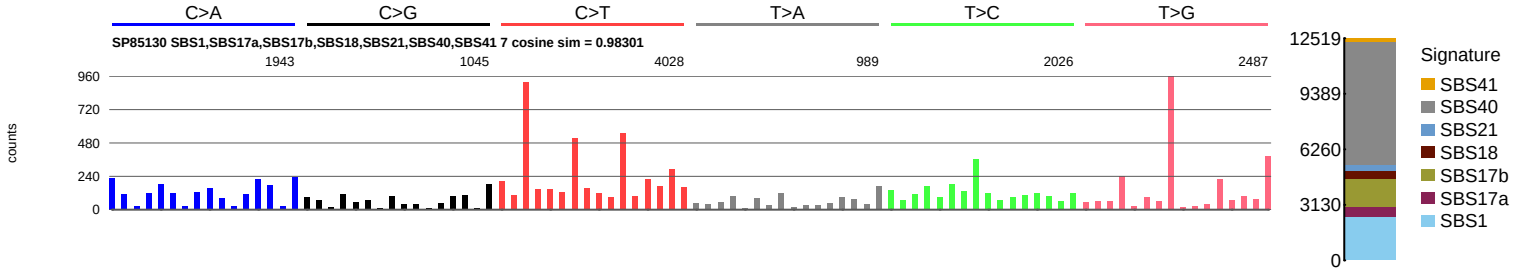

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 8 cosine sim = 0.98308
1954
1031
4198
1005
1841
2486
counts
960
720
480
240
0
12519
9389
6260
3130
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS1

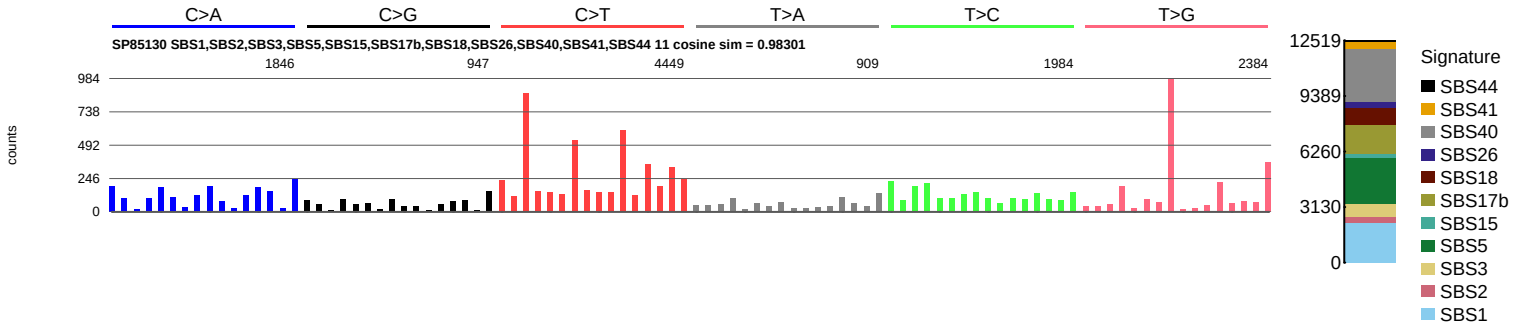

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41,SBS44 12 cosine sim = 0.98307
1843
945
4459
906
1987
2383
counts
984
738
492
246
0
12519
9389
6260
3130
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS15
SBS5
SBS3
SBS2
SBS1

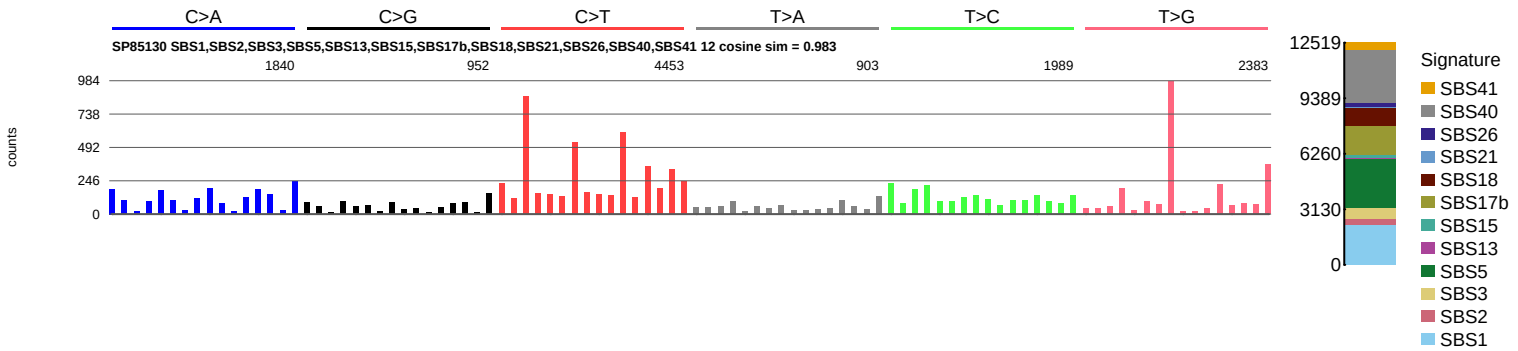

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41,SBS44 13 cosine sim = 0.98305
1843
950
4458
903
1987
2380
counts
984
738
492
246
0
12519
9389
6260
3130
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1

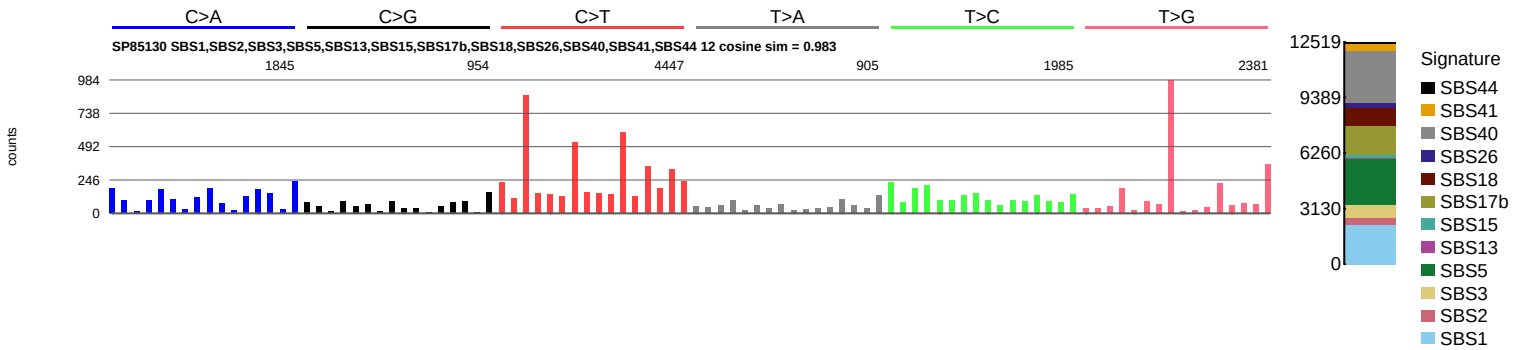

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS17a,SBS17b,SBS40 5 cosine sim = 0.98305
1760
1108
4339
984
1805
2526
counts
968
726
484
242
0
12519
9389
6260
3130
0
Signature
SBS40
SBS17b
SBS17a
SBS2
SBS1

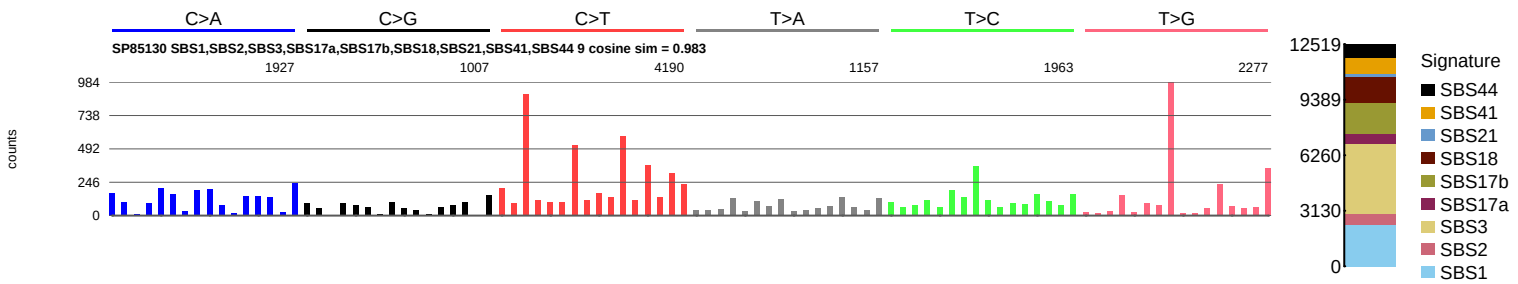

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 10 cosine sim = 0.98304
1925
1008
4194
1158
1959
2277
counts
984
738
492
246
0
12519
9389
6260
3130
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1

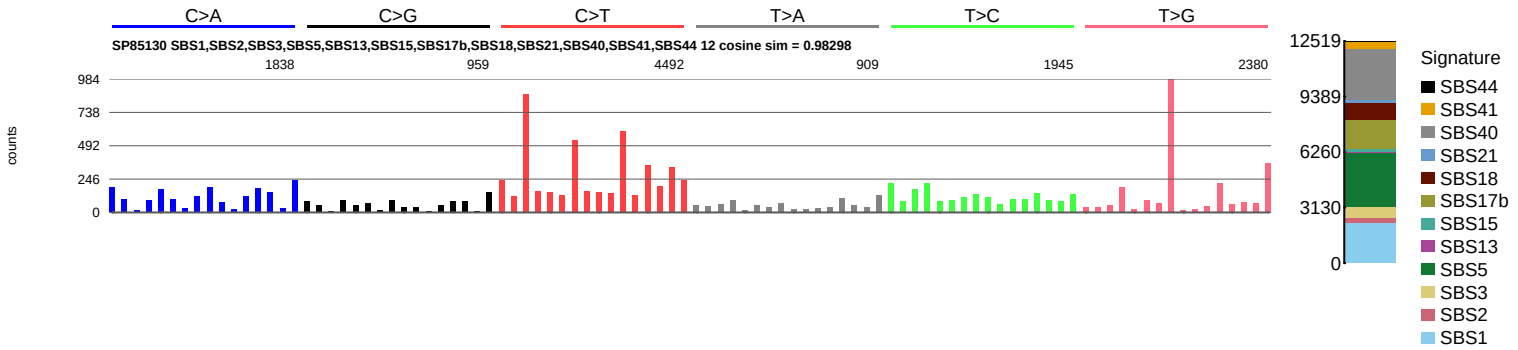

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 7 cosine sim = 0.98301
1943
1045
4028
989
2026
2487
counts
960
720
480
240
0
12519
9389
6260
3130
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS1

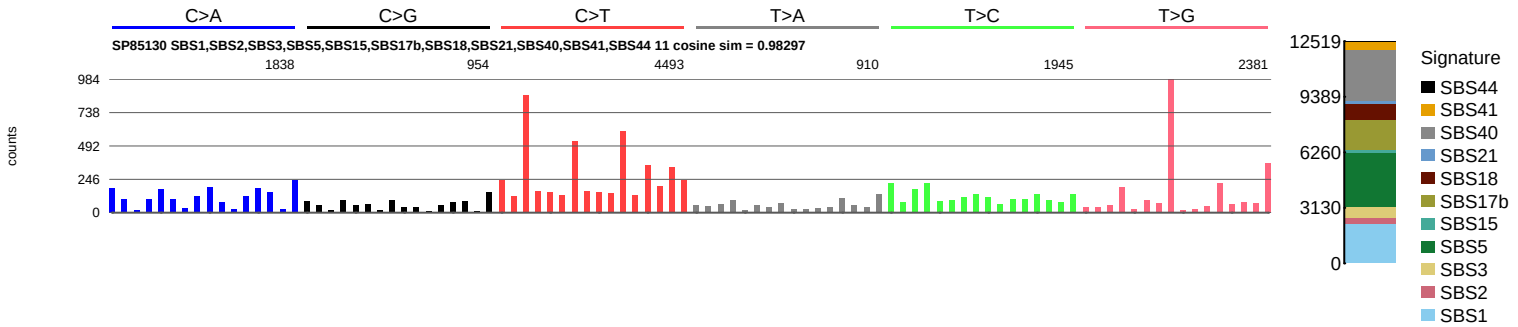

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 11 cosine sim = 0.98301
1846
947
4449
909
1984
2384
counts
984
738
492
246
0
12519
9389
6260
3130
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS15
SBS5
SBS3
SBS2
SBS1

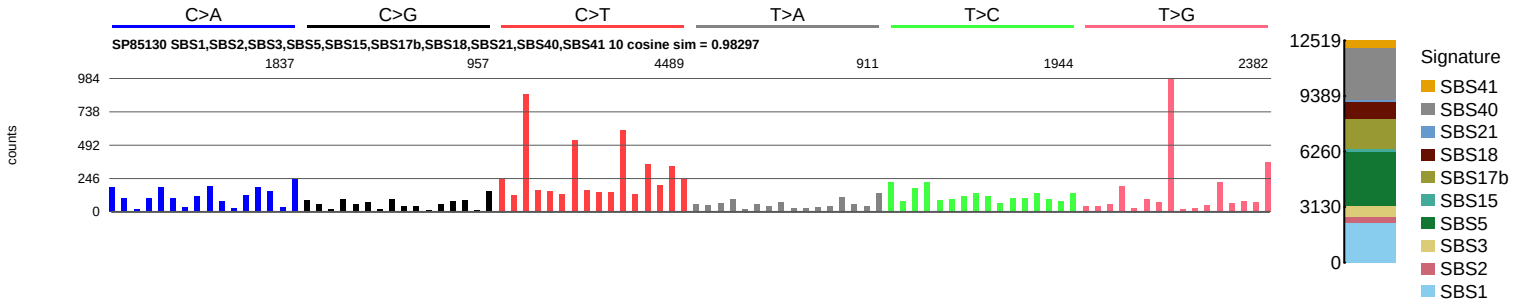

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41 12 cosine sim = 0.983
1840
952
4453
903
1989
2383
counts
984
738
492
246
0
12519
9389
6260
3130
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1

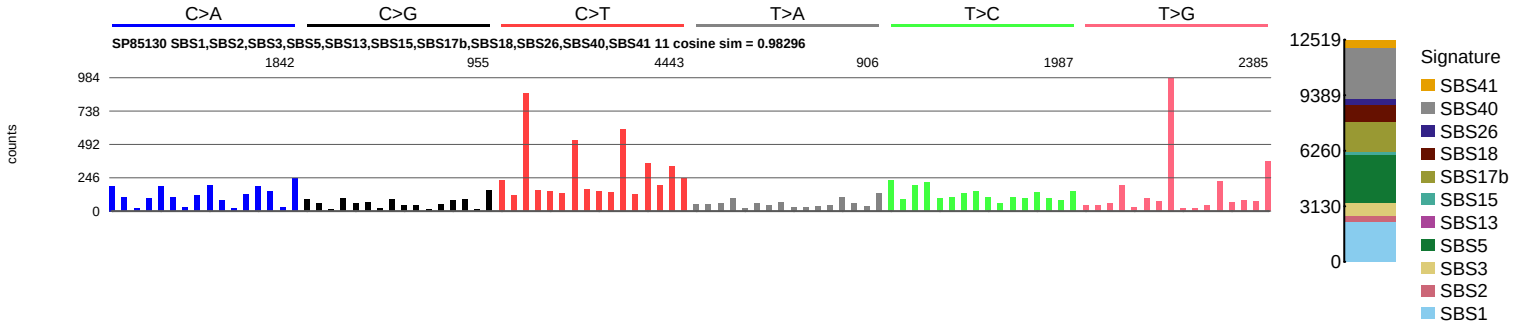

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 12 cosine sim = 0.983
1845
954
4447
905
1985
2381
counts
984
738
492
246
0
12519
9389
6260
3130
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1

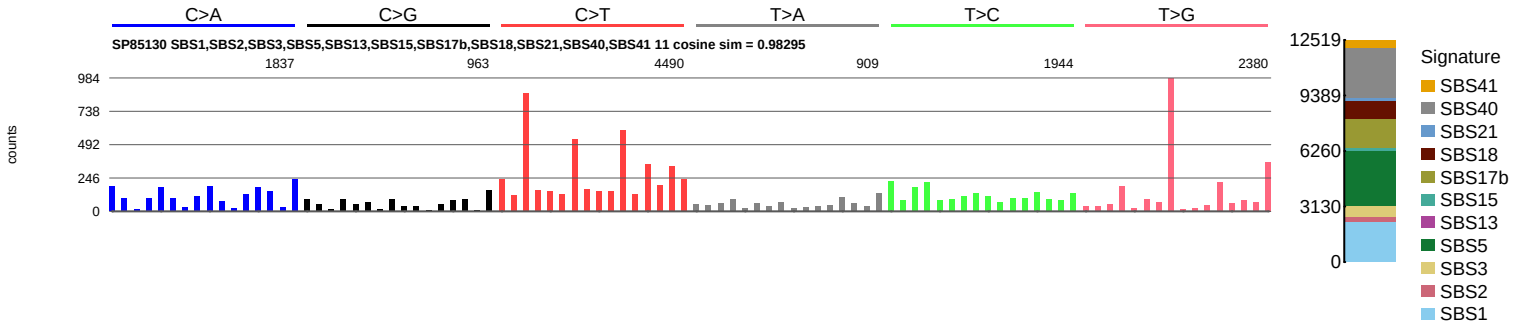

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 9 cosine sim = 0.983
1927
1007
4190
1157
1963
2277
counts
984
738
492
246
0
12519
9389
6260
3130
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1

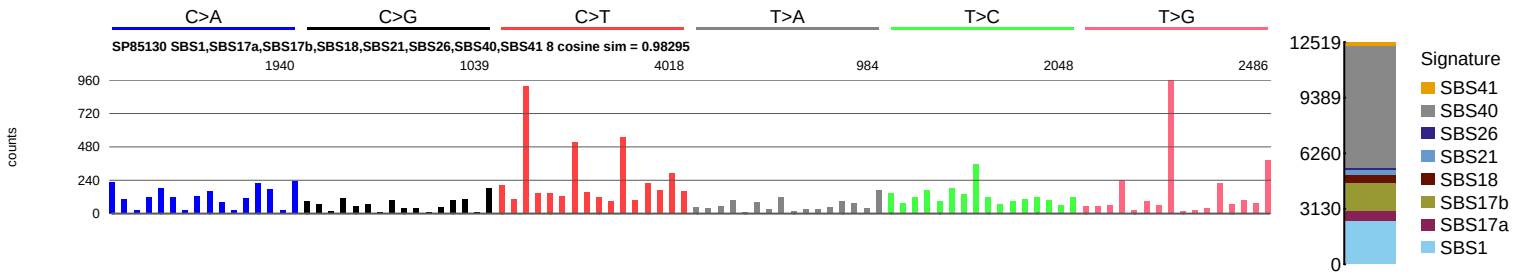

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 12 cosine sim = 0.98298
1838
959
4492
909
1945
2380
counts
984
738
492
246
0
12519
9389
6260
3130
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1

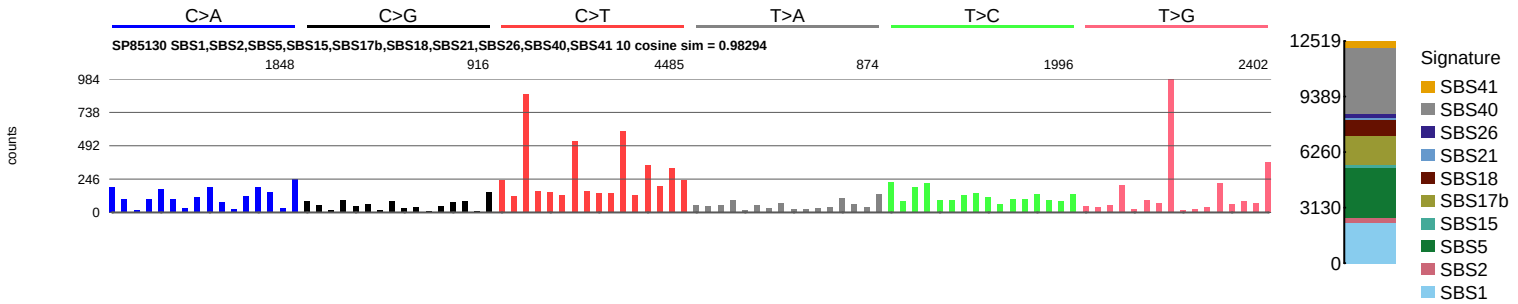

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 11 cosine sim = 0.98297
1838
954
4493
910
1945
2381
counts
984
738
492
246
0
12519
9389
6260
3130
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS15
SBS5
SBS3
SBS2
SBS1

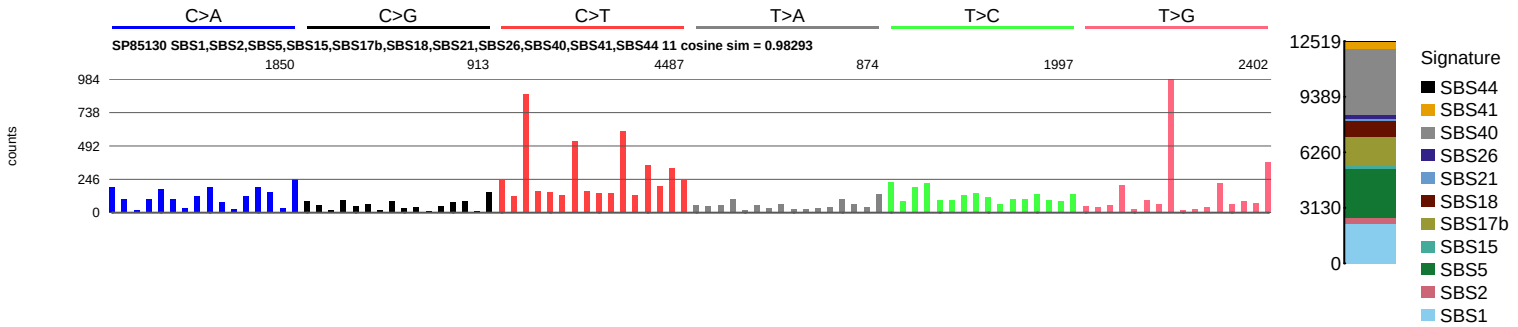

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17b,SBS18,SBS21,SBS40,SBS41 10 cosine sim = 0.98297
1837
957
4489
911
1944
2382
counts
984
738
492
246
0
12519
9389
6260
3130
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS15
SBS5
SBS3
SBS2
SBS1

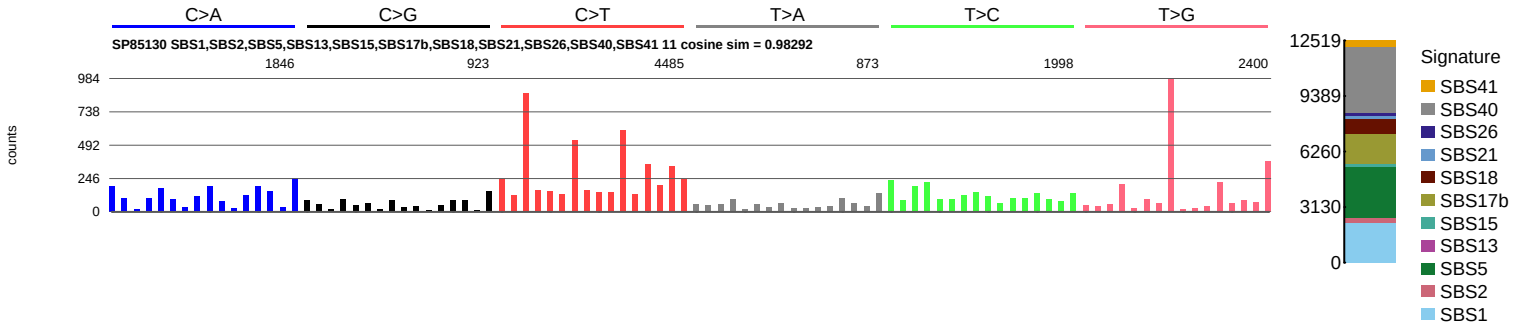

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS26,SBS40,SBS41 11 cosine sim = 0.98296
1842
955
4443
906
1987
2385
counts
984
738
492
246
0
12519
9389
6260
3130
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1

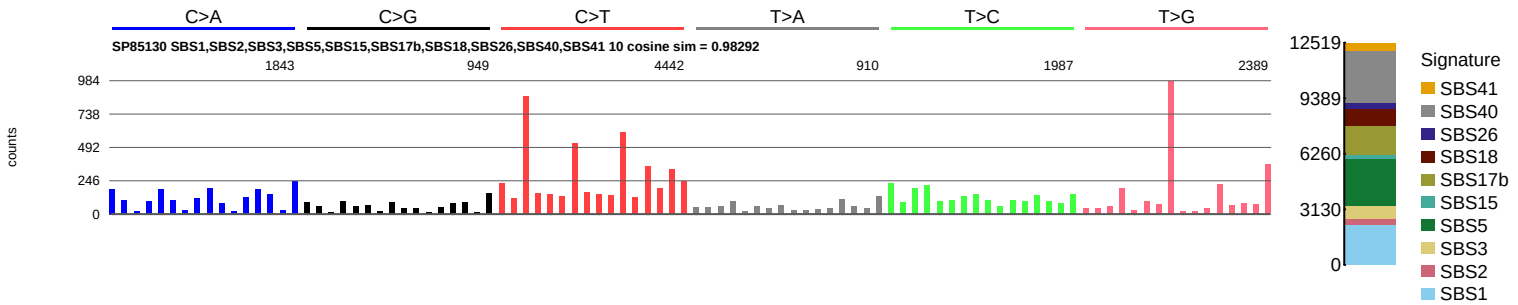

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS40,SBS41 11 cosine sim = 0.98295
1837
963
4490
909
1944
2380
counts
984
738
492
246
0
12519
9389
6260
3130
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1

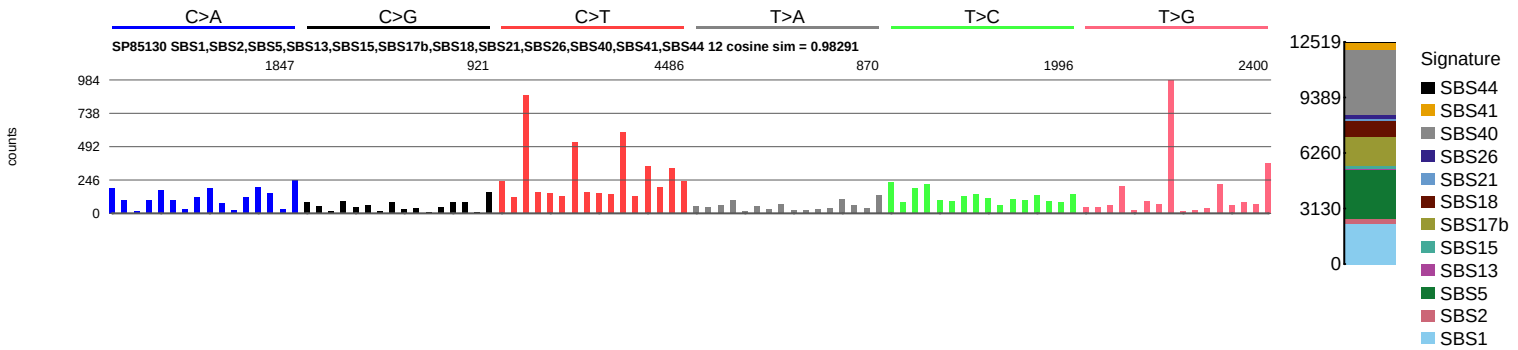

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41 8 cosine sim = 0.98295
1940
1039
4018
984
2048
2486
counts
960
720
480
240
0
12519
9389
6260
3130
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS1

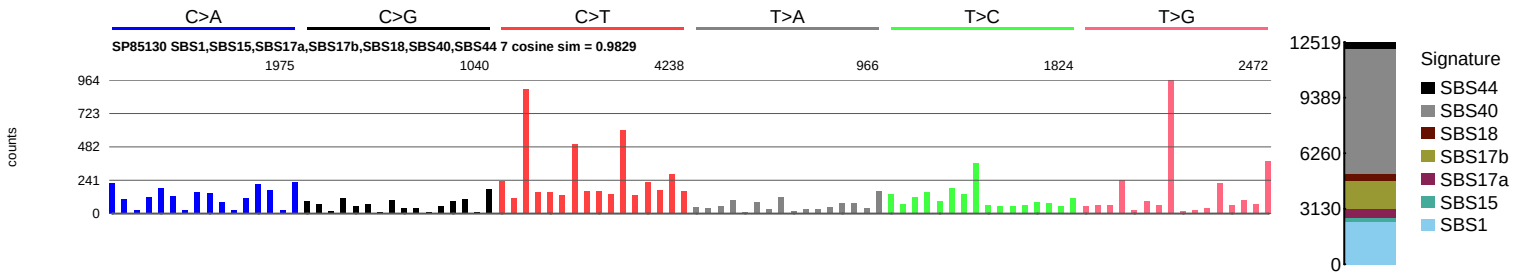

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS5,SBS15,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41 10 cosine sim = 0.98294
1848
916
4485
874
1996
2402
counts
984
738
492
246
0
12519
9389
6260
3130
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS15
SBS5
SBS2
SBS1

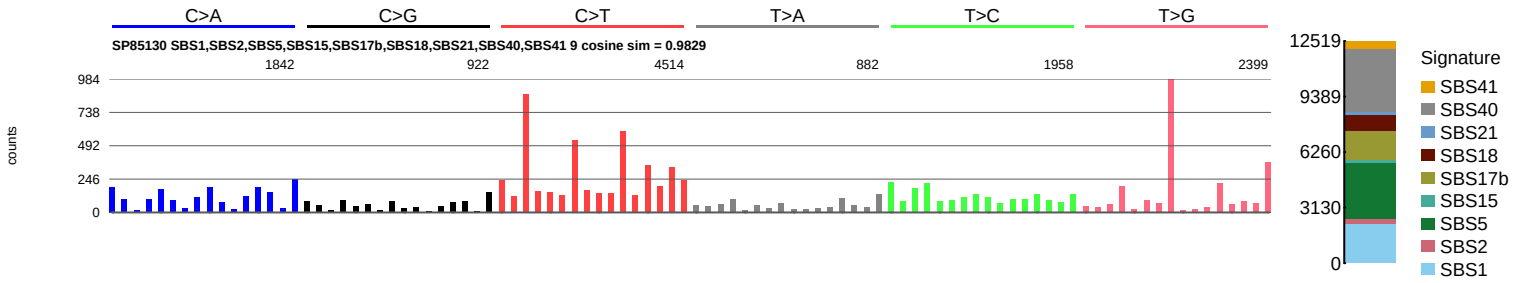

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS5,SBS15,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41,SBS44 11 cosine sim = 0.98293
1850
913
4487
874
1997
2402
counts
984
738
492
246
0
12519
9389
6260
3130
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS15
SBS5
SBS2
SBS1

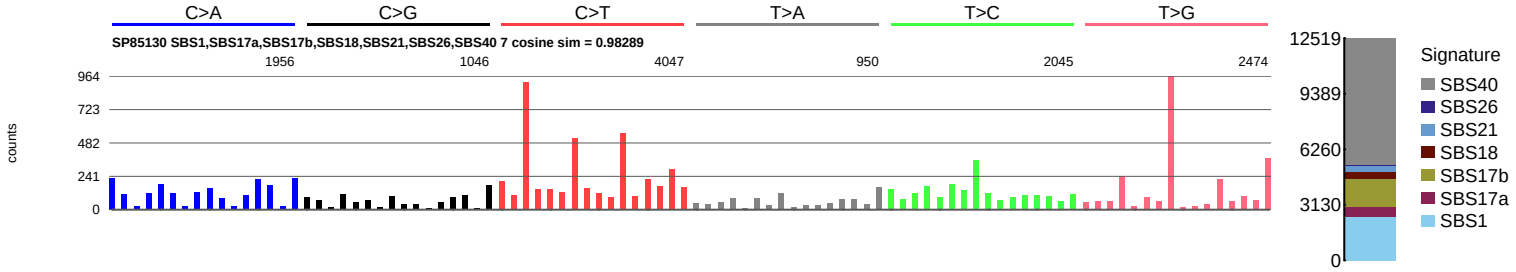

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41 11 cosine sim = 0.98292
1846
923
4485
873
1998
2400
counts
984
738
492
246
0
12519
9389
6260
3130
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS2
SBS1

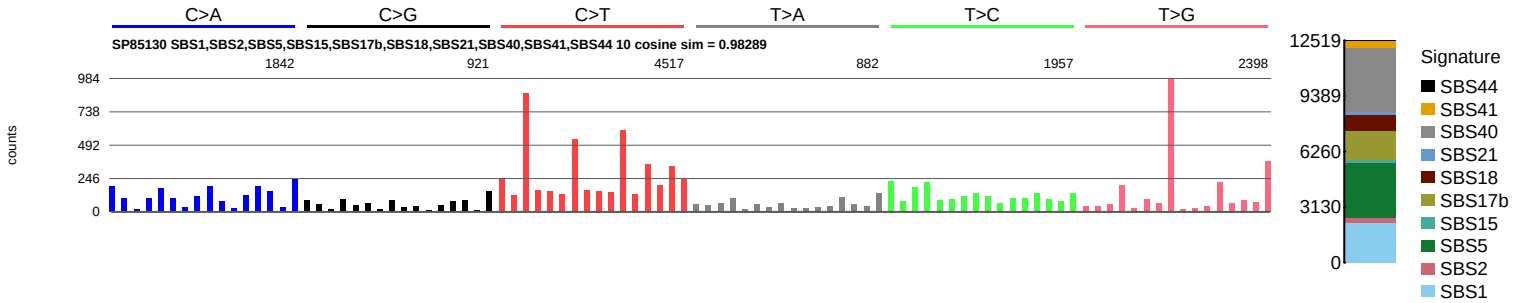

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17b,SBS18,SBS26,SBS40,SBS41 10 cosine sim = 0.98292
1843
949
4442
910
1987
2389
counts
984
738
492
246
0
12519
9389
6260
3130
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS15
SBS5
SBS3
SBS2
SBS1

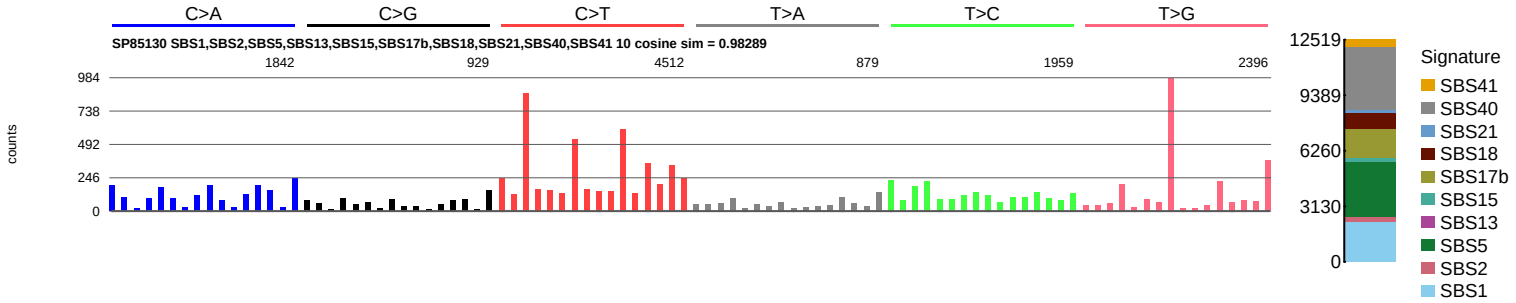

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41,SBS44 12 cosine sim = 0.98291
1847
921
4486
870
1996
2400
counts
984
738
492
246
0
12519
9389
6260
3130
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS2
SBS1

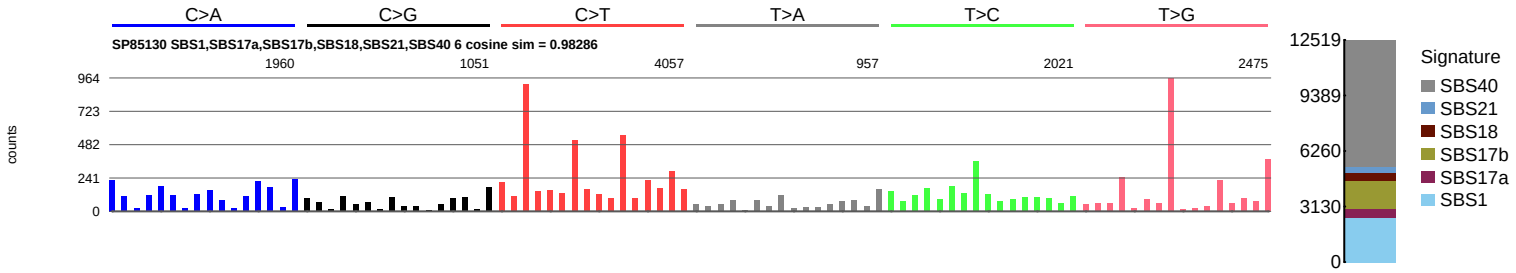

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS44 7 cosine sim = 0.9829
1975
1040
4238
966
1824
2472
counts
964
723
482
241
0
12519
9389
6260
3130
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS1

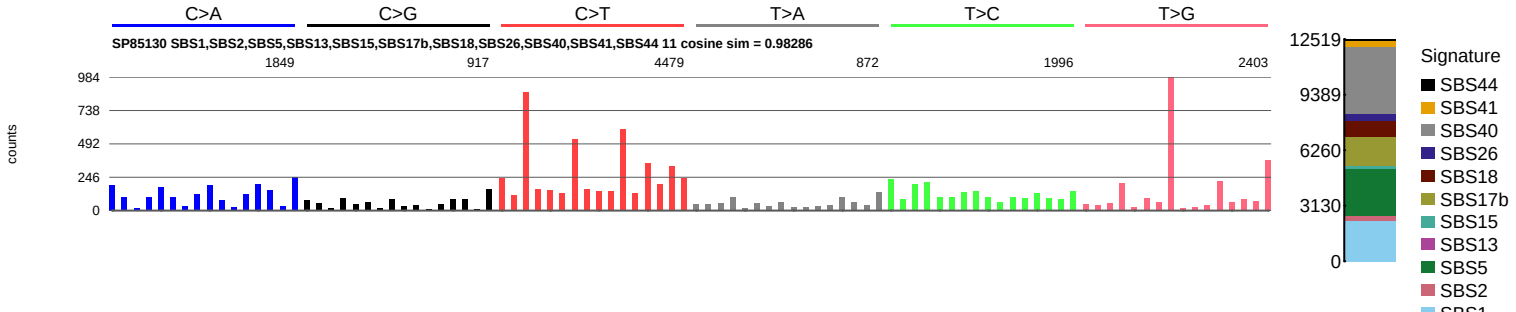

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS5,SBS15,SBS17b,SBS18,SBS21,SBS40,SBS41 9 cosine sim = 0.9829
1842
922
4514
882
1958
2399
counts
984
738
492
246
0
12519
9389
6260
3130
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS15
SBS5
SBS2
SBS1

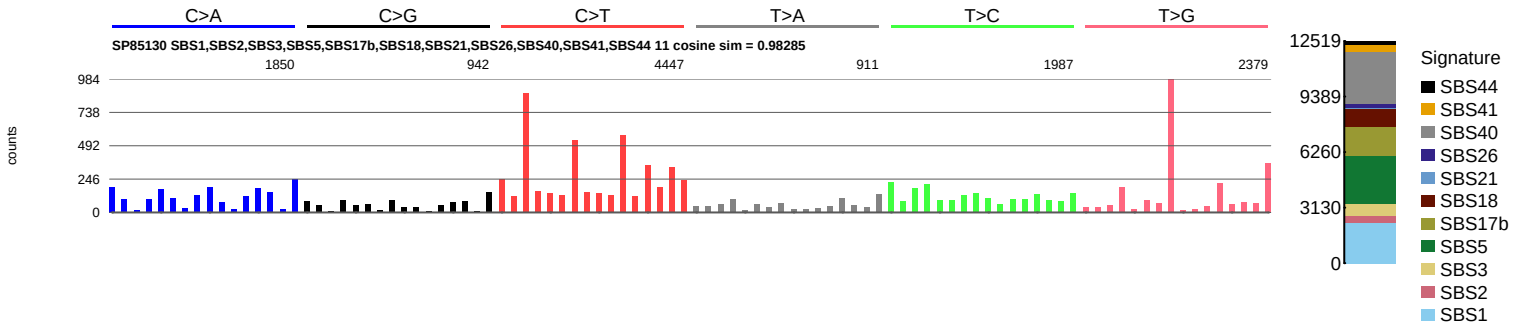

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40 7 cosine sim = 0.98289
1956
1046
4047
950
2045
2474
counts
964
723
482
241
0
12519
9389
6260
3130
0
Signature
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS5,SBS15,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 10 cosine sim = 0.98289
1842
921
4517
882
1957
2398
counts
984
738
492
246
0
12519
9389
6260
3130
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS15
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS40,SBS41 10 cosine sim = 0.98289
1842
929
4512
879
1959
2396
counts
984
738
492
246
0
12519
9389
6260
3130
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS17a,SBS17b,SBS18,SBS21,SBS40 6 cosine sim = 0.98286
1960
1051
4057
957
2021
2475
counts
964
723
482
241
0
12519
9389
6260
3130
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 11 cosine sim = 0.98286
1849
917
4479
872
1996
2403
counts
984
738
492
246
0
12519
9389
6260
3130
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS2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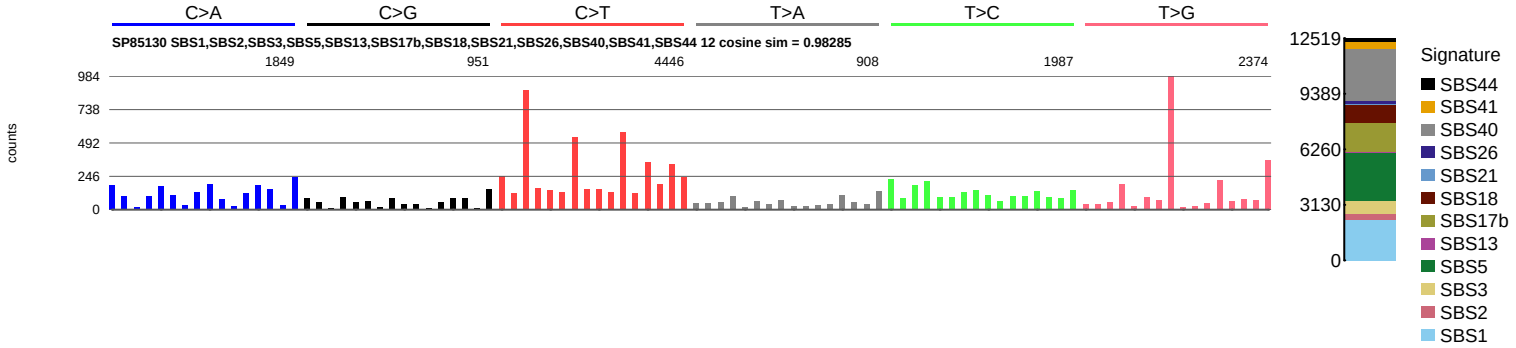
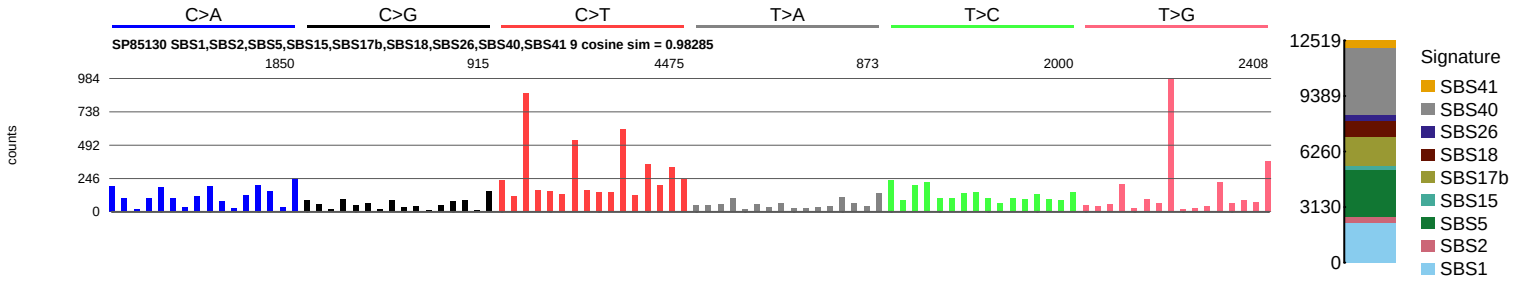
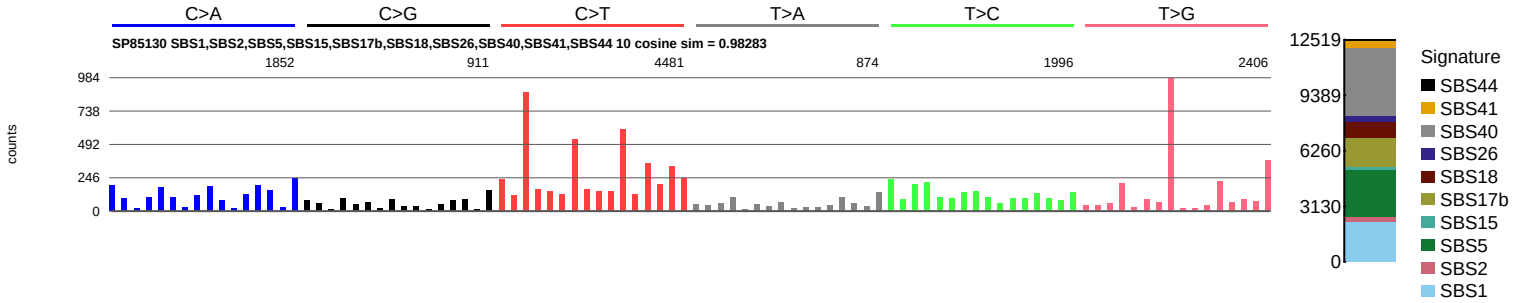
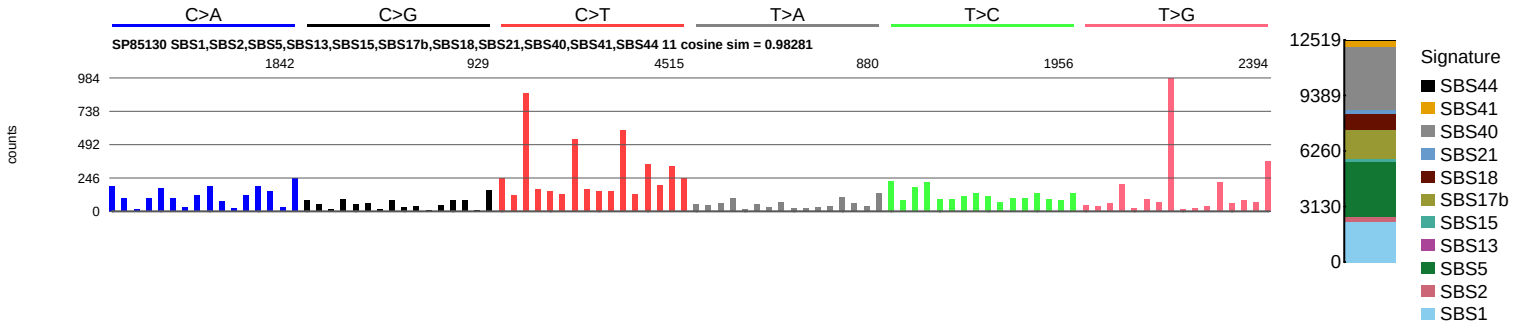
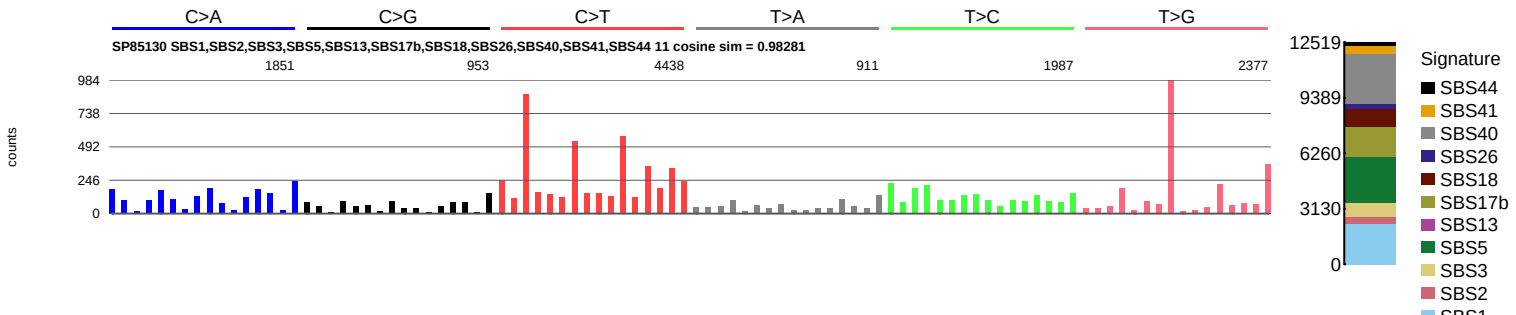
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS5,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41,SBS44 11 cosine sim = 0.98285
1850
942
4447
911
1987
2379
counts
984
738
492
246
0
12519
9389
6260
3130
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS5
SBS3
SBS2
SBS1

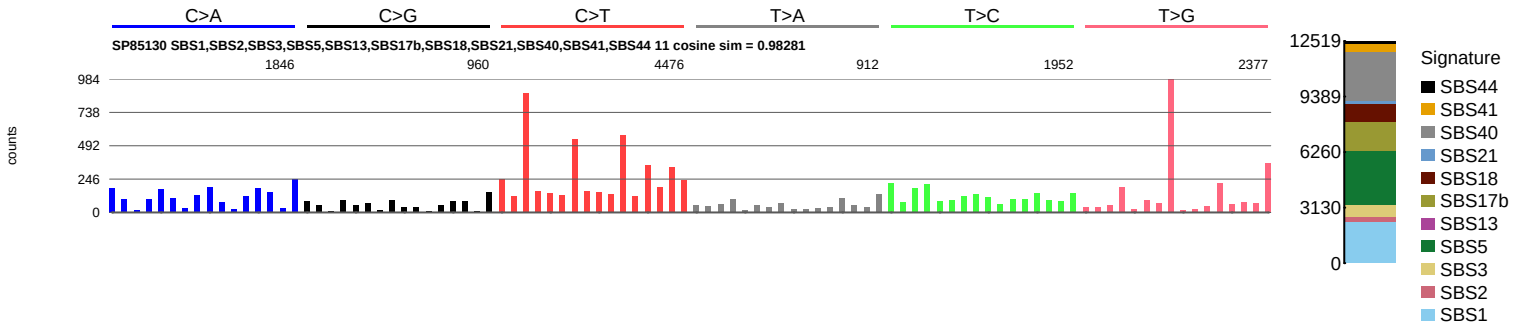

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41,SBS44 12 cosine sim = 0.98285
1849
951
4446
908
1987
2374
counts
984
738
492
246
0
12519
9389
6260
3130
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS13
SBS5
SBS3
SBS2
SBS1

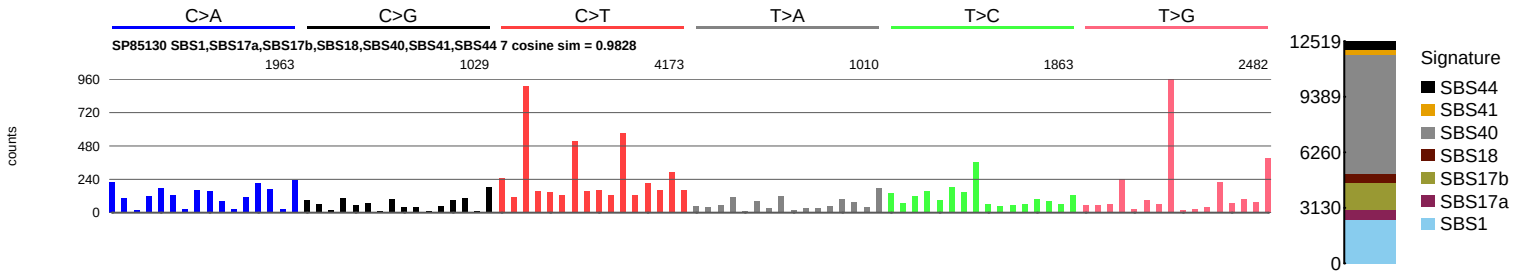

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS5,SBS15,SBS17b,SBS18,SBS26,SBS40,SBS41 9 cosine sim = 0.98285
1850
915
4475
873
2000
2408
counts
984
738
492
246
0
12519
9389
6260
3130
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS15
SBS5
SBS2
SBS1

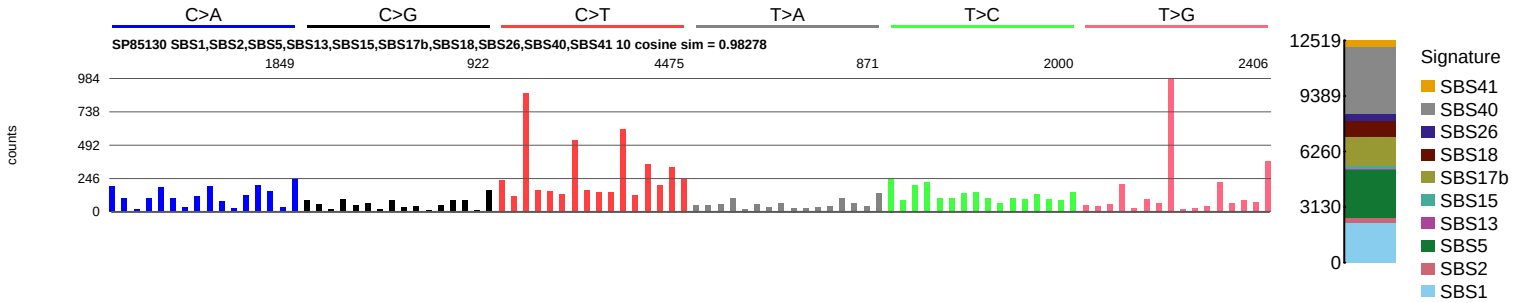

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS5,SBS15,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.98283
1852
911
4481
874
1996
2406
counts
984
738
492
246
0
12519
9389
6260
3130
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS15
SBS5
SBS2
SBS1

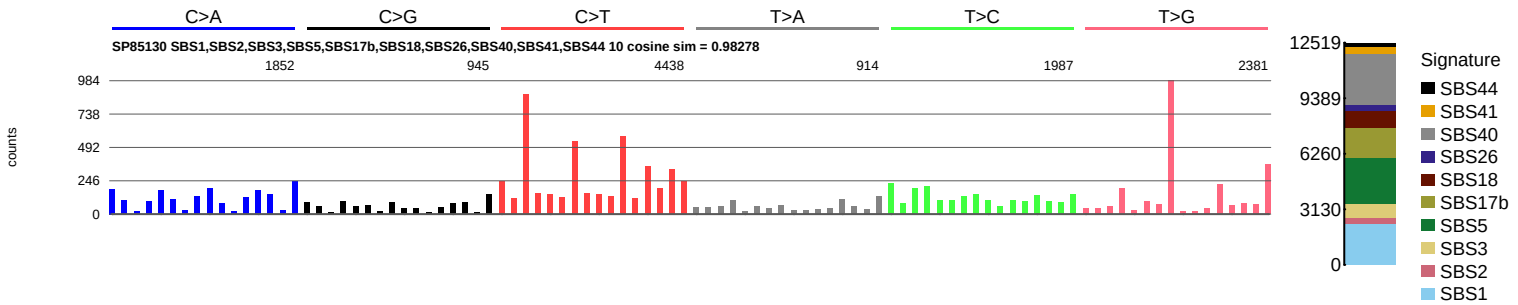

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 11 cosine sim = 0.98281
1842
929
4515
880
1956
2394
counts
984
738
492
246
0
12519
9389
6260
3130
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS2
SBS1

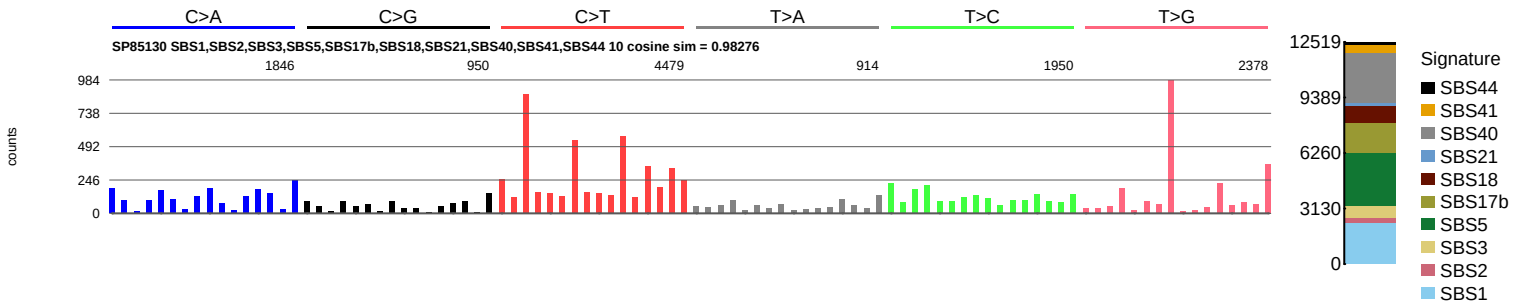

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 11 cosine sim = 0.98281
1851
953
4438
911
1987
2377
counts
984
738
492
246
0
12519
9389
6260
3130
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS13
SBS5
SBS3
SBS2

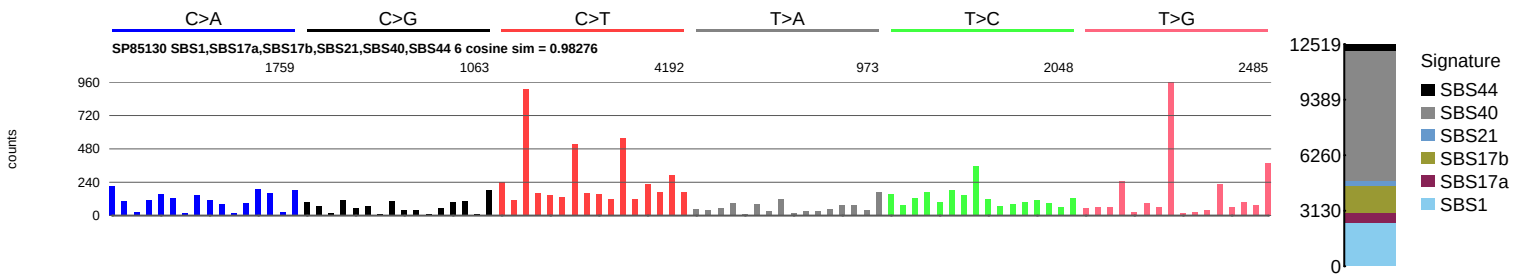

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 11 cosine sim = 0.98281
1846
960
4476
912
1952
2377
counts
984
738
492
246
0
12519
9389
6260
3130
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS13
SBS5
SBS3
SBS2
SBS1

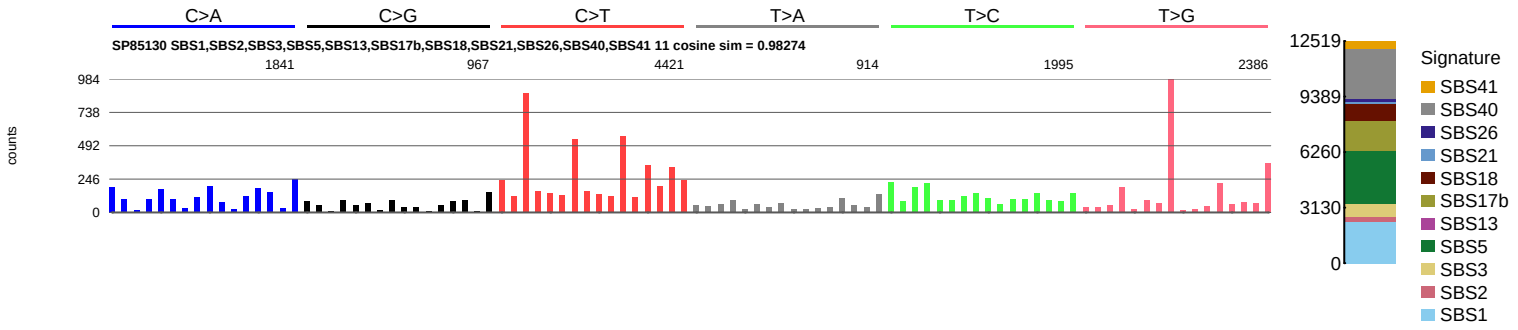

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 7 cosine sim = 0.9828
1963
1029
4173
1010
1863
2482
counts
960
720
480
240
0
12519
9389
6260
3130
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS1

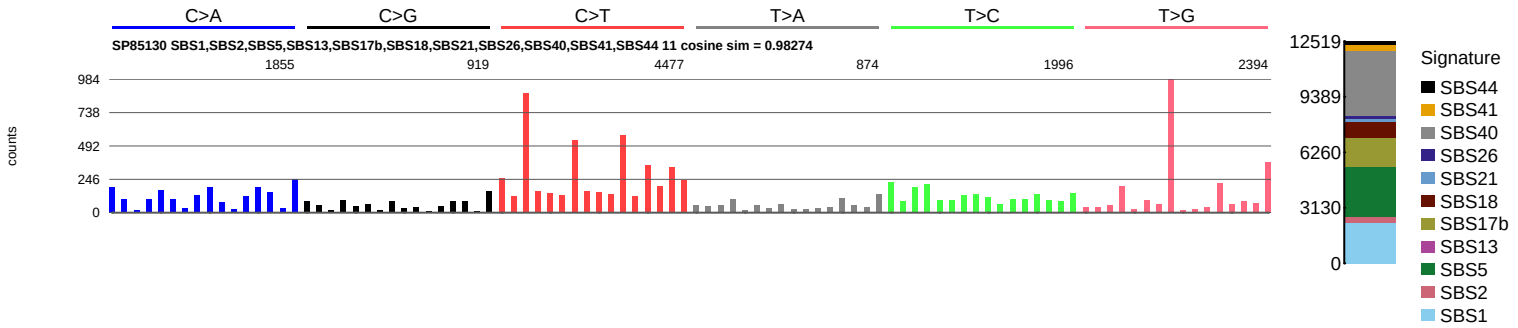

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS26,SBS40,SBS41 10 cosine sim = 0.98278
1849
922
4475
871
2000
2406
counts
984
738
492
246
0
12519
9389
6260
3130
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS2
SBS1

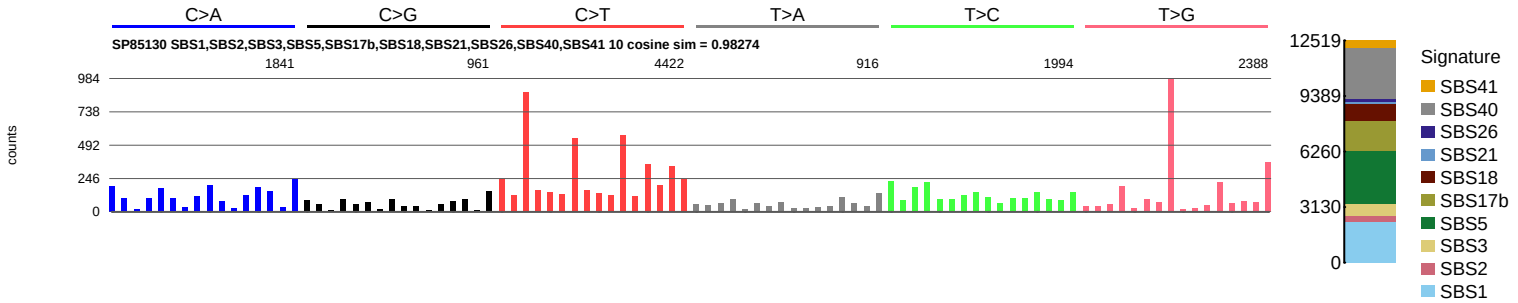

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS5,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.98278
1852
945
4438
914
1987
2381
counts
984
738
492
246
0
12519
9389
6260
3130
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS5
SBS3
SBS2
SBS1

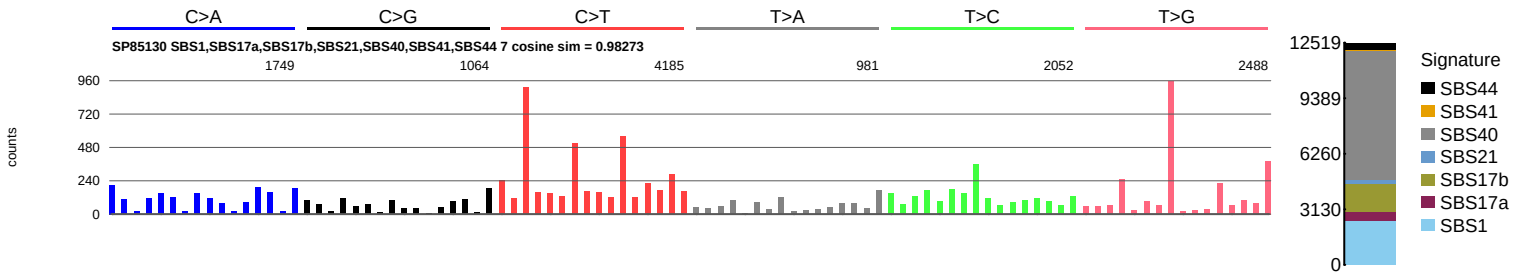

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS5,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 10 cosine sim = 0.98276
1846
950
4479
914
1950
2378
counts
984
738
492
246
0
12519
9389
6260
3130
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS5
SBS3
SBS2
SBS1

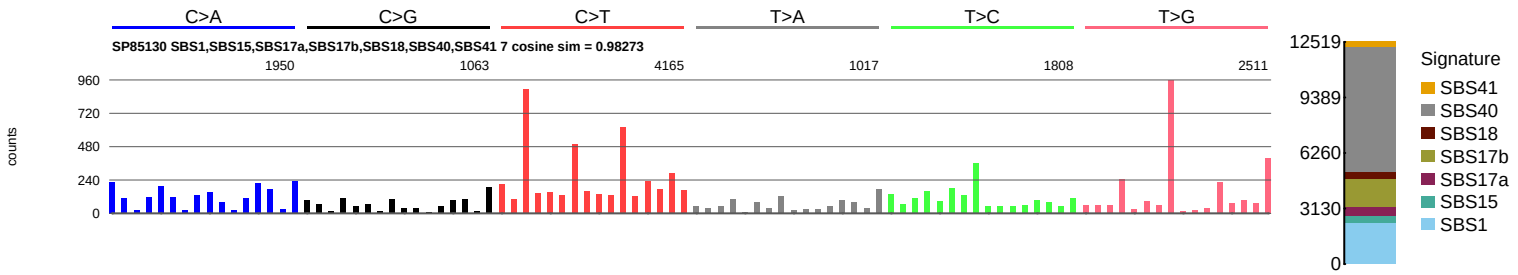

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS17a,SBS17b,SBS21,SBS40,SBS44 6 cosine sim = 0.98276
1759
1063
4192
973
2048
2485
counts
960
720
480
240
0
12519
9389
6260
3130
0
Signature
SBS44
SBS40
SBS21
SBS17b
SBS17a
SBS1

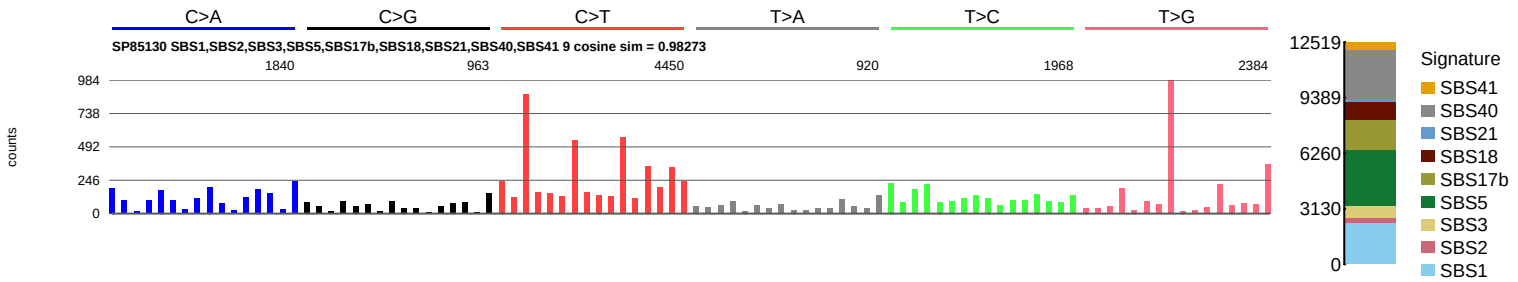

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41 11 cosine sim = 0.98274
1841
967
4421
914
1995
2386
counts
984
738
492
246
0
12519
9389
6260
3130
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS13
SBS5
SBS3
SBS2
SBS1

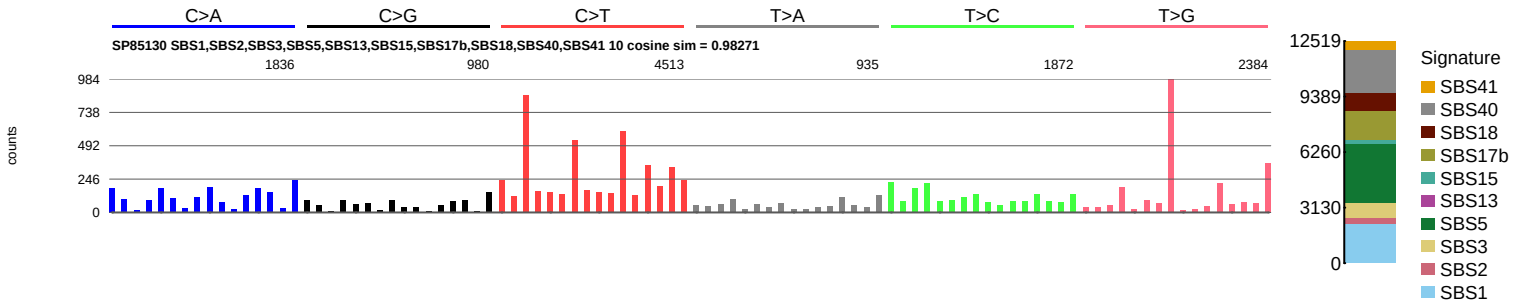

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS5,SBS13,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41,SBS44 11 cosine sim = 0.98274
1855
919
4477
874
1996
2394
counts
984
738
492
246
0
12519
9389
6260
3130
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS13
SBS5
SBS2
SBS1

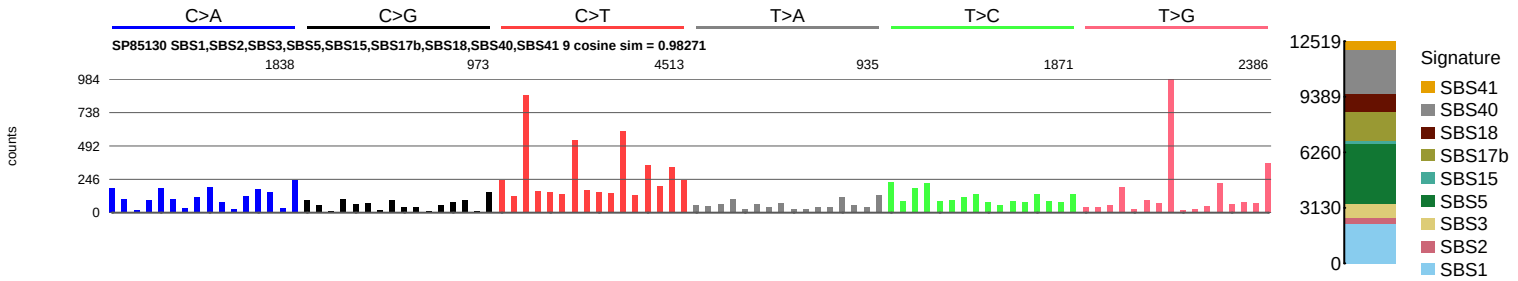

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS5,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41 10 cosine sim = 0.98274
1841
961
4422
916
1994
2388
counts
984
738
492
246
0
12519
9389
6260
3130
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS5
SBS3
SBS2
SBS1

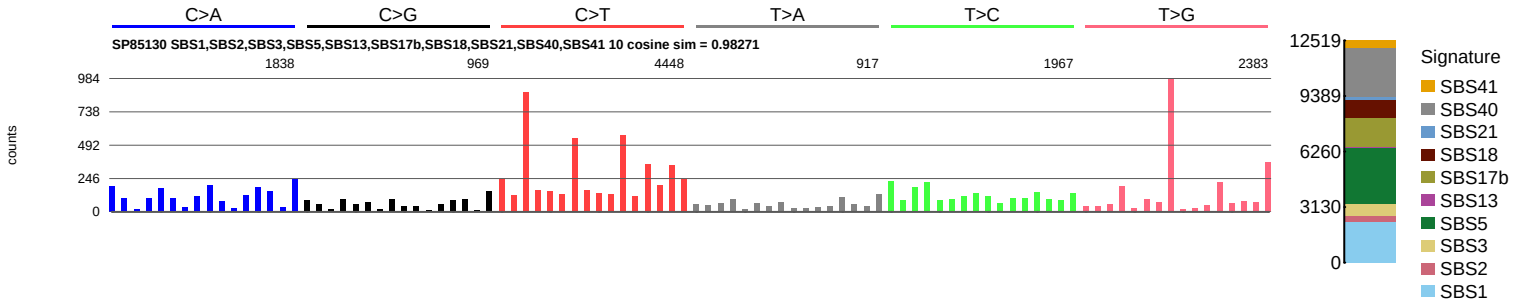

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS17a,SBS17b,SBS21,SBS40,SBS41,SBS44 7 cosine sim = 0.98273
1749
1064
4185
981
2052
2488
counts
960
720
480
240
0
12519
9389
6260
3130
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS1

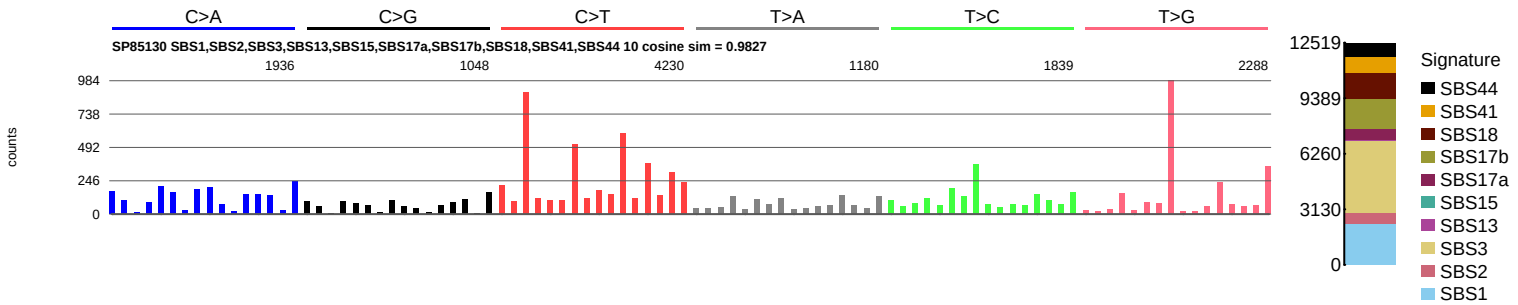

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41 7 cosine sim = 0.98273
1950
1063
4165
1017
1808
2511
counts
960
720
480
240
0
12519
9389
6260
3130
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS1

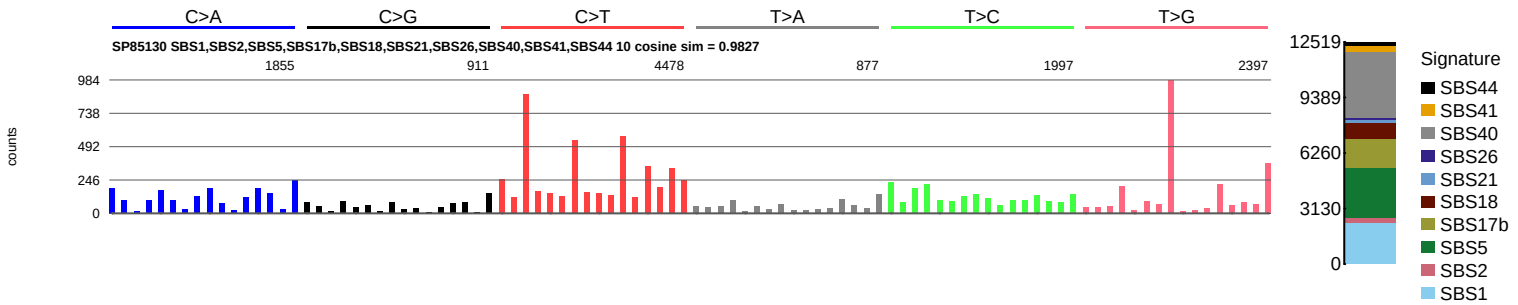

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS5,SBS17b,SBS18,SBS21,SBS40,SBS41 9 cosine sim = 0.98273
1840
963
4450
920
1968
2384
counts
984
738
492
246
0
12519
9389
6260
3130
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS5
SBS3
SBS2
SBS1

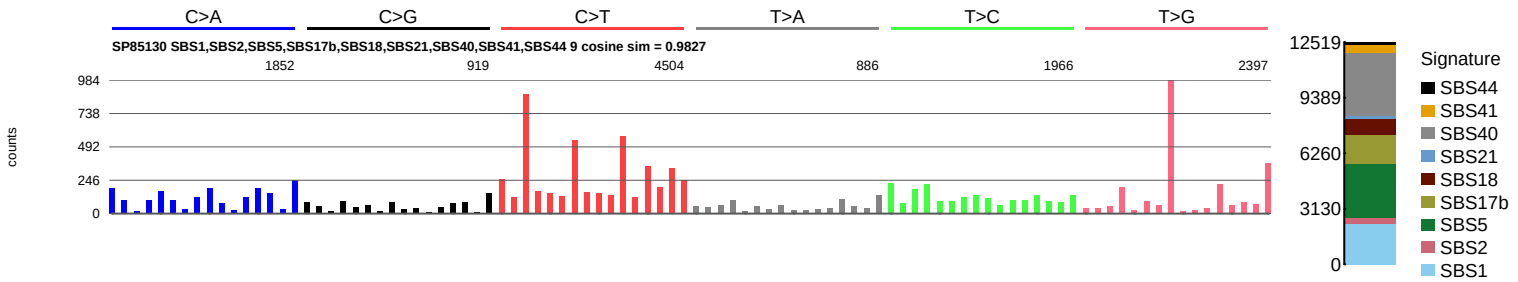

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS40,SBS41 10 cosine sim = 0.98271
1836
980
4513
935
1872
2384
counts
984
738
492
246
0
12519
9389
6260
3130
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17b,SBS18,SBS40,SBS41 9 cosine sim = 0.98271
1838
973
4513
935
1871
2386
counts
984
738
492
246
0
12519
9389
6260
3130
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS15
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS18,SBS21,SBS40,SBS41 10 cosine sim = 0.98271
1838
969
4448
917
1967
2383
counts
984
738
492
246
0
12519
9389
6260
3130
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS41,SBS44 10 cosine sim = 0.9827
1936
1048
4230
1180
1839
2288
counts
984
738
492
246
0
12519
9389
6260
3130
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS5,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.9827
1855
911
4478
877
1997
2397
counts
984
738
492
246
0
12519
9389
6260
3130
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS5
SBS2
SBS1


C>A
C>G
C>T
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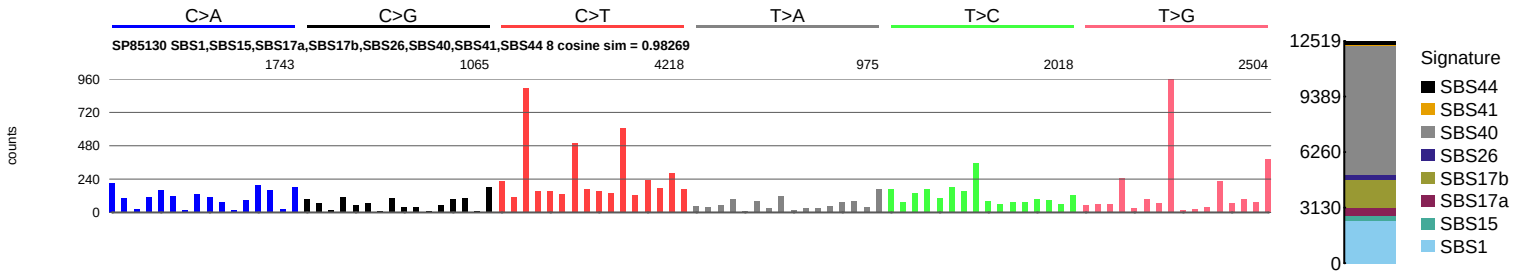
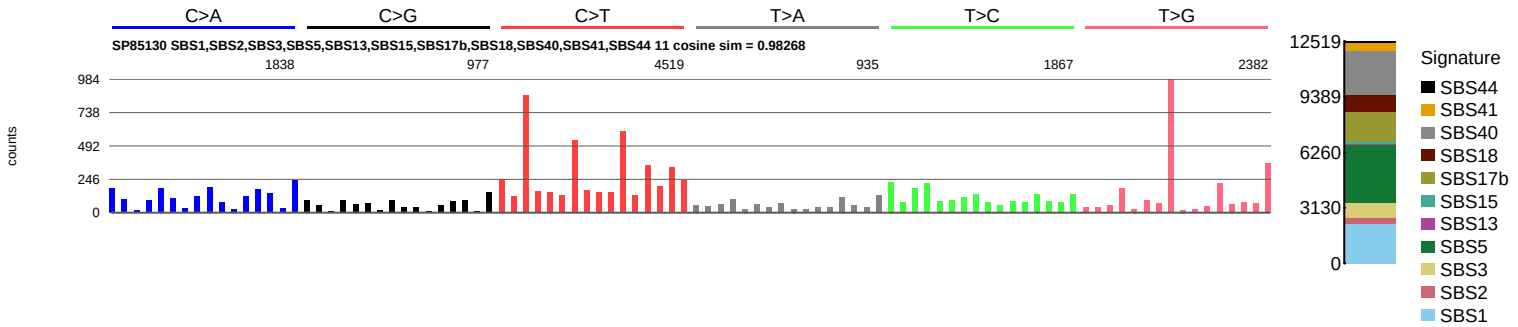
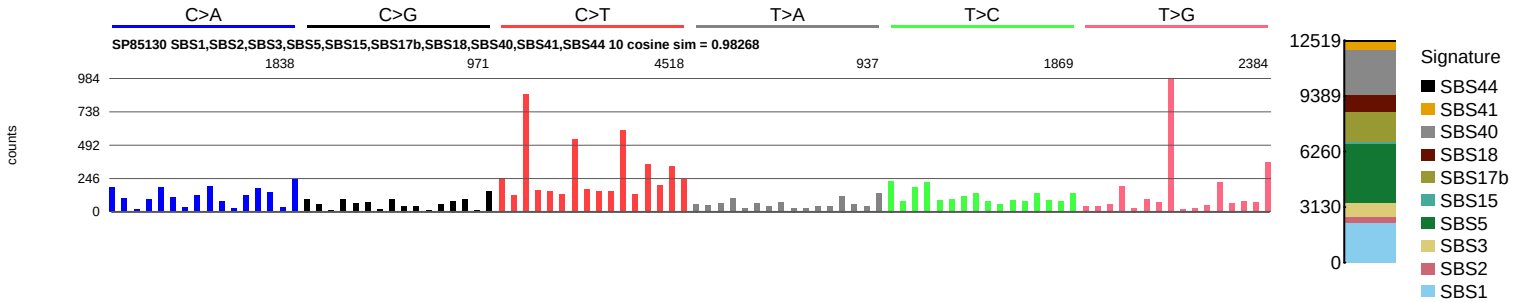
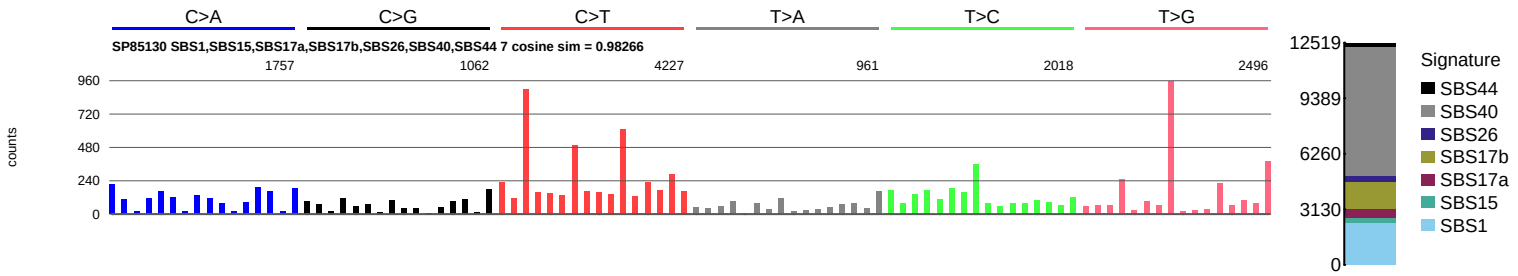
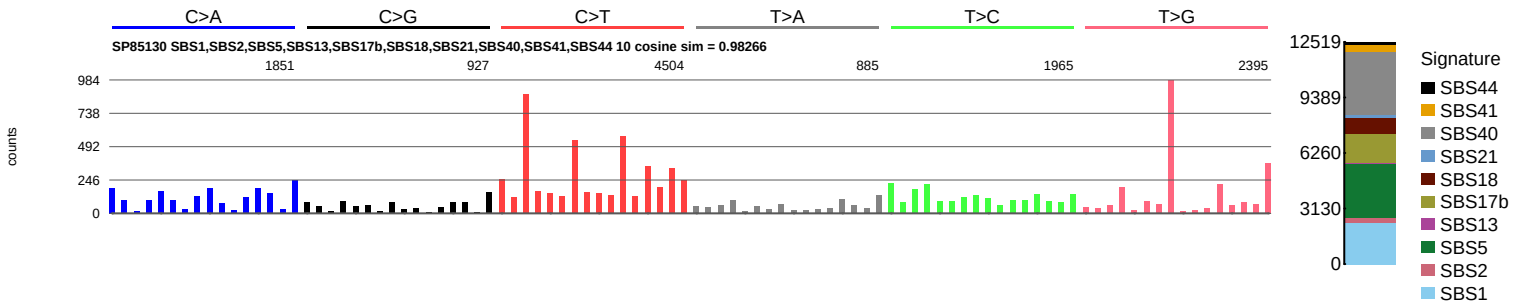
T>C
T>G
SP85130 SBS1,SBS2,SBS5,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 9 cosine sim = 0.9827
1852
919
4504
886
1966
2397
counts
984
738
492
246
0
12519
9389
6260
3130
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS5
SBS2
SBS1

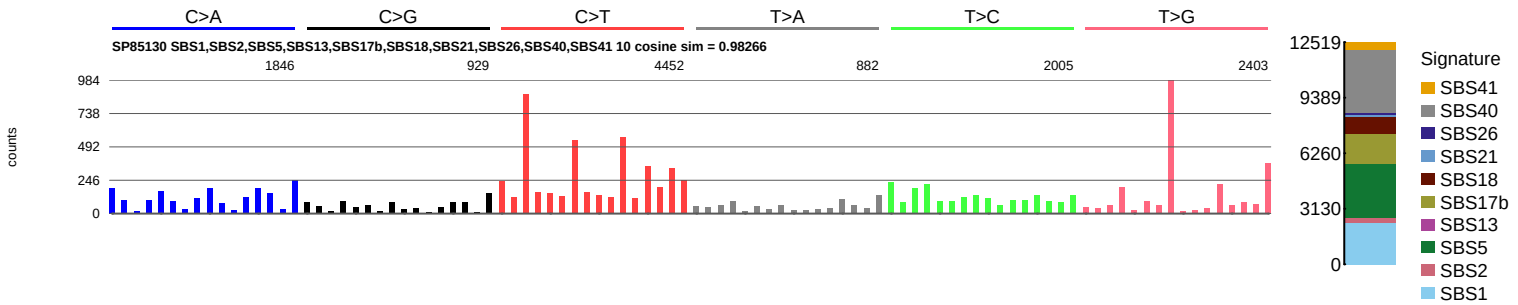

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS15,SBS17a,SBS17b,SBS26,SBS40,SBS41,SBS44 8 cosine sim = 0.98269
1743
1065
4218
975
2018
2504
counts
960
720
480
240
0
12519
9389
6260
3130
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS1

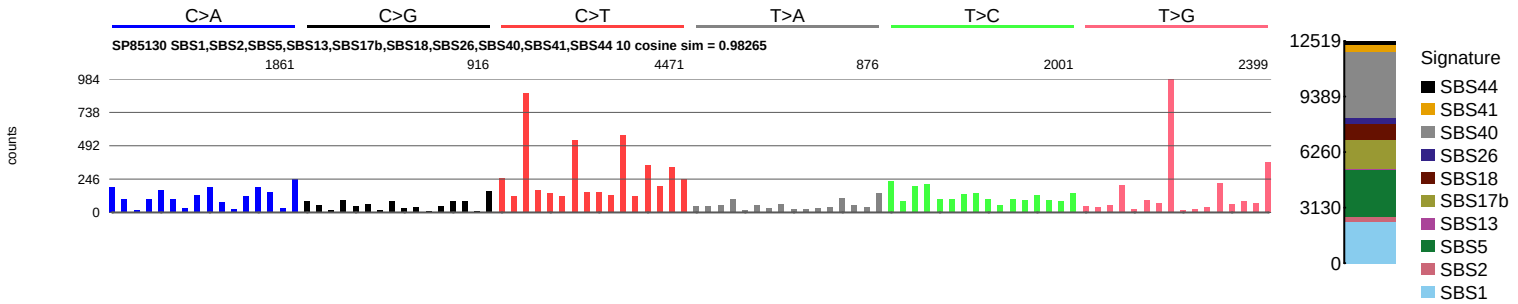

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS40,SBS41,SBS44 11 cosine sim = 0.98268
1838
977
4519
935
1867
2382
counts
984
738
492
246
0
12519
9389
6260
3130
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1

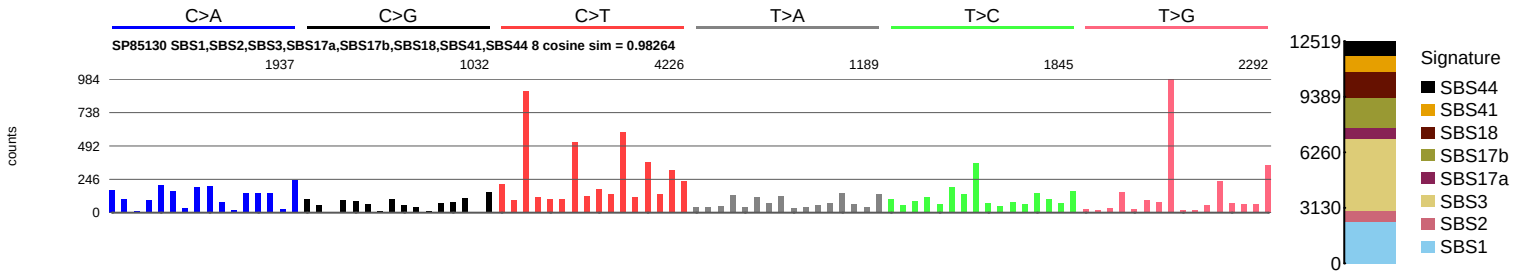

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17b,SBS18,SBS40,SBS41,SBS44 10 cosine sim = 0.98268
1838
971
4518
937
1869
2384
counts
984
738
492
246
0
12519
9389
6260
3130
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS15
SBS5
SBS3
SBS2
SBS1

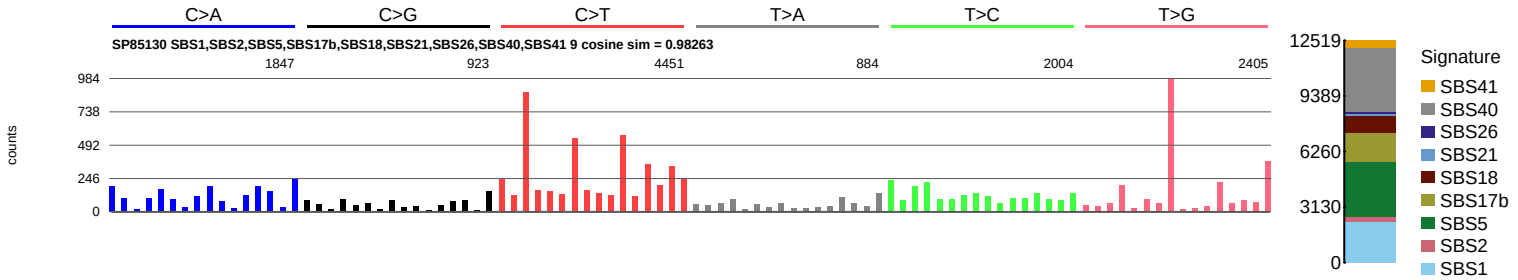

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS15,SBS17a,SBS17b,SBS26,SBS40,SBS44 7 cosine sim = 0.98266
1757
1062
4227
961
2018
2496
counts
960
720
480
240
0
12519
9389
6260
3130
0
Signature
SBS44
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS1

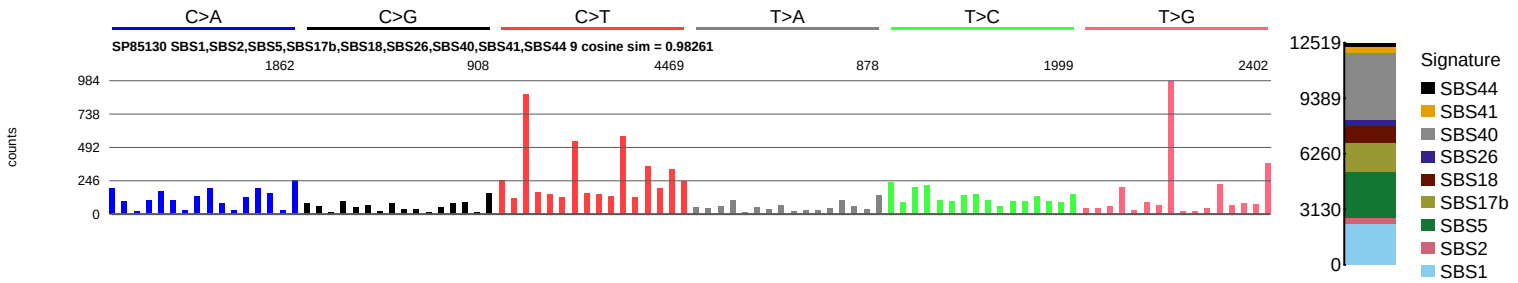

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS5,SBS13,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 10 cosine sim = 0.98266
1851
927
4504
885
1965
2395
counts
984
738
492
246
0
12519
9389
6260
3130
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS13
SBS5
SBS2
SBS1

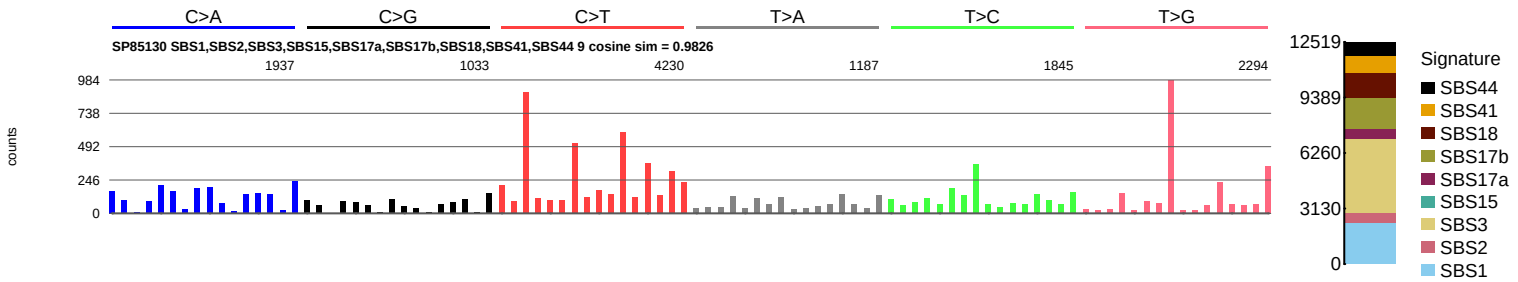

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS5,SBS13,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41 10 cosine sim = 0.98266
1846
929
4452
882
2005
2403
counts
984
738
492
246
0
12519
9389
6260
3130
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS13
SBS5
SBS2
SBS1

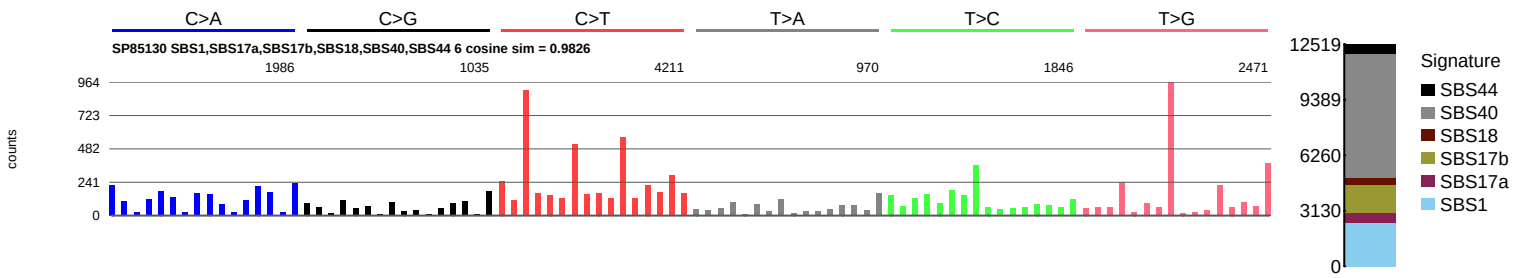

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS5,SBS13,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.98265
1861
916
4471
876
2001
2399
counts
984
738
492
246
0
12519
9389
6260
3130
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS13
SBS5
SBS2
SBS1

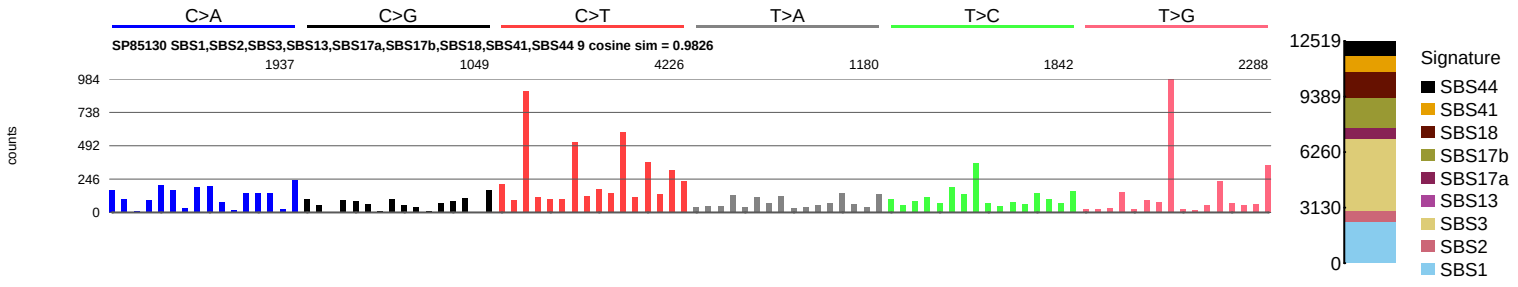

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS41,SBS44 8 cosine sim = 0.98264
1937
1032
4226
1189
1845
2292
counts
984
738
492
246
0
12519
9389
6260
3130
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1

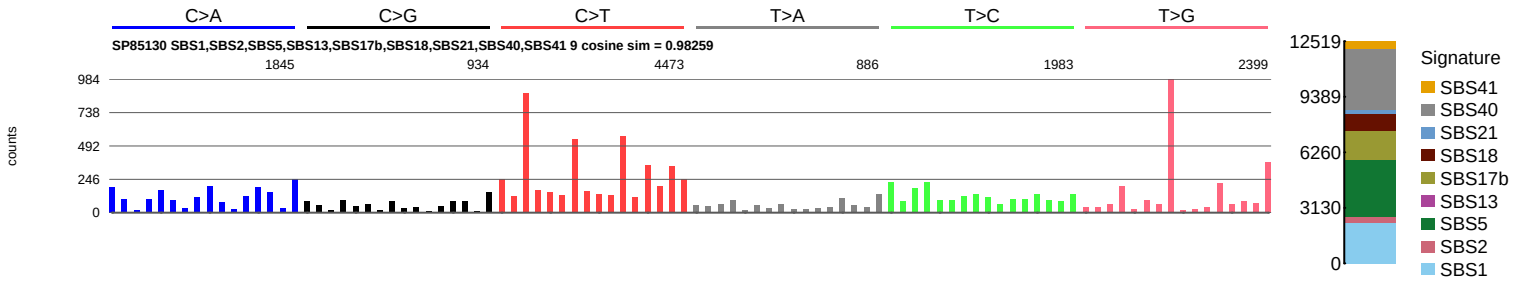

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS5,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41 9 cosine sim = 0.98263
1847
923
4451
884
2004
2405
counts
984
738
492
246
0
12519
9389
6260
3130
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS5
SBS2
SBS1

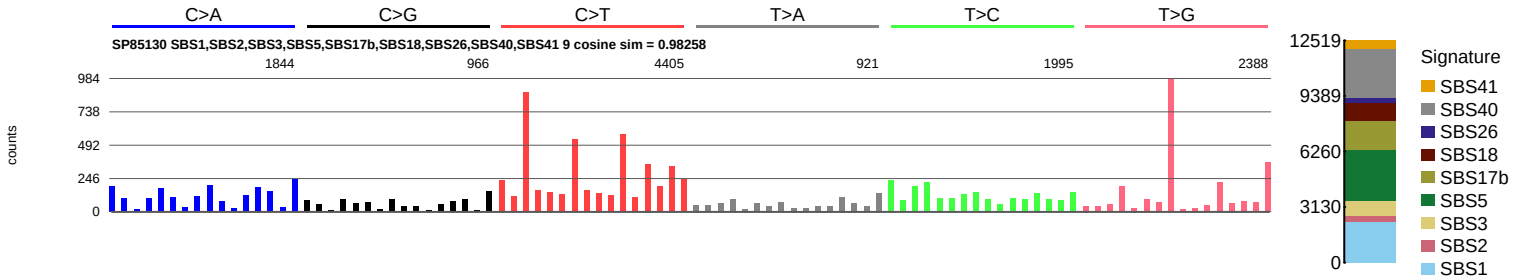

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS5,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 9 cosine sim = 0.98261
1862
908
4469
878
1999
2402
counts
984
738
492
246
0
12519
9389
6260
3130
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS5
SBS2
SBS1

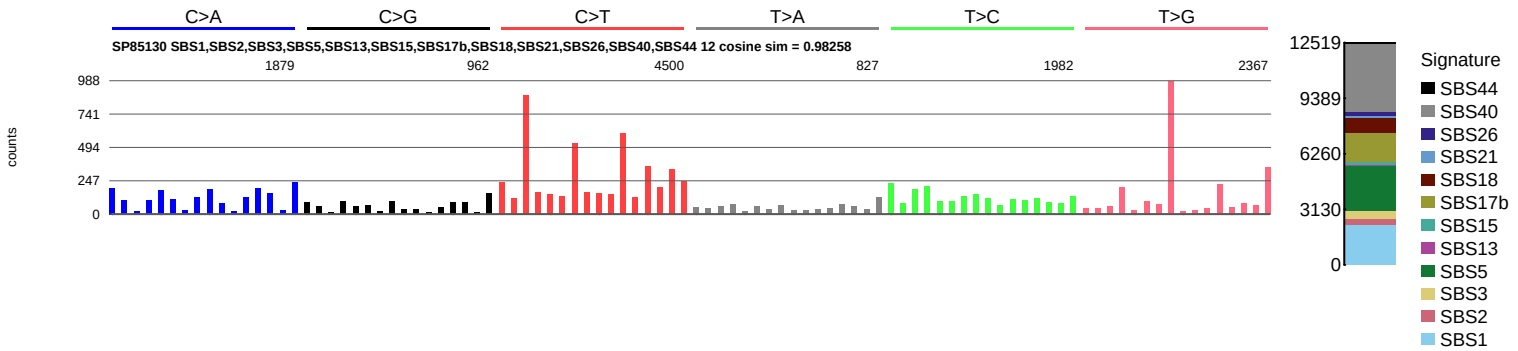

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS41,SBS44 9 cosine sim = 0.9826
1937
1033
4230
1187
1845
2294
counts
984
738
492
246
0
12519
9389
6260
3130
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1

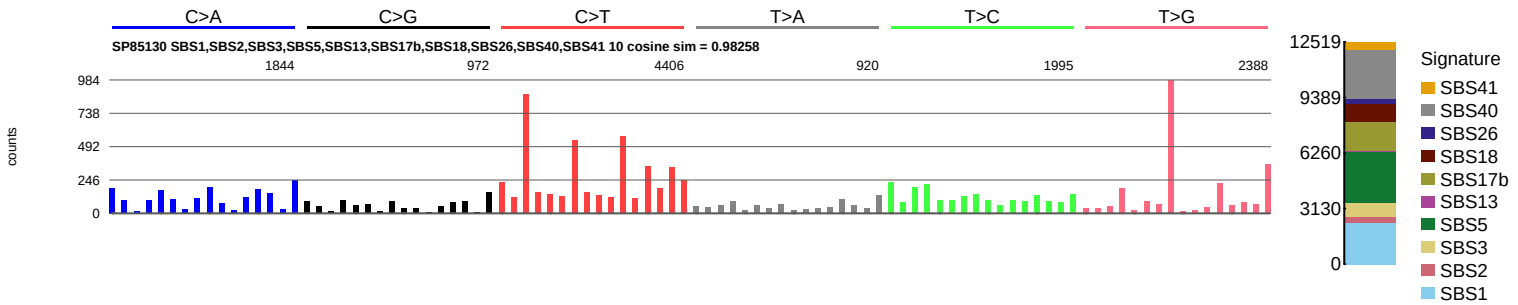

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS17a,SBS17b,SBS18,SBS40,SBS44 6 cosine sim = 0.9826
1986
1035
4211
970
1846
2471
counts
964
723
482
241
0
12519
9389
6260
3130
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS1

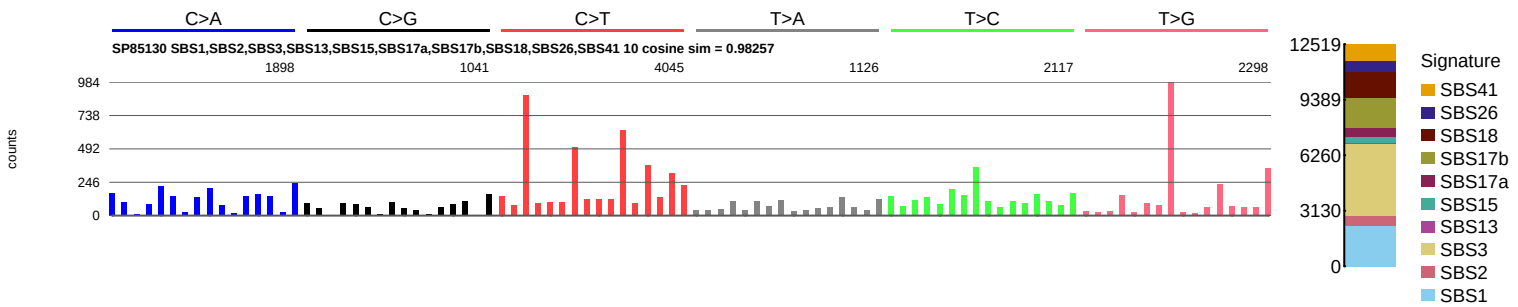

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS41,SBS44 9 cosine sim = 0.9826
1937
1049
4226
1180
1842
2288
counts
984
738
492
246
0
12519
9389
6260
3130
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1

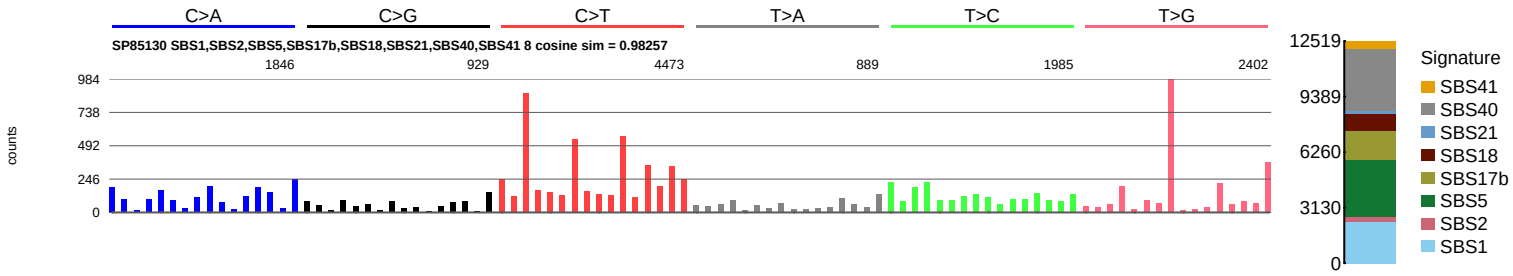

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS5,SBS13,SBS17b,SBS18,SBS21,SBS40,SBS41 9 cosine sim = 0.98259
1845
934
4473
886
1983
2399
counts
984
738
492
246
0
12519
9389
6260
3130
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS13
SBS5
SBS2
SBS1

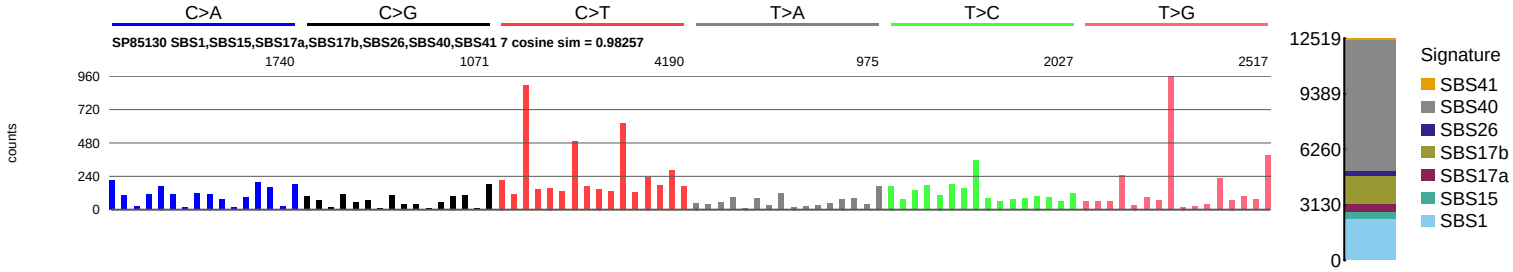

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS5,SBS17b,SBS18,SBS26,SBS40,SBS41 9 cosine sim = 0.98258
1844
966
4405
921
1995
2388
counts
984
738
492
246
0
12519
9389
6260
3130
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS5
SBS3
SBS2
SBS1

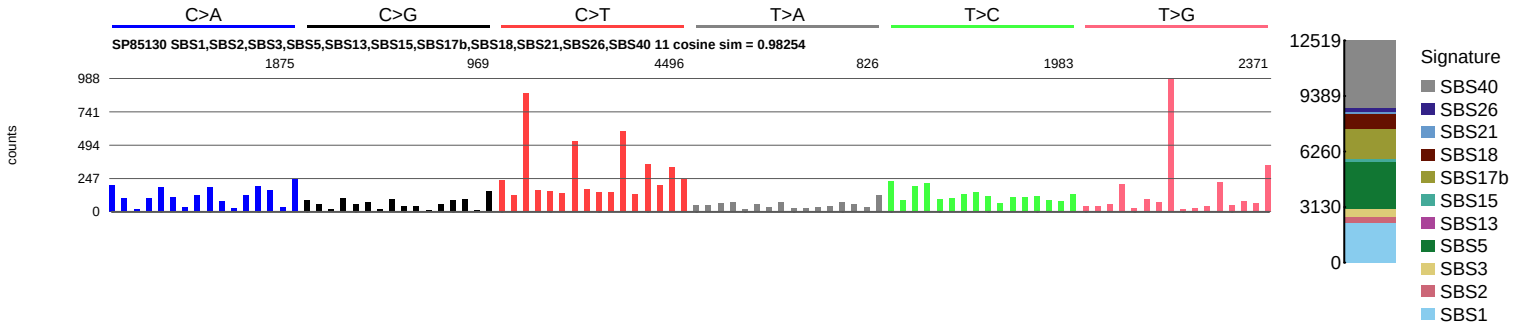

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS44 12 cosine sim = 0.98258
1879
962
4500
827
1982
2367
counts
988
741
494
247
0
12519
9389
6260
3130
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS18,SBS26,SBS40,SBS41 10 cosine sim = 0.98258
1844
972
4406
920
1995
2388
counts
984
738
492
246
0
12519
9389
6260
3130
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41 10 cosine sim = 0.98257
1898
1041
4045
1126
2117
2298
counts
984
738
492
246
0
12519
9389
6260
3130
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS5,SBS17b,SBS18,SBS21,SBS40,SBS41 8 cosine sim = 0.98257
1846
929
4473
889
1985
2402
counts
984
738
492
246
0
12519
9389
6260
3130
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS15,SBS17a,SBS17b,SBS26,SBS40,SBS41 7 cosine sim = 0.98257
1740
1071
4190
975
2027
2517
counts
960
720
480
240
0
12519
9389
6260
3130
0
Signature
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS26,SBS40 11 cosine sim = 0.98254
1875
969
4496
826
1983
2371
counts
988
741
494
247
0
12519
9389
6260
3130
0
Signature
SBS40
SBS26
SBS21
SBS18
SBS17b
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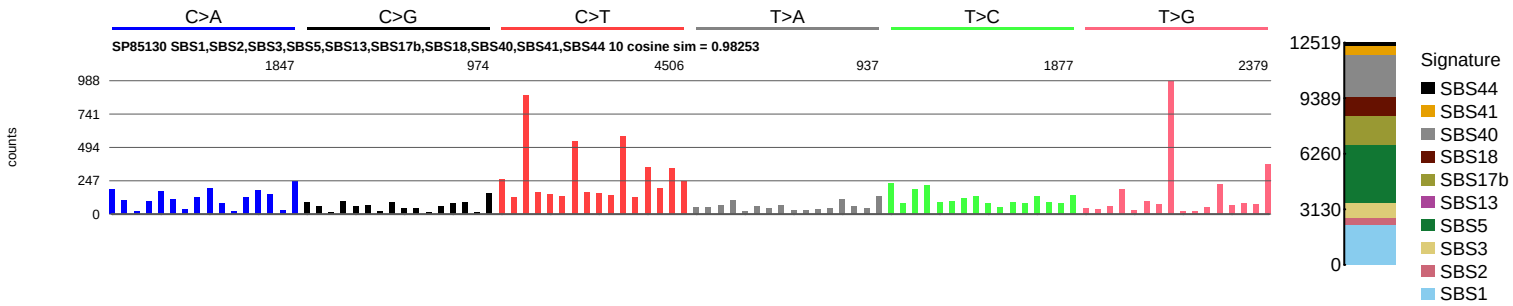
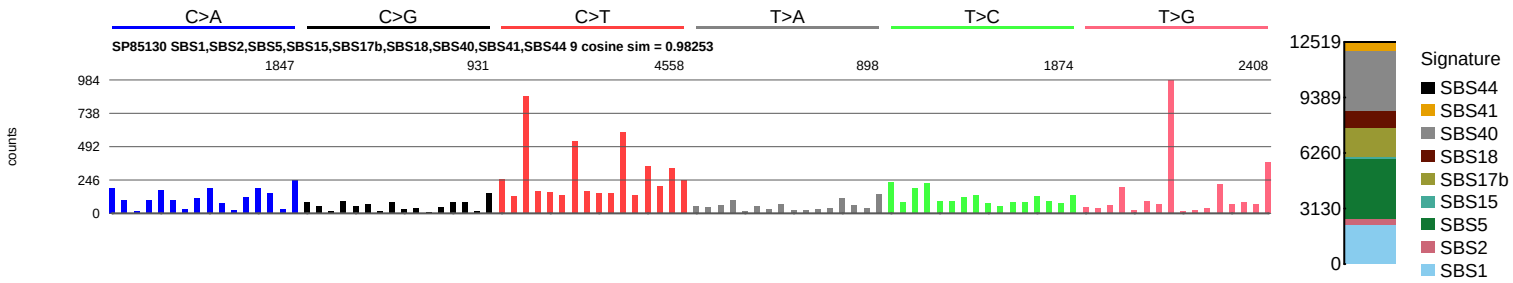
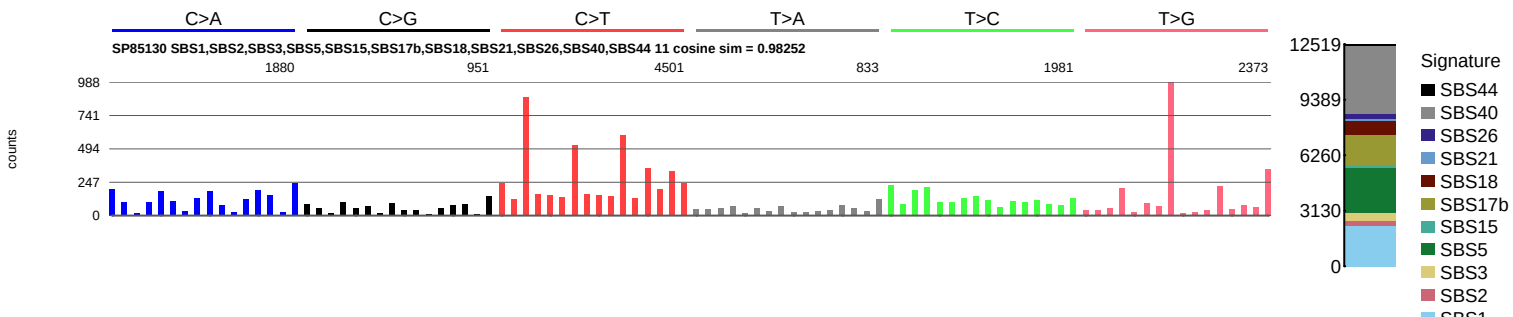
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS18,SBS40,SBS41,SBS44 10 cosine sim = 0.98253
1847
974
4506
937
1877
2379
counts
988
741
494
247
0
12519
9389
6260
3130
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS5,SBS15,SBS17b,SBS18,SBS40,SBS41,SBS44 9 cosine sim = 0.98253
1847
931
4558
898
1874
2408
counts
984
738
492
246
0
12519
9389
6260
3130
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS15
SBS5
SBS2
SBS1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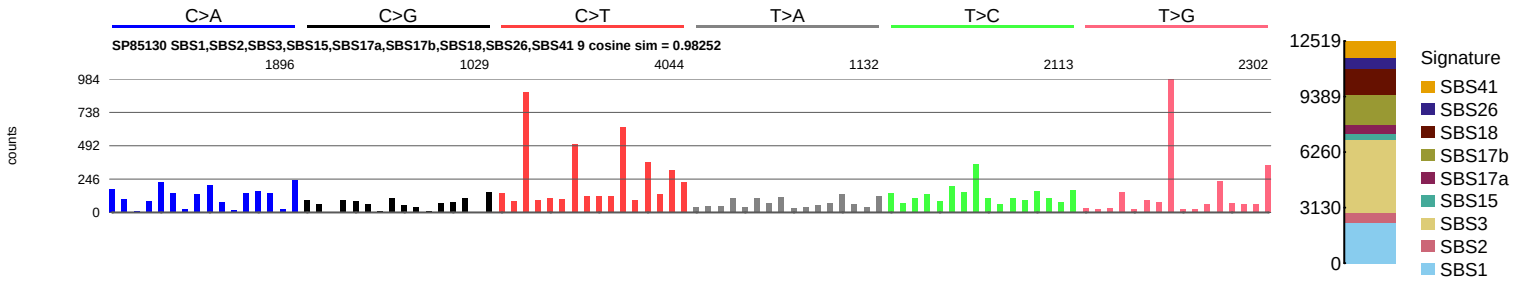
T>G
SP85130 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS44 11 cosine sim = 0.98252
1880
951
4501
833
1981
2373
counts
988
741
494
247
0
12519
9389
6260
3130
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS15
SBS5
SBS3
SBS2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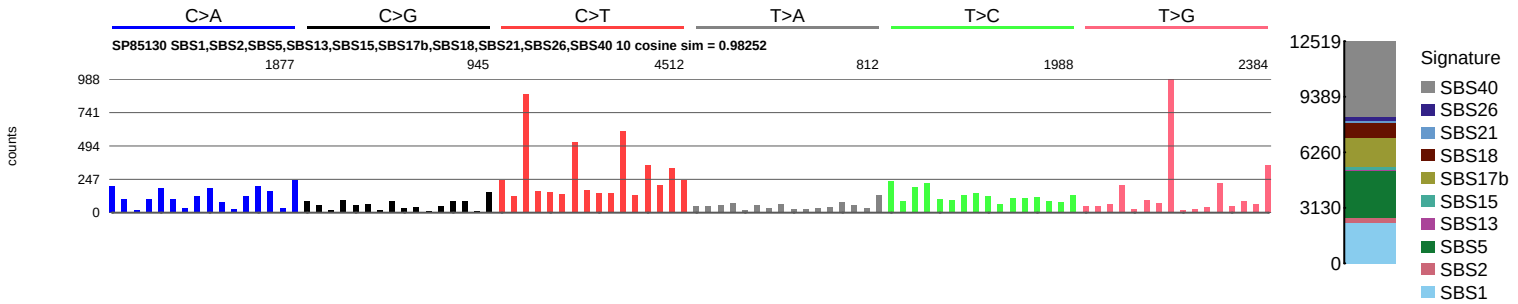
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41 9 cosine sim = 0.98252
1896
1029
4044
1132
2113
2302
counts
984
738
492
246
0
12519
9389
6260
3130
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1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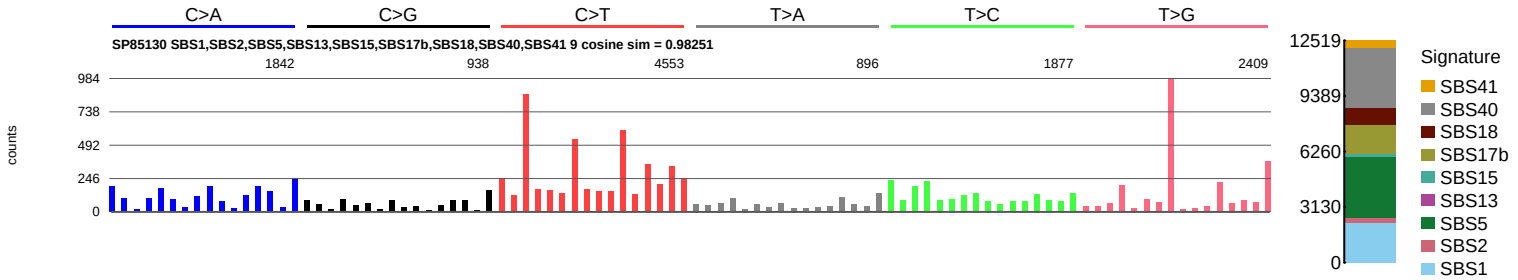
T>C
T>G
SP85130 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS26,SBS40 10 cosine sim = 0.98252
1877
945
4512
812
1988
2384
counts
988
741
494
247
0
12519
9389
6260
3130
0
Signature
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS2
SBS1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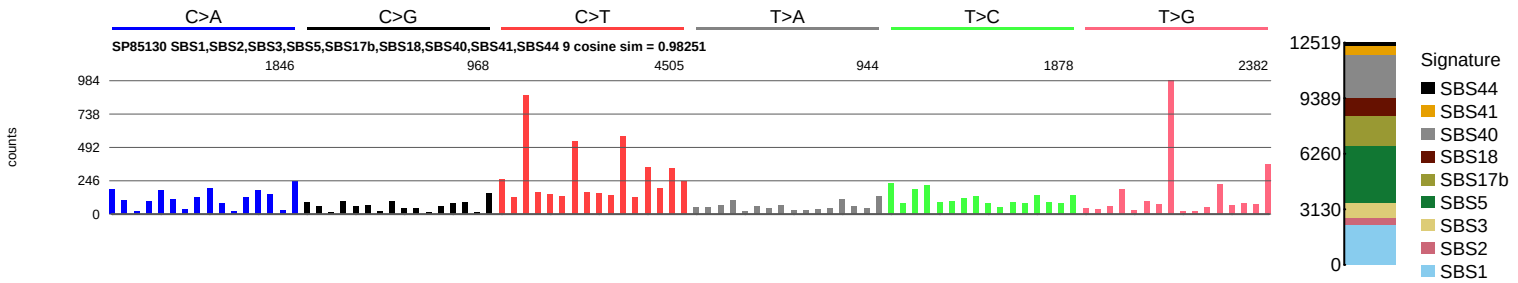
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS40,SBS41 9 cosine sim = 0.98251
1842
938
4553
896
1877
2409
counts
984
738
492
246
0
12519
9389
6260
3130
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS2
SBS1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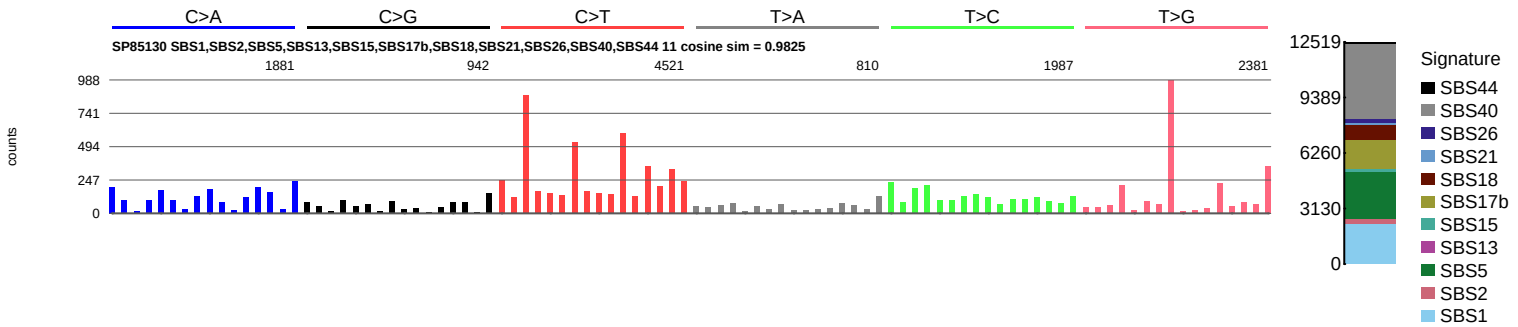
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS5,SBS17b,SBS18,SBS40,SBS41,SBS44 9 cosine sim = 0.98251
1846
968
4505
944
1878
2382
counts
984
738
492
246
0
12519
9389
6260
3130
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS5
SBS3
SBS2
SBS1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C>G
C>T
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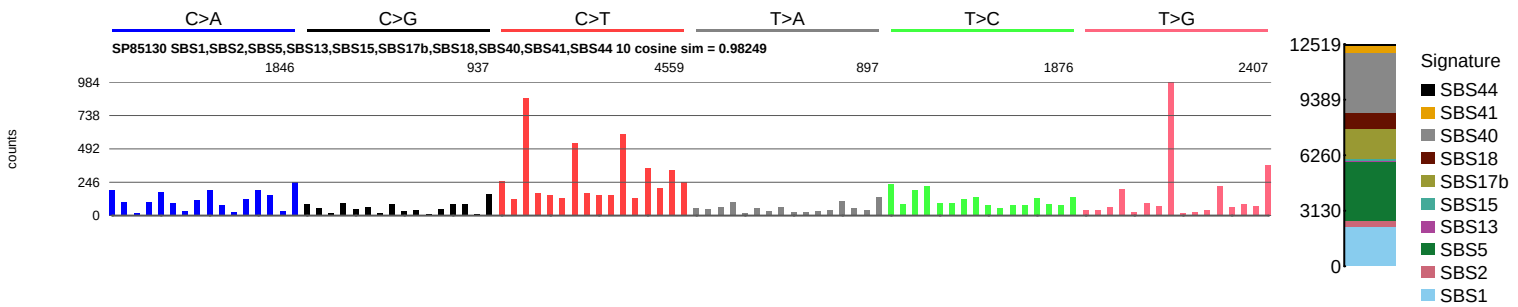
T>C
T>G
SP85130 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS44 11 cosine sim = 0.9825
1881
942
4521
810
1987
2381
counts
988
741
494
247
0
12519
9389
6260
3130
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS2
SBS1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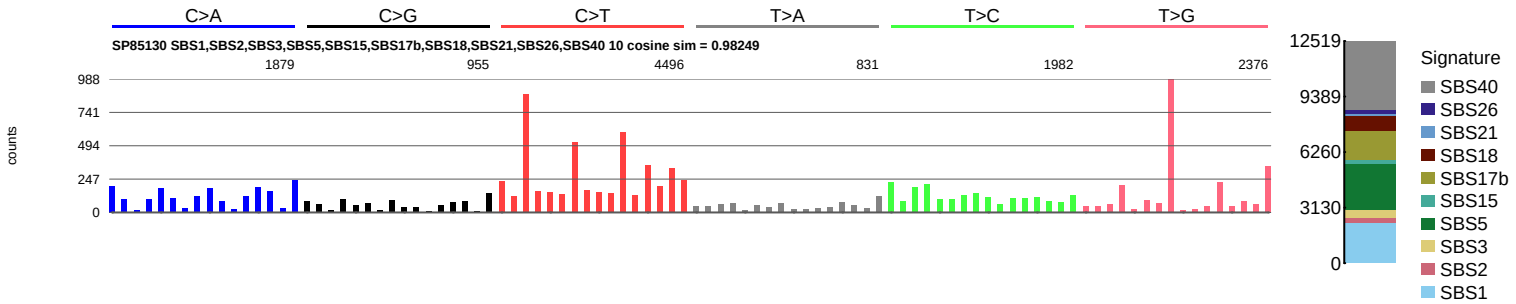
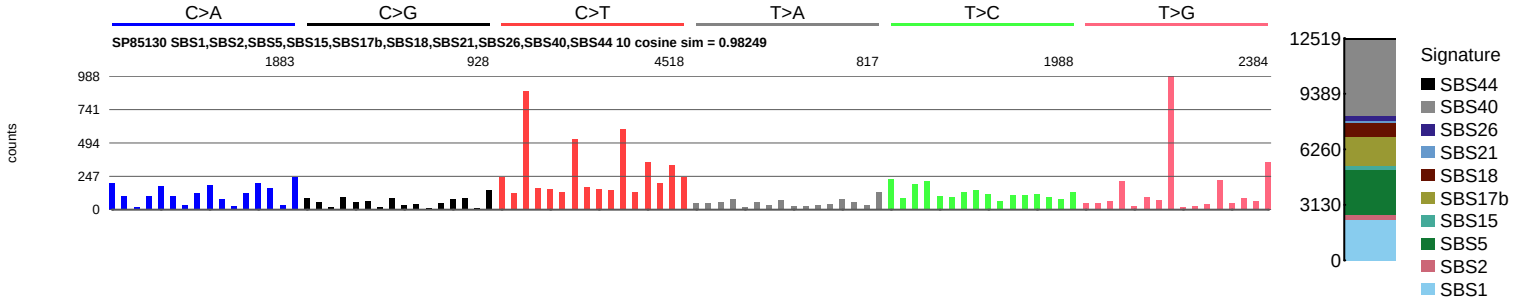
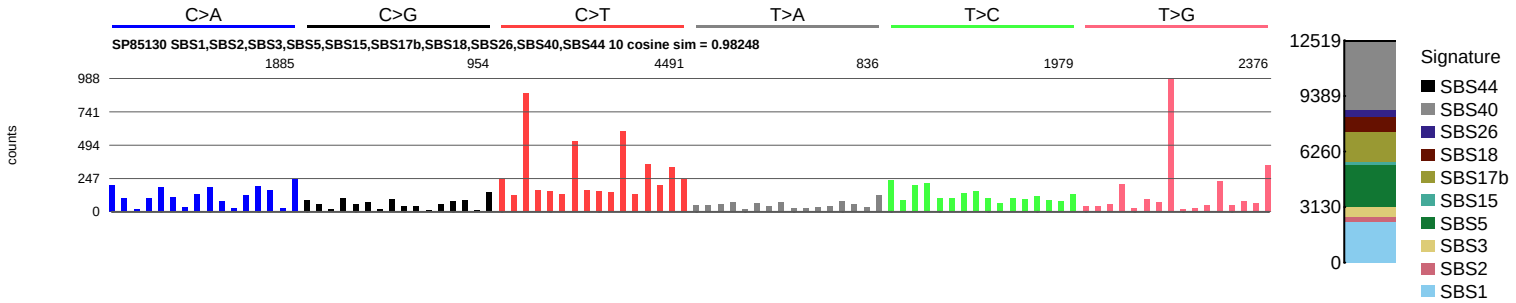
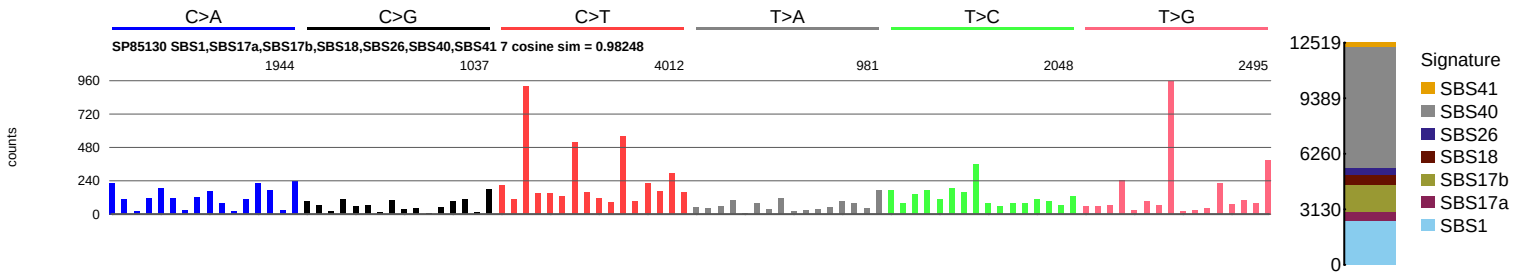
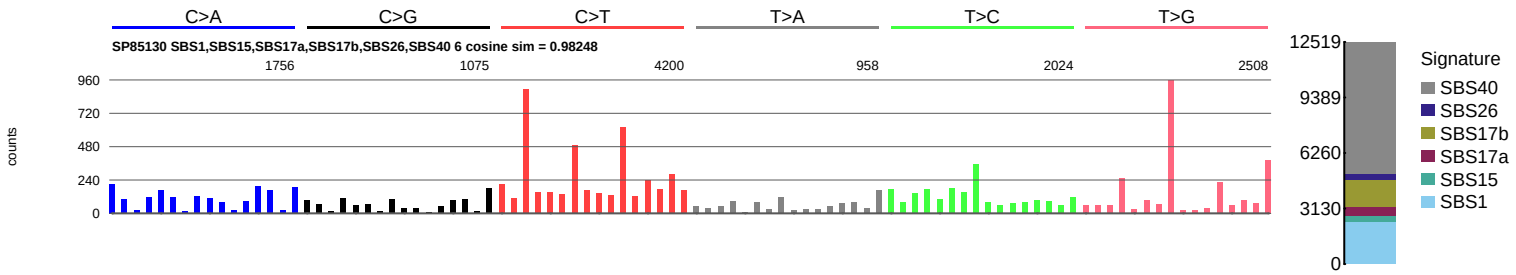
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS40,SBS41,SBS44 10 cosine sim = 0.98249
1846
937
4559
897
1876
2407
counts
984
738
492
246
0
12519
9389
6260
3130
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS2
SBS1

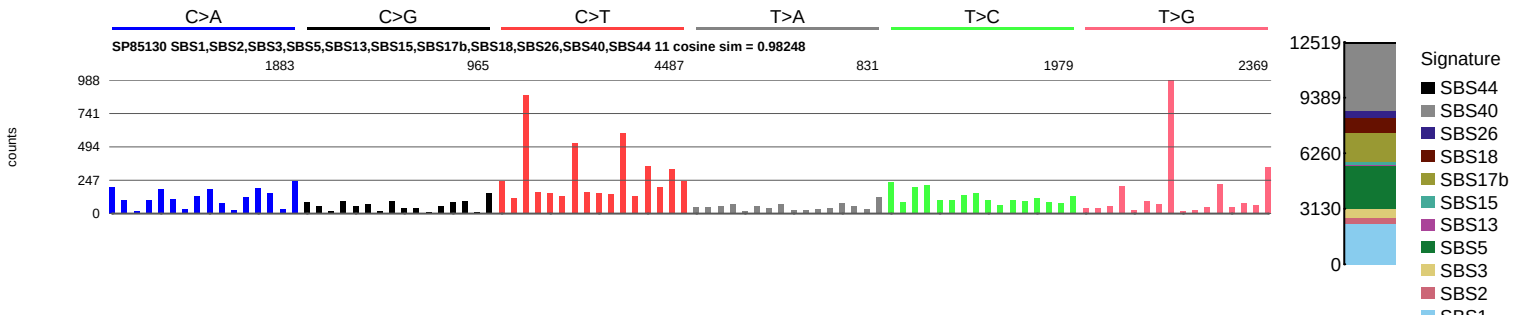

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17b,SBS18,SBS21,SBS26,SBS40 10 cosine sim = 0.98249
1879
955
4496
831
1982
2376
counts
988
741
494
247
0
12519
9389
6260
3130
0
Signature
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS15
SBS5
SBS3
SBS2
SBS1

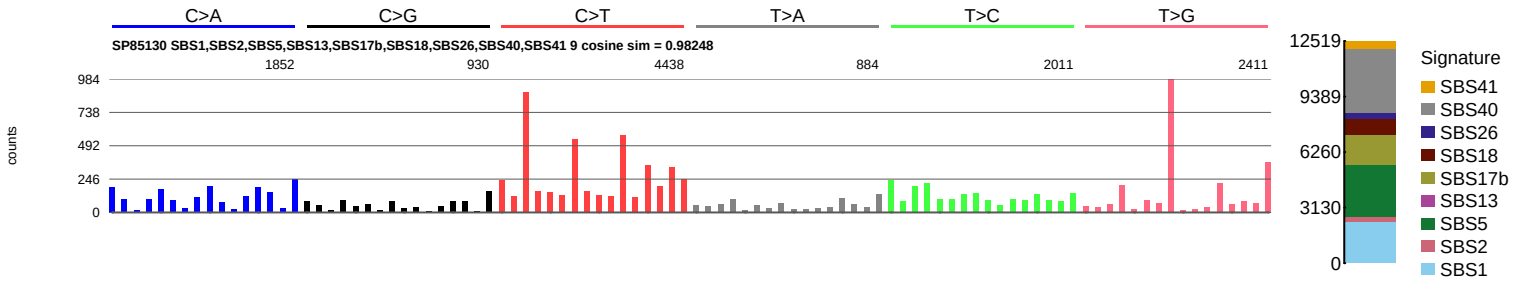

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS5,SBS15,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS44 10 cosine sim = 0.98249
1883
928
4518
817
1988
2384
counts
988
741
494
247
0
12519
9389
6260
3130
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS15
SBS5
SBS2
SBS1

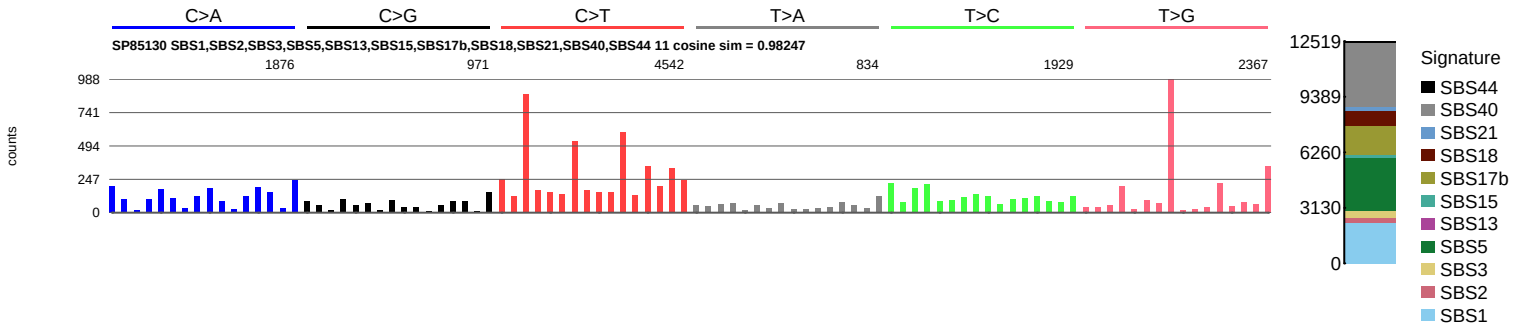

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17b,SBS18,SBS26,SBS40,SBS44 10 cosine sim = 0.98248
1885
954
4491
836
1979
2376
counts
988
741
494
247
0
12519
9389
6260
3130
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS15
SBS5
SBS3
SBS2
SBS1

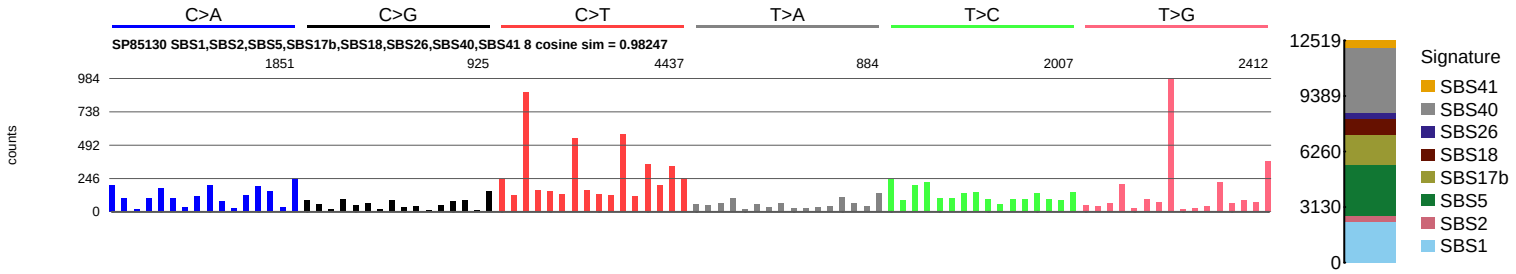

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 7 cosine sim = 0.98248
1944
1037
4012
981
2048
2495
counts
960
720
480
240
0
12519
9389
6260
3130
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS15,SBS17a,SBS17b,SBS26,SBS40 6 cosine sim = 0.98248
1756
1075
4200
958
2024
2508
counts
960
720
480
240
0
12519
9389
6260
3130
0
Signature
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS26,SBS40,SBS44 11 cosine sim = 0.98248
1883
965
4487
831
1979
2369
counts
988
741
494
247
0
12519
9389
6260
3130
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS5,SBS13,SBS17b,SBS18,SBS26,SBS40,SBS41 9 cosine sim = 0.98248
1852
930
4438
884
2011
2411
counts
984
738
492
246
0
12519
9389
6260
3130
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS40,SBS44 11 cosine sim = 0.98247
1876
971
4542
834
1929
2367
counts
988
741
494
247
0
12519
9389
6260
3130
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1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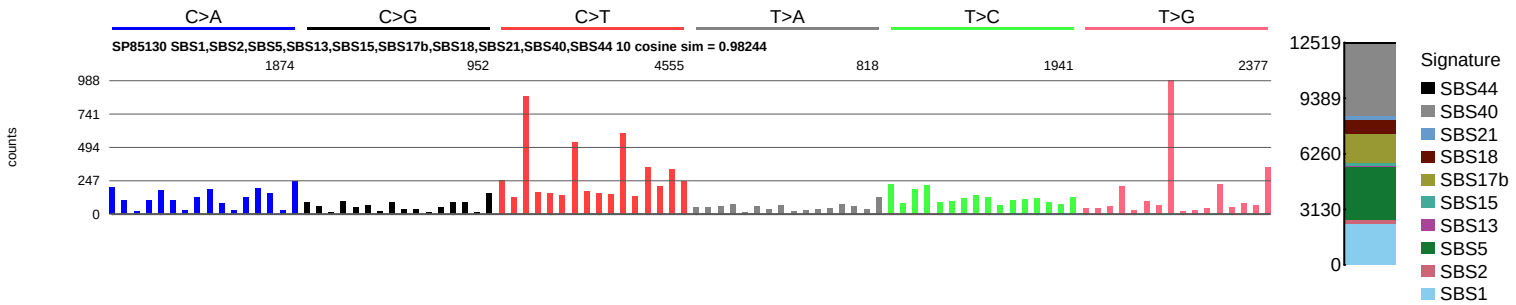
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS5,SBS17b,SBS18,SBS26,SBS40,SBS41 8 cosine sim = 0.98247
1851
925
4437
884
2007
2412
counts
984
738
492
246
0
12519
9389
6260
3130
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS5
SBS2
SBS1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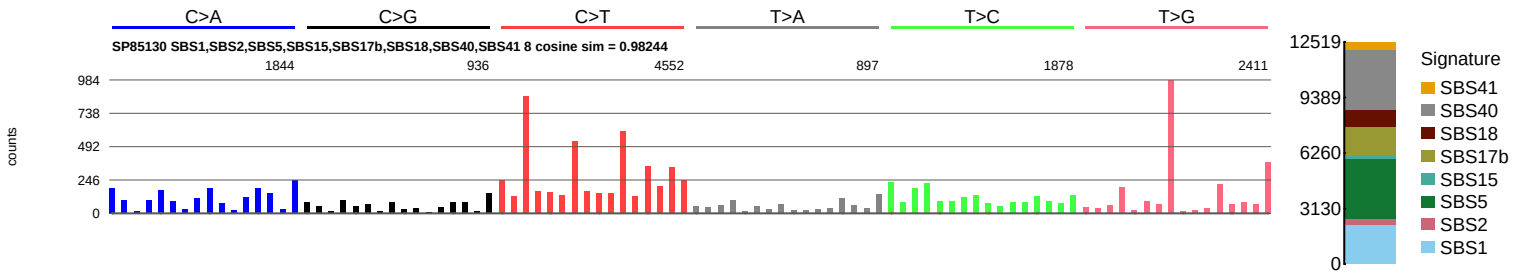
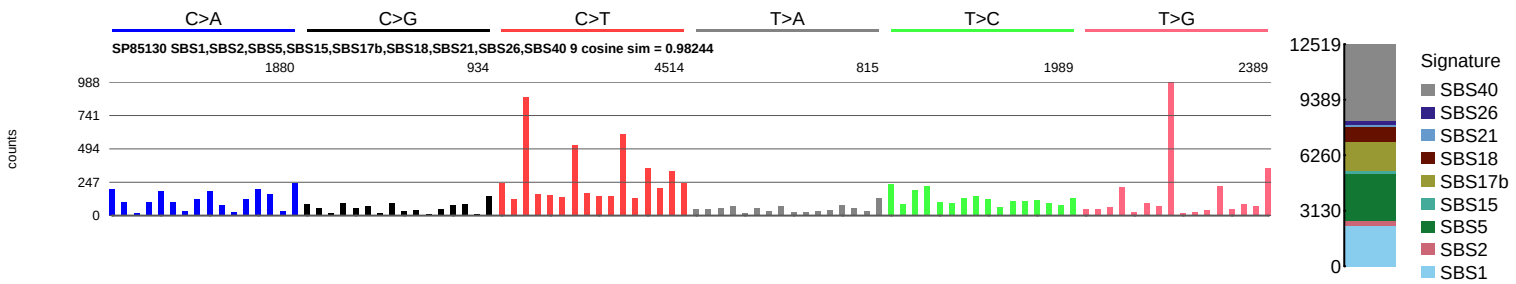
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS40,SBS44 10 cosine sim = 0.98244
1874
952
4555
818
1941
2377
counts
988
741
494
247
0
12519
9389
6260
3130
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS5,SBS15,SBS17b,SBS18,SBS40,SBS41 8 cosine sim = 0.98244
1844
936
4552
897
1878
2411
counts
984
738
492
246
0
12519
9389
6260
3130
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS15
SBS5
SBS2
SBS1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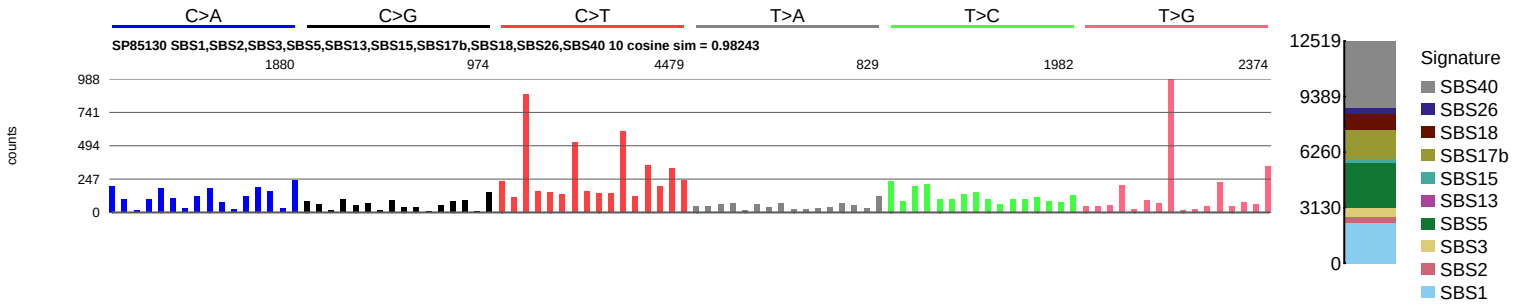
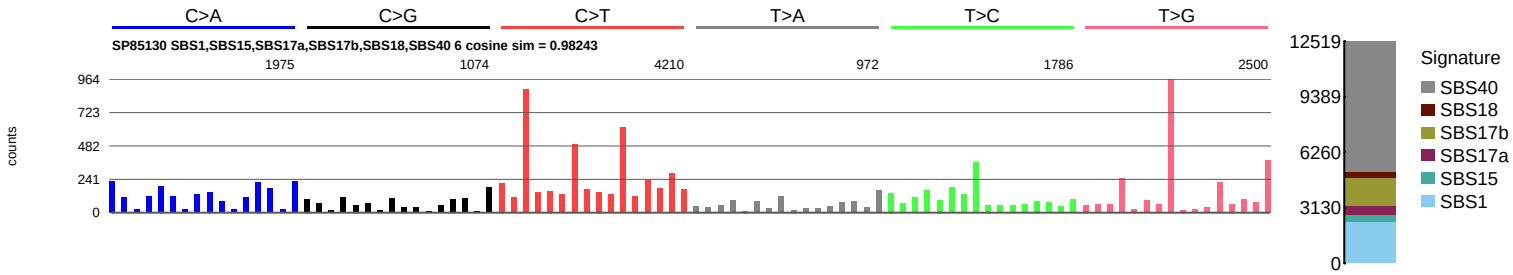
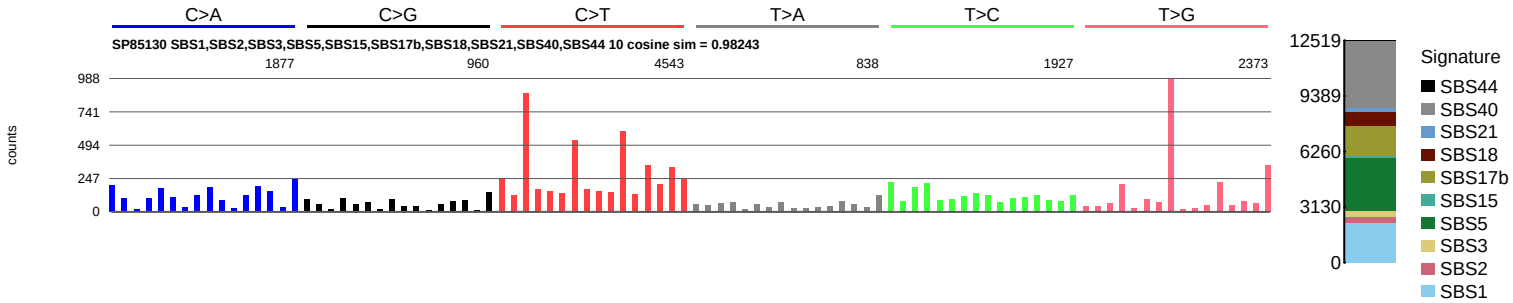
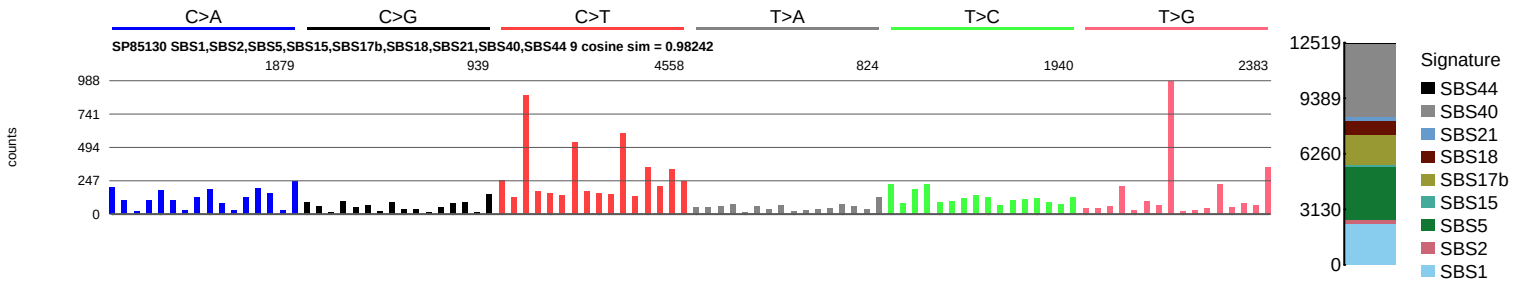
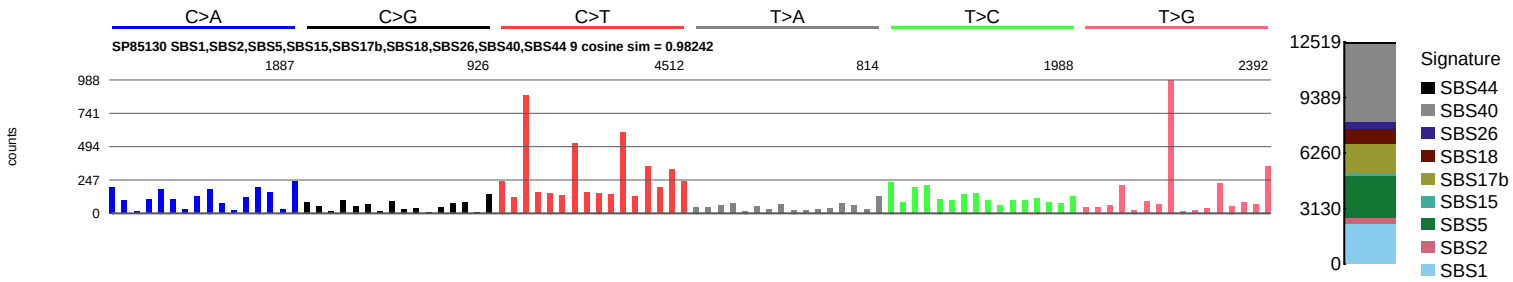
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS5,SBS15,SBS17b,SBS18,SBS21,SBS26,SBS40 9 cosine sim = 0.98244
1880
934
4514
815
1989
2389
counts
988
741
494
247
0
12519
9389
6260
3130
0
Signature
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS15
SBS5
SBS2
SBS1

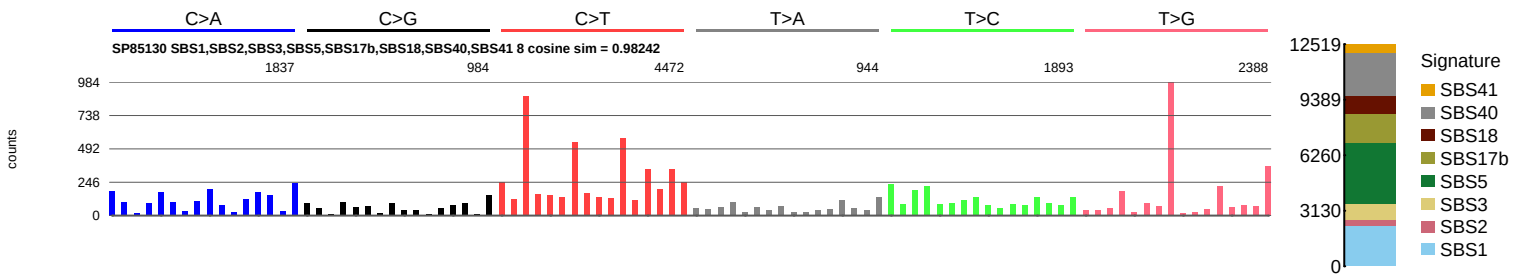

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS26,SBS40 10 cosine sim = 0.98243
1880
974
4479
829
1982
2374
counts
988
741
494
247
0
12519
9389
6260
3130
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS40 6 cosine sim = 0.98243
1975
1074
4210
972
1786
2500
counts
964
723
482
241
0
12519
9389
6260
3130
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17b,SBS18,SBS21,SBS40,SBS44 10 cosine sim = 0.98243
1877
960
4543
838
1927
2373
counts
988
741
494
247
0
12519
9389
6260
3130
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS15
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS5,SBS15,SBS17b,SBS18,SBS21,SBS40,SBS44 9 cosine sim = 0.98242
1879
939
4558
824
1940
2383
counts
988
741
494
247
0
12519
9389
6260
3130
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS15
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS5,SBS15,SBS17b,SBS18,SBS26,SBS40,SBS44 9 cosine sim = 0.98242
1887
926
4512
814
1988
2392
counts
988
741
494
247
0
12519
9389
6260
3130
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS15
SBS5
SBS2
SBS1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C>G
C>T
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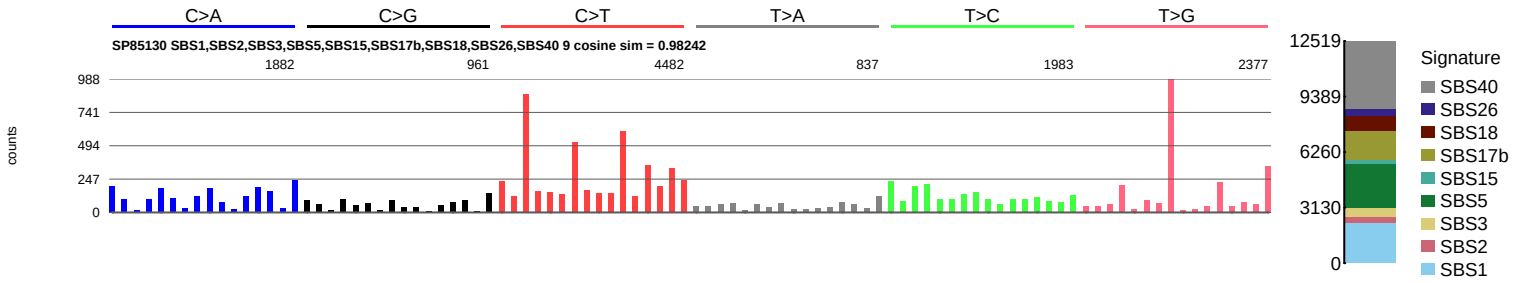
T>G
SP85130 SBS1,SBS2,SBS3,SBS5,SBS17b,SBS18,SBS40,SBS41 8 cosine sim = 0.98242
1837
984
4472
944
1893
2388
counts
984
738
492
246
0
12519
9389
6260
3130
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS5
SBS3
SBS2
SBS1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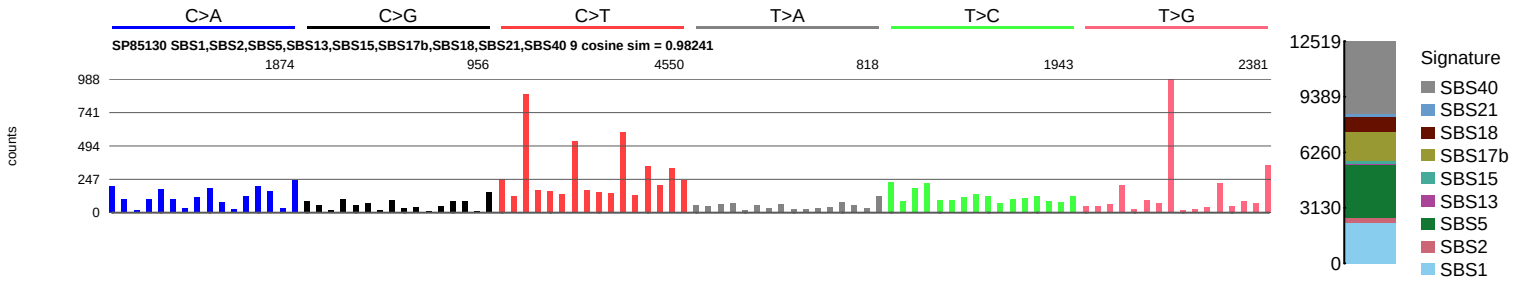
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17b,SBS18,SBS26,SBS40 9 cosine sim = 0.98242
1882
961
4482
837
1983
2377
counts
988
741
494
247
0
12519
9389
6260
3130
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS15
SBS5
SBS3
SBS2
SBS1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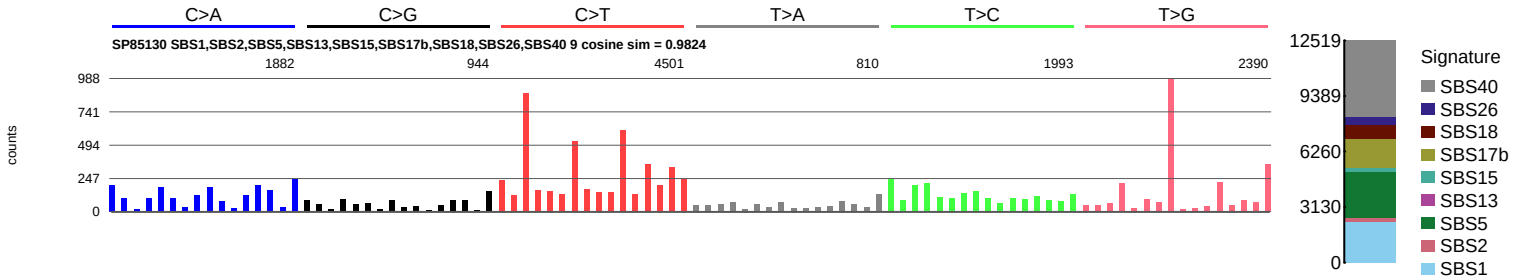
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS40 9 cosine sim = 0.98241
1874
956
4550
818
1943
2381
counts
988
741
494
247
0
12519
9389
6260
3130
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS2
SBS1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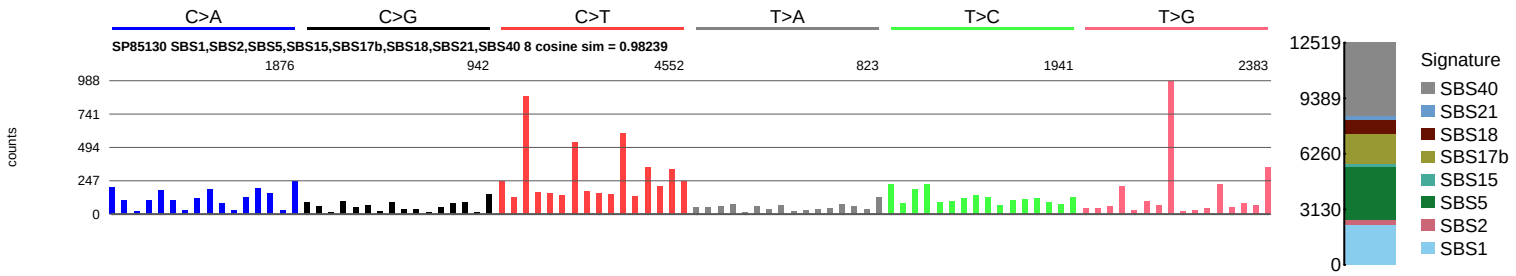
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS26,SBS40 9 cosine sim = 0.9824
1882
944
4501
810
1993
2390
counts
988
741
494
247
0
12519
9389
6260
3130
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS2
SBS1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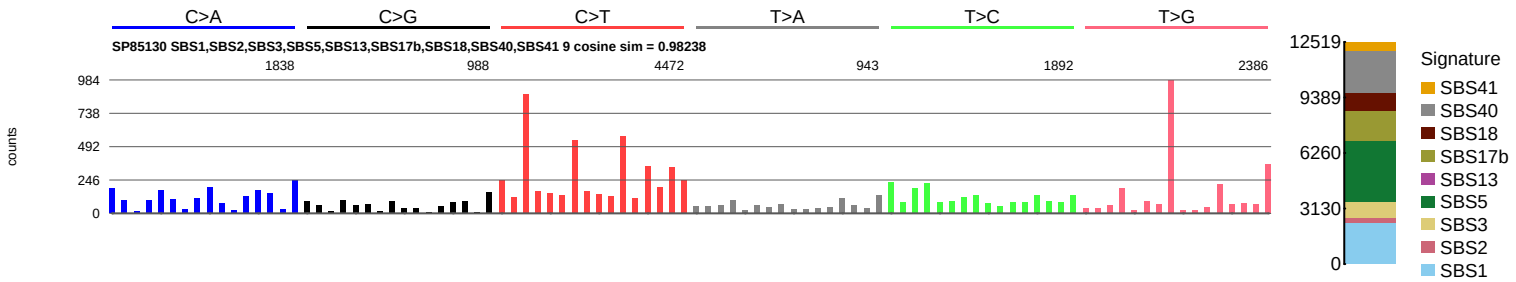
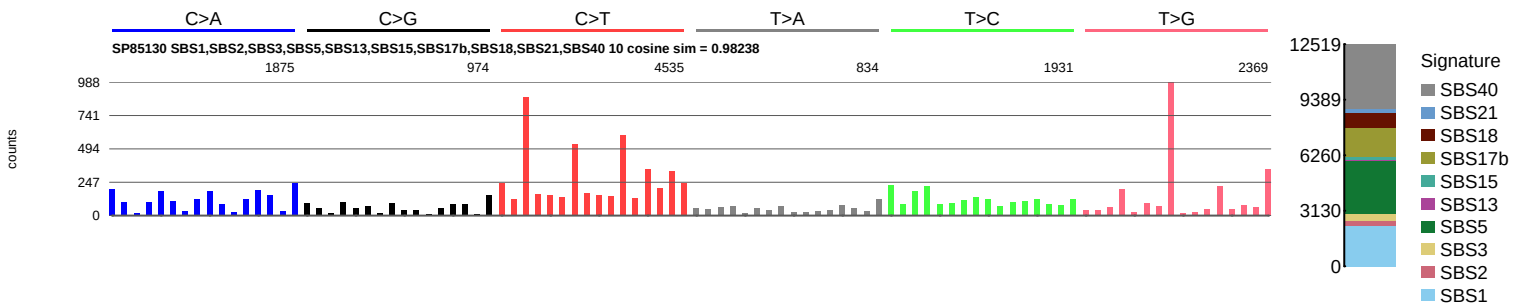
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS5,SBS15,SBS17b,SBS18,SBS21,SBS40 8 cosine sim = 0.98239
1876
942
4552
823
1941
2383
counts
988
741
494
247
0
12519
9389
6260
3130
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS15
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS18,SBS40,SBS41 9 cosine sim = 0.98238
1838
988
4472
943
1892
2386
counts
984
738
492
246
0
12519
9389
6260
3130
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS13
SBS5
SBS3
SBS2
SBS1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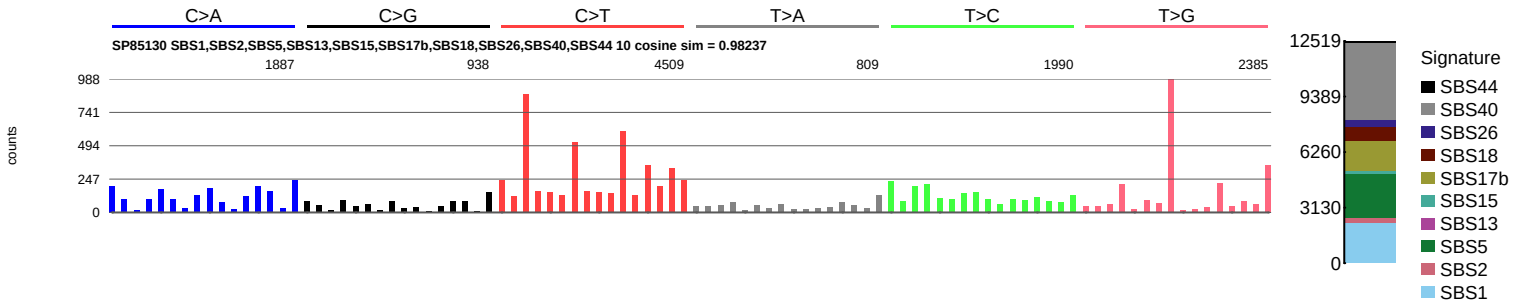
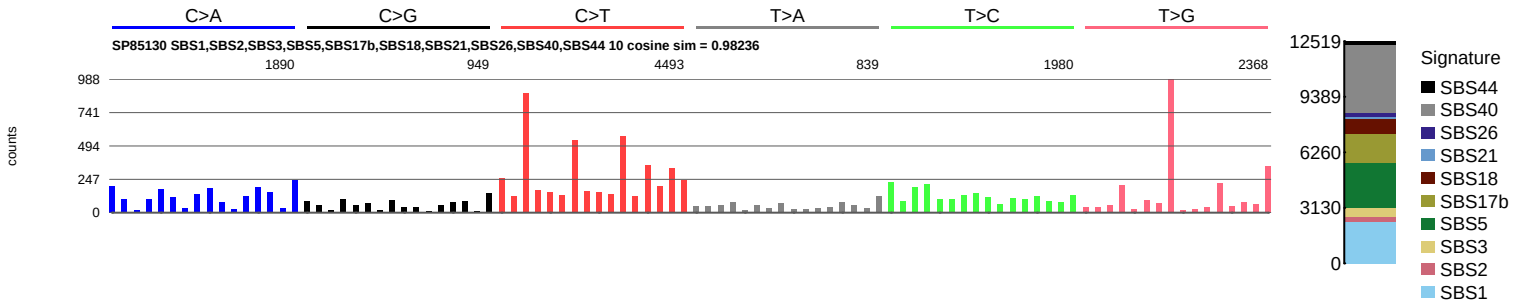
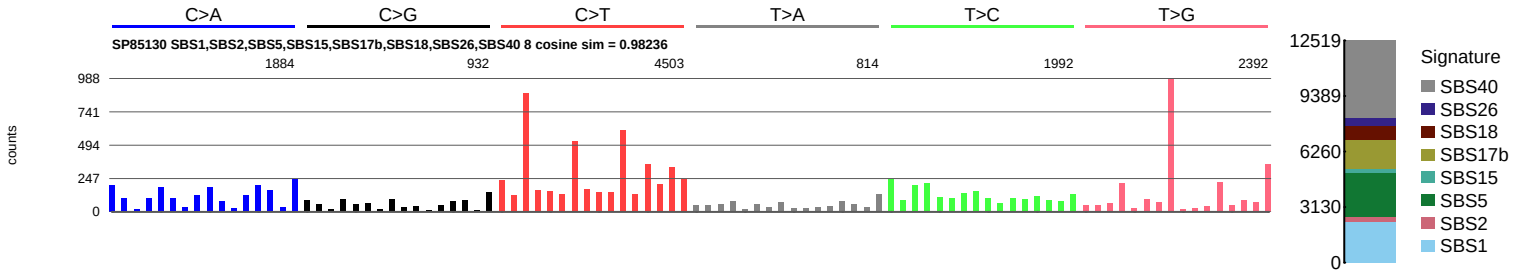
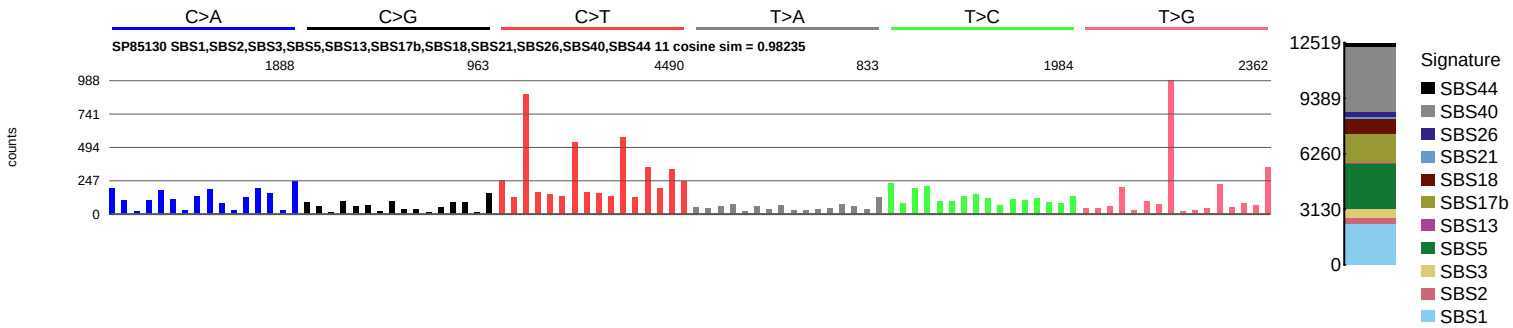
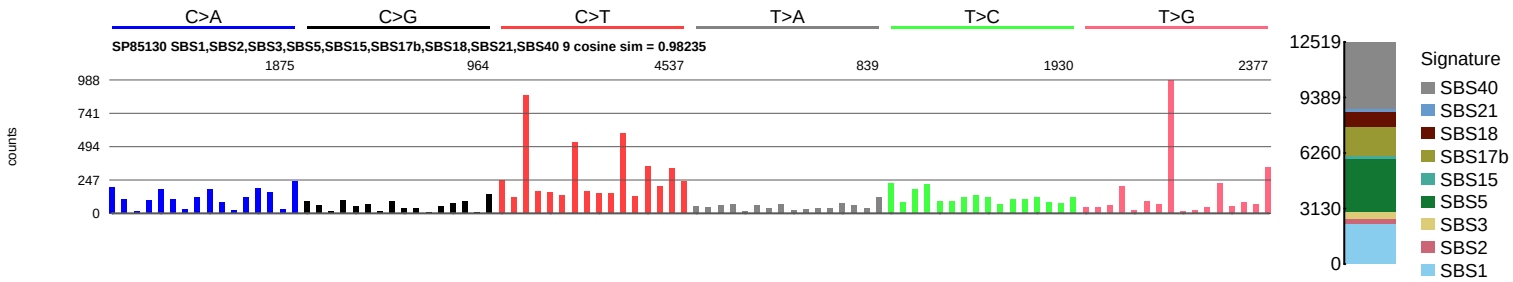
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS40 10 cosine sim = 0.98238
1875
974
4535
834
1931
2369
counts
988
741
494
247
0
12519
9389
6260
3130
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1

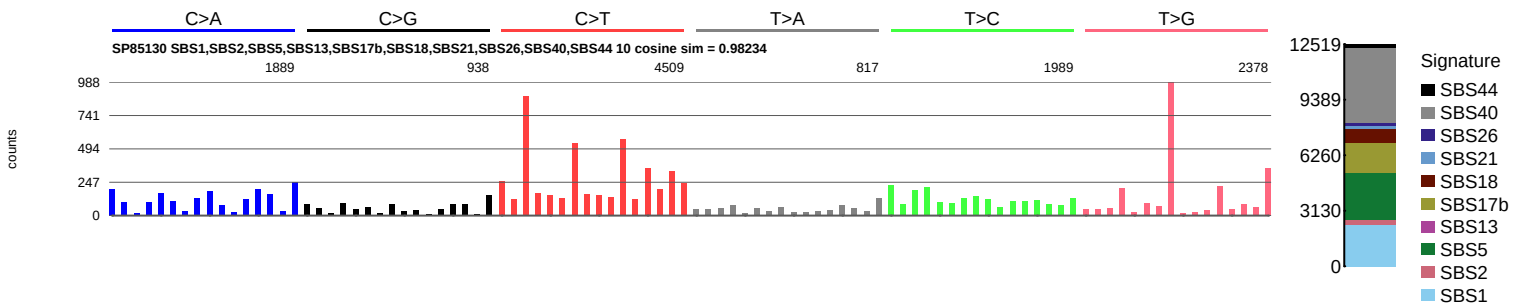

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS26,SBS40,SBS44 10 cosine sim = 0.98237
1887
938
4509
809
1990
2385
counts
988
741
494
247
0
12519
9389
6260
3130
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS2
SBS1

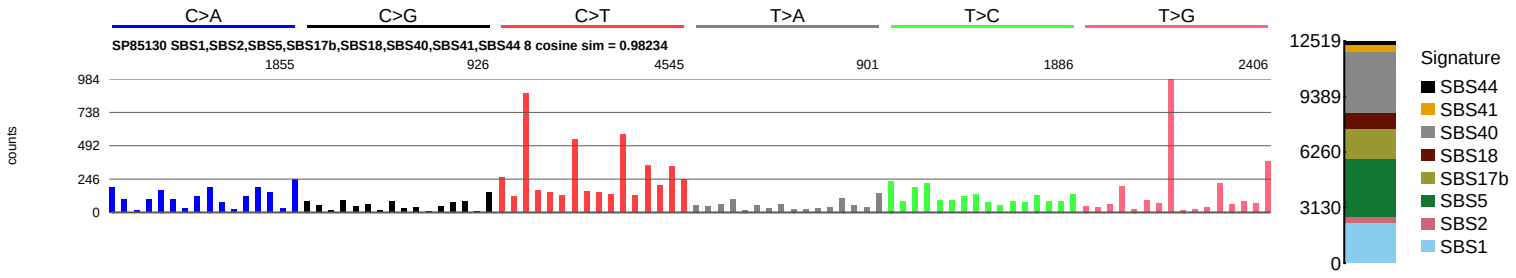

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS5,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS44 10 cosine sim = 0.98236
1890
949
4493
839
1980
2368
counts
988
741
494
247
0
12519
9389
6260
3130
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS5
SBS3
SBS2
SBS1

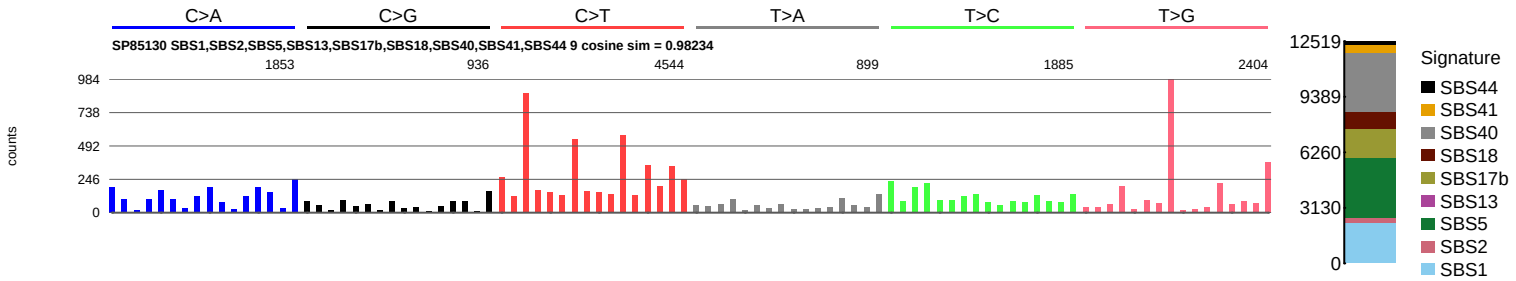

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS5,SBS15,SBS17b,SBS18,SBS26,SBS40 8 cosine sim = 0.98236
1884
932
4503
814
1992
2392
counts
988
741
494
247
0
12519
9389
6260
3130
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS15
SBS5
SBS2
SBS1

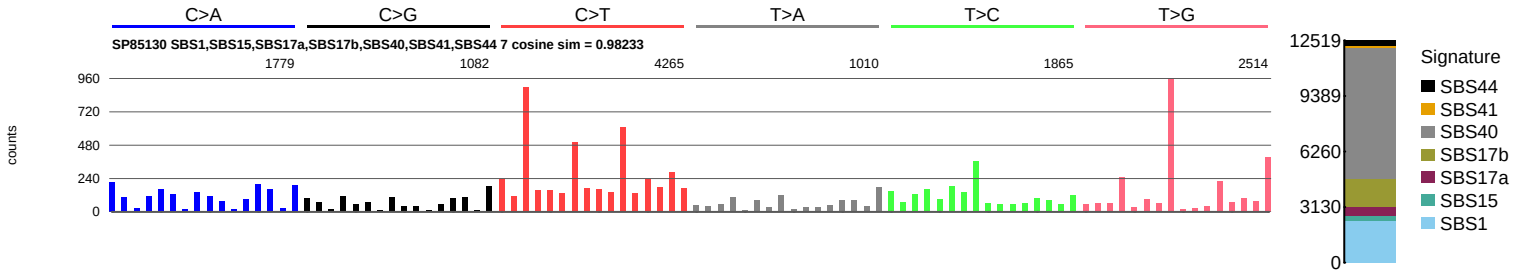

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS44 11 cosine sim = 0.98235
1888
963
4490
833
1984
2362
counts
988
741
494
247
0
12519
9389
6260
3130
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS13
SBS5
SBS3
SBS2
SBS1

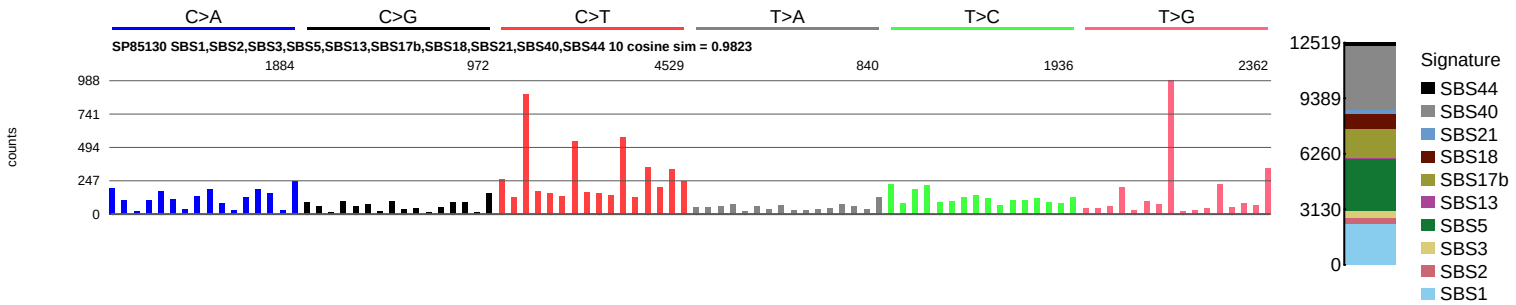

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17b,SBS18,SBS21,SBS40 9 cosine sim = 0.98235
1875
964
4537
839
1930
2377
counts
988
741
494
247
0
12519
9389
6260
3130
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS15
SBS5
SBS3
SBS2
SBS1

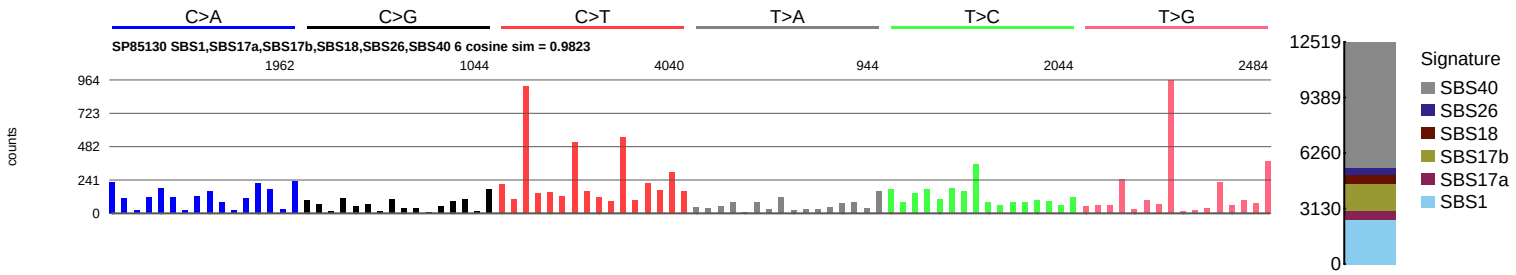

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS5,SBS13,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS44 10 cosine sim = 0.98234
1889
938
4509
817
1989
2378
counts
988
741
494
247
0
12519
9389
6260
3130
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS13
SBS5
SBS2
SBS1

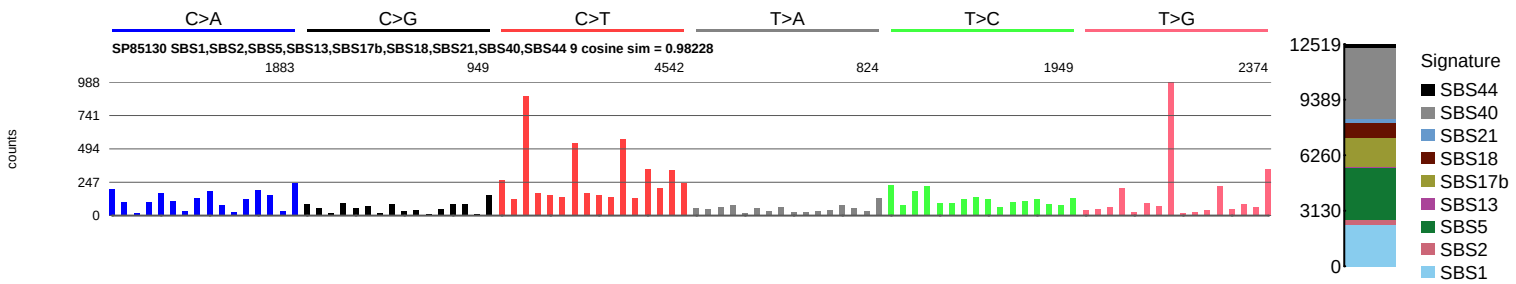

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS5,SBS17b,SBS18,SBS40,SBS41,SBS44 8 cosine sim = 0.98234
1855
926
4545
901
1886
2406
counts
984
738
492
246
0
12519
9389
6260
3130
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS5
SBS2
SBS1

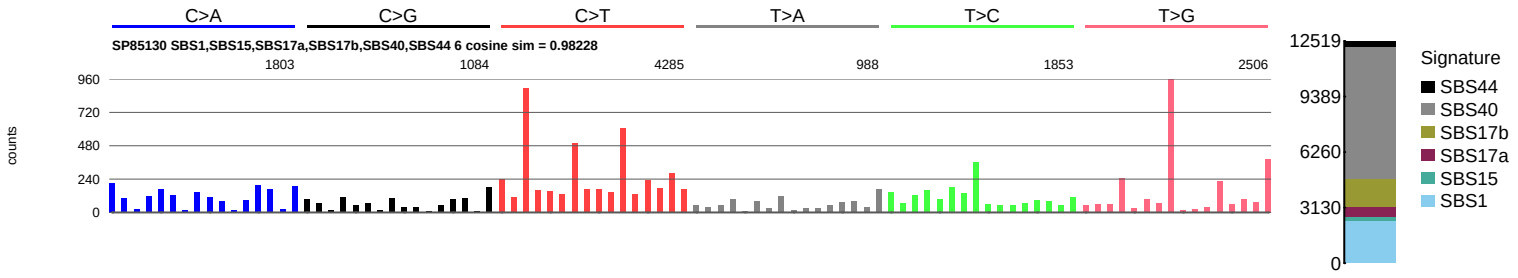

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS5,SBS13,SBS17b,SBS18,SBS40,SBS41,SBS44 9 cosine sim = 0.98234
1853
936
4544
899
1885
2404
counts
984
738
492
246
0
12519
9389
6260
3130
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS13
SBS5
SBS2
SBS1

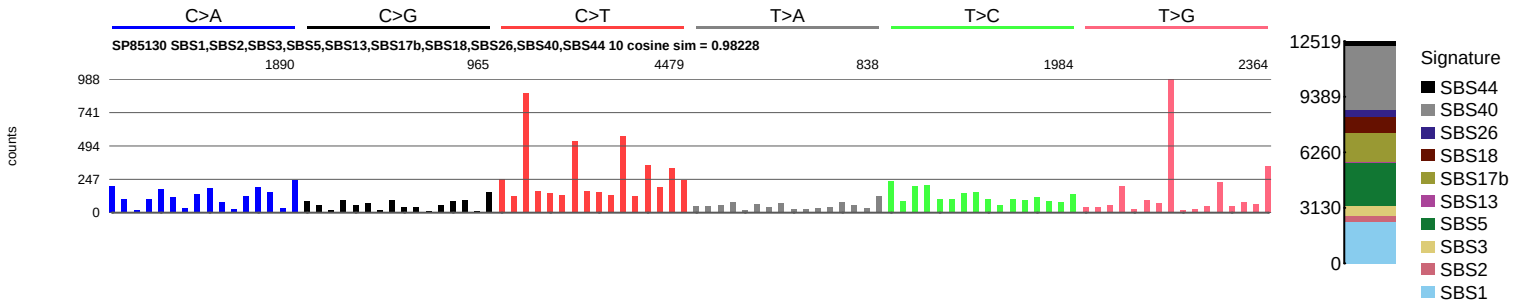

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS15,SBS17a,SBS17b,SBS40,SBS41,SBS44 7 cosine sim = 0.98233
1779
1082
4265
1010
1865
2514
counts
960
720
480
240
0
12519
9389
6260
3130
0
Signature
SBS44
SBS41
SBS40
SBS17b
SBS17a
SBS15
SBS1

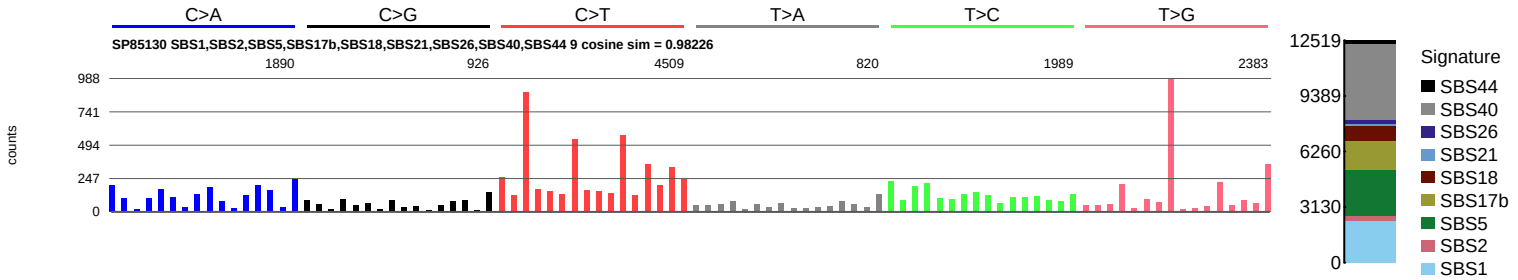

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS18,SBS21,SBS40,SBS44 10 cosine sim = 0.9823
1884
972
4529
840
1936
2362
counts
988
741
494
247
0
12519
9389
6260
3130
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS13
SBS5
SBS3
SBS2
SBS1

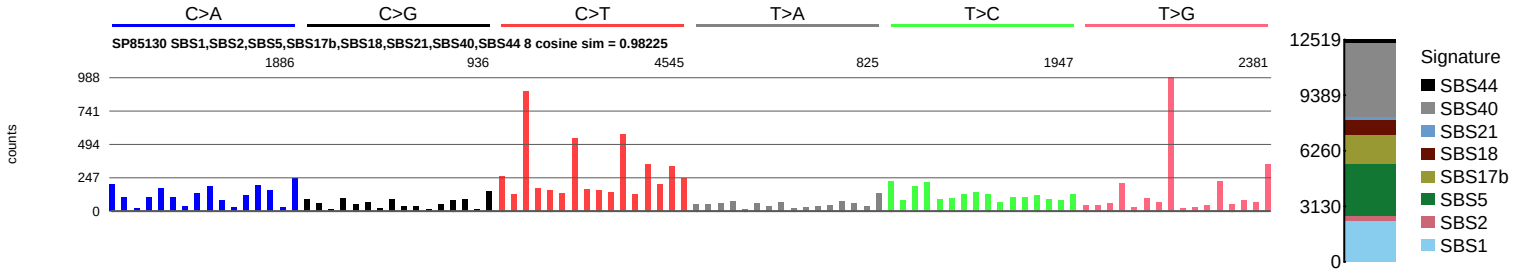

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS17a,SBS17b,SBS18,SBS26,SBS40 6 cosine sim = 0.9823
1962
1044
4040
944
2044
2484
counts
964
723
482
241
0
12519
9389
6260
3130
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS1

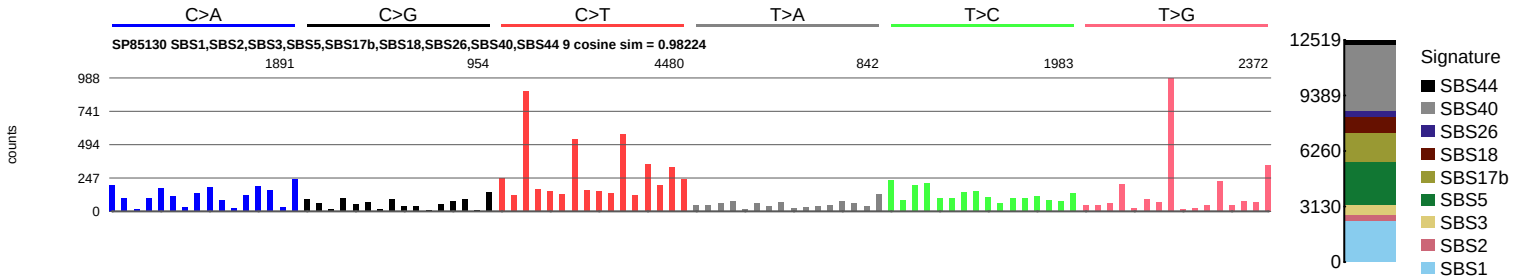

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS5,SBS13,SBS17b,SBS18,SBS21,SBS40,SBS44 9 cosine sim = 0.98228
1883
949
4542
824
1949
2374
counts
988
741
494
247
0
12519
9389
6260
3130
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS13
SBS5
SBS2
SBS1

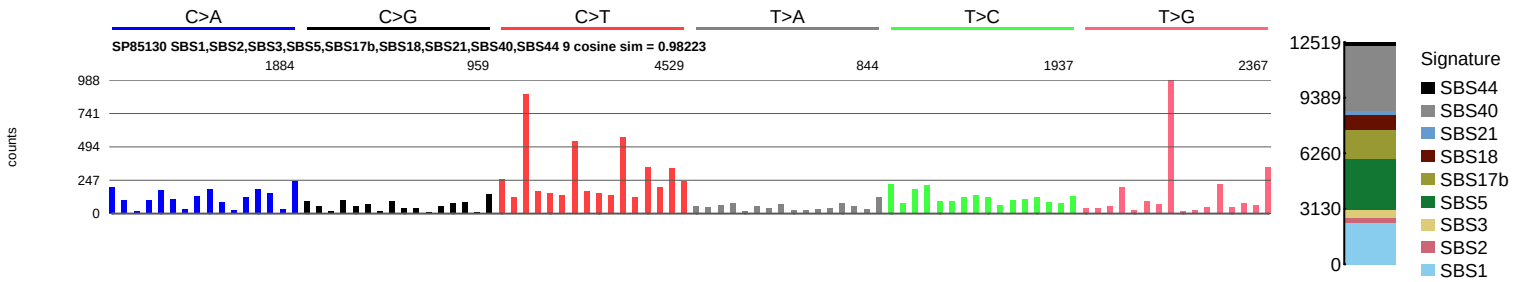

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS15,SBS17a,SBS17b,SBS40,SBS44 6 cosine sim = 0.98228
1803
1084
4285
988
1853
2506
counts
960
720
480
240
0
12519
9389
6260
3130
0
Signature
SBS44
SBS40
SBS17b
SBS17a
SBS15
SBS1

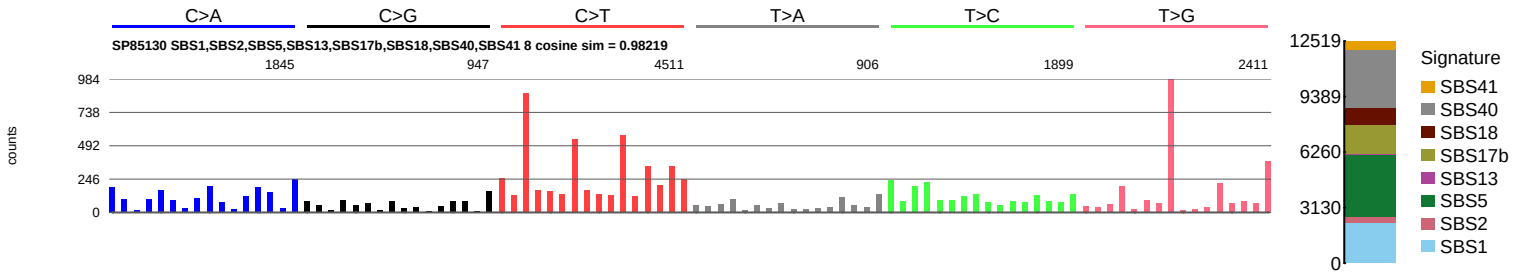

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS18,SBS26,SBS40,SBS44 10 cosine sim = 0.98228
1890
965
4479
838
1984
2364
counts
988
741
494
247
0
12519
9389
6260
3130
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS5,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS44 9 cosine sim = 0.98226
1890
926
4509
820
1989
2383
counts
988
741
494
247
0
12519
9389
6260
3130
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS5,SBS17b,SBS18,SBS21,SBS40,SBS44 8 cosine sim = 0.98225
1886
936
4545
825
1947
2381
counts
988
741
494
247
0
12519
9389
6260
3130
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS5,SBS17b,SBS18,SBS26,SBS40,SBS44 9 cosine sim = 0.98224
1891
954
4480
842
1983
2372
counts
988
741
494
247
0
12519
9389
6260
3130
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS5,SBS17b,SBS18,SBS21,SBS40,SBS44 9 cosine sim = 0.98223
1884
959
4529
844
1937
2367
counts
988
741
494
247
0
12519
9389
6260
3130
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS5
SBS3
SBS2
SBS1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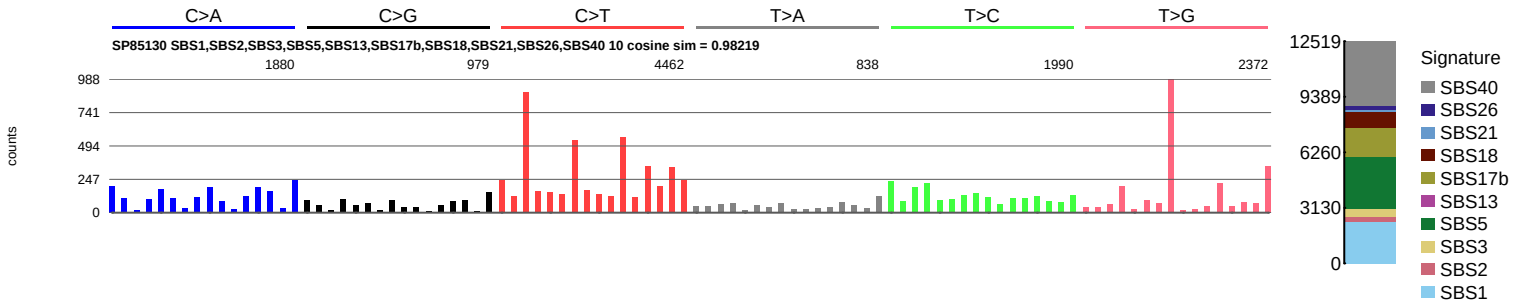
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS5,SBS13,SBS17b,SBS18,SBS40,SBS41 8 cosine sim = 0.98219
1845
947
4511
906
1899
2411
counts
984
738
492
246
0
12519
9389
6260
3130
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS13
SBS5
SBS2
SBS1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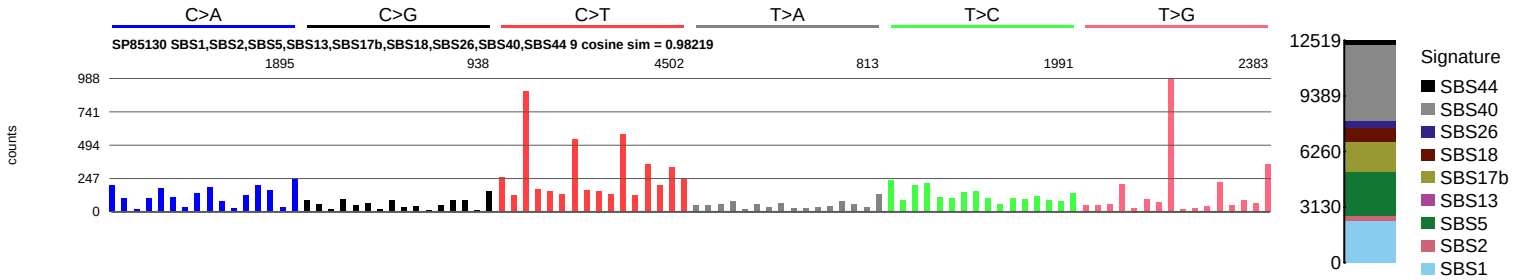
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS18,SBS21,SBS26,SBS40 10 cosine sim = 0.98219
1880
979
4462
838
1990
2372
counts
988
741
494
247
0
12519
9389
6260
3130
0
Signature
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS13
SBS5
SBS3
SBS2
SBS1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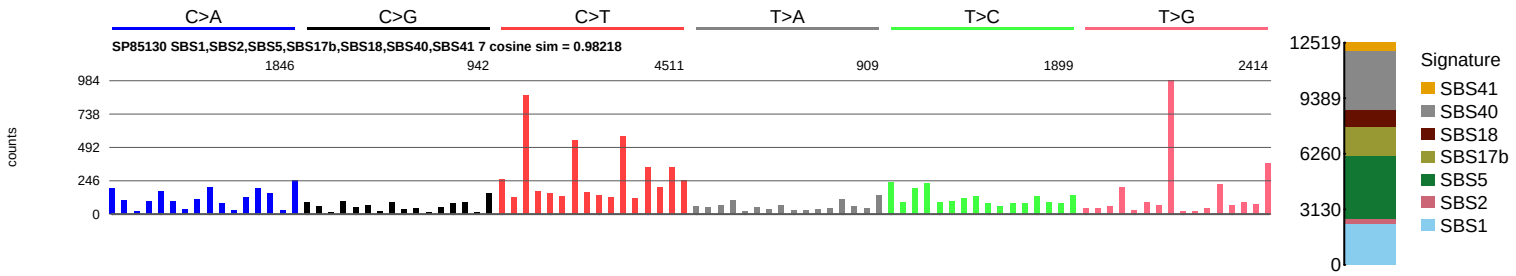
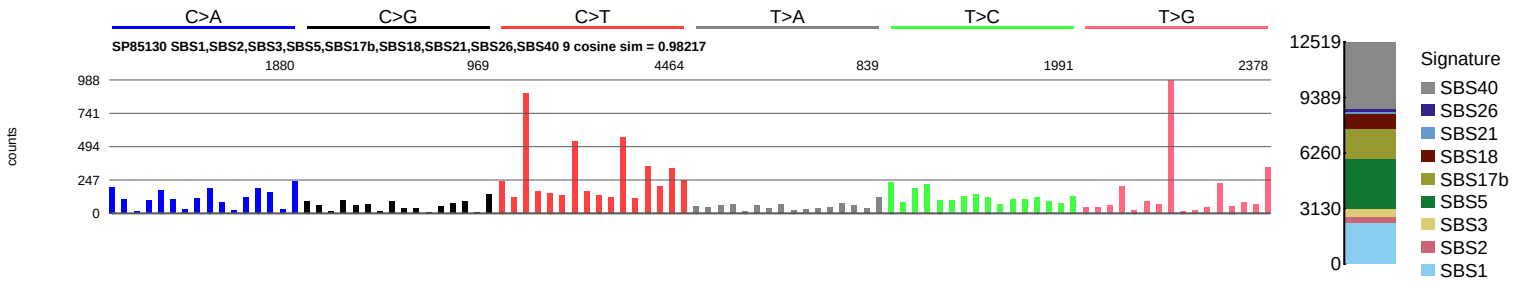
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS5,SBS13,SBS17b,SBS18,SBS26,SBS40,SBS44 9 cosine sim = 0.98219
1895
938
4502
813
1991
2383
counts
988
741
494
247
0
12519
9389
6260
3130
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS5,SBS17b,SBS18,SBS40,SBS41 7 cosine sim = 0.98218
1846
942
4511
909
1899
2414
counts
984
738
492
246
0
12519
9389
6260
3130
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS5
SBS2
SBS1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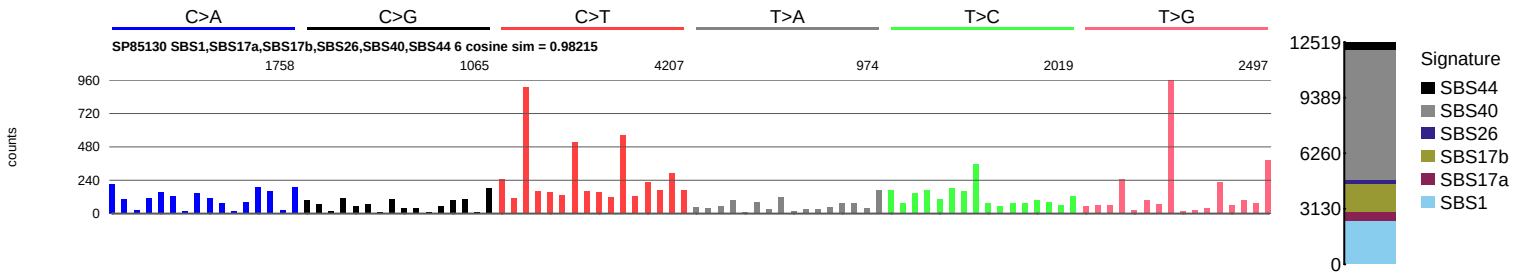
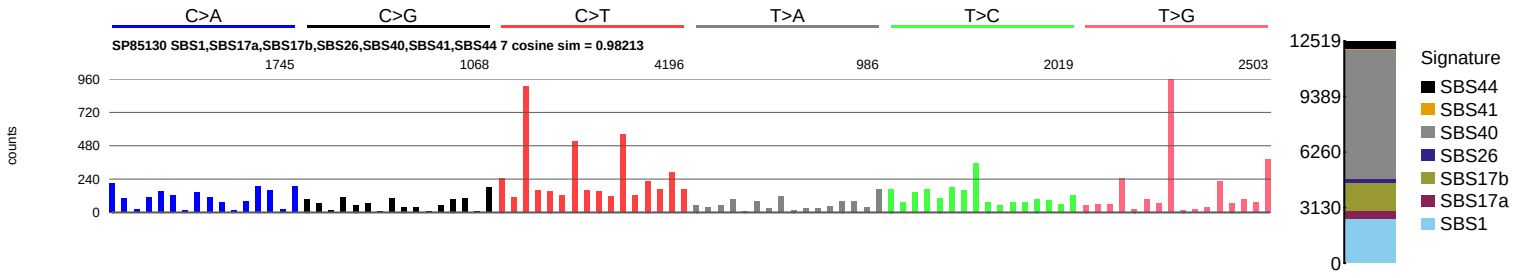
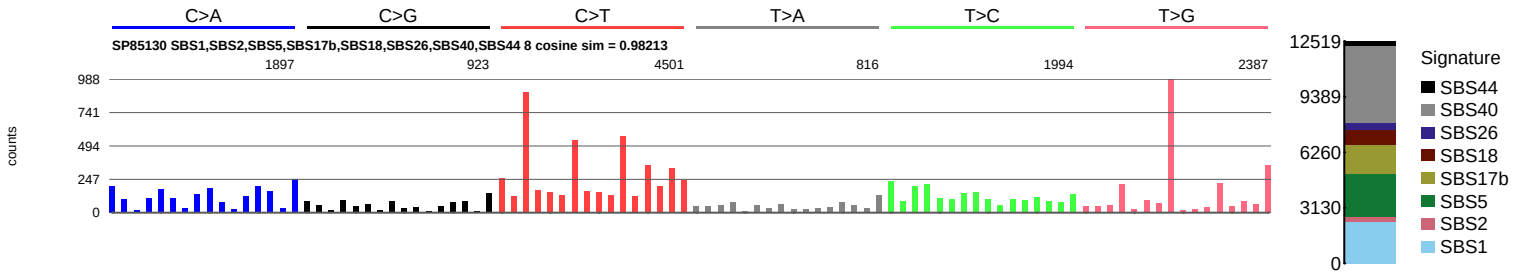
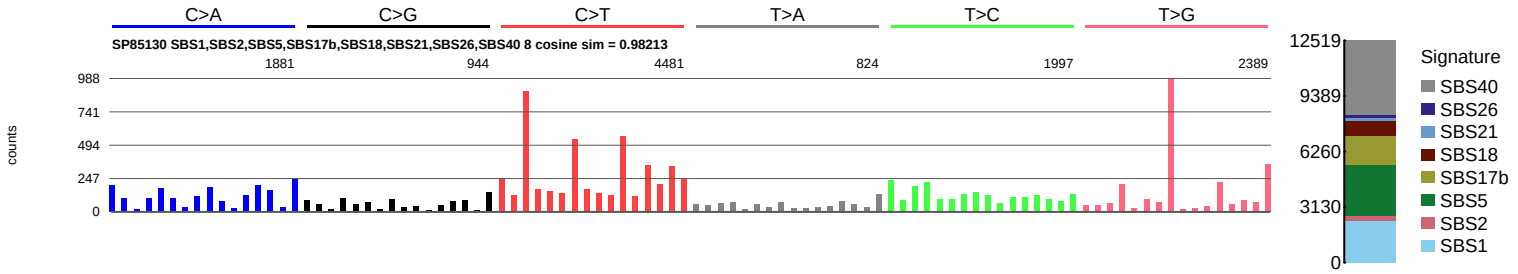
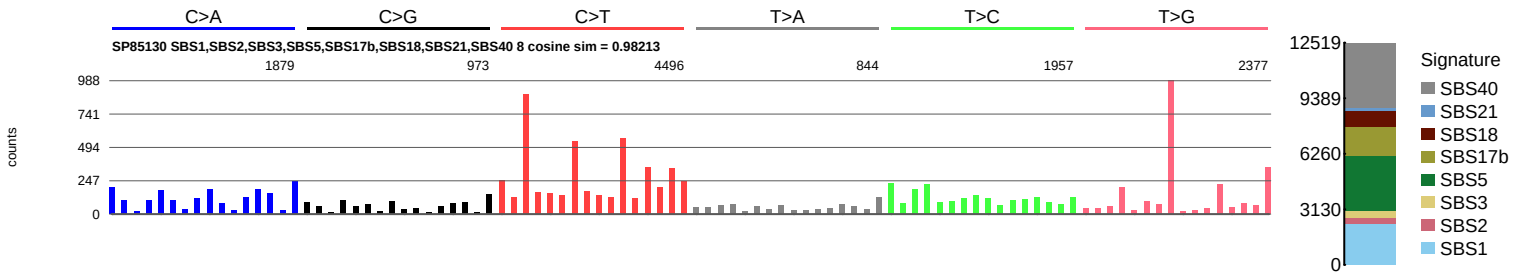
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS5,SBS17b,SBS18,SBS21,SBS26,SBS40 9 cosine sim = 0.98217
1880
969
4464
839
1991
2378
counts
988
741
494
247
0
12519
9389
6260
3130
0
Signature
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS5
SBS3
SBS2
SBS1

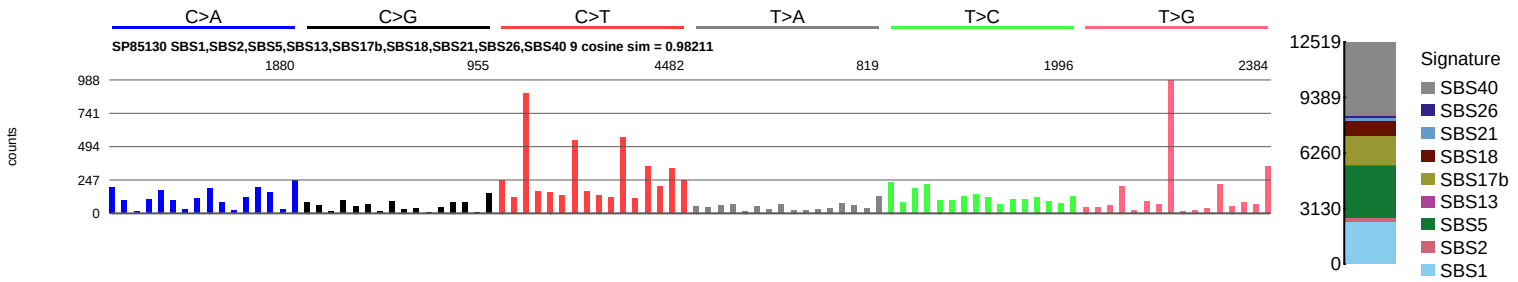

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS17a,SBS17b,SBS26,SBS40,SBS44 6 cosine sim = 0.98215
1758
1065
4207
974
2019
2497
counts
960
720
480
240
0
12519
9389
6260
3130
0
Signature
SBS44
SBS40
SBS26
SBS17b
SBS17a
SBS1

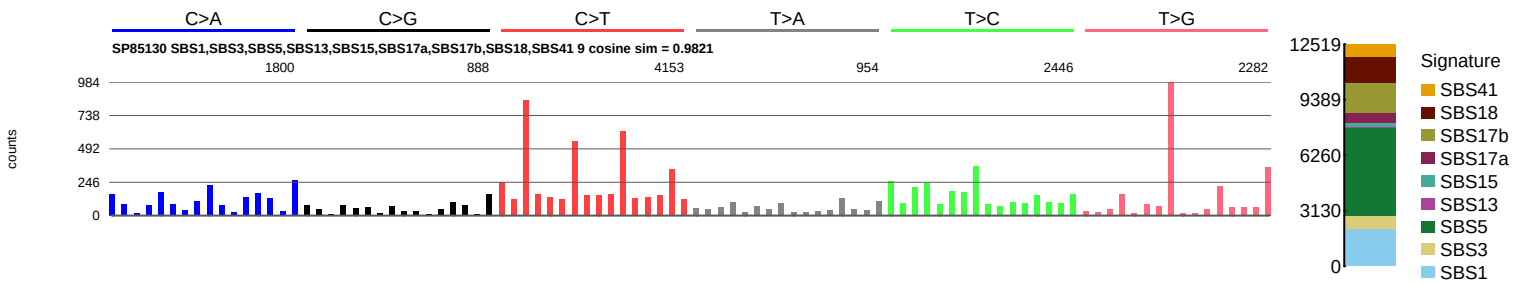

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS17a,SBS17b,SBS26,SBS40,SBS41,SBS44 7 cosine sim = 0.98213
1745
1068
4196
986
2019
2503
counts
960
720
480
240
0
12519
9389
6260
3130
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS1

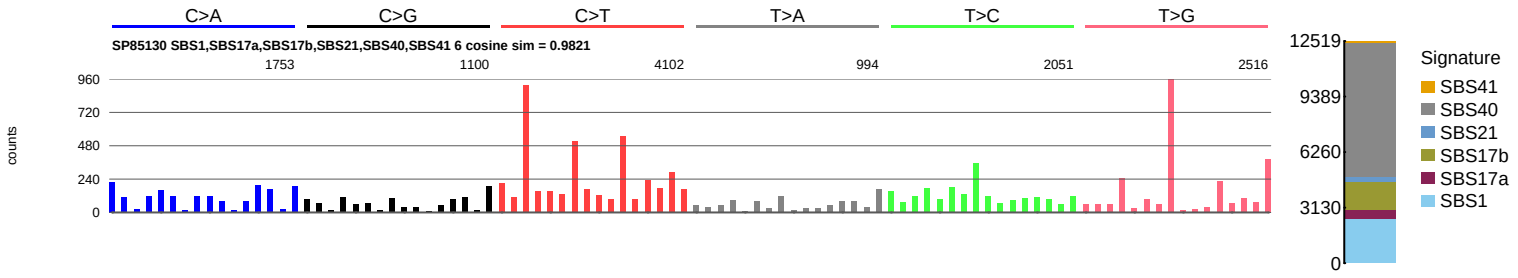

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS5,SBS17b,SBS18,SBS26,SBS40,SBS44 8 cosine sim = 0.98213
1897
923
4501
816
1994
2387
counts
988
741
494
247
0
12519
9389
6260
3130
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS5
SBS2
SBS1

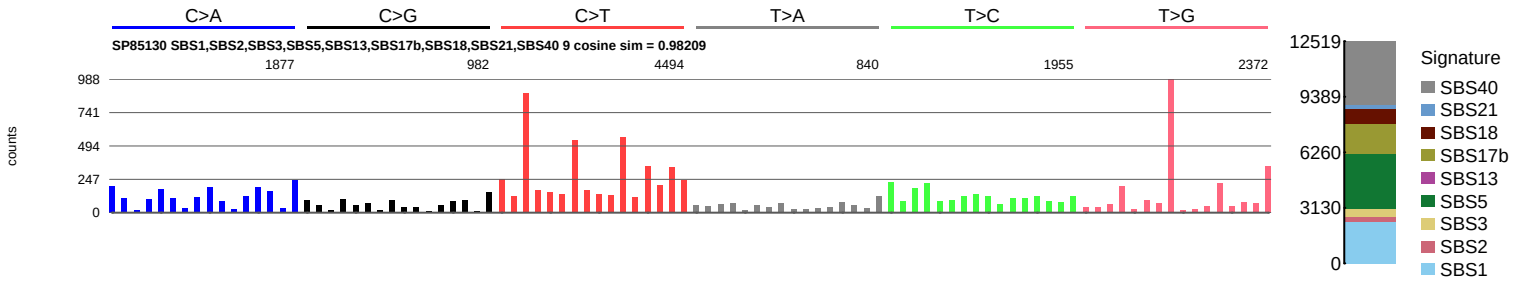

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS5,SBS17b,SBS18,SBS21,SBS26,SBS40 8 cosine sim = 0.98213
1881
944
4481
824
1997
2389
counts
988
741
494
247
0
12519
9389
6260
3130
0
Signature
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS5
SBS2
SBS1

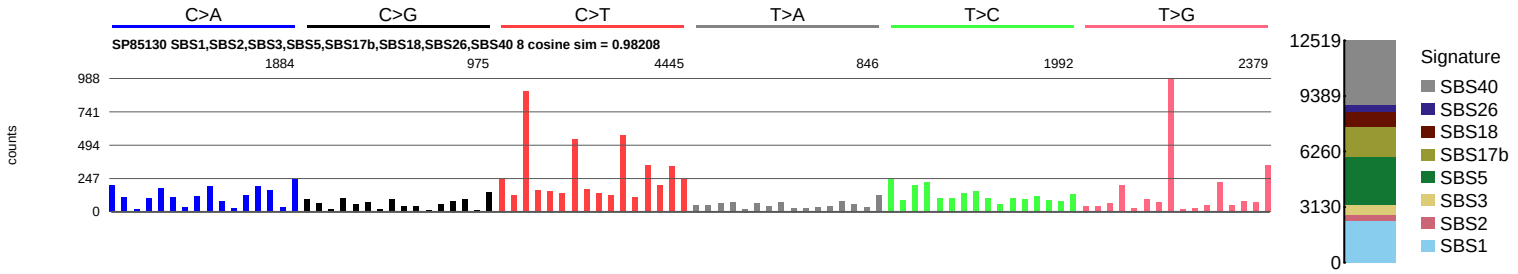

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS5,SBS17b,SBS18,SBS21,SBS40 8 cosine sim = 0.98213
1879
973
4496
844
1957
2377
counts
988
741
494
247
0
12519
9389
6260
3130
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS5
SBS3
SBS2
SBS1

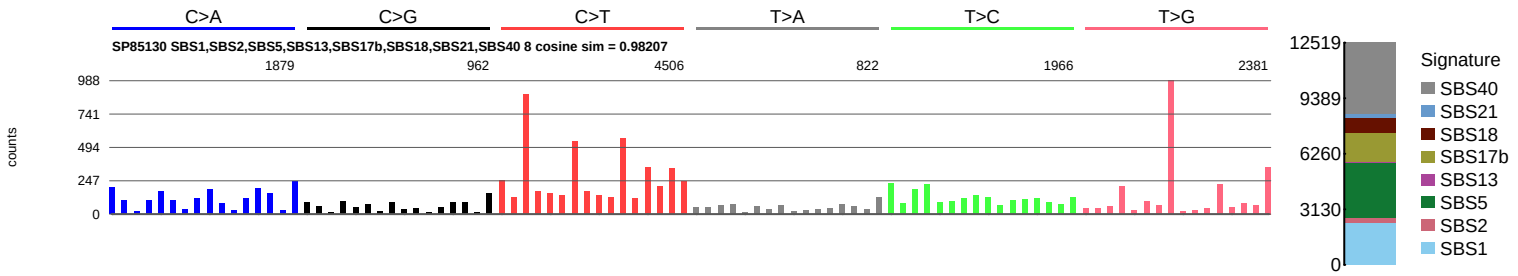

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS5,SBS13,SBS17b,SBS18,SBS21,SBS26,SBS40 9 cosine sim = 0.98211
1880
955
4482
819
1996
2384
counts
988
741
494
247
0
12519
9389
6260
3130
0
Signature
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS13
SBS5
SBS2
SBS1

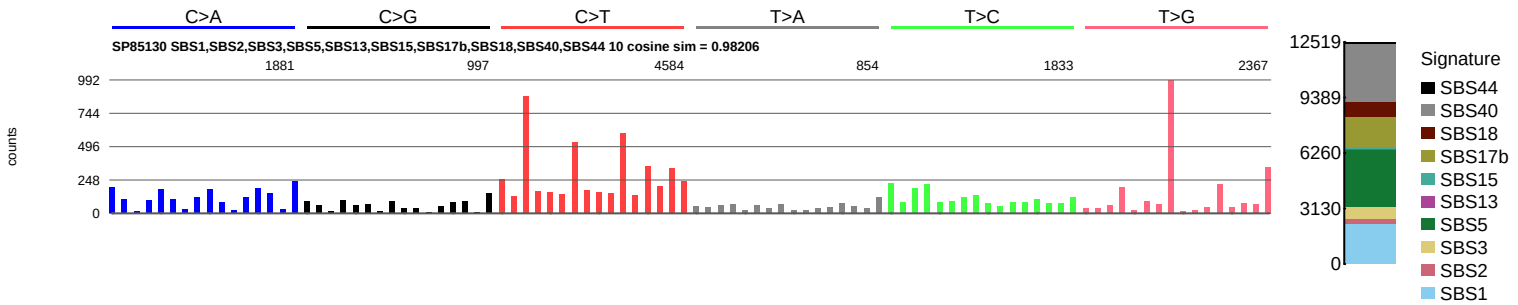

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS41 9 cosine sim = 0.9821
1800
888
4153
954
2446
2282
counts
984
738
492
246
0
12519
9389
6260
3130
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS1

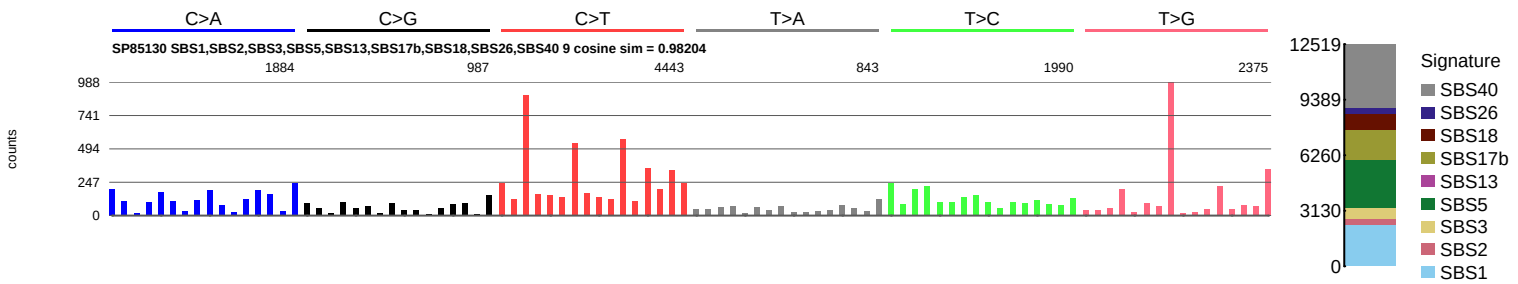

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS17a,SBS17b,SBS21,SBS40,SBS41 6 cosine sim = 0.9821
1753
1100
4102
994
2051
2516
counts
960
720
480
240
0
12519
9389
6260
3130
0
Signature
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS1

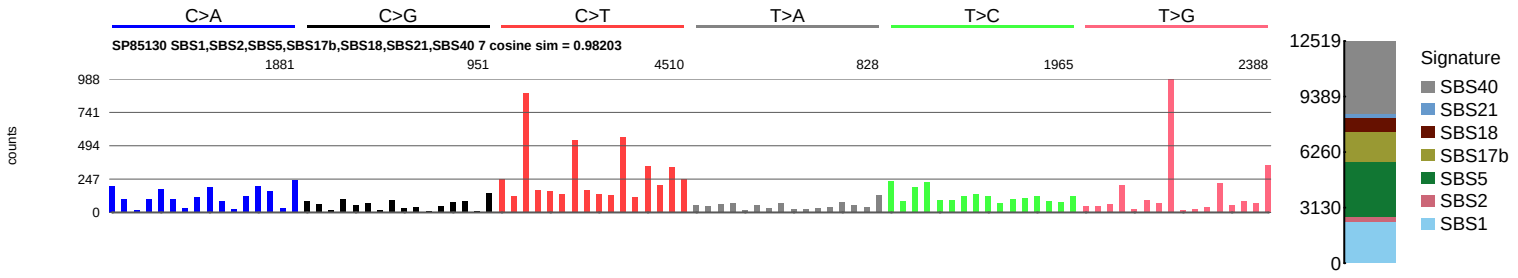

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS18,SBS21,SBS40 9 cosine sim = 0.98209
1877
982
4494
840
1955
2372
counts
988
741
494
247
0
12519
9389
6260
3130
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS13
SBS5
SBS3
SBS2
SBS1

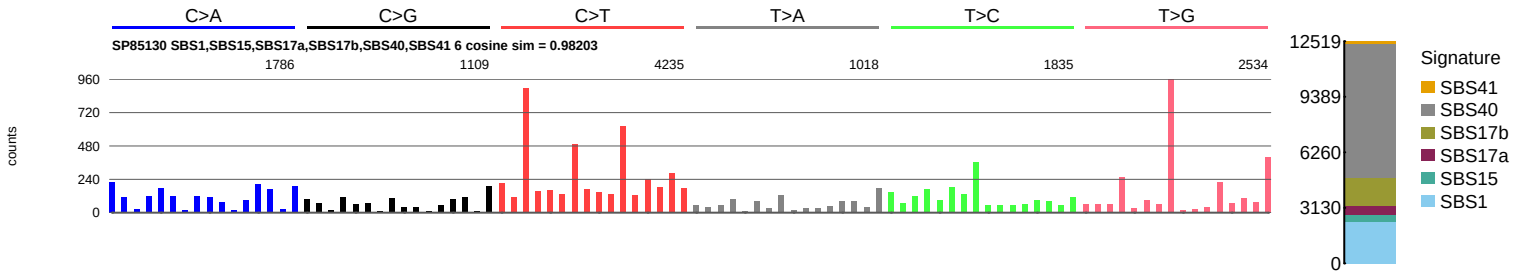

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS5,SBS17b,SBS18,SBS26,SBS40 8 cosine sim = 0.98208
1884
975
4445
846
1992
2379
counts
988
741
494
247
0
12519
9389
6260
3130
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS5
SBS3
SBS2
SBS1

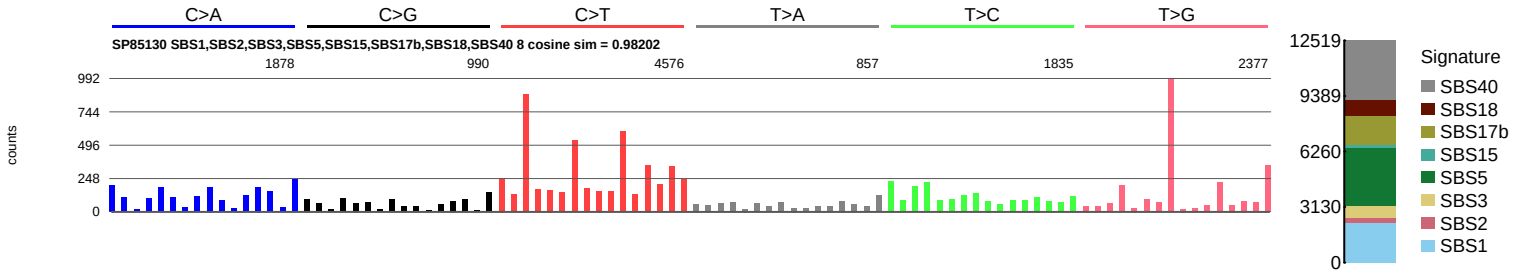

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS5,SBS13,SBS17b,SBS18,SBS21,SBS40 8 cosine sim = 0.98207
1879
962
4506
822
1966
2381
counts
988
741
494
247
0
12519
9389
6260
3130
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS13
SBS5
SBS2
SBS1

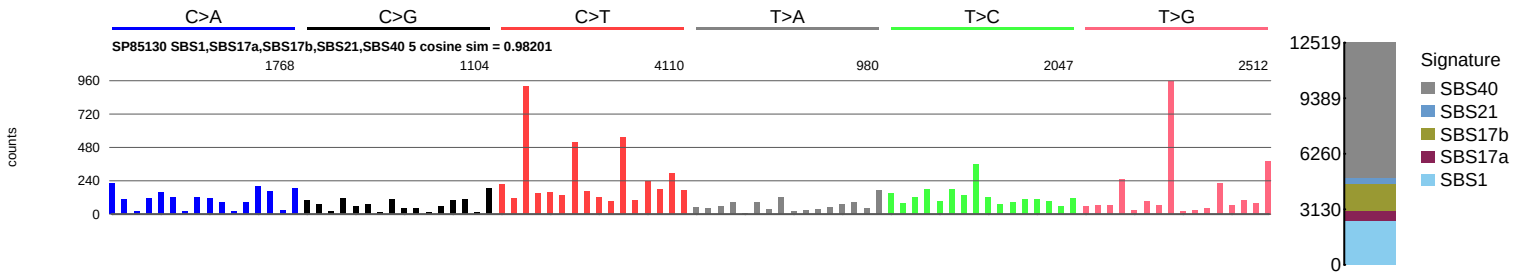

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS40,SBS44 10 cosine sim = 0.98206
1881
997
4584
854
1833
2367
counts
992
744
496
248
0
12519
9389
6260
3130
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1

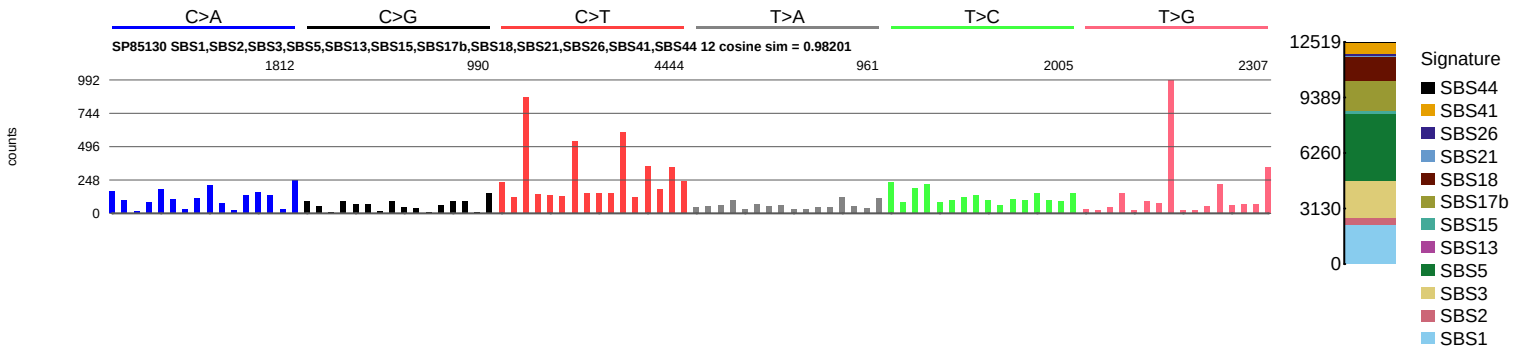

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS18,SBS26,SBS40 9 cosine sim = 0.98204
1884
987
4443
843
1990
2375
counts
988
741
494
247
0
12519
9389
6260
3130
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS5,SBS17b,SBS18,SBS21,SBS40 7 cosine sim = 0.98203
1881
951
4510
828
1965
2388
988
741
494
247
0
counts
12519
9389
6260
3130
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS15,SBS17a,SBS17b,SBS40,SBS41 6 cosine sim = 0.98203
1786
1109
4235
1018
1835
2534
960
720
480
240
0
counts
12519
9389
6260
3130
0
Signature
SBS41
SBS40
SBS17b
SBS17a
SBS15
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17b,SBS18,SBS40 8 cosine sim = 0.98202
1878
990
4576
857
1835
2377
992
744
496
248
0
counts
12519
9389
6260
3130
0
Signature
SBS40
SBS18
SBS17b
SBS15
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS17a,SBS17b,SBS21,SBS40 5 cosine sim = 0.98201
1768
1104
4110
980
2047
2512
960
720
480
240
0
counts
12519
9389
6260
3130
0
Signature
SBS40
SBS21
SBS17b
SBS17a
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS26,SBS41,SBS44 12 cosine sim = 0.98201
1812
990
4444
961
2005
2307
992
744
496
248
0
counts
12519
9389
6260
3130
0
Signature
SBS44
SBS41
SBS26
SBS21
SBS18
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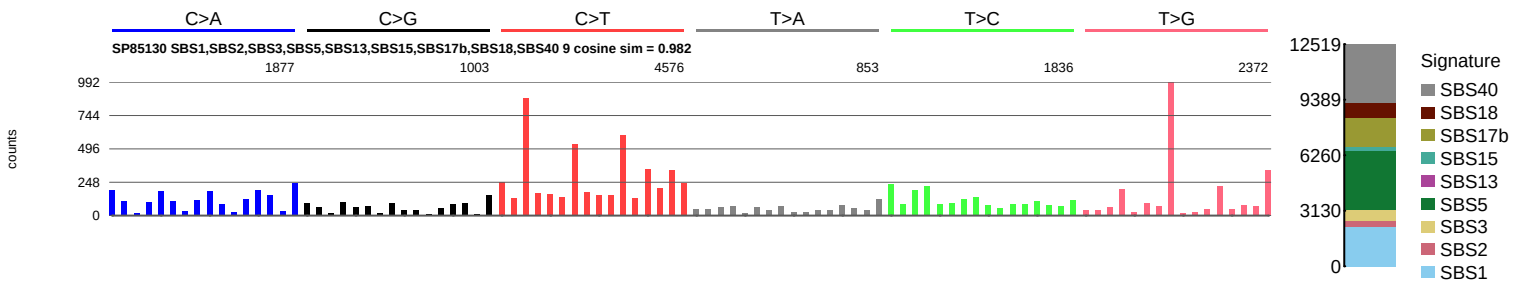
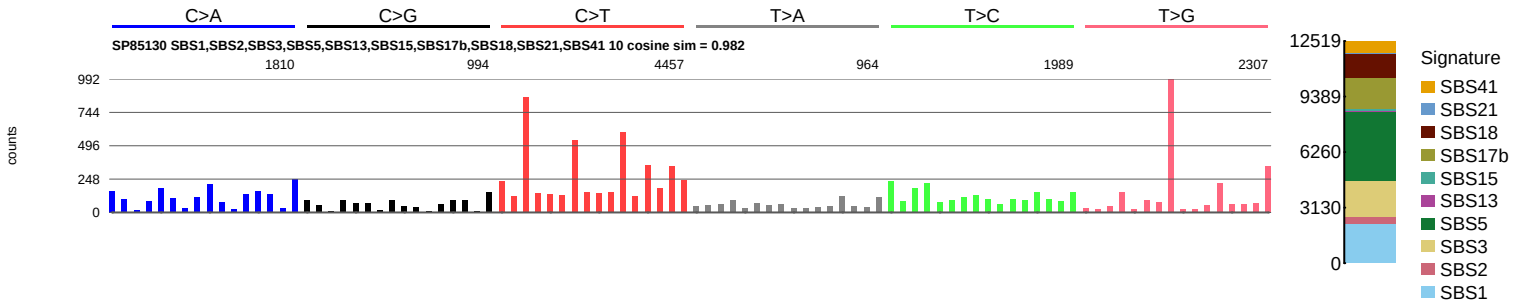
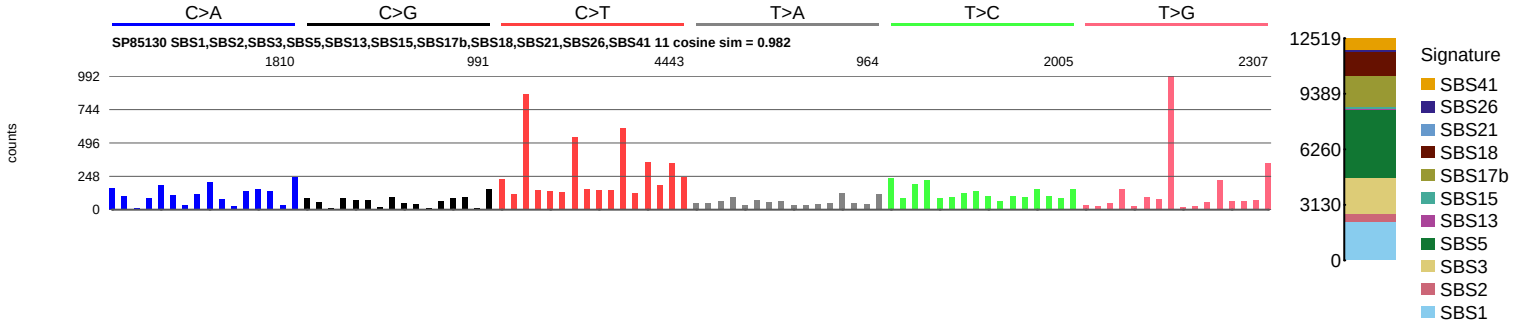
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS40 9 cosine sim = 0.982
1877
1003
4576
853
1836
2372
992
744
496
248
0
counts
12519
9389
6260
3130
0
Signature
SBS40
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS41 10 cosine sim = 0.982
1810
994
4457
964
1989
2307
counts
992
744
496
248
0
12519
9389
6260
3130
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1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C>G
C>T
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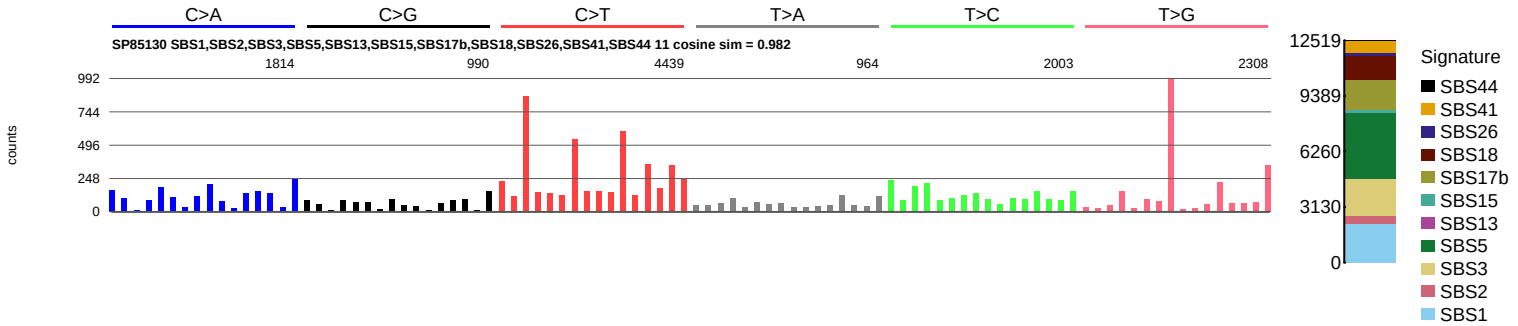
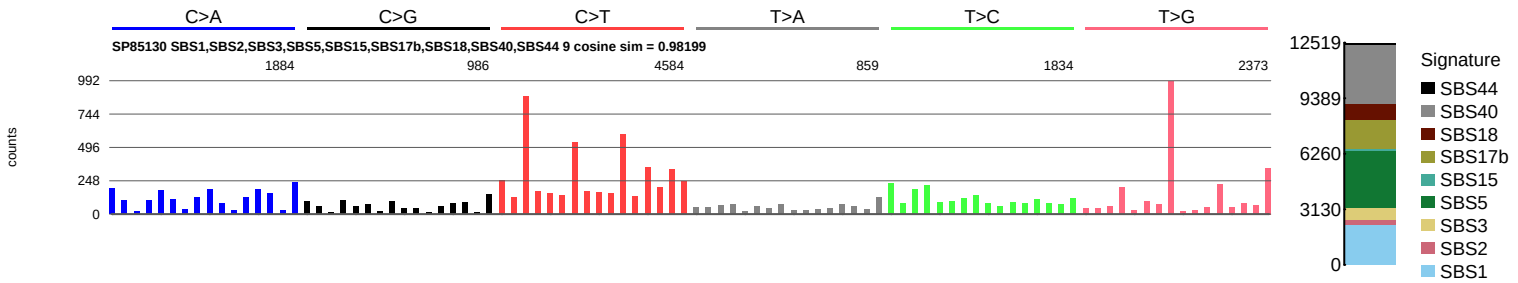
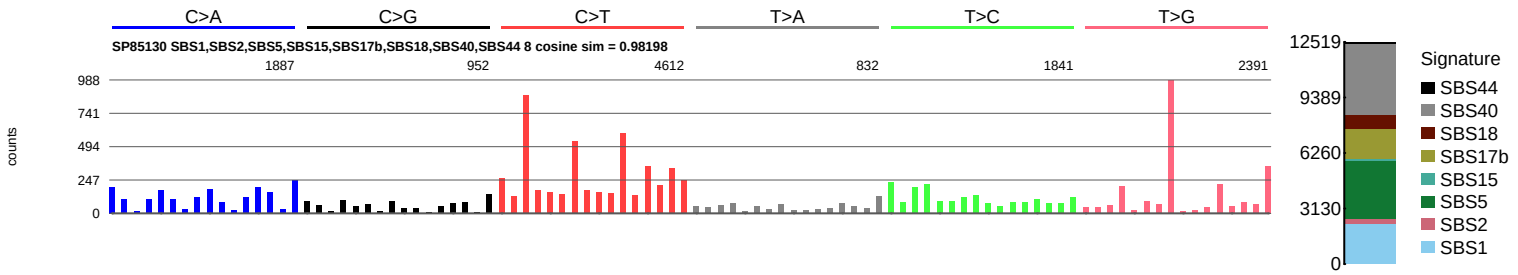
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS26,SBS41 11 cosine sim = 0.982
1810
991
4443
964
2005
2307
counts
992
744
496
248
0
12519
9389
6260
3130
0
Signature
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS26,SBS41,SBS44 11 cosine sim = 0.982
1814
990
4439
964
2003
2308
counts
992
744
496
248
0
12519
9389
6260
3130
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17b,SBS18,SBS40,SBS44 9 cosine sim = 0.98199
1884
986
4584
859
1834
2373
counts
992
744
496
248
0
12519
9389
6260
3130
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS15
SBS5
SBS3
SBS2
SBS1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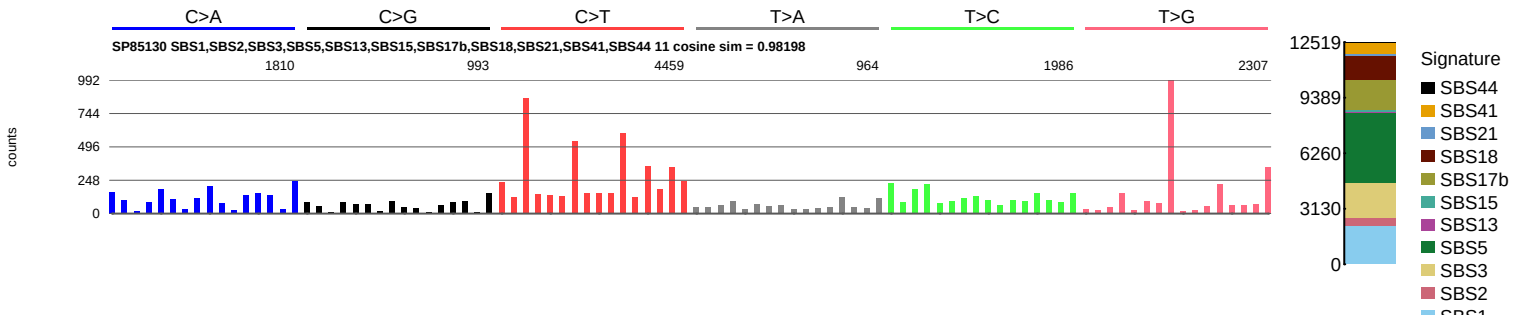
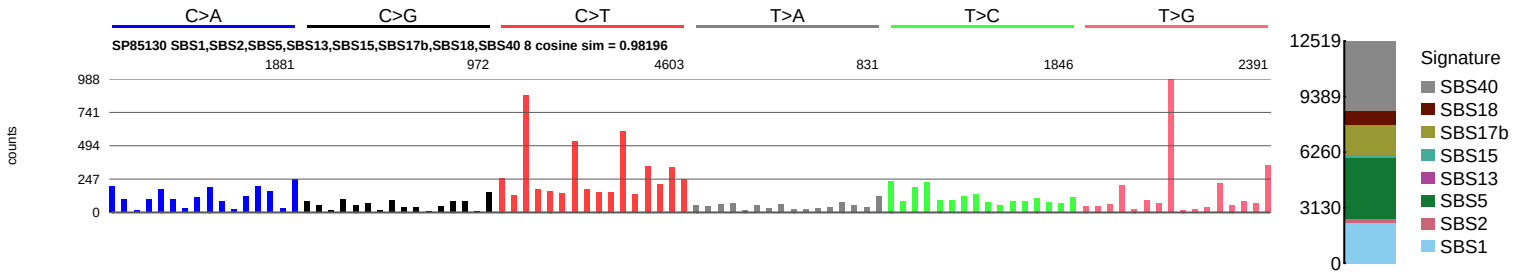
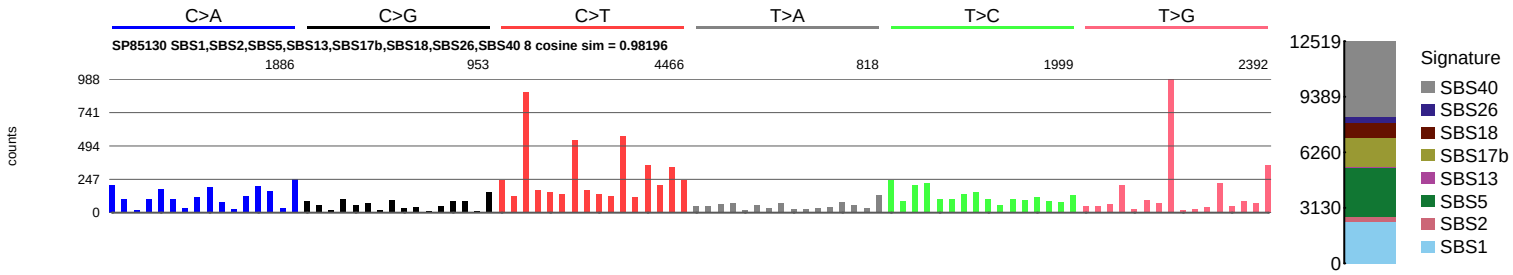
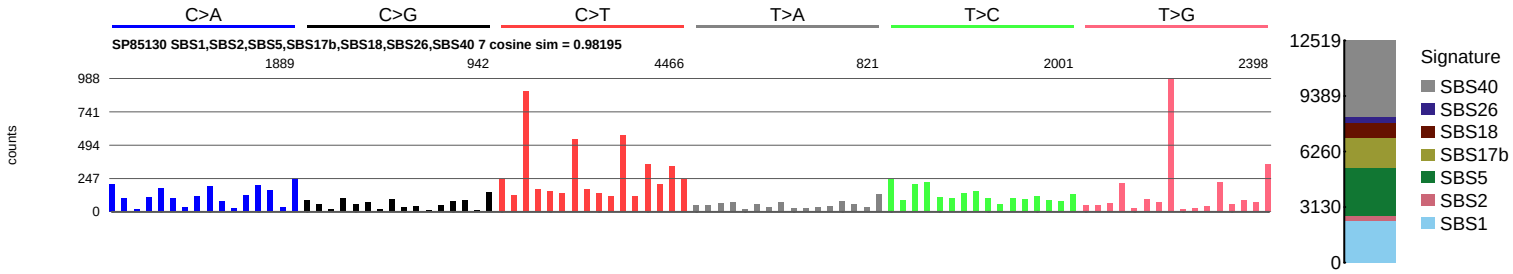
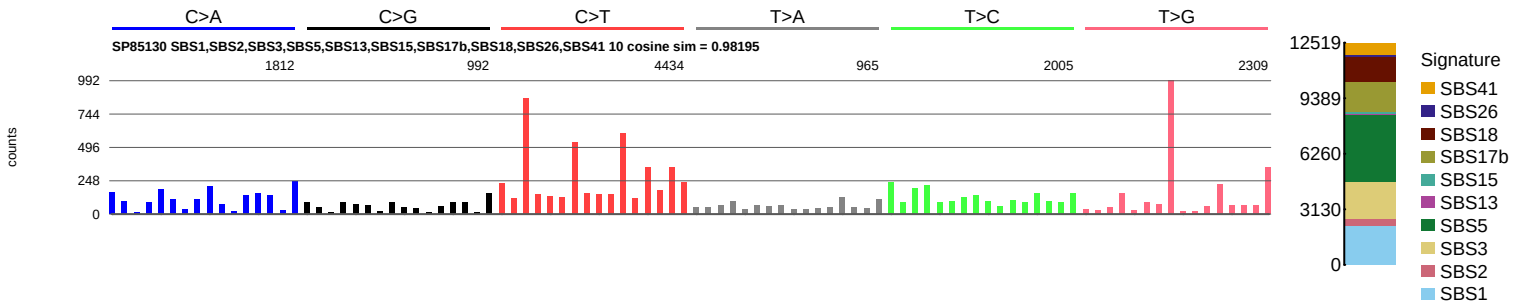
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS5,SBS15,SBS17b,SBS18,SBS40,SBS44 8 cosine sim = 0.98198
1887
952
4612
832
1841
2391
counts
988
741
494
247
0
12519
9389
6260
3130
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS15
SBS5
SBS2
SBS1

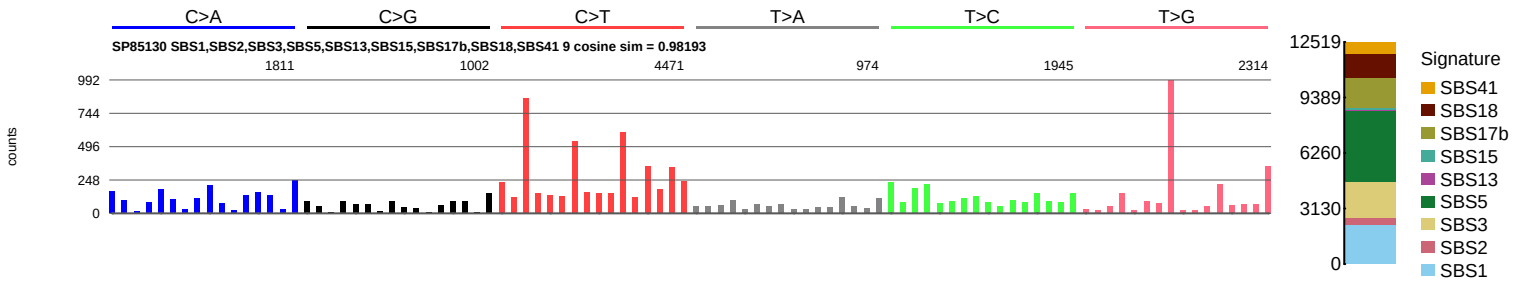

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS41,SBS44 11 cosine sim = 0.98198
1810
993
4459
964
1986
2307
counts
992
744
496
248
0
12519
9389
6260
3130
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS40 8 cosine sim = 0.98196
1881
972
4603
831
1846
2391
counts
988
741
494
247
0
12519
9389
6260
3130
0
Signature
SBS40
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS5,SBS13,SBS17b,SBS18,SBS26,SBS40 8 cosine sim = 0.98196
1886
953
4466
818
1999
2392
counts
988
741
494
247
0
12519
9389
6260
3130
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS5,SBS17b,SBS18,SBS26,SBS40 7 cosine sim = 0.98195
1889
942
4466
821
2001
2398
counts
988
741
494
247
0
12519
9389
6260
3130
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS26,SBS41 10 cosine sim = 0.98195
1812
992
4434
965
2005
2309
counts
992
744
496
248
0
12519
9389
6260
3130
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS41 9 cosine sim = 0.98193
1811
1002
4471
974
1945
2314
counts
992
744
496
248
0
12519
9389
6260
3130
0
Signature
SBS41
SBS18
SBS17b
SBS15
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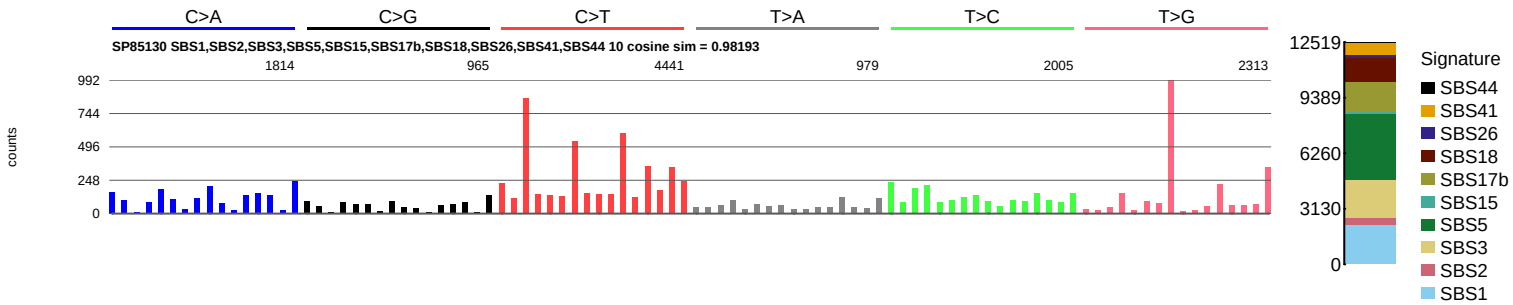
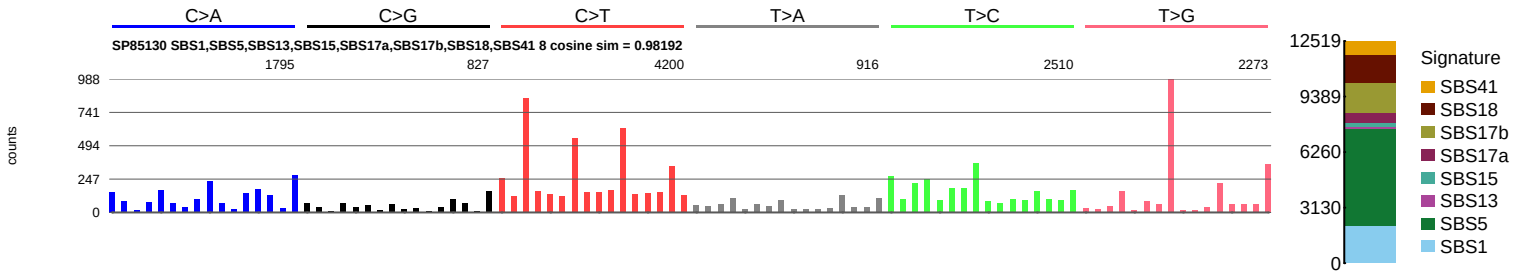
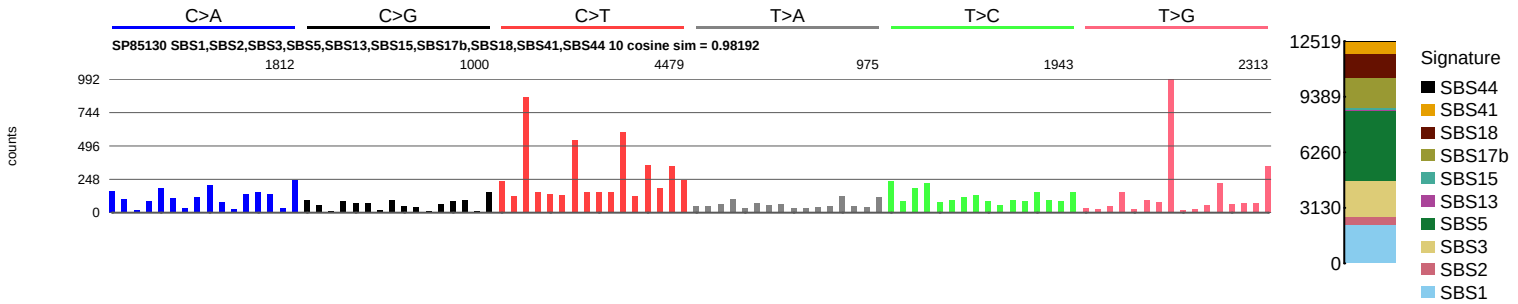
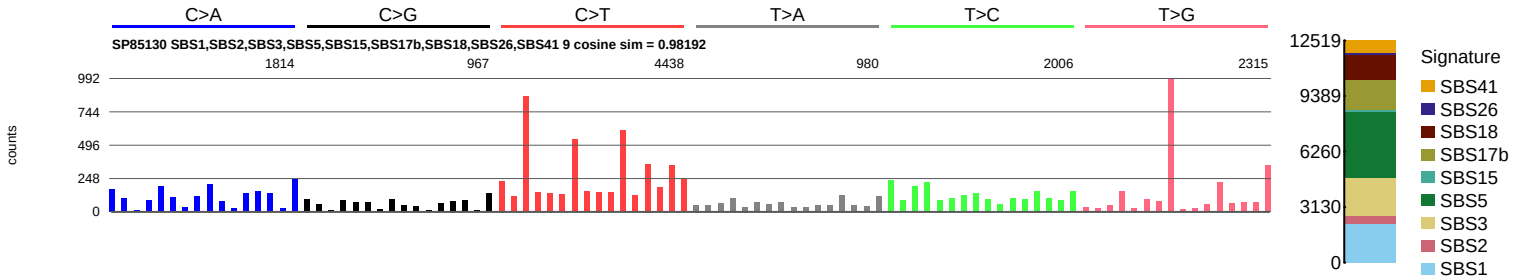
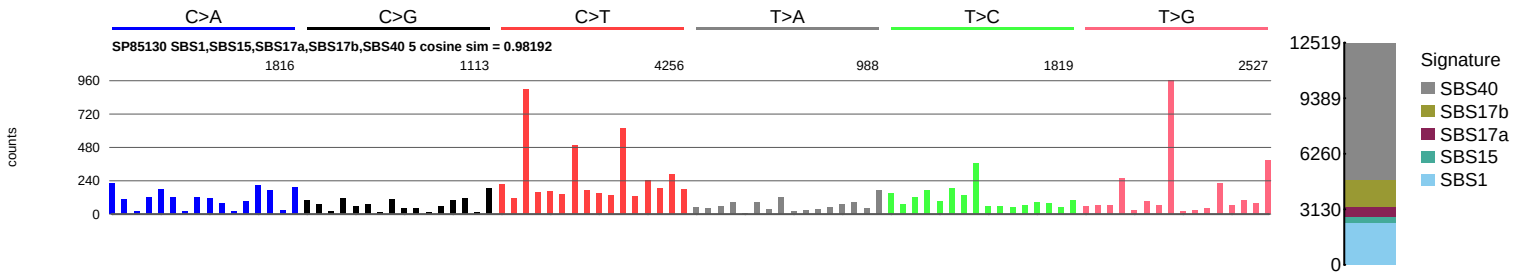
SBS3
SBS2
SBS1

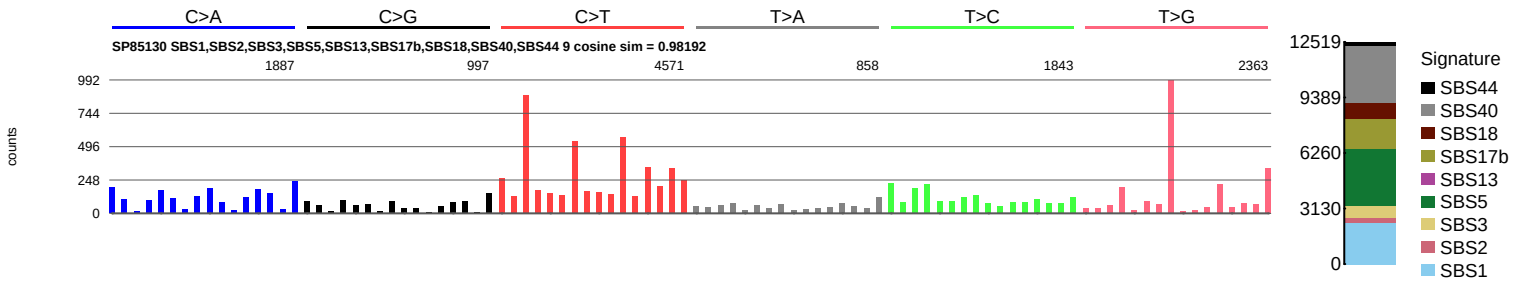

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17b,SBS18,SBS26,SBS41,SBS44 10 cosine sim = 0.98193
1814
965
4441
979
2005
2313
counts
992
744
496
248
0
12519
9389
6260
3130
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS15
SBS5
SBS3
SBS2
SBS1

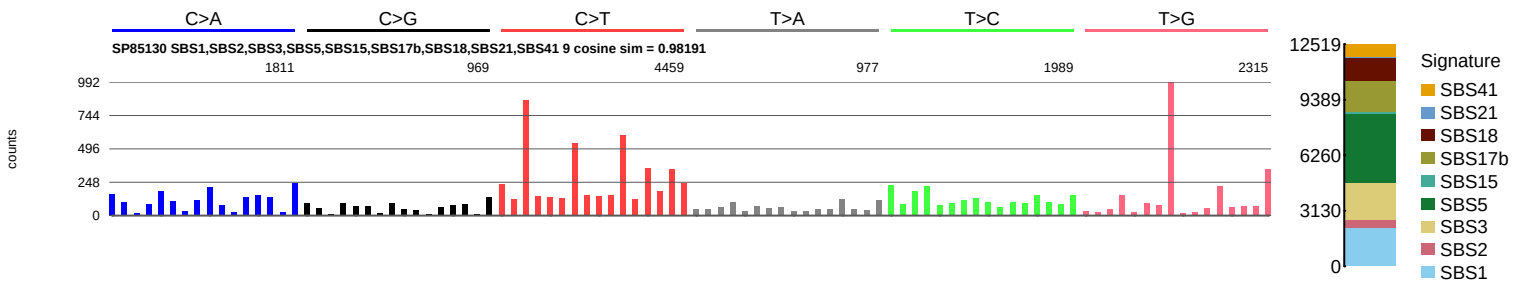

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS41 8 cosine sim = 0.98192
1795
827
4200
916
2510
2273
counts
988
741
494
247
0
12519
9389
6260
3130
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS1

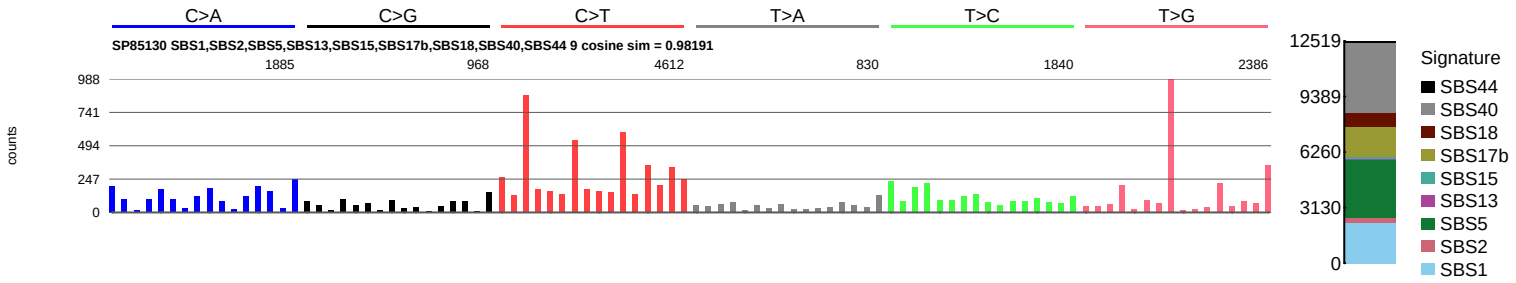

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS41,SBS44 10 cosine sim = 0.98192
1812
1000
4479
975
1943
2313
counts
992
744
496
248
0
12519
9389
6260
3130
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1

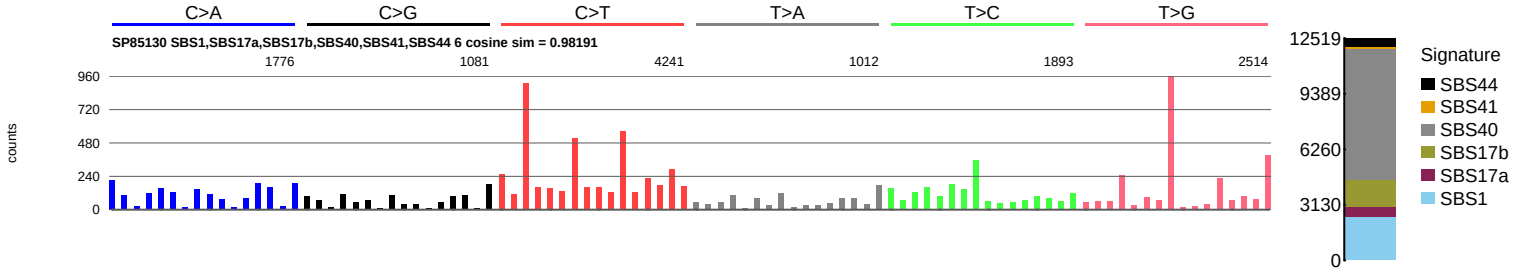

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17b,SBS18,SBS26,SBS41 9 cosine sim = 0.98192
1814
967
4438
980
2006
2315
counts
992
744
496
248
0
12519
9389
6260
3130
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS15
SBS5
SBS3
SBS2
SBS1

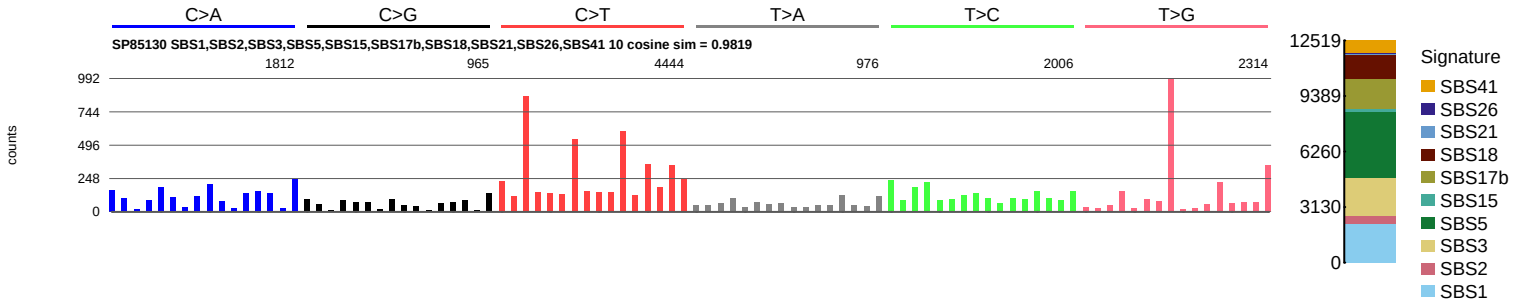

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS15,SBS17a,SBS17b,SBS40 5 cosine sim = 0.98192
1816
1113
4256
988
1819
2527
counts
960
720
480
240
0
12519
9389
6260
3130
0
Signature
SBS40
SBS17b
SBS17a
SBS15
SBS1

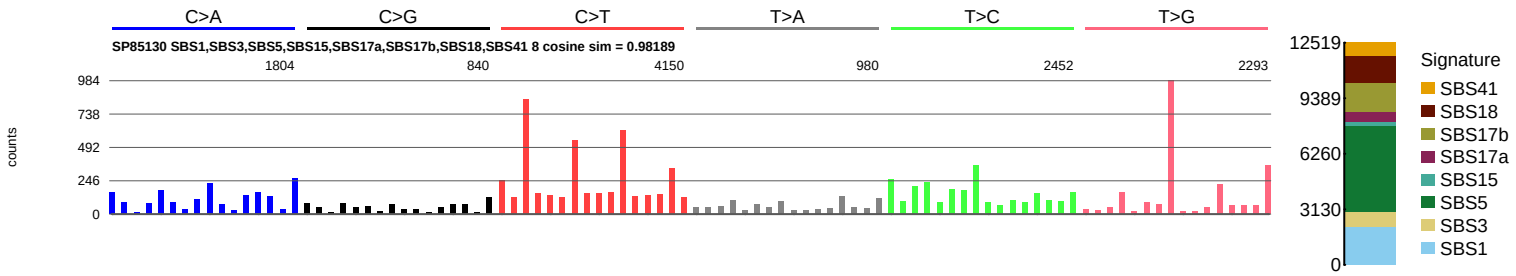

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS18,SBS40,SBS44 9 cosine sim = 0.98192
1887
997
4571
858
1843
2363
counts
992
744
496
248
0
12519
9389
6260
3130
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS13
SBS5
SBS3
SBS2
SBS1

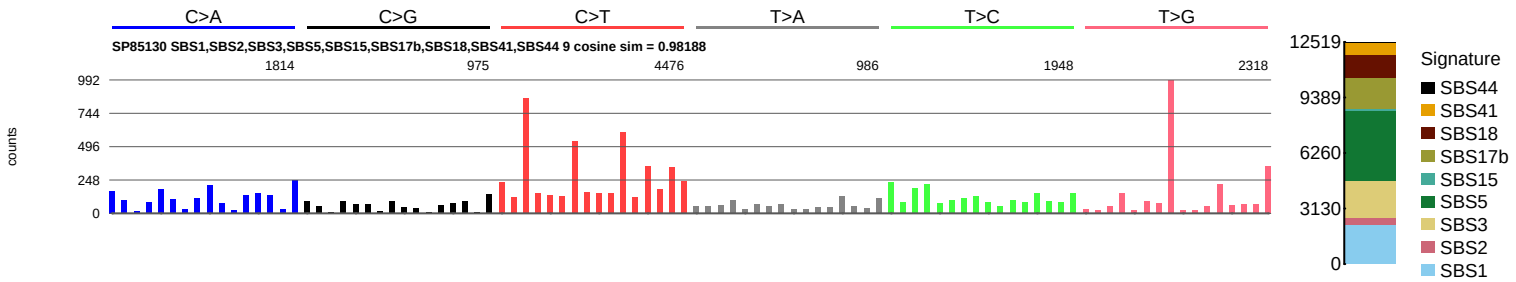

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17b,SBS18,SBS21,SBS41 9 cosine sim = 0.98191
1811
969
4459
977
1989
2315
counts
992
744
496
248
0
12519
9389
6260
3130
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS15
SBS5
SBS3
SBS2
SBS1

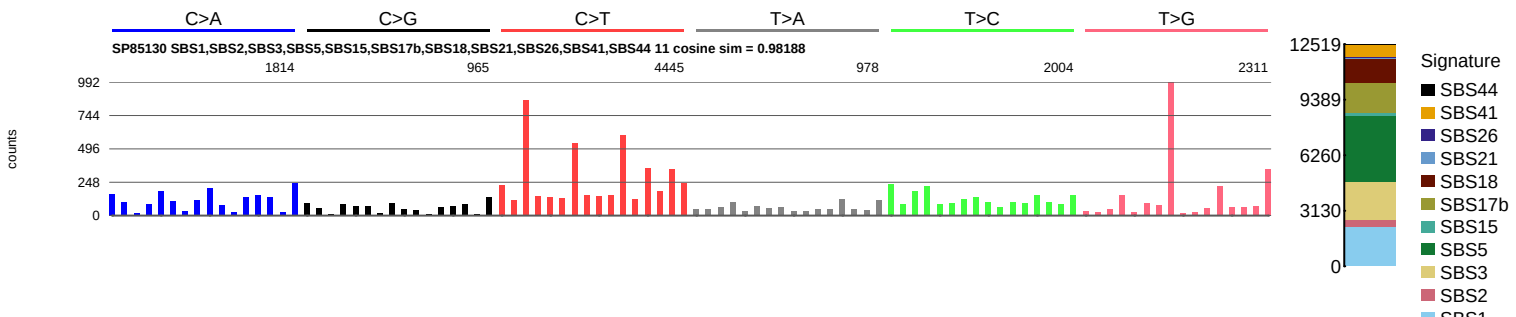

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS40,SBS44 9 cosine sim = 0.98191
1885
968
4612
830
1840
2386
988
741
494
247
0
counts
12519
9389
6260
3130
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS2
SBS1

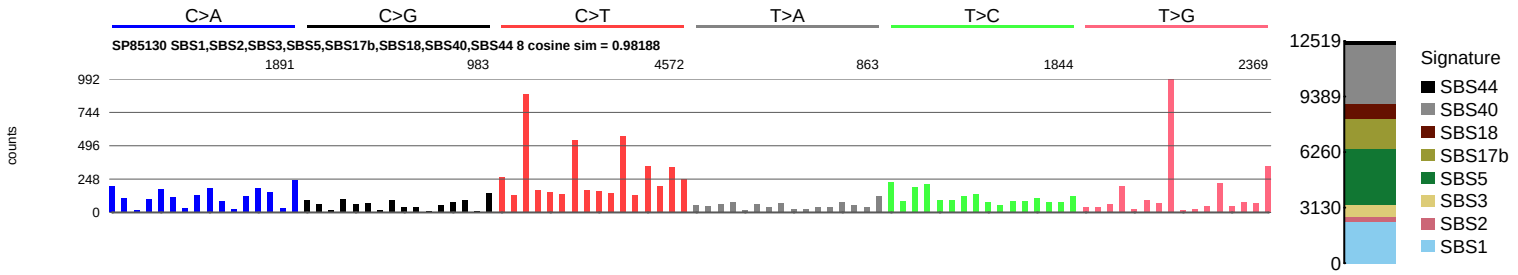

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS17a,SBS17b,SBS40,SBS41,SBS44 6 cosine sim = 0.98191
1776
1081
4241
1012
1893
2514
960
720
480
240
0
counts
12519
9389
6260
3130
0
Signature
SBS44
SBS41
SBS40
SBS17b
SBS17a
SBS1

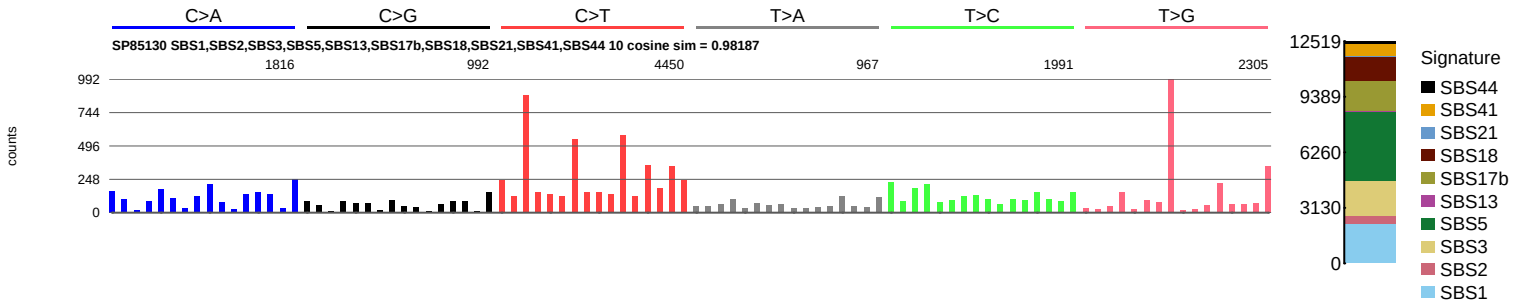

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17b,SBS18,SBS21,SBS26,SBS41 10 cosine sim = 0.9819
1812
965
4444
976
2006
2314
992
744
496
248
0
counts
12519
9389
6260
3130
0
Signature
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS15
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS41 8 cosine sim = 0.98189
1804
840
4150
980
2452
2293
984
738
492
246
0
counts
12519
9389
6260
3130
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17b,SBS18,SBS41,SBS44 9 cosine sim = 0.98188
1814
975
4476
986
1948
2318
992
744
496
248
0
counts
12519
9389
6260
3130
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS15
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17b,SBS18,SBS21,SBS26,SBS41,SBS44 11 cosine sim = 0.98188
1814
965
4445
978
2004
2311
992
744
496
248
0
counts
12519
9389
6260
3130
0
Signature
SBS44
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS15
SBS5
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS5,SBS17b,SBS18,SBS40,SBS44 8 cosine sim = 0.98188
1891
983
4572
863
1844
2369
counts
992
744
496
248
0
12519
9389
6260
3130
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS5
SBS3
SBS2
SBS1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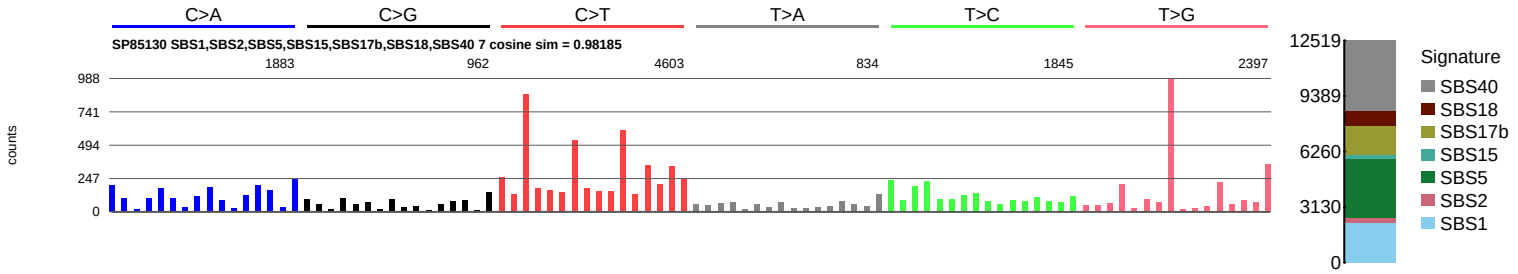
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS18,SBS21,SBS41,SBS44 10 cosine sim = 0.98187
1816
992
4450
967
1991
2305
counts
992
744
496
248
0
12519
9389
6260
3130
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS13
SBS5
SBS3
SBS2
SBS1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C>G
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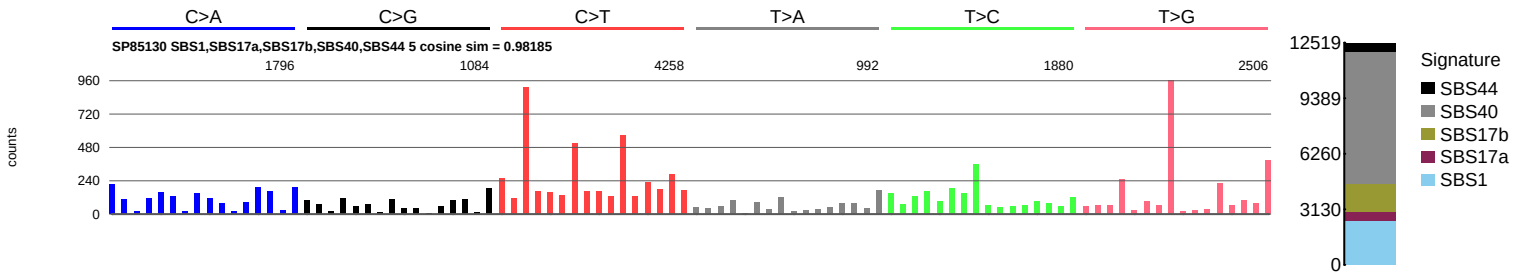
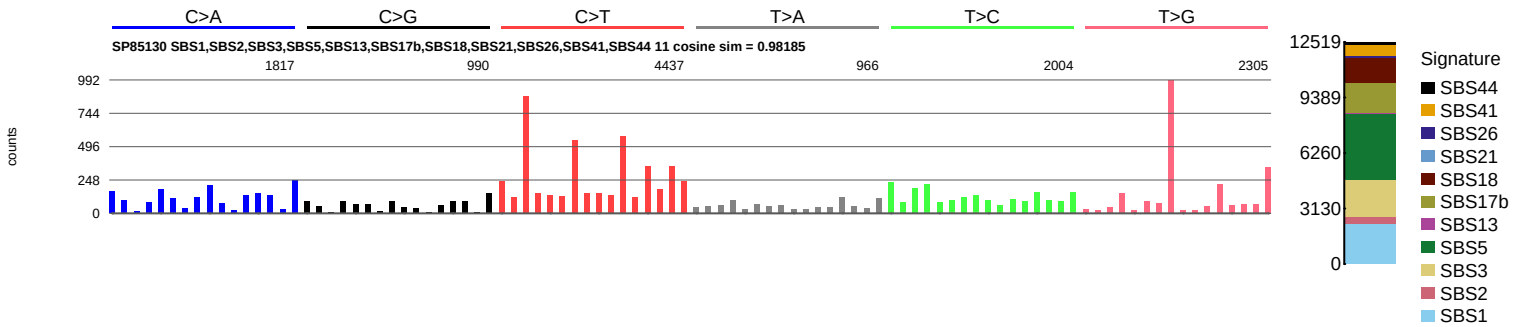
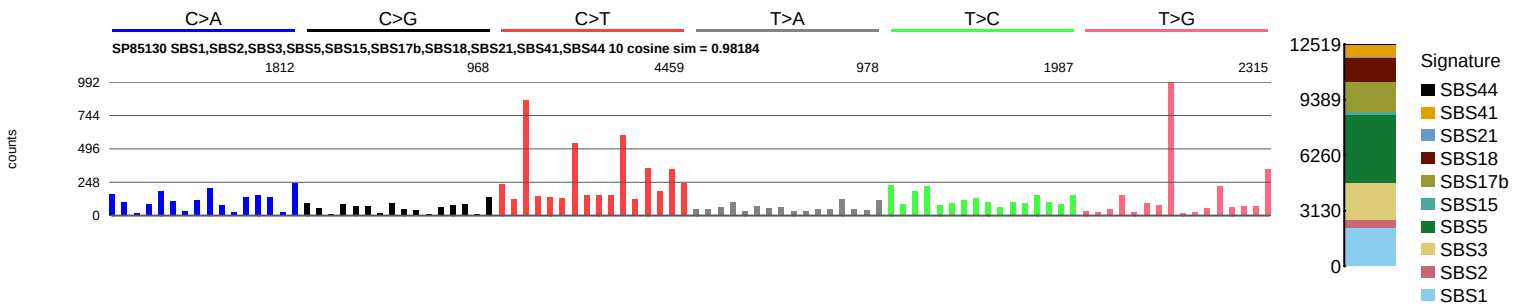
SP85130 SBS1,SBS2,SBS5,SBS15,SBS17b,SBS18,SBS40 7 cosine sim = 0.98185
1883
962
4603
834
1845
2397
counts
988
741
494
247
0
12519
9389
6260
3130
0
Signature
SBS40
SBS18
SBS17b
SBS15
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS17a,SBS17b,SBS40,SBS44 5 cosine sim = 0.98185
1796
1084
4258
992
1880
2506
counts
960
720
480
240
0
12519
9389
6260
3130
0
Signature
SBS44
SBS40
SBS17b
SBS17a
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS18,SBS21,SBS26,SBS41,SBS44 11 cosine sim = 0.98185
1817
990
4437
966
2004
2305
counts
992
744
496
248
0
12519
9389
6260
3130
0
Signature
SBS44
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS13
SBS5
SBS3
SBS2
SBS1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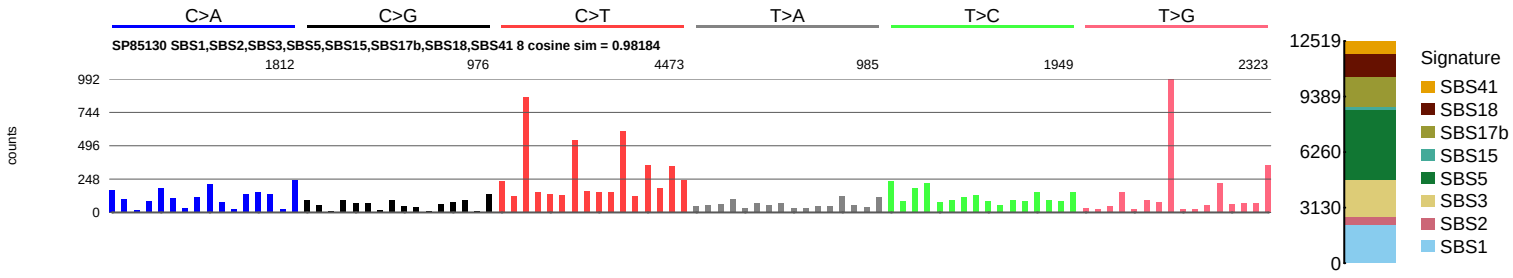
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17b,SBS18,SBS21,SBS41,SBS44 10 cosine sim = 0.98184
1812
968
4459
978
1987
2315
counts
992
744
496
248
0
12519
9389
6260
3130
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS15
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17b,SBS18,SBS41 8 cosine sim = 0.98184
1812
976
4473
985
1949
2323
counts
992
744
496
248
0
12519
9389
6260
3130
0
Signature
SBS41
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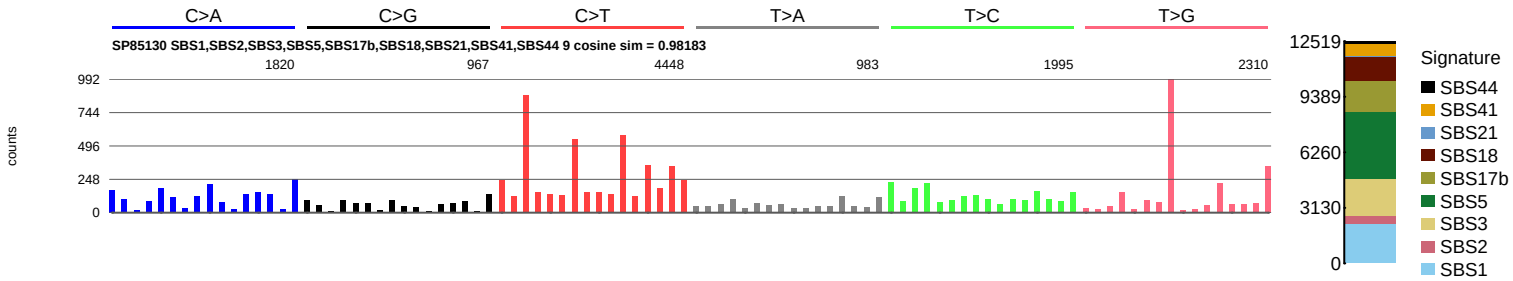
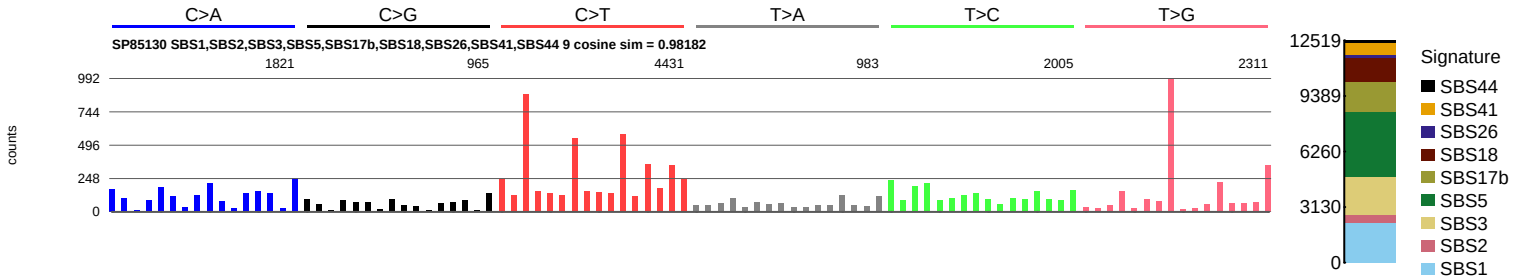
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS5,SBS17b,SBS18,SBS21,SBS41,SBS44 9 cosine sim = 0.98183
1820
967
4448
983
1995
2310
counts
992
744
496
248
0
12519
9389
6260
3130
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS5,SBS17b,SBS18,SBS26,SBS41,SBS44 9 cosine sim = 0.98182
1821
965
4431
983
2005
2311
counts
992
744
496
248
0
12519
9389
6260
3130
0
Signature
SBS44
SBS41
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SBS17b
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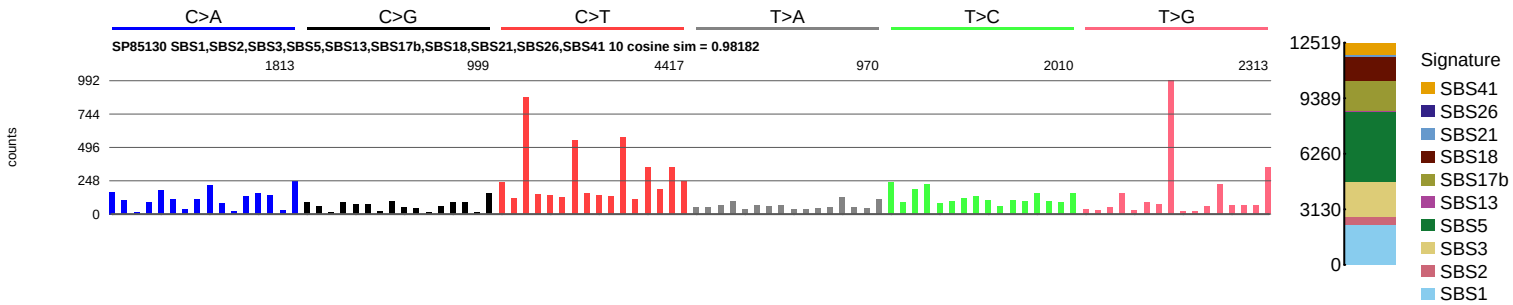
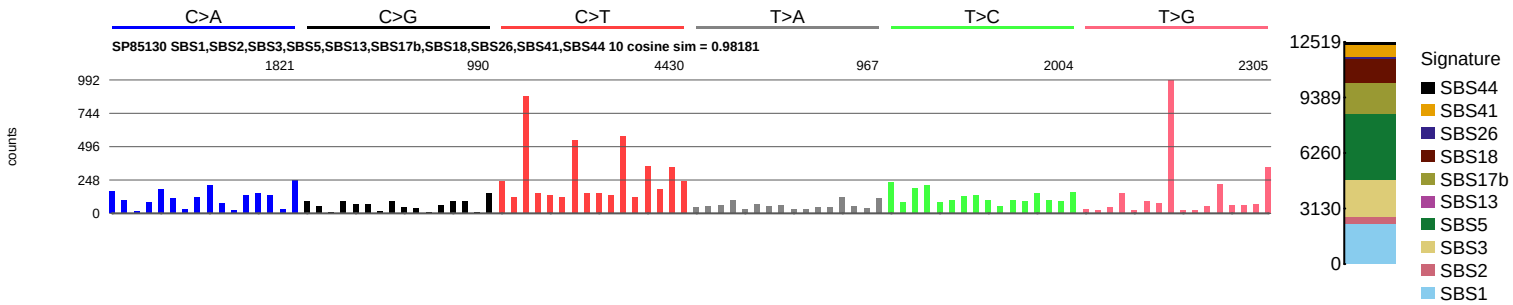
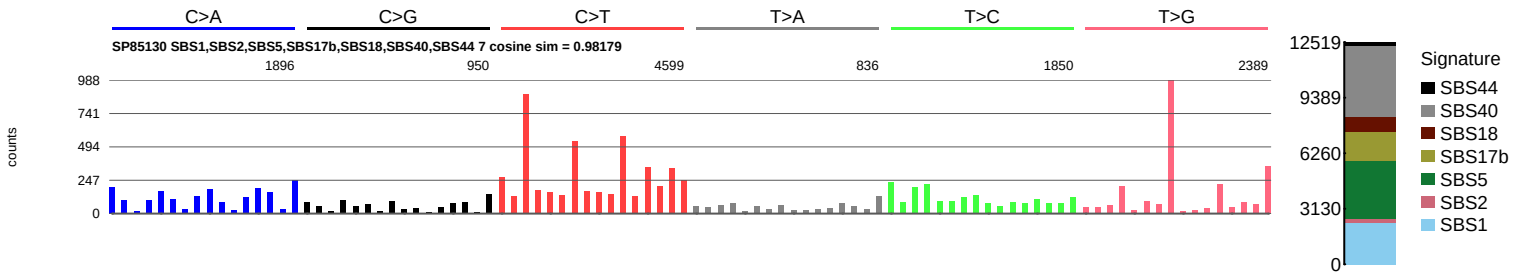
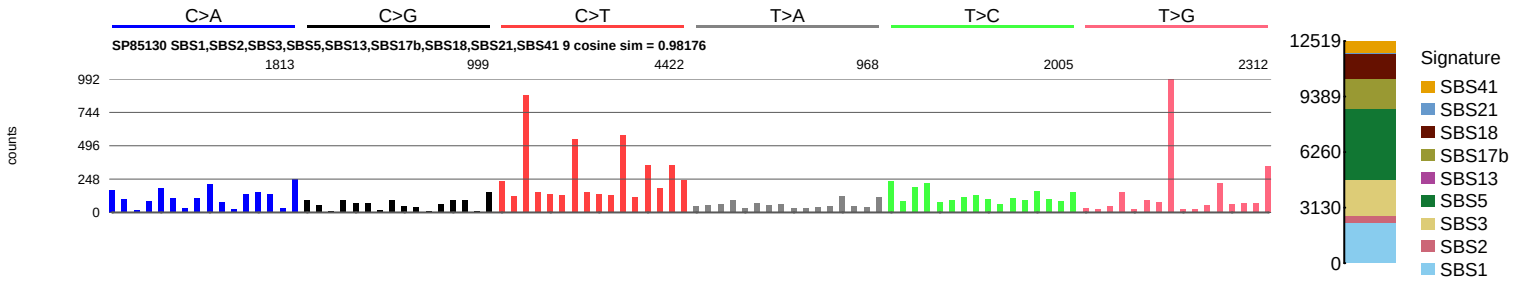
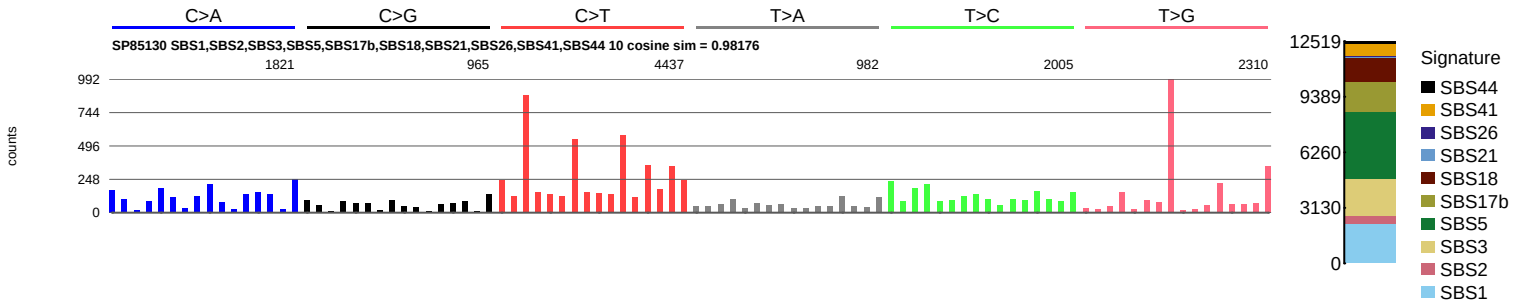
SBS3
SBS2
SBS1

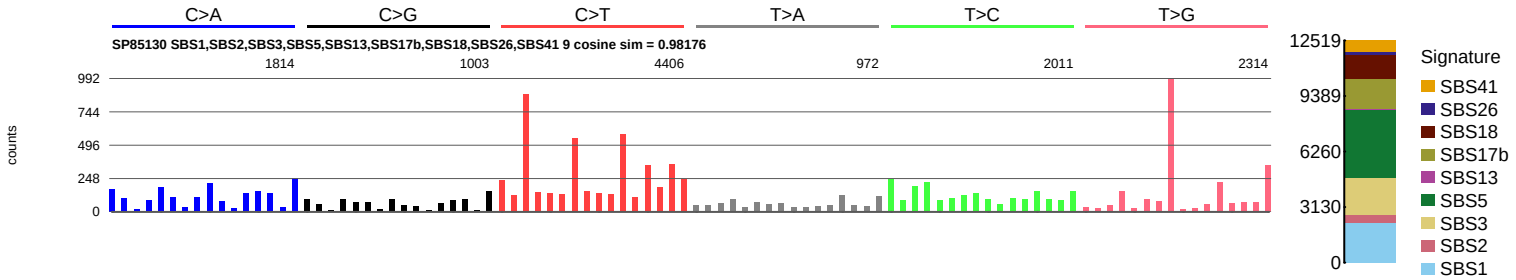

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS18,SBS21,SBS26,SBS41 10 cosine sim = 0.98182
1813
999
4417
970
2010
2313
counts
992
744
496
248
0
12519
9389
6260
3130
0
Signature
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS13
SBS5
SBS3
SBS2
SBS1

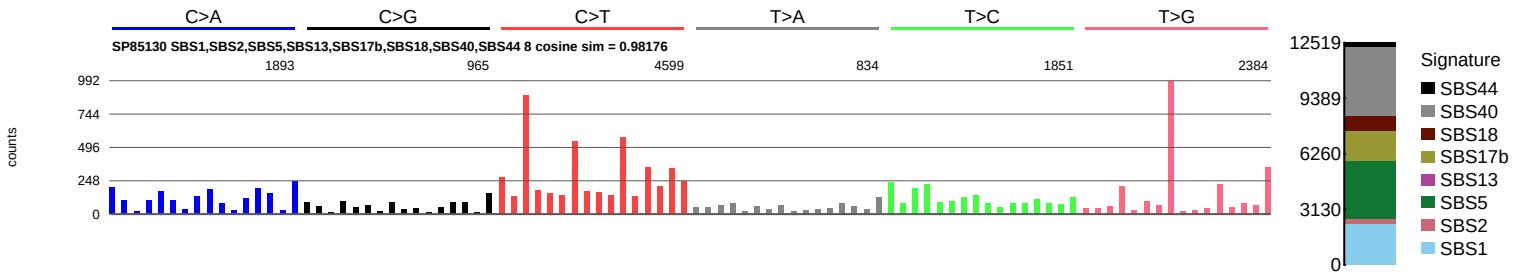

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS18,SBS26,SBS41,SBS44 10 cosine sim = 0.98181
1821
990
4430
967
2004
2305
counts
992
744
496
248
0
12519
9389
6260
3130
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS13
SBS5
SBS3
SBS2
SBS1

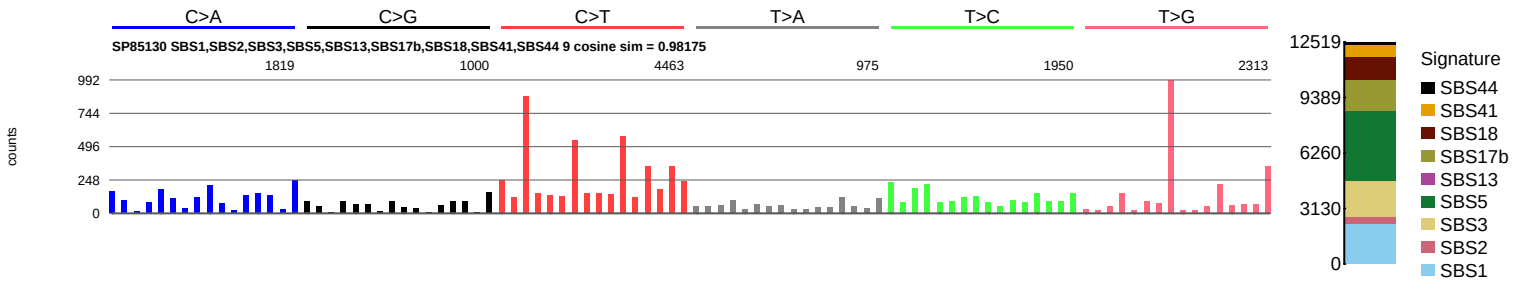

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS5,SBS17b,SBS18,SBS40,SBS44 7 cosine sim = 0.98179
1896
950
4599
836
1850
2389
counts
988
741
494
247
0
12519
9389
6260
3130
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS18,SBS21,SBS41 9 cosine sim = 0.98176
1813
999
4422
968
2005
2312
counts
992
744
496
248
0
12519
9389
6260
3130
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS5,SBS17b,SBS18,SBS21,SBS26,SBS41,SBS44 10 cosine sim = 0.98176
1821
965
4437
982
2005
2310
counts
992
744
496
248
0
12519
9389
6260
3130
0
Signature
SBS44
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS18,SBS26,SBS41 9 cosine sim = 0.98176
1814
1003
4406
972
2011
2314
counts
992
744
496
248
0
12519
9389
6260
3130
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS5,SBS13,SBS17b,SBS18,SBS40,SBS44 8 cosine sim = 0.98176
1893
965
4599
834
1851
2384
counts
992
744
496
248
0
12519
9389
6260
3130
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS13
SBS5
SBS2
SBS1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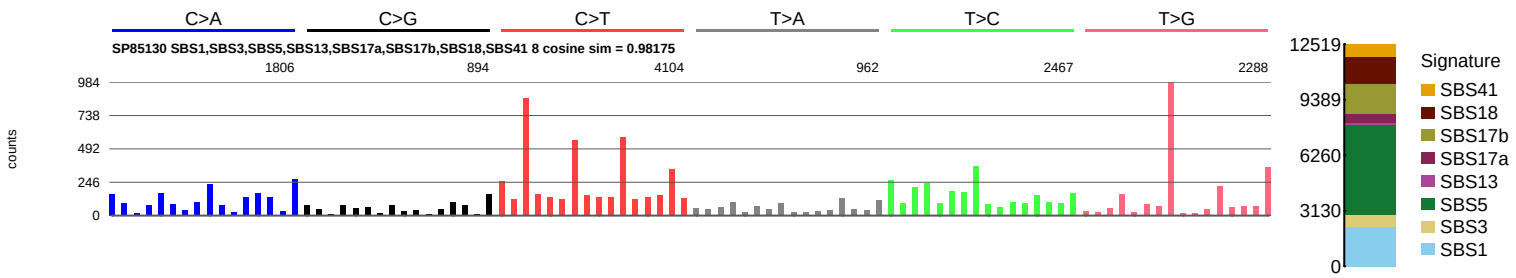
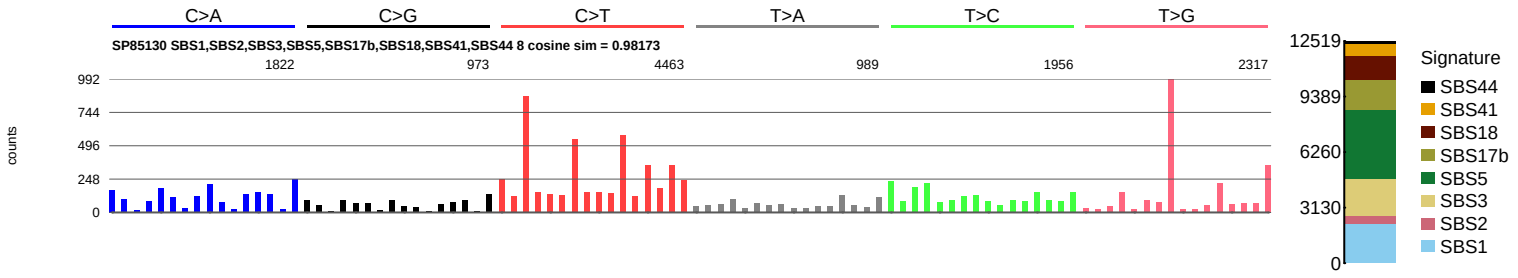
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS18,SBS41,SBS44 9 cosine sim = 0.98175
1819
1000
4463
975
1950
2313
counts
992
744
496
248
0
12519
9389
6260
3130
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS41 8 cosine sim = 0.98175
1806
894
4104
962
2467
2288
counts
984
738
492
246
0
12519
9389
6260
3130
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS5,SBS17b,SBS18,SBS41,SBS44 8 cosine sim = 0.98173
1822
973
4463
989
1956
2317
counts
992
744
496
248
0
12519
9389
6260
3130
0
Signature
SBS44
SBS41
SBS18
SBS17b
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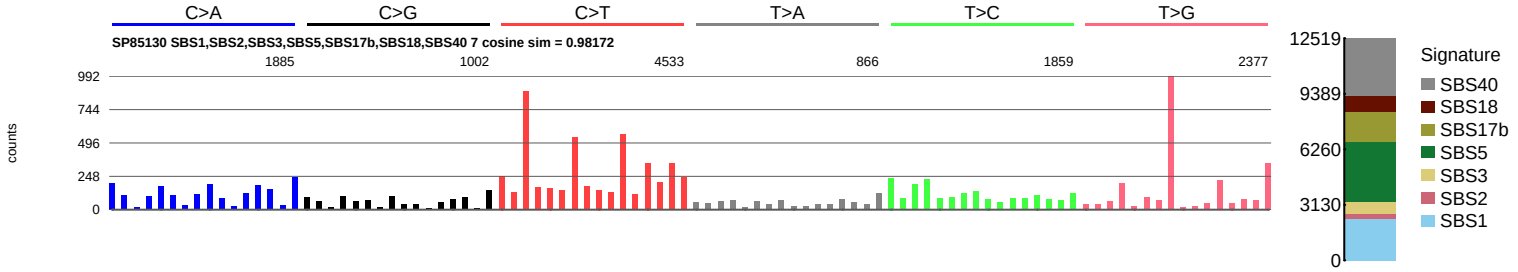
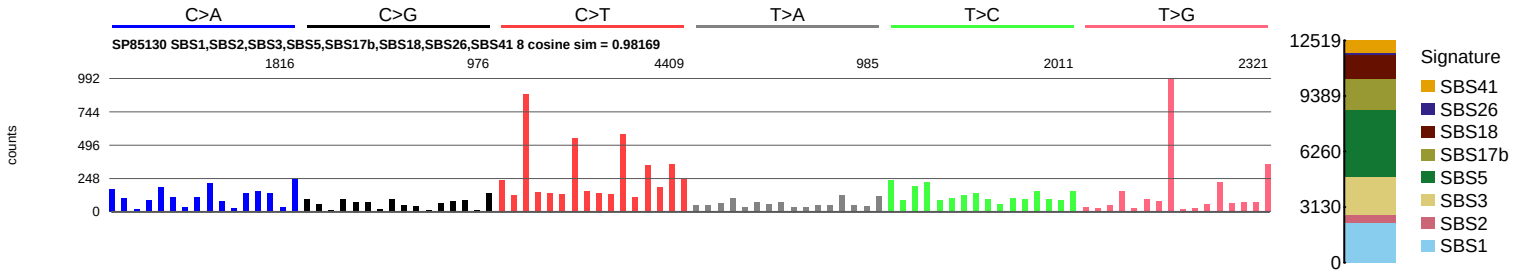
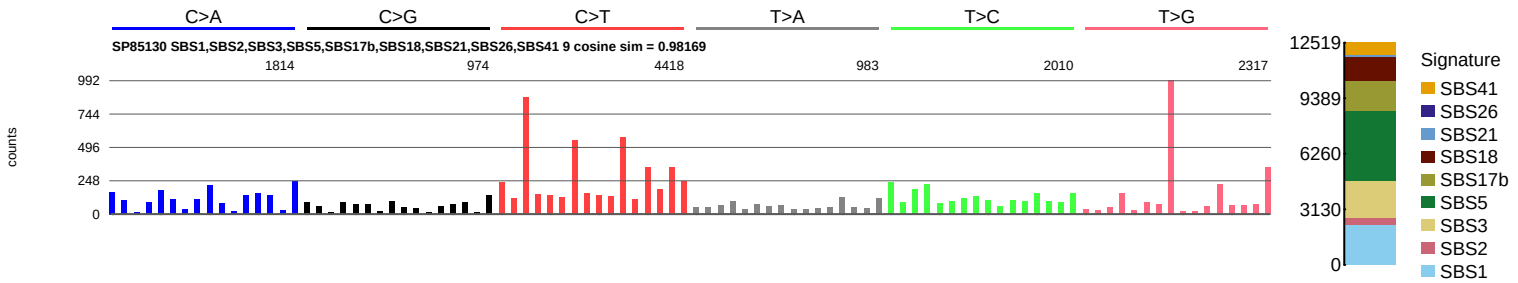
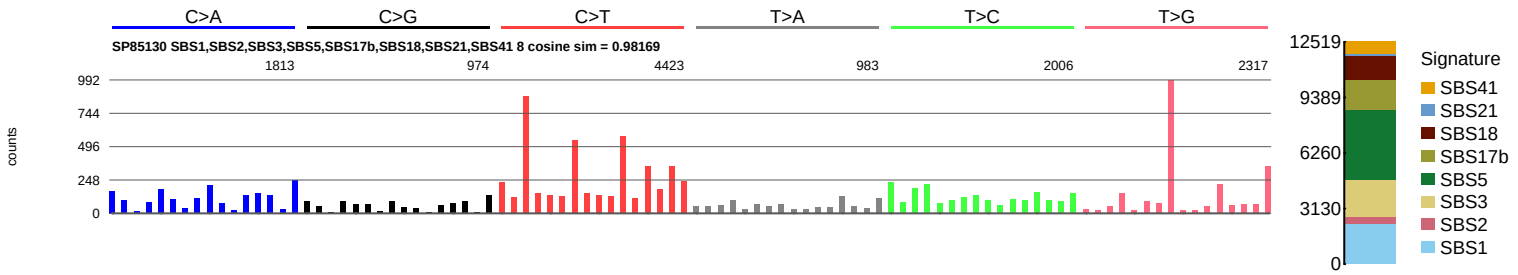
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS5,SBS17b,SBS18,SBS40 7 cosine sim = 0.98172
1885
1002
4533
866
1859
2377
counts
992
744
496
248
0
12519
9389
6260
3130
0
Signature
SBS40
SBS18
SBS17b
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS5,SBS17b,SBS18,SBS26,SBS41 8 cosine sim = 0.98169
1816
976
4409
985
2011
2321
counts
992
744
496
248
0
12519
9389
6260
3130
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS5,SBS17b,SBS18,SBS21,SBS26,SBS41 9 cosine sim = 0.98169
1814
974
4418
983
2010
2317
counts
992
744
496
248
0
12519
9389
6260
3130
0
Signature
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS5,SBS17b,SBS18,SBS21,SBS41 8 cosine sim = 0.98169
1813
974
4423
983
2006
2317
counts
992
744
496
248
0
12519
9389
6260
3130
0
Signature
SBS41
SBS21
SBS18
SBS17b
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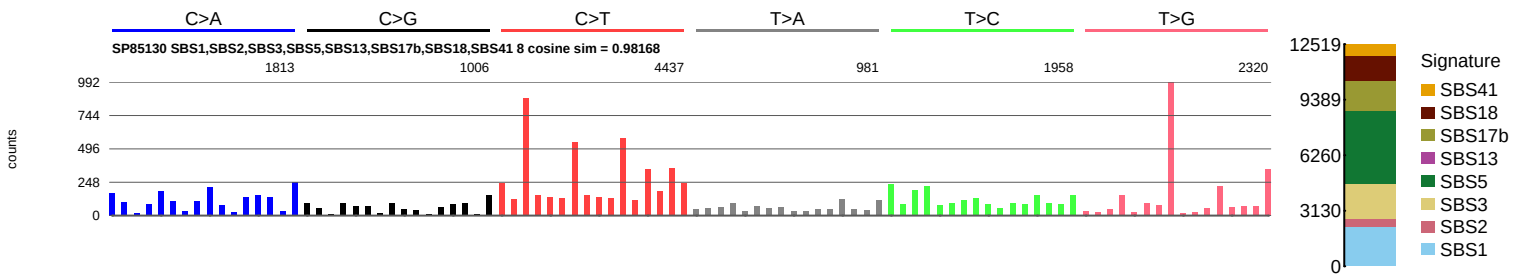
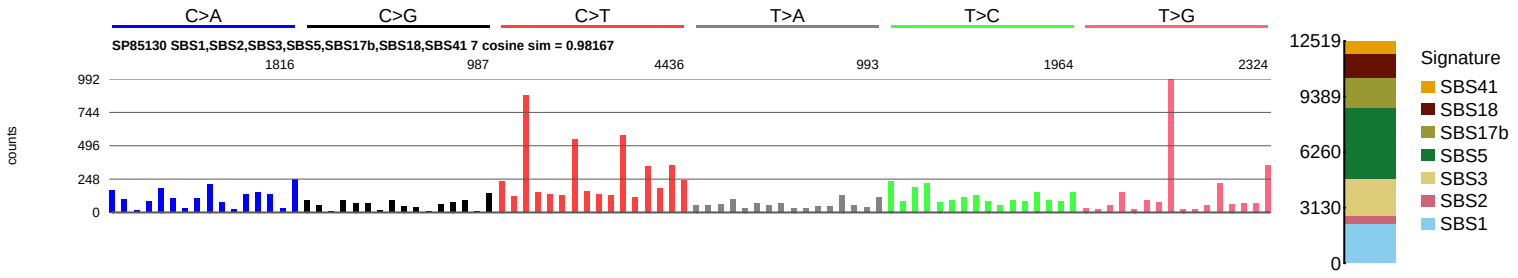
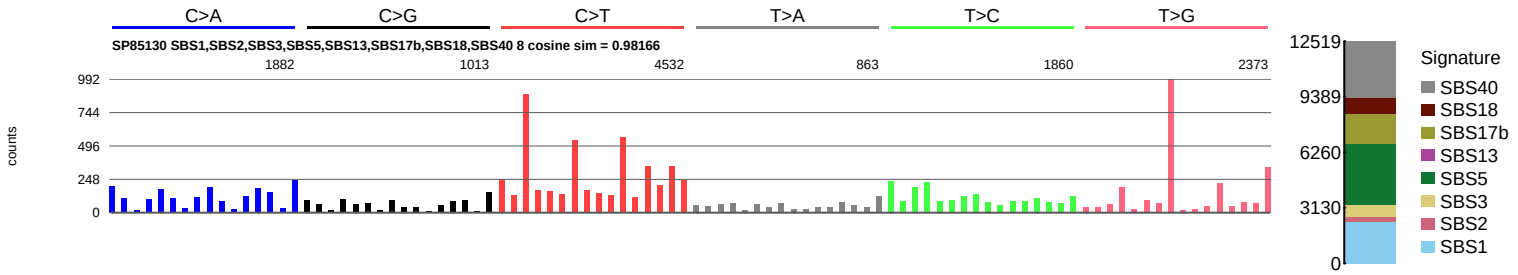
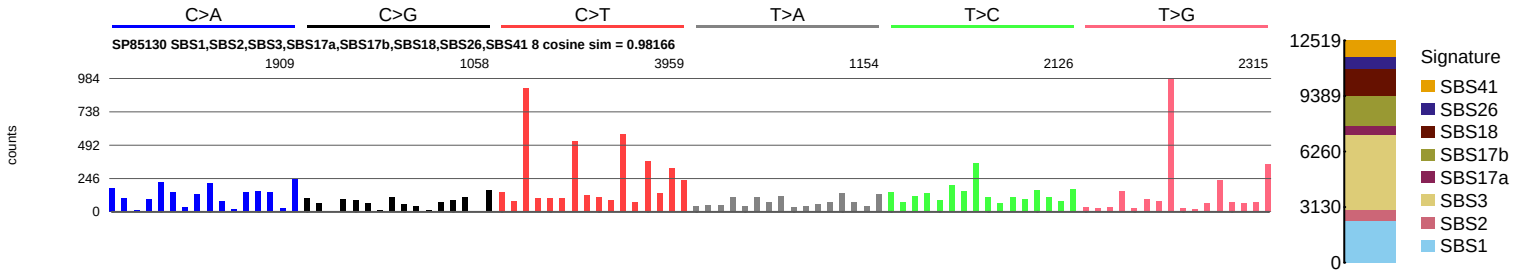
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS18,SBS41 8 cosine sim = 0.98168
1813
1006
4437
981
1958
2320
counts
992
744
496
248
0
12519
9389
6260
3130
0
Signature
SBS41
SBS18
SBS17b
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS5,SBS17b,SBS18,SBS41 7 cosine sim = 0.98167
1816
987
4436
993
1964
2324
counts
992
744
496
248
0
12519
9389
6260
3130
0
Signature
SBS41
SBS18
SBS17b
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS18,SBS40 8 cosine sim = 0.98166
1882
1013
4532
863
1860
2373
counts
992
744
496
248
0
12519
9389
6260
3130
0
Signature
SBS40
SBS18
SBS17b
SBS13
SBS5
SBS3
SBS2
SBS1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C>G
C>T
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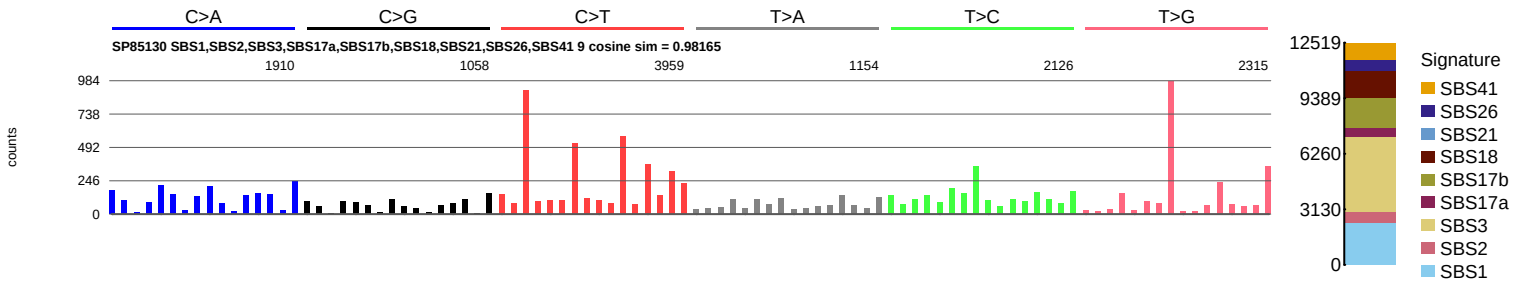
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS26,SBS41 8 cosine sim = 0.98166
1909
1058
3959
1154
2126
2315
counts
984
738
492
246
0
12519
9389
6260
3130
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1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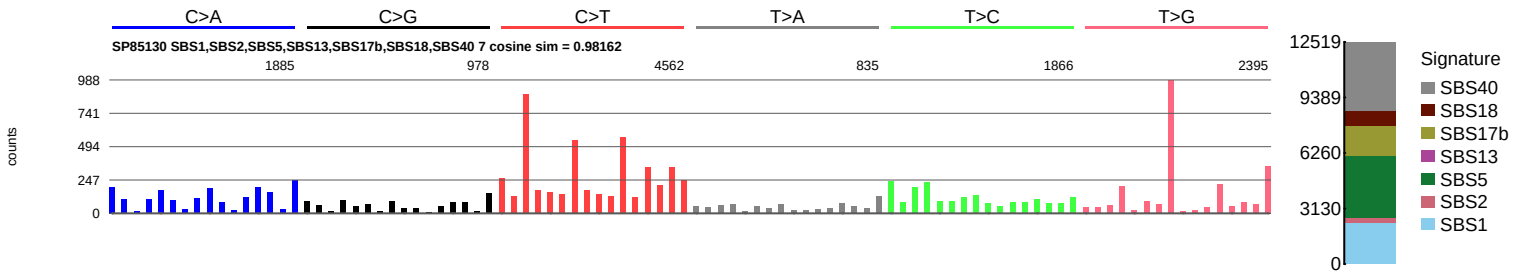
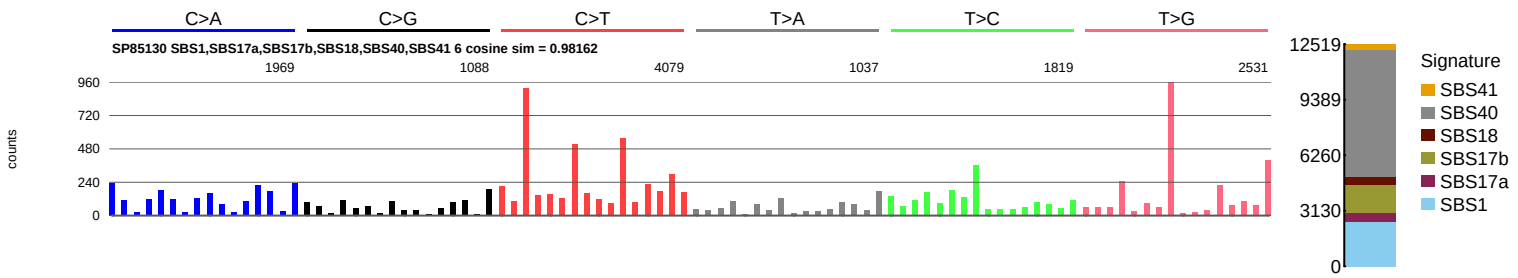
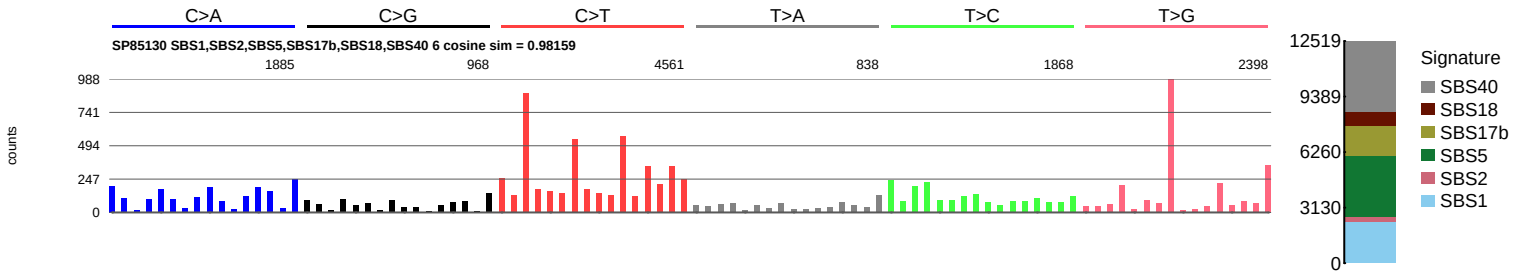
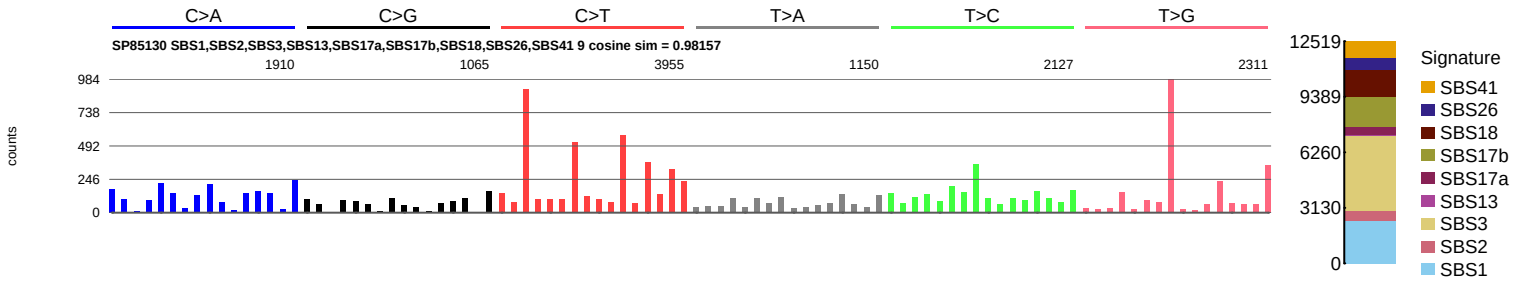
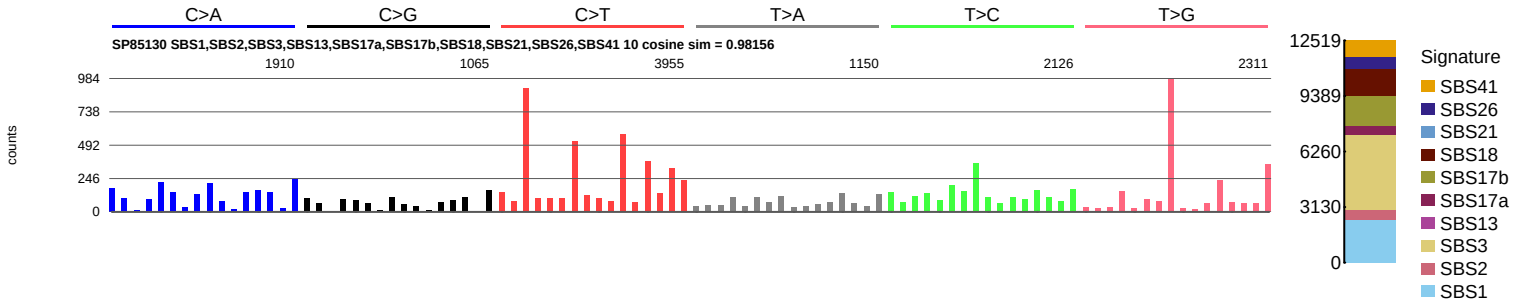
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41 9 cosine sim = 0.98165
1910
1058
3959
1154
2126
2315
counts
984
738
492
246
0
12519
9389
6260
3130
0
Signature
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1

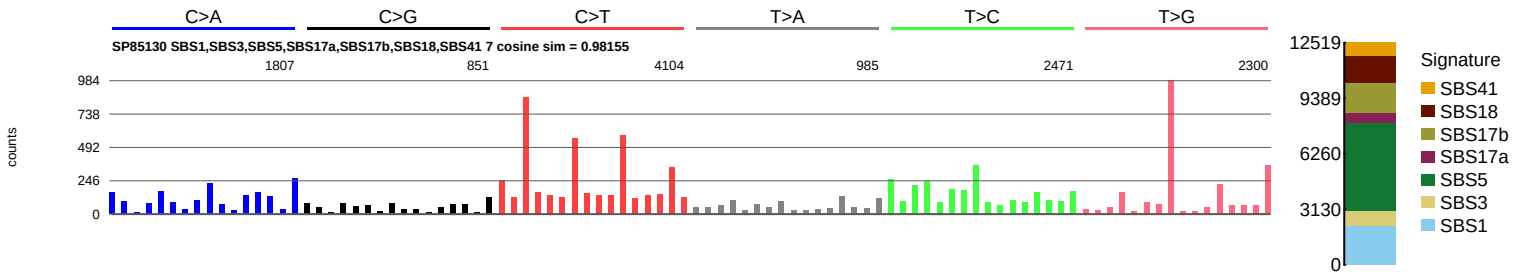

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS5,SBS13,SBS17b,SBS18,SBS40 7 cosine sim = 0.98162
1885
978
4562
835
1866
2395
counts
988
741
494
247
0
12519
9389
6260
3130
0
Signature
SBS40
SBS18
SBS17b
SBS13
SBS5
SBS2
SBS1

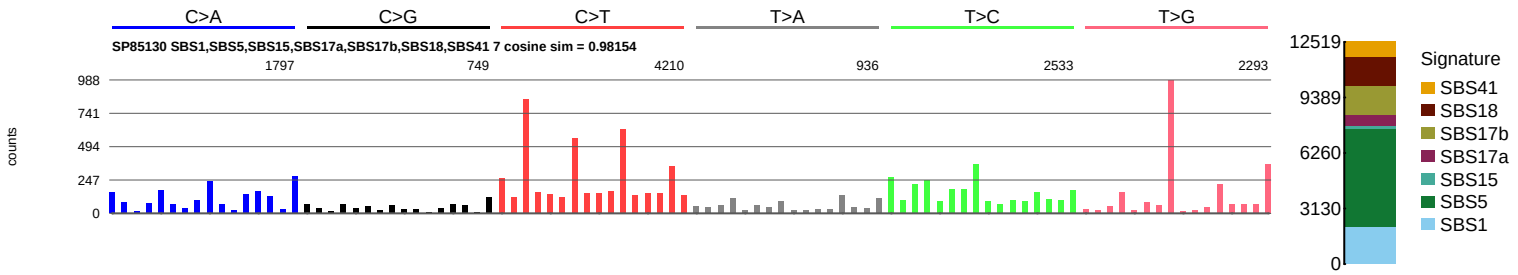

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS17a,SBS17b,SBS18,SBS40,SBS41 6 cosine sim = 0.98162
1969
1088
4079
1037
1819
2531
counts
960
720
480
240
0
12519
9389
6260
3130
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS1

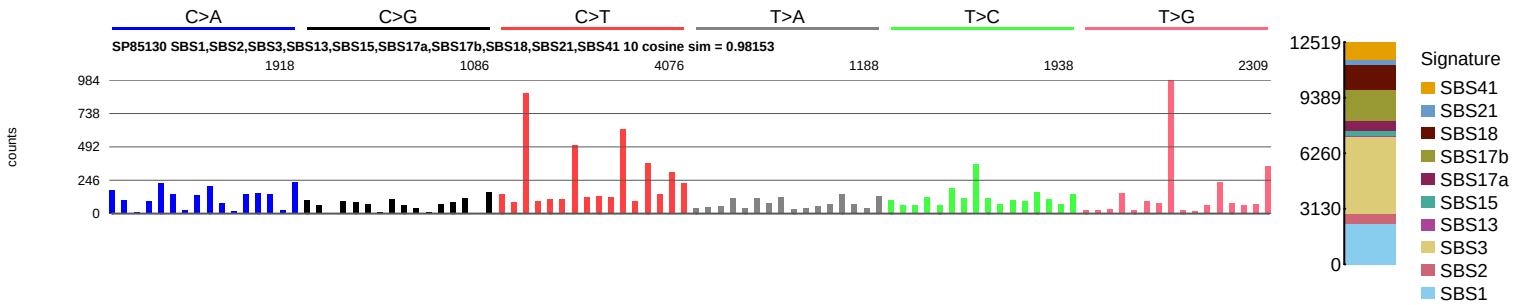

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS5,SBS17b,SBS18,SBS40 6 cosine sim = 0.98159
1885
968
4561
838
1868
2398
counts
988
741
494
247
0
12519
9389
6260
3130
0
Signature
SBS40
SBS18
SBS17b
SBS5
SBS2
SBS1

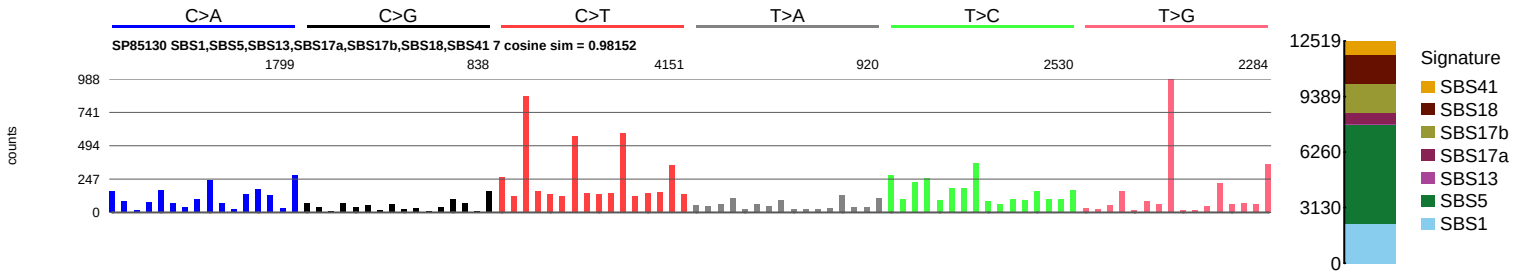

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS41 9 cosine sim = 0.98157
1910
1065
3955
1150
2127
2311
counts
984
738
492
246
0
12519
9389
6260
3130
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1

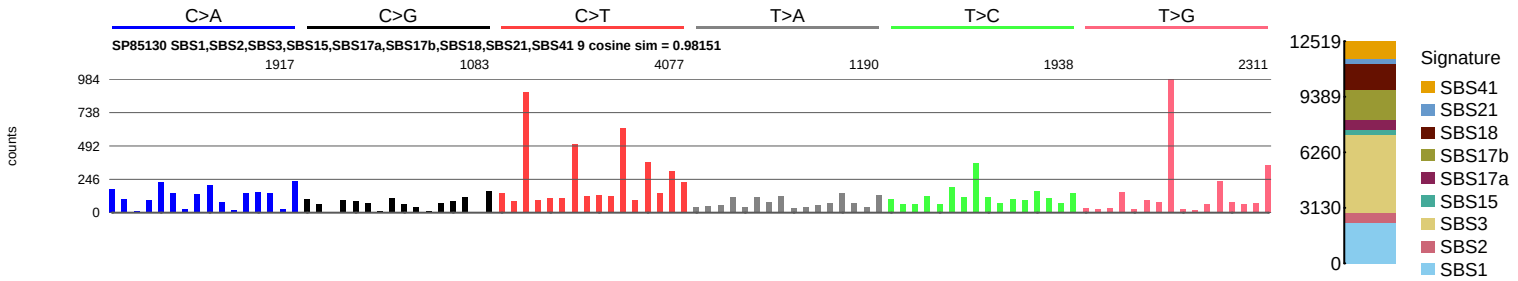

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41 10 cosine sim = 0.98156
1910
1065
3955
1150
2126
2311
counts
984
738
492
246
0
12519
9389
6260
3130
0
Signature
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1

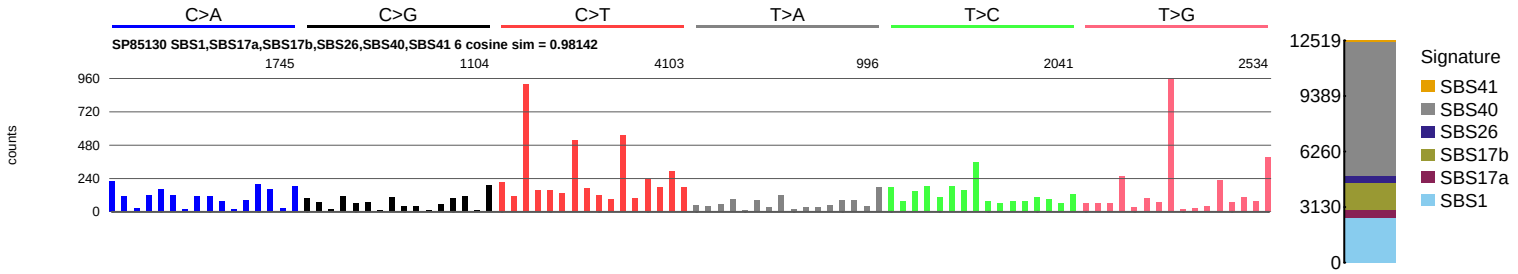

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS41 7 cosine sim = 0.98155
1807
851
4104
985
2471
2300
counts
984
738
492
246
0
12519
9389
6260
3130
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS1

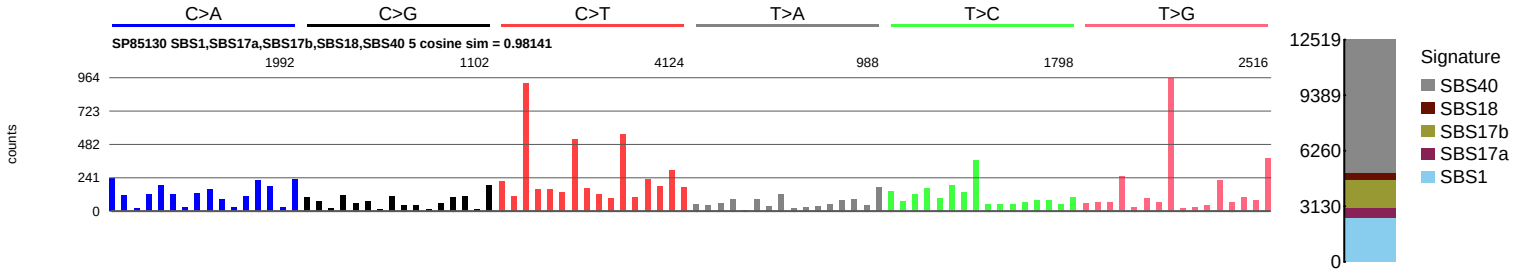

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS41 7 cosine sim = 0.98154
1797
749
4210
936
2533
2293
counts
988
741
494
247
0
12519
9389
6260
3130
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS1

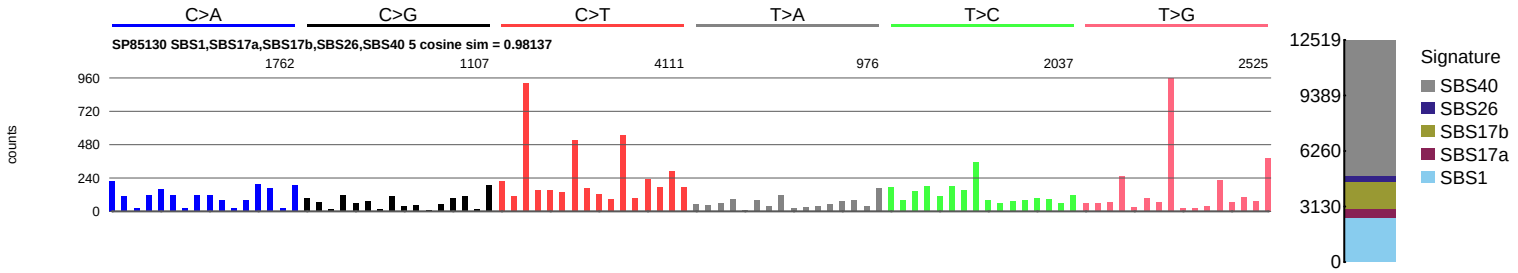

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 10 cosine sim = 0.98153
1918
1086
4076
1188
1938
2309
counts
984
738
492
246
0
12519
9389
6260
3130
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

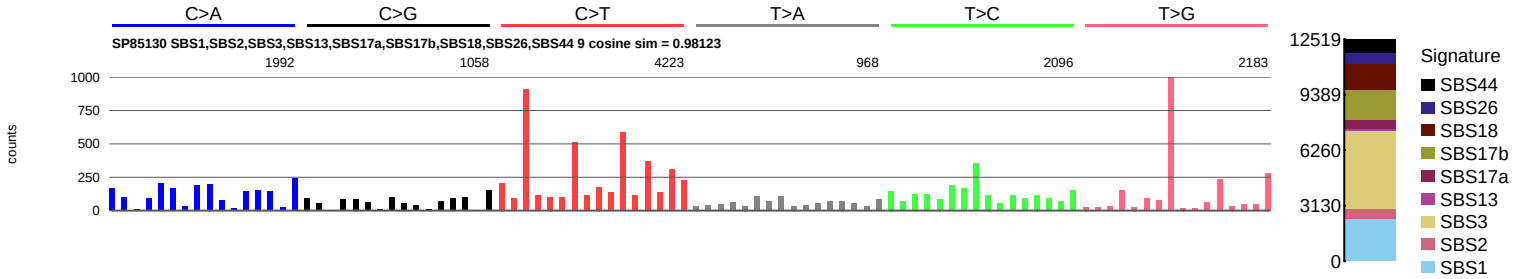

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS41 7 cosine sim = 0.98152
1799
838
4151
920
2530
2284
988
741
494
247
0
counts
12519
9389
6260
3130
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS1

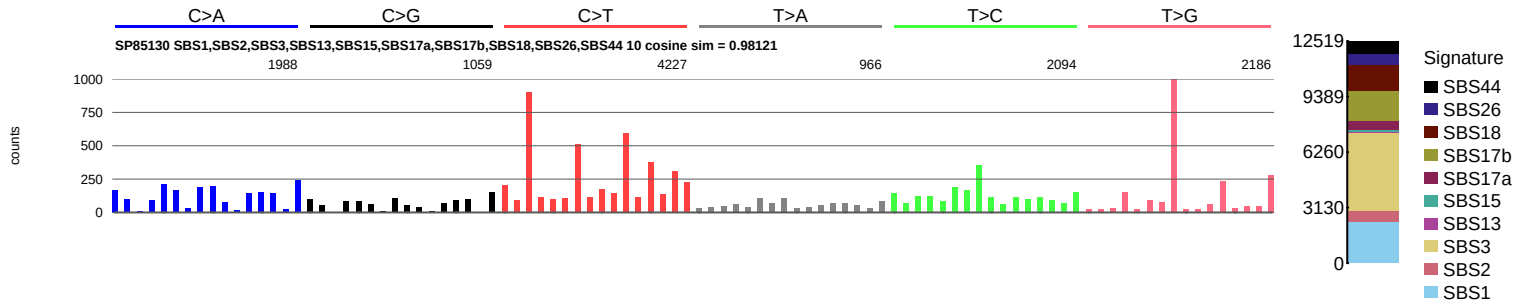

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 9 cosine sim = 0.98151
1917
1083
4077
1190
1938
2311
984
738
492
246
0
counts
12519
9389
6260
3130
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1

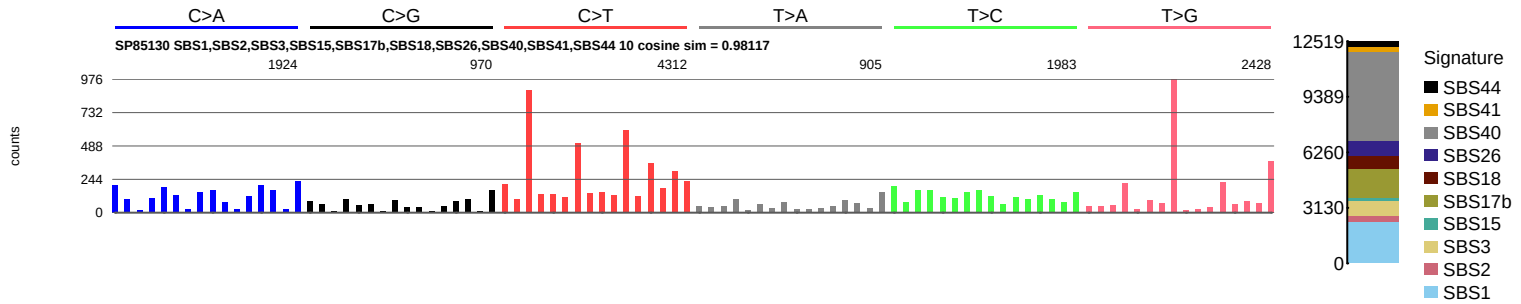

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS17a,SBS17b,SBS26,SBS40,SBS41 6 cosine sim = 0.98142
1745
1104
4103
996
2041
2534
960
720
480
240
0
counts
12519
9389
6260
3130
0
Signature
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS1

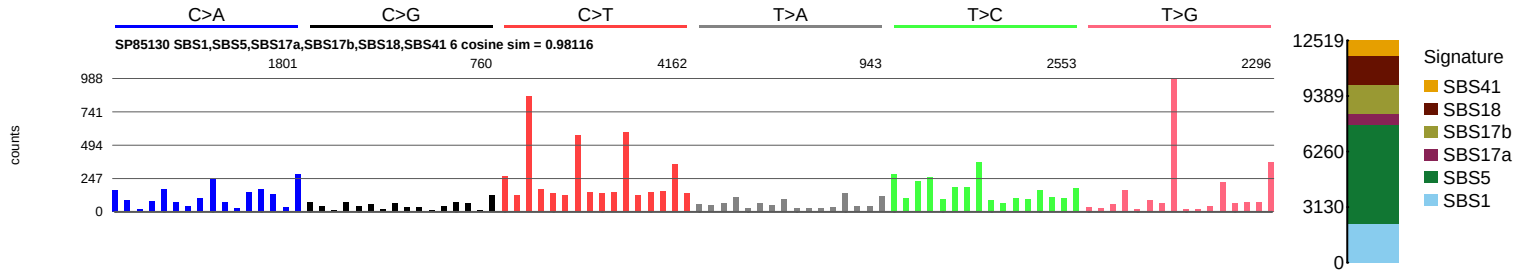

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS17a,SBS17b,SBS18,SBS40 5 cosine sim = 0.98141
1992
1102
4124
988
1798
2516
964
723
482
241
0
counts
12519
9389
6260
3130
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS1

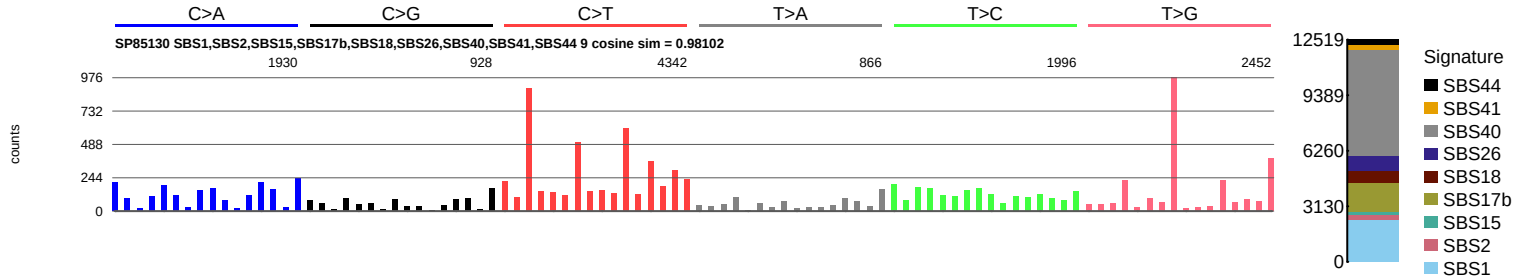

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS17a,SBS17b,SBS26,SBS40 5 cosine sim = 0.98137
1762
1107
4111
976
2037
2525
960
720
480
240
0
counts
12519
9389
6260
3130
0
Signature
SBS40
SBS26
SBS17b
SBS17a
SBS1

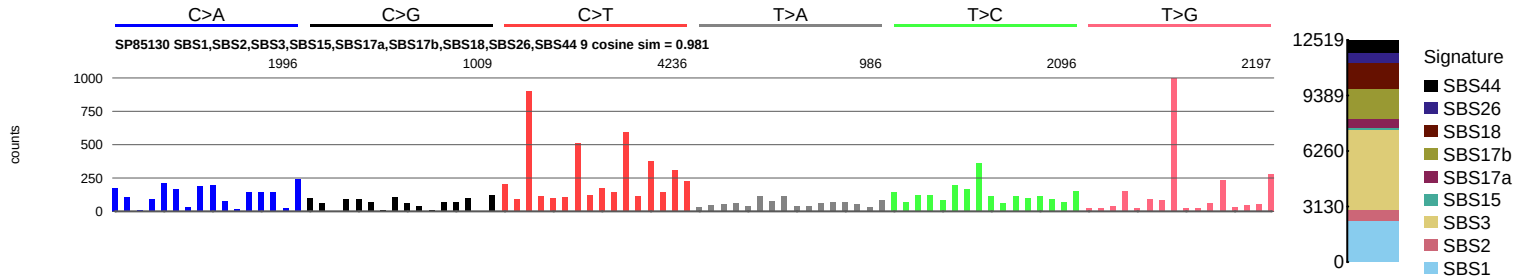

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS44 9 cosine sim = 0.98123
1992
1058
4223
968
2096
2183
1000
750
500
250
0
counts
12519
9389
6260
3130
0
Signature
SBS44
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1

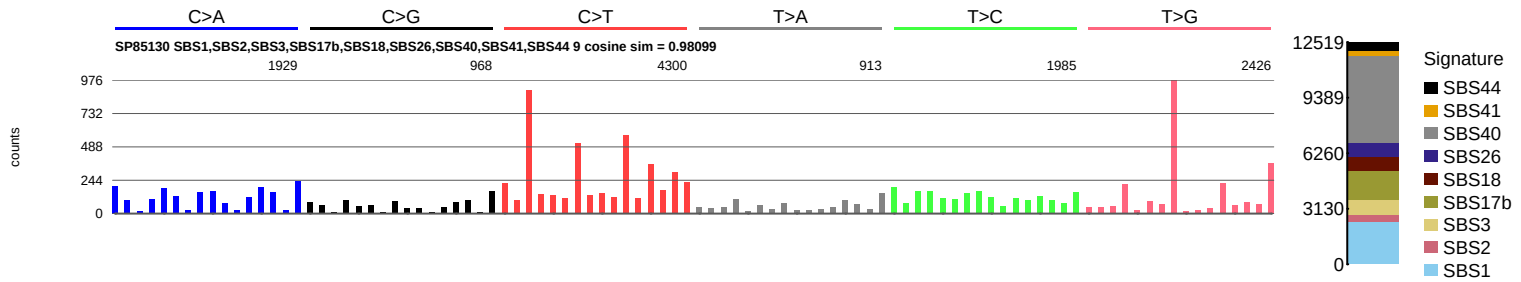

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS44 10 cosine sim = 0.98121
1988
1059
4227
966
2094
2186
counts
1000
750
500
250
0
12519
9389
6260
3130
0
Signature
SBS44
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

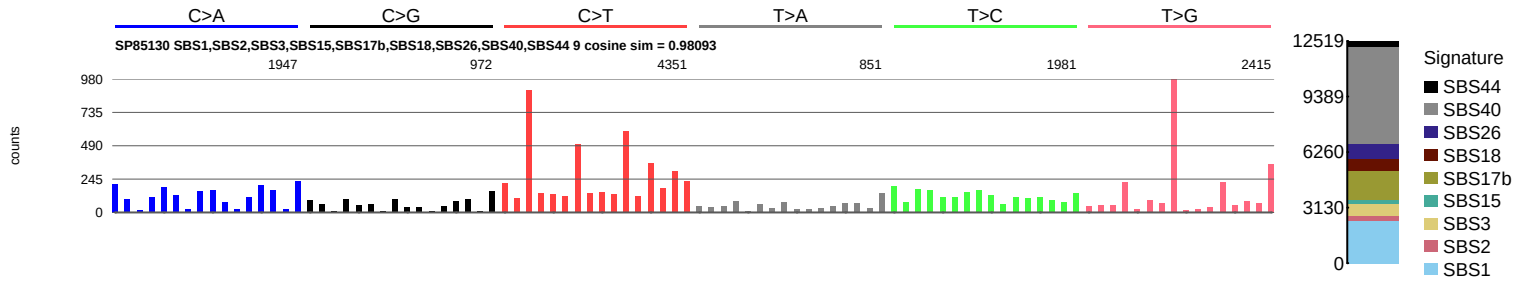

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS15,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.98117
1924
970
4312
905
1983
2428
counts
976
732
488
244
0
12519
9389
6260
3130
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS15
SBS3
SBS2
SBS1

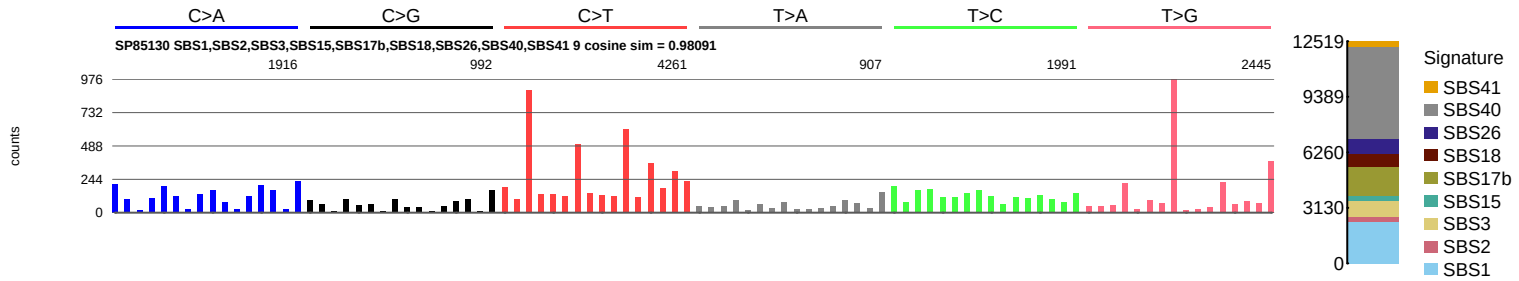

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS5,SBS17a,SBS17b,SBS18,SBS41 6 cosine sim = 0.98116
1801
760
4162
943
2553
2296
counts
988
741
494
247
0
12519
9389
6260
3130
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS5
SBS1

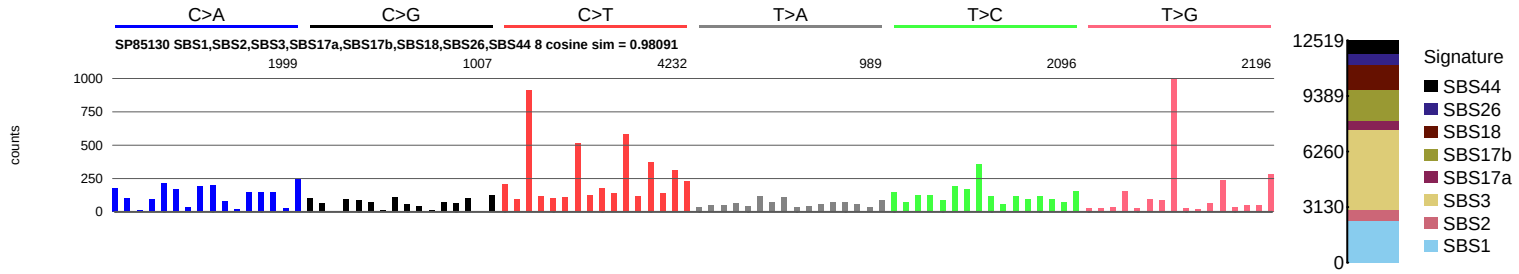

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS15,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 9 cosine sim = 0.98102
1930
928
4342
866
1996
2452
counts
976
732
488
244
0
12519
9389
6260
3130
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS15
SBS2
SBS1

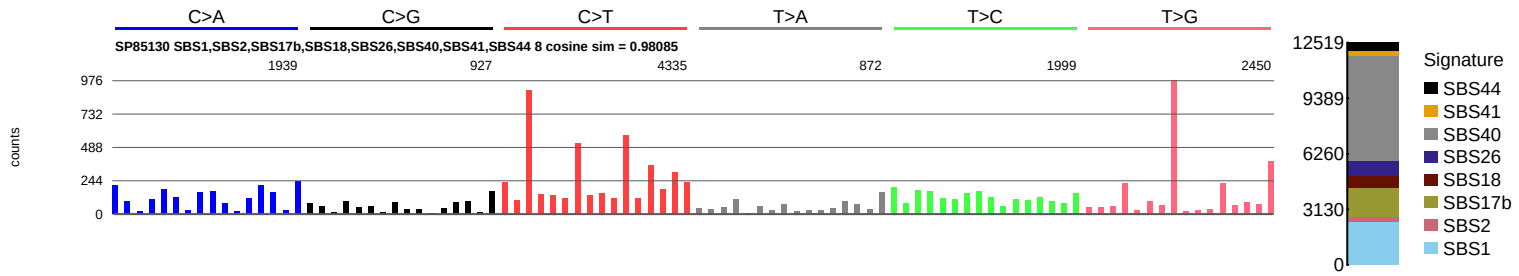

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS44 9 cosine sim = 0.981
1996
1009
4236
986
2096
2197
counts
1000
750
500
250
0
12519
9389
6260
3130
0
Signature
SBS44
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 9 cosine sim = 0.98099
1929
968
4300
913
1985
2426
counts
976
732
488
244
0
12519
9389
6260
3130
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS15,SBS17b,SBS18,SBS26,SBS40,SBS44 9 cosine sim = 0.98093
1947
972
4351
851
1981
2415
counts
980
735
490
245
0
12519
9389
6260
3130
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS15
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS15,SBS17b,SBS18,SBS26,SBS40,SBS41 9 cosine sim = 0.98091
1916
992
4261
907
1991
2445
counts
976
732
488
244
0
12519
9389
6260
3130
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS15
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS26,SBS44 8 cosine sim = 0.98091
1999
1007
4232
989
2096
2196
counts
1000
750
500
250
0
12519
9389
6260
3130
0
Signature
SBS44
SBS26
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1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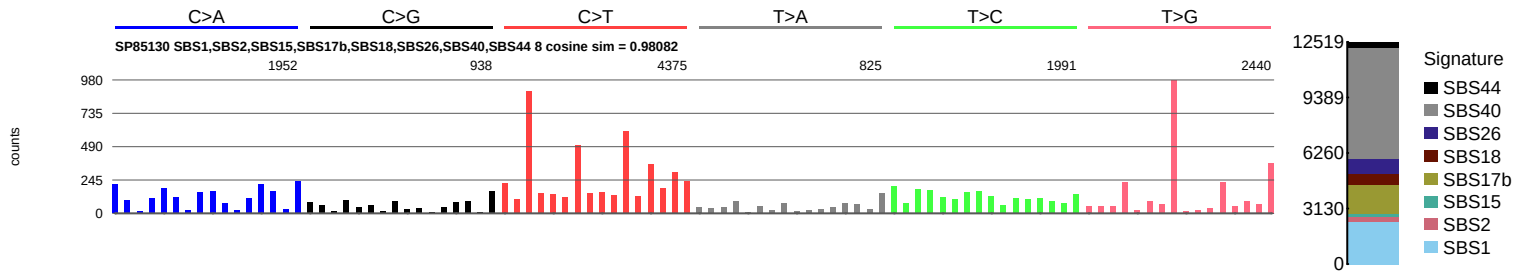
SP85130 SBS1,SBS2,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 8 cosine sim = 0.98085
1939
927
4335
872
1999
2450
counts
976
732
488
244
0
12519
9389
6260
3130
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS2
SBS1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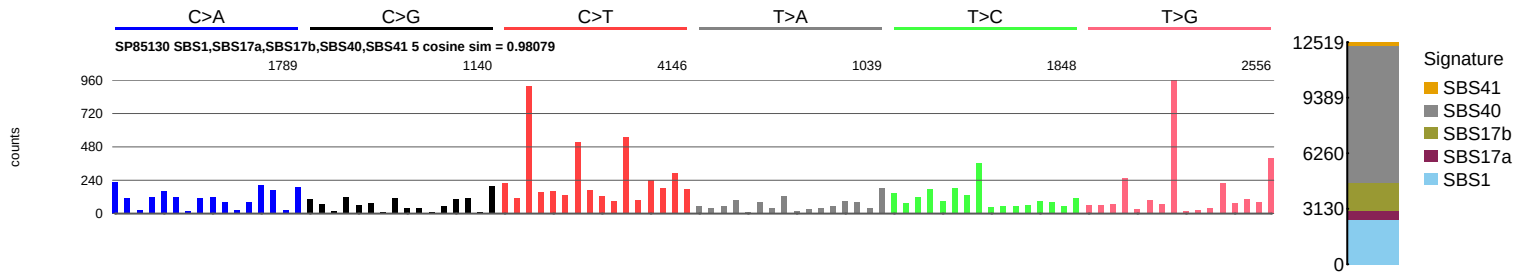
SP85130 SBS1,SBS2,SBS15,SBS17b,SBS18,SBS26,SBS40,SBS44 8 cosine sim = 0.98082
1952
938
4375
825
1991
2440
counts
980
735
490
245
0
12519
9389
6260
3130
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS15
SBS2
SBS1


C>A
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C>T
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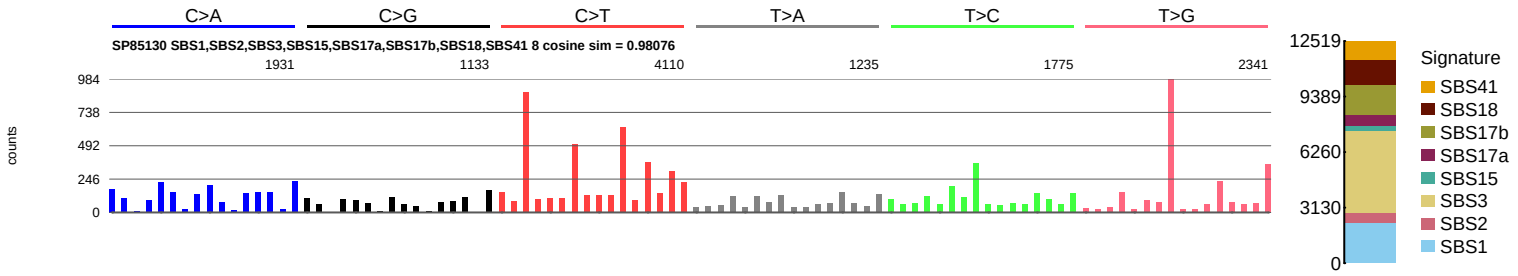
T>G
SP85130 SBS1,SBS17a,SBS17b,SBS40,SBS41 5 cosine sim = 0.98079
1789
1140
4146
1039
1848
2556
counts
960
720
480
240
0
12519
9389
6260
3130
0
Signature
SBS41
SBS40
SBS17b
SBS17a
SBS1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C>T
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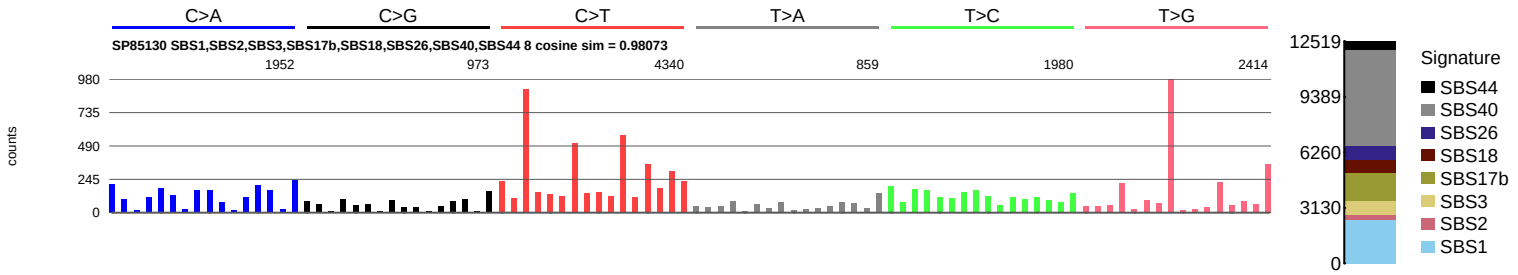
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS41 8 cosine sim = 0.98076
1931
1133
4110
1235
1775
2341
counts
984
738
492
246
0
12519
9389
6260
3130
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1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C>G
C>T
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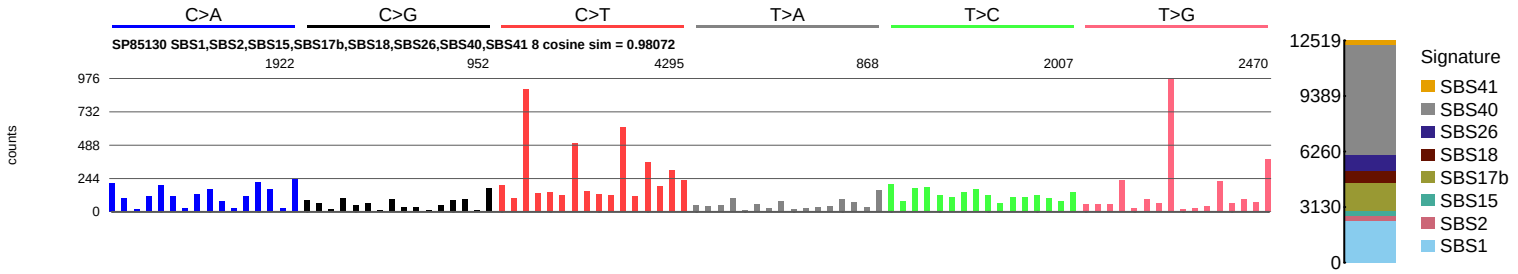
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS17b,SBS18,SBS26,SBS40,SBS44 8 cosine sim = 0.98073
1952
973
4340
859
1980
2414
counts
980
735
490
245
0
12519
9389
6260
3130
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS3
SBS2
SBS1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C>G
C>T
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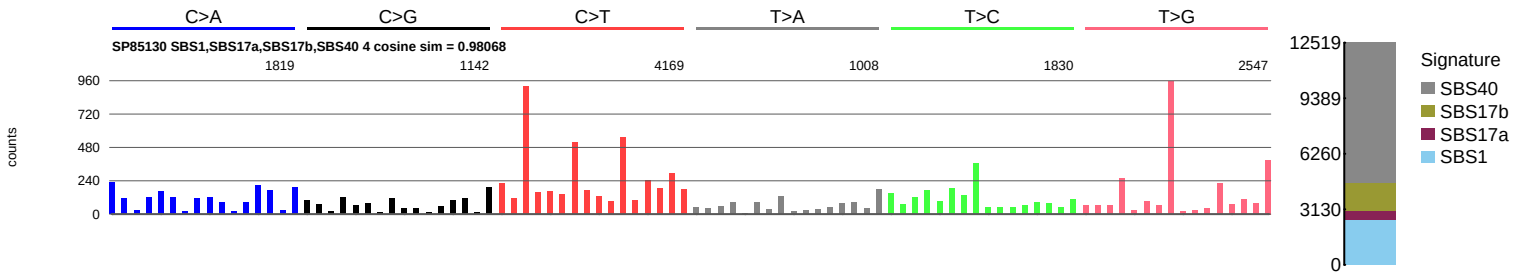
T>C
T>G
SP85130 SBS1,SBS2,SBS15,SBS17b,SBS18,SBS26,SBS40,SBS41 8 cosine sim = 0.98072
1922
952
4295
868
2007
2470
counts
976
732
488
244
0
12519
9389
6260
3130
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS15
SBS2
SBS1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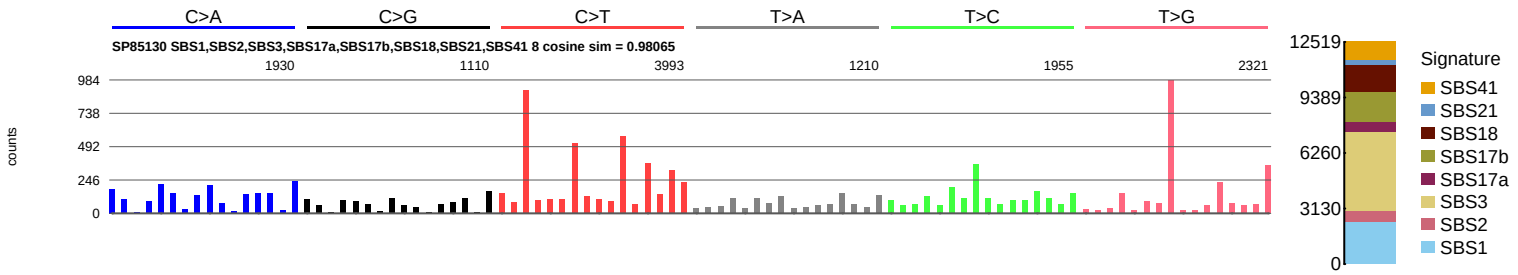
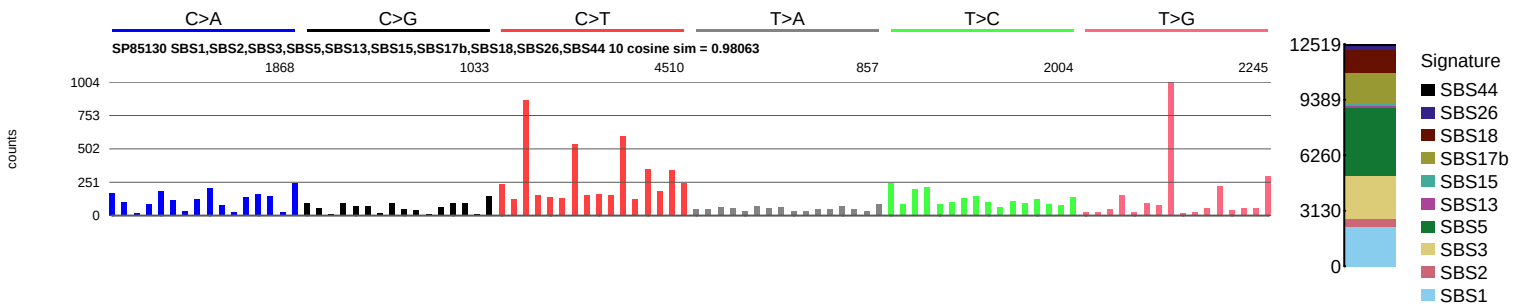
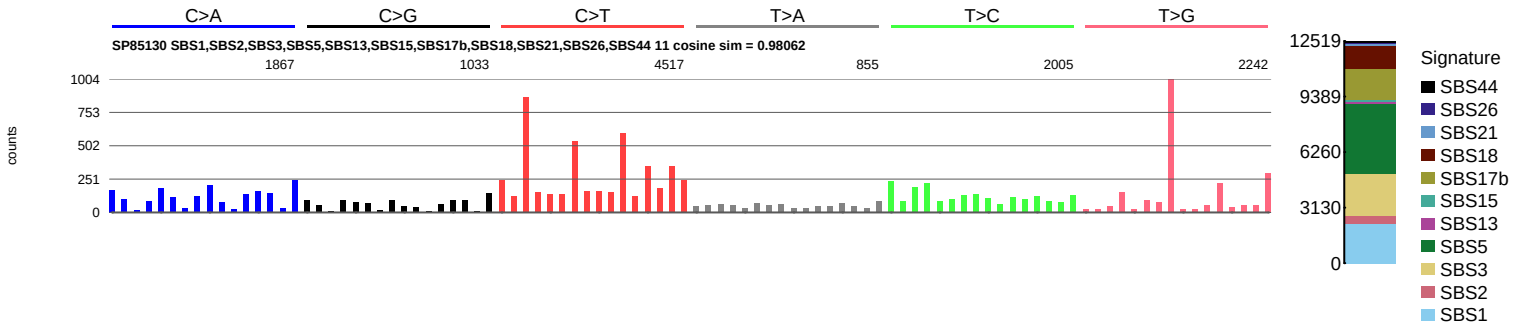
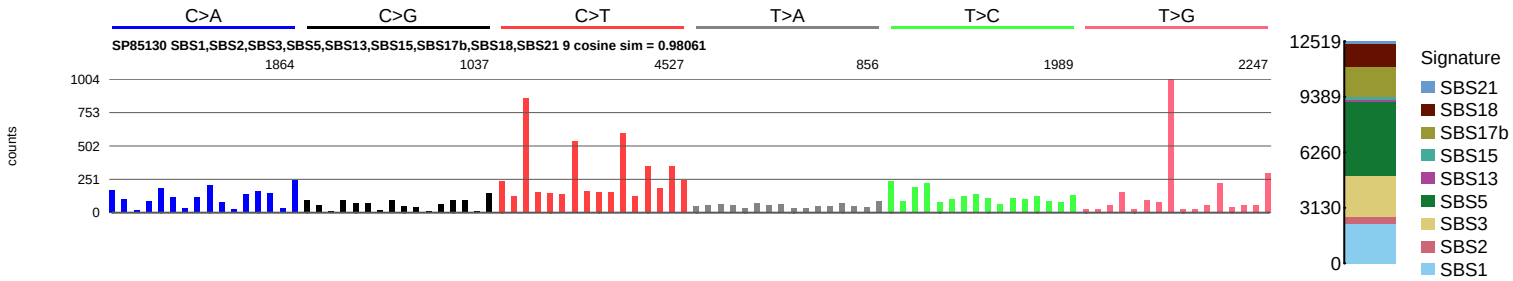
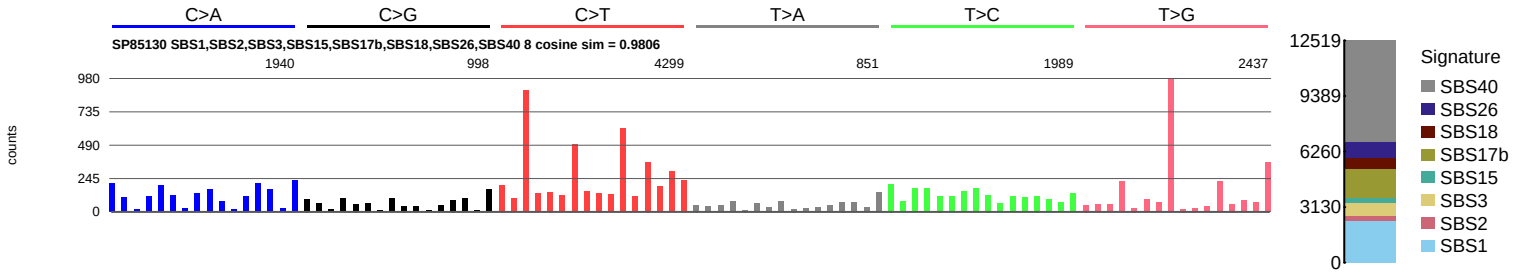
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS17a,SBS17b,SBS40 4 cosine sim = 0.98068
1819
1142
4169
1008
1830
2547
counts
960
720
480
240
0
12519
9389
6260
3130
0
Signature
SBS40
SBS17b
SBS17a
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS41 8 cosine sim = 0.98065
1930
1110
3993
1210
1955
2321
counts
984
738
492
246
0
12519
9389
6260
3130
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS26,SBS44 10 cosine sim = 0.98063
1868
1033
4510
857
2004
2245
counts
1004
753
502
251
0
12519
9389
6260
3130
0
Signature
SBS44
SBS26
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS26,SBS44 11 cosine sim = 0.98062
1867
1033
4517
855
2005
2242
counts
1004
753
502
251
0
12519
9389
6260
3130
0
Signature
SBS44
SBS26
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS21 9 cosine sim = 0.98061
1864
1037
4527
856
1989
2247
counts
1004
753
502
251
0
12519
9389
6260
3130
0
Signature
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1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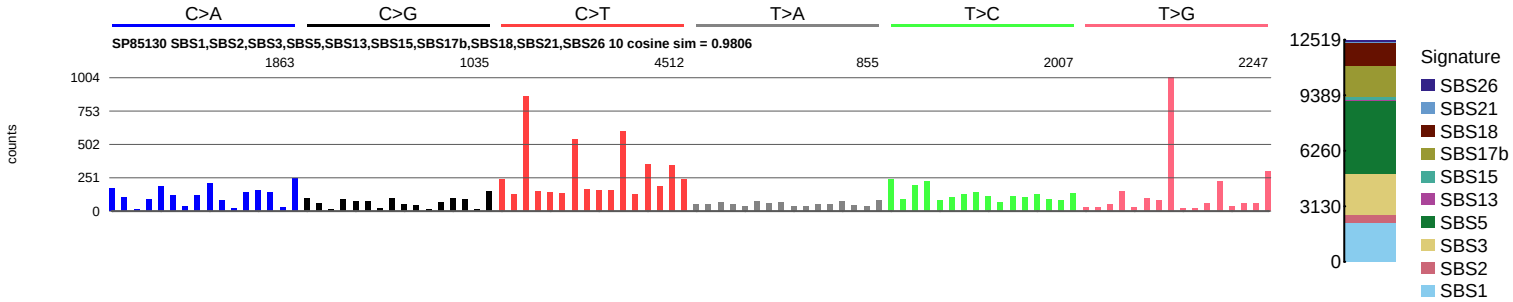
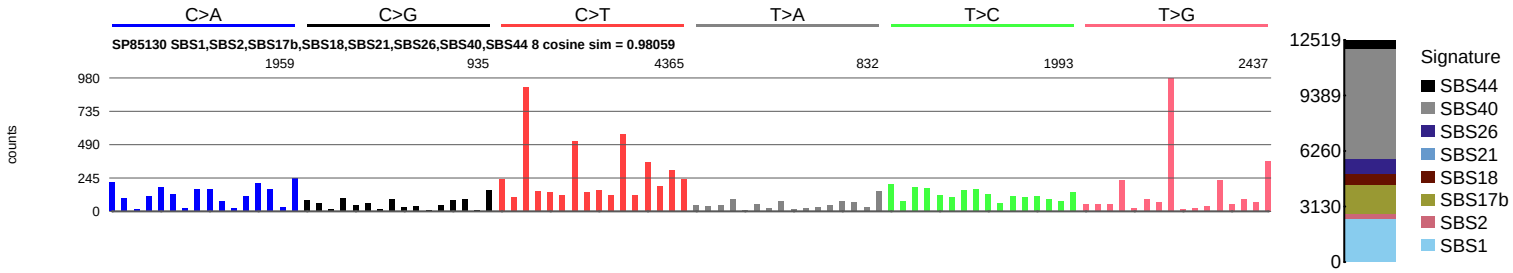
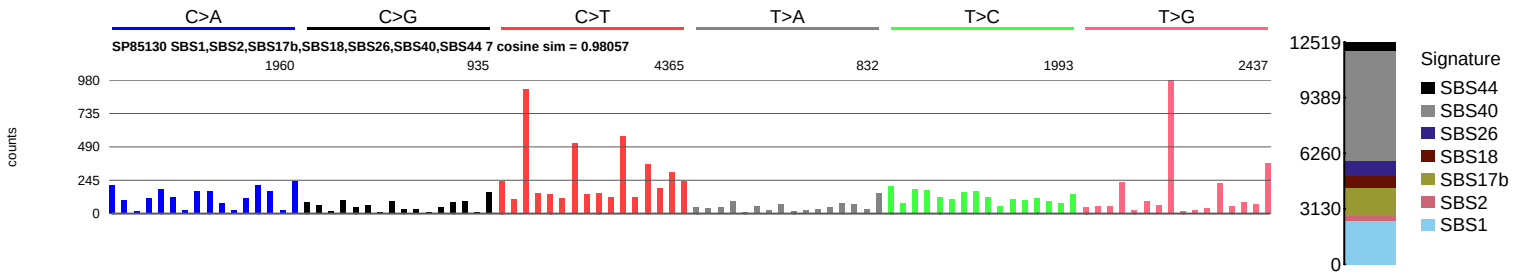
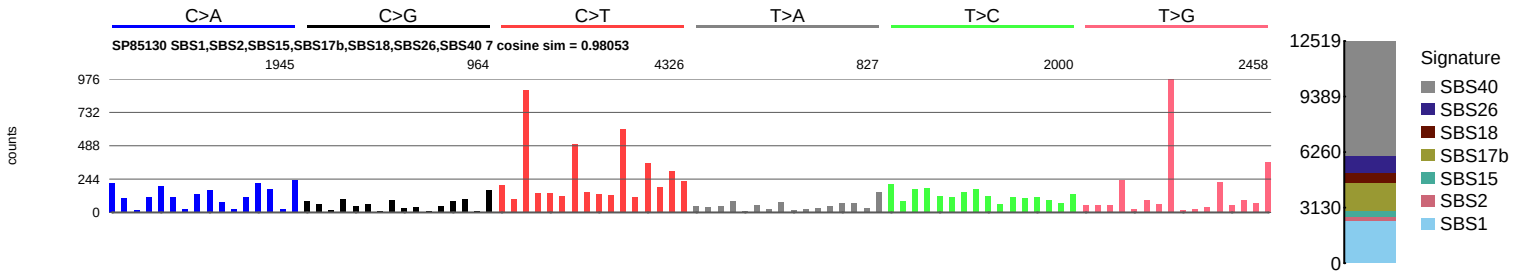
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS15,SBS17b,SBS18,SBS26,SBS40 8 cosine sim = 0.9806
1940
998
4299
851
1989
2437
counts
980
735
490
245
0
12519
9389
6260
3130
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS15
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS26 10 cosine sim = 0.9806
1863
1035
4512
855
2007
2247
counts
1004
753
502
251
0
12519
9389
6260
3130
0
Signature
SBS26
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS44 8 cosine sim = 0.98059
1959
935
4365
832
1993
2437
counts
980
735
490
245
0
12519
9389
6260
3130
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS17b,SBS18,SBS26,SBS40,SBS44 7 cosine sim = 0.98057
1960
935
4365
832
1993
2437
counts
980
735
490
245
0
12519
9389
6260
3130
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS2
SBS1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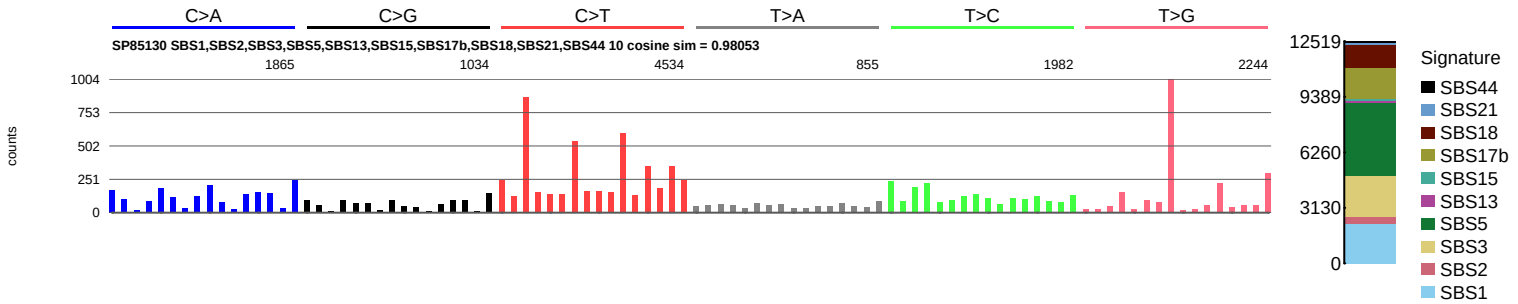
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS15,SBS17b,SBS18,SBS26,SBS40 7 cosine sim = 0.98053
1945
964
4326
827
2000
2458
counts
976
732
488
244
0
12519
9389
6260
3130
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS15
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS44 10 cosine sim = 0.98053
1865
1034
4534
855
1982
2244
counts
1004
753
502
251
0
12519
9389
6260
3130
0
Signature
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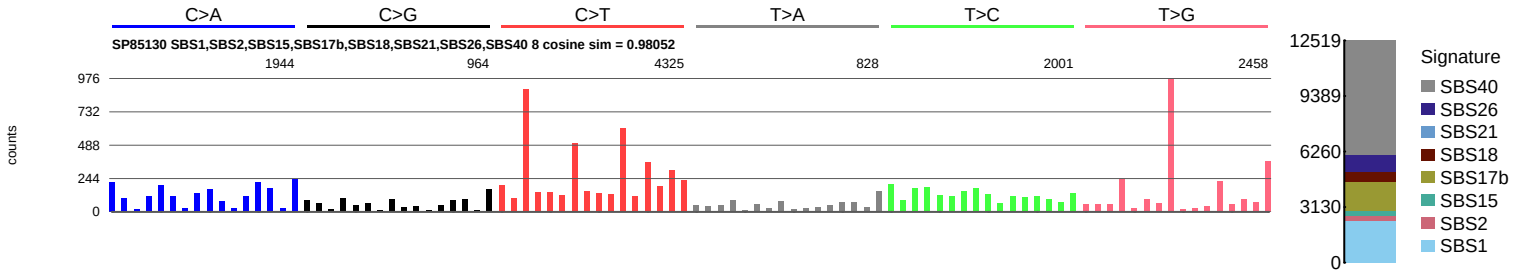
SBS2
SBS1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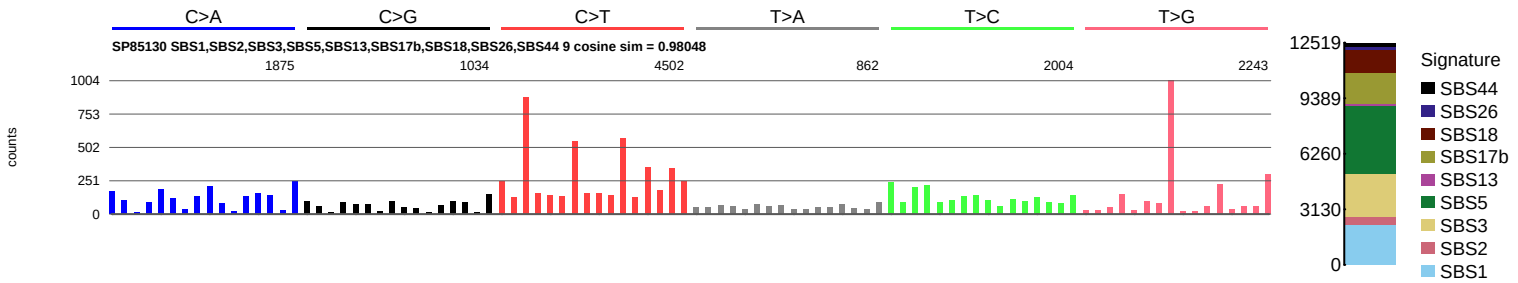
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS15,SBS17b,SBS18,SBS21,SBS26,SBS40 8 cosine sim = 0.98052
1944
964
4325
828
2001
2458
counts
976
732
488
244
0
12519
9389
6260
3130
0
Signature
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS18,SBS26,SBS44 9 cosine sim = 0.98048
1875
1034
4502
862
2004
2243
counts
1004
753
502
251
0
12519
9389
6260
3130
0
Signature
SBS44
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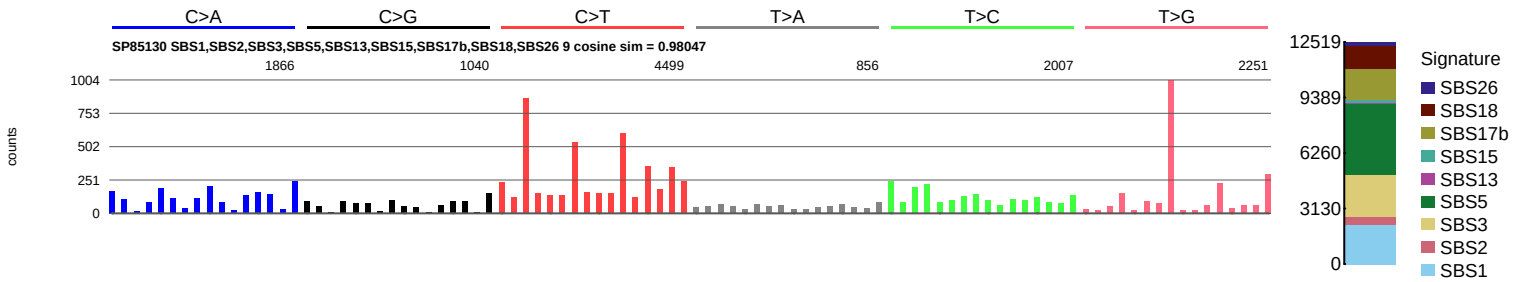
SBS2
SBS1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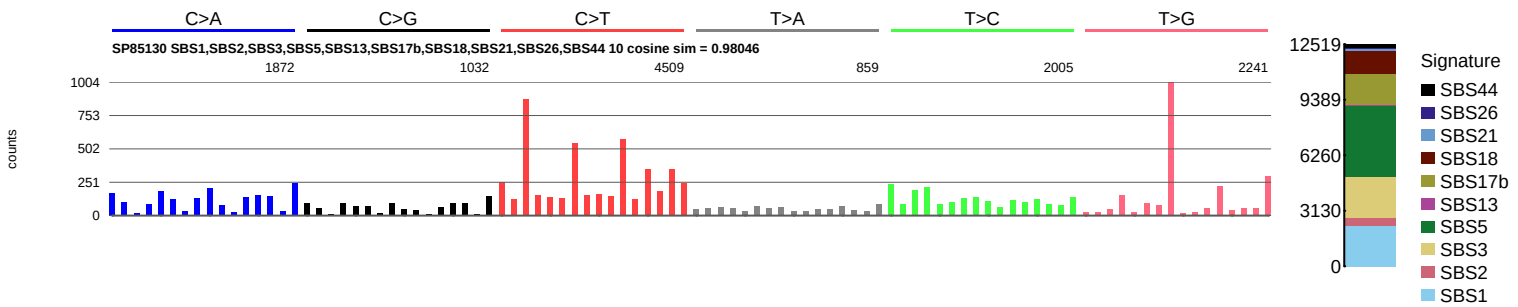
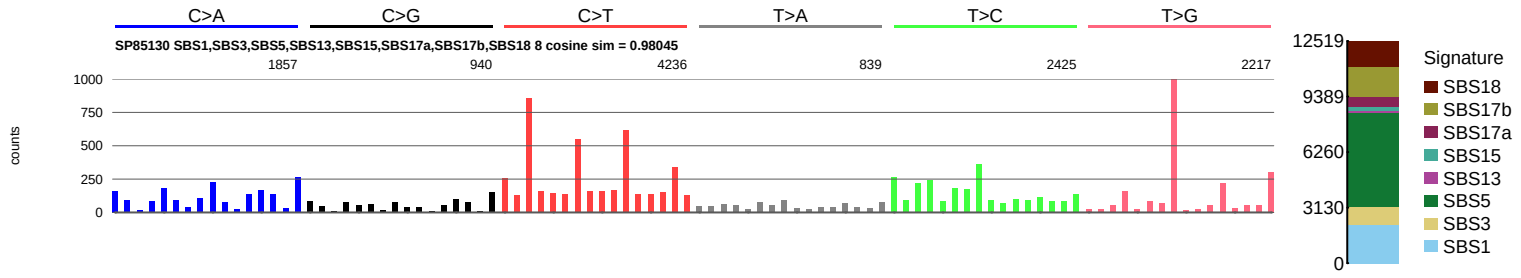
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS26 9 cosine sim = 0.98047
1866
1040
4499
856
2007
2251
counts
1004
753
502
251
0
12519
9389
6260
3130
0
Signature
SBS26
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS18,SBS21,SBS26,SBS44 10 cosine sim = 0.98046
1872
1032
4509
859
2005
2241
counts
1004
753
502
251
0
12519
9389
6260
3130
0
Signature
SBS44
SBS26
SBS21
SBS18
SBS17b
SBS13
SBS5
SBS3
SBS2
SBS1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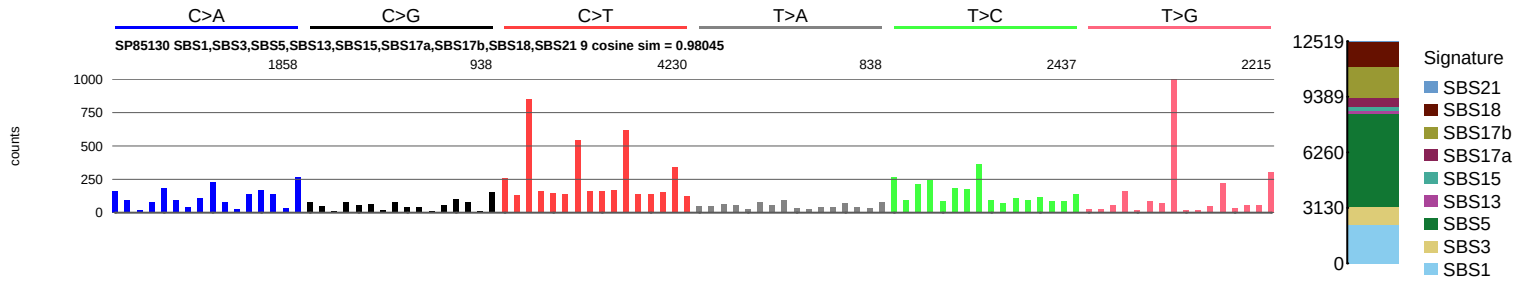
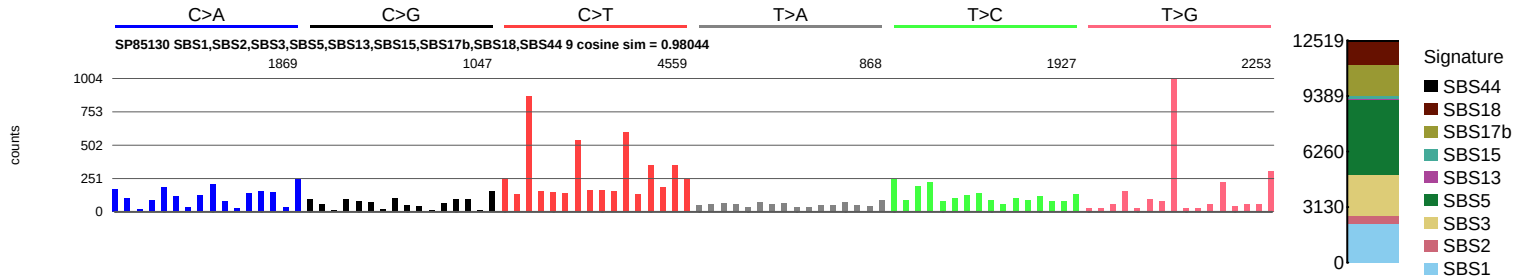
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18 8 cosine sim = 0.98045
1857
940
4236
839
2425
2217
counts
1000
750
500
250
0
12519
9389
6260
3130
0
Signature
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21 9 cosine sim = 0.98045
1858
938
4230
838
2437
2215
counts
1000
750
500
250
0
12519
9389
6260
3130
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS44 9 cosine sim = 0.98044
1869
1047
4559
868
1927
2253
counts
1004
753
502
251
0
12519
9389
6260
3130
0
Signature
SBS44
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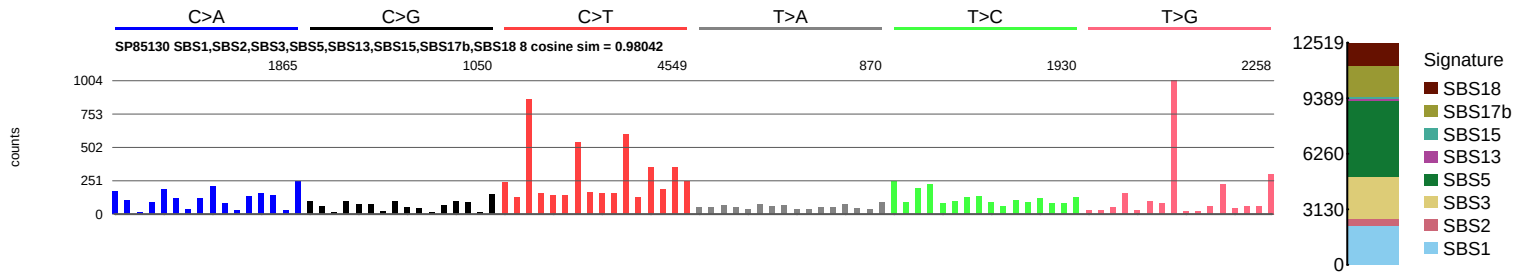
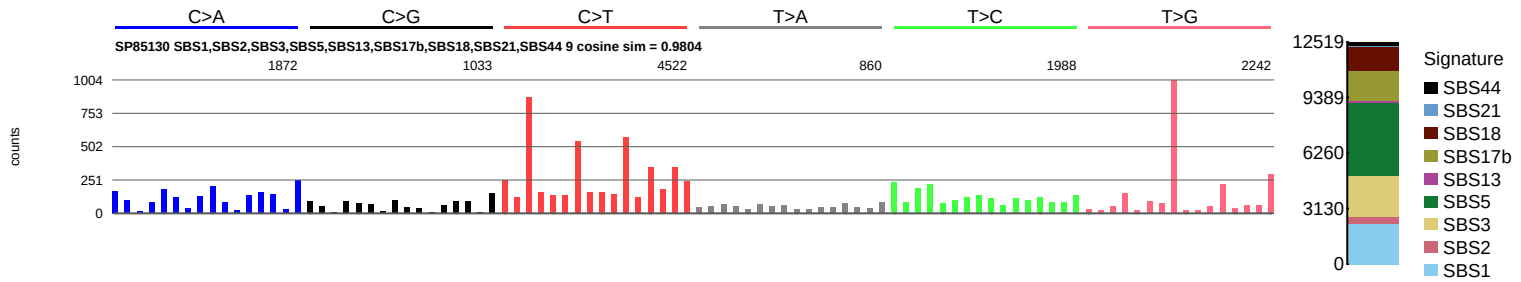
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS18 8 cosine sim = 0.98042
1865
1050
4549
870
1930
2258
counts
1004
753
502
251
0
12519
9389
6260
3130
0
Signature
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1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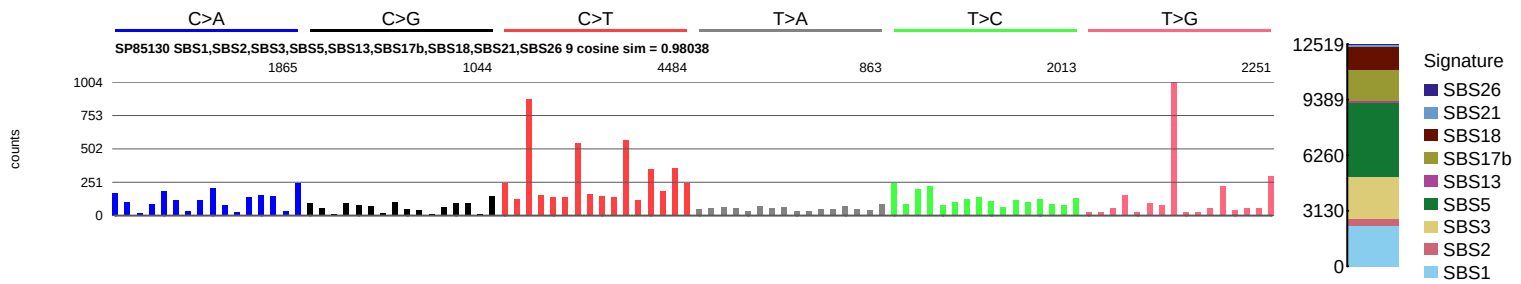
T>G
SP85130 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS18,SBS21,SBS44 9 cosine sim = 0.9804
1872
1033
4522
860
1988
2242
counts
1004
753
502
251
0
12519
9389
6260
3130
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS13
SBS5
SBS3
SBS2
SBS1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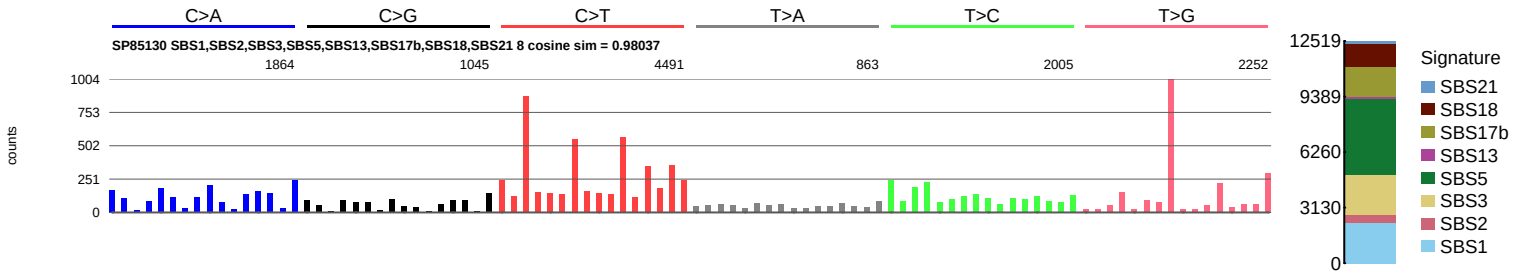
T>G
SP85130 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS18,SBS21,SBS26 9 cosine sim = 0.98038
1865
1044
4484
863
2013
2251
counts
1004
753
502
251
0
12519
9389
6260
3130
0
Signature
SBS26
SBS21
SBS18
SBS17b
SBS13
SBS5
SBS3
SBS2
SBS1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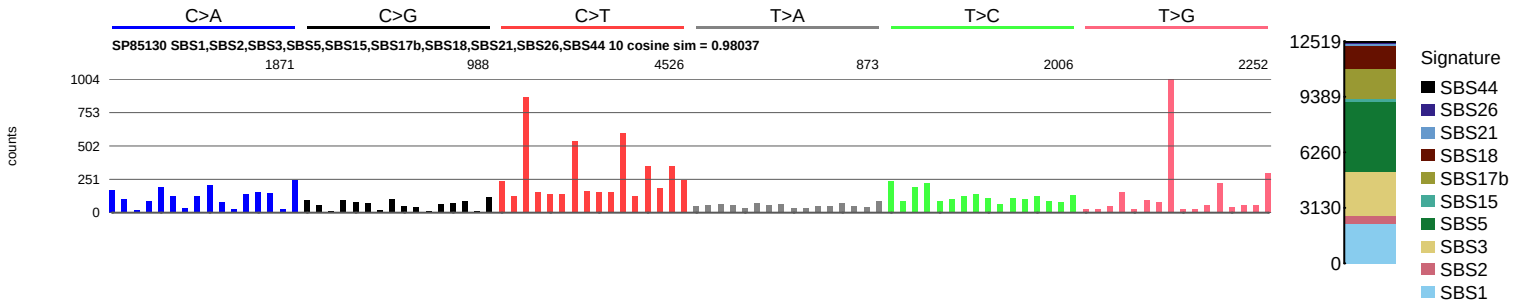
SP85130 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS18,SBS21 8 cosine sim = 0.98037
1864
1045
4491
863
2005
2252
counts
1004
753
502
251
0
12519
9389
6260
3130
0
Signature
SBS21
SBS18
SBS17b
SBS13
SBS5
SBS3
SBS2
SBS1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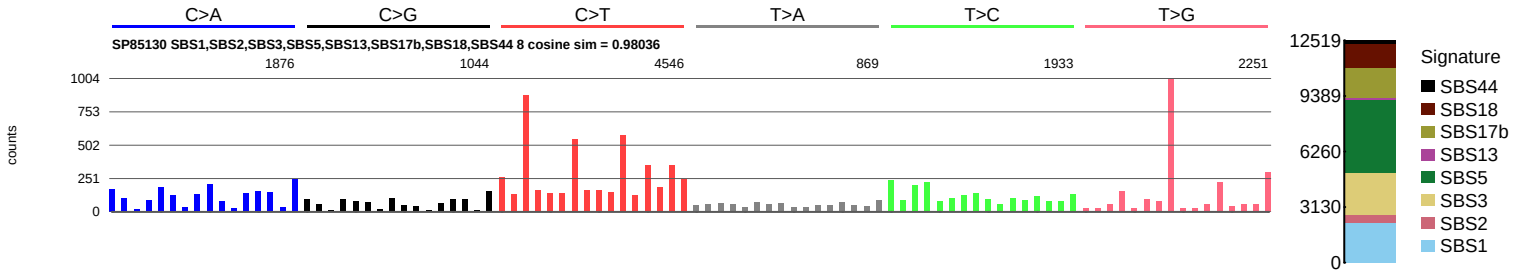
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17b,SBS18,SBS21,SBS26,SBS44 10 cosine sim = 0.98037
1871
988
4526
873
2006
2252
counts
1004
753
502
251
0
12519
9389
6260
3130
0
Signature
SBS44
SBS26
SBS21
SBS18
SBS17b
SBS15
SBS5
SBS3
SBS2
SBS1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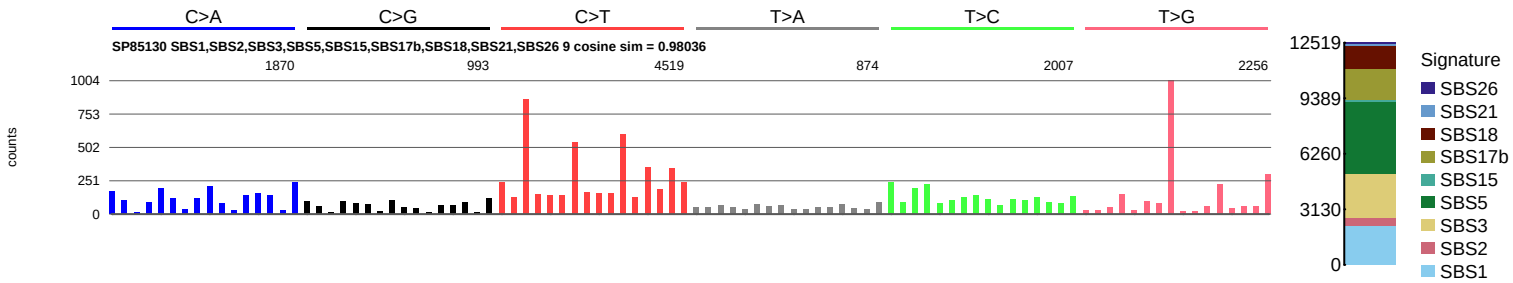
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS18,SBS44 8 cosine sim = 0.98036
1876
1044
4546
869
1933
2251
counts
1004
753
502
251
0
12519
9389
6260
3130
0
Signature
SBS44
SBS18
SBS17b
SBS13
SBS5
SBS3
SBS2
SBS1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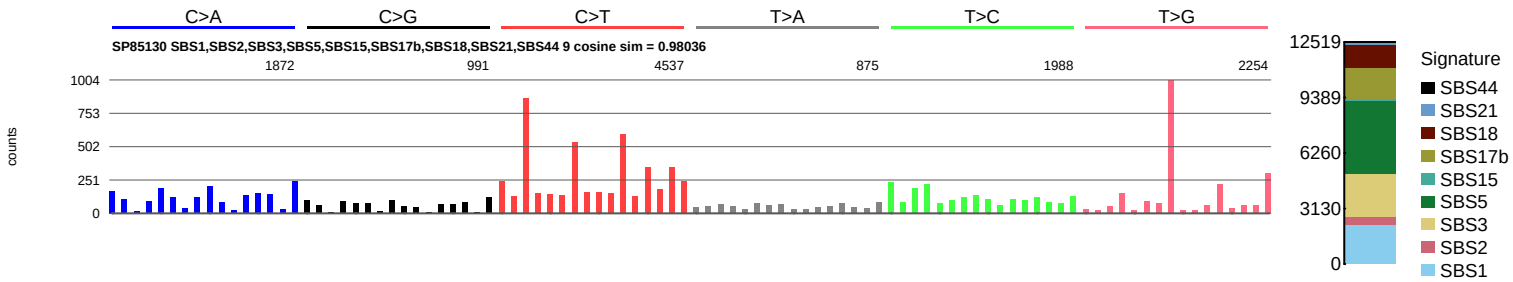
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17b,SBS18,SBS21,SBS26 9 cosine sim = 0.98036
1870
993
4519
874
2007
2256
counts
1004
753
502
251
0
12519
9389
6260
3130
0
Signature
SBS26
SBS21
SBS18
SBS17b
SBS15
SBS5
SBS3
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17b,SBS18,SBS21,SBS44 9 cosine sim = 0.98036
1872
991
4537
875
1988
2254
counts
1004
753
502
251
0
12519
9389
6260
3130
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS15
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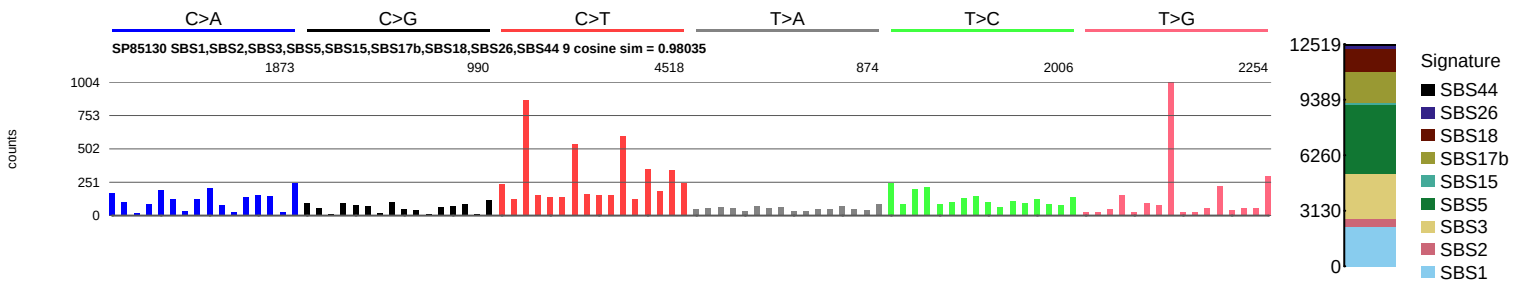
SBS3
SBS2
SBS1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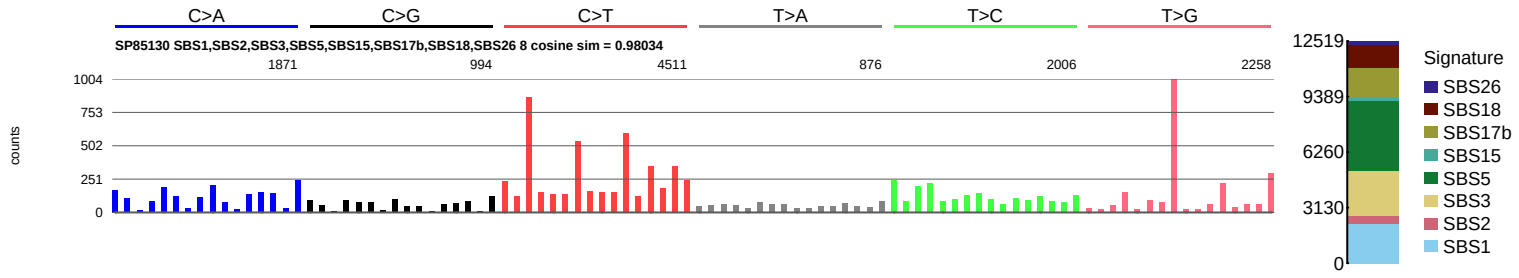
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17b,SBS18,SBS26,SBS44 9 cosine sim = 0.98035
1873
990
4518
874
2006
2254
counts
1004
753
502
251
0
12519
9389
6260
3130
0
Signature
SBS44
SBS26
SBS18
SBS17b
SBS15
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17b,SBS18,SBS26 8 cosine sim = 0.98034
1871
994
4511
876
2006
2258
counts
1004
753
502
251
0
12519
9389
6260
3130
0
Signature
SBS26
SBS18
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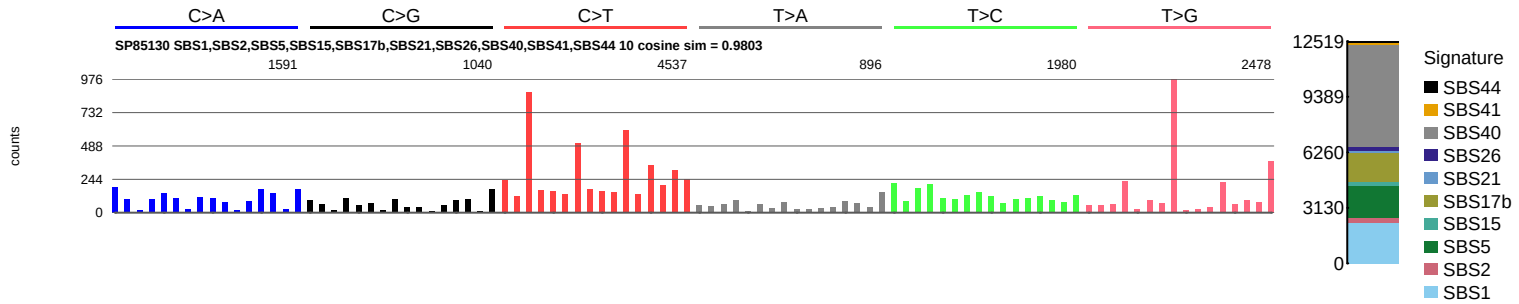
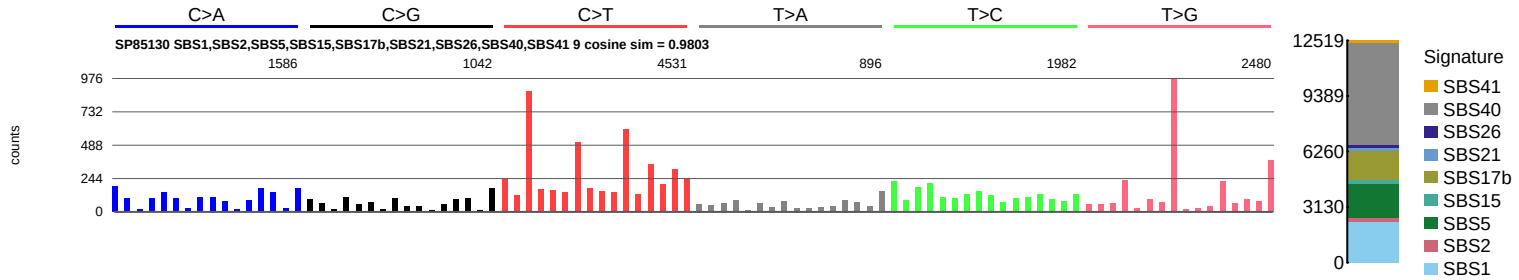
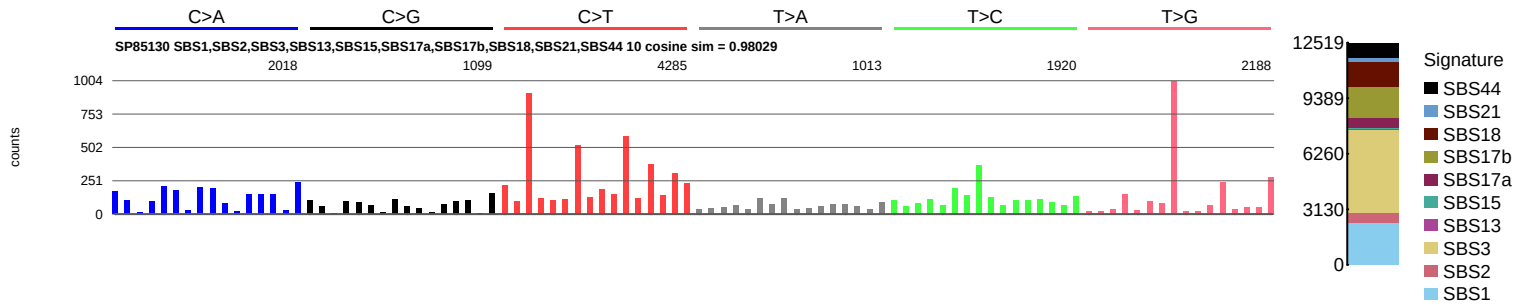
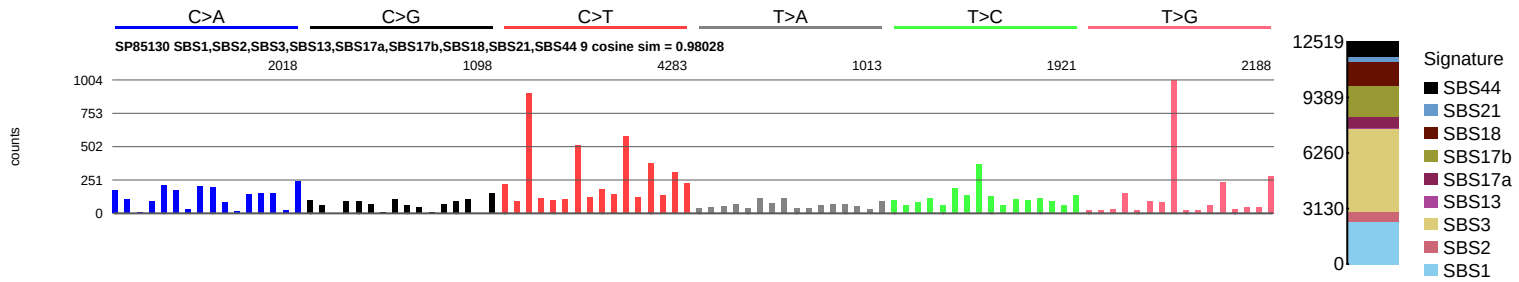
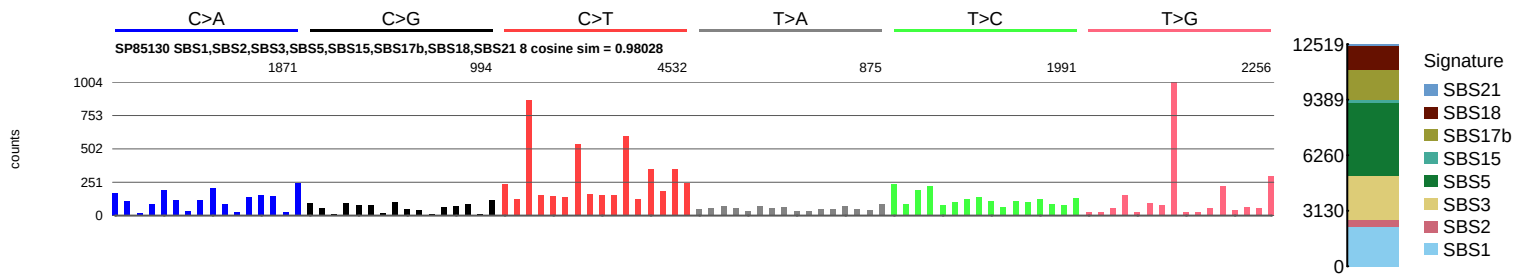
SBS2
SBS1

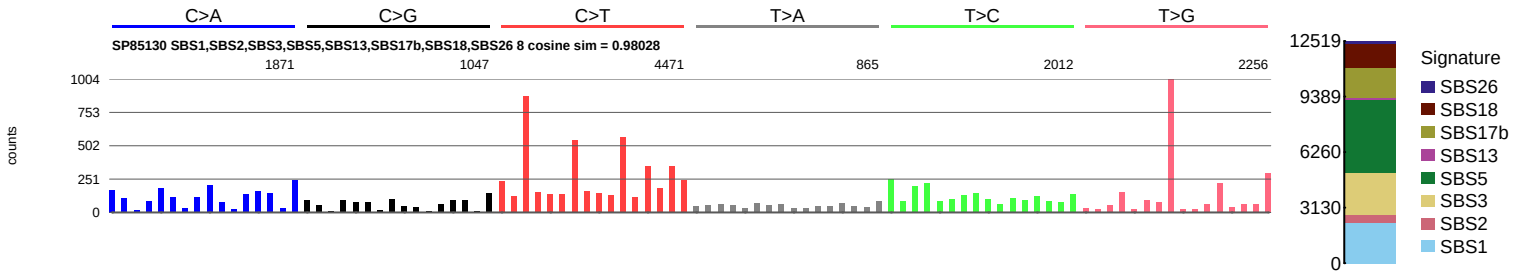

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS5,SBS15,SBS17b,SBS21,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.9803
1591
1040
4537
896
1980
2478
counts
976
732
488
244
0
12519
9389
6260
3130
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS17b
SBS15
SBS5
SBS2
SBS1

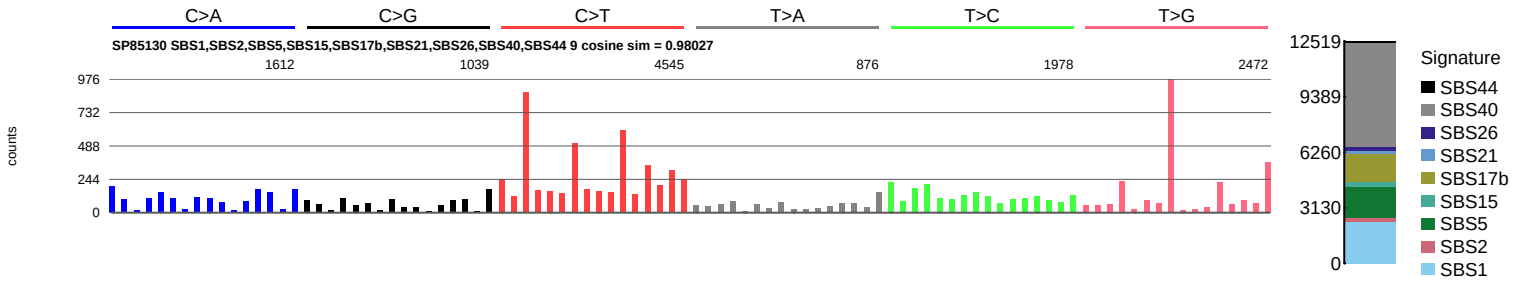

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS5,SBS15,SBS17b,SBS21,SBS26,SBS40,SBS41 9 cosine sim = 0.9803
1586
1042
4531
896
1982
2480
counts
976
732
488
244
0
12519
9389
6260
3130
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS17b
SBS15
SBS5
SBS2
SBS1

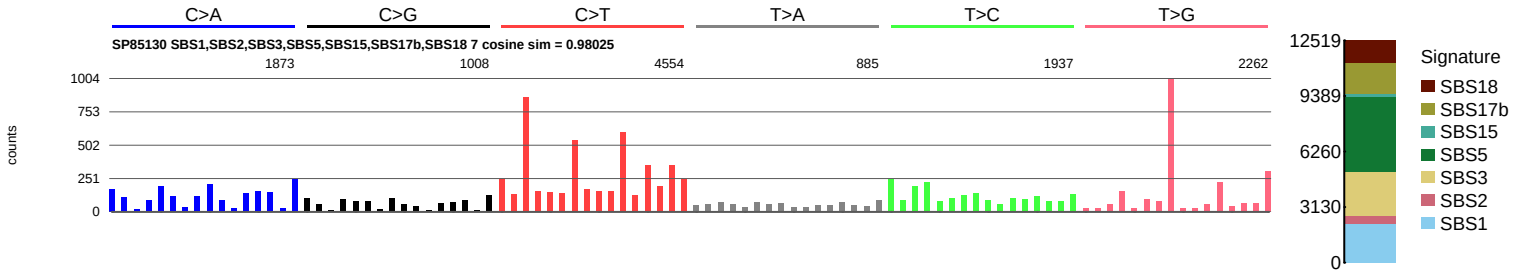

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS44 10 cosine sim = 0.98029
2018
1099
4285
1013
1920
2188
counts
1004
753
502
251
0
12519
9389
6260
3130
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS44 9 cosine sim = 0.98028
2018
1098
4283
1013
1921
2188
counts
1004
753
502
251
0
12519
9389
6260
3130
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17b,SBS18,SBS21 8 cosine sim = 0.98028
1871
994
4532
875
1991
2256
counts
1004
753
502
251
0
12519
9389
6260
3130
0
Signature
SBS21
SBS18
SBS17b
SBS15
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS18,SBS26 8 cosine sim = 0.98028
1871
1047
4471
865
2012
2256
counts
1004
753
502
251
0
12519
9389
6260
3130
0
Signature
SBS26
SBS18
SBS17b
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS5,SBS15,SBS17b,SBS21,SBS26,SBS40,SBS44 9 cosine sim = 0.98027
1612
1039
4545
876
1978
2472
counts
976
732
488
244
0
12519
9389
6260
3130
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS17b
SBS15
SBS5
SBS2
SBS1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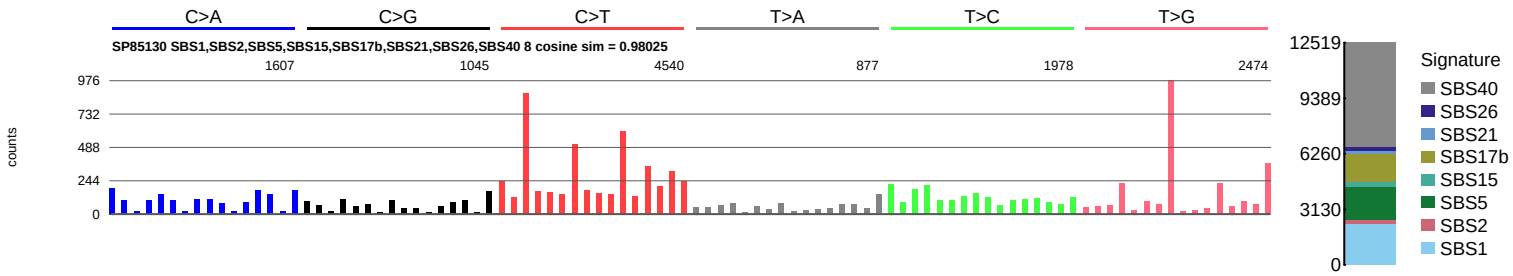
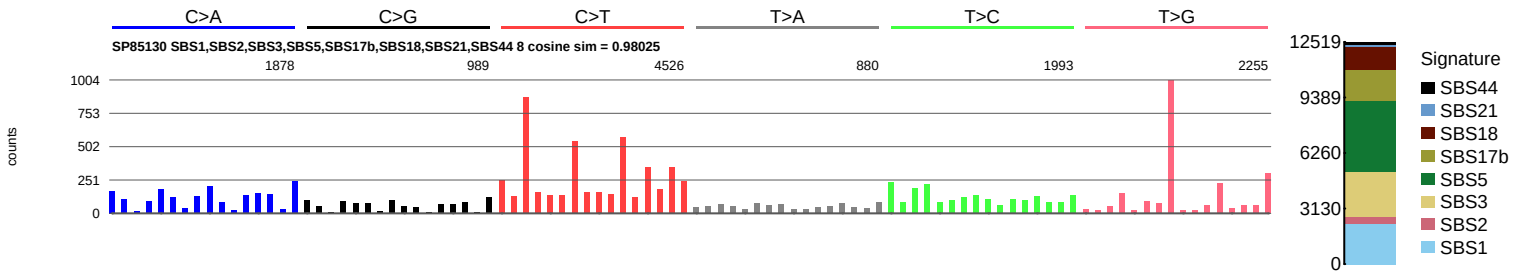
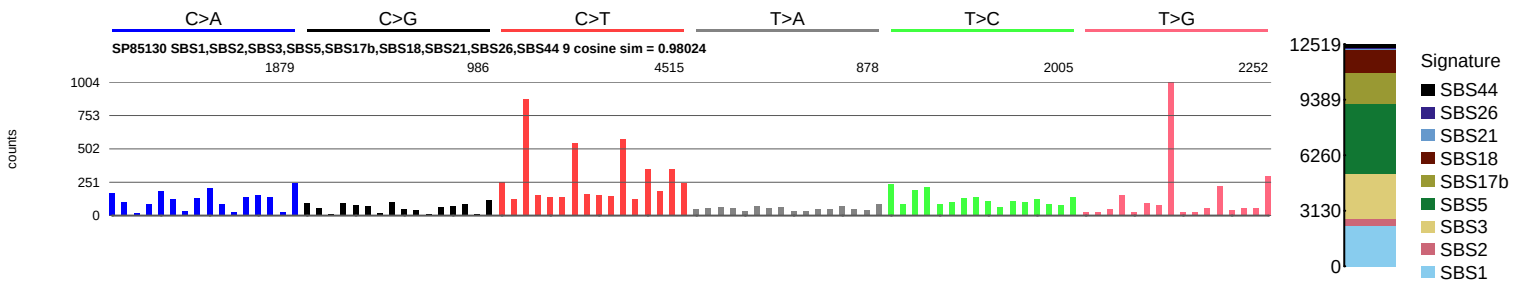
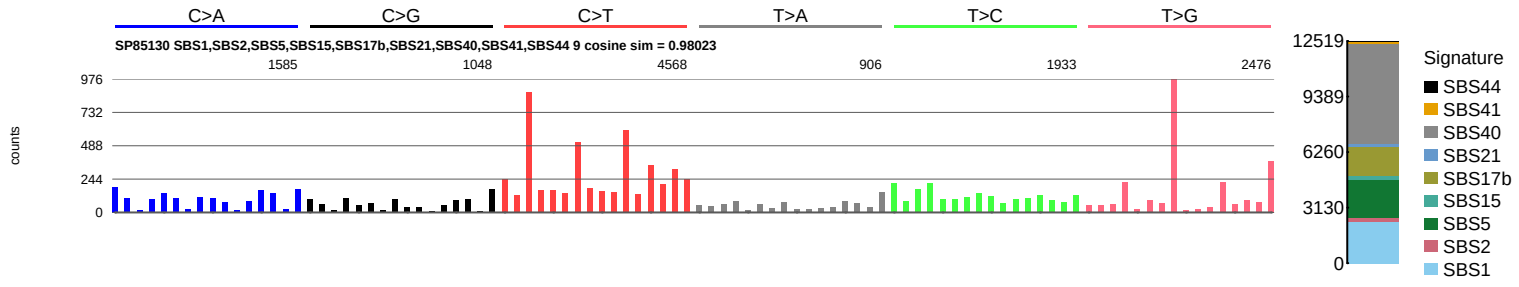
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17b,SBS18 7 cosine sim = 0.98025
1873
1008
4554
885
1937
2262
counts
1004
753
502
251
0
12519
9389
6260
3130
0
Signature
SBS18
SBS17b
SBS15
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS5,SBS15,SBS17b,SBS21,SBS26,SBS40 8 cosine sim = 0.98025
1607
1045
4540
877
1978
2474
counts
976
732
488
244
0
12519
9389
6260
3130
0
Signature
SBS40
SBS26
SBS21
SBS17b
SBS15
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS5,SBS17b,SBS18,SBS21,SBS44 8 cosine sim = 0.98025
1878
989
4526
880
1993
2255
counts
1004
753
502
251
0
12519
9389
6260
3130
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS5,SBS17b,SBS18,SBS21,SBS26,SBS44 9 cosine sim = 0.98024
1879
986
4515
878
2005
2252
counts
1004
753
502
251
0
12519
9389
6260
3130
0
Signature
SBS44
SBS26
SBS21
SBS18
SBS17b
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS5,SBS15,SBS17b,SBS21,SBS40,SBS41,SBS44 9 cosine sim = 0.98023
1585
1048
4568
906
1933
2476
counts
976
732
488
244
0
12519
9389
6260
3130
0
Signature
SBS44
SBS41
SBS40
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SBS17b
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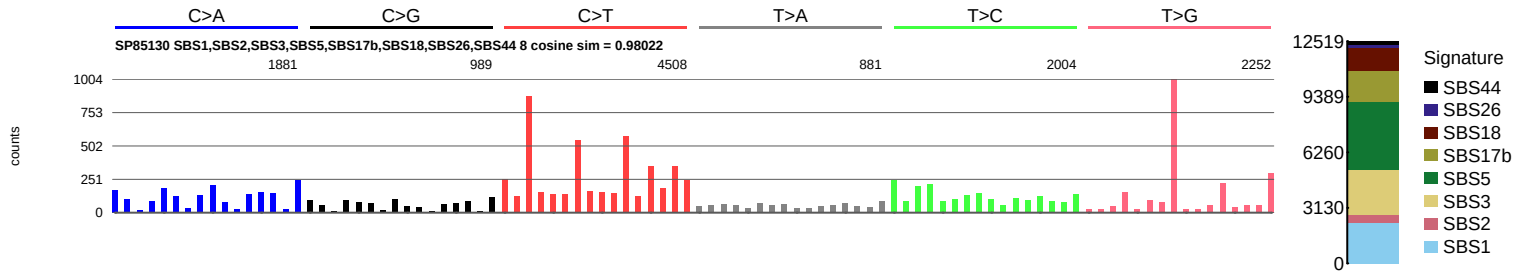
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS5,SBS17b,SBS18,SBS26,SBS44 8 cosine sim = 0.98022
1881
989
4508
881
2004
2252
counts
1004
753
502
251
0
12519
9389
6260
3130
0
Signature
SBS44
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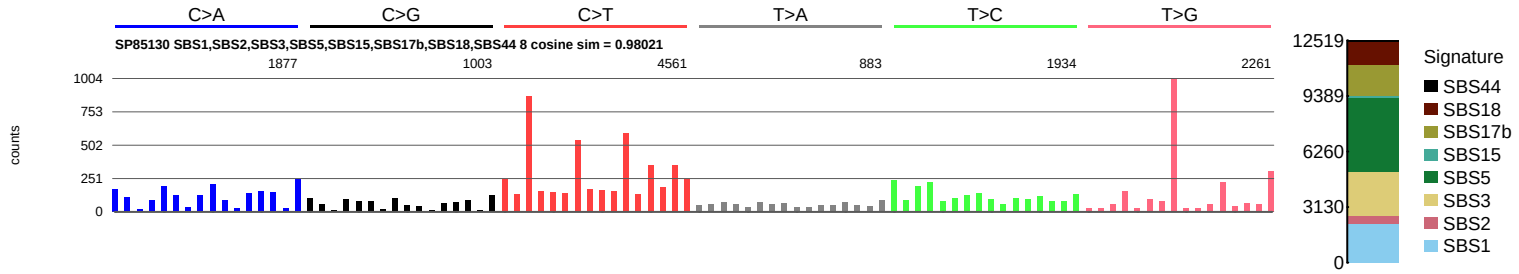
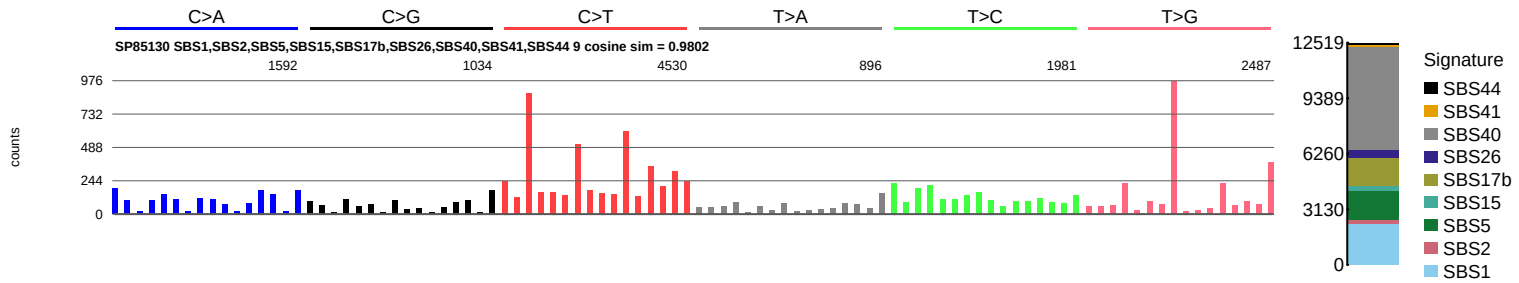
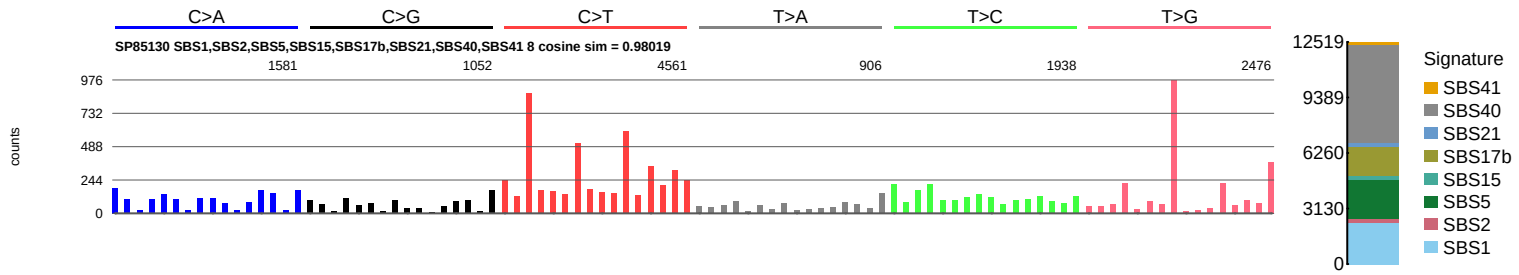
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17b,SBS18,SBS44 8 cosine sim = 0.98021
1877
1003
4561
883
1934
2261
counts
1004
753
502
251
0
12519
9389
6260
3130
0
Signature
SBS44
SBS18
SBS17b
SBS15
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS5,SBS15,SBS17b,SBS26,SBS40,SBS41,SBS44 9 cosine sim = 0.9802
1592
1034
4530
896
1981
2487
counts
976
732
488
244
0
12519
9389
6260
3130
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS17b
SBS15
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS5,SBS15,SBS17b,SBS21,SBS40,SBS41 8 cosine sim = 0.98019
1581
1052
4561
906
1938
2476
counts
976
732
488
244
0
12519
9389
6260
3130
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SBS41
SBS40
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SBS17b
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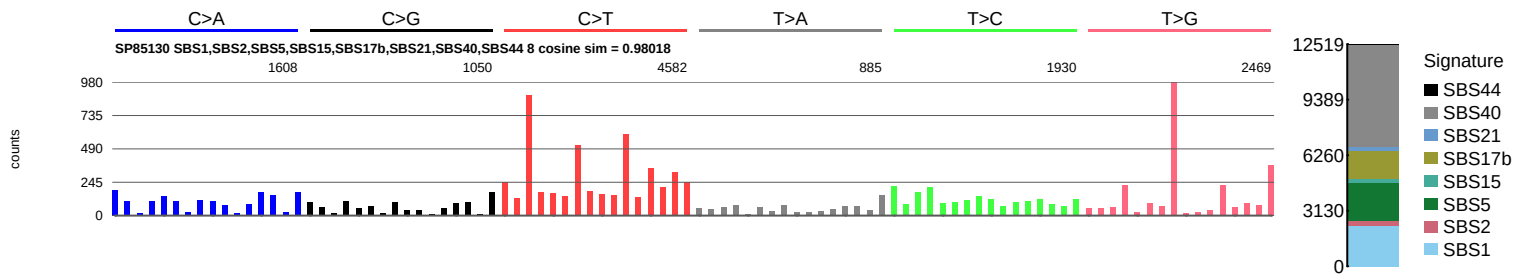
SBS5
SBS2
SBS1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T>G
SP85130 SBS1,SBS2,SBS5,SBS15,SBS17b,SBS21,SBS40,SBS44 8 cosine sim = 0.98018
1608
1050
4582
885
1930
2469
counts
980
735
490
245
0
12519
9389
6260
3130
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Signature
SBS44
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SBS17b
SBS15
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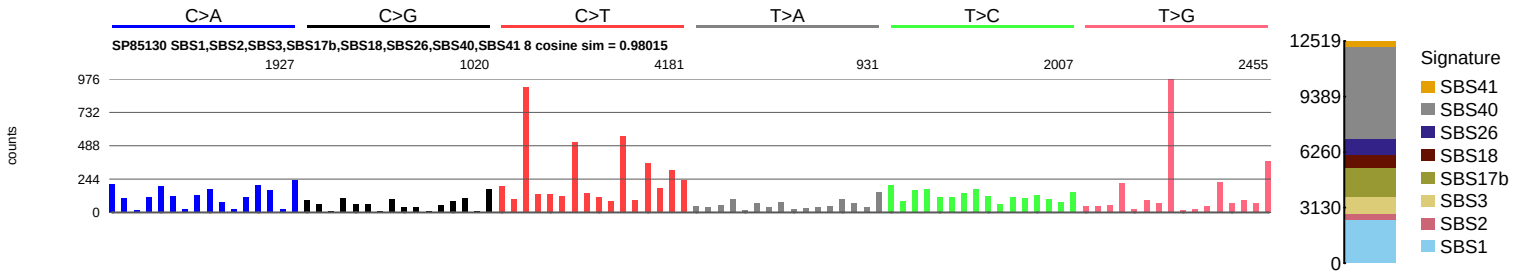
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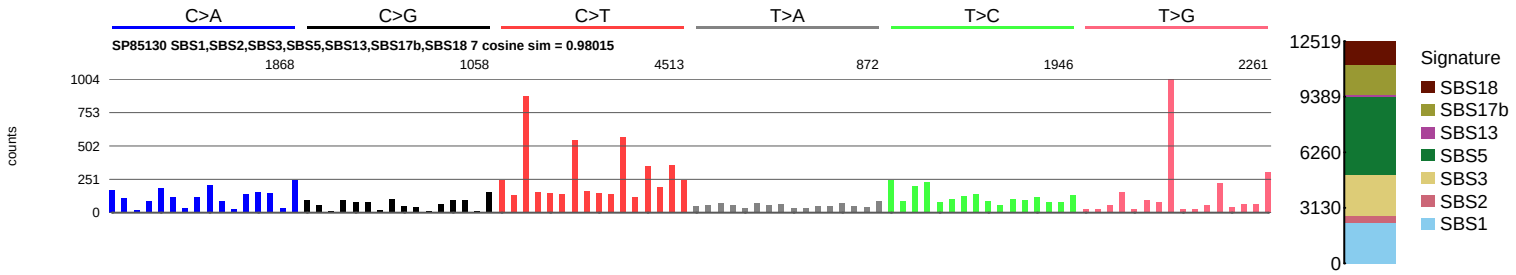
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS17b,SBS18,SBS26,SBS40,SBS41 8 cosine sim = 0.98015
1927
1020
4181
931
2007
2455
counts
976
732
488
244
0
12519
9389
6260
3130
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS3
SBS2
SBS1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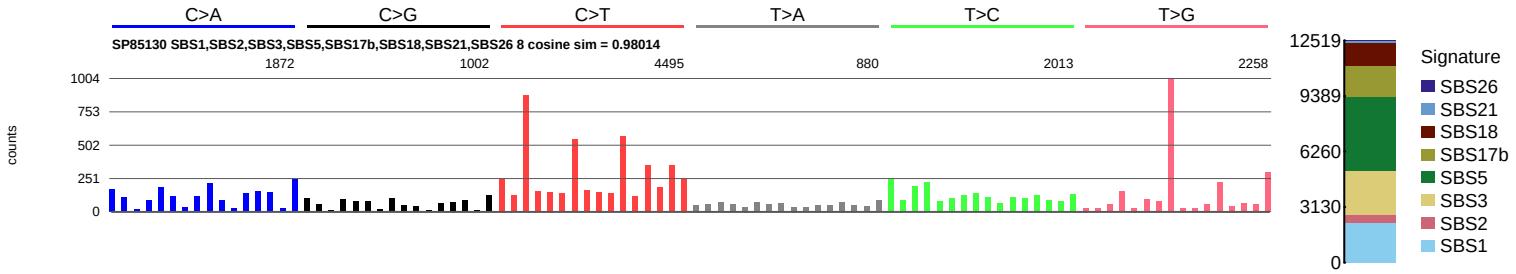
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS18 7 cosine sim = 0.98015
1868
1058
4513
872
1946
2261
counts
1004
753
502
251
0
12519
9389
6260
3130
0
Signature
SBS18
SBS17b
SBS13
SBS5
SBS3
SBS2
SBS1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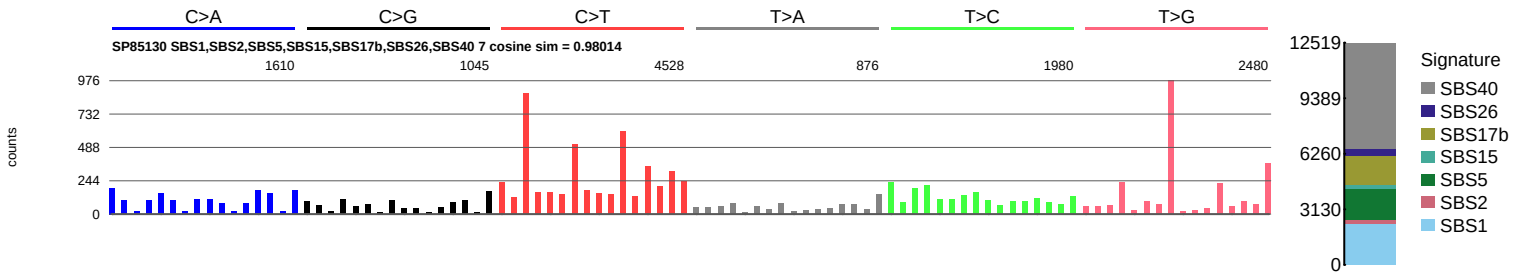
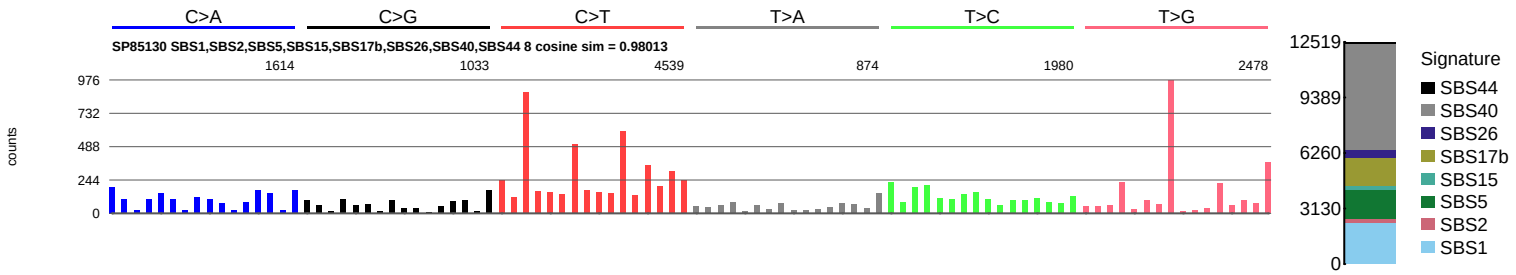
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS5,SBS17b,SBS18,SBS21,SBS26 8 cosine sim = 0.98014
1872
1002
4495
880
2013
2258
counts
1004
753
502
251
0
12519
9389
6260
3130
0
Signature
SBS26
SBS21
SBS18
SBS17b
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS5,SBS15,SBS17b,SBS26,SBS40 7 cosine sim = 0.98014
1610
1045
4528
876
1980
2480
counts
976
732
488
244
0
12519
9389
6260
3130
0
Signature
SBS40
SBS26
SBS17b
SBS15
SBS5
SBS2
SBS1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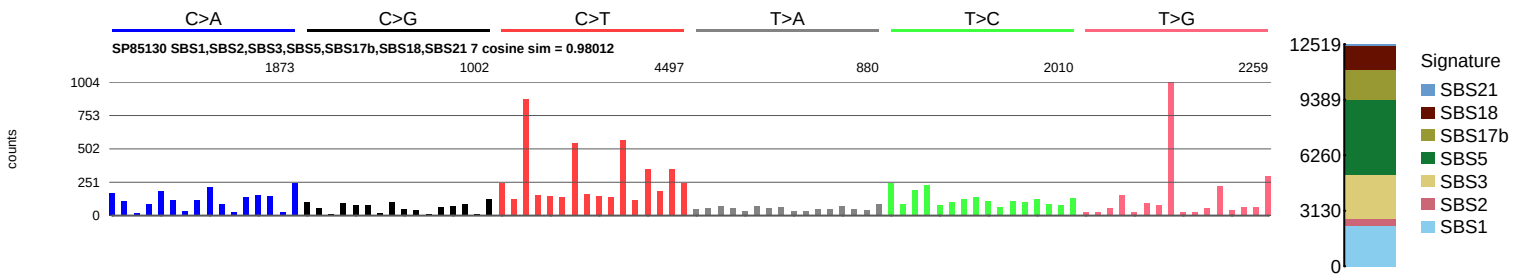
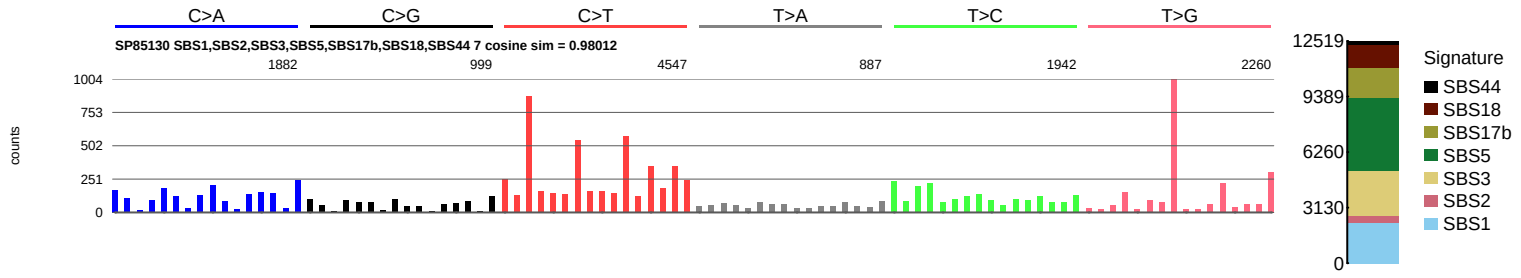
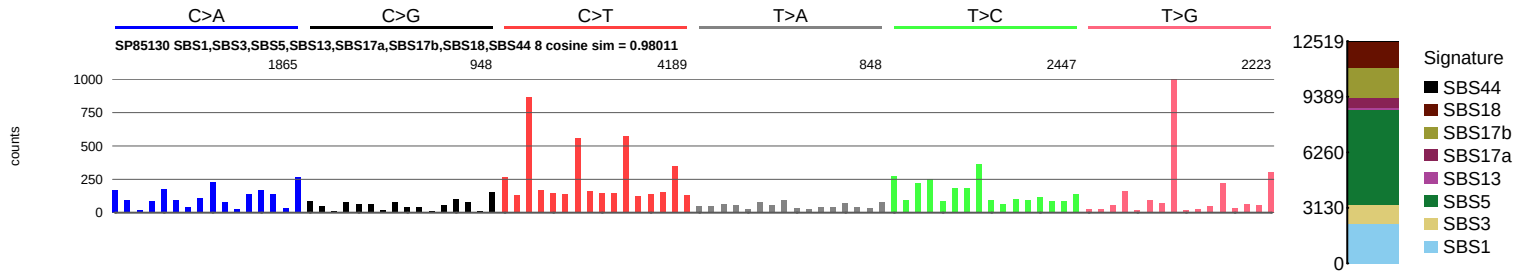
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS5,SBS15,SBS17b,SBS26,SBS40,SBS44 8 cosine sim = 0.98013
1614
1033
4539
874
1980
2478
counts
976
732
488
244
0
12519
9389
6260
3130
0
Signature
SBS44
SBS40
SBS26
SBS17b
SBS15
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS5,SBS17b,SBS18,SBS21 7 cosine sim = 0.98012
1873
1002
4497
880
2010
2259
counts
1004
753
502
251
0
12519
9389
6260
3130
0
Signature
SBS21
SBS18
SBS17b
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS5,SBS17b,SBS18,SBS44 7 cosine sim = 0.98012
1882
999
4547
887
1942
2260
counts
1004
753
502
251
0
12519
9389
6260
3130
0
Signature
SBS44
SBS18
SBS17b
SBS5
SBS3
SBS2
SBS1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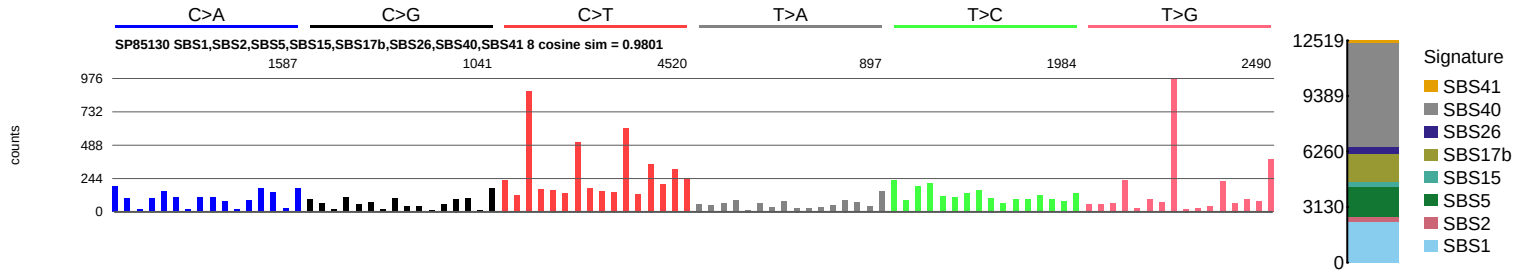
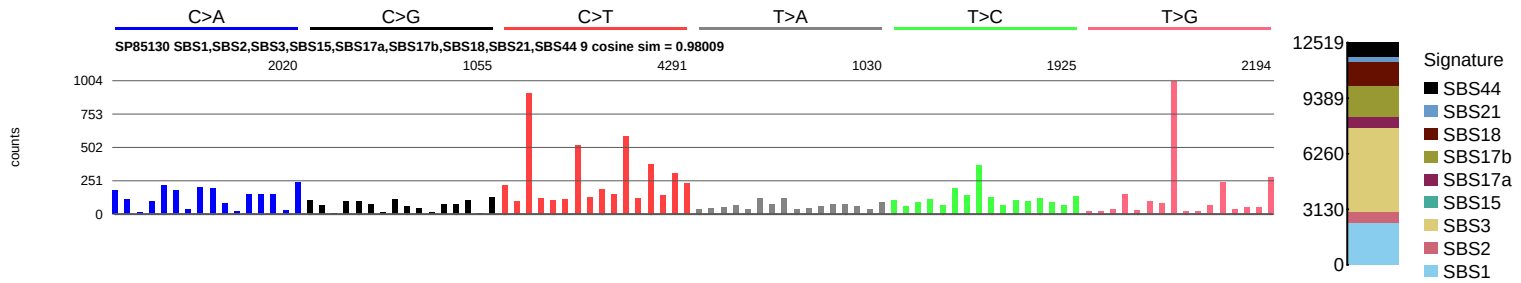
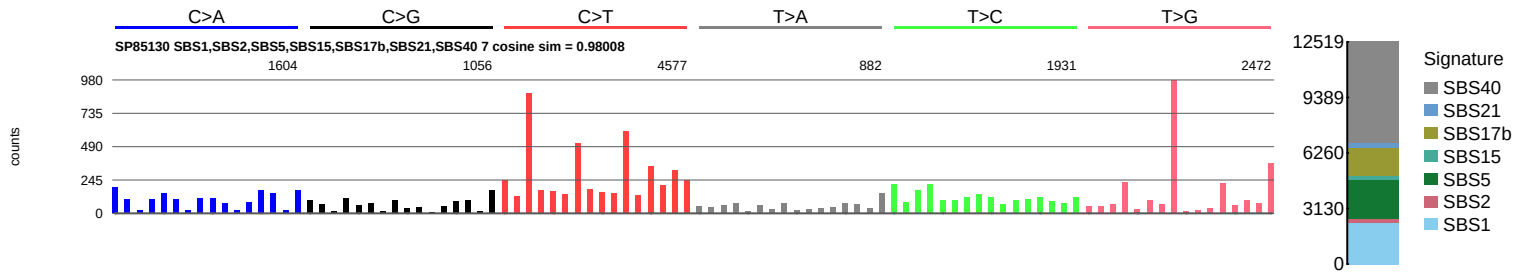
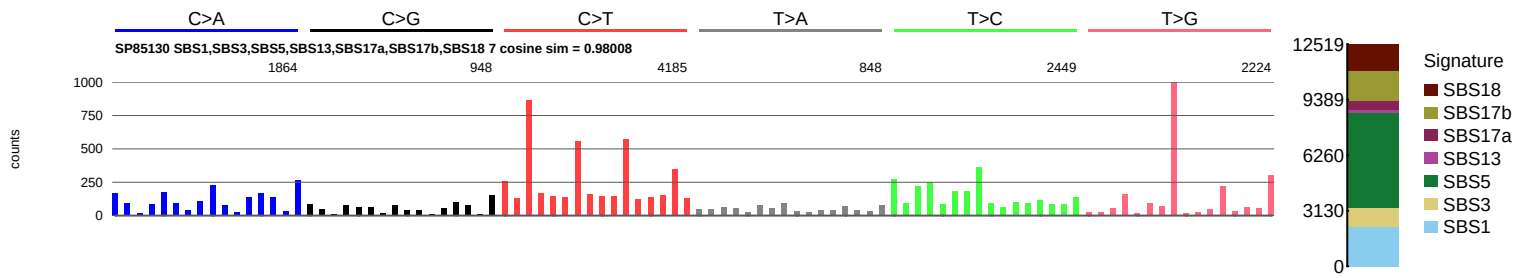
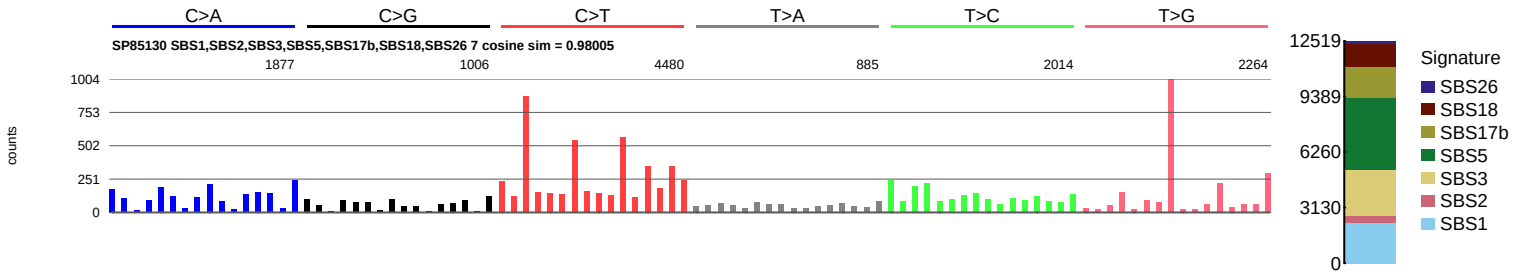
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS44 8 cosine sim = 0.98011
1865
948
4189
848
2447
2223
counts
1000
750
500
250
0
12519
9389
6260
3130
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS1

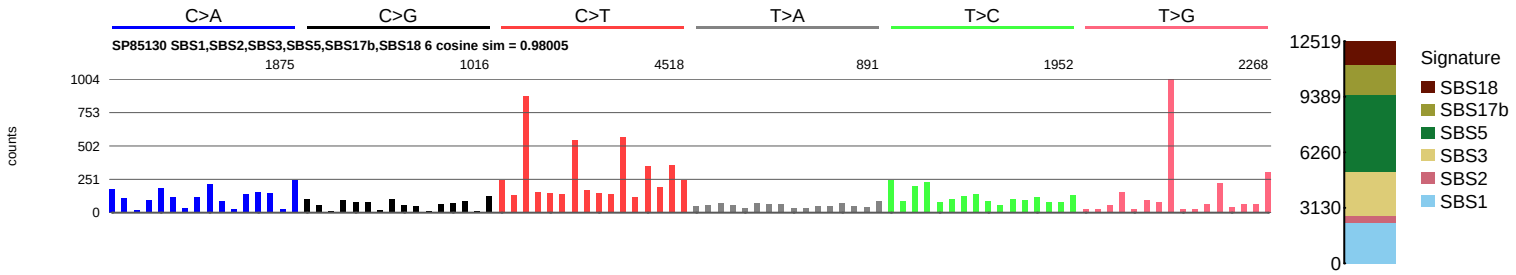

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS5,SBS15,SBS17b,SBS26,SBS40,SBS41 8 cosine sim = 0.9801
1587
1041
4520
897
1984
2490
counts
976
732
488
244
0
12519
9389
6260
3130
0
Signature
SBS41
SBS40
SBS26
SBS17b
SBS15
SBS5
SBS2
SBS1

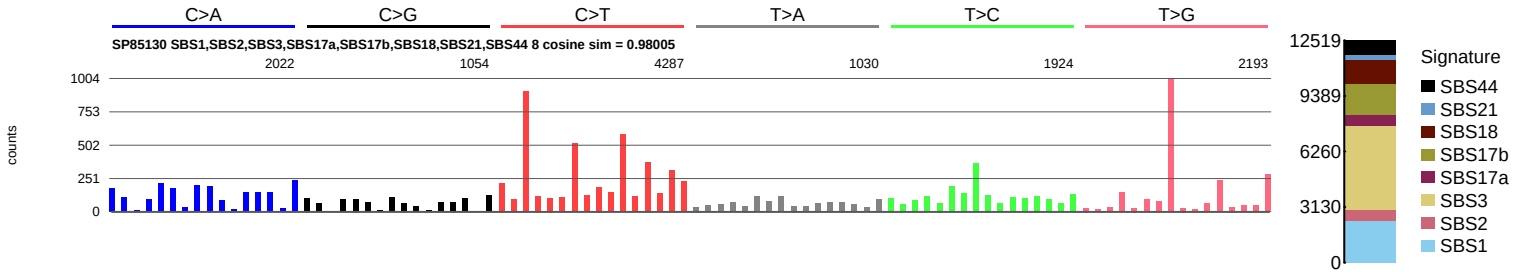

C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS44 9 cosine sim = 0.98009
2020
1055
4291
1030
1925
2194
counts
1004
753
502
251
0
12519
9389
6260
3130
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS5,SBS15,SBS17b,SBS21,SBS40 7 cosine sim = 0.98008
1604
1056
4577
882
1931
2472
counts
980
735
490
245
0
12519
9389
6260
3130
0
Signature
SBS40
SBS21
SBS17b
SBS15
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18 7 cosine sim = 0.98008
1864
948
4185
848
2449
2224
counts
1000
750
500
250
0
12519
9389
6260
3130
0
Signature
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS5,SBS17b,SBS18,SBS26 7 cosine sim = 0.98005
1877
1006
4480
885
2014
2264
counts
1004
753
502
251
0
12519
9389
6260
3130
0
Signature
SBS26
SBS18
SBS17b
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS5,SBS17b,SBS18 6 cosine sim = 0.98005
1875
1016
4518
891
1952
2268
counts
1004
753
502
251
0
12519
9389
6260
3130
0
Signature
SBS18
SBS17b
SBS5
SBS3
SBS2
SBS1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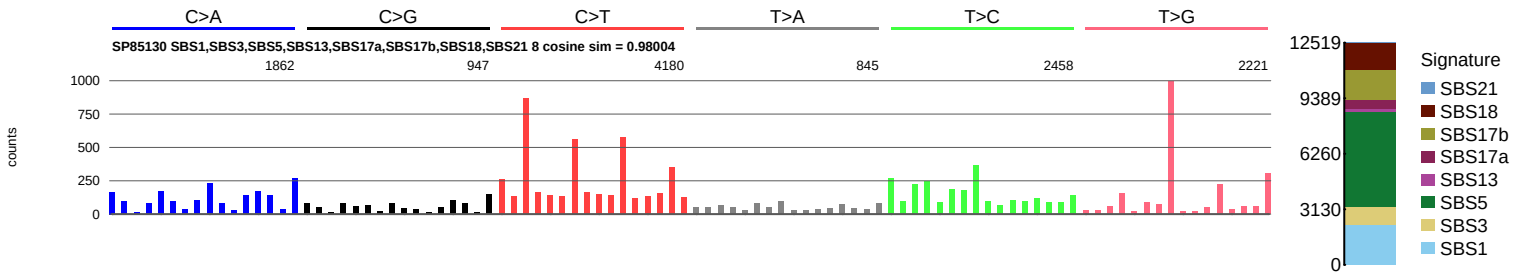
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS44 8 cosine sim = 0.98005
2022
1054
4287
1030
1924
2193
counts
1004
753
502
251
0
12519
9389
6260
3130
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1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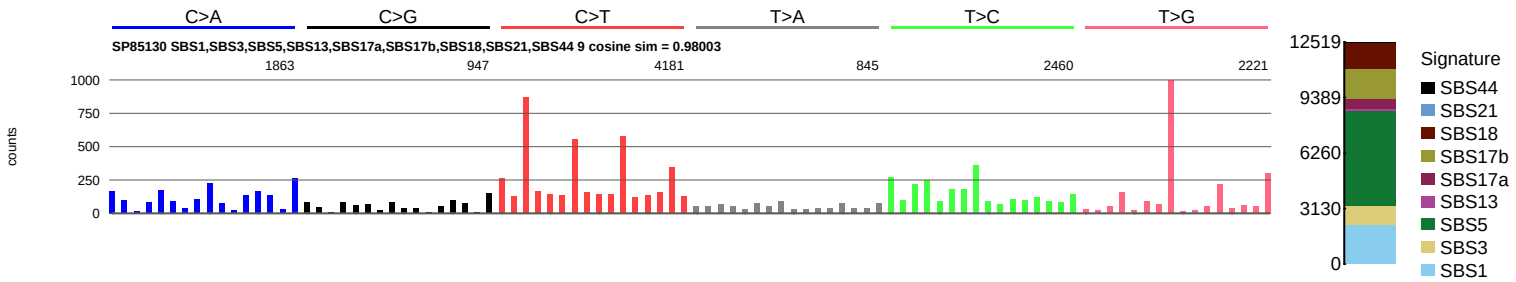
C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21 8 cosine sim = 0.98004
1862
947
4180
845
2458
2221
counts
1000
750
500
250
0
12519
9389
6260
3130
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS1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C>G
C>T
T>A
T>C
T>G
SP85130 SBS1,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS44 9 cosine sim = 0.98003
1863
947
4181
845
2460
2221
counts
1000
750
500
250
0
12519
9389
6260
3130
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS1